

Roar Elias Georgsen

Building Smarter Environments for Small Teams

A Model-Based Approach to Dependable Smart
City Infrastructure in Very Small Entities

**Dissertation for the
degree of PhD**
Applied Micro-and
Nanosystems

To the evaluation committee
2026

For teachers everywhere, who see what is not yet there.

And in memory of two who did:

Tore Hagen and **Michael Samuda**

Preface

The intellectual origins of this thesis can be traced, with some precision, to the library of Skagerak International School in Sandefjord, Norway, during the autumn term of 1999. This is either a testament to the value of patient curiosity or a cautionary tale about what happens when a teenager picks up the wrong book. The reader may judge for themselves by the final chapter.

Most doctoral candidates arrive at university as a natural continuation of family tradition. Their childhood homes contained bookshelves that groaned under the weight of accumulated knowledge, and dinner table conversations touched on epistemology between courses. My home was not like that. I am the first in my family to attend university. I grew up in a loving household, but it was not one with many books, and the prevailing mode of inquiry was practical rather than philosophical. My mother has been my greatest supporter throughout every stage of this journey, but her eyes still glaze over whenever I try to explain anything abstract. My parents are people who fix things, build things, and get things done, and I suspect they have spent the better part of three decades wondering where they went wrong.

In 1997, Ms Elisabeth Norr, a teacher who saw potential where others might reasonably have seen a teenager with more energy than direction, awarded me a scholarship to attend Skagerak and its International Baccalaureate programme. The IB, and in particular its Theory of Knowledge component, opened a door I had not known existed. For the first time, I encountered the idea that the way we come to know things is itself worth examining, and that the frameworks we use to make sense of the world are not given but constructed. This realisation has, in hindsight, proven both formative and financially costly.

It was in the Skagerak library that I found Richard Dawkins's *The Selfish Gene* (1976). The book contained a reference to a simple computer programme that simulated evolution, and I was captivated by the idea that the astonishing complexity of the living world could emerge from simple mechanistic rules operating over time. Towards the end of that book, Dawkins speculates about whether the same logic might apply to ideas and culture. Ms Vivian Hall, who served as both English teacher and librarian at Skagerak, kindly ordered Susan Blackmore's *The Meme Machine* (1999) at my request, and that book pushed the speculation further. Inspired by what I had read, I turned to the late 90s search engine Ask Jeeves, a digital butler who, unlike most butlers, was free of charge and only occasionally useful. A search led me to Hazlehurst and Hutchins (1998), a paper demonstrating how simple cognitive agents can develop shared representational conventions through coordinated action in a shared world, without requiring those agents to possess the answers in advance. That idea, that structure and meaning can emerge from interaction rather than innate endowment, planted the seed that has, after a truly unreasonable germination period,

produced the thesis you are now reading. Twenty-seven years later, I find that this insight describes not only the theoretical core of my doctoral work but also, with uncomfortable accuracy, the process by which the doctoral work itself came into being.

The path from that library to this thesis was not direct. Inspired by the questions I had encountered, I began a joint degree in philosophy and psychology. I applied to Oxford and, having grown up without the social capital that prepares one for that particular style of interview, failed in a manner that was both comprehensive and educational. I went instead to Durham University, which turned out to be an excellent decision for reasons I could not have appreciated at the time.

It was at Durham that I picked up Andy Clark's *Being There* (1997), which led me to Rodney Brooks's *Flesh and Machines* (2002). Brooks had built robots at MIT that navigated real corridors without symbolic representations or deliberate programming: they just coupled to their environment, and it worked. Clark explained why: cognition does not stop at the skull, and sometimes the smart thing to do is let the environment do the thinking. Through Brooks, I learned about the work of his doctoral student Cynthia Breazeal. Her *Designing Sociable Robots* (2002) was, to me, the obvious next step: if a robot can navigate a physical space by coupling directly to its sensory environment without an internal map, then navigating a social space might work the same way. Kismet did not model human intentions symbolically. It coupled to faces, voices, and turn-taking rhythms through layered behavioural systems, and something that looked remarkably like infant-caregiver interaction emerged. I wanted in.

Clark was lecturing on an interdisciplinary Master's in cognitive science at Bristol, but I did not have the academic background to enrol. Philosophy and psychology, it turned out, were subjects where the truly interesting work required considerably more mathematics than I possessed. The cool stuff, as I thought of it then and still think of it now, demanded fluency in a language I had not yet learned. So I went back to school, this time to acquire the mathematics and physics I needed to become an engineer, what I affectionately like to think of as the Oompa Loompas of science. Being a single father with limited resources meant that this process took rather longer than a conventional academic trajectory would suggest. The phrase "non-traditional student" does not quite capture the experience, but it will have to do.

Just before starting my Master's programme in systems engineering, I discovered that my particular combination of subjects did not, in fact, qualify for the Bachelor of Engineering in computer science that I had believed myself to possess. The administrative staff at USN, whose resourcefulness in navigating bureaucratic impossibilities deserves its own thesis, found a way for me to officially complete my Bachelor's degree a full year into the Master's. I have no idea how they managed it, and I have learned not to ask.

Engineering led me to Aiwell in 2016, a company small enough that everyone knows when the coffee runs out. The place runs on creative joy, mechanical ingenuity, and a cheerful disregard for the concept of a plan. I arrived for a three-year industrial Master's in systems engineering at USN, structured around a week of intensive courses followed by ten weeks of working as an engineer at Aiwell. I would spend a week immersed in requirements management frameworks designed for organisations with hundreds of engineers and formal change-control boards, and then return to Aiwell on Monday morning, where someone had already built a prototype of something nobody had formally requested, and it was brilliant. Every method and framework I encountered in class had to answer a simple question on Monday morning: Can this survive a team of three engineers without killing the thing that makes them good? Aiwell did not need to become lean and agile. Large companies aspire to that. Aiwell needed structure, but a structure that did not crush its creative soul.

While working on the Master's thesis, I picked up Clark's *Surfing Uncertainty* (2016), and a sailing metaphor in its opening chapters hit me as apt for what I was trying to do at Aiwell: a crew adapting to wind and waves, acting as one, the intelligence distributed across both the people and the vessel itself. The metaphor turned out to come from Hutchins, the co-author of the very first academic paper I had read back in the Skagerak library in 1999. I am not in the habit of attributing significance to coincidences, but that one got my attention.

My Master's thesis supervisor was Professor Cecilia Haskins, who took an interest in my work, which proved decisive for everything that followed. Having secured the Master's degree, I knew I wanted to continue, but I was tired, and the temptation to rest was considerable. Cecilia did not allow it. She is a formidable presence despite her short stature. At my graduation ceremony, she cornered Asle Johnsen, the owner of Aiwell, and did not let him go until he agreed to support a PhD. I was standing right there. It involved persuasion and the kind of sustained pressure that geologists associate with tectonic forces. She was right to push. Had I stopped to rest, I would probably not have started again. Despite being nominally retired, a term that in Cecilia's case should be understood loosely, she remained as my second supervisor throughout the doctoral project, not paying much attention to the "second" part.

Ten years at Aiwell taught me two things. The first was that a good standard genuinely helps: it gives a small team structure, a shared vocabulary, and a way to show clients that you know what you are doing. The second was that a good standard is not enough. You can document every process, define every role, and tick every box, and your team will still ignore the alert that matters because they are busy with yesterday's crisis and the alert is buried in an email thread that nobody reads. The standard tells you what to do. It has nothing to say about making people actually do it. The question of what kind of structure could fill that gap became the subject of this thesis.

The result, twenty-seven years after a teenager typed a question into Ask Jeeves, is the document you are now reading. The intellectual thread that runs from Hazlehurst and Hutchins through Clark's extended mind, Brooks's situated robots, Breazeal's sociable ones, distributed cognition, active inference, and systems engineering to the concrete problem of a small company building smart city infrastructure is, I hope, visible in the pages that follow. If it is not, the fault lies with the author and not with the twenty-seven years of gestation.

To my supervisors, who have shown extraordinary patience with a candidate whose timeline more closely resembles a geological process than an academic one: thank you, and you are welcome for the relief.

Acknowledgements

Professor Geir Myrdahl Køien, my principal supervisor, combined deep technical knowledge of security engineering with an instinct for when a doctoral candidate was overcomplicating things, which was often. His willingness to engage seriously with theoretical frameworks far from his own starting point, and to insist that they earn their place through explanatory power rather than novelty, shaped every chapter of this work. Professor Cecilia Haskins, my co-supervisor, brought the systems engineering rigour required for the thesis, a rigour that the candidate sometimes resisted. Her ability to see structural problems in an argument before I had finished making it saved me from more than one extended detour. Together, they supervised a project that changed direction, expanded its theoretical ambitions, and took considerably longer than anyone had planned for, with a steadiness that I can only properly appreciate now that it is finished.

This thesis would not exist without Aiwell AS and the people who made it possible to conduct six years of embedded research in a working engineering company. Asle Ingmar Johnsen, the company's owner, took a genuine risk in committing a small firm's resources to a doctoral project whose practical returns were, at the outset, uncertain at best. Tore Johnsen, as CEO, provided the operational space and trust that allowed the research to proceed alongside commercial delivery, and participated directly in the case studies with a seriousness that went well beyond obligation. The employees at Aiwell, who played card games against threat models, debated requirements with language models, and tolerated a researcher documenting their working lives for the better part of six years, deserve more thanks than a paragraph can convey. They are the reason the case studies contain real data rather than thought experiments.

The University of Oslo and the Norwegian University of Science and Technology (NTNU) generously allowed me to take doctoral courses as a visiting candidate from the University of South-Eastern Norway (USN). I am particularly grateful to Professor Bjarne Helvik at NTNU, who, when the COVID-19 lockdowns reduced the audience for his dependability seminar to two doctoral candidates, conducted it via video conference with the same thoroughness he would have brought to a full lecture hall. That commitment to teaching under absurd circumstances made a real difference to the theoretical foundations of this work.

Finally, my daughter's grandparents provided the essential and entirely unglamorous infrastructure without which academic conferences and doctoral courses would have been logistically impossible. Childcare is not typically listed among the enabling conditions for doctoral research, but it should be.

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Sammendrag

Denne avhandlingen undersøker hvordan svært små virksomheter (VSE-er), organisasjoner med opptil 25 ansatte, kan levere pålitelig smart infrastruktur til tross for begrensede ressurser, verktøy og kompetanse. VSE-er utgjør over 92 % av europeiske virksomheter og dominerer infrastrukturens leverandørkjede, men er strukturelt dårlig tilpasset de sikkerhets-, pålitelighets- og integrasjonskravene som moderne cyber-fysiske systemer stiller.

Avhandlingen inkluderer en formell modell for kompleksitet og resiliens, der kompleksitet defineres som mulig distinkt opsjonalitet og resiliens som mobiliserbar variasjon under tidspress. En utvidet modell for målrettede menneskelige aktivitetssystemer (PHAS) integrerer frienergiprinsippet fra biologi og introduserer to konstrukter: oppmerksomhetsregimer, som retter kollektivt fokus mot kritiske signaler, og praksismønstre, som opprettholder engasjement gjennom regelmessig, strukturert samhandling. Fjorten testbare hypoteser og et rammeverk for utforming av intervensjoner utledes fra den formelle analysen.

Fire industrielle casestudier gjennomført over seks år med en norsk VSE evaluerer komplementære digitale ingeniørintervensjoner: modellbasert pålitelighets- og sikkerhetsvalidering, spillifisert trusselmodellering, LLM-assistert kravgjennomgang og miljødesignet cybersikkerhetspraksis. Kryssanalyse av casene finner konvergerende støtte for fire påstander: designet kognitiv reserve overgår erfaringsbasert investering, modelleringsmønstre reduserer beslutningsbyrden uten å svekke resiliens, maskinlesbar sporbarhet reduserer støy på tvers av grensesnitt, og miljømessig synlighet opprettholder forbedring mer effektivt enn etterlevelsespålegg.

Nøkkelord: svært små virksomheter, smarte byer, digital ingeniørpraksis, modellbasert ingeniørarbeid, resiliens, systemsikkerhet, frienergiprinsippet, aktiv inferens, målrettede menneskelige aktivitetssystemer

Abstract

This thesis investigates how Very Small Entities (VSEs), organisations up to 25 employees, can deliver dependable smart city infrastructure despite severe constraints in capacity, tooling, and expertise. VSEs constitute over 92% of European enterprises and dominate the infrastructure supply chain, yet they are structurally mismatched with the security, reliability, and integration demands of modern cyber-physical systems.

The thesis develops a formal model of complexity and resilience, defining complexity as feasible distinct optionality and resilience as mobilisable variety under time pressure. An extended Purposeful Human Activity Systems (PHAS) model integrates the Free-Energy Principle from computational biology, introducing two constructs: *Regimes of Attention*, which channel collective focus towards critical signals, and *Patterned Practices*, which sustain engagement through regular, structured interaction. Fourteen testable hypotheses and a leverage points framework for intervention design are derived from the formal analysis.

Four industrial case studies conducted over six years with a Norwegian VSE evaluate complementary digital engineering interventions: model-based reliability and security validation, gamified threat modelling, LLM-assisted requirements review, and environmentally designed cybersecurity practices. Cross-case analysis finds convergent support for four claims: designed cognitive reserve outperforms experience investment, modelling patterns reduce decision burden without reducing resilience, machine-readable traceability reduces cross-boundary noise, and environmental salience sustains improvement more effectively than compliance mandates.

Keywords: Very Small Entities, smart city, digital engineering, model-based engineering, resilience, systems security, Free-Energy Principle, active inference, Purposeful Human Activity Systems

List of papers

Paper	Title	Venue
I	Validating Reliability and Security Requirements in Public Sector Infrastructure Built by Small Companies	Journal of ICT Standardization
II	Very Small Entities (VSEs): Outsourcing Risk to the Supply Chain Is Placing Systems Security Engineering on a Clay Foundation, but Playing Games May Help	Insight (INCOSE)
III	Serious Games with SysML: Gamifying Threat Modelling in a Small Business Setting	INCOSE International Symposium
IV	Beyond Code Assistance with GPT-4: Leveraging GitHub Copilot and ChatGPT for Peer Review in VSE Engineering	Norwegian ICT Conference for Research and Education
V	Navigating Uncertainty: Guiding Attention in Purposeful Human Activity Systems	Systems Engineering

Papers are numbered in order of submission. Paper I was accepted and presented at the WPC 2020 (*Wireless Personal Communication*) conference, but experienced severe delays in publication due to administrative backlog at the journal. It was subsequently withdrawn and resubmitted to the *Journal of ICT Standardization*, where it was published in 2025. Work on Paper V was conducted throughout the period of study but was finalised and published in January 2026.

Papers Summary

Paper I: Validating Reliability and Security Requirements in Public Sector Infrastructure Built by Small Companies

Journal of ICT Standardization

Paper I demonstrates how a SysML system architecture model can function as a shared boundary object that integrates reliability, security, and cost analysis across organisational boundaries. The case study examines a 6LoWPAN/CoAP wireless siphonic drainage system for urban flood mitigation, developed by a small Norwegian company in collaboration with subcontractors for PCB design and firmware. Because no single organisation possessed a complete view of the system's security and reliability properties, the paper shows how an integrated model can bridge this gap by making cross-boundary dependencies visible and traceable.

Stakeholders played the *Elevation of Privilege* (EoP) card game against the SysML model to surface STRIDE cybersecurity threats, which were then formalised as traceable requirements linked to specific components, interfaces, and mitigation options. A separate parametric reliability and availability model provided quantitative sensitivity metrics for cost-effective mitigation selection. Non-expert participants discovered protocol-level vulnerabilities during gameplay without specialist assistance, demonstrating that competence embedded in models and structured activities can compensate for limited security expertise. The SysML model also served as a machine-readable traceability artefact, reducing communication noise across the supply chain's organisational boundaries.

This paper was originally accepted and presented at the *WPC 2020* conference but was withdrawn due to publication delays and resubmitted to the *Journal of ICT Standardization*, where it was published in 2025.

Paper II: Very Small Entities (VSEs): Outsourcing Risk to the Supply Chain Is Placing Systems Security Engineering on a Clay Foundation, but Playing Games May Help

INCOSE Insight

Paper II examines the systemic security risk posed when complex public-sector infrastructure relies on networks of Very Small Entities (VSEs) with limited security expertise and insufficient resources for formal engineering. In a globalised economy based on outsourcing and risk-sharing, most engineering activities are carried out by the smallest companies, even on large and complex projects. The paper argues that contractual risk transfer pushes liability to the actors least equipped to manage it, creating a structurally fragile foundation for critical infrastructure.

Drawing on the INCOSE Future of Systems Engineering (FuSE) initiative, the paper argues that systems security engineering must be reframed for VSE contexts by leveraging their natural strengths as dynamic social organisations rather than imposing heavyweight controls. It proposes combining agile model-based systems engineering (AMBSE) with a Zero-Trust mindset and gamified stakeholder engagement, illustrated using the *Elevation of Privilege* card game within a SysML threat model. The paper is a conceptual contribution that diagnoses the structural problem and articulates design principles for interventions suited to VSE constraints.

Paper III: Serious Games with SysML: Gamifying Threat Modelling in a Small Business Setting

INCOSE International Symposium

Paper III investigates how gamification can lower the threshold for structured threat modelling in a small company. The paper presents a Grounded Theory Method (GTM) analysis of workshops in which cross-functional teams used the Elevation of Privilege (EoP) card game to surface STRIDE-related threats against a SysML system model. Workshop recordings, observer notes, and post-session interviews were coded to identify the mechanisms through which the game format supported engagement and learning.

The analysis reveals that the tactile, rule-based format attracted technicians and managers who would otherwise avoid security discussions, that narrating threats in everyday language before translating them into technical terms rapidly created a shared vocabulary, and that repeated cycles of card play prompted the team to probe increasingly complex system interactions. These outcomes were linked to specific game mechanics: timed rounds sustained momentum, collaborative scoring underscored collective responsibility, and handling physical cards anchored abstract threats in concrete artefacts. The findings illustrate how a regular, structured interaction can channel collective attention towards critical signals, build shared understanding, and reduce the decision burden on individual participants without requiring specialist security expertise.

Paper IV: Beyond Code Assistance with GPT-4: Leveraging GitHub Copilot and ChatGPT for Peer Review in VSE Engineering

Norwegian ICT Conference for Research and Education

Paper IV explores how large language models (LLMs) can serve as cognitive scaffolding for requirements engineering in a small company that lacks in-house expertise for systematic peer review. Engineers at the case company used ChatGPT and GitHub Copilot to construct a GitHub Action that, whenever a repository issue was labelled as a requirement, automatically invoked the GPT-4 API. The API was given access to current project files, a description of the system context, and reference materials from the INCOSE *Systems Engineering Handbook* and the *Systems Engineering Body of Knowledge* (SEBoK). The model then generated a structured review, posted as a comment in the issue discussion thread, and scored each requirement on clarity, conciseness, testability, and traceability.

Engineers interacted with the AI in ordinary GitHub discussions, requesting re-evaluation after edits and, at times, inviting the model into design debates. Logs captured quantitative metrics (review duration, revision counts, and score distributions) over several cycles, and chat transcripts and semi-structured interviews were coded using GTM to analyse stakeholder perceptions. The study found that AI-assisted reviews immediately and positively influenced the engineering process, with fewer revisions and reduced time spent on each requirement. Participants highlighted the complementary nature of human intelligence and AI, where the LLM augmented human judgment through dialogue. The paper demonstrates that competence embedded in tools and automated workflows can substitute for scarce specialist capacity, even when relying solely on publicly available, low-cost tools with no custom training.

Paper V: Navigating Uncertainty: Guiding Attention in Purposeful Human Activity Systems

Systems Engineering

Paper V extends the Purposeful Human Activity Systems (PHAS) framework by integrating the Free-Energy Principle (FEP) from computational biology. The paper presents a formal model of human activity systems as active inference agents that sense, interpret, and act on signals in their environment by minimising surprise. This extension provides a mechanistic account of how attention is weighted among competing signals, grounded in the precision-weighting mechanism of active inference. Two practical constructs are derived from the formal model: *Regimes of Attention*, which are environmental features that naturally channel collective focus towards critical information, and *Patterned Practices*, which are the regular, often ritualised interactions through which agents engage with these features and internalise the knowledge they convey. The paper derives a structured set of inquiry questions from the formal framework, guiding engineers to specify which cues matter, how people interact with them, what sustains engagement, whose values are served, and what evidence will confirm progress.

The framework is illustrated through an engineering case study. Automated code-scanning and threat-intelligence alerts were being overlooked because they competed with more salient operational demands. To elevate these signals, the team redirected high-severity warnings to a shared messaging channel and coupled them with a colour-changing smart bulb on the workshop floor that turned red whenever critical action was required, functioning as a Regime of Attention. To convert heightened awareness into consistent follow-through, the engineers introduced reinforcing Patterned Practices: regularly scheduled collaborative *Elevation of Privilege* game sessions paired with brief patch-triage huddles whenever the bulb activated. Over two years, 90 vulnerabilities were detected across four repositories, of which 88 were fixable and were patched within 24 hours of their CVEs being publicly listed. This pattern is consistent with attention being reliably captured at the moment of need and with shared expectations that next-day remediation is the default.

Work on this paper was conducted throughout the period of the thesis project, but was finalised and published in January 2026.

List of case studies

Case	Paper	System of interest	Period
A	I	6LoWPAN/CoAP siphonic drainage system	2020
B	III	Same siphonic drainage system (supply chain perspective)	2021
C	IV	Requirements engineering workflow	2022–2023
D	V	Software repositories with known vulnerability exposure	2022–2024

All four case studies were conducted at or in close collaboration with Aiwel AS, a Norwegian VSE developing outdoor building and facility automation products. Some of the published papers anonymised the company.

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Abbreviations

6LoWPAN – IPv6 over Low-Power Wireless Personal Area Networks
AI – Artificial Intelligence
AMBSE – Agile Model-Based Systems Engineering
API – Application Programming Interface
AR – Action Research
BAE – Bygg, Anlegg og Eiendom (Building, Civil Engineering and Real Estate)
BIM – Building Information Modelling
CI/CD – Continuous Integration/Continuous Deployment
CoAP – Constrained Application Protocol
CPS – Cyber-Physical System
CVE – Common Vulnerabilities and Exposures
CWE – Common Weakness Enumeration
DE – Digital Engineering
EAI – Extended Active Inference
EoP – Elevation of Privilege
FEP – Free-Energy Principle
GTM – Grounded Theory Method
HAS – Human Activity System
IaL – Industry-as-Laboratory
ICT – Information and Communication Technology
IoT – Internet of Things
LLM – Large Language Model
MBE – Model-Based Engineering
MBSE – Model-Based Systems Engineering
PHAS – Purposeful Human Activity System
RAMS – Reliability, Availability, Maintainability, Safety
SCA – Software Composition Analysis
SEBoK – Systems Engineering Body of Knowledge
SME – Small and Medium Enterprise
SoS – System of Systems
STRIDE – Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, Elevation of Privilege
SysML – Systems Modeling Language
VSE – Very Small Entity

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1 Introduction

Cities concentrate people, capital, services, and risk. By 2050, at least two-thirds of the global population, approximately 6.3 billion people, will live in urban areas, amounting to an influx of 200,000 people arriving every day (DESA-EN 2019). This rapid densification places unprecedented strain on the physical and digital fabrics that sustain urban life. The Intergovernmental Panel on Climate Change (IPCC) warns that more frequent severe weather events, such as heatwaves, flooding and heavy rainfall, will increasingly put pressure on already stretched systems (IPCC 2018, 2022). The financial implications are staggering. The United Nations Environment Programme (UNEP) projects that global spending on water infrastructure alone could reach EUR 23 trillion by 2030 (UNEP 2016). Consequently, the 2030 Agenda for Sustainable Development dedicates Goal 11 to “*Make cities and human settlements inclusive, safe, resilient and sustainable*” (UN General Assembly 2015).

Governments worldwide have responded to these pressures with substantial investment. The United States White House Smart Cities Initiative channelled over USD 160 million into projects targeting congestion, public safety, employment, and climate adaptation (The White House 2015). Canada’s Smart Cities Challenge encouraged municipalities to pursue data-driven quality-of-life improvements (Infrastructure Canada 2017). The European Innovation Partnership on Smart Cities and Communities promoted integrated solutions spanning energy, transport, and information and communication technology (European Commission 2013). In Norway, the National Roadmap for Smart and Sustainable Cities and Communities is committed to using digital tools to improve community liveability (KS 2019). These programmes align with the United Nations Sustainable Development Goals, linking technological innovation to sustainable urban development (UN General Assembly 2015). Yet despite this global momentum, the transition from an aspirational framework to a functioning urban system remains fraught with difficulty.

Smart city work is embedded within messy socio-technical systems and systems-of-systems characterised by distributed ownership, asynchronous lifecycles, emergent behaviour, and limited central authority (Hause et al. 2023; Russell et al. 2023). A fundamental tension exists between the drive for optimisation and the need for robustness. A smart city is not necessarily a resilient city, and the goals and strategies of the two concepts may conflict or overlap (Zhu et al. 2020). Smart strategies typically prioritise efficiency. They often lack the redundancy and buffers that resilient strategies require to absorb shocks. Resilience depends on more than hardware and software. It depends on coordination, shared understanding, and the ability to anticipate and absorb shocks while maintaining core capabilities.

This tension is exacerbated by specific failure patterns inherent in technocratic urbanism. The most pervasive is technological solutionism, the tendency to deploy technology in search of a problem rather than in response to a verified human or environmental need (Morozov 2013; Prakash 2025). Initiatives often result in sterile environments that fail to attract residents because they lack the organic evolutionary layers of a real city. Furthermore, there is a frequent governance gap. Cities function as a *collaborative system-of-systems*, where constituents are operationally independent (INCOSE Systems of Systems Working Group 2018). Yet smart city initiatives often treat the city as a *directed system-of-systems*. They impose central command-and-control mechanisms that clash with the autonomy of local agencies and the democratic process.

Most critically for this thesis, there is a dangerous misalignment of lifecycles. Information and communication technology (ICT) firms operate on product cycles measured in months or single-digit years. In contrast, the built environment operates on property cycles spanning decades (Huh et al. 2024). Infrastructure designed to last eighty years is now being embedded with sensors and controllers that may become obsolete or insecure within five years. The resulting maintenance debt and security vulnerability create a resilience gap. If a city's digital nervous system fails due to outdated systems or a cyberattack, the city's physical body becomes vulnerable. As urban infrastructure becomes increasingly cyber-physical, the digital layer meant to provide intelligence also introduces new vectors for instability. The integration of ICT into physical substrates creates a landscape where a digital vulnerability in a sensor network can translate into physical consequences, such as flooded streets or contaminated water.

The actors who physically build and maintain this infrastructure are overwhelmingly very small entities (VSEs), organisations with up to 25 employees (ISO/IEC 2016), that dominate construction and infrastructure supply chains across Europe and in Norway in particular (Statistics Norway 2020b). These firms face the full force of the challenges described above, yet they typically lack the capacity, tooling, and formal engineering practices needed to address them. This thesis examines how VSEs can contribute to dependable smart city infrastructure under these conditions. It argues that the relevant intervention target is not the knowledge inside individual heads but the *design of the environment* in which small teams work: their tools, their monitoring systems, their shared practices, and the information structures that connect them to each other and to the wider supply chain. The thesis proposes utilising digital engineering (DE), grounded in model-based engineering (MBE), to build that environment, bridging the gap between the limited resources of small teams and the escalating complexity of modern infrastructure.

1.1 Problem Statement

The central problem addressed by this research is the systemic fragility introduced into critical infrastructure when the primary actors responsible for its creation, very small entities (VSEs), lack the capacity, tools, and incentives to practise rigorous systems security and reliability engineering. The thesis argues that this fragility is not primarily a deficit of knowledge or skill. It is a deficit of *environmental design*: the information environments in which VSEs operate do not make the right signals salient, do not make the right actions easy, and do not sustain collective attention on the quality attributes that matter most for long-term dependability.

1.1.1 The Dominance of the Very Small Entity

Very small entities (VSEs), defined as organisations consisting of up to 25 people (ISO/IEC 2016), comprise the vast majority of the infrastructure supply chain (Statistics Norway 2020b). More than 92% of European enterprises are micro-enterprises with fewer than ten employees (OECD 2017). In the ICT sector specifically, 85% of companies have ten or fewer employees (Larrucea et al. 2016). Norway's construction and infrastructure sector is the largest on the mainland and the most significant overall outside the oil and gas industry (Bygballe et al. 2019). When work on this thesis began in 2020, this sector comprised 67,908 registered entities. However, of these, only 36 companies employed more than 250 people, and most had fewer than ten employees (Statistics Norway 2020b).

This structural dominance presents a unique challenge. VSEs globally drive a disproportionate share of innovation and R&D investment despite operating with thin margins and scarce engineering capacity (Ptack 2017; Robinson 2018). In Norway, VSEs account for roughly 30% of R&D spending, and derive 20 to 30% of their revenue from new product development, compared to only 9% for large companies (Statistics Norway 2020a, 2020c). However, VSEs in the infrastructure sector have historically focused on specialised local delivery rather than systemic innovation (Krogstad et al. 2022).

Historically, the dominance of VSEs in infrastructure supply chains posed few systemic problems. Projects were smaller, technologies were stable, risk was partitioned by physical distance and distinct disciplines, and their interactions were physical and observable. Today, climate pressure, regulatory obligations, cyber-physical integration, and tighter interdependence increase system complexity and exposure. Small teams must now comply with formal integration standards, manage software supply chain risks, and deliver verifiable evidence of reliability and security. The cumulative effect on VSEs is rising administrative overhead and a growing gap between what they can deliver informally and what projects now require.

The challenge of reaching these firms with appropriate engineering standards is well documented. A 29-country survey found that VSEs adopting ISO/IEC 29110 reported 88% satisfaction and 64% productivity improvement within six months (Laporte and Miranda 2020), yet global adoption rates remain low. When VSEs do adopt tailored standards, measured rework rates of 8.5% to 18% demonstrate that small teams can match the output quality of mature organisations (Laporte et al. 2018). The difficulty lies not in capability but in the conditions under which that capability can be mobilised.

1.1.2 The Clay Foundation of Security and Reliability Engineering

Urban infrastructure has become cyber-physical. In response, project owners, end-users, and prime contractors are imposing new integration, assurance, and cybersecurity obligations on the VSE subcontractor network. These obligations are typically enforced through complex contractual flow-down clauses. However, where sector regulation is weak or uneven, risk becomes latent. In Norwegian water infrastructure, for example, the regulatory text governing ICT security comprises only 96 words, compared with 9,826 words for the power grid. Enforcement is based mainly on self-reporting (Seglsten 2021).

This regulatory imbalance creates an environment in which security is often treated as a paper exercise. Large prime contractors may shift liability to subcontractors who lack the legal and technical sophistication to understand the risks they are assuming. This creates a clay foundation for systems security and reliability engineering. VSEs operate under intense delivery pressure (Ccallo et al. 2024). They prioritise immediate functionality over abstract quality attributes such as security or long-term maintainability (Sanchez-Gordon et al. 2017). When formal requirements are imposed without appropriate support or tooling, the typical response is performative compliance rather than substantive change, with limited lasting improvement in engineering practice (Muñoz et al. 2020). The consequences of this dynamic can be severe. Laporte et al. (2023) document a case in which a non-profit organisation, lacking experience in specifying software quality characteristics, invested USD 90,000 in a project that partially failed because neither customer nor supplier applied the engineering practices codified in available standards. The customer did not know what it did not know, and the information environment provided no mechanism to make this gap visible.

The theoretical framework developed in Chapter 3 provides a structural explanation for this pattern. In the predictive processing account of cognition (Clark 2016), agents allocate attention by adjusting precision estimates across competing signals: high-precision signals dominate inference and action selection, while noisy or ambiguous signals are attenuated. For a VSE under delivery pressure, the immediate demands of functionality and schedule arrive as high-precision signals with clear consequences and short feedback loops. Security and reliability, by contrast, arrive as diffuse, long-horizon signals whose consequences are uncertain and temporally remote. The deprioritisation of security is not a failure of individual

diligence. It is the *rational response* of an inference system operating in an information environment where the wrong signals are salient.

The public sector struggles to hire and retain engineers with the necessary digital competence to oversee these complex systems. This leads to a greater reliance on rigid contract requirements and external consultants to manage risk. Fragmented supply chains and contractual risk-sharing arrangements compound this dynamic. Such arrangements often shift responsibility and accountability downwards to the smallest suppliers, who are the least equipped to manage systemic risk. Ultimately, this reduces the system's actual reliability.

1.1.3 The Failure of Traditional Systems Engineering in VSEs

Smart city tooling and literature have matured but remain fragmented and often technology-led. They contain gaps in governance, failures, and limited empirical testing (Anthopoulos 2015; Shelton et al. 2015; Kirimtati et al. 2020; Zhao et al. 2021). Non-technical risks are underrepresented in the literature despite their system-level impact (Ramirez Lopez et al. 2020; Zhu et al. 2020). Dependability depends on social coordination as much as it does on devices and code. Digital engineering (DE) has the potential to integrate people, processes, and technology in a lifecycle framework. Model-based engineering can provide shared system models to support cross-disciplinary work, reduce ambiguity, and connect design intent to verification and operations. Agile practices such as DevOps and DevSecOps can extend these models into continuous delivery and assurance, and couple automated tests with pipelines that integrate threat intelligence and compliance checks.

However, adopting new tools and methods in a VSE context is notoriously difficult. Tooling can be costly and complex. It often requires steep learning curves that small teams cannot afford. Skills are scarce. Legacy assets constrain change. Traditional systems engineering frameworks are often viewed by VSEs as overly bureaucratic and rigid. They are designed for large aerospace or defence programmes rather than agile infrastructure projects. Honour (2018) traces this misalignment to a historical shift in which systems engineering moved from first-principles technical analysis to process-defined management control, a shift that improved performance on routine projects but actively harmed it on the most complex ones.

This thesis argues that the failure of traditional frameworks in VSE contexts is not merely practical but *structural*. Traditional approaches assume that engineering competence resides in individual expertise: if the right people can be trained, the right processes will follow. Chapter 3 establishes that the relevant cognitive resources of an engineering team are not confined to individual heads but are distributed across people, tools, and the team's shared environment. An engineering team is an extended embodied mind in the sense of Clark

(2016) and Hutchins (1995), a framing that Rao and Francis (2022) have recently articulated for the systems engineering community through the 4E cognition framework. Monitoring systems, shared models, and structured practices participate directly in the team's inferential dynamics. When this perspective is adopted, the target of the intervention shifts. The question is not how to train individuals to carry out formal processes despite resource constraints. The question is how to design the team's working environment so that dependability-relevant signals are salient, dependability-relevant actions are easy, and shared understanding is sustained through practices that the team finds valuable rather than burdensome.

The approach has a precise precedent. In the late 1980s, Brooks (1991) demonstrated that mobile robots could navigate complex, unstructured environments with far less internal machinery than classical artificial intelligence assumed was necessary. Classical AI had pursued increasingly elaborate internal representations, attempting to model the world exhaustively before acting, and the resulting systems were brittle and slow. Brooks reversed the design logic. His subsumption architecture coupled simple behavioural modules directly to sensors and actuators, exploiting the structure of the physical environment rather than attempting to replicate it internally. The robots were simpler than their predecessors yet more resilient. The system's intelligence emerged from the coupling between the agent and the environment, not from the agent's sophistication alone. The science that explained why this worked, subsequently developed by Clark (1997, 2016) and the embodied cognition tradition, was anything but simple. The critical insight was that precise science about the relationship between the agent and the environment produced simpler, more robust agents, not more complex ones.

This thesis applies the same logic to engineering organisations. Traditional systems engineering responds to rising environmental complexity by adding internal complexity: more processes, more documentation, more governance layers. The thesis proposes, instead, using a formal understanding of how teams interact with their environments to design simpler, more resilient working practices. The formal model developed in Chapters 3 and 4 serves as the scientific basis for designing interventions that VSEs can adopt in an afternoon, such as a smart bulb, a card game, or an automated review pipeline. The interventions are simple. The science that explains why they work is *not*.

The central problem this thesis addresses is therefore how small teams can effectively and efficiently integrate DE practices to enhance the reliability, security, and resilience of their outputs, not by adding procedural overhead but by redesigning their cognitive environment on the basis of precise scientific understanding of how that environment shapes collective behaviour.

1.2 Scope

This thesis focuses on mature VSEs that currently practise little or no DE and require a path that aligns with their constraints. The distinction of maturity is important. These are not early-stage start-ups but established firms with existing procedures (read *habits*), products and customers. The aim is not to showcase a fully mature DE implementation but to introduce a practical model that supports incremental adoption.

The thesis examines how DE practices rooted in MBE can improve the reliability, security, and overall resilience of cyber-physical and socio-technical public infrastructure that VSE networks build and maintain. The emphasis is on concrete practices that create leverage in small teams. These include model-based requirements and interfaces, architecture viewpoints, threat modelling as a routine activity, lightweight reliability modelling, traceability across code and configuration, and automation that turns process intent into repeatable action.

The scope encompasses both the technical artefacts of engineering, such as code and models, and the social practices that sustain the engineering system. The thesis treats these as inseparable: tools and practices together constitute the designed cognitive reserve that extends the team's inferential capacity beyond what any individual member can achieve alone. The formal model that underpins the thesis is not itself within the scope of what VSEs are expected to adopt. Its role is to provide the scientific basis for intervention design, ensuring that the recommended practices are principled rather than anecdotal and that their boundary conditions are stated rather than discovered through failure.

1.3 Research Questions

The following research questions guided project activities:

RQ1: Given the dominance of Very Small Entities (VSEs) in the infrastructure supply network and the knowledge gap in public technical organisations, what are the new dependability challenges in building and operating resilient interdependent smart city infrastructure?

RQ2: What are the key requirements and associated metrics for integrating digital engineering (DE) practices in the design of resilient smart city infrastructure?

RQ3: How can system architecture modelling patterns help develop resilient smart city infrastructure?

Here, the term “modelling patterns” is used in a broad sense. It includes reusable components and templates, as well as patterned practices that guide the modelling process itself. The distinction matters because the thesis demonstrates that it is often the *practice*, the regular, structured interaction through which engineers engage with their models, that produces the sustained improvement, not the model artefact in isolation.

1.4 Motivation and Context

This work grew out of an embedded role at Norwegian VSE Aiwel AS, which designs and supplies facility automation equipment for outdoor infrastructure. During the period 2016 to 2019, previously outsourced engineering work was moved in-house, which exposed practical constraints in processes, tools, and staffing. At the same time, the growing adoption of formal integration standards such as NS 3935 (Standard Norge 2019) increased expectations for interface descriptions, versioning, test protocols, and documented coordination across software and hardware. Larger firms absorbed these obligations with existing processes and staff. Smaller companies had to find ways to meet new requirements within their limited resources without losing their agility.

The immediate gaps were:

1. Limited support for system integration across partner organisations.
2. Weak version control and traceability between system models and implementation artefacts.
3. Difficulty translating textual requirements into objective, verifiable tests.
4. Ad hoc, or absent, security threat modelling.
5. Reliability assessment that rarely progressed beyond qualitative judgement.

Each of these gaps represents a failure of the team’s information environment rather than an individual’s competence. The engineers at Aiwel possessed deep domain knowledge of their products and markets. What they lacked was an environment that made the right engineering information salient at the right time, in a form that connected to their existing workflows and supported action rather than imposing additional bureaucratic burden.

The proposed response in this thesis is a tailored set of DE practices that VSEs can adopt with modest cost and reasonable training effort. The key design principle is to structure work around models that connect to the rest of the toolchain. That enables small teams to reuse cognitive effort, automate checks, and make evidence visible to partners and clients. In the theoretical vocabulary developed in Chapter 3, the principle is to build designed cognitive reserve: competence embedded in tools and practices that participate directly in the team's inferential dynamics, extending the team's capacity to detect, decide, and act beyond what individual expertise alone can support.

1.5 Conceptual Foundations

The thesis frames VSEs as *purposeful human activity systems* (PHAS), a construct developed by Calvo-Amodio and Rousseau (2019) that models organisations as purposeful systems whose behaviour emerges from the interaction of purpose, boundaries, relationships, and context. Drawing on Rousseau's (2017, 2018b, 2018a, 2018c) systems principles, the thesis takes as a starting constraint the principle of *complexity dominance*: in interactions between subsystems of unequal complexity, the simpler subsystem undergoes a proportionally greater degree of submergence. A VSE confronting a complex cyber-physical system-of-systems is, by structural necessity, the simpler partner. The team cannot build a model as complex as the world it must navigate. Calvo-Amodio (2019) demonstrates that engineering management tools begin to fail when the complexity of engineered systems exceeds a threshold, and proposes operational principles and metrics as activity drivers within the PHAS framework. This provides the organisational lens through which the thesis extends via formal modelling.

This constraint is formally unavoidable, but it need not be paralysing. The thesis develops two complementary lines of response.

The first draws on the embodied and extended cognition tradition (Clark 1997, 2016; Hutchins 1995) to establish that the cognitive resources of an engineering team are distributed across people, tools, and environment. When this perspective is adopted, the intervention target shifts from training individuals to designing the team's working environment.

The second draws on *active inference* and the *Free-Energy Principle* (FEP) from computational biology (Friston 2010) to provide a mechanistic account of how attention is allocated among competing signals. The extension to social cognition, showing how shared expectations emerge from individual interactions, draws on the Thinking Through Other Minds framework (Veissière et al. 2020). Two practical constructs, *Regimes of Attention* and *Patterned Practices* (Ramstead, Veissière and Kirmayer 2016), are re-purposed from this

tradition as prescriptive tools for directing attention in engineering organisations. Chapter 3 develops these foundations in full.

To operationalise the framework, the thesis employs Dennett's (1971) concept of *loans of intelligence* as a methodological discipline: each use of the intentional stance is marked as a theoretical loan, and each formal equation that provides a mechanistic underpinning counts as a partial repayment. This discipline ensures transparency about what the model explains and what it assumes.

1.6 Methodology

The research followed a two-pronged approach. First, an analytical modelling strategy was developed, and formal models were built and analysed: an extended active-inference model of PHAS and a separate formal model of complexity and resilience. Assumptions, validation strategy, and limits are stated with each model.

Second, a qualitative case-study approach combined *industry-as-laboratory* and *Action Research* (AR) with *Grounded Theory Method* (GTM), introducing interventions in live projects and iterating based on effects.

The methodological choice aligns with the domain's constraints. In public infrastructure, and especially within VSEs, controlled experiments are rare and slow. Small firms cannot run parallel process variants at scale. A design-oriented approach that iterates interventions and evaluates effects in context offers a feasible path to knowledge.

1.6.1 Impact of COVID-19 on the Research Design

The onset of the COVID-19 pandemic initially drove the methodological approach and research design. The project began shortly before the shutdown, and restrictions limited access to traditional data-collection methodologies and stakeholder workshops.

In response, the early phase of the work focused on building precise theoretical models grounded in the literature and practical experience, supported by simulation and targeted prototyping. The modelling portfolio explored SysML variants, Systemigrams, state-based and stochastic models, agent-based models with finite-state machines, stock-and-flow models, and active inference agents.

When restrictions eased, the models informed applied work in subcontracted projects at Aiwell. That sequence shaped the pattern catalogue and the measurement focus adopted later in the thesis.

1.7 Thesis Structure

The thesis comprises this extended introduction and five appended papers. The extended introduction is organised as follows:

Chapter 2, Background, frames smart city infrastructure as a socio-technical, cyber-physical system-of-systems, positions VSEs as key technical implementers, and outlines enabling engineering practices and governance mechanisms.

Chapter 3, Theoretical Foundations, builds a single progressive argument from models and compression through scientific systems principles, loans of intelligence, and the intentional stance, to the extended and embodied mind traditions, the PHAS framework, and the free-energy principle with extended active inference. The chapter establishes that an engineering team is an extended embodied mind whose cognitive resources are distributed across people, tools, and environment, and that the formal model provides the mathematical syntax for converting this insight into testable claims.

Chapter 4, Research Design and Methodology, specifies the formal PHAS-EAI model (which defines complexity as feasible distinct optionality, resilience as mobilisable variety under time pressure, and operational functional information as the effective cost of choice) and the qualitative case-study strand, which combines industry-as-laboratory, AR, and GTM.

Chapter 5, Key Findings from Model Analysis, derives fourteen testable hypotheses (H1 to H14) from seven thematic results: burst dynamics, the bounded VSE advantage, time pressure, functional information, attention and authority structures, empirical alignment, and joint intervention dominance. The formal model provides the mathematical formalisation of what Clark (2016) describes as surfing uncertainty: the ability of a well-designed system to stay responsive under conditions that exceed its modelling capacity.

Chapter 6, Key Findings from Case Studies, maps four industrial cases (A to D) conducted with Aiwell AS onto the model hypotheses and presents a cross-case analysis of convergent and divergent findings. Each case demonstrates environmental restructuring for an extended embodied mind: navigable models, game-mediated interaction cycles, automated precision-weighted feedback, and ambient attention signals.

Chapter 7, Summary, Discussion and Conclusions, addresses the three research questions, presents contributions, and discusses scalability boundaries, the loans of intelligence repayment, and directions for future work.

The five papers are appended after the references. Three appendices follow the papers. Appendix A1 contains the full formal model analysis, including definitions, theorems with proofs, and links to executable simulation code. Appendix A2 documents the agent-based simulation of collective active inference, comprising the reproduction of the Kaufmann, Gupta and Taylor (2021) model and the PHAS-EAI extensions that operationalise designed cognitive reserve, Regimes of Attention, dynamic shared expectations, Patterned Practices, and disturbance episodes in a computational testbed. Appendix A3 documents the VSE Systems Engineering plugin, an open-source tool that operationalises the four design requirements from Table 24 as a full-lifecycle companion for the Claude Code development environment. Appendix A4 presents the data management plan.

1.8 Terminology Note

In this thesis, *digital engineering* (DE) refers to a digital-first approach that integrates modelling, DevOps/DevSecOps, communication tracking, analysis, and automation into a cohesive lifecycle, ensuring artefacts remain machine-readable and enabling automation to reduce human workload. *Model-based engineering* (MBE) is used here in the same sense as *model-based systems engineering* (MBSE). However, the *systems* qualifier is omitted to emphasise application beyond what is typically referred to as systems engineering in the literature. This includes disciplines such as threat modelling, reliability modelling, and *building information modelling* (BIM). Where sources or standards use the term MBSE, the original wording is retained in direct quotations. Throughout the thesis, the term *modelling patterns* is used broadly to encompass both reusable components and patterned practices that shape the modelling process.

2 Background

Chapter 2 establishes the domain and practice context for the thesis. It frames smart city infrastructure as a socio-technical, cyber-physical system-of-systems in which tight interdependence and long asset lifetimes give rise to characteristic failure modes. It then positions VSEs as key technical implementers in the supply network and explains how procurement, regulation, and digitalisation shape their constraints. The chapter closes by outlining enabling engineering practices and governance mechanisms, including systems engineering, model-based engineering, standards, and accessible security practices.

Of the governance instruments surveyed, ISO/IEC 29110 occupies a distinctive position as the international lifecycle standard expressly designed for entities with 25 employees or fewer, and the chapter returns to it when examining how process discipline can be scaled without paralysing a small organisation. This context underpins the arguments and case material developed in Papers I to V.

2.1 The Smart City System

Smart city programmes are often presented as technology initiatives, yet their success depends on the interplay between infrastructure, governance, suppliers, and citizens. In this thesis, a smart city is treated as an engineered service ecosystem built on interdependent urban infrastructure, where digital components increasingly mediate physical services. The following sections summarise how smart city thinking has evolved, the principal drivers of complexity, and the socio-technical structures that shape failure and resilience.

2.1.1 The Evolution of Urban Management

The term smart city does not denote a single technology, supplier, or governance model. It reflects an evolving set of urban management ideas shaped by advances in data collection, communications, and automation. Contemporary smart city discourse increasingly links optimisation with resilience and sustainability, recognising the socio-technical nature of city services.

Early twentieth-century reformers argued that better data would make cities run more effectively (Fairfield 1994; Ford 1913; Shelton, Zook, and Wiig 2015). By the 1990s, the “smart city” label crystallised from earlier “digital” and “intelligent” city ideas, marking a shift from connectivity projects to broader, cross-domain ambitions (Anthopoulos 2015). The rapid diffusion of information and communication technology (ICT) in the 2000s, including cheap

sensors, data platforms, and automation, made city-scale monitoring and optimisation technically viable, fuelling pilots and integrated programmes (Kirimtat et al. 2020).

By the 2010s, the literature began to note technology-led growth and conceptual fragmentation, prompting calls to integrate governance, participation, and theory-building rather than rely solely on toolchains (Zhao et al. 2021). Parallel literatures evolved with limited exchange: smart city research focused on ICT-enabled optimisation, while resilient city research drew on sustainability and risk-management traditions (Zhu et al. 2020). At the same time, risk reviews began to highlight privacy, cybersecurity, and operational exposures arising from ICT-heavy deployments (Al Sharif and Pokharel 2022). Late-2010s to early-2020s guidance increasingly positioned “smart” programmes within sustainability and resilience agendas aligned with the UN Sustainable Development Goals (UN General Assembly 2015), reframing success in terms of climate adaptation, equity, and capability building rather than efficiency alone (Ramirez Lopez and Grijalba Castro 2020). The INCOSE Smart City Initiative echoes this shift, promoting a systems view of urban environments that integrates governance, societal needs, and technology, emphasising cross-disciplinary collaboration and lifecycle thinking to manage the interdependencies and complexity inherent in smart city delivery (Russell et al. 2023).

2.1.2 Drivers of Complexity: Urbanisation, Climate and Technology

The urbanisation, climate, and investment pressures introduced in Chapter 1 do not act in isolation. Multiple global trends converge to create a volatile operating environment for organisations delivering urban infrastructure. These trends increase interdependence, amplify uncertainty, and heighten the consequences of local failures across interconnected services.

Growth often outpaces infrastructure renewal (Eurostat 2022; World Bank 2021), and ageing populations increase demand for accessible, reliable services, as many countries will see the proportion of people over 65 double by mid-century (OECD 2020; U.S. Census Bureau 2020).

Digitalisation, automation, and the Internet of Things (IoT) enable real-time monitoring and control, yet they also increase coupling between systems and expand the cyber-attack surface. Rapidly evolving standards and protocols create compliance pressure and integration churn across supply networks (ENISA 2023; OECD 2021). Climate change amplifies risks to urban systems. The Intergovernmental Panel on Climate Change warns that more frequent heatwaves, heavy rainfall, and flooding will stress water, energy, and transport networks (IPCC 2018, 2022). The United Nations Environment Programme (UNEP) estimates that global investment needs for water infrastructure alone could reach EUR 23

trillion by 2030 (UNEP 2016, 2023). These pressures intersect with environmental targets and regulations that demand both adaptation and mitigation.

Economic and regulatory restructuring further shapes the operating environment. Public infrastructure projects are becoming larger and more complex, with increasing consolidation among main contractors (Oslo Economics 2020; Statens Vegvesen 2015). Small companies remain critical for specialist engineering and delivery, yet face greater administrative and technical burdens as formal requirements proliferate. Risk-sharing contracts, increasingly adopted in supply chains, aim to distribute uncertainty between parties while also shifting responsibility for compliance, safety, and performance onto smaller suppliers (Figueiredo, Gutenberg, and Sbragia 2007; Ghadge et al. 2017).

Collectively, these drivers create a landscape of converging pressures. The interplay of urban expansion, technological volatility, climate instability, and economic consolidation significantly amplifies the complexity of delivering public infrastructure. Addressing these challenges requires resilient, coordinated strategies that go beyond high-level planning. Crucially, these approaches must account for the operational realities of small suppliers, as successful project execution often depends on their capacity to manage escalating demands.

2.1.3 The City as a Collaborative System-of-Systems

To accurately model the complexity of modern urban environments, it is necessary to move beyond purely technological definitions. The Systems Engineering Body of Knowledge (SEBoK) defines socio-technical systems as those operating at the intersection of social and technical domains (SEBoK Editorial Board 2024; Kroes et al. 2006). This concept posits that complex systems are formed through the interaction of technical components, such as hardware and software, with social elements, including people, communities, and organisations. A critical insight from this field is that optimising the technical aspect alone does not guarantee the success of the *whole* (Trist 1981). Instead, the social and technical layers must be *jointly* optimised to ensure the system functions effectively within its human context.

When applied at the scale of a city, this socio-technical perspective aligns with the systems engineering concept of a system-of-systems (SoS), a collection of independent systems integrated to deliver capabilities that are not inherent in the constituent systems (INCOSE Systems of Systems Working Group 2018). These capabilities result from emergent behaviours and properties that arise from the interaction of the parts and cannot be achieved by any single constituent acting alone.

A defining characteristic of an SoS is the independence of its components. Maier (1998) notes that constituent systems retain operational independence, meaning they can function autonomously if separated from the group. They also possess managerial independence, allowing them to make decisions about their own funding, governance, and development without consulting the wider system (INCOSE Systems of Systems Working Group 2018). This independence creates a challenging environment for city planners. Constituent systems may upgrade their technology, change their operational priorities, or even withdraw from the city network entirely if their private goals conflict with the collective objectives.

To manage these dynamics, it is essential to distinguish among the different governance structures within an SoS. A taxonomy developed by Maier (1998) and extended by Dahmann and Baldwin (2008) categorises these systems based on the degree of central control they exhibit (INCOSE Systems of Systems Working Group 2018).

A *directed* SoS is a system built and managed to fulfil specific purposes. In this model, a central authority manages operations to ensure that high-level goals are met. Although the constituent systems retain the ability to operate independently, they accept that their normal operational mode is subordinate to the central goals (INCOSE Systems of Systems Working Group 2018). An example of this is a metropolitan transport network in which independent bus and rail operators must strictly adhere to centrally mandated timetables and ticketing standards to operate.

In contrast, a *collaborative* SoS comprises constituent systems that voluntarily choose to participate to fulfil a central purpose. Crucially, there is no overall directing authority (INCOSE Systems of Systems Working Group 2018). The constituents negotiate standards and working practices individually to evolve their roles. An electrical grid often functions as a collaborative SoS, in which power generators and distributors agree to adhere to transmission protocols while remaining financially and operationally autonomous.

The friction observed in many smart city projects often stems from a failure to recognise the city as a socio-technical collaborative SoS. Planners frequently attempt to implement a directed SoS architecture, assuming they can command independent agencies and citizens as if they were subordinate components. However, in a socio-technical environment without a central director, success depends on negotiation and voluntary alignment rather than on command-and-control.

2.1.4 Why Smart Cities Fail

Shelton, Zook, and Wiig (2015) introduced the concept of the “*actually existing smart city*” to capture the gap between idealised visions and the grounded reality of implementation. While

technology vendors and governments present smart cities as engines for sustainability, economic growth, and urban efficiency, real projects often emerge through uneven processes shaped by local politics, resource constraints, and competing interests.

The gap between the idealised vision and the actually existing smart city suggests that funding and political will alone are insufficient to guarantee success. Consequently, many projects falter not due to a lack of *ambition* but because they fall victim to specific *structural* and social pitfalls (Shelton, Zook, and Wiig 2015). The following sections detail five critical failure patterns that frequently undermine these initiatives.

2.1.4.1 Technological Solutionism (The “Tech-Led” Trap)

The most pervasive pattern of failure is the “solution looking for a problem” dynamic, where technology is prioritised over actual human needs (Prakash 2025). This occurs when cities or vendors treat the city as a technology programme rather than a social enterprise. Successful cities focus on outcomes such as liveability and safety, whereas failed initiatives often focus on hardware deployment. In Toronto, Sidewalk Labs was criticised for starting with “cool tech” and trying to convince residents they needed it, rather than identifying existing needs first (Filion, Moos, and Sands 2023; Olmstead 2025; Wang 2021). In Songdo, developers installed expensive telepresence systems in apartments, which residents largely ignored, preferring to use their own smartphones (Peyrard and Gelézeau 2020). This exemplifies the risk of “technological chauvinism,” the belief that technology is the answer to all questions (Bartmanski et al. 2023; Huh et al. 2024; Jung 2024; Peyrard and Gelézeau 2020; Wang 2021). A common governance mistake is assigning smart city leadership to IT departments. While proficient at technology, IT leaders often lack the multidisciplinary authority to manage complex social and urban domains, leading to projects that are technically functional but socially irrelevant (Reichental 2020).

2.1.4.2 The Governance Gap: “Directed” vs. “Collaborative” Friction

In the context of systems engineering theory, a city functions as a collaborative SoS, where constituent components (such as utilities, citizens, and agencies) voluntarily choose to participate to fulfil the system’s central purpose (INCOSE Systems of Systems Working Group 2018). Crucially, there is no overall directing authority, meaning constituents negotiate individually to evolve their roles rather than following orders. Failures occur when vendors attempt to treat the city as a *directed* SoS, a structure built and managed to fulfil specific purposes through central management (INCOSE Systems of Systems Working Group 2018). This imposes a top-down, command-and-control approach that erroneously expects operationally independent constituents to subordinate their actions to the central goals.

Sidewalk Labs failed in Toronto because it attempted to bypass the “messy” democratic bureaucracy. The company proposed a “Public Administrator” to sidestep municipal responsibilities, misreading the city’s political dynamics and provoking resistance against what was perceived as a “monopolistic” claim to development (Filion, Moos, and Sands 2023; Olmstead 2025; Wang 2021). More broadly, smart cities require cross-departmental integration, yet many initiatives fail because they operate in silos, lacking a unified vision and governance structure (Herrera 2025; Reichental 2020). In New York, the lack of coordination regarding “outcome-based codes” and zoning created friction between the desire for flexible, data-driven planning and rigid state regulations (Artyushina 2023).

2.1.4.3 The “Ghost City” Syndrome (Greenfield vs. Brownfield)

There is a stark contrast between the theoretical perfection of greenfield cities (built from scratch) and their lived reality. Projects like Masdar (UAE), Lavasa (India), and Ordos (China) failed to attract populations because they did not grow organically. Masdar City was planned for 50,000 residents but houses only a fraction of that (primarily students), failing to become a vibrant community (D'Eramo 2021). Similarly, Ordos was built for millions but remained largely empty because the housing was too expensive for the local population (Euklidiadas 2020). These cities are often described as “cookie-cutter urbanism,” sterile environments that lack the cultural and historical depth people crave. Without the messy, organic complexity of a real city, these projects become “ghost towns” or investment vehicles rather than living communities (Euklidiadas 2020).

2.1.4.4 Economic and Lifecycle Misalignment

A critical, often-overlooked cause of failure is the clash between the economic models of the technology sector and those of the built environment. ICT firms operate on short product lifecycles (software updates every few years or months), whereas real estate developers and city planners operate on property cycles spanning decades (Huh et al. 2024). This misalignment makes it structurally difficult for the “smart” layer to lead the “built” layer, as the physical infrastructure is designed to last 80 to 100 years, while the embedded technology may become obsolete within a few years (INCOSE Infrastructure Working Group 2012). Many cities fail to account for the long-term maintenance costs of smart infrastructure. They secure funding for the “ribbon-cutting” (construction) but lack the budget for the 15 to 20 years of maintenance required to keep sensors and networks operational. This leads to technical debt and eventual obsolescence (Carrier 2021). Some projects, like Lavasa, collapsed entirely due to financial bankruptcy when they could not sustain the massive upfront capital investment required before residents arrived to pay taxes or fees (Euklidiadas 2020).

2.1.4.5 The Trust Deficit: Surveillance and Data Privacy

The aggressive collection of data often triggers a backlash from citizens who view smart cities as instruments of surveillance capitalism (Wang 2021). In Masdar, the proposal of an energy currency (“ergos”) to monitor and limit individual consumption, combined with a central tower that glows red when the city over-consumes, created a disciplinary, prison-like atmosphere (D’Eramo 2021; Filion, Moos, and Sands 2023). The Sidewalk Toronto project collapsed partly because citizens refused to accept the trade-off of privacy for convenience. The Yellow Book leak revealed plans for a social credit system and private policing, which destroyed public trust (Artyushina 2023; Filion, Moos, and Sands 2023). Beyond privacy, connecting critical infrastructure to the internet introduces cybersecurity vulnerabilities (Habibzadeh et al. 2019; Walker-Roberts et al. 2020). Without proper segmentation, a hack on a legacy HVAC system in a retrofitted building can enable attackers to pivot to critical control systems, leading to ransomware attacks or permanent denial-of-service (PDoS) (Ahmed and Zhou 2020; Habibzadeh et al. 2019; Walker-Roberts et al. 2020).

Implementing smart city infrastructure requires precise service orchestration rather than merely aggregating components. Investing in advanced hardware does not automatically result in a coherent urban ecosystem. When operational silos refuse to align with central governance strategies, or when a lack of transparency generates resistance among end users, the system fails to meet its objectives. These initiatives flounder because they emphasise the deployment of instruments and sensors while neglecting the human and organisational dynamics required for operational cohesion.

Table 1 Summary of smart city failure patterns.

Pattern	Description	Case Example
Technological Solutionism	Technology is deployed without identifying a user need.	Songdo: Residents ignoring smart home tech (Peyrard & Gelézéau, 2020).
Governance Friction	Attempting to command independent agencies or bypass democracy.	Toronto: Sidewalk Labs bypassing city hall (Filion et al., 2023).
The Ghost City	Greenfield projects that fail to attract residents due to a lack of organic culture or affordability.	Masdar/Ordos: Empty streets, missed targets (D’Eramo, 2021; Eukliadias, 2020; Pandita et al., 2024).
Lifecycle Mismatch	Short-term tech vs long-term infrastructure; failure to fund maintenance.	Songdo: Tech obsolescence and maintenance debt (Huh et al., 2024).

Pattern	Description	Case Example
The Panopticon	Excessive surveillance is eroding trust and creating a chilling effect on public life.	Masdar: Energy rationing and monitoring (D'Eramo, 2021).

2.1.5 The City as a Cyber-Physical System

2.1.5.1 Physical Consequence and the Inversion of Priorities

One characteristic that distinguishes a standard ICT system from a city as a *cyber-physical system* (CPS) is the tangible consequences of failure. In a traditional cyber environment, a system failure or malicious attack results primarily in the loss of digital information or financial assets. In contrast, a CPS integrates computational algorithms with physical processes, in which the cyber system controls physical actuators, such as valves, traffic signals, and pumps. Failure in this domain can result in damage to physical property, environmental disasters, or loss of human life (Ahmed and Zhou 2020).

This shift in consequence demands an inversion of the established ICT security paradigm. Standard information security models rely on the “CIA triad”, which prioritises confidentiality, followed by integrity and availability. For a resilient smart city, this hierarchy must be reversed to AIC (Ahmed and Zhou 2020). *Availability* is paramount because the system governs critical infrastructure essential for survival. Systems such as power grids or water treatment plants must remain operational at all costs. The option to shut down and reboot, acceptable in ICT, is often dangerous in a city context. *Integrity* is the second priority because accurate data is essential for physical control. If an attacker falsifies sensor readings to indicate that a tank is empty when it is full, the system may overfill it, causing a rupture. While confidentiality remains a concern, it is often the least critical factor when weighed against the immediate physical safety of the population (Ahmed and Zhou 2020).

2.1.5.2 The Fundamental Attributes of a Resilient CPS

To reconcile the tension between the optimised nature of a smart city and the robust nature of a resilient one, it is necessary to define what is meant by dependability. Dependability is a general concept that encompasses several attributes that are often in conflict (Avizienis et al. 2004). The first attribute is *availability*, defined as the readiness to provide correct service. In a resilient smart city (RSC), this means that city services such as lighting and transport must remain accessible when needed, despite cyberattacks or physical disruptions (Avizienis et al. 2004; Ahmed and Zhou 2020).

Safety is defined as the absence of catastrophic consequences to the user(s) and the environment. In a CPS, safety must be integrated with *security*, as a cyber-attack on a traffic

system poses a safety hazard rather than just a data breach (Avizienis et al. 2004; Friedberg et al. 2017). Closely linked is *integrity*, defined as the absence of improper alterations to the system state. This is vital for the smart city, which relies on data-driven precision. If sensor data is spoofed, such as via GPS spoofing or false smart meter readings, the optimisation algorithms will fail and potentially cause physical harm (Avizienis et al. 2004; Ahmed and Zhou 2020; Friedberg et al. 2017). Finally, *security* encompasses the *concurrent existence of availability for authorised users, confidentiality, and integrity*. The RSC must treat security not as a bolt-on feature but as a safety requirement (Avizienis et al. 2004; Friedberg et al. 2017).

2.1.5.3 Integration and Aggregation Risks

A smart city functions as an aggregated system, which is a complex entity composed of multiple cooperating CPS. In this architecture, distinct systems, such as the energy grid and the transport network, must continuously communicate to coordinate operations (Eichmann et al. 2024). While this high level of interconnectivity drives the efficiency required for a smart city, it simultaneously expands the attack surface and introduces systemic fragility. The integration of diverse subsystems creates pathways for cascading failures, in which a vulnerability in a non-critical component can compromise the entire network. A security flaw in a seemingly trivial device, such as a smart toaster or a building HVAC system, can serve as an entry point for attackers to move laterally into critical control systems (Habibzadeh et al. 2019).

Compounding this risk is the reality that most cities are “brownfields” rather than greenfield developments. This means that new smart technology is frequently bolted onto legacy infrastructure that was never designed for modern connectivity. These legacy systems, such as 20-year-old water pumps or industrial controllers, typically lack basic security features, including encryption and authentication (Habibzadeh et al. 2019). When these insecure legacy components are integrated with modern networks, they become the weakest link in the hybrid system. This brittle integration means that the resilience of the entire aggregated system is often capped by the security limitations of its oldest hardware.

2.1.6 The Smart City and The Resilient City

Although these terms are frequently conflated in policy discourse, the academic literature identifies the *smart city* (SC) and the *resilient city* (RC) as distinct paradigms with different objectives, temporal orientations, technological assumptions, and governance logics. Zhu et al. (2020), in a scientometric analysis of more than 3,100 articles, found that the two literatures had developed largely in isolation, with only 25 of the surveyed publications addressing both concepts. Understanding the tensions between them is essential for this thesis, because VSEs building smart city infrastructure operate at the intersection: they deliver the digitally connected components on which smart city promises depend, yet the

infrastructure they build must also withstand the disturbances that resilient city thinking foregrounds.

2.1.6.1 Objectives and Temporal Orientation

The primary distinction lies in what each paradigm seeks to achieve and over what timescale. The smart city is characterised as an active process aimed at promoting efficiency, creativity, and convenience through the application of information and communication technologies (Zhu et al. 2020). Its central question is how to make the city smarter, and its dominant orientation is towards the present: monitoring and responding to urban flows in real time, treating the city as a calculable system amenable to continuous optimisation (Kitchin 2014). Kitchin distinguishes a narrow techno-instrumental vision of the smart city, centred on pervasive computing and sensor networks, from a broader knowledge-economy vision emphasising innovation and human capital, but notes that both share an underlying commitment to data-driven operational control.

The resilient city, by contrast, is oriented towards long-term adaptive capacity. Its central question is not how to optimise performance but how to maintain or restore desired functions in the face of disturbance (Meerow, Newell, and Stults 2016). Meerow, Newell, and Stults (2016), reviewing 25 definitions of urban resilience accumulated over four decades, identify six conceptual tensions that run through the literature, including whether resilience denotes a return to a single equilibrium state or the capacity to navigate between multiple states, whether cities should be adapted to specific known threats or maintain general adaptive capacity for unforeseen challenges, and over what timescale recovery should be evaluated. Their proposed definition emphasises that urban systems comprise both socio-ecological and socio-technical networks, and that resilience encompasses not only persistence but also adaptation and transformation. This stands in contrast to the smart city paradigm, which typically assumes stable operating conditions that can be monitored and adjusted incrementally.

The temporal contrast carries practical consequences. Ottenburger and Ufer (2023) describe a mechanism they term “digital massification,” whereby the steady proliferation of IoT devices and market participants within physically finite infrastructure accumulates latent systemic risk. Crucially, this accumulation is incremental: minor degradations in performance, such as increased latency, network congestion, and reduced service reliability, are socially normalised before they cross a threshold into systemic failure. The authors term this gradual process “risk transition” and argue that it occupies a blind spot in both societal awareness and long-term governance. Where the smart city paradigm treats each new device deployment as an improvement in capability, the resilient city paradigm asks whether the aggregate effect of many such deployments erodes the adaptive margin the system needs to absorb shocks. Zhu et al. (2020) frame this difference in terms of driving force: the smart city

is driven by the pursuit of technological advancement and economic growth, whereas the resilient city is driven by the need for disaster prevention and long-term hazard mitigation.

2.1.6.2 Technology and Governance

Technology occupies fundamentally different positions in the two paradigms. For the smart city, ICT is the core enabling component. Big data analytics, the *Internet of Things* (IoT), and cloud computing are treated as instruments for achieving urban intelligence, and the literature is dominated by work on their adoption, integration, and evolution (Zhu et al. 2020). For the resilient city, technology is simultaneously an enabler and a source of vulnerability. Kitchin (2014) describes smart city environments as spaces in which software and urban spatiality become mutually constituted, such that software failure renders the space itself inoperative. When two highly complex open systems, the city and its information infrastructure, are tightly interwoven, the result is not merely an addition of risks but a compounding of failure modes. Ottenburger and Ufer (2023) illustrate this with the example of smart grid infrastructure, in which power distribution and data communication share physical nodes, so that a fault in one subsystem can cascade into the other. Zhu et al. (2020) report that while smart city technology can support resilient city goals, for example, through disaster warning systems and data-driven risk assessment, it can also undermine resilience by creating additional entry points for cyberattack, increasing data-stream dependency, and expanding the attack surface of critical infrastructure.

The governance implications of this tension are significant. Kitchin (2014) identifies “solutionism” as the governing logic of the smart city: complex social and institutional challenges are reframed as technical problems amenable to computational solutions, and the resulting governance tends towards centralised, technocratic, data-driven command. Smart city development is frequently driven by vendor push rather than city-government pull, with major technology corporations promoting platform adoption, thereby creating long-term dependence and constraining future adaptive capacity. This corporate path dependency is antithetical to resilience thinking, which requires the capacity to change course in response to unforeseen conditions.

Resilient governance, by contrast, demands collaborative, multi-stakeholder approaches that extend beyond the deployment of technology. Harakan et al. (2025), examining inter-agency collaboration in urban fire resilience in Makassar, Indonesia, document how integrated information management systems and real-time data sharing are necessary for effective disaster response, yet Makassar’s governance struggles precisely because these systems remain fragmented across agencies. Their central finding is that differing organisational cultures, inadequate coordination mechanisms, and challenges in resource distribution undermine collaboration even when the technical tools exist. No single agency can independently manage all dimensions of disaster response. Harakan et al. (2025) present urban safety resilience as comprising three dimensions: facility safety resilience (physical

infrastructure and technical systems), community safety resilience (social networks, awareness, and participation), and safety management resilience (institutional coordination, policy frameworks, and resource allocation). This tripartite framing reinforces the argument that resilience is irreducible to technological infrastructure alone and demands social and institutional capacity in equal measure.

Meerow, Newell, and Stults (2016) argue that urban resilience can function as a “boundary object” that enables collaboration across disciplines without requiring definitional agreement. Zhu et al. (2020) conclude that the two paradigms can be complementary through what they term the resilient smart city, provided that technology deployment is accompanied by governance architectures designed for adaptive capacity rather than optimisation alone.

For VSEs that build components of resilient smart city infrastructure, this framing has a direct implication: the systems they deliver must be engineered not only for performance under normal conditions but also for dependability under disturbances anticipated by resilient city thinking. That challenge, and how digital engineering practices can help meet it, is the subject of this thesis.

Table 2 Comparison of smart city and resilient city paradigms.

Feature	Smart City (SC)	Resilient City (RC)
Primary goal	Efficiency and optimisation (Zhu et al., 2020)	Adaptation and recovery (Meerow, Newell and Stults, 2016; Zhu et al., 2020)
Key driver	ICT, IoT, economic growth (Zhu et al., 2020)	Climate change, disaster mitigation, long-term adaptive capacity (Meerow et al., 2016; Zhu et al., 2020)
Technology role	Core enabling component (Zhu et al., 2020)	Enabler and vulnerability vector (Kitchin, 2014; Ottenburger and Ufer, 2023)
Risk profile	Increases attack surface; accumulates latent risk through digital massification (Ottenburger and Ufer, 2023)	Mitigates vulnerability through adaptive capacity and institutional coordination (Harakan et al., 2025; Meerow et al., 2016)
Governance	Technocratic, vendor-led, solutionist (Kitchin, 2014)	Collaborative, multi-stakeholder, multi-dimensional (Harakan et al., 2025; Meerow et al., 2016)
Temporal orientation	Real-time monitoring and optimisation (Kitchin, 2014)	Long-term anticipatory adaptation (Meerow et al., 2016; Ottenburger and Ufer, 2023)

2.1.7 Building the smart city: Very Small Entities (VSEs)

2.1.7.1 The Hidden Majority: Ubiquity and Precarity

Multinational corporations and large-scale projects often shape the standards and expectations of the global engineering landscape, yet this view obscures the industry's actual composition. The vast majority of enterprises are *very small entities* (VSEs), defined by ISO/IEC 29110 as organisations of up to 25 people (ISO/IEC 2016). As Figure 1 illustrates, micro-enterprises with fewer than ten employees account for between 57% of enterprises in the USA and 94% in France and Italy, with most European countries clustering above 90%. In Europe as a whole, micro-enterprises account for 92% of all enterprises, and small and medium-sized enterprises (SMEs) collectively account for 99.8% (Eurostat 2025; Moll 2013; OECD 2024). The economic weight of these firms is substantial: micro-firms alone account for 31% of total OECD employment and 21% of value added in the business sector, and for every 10 jobs created by large firms, SMEs generate 16 additional jobs (OECD 2025). Table 3 summarises these figures.

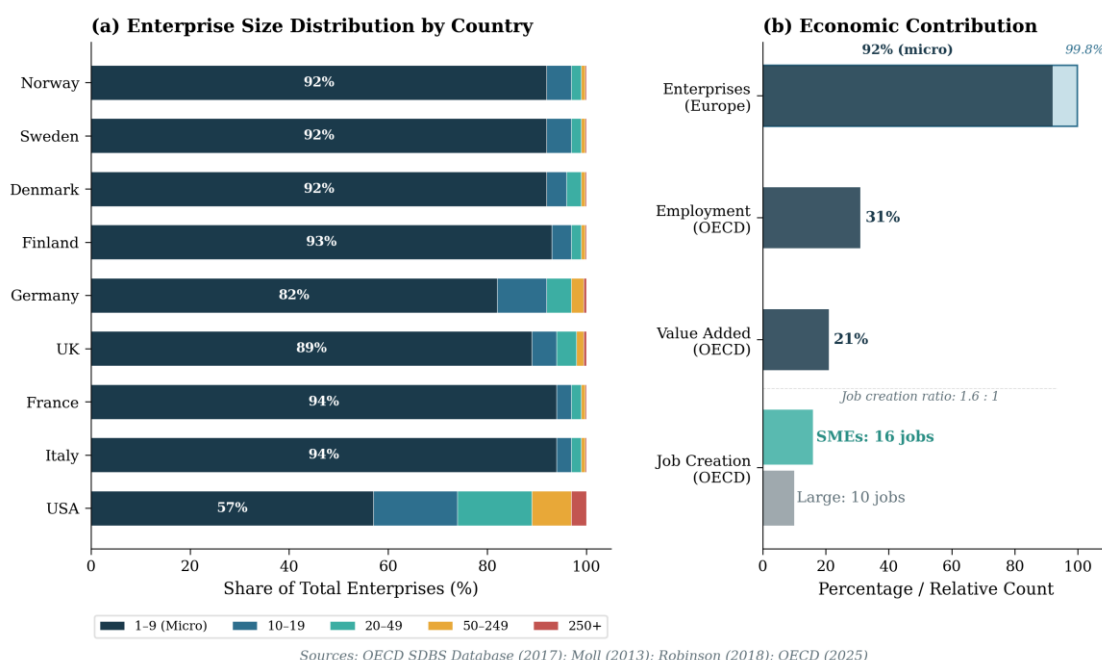


Figure 1 Enterprise size distribution and economic contribution of VSEs/SMEs

Panel (a) shows the share of total enterprises by employee size class across selected OECD countries. Panel (b) summarises key economic contribution metrics for micro-enterprises and SMEs, including a 1.6:1 job creation advantage over large firms.

Table 3 Economic contribution and prevalence of VSEs and SMEs.

Indicator	Region / Scope	Value	Source
Micro-enterprises (<9 emp.): Share of total enterprises	Europe	92%	Moll, 2013
Micro-enterprises (<9 emp.): Share of total enterprises	USA	57%	Robinson, 2018
Micro-enterprises (<9 emp.): Share of total employment	OECD	31%	OECD, 2025
Micro-enterprises (<9 emp.): Share of value added	OECD	21%	OECD, 2025
SMEs: Share of total companies	Europe	99.8%	Moll, 2013
SMEs: Job creation ratio (vs large firms)	OECD	1.6:1	OECD, 2025

Eurostat conducted a pilot project involving Norway to assess micro high-growth enterprises, defined as enterprises with fewer than 10 employees that grow by at least 3.31 employees over three years (Eurostat 2020). The results reveal a distinctive Norwegian pattern, visualised in Figure 2. Norway, along with Denmark and the Netherlands, recorded employee shares well above 20% in both the 1 to 4 and 5 to 9 employee size classes for high-growth firms, and Norway has one of the highest rates of high-growth enterprises in the 5 to 9 employee size class among European countries (approximately 15%). However, two features distinguish Norway from its peers. First, approximately 40% of these high-growth micro-enterprises in Norway are “dependent”, meaning they belong to a larger enterprise group or supply network rather than being independent start-ups. This is roughly four times the rate observed in the Netherlands or Finland, where fewer than 10% are dependent (Eurostat 2020). Second, Norway has a comparatively low share of young enterprises (five years old or less) among its high-growth firms. Approximately 20% of high-growth enterprises in Norway’s business economy were young, whereas in countries such as Bulgaria and Estonia, the share ranged from 35% to 40%. The same pattern holds in the information and communication (ICT) sector, where slightly over 20% of Norwegian high-growth enterprises were young, well below the European average of 37% (Eurostat 2020). In the building, civil engineering and real estate (BAE) sector, 27% of companies existing in 2018 had been established in the preceding six years, while 16% of companies existing in 2012 had been liquidated by 2018 (Bygballe et al. 2019). Table 4 provides the underlying data.

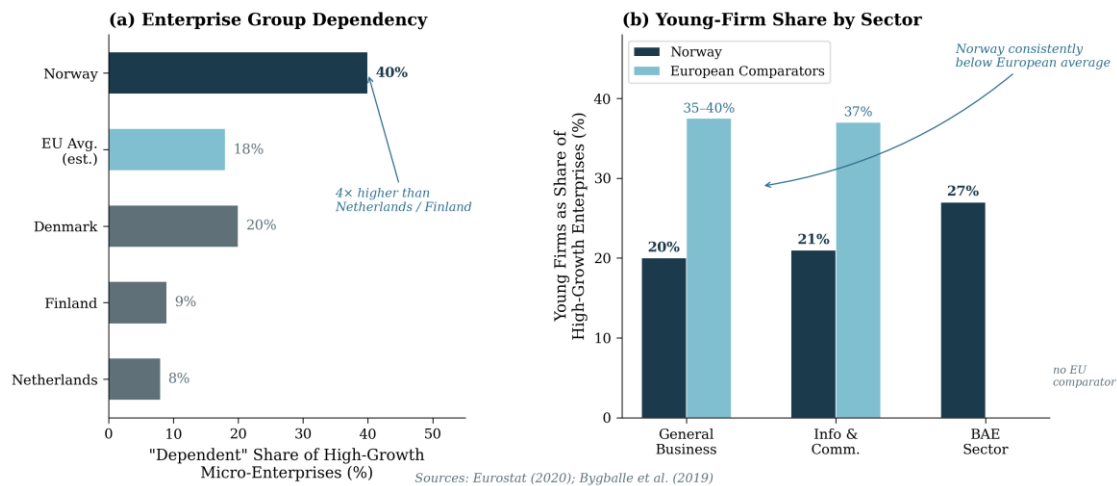


Figure 2 Characteristics of high-growth micro-enterprises in Norway

Panel (a) shows the share of high-growth micro-enterprises that are “dependent” (belonging to a larger enterprise group), comparing Norway with selected European countries. Panel (b) compares the share of young firms (five years old or less) among high-growth enterprises, by sector. Norway is consistently below European comparators.

Table 4 Characteristics of high-growth and young micro-enterprises in Norway.

Characteristic	Norway	Comparison / Note	Source
Employee share (high-growth)	>20%	Similar to Denmark & Netherlands	Eurostat, 2020
Prevalence (5–9 size class)	~15%	Highest among European countries	Eurostat, 2020
Dependent status	~40%	vs <10% in Netherlands/Finland	Eurostat, 2020
Young firms (general business)	~20%	vs 35–40% in Bulgaria/Estonia (≤5 yr)	Eurostat, 2020
Young firms (Info & Comm)	>20%	Well below European avg of 37%	Eurostat, 2020
Young firms (BAE sector)	27%	Established in preceding 6 years (2012–2018)	Bygballe et al., 2019

The BAE sector in Norway accounted for approximately 16% of total value creation in 2018 (Bygballe et al. 2019), the most recent year for which data were available when work on this thesis began in 2020. The BAE sector is the Norwegian mainland’s largest industry,

surpassed in the overall economy only by the offshore oil and gas sector (Bygballe et al. 2019). Figure 3 conveys the scale and structural characteristics of this sector. At the start of this project, the construction sector alone comprised 67,908 registered entities (Statistics Norway 2020b). Including companies that manage properties, the BAE sector as a whole comprised 93,000 active companies in 2018, accounting for 25% of all employment in Norway (Bygballe et al. 2019). The size distribution is extreme: only 36 of these companies employed more than 250 people, corresponding to a ratio of 1:2,582, and most had fewer than 10 employees (Statistics Norway 2020b). As Figure 3(c) shows, this is also a sector with considerable churn: 27% of firms in 2018 were established in the preceding six years, and 16% of 2012 firms were liquidated by 2018.

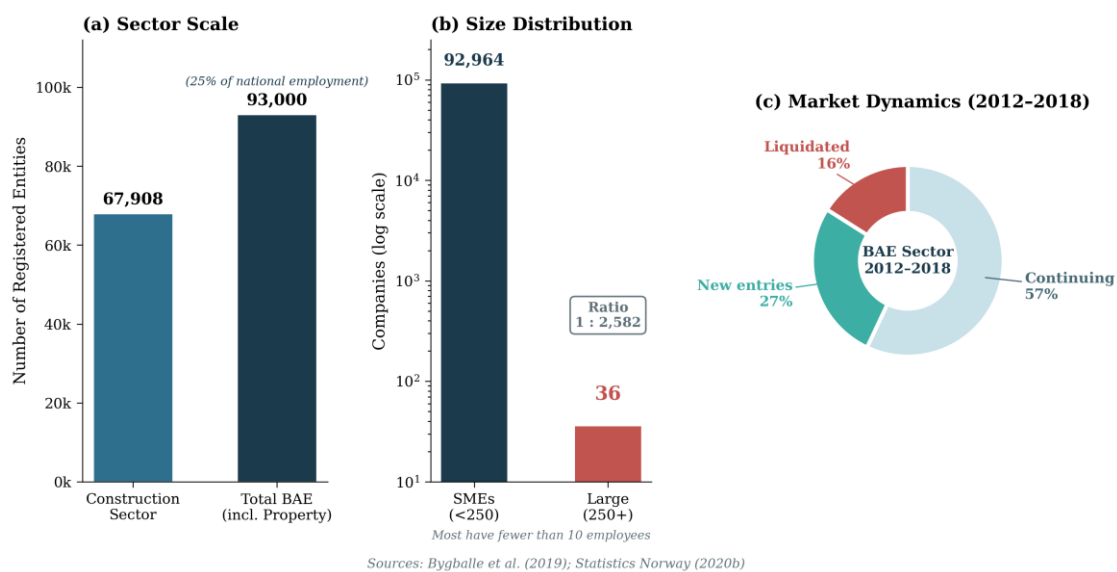


Figure 3 The Norwegian BAE sector

Taken together, these data paint a picture of a BAE sector dominated by very small, often dependent firms operating in a volatile market. Norway's high-growth micro-enterprises are less likely to be young independent start-ups than their European counterparts and more likely to be embedded within larger enterprise groups and supplier networks, creating supply-chain dependencies that compound the inherent resource constraints of the VSE. In the European Union more broadly, young firms account for approximately 25% to 30% of micro high-growth enterprises (Eurostat 2020), confirming that dynamism and precarity coexist across the continent.

VSEs drive innovation (Larrucea et al. 2016; OECD 2025) yet remain uniquely vulnerable (Muñoz et al. 2021). The very small engineering organisation is a distinct species with specific operational characteristics and economic constraints (Larrucea et al. 2016; Tuape et al. 2021). Traditional management theories developed for large hierarchies often fail in this

context because they assume abundant resources (O'Connor and Laporte 2011; Varkoi 2010). A 29-country survey confirms the scale of the challenge: the vast majority of VSEs worldwide remain without structured process support, even though 88% of those that adopted ISO/IEC 29110 reported satisfaction and 64% observed productivity gains within six months (Laporte and Miranda 2020).

2.1.7.2 Internal Constraints and Resource Scarcity

Resource scarcity fundamentally defines the operational reality of the VSE and distinguishes it from larger counterparts. Large enterprises possess financial buffers and staff capacity that allow them to absorb project delays or market fluctuations. In contrast, VSEs operate in a state of financial precarity, in which cash flow dictates daily decision-making (Davila and Pessoa 2015). This financial pressure forces a short-term focus on survival and immediate project delivery rather than strategic optimisation or process improvement (O'Connor and Laporte 2017). Consequently, these entities often view the adoption of formal engineering standards or quality assurance frameworks as an overhead they cannot afford (Sanchez-Gordon, O'Connor, and Colomo-Palacios 2015).

The scarcity of human resources further compounds fragility. VSEs rarely employ specialists for support functions such as quality assurance, legal review, or procurement (Larrucea et al. 2016). Instead, employees must function as generalists, switching contexts frequently throughout the workday (Sanchez-Gordon et al. 2017). A single engineer often serves as the project manager, requirements analyst, developer, and tester (Larrucea et al. 2016). This versatility fosters a highly adaptive workforce but creates significant operational risks. The lack of specialised roles means that critical tasks such as security auditing or requirements traceability may receive insufficient attention (Laporte et al. 2018). Furthermore, this reliance on multi-role individuals creates single points of failure, as the illness or departure of a single employee can halt an entire project (Gunawan and Budiardjo 2021). The organisation relies on the availability of specific individuals rather than on the system's capability (Simpson 2024). Empirical evidence from SME owner-operators confirms this vulnerability: none of the entrepreneurs studied held key person insurance or maintained succession plans, leaving each business entirely reliant on the owner's presence and health (Moschella, Boulianne, and Magnan 2023).

2.1.7.3 Reliance on Informal Processes

VSEs characteristically rely on informal communication and tacit knowledge rather than documented procedures, with management typically occurring through direct monitoring and ad hoc conversations rather than through formal governance structures (Larrucea et al. 2016). This informality allows VSEs to make rapid decisions and adapt quickly to changing customer requirements without the bureaucratic burden found in larger organisations (Muñoz, Mejía, and Laporte 2019). Research on SME owner-operators suggests that such

informality, while lacking documentation, can be deliberate and fully integrated within the fabric of the organisation (Moschella, Boulianne, and Magnan 2023). Their flexibility is a primary competitive advantage, enabling small entities to serve niche markets that larger firms ignore.

However, reliance on undocumented processes creates significant long-term liabilities. Critical process knowledge resides exclusively in staff's minds rather than in organisational assets (O'Connor and Laporte 2017). When personnel turnover occurs, the organisation suffers an immediate and often unrecoverable loss of intellectual property and process capabilities. This phenomenon leads to inconsistent output quality because there is no standard baseline for new employees to follow (Bucena and Kirikova 2017). Furthermore, as projects scale or regulatory requirements increase, the inability to prove compliance through documentation excludes VSEs from higher-tier value chains. The "hero culture" often prevalent in VSEs, in which individuals save projects through extraordinary effort rather than through reliable processes, ultimately inhibits the organisation's maturation into a stable system capable of predictable delivery (O'Connor and Laporte 2011; Sanchez-Gordon et al. 2015).

Gap analyses confirm this pattern empirically: across eight Mexican VSEs, none had established verification and validation or traceability practices, and at most one had any form of change management before intervention (Galván-Cruz et al. 2021). Longitudinal evidence, however, shows that formalisation need not sacrifice agility. The CSiT case in Montreal demonstrates that a VSE can maintain auditable processes over five years through tiered process groups tailored to project size, passing a third-party audit with zero non-conformances (Laporte et al. 2017; Tremblay et al. 2019), though such outcomes require sustained commitment and the type of structured support described in Section 2.2.3.

2.1.7.4 The Norwegian Context: The Strategic Shift

The Norwegian public sector has fundamentally altered the engineering landscape through changes in procurement strategies. Historically, public technical organisations acted as "Master Builders" that possessed deep internal competence and directed detailed execution (Oslo Economics 2015). This model allowed VSEs to interact directly with a knowledgeable client who understood technical constraints. However, the sector has transitioned toward design-build contracts known as *Totalentreprise* (Krogstad and Meland 2022). In this model, the public client defines high-level functional requirements and transfers responsibility for design, execution, and risk to a prime contractor.

This shift demotes the VSE from a collaborative partner of the public client to a commoditised subcontractor buried deep within the supply chain. The adoption of international standards is no longer a tool for internal improvement, but a mandatory gatekeeping mechanism imposed

by prime contractors (Krogstad and Meland 2022; Oslo Economics 2015). The industry's focus has shifted from strict specifications of what to build to open-ended descriptions of how to function. VSEs now face vague functional requirements that demand sophisticated engineering calculations and risk assessments, which often exceed their capacity (Laporte et al. 2018). The VSE must now interpret complex functional needs without direct access to the end-user (Krogstad and Meland 2022). This separation creates a vacuum where the VSE bears the cost of ambiguity. They must deliver complex systems without the administrative machinery required to manage the associated contractual risks.

2.1.7.5 Contractual Pressures and Risk Transfer

Public clients increasingly bundle infrastructure needs into “jumbo” contracts that often exceed 1 billion NOK (Oslo Economics 2020). While VSEs and SMEs account for 98 per cent of the supplier market, they account for only 60 per cent of public purchasing volume due to consolidation trends (Oslo Economics 2020). These large contract sizes effectively exclude VSEs from prime-contractor roles. Instead, they are forced into positions as sub-suppliers with little commercial influence. They are buffered from the public client, who ultimately owns the project risk and rarely receives the protections afforded to the prime contractor (Oslo Economics 2015).

Prime contractors utilise tiered subcontractor networks to distribute and absorb financial risk. VSEs perform physical engineering tasks under open-ended requirements that were originally designed to foster innovation at the top level (Krogstad and Meland 2022). A requirement such as “ensure drainage for a 200-year flood” forces a small excavation firm to engage in complex hydrological risk management (Oslo Economics 2015). This creates a dangerous competency gap in which the administrative and legal demands of the contract outstrip the small firm's operational capacity (Laporte et al. 2018). Simultaneously, the public sector risks becoming a “Hollow State” or “Master Purchaser” (Krogstad and Meland 2022). As they retain responsibility for safety but lose the competence to verify technical details, VSEs may be pressured by prime contractors to deliver a minimum level of viability to protect thin margins (Oslo Economics 2015).

Deployment packages developed for ISO/IEC 29110 aim to reduce this barrier by providing lightweight templates and examples that translate standard requirements into actionable artefacts (Houde, Laporte, and Blondelle 2016). The INCOSE VSE Working Group developed a comprehensive set of deployment packages spanning requirements engineering, architecture design, integration and testing, and project management (Ptack 2017). Section 2.2.3 examines these mechanisms in detail.

2.1.7.6 The Digital Capability Gap

Modern infrastructure increasingly relies on digital integration. The Norwegian construction and engineering industry aims to achieve comprehensive digitisation through *building information modelling* (BIM) and *digital twins* (Krogstad and Meland 2022). However, a profound digital divide exists between large prime contractors and the VSEs that perform the work (Byggenæringens Landsforening 2020). Implementing advanced digital workflows, such as continuous integration/continuous deployment (CI/CD) or model-based systems engineering (MBSE), is particularly challenging for VSEs due to limited financial resources and technical expertise (Ccallo and Quispe-Quispe 2024).

This capability gap results in significant data loss at organisational interfaces. A VSE often lacks the resources to maintain the semantic integrity of data required for a digital twin (Revell et al. 2023). They frequently resort to manual data entry or incompatible software tools, severing the digital thread. Complex methodologies such as MBSE often entail training and licensing overheads that outweigh their perceived benefits for small-scale projects (Revell et al. 2023). Additionally, modern infrastructure projects now impose cybersecurity requirements that extend to physical systems. This introduces a layer of complexity orthogonal to traditional trade skills, which VSEs are rarely equipped to manage (Simpson 2024). This gap extends to data protection: VSEs accumulate “privacy debt” as the resources required for privacy-by-design consistently exceed their capacity (Santamaria, Larrucea, and Fernandez-Gauna 2023). Research on critical infrastructure organisations suggests that integrating security into development workflows demands substantial organisational changes (Ramaj et al. 2024), changes that small teams struggle to sustain (Simpson 2024). The result is a fragmented digital landscape where the smart city vision breaks down at the level of the smallest contributors. Sections 2.2.2 and 2.2.5 examine model-based engineering and DevOps approaches that may help close this gap.

2.1.7.7 Human Factors and Motivation

In VSEs, the operational process is often indistinguishable from the individuals who execute it. This makes human factors, such as motivation, communication style, and organisational culture, the primary determinants of project success or failure (Sanchez-Gordon et al. 2017). Unlike large enterprises that rely on structural governance to maintain continuity, VSEs depend on tacit agreements and shared mental models among their staff. Consequently, the introduction of formal integration processes often encounters psychological resistance rather than technical impediments. Practitioners in these environments frequently view “best practices” with deep scepticism, interpreting them as bureaucratic impediments that stifle the agility required to survive in a competitive market (Larrucea et al. 2016).

This resistance stems from a professional identity rooted in flexibility and autonomy. Staff often believe that heavy documentation and rigid workflows are characteristic of large, slow-

moving hierarchies and offer little value in their specific context (O'Connor and Laporte 2011). As a result, the motivation to adopt new integration methods rarely arises from a desire for strategic maturity in the abstract. Instead, it is typically reactive and driven by immediate survival needs or the explicit demands of a key customer (Muñoz, Mejía, and Laporte 2020). Furthermore, the learning culture within VSEs rejects the abstract training modules common in corporate integration programmes. These entities prefer practical, self-directed learning and peer-to-peer mentoring (Tuape et al. 2021). Integration efforts that fail to account for this preference often collapse because they alienate the workforce.

Internal motivation typically arises when the psychological toll of chaotic operations becomes unsustainable. Staff seek to reduce the stress of constant firefighting and the fatigue associated with the “hero culture” (Sanchez-Gordon et al. 2015). A critical tipping point often occurs when the business owner recognises that the entity cannot grow beyond their personal supervision without formalising processes (Laporte et al. 2018). The owner is motivated by the need to install a “corporate memory” that allows the business to function independently of specific individuals (O'Connor and Laporte 2011). Research indicates that, beyond mere survival, VSEs are motivated by a sense of professional pride and a specific desire to improve product quality (Muñoz, Mejía, and Laporte 2020). VSEs view the technical quality of their output as their “best letter of introduction” and a direct reflection of their craftsmanship (Muñoz et al. 2019). Consequently, the adoption of standards is often driven by the desire to validate this professionalism and gain recognition as a reliable partner in the supply chain (Muñoz, Mejía, and Laporte 2020). Chapter 3 returns to these dynamics through a formal model of attention and shared expectations.

Taken together, these constraints form a “clay foundation”: a condition in which the security and reliability of public infrastructure ultimately rest on the capacity of the smallest, least-resourced participants in the supply chain, organisations whose professional pride and technical skill are real but whose structural fragility leaves them, and the systems they build, exposed to failures that no individual competence can reliably prevent.

2.2 Enabling VSEs to Build Resilient Systems

2.2.1 The Value of Systems Engineering Under Complexity

Systems engineering is the coordinated set of technical and managerial activities that deliver and sustain functional systems across their life cycle. It frames problems at the system level, integrates diverse disciplines, and aligns design choices with stakeholder needs and operational context. Quantitative evidence for the benefits of systems engineering remains limited relative to the large number of qualitative case studies available. However, despite their scarcity, the available results are robust and consistent across industries.

Table 5 summarises the results from a survey by Elm and Goldenson (2012), showing the correlations between project performance and different systems engineering capability areas. The highest correlations are found in project planning (+0.46), requirements development and management (+0.44), verification (+0.43), and product architecture (+0.41). These activities occur early in the life cycle and yield the greatest return because they reduce the likelihood of costly late rework.

Table 5 Systems engineering capability and project performance

SE Capability Area	Gamma Correlation
Total deployed SE	+0.49 ⇒ Very strong positive
Project planning	+0.46 ⇒ Very strong positive
Requirements development and management	+0.44 ⇒ Very strong positive
Verification	+0.43 ⇒ Very strong positive
Product architecture	+0.41 ⇒ Very strong positive
Configuration management	+0.38 ⇒ Strong positive
Trade studies	+0.38 ⇒ Strong positive
Project monitoring and control	+0.38 ⇒ Strong positive
Product integration	+0.33 ⇒ Strong positive
Validation	+0.33 ⇒ Strong positive
Risk management	+0.21 ⇒ Moderate positive
Integrated product team utilisation	+0.18 ⇒ Weak positive

Source: (Elm and Goldenson, 2012)

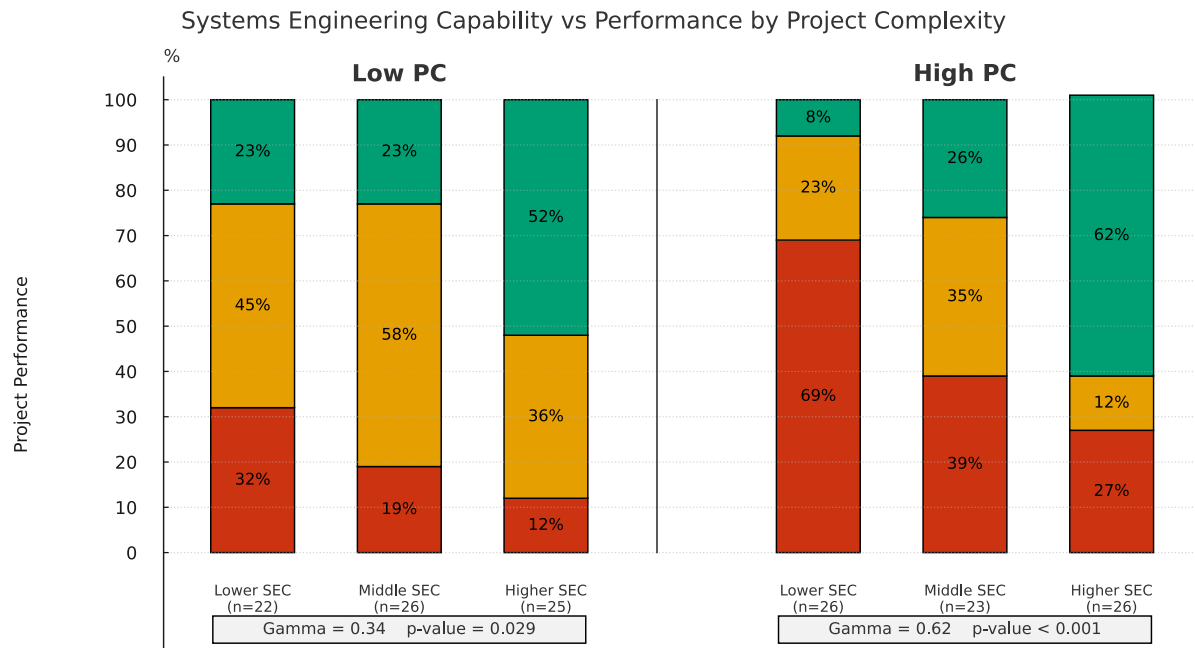


Figure 4 Project performance versus systems engineering capability for low- and high-challenge projects

Figure 4 from the same study further illustrates how overall systems engineering capability influences project performance, with results split between low- and high-challenge projects. For low-challenge projects, increasing systems engineering capability raises the proportion of higher-performing projects from 23% to 52%, while the proportion of lower-performing projects falls from 32% to 12%. For high-challenge projects, the proportion of higher-performing projects rises from 8% to 62% as capability increases, while the share of lower-performing projects falls from 69% to 27%. This illustrates that systems engineering delivers the greatest business value in complex, technical environments, where disciplined integration, verification, and architecture work are essential.

Table 6 Project performance versus experience

Success Measure	Samples	R ²	Significant
Cost compliance	73	0.246	Yes
Schedule compliance	70	0.24	Yes
Overall success	80	0.308	Yes
Technical quality	34	0.122	Marginal

Source: (Elm and Goldenson, 2012).

Table 7 Systems engineering effort, cost overrun, and return on investment

SE Effort (% of cost)	Average Cost Overrun	ROI (cost reduction per \$ reallocated to SE)
0%	53%	7.0
5%	24%	4.6
7.2% (median)	15%	3.5
10%	7%	2.1
15%	3%	-0.3
20%	10%	-2.8

Source: (Honour, 2013).

Honour (2013) analysed 48 programmes from 16 organisations, plus 44 prior surveys. He identified a median return of approximately 3.5:1 for systems engineering investments, with a potential return of up to 7:1 for projects that initially applied little or no systems engineering. The optimal total systems engineering effort was about 14-15% of the total project effort. Below this threshold, programmes typically incurred cost and schedule overruns. Beyond it, additional systems engineering effort yielded diminishing returns.

Ferrer and Rodríguez (2022) studied 70 large infrastructure design projects and found that dedicating a modest share of effort to systems engineering was correlated with improved cost outcomes. Their model suggests an optimum of around 6.5% of total effort, which is lower than Honour (2013), but none of the projects analysed by Ferrer and Rodríguez (2022) exceeded 6.0%, implying underinvestment.

A closer examination of the data supports the case that VSEs, in particular, will benefit from adopting systems engineering. Table 8 and Figure 5 use data from the Elm and Goldenson (2012) survey to illustrate why VSEs excel at low-complexity projects but lose their advantage as the challenge increases.

Table 8 Correlation of project performance with experience and project challenge

Driver	Gamma Correlation
Project challenge	-0.26 ⇒ Moderate negative
Prior experience	+0.36 ⇒ Strong positive

Source: (Elm and Goldenson, 2012).

Overall, prior experience correlates strongly with performance (gamma +0.36), whereas project challenge correlates negatively (gamma -0.26).

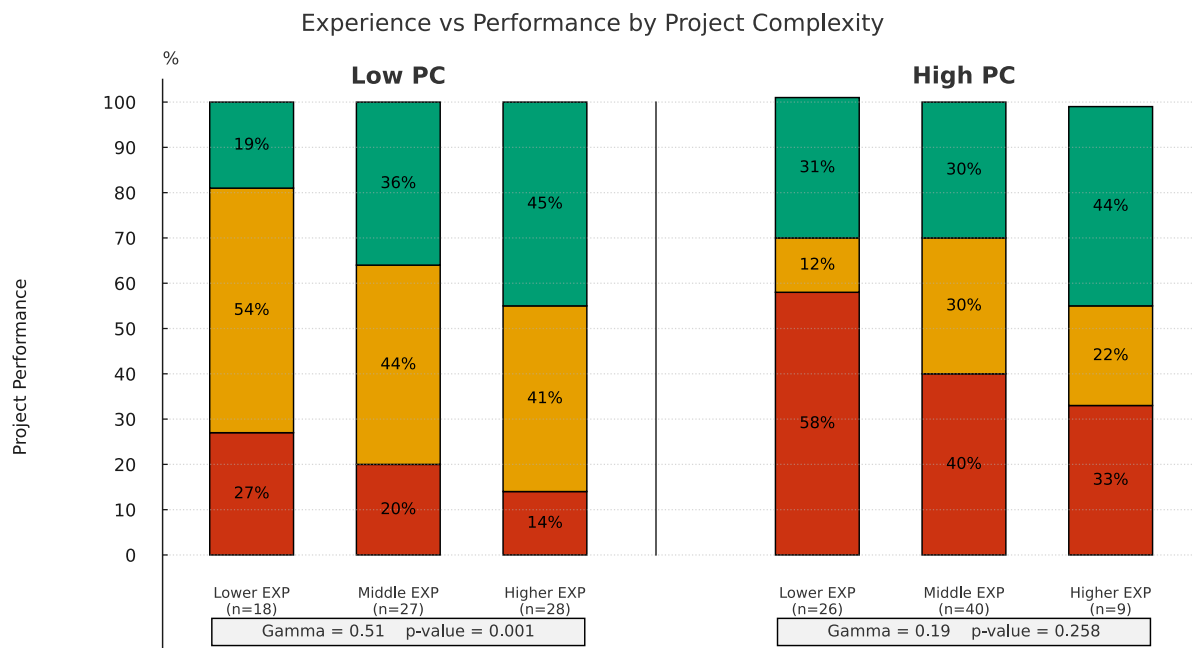


Figure 5 Project performance versus prior experience under low and high complexity

Source: (Elm and Goldenson, 2012).

In low-complexity situations, experience translates directly into outcomes. Figure 5 shows high-performing results rising from 19% to 45% across the experience bands, while low-performing results decrease from 27% to 14%. This is where VSEs thrive. Short communication paths and shared tacit knowledge enable experienced individuals to act quickly without requiring extensive coordination. Under high complexity, the gradient flattens. High-performing results rise only from 31% to 44%, while low-performing results drop from 58% to 33%. This indicates that experience primarily reduces failures rather than produces excellence, as most of the gains shift projects from low to medium performance. Furthermore, variance remains high even among the most experienced group, leaving one-third of projects in the low tier. Small firms feel these residual risks more acutely because buffers are limited. Once complexity dominates, formal systems engineering becomes the decisive factor.

2.2.2 Model-Based Engineering (MBE)

The complexity of modern critical infrastructure demands a fundamental shift in how engineers capture, analyse and communicate system information. This section defines the core principles of the model-based approach, examines the evidence for its utility, and justifies extending the scope beyond traditional model-based systems engineering (MBSE) to a broader model-based engineering (MBE) framework suitable for the smart city context.

2.2.2.1 The “Model-First” Paradigm

The International Council on Systems Engineering (INCOSE) defines MBSE as *“the formalized application of modeling to support system requirements, design, analysis, verification and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases”* (INCOSE 2023). This definition marks a departure from traditional engineering practices, in which models are often used merely as illustrations in text-based specifications or to support specific tasks.

Madni and Sievers (2018) provide the critical distinction between “engineering with models” and true MBSE. “Engineering with models” has been a common practice for decades. Engineers frequently generate diagrams, CAD drawings and spreadsheets to support specific tasks. However, these artefacts typically exist in isolation. They rely on inconsistent assumptions and disconnected underlying semantics. When a requirement changes in the text document, the engineer must manually update the diagram, the spreadsheet and the test plan. This manual synchronisation is error-prone and scales poorly.

In contrast, MBSE adopts a holistic approach where the evolving system model serves as the “sole source of truth” (Madni and Sievers 2018). The model acts as a primary repository of knowledge. Documents, views and diagrams are secondary artefacts generated directly from this repository. If an engineer changes a system attribute in one view, that change propagates automatically to all other views. This ensures consistency and traceability throughout the life cycle.

It is crucial to distinguish the methodology from the tool or the language. Estefan and Weilkiens (2022) clarify that while the Systems Modeling Language (SysML) is the de facto standard notation for MBSE, using SysML does not guarantee a model-based approach. An engineer can draw static SysML diagrams in PowerPoint, which remains a document-centric practice. Conversely, one can practise MBSE using other languages or databases. The defining characteristic is not the notation but the model’s centrality in driving engineering activities. Topper and Horner (2013) emphasise that the primary value of this approach lies not just in documentation but in managing complexity that exceeds human cognitive limits. The model enables rigorous analysis and stakeholder communication that static documents cannot support.

2.2.2.2 State of the Art and Evidence

While the theoretical benefits of MBSE are well documented, empirical evidence paints a more nuanced picture. Henderson and Salado (2021) conducted a systematic review of the literature to assess the value of MBSE. Their findings indicate that, although perceived benefits are high, empirical evidence remains scarce. Two-thirds of the benefits claimed in

the literature rely on perceptions rather than measured data. This evidence gap suggests that organisations, particularly resource-constrained VSEs, should adopt MBSE for its capacity to manage complexity rather than for guaranteed, immediate efficiency gains.

Furthermore, the barriers to adoption are often non-technical. Huldt and Stenius (2019) surveyed the state of practice and found that while modelling languages and tools have matured, organisational structures and culture remain significant hurdles. Implementing MBSE requires a fundamental change in how teams collaborate. This cultural shift is particularly acute for VSEs that lack the hierarchy to enforce new processes.

Despite these challenges, specific technical benefits are well-supported. Cederbladh, Cicchetti, and Suryadevara (2024) highlight the value of early verification and validation (V&V). By simulating system behaviour in the model phase, engineers can identify design flaws before physical implementation begins. For a VSE operating with limited financial buffers, this ability to detect errors early reduces the risk of costly rework during construction or deployment.

2.2.2.3 Broadening the Scope: From MBSE to MBE

Traditional MBSE literature often focuses on the aerospace and defence sectors, emphasising functional architecture and requirements. However, the smart city context requires a broader perspective. Teper et al. (2025) argue that infrastructure and cyber-physical systems have distinct modelling needs that differ from pure avionics or software systems. A smart city project does not exist in a vacuum. It interacts with physical geography, civil structures, and a hostile cyber-threat landscape.

Therefore, this thesis adopts the term model-based engineering (MBE) to denote a convergent approach that integrates three distinct modelling domains:

1. **Functional Logic:** This corresponds to traditional MBSE, typically to define system behaviour, requirements and interfaces.
2. **Physical and Spatial Context:** A smart city system is inseparable from its built environment. Michael et al. (2025) argue that systems engineering must integrate with building information modelling (BIM) and spatial data. The “system” includes the concrete, the sensors and the spatial constraints of the urban fabric.
3. **Safety and Security Context and Attributes:** Safety and security cannot be treated as checklists appended at the end of the project. Khan et al. (2017) and Bakirtzis et

al. (2018) demonstrate that failure modes and weaknesses must be modelled in context.

Techniques such as STRIDE allow engineers to create threat models that map directly to the system architecture. MBE acts as the umbrella for these disciplines. It enables the technical implementer to model the full reality of their component within its broader system-of-systems context. The model defines how the component is intended to function, where it fits physically, potential failure modes and weak points, and how attackers might compromise it.

2.2.2.4 Emerging Trends and Enablers

Several technological advancements are lowering the barrier to entry for MBE, making it more accessible to smaller entities.

Digital Twins and TwinOps: The model does not become obsolete once the system is built. Michael et al. (2025) and Hugues et al. (2022) describe “TwinOps” as the integration of MBE with DevOps. In this workflow, the design models evolve into a Digital Twin connected to the operational system. This allows VSEs to test software updates against a high-fidelity virtual model before deploying them to critical city infrastructure.

SysML v2 and Textual Modelling: A critical evolution in the field is the transition to SysML v2. Estefan and Weilkens (2022) note that, unlike the vendor-locked files of the previous generation, SysML v2 supports a standard textual notation. This shift makes system models “machine-readable” and “diff-able”. Michael et al. (2025) highlight that this allows models to sit in standard Git repositories alongside software code. This integration enables true DevOps workflows, where architectural changes can trigger automated analysis pipelines.

Generative AI: The rise of large language models (LLMs) offers a new frontier for engineering productivity. Hou et al. (2024) review the impact of LLMs on software engineering, noting their ability to generate boilerplate code, test cases and requirements. For MBE, the move to textual modelling (SysML v2) acts as a force multiplier for AI adoption. LLMs can parse and generate text-based models far more effectively than graphical diagrams. This capability lowers the barrier to entry, enabling VSEs to use AI to draft initial architectures or traceability matrices, effectively augmenting the limited engineering capacity of small teams.

2.2.2.5 Methodologies for Implementation

Selecting the right methodology is as critical as selecting the tool. The literature offers several established frameworks, each with different levels of rigour.

- **OOSEM (Object-Oriented Systems Engineering Method):** Described by Estefan and Weilkiens (2022), OOSEM is a top-down, rigorous methodology often associated with large defence contractors. It provides comprehensive support for specification, analysis and verification. However, its heavy process overhead often requires significant tailoring to be viable for VSEs.
- **OPM (Object-Process Methodology):** Di Maio et al. (2021) highlight OPM as an ISO standard (ISO 19450) that combines structure and behaviour into a single view. This reduces the cognitive load of maintaining multiple diagram types, making it an attractive option for smaller teams seeking simplicity without sacrificing formality.
- **SYSMOD (Systems Modeling Toolbox):** Weilkiens et al. (2016) present SYSMOD as a pragmatic, heuristic-driven approach. Unlike the rigid flows of OOSEM, SYSMOD offers a “toolbox” of methods that engineers can select based on project needs. It emphasises defining the system context and requirements first, providing a practical entry point for VSEs.
- **ARCADIA:** Bonnet et al. (2015) discuss the ARCADIA method, which enforces a strict separation between the “Operational Analysis” (what the user needs) and the “Physical Architecture” (how the system is built). While powerful, it is tightly coupled with the Capella tool, which may limit the flexibility required by some VSEs.
- **MBSE Grid:** Morkevicius et al. (2017) introduce the MBSE Grid as a simplified framework organised as a matrix of abstraction layers and domains. It specifically answers the questions “Where do I start?” and “What do I model next?”. This framework is particularly relevant for VSEs adopting MBE for the first time, as it provides a clear, incremental roadmap rather than an overwhelming set of abstract concepts.

By adopting a right-sized methodology, such as the MBSE Grid or SYSMOD, and leveraging enablers such as SysML v2 and AI, VSEs can transition from ad hoc documentation to true model-based engineering. This transition equips them to handle the multi-domain complexity of the smart city without succumbing to the administrative burden of traditional systems engineering.

2.2.3 Standardising the Process

To function effectively as a technical implementer within a complex smart city supply chain, the VSE must adopt a level of process discipline that ensures interoperability without

inducing organisational paralysis. Several layers of standardisation bear on this problem simultaneously: international lifecycle standards govern how the VSE conducts its engineering, information management standards govern the data it must deliver, and national domain standards govern the integration, commissioning, and long-term operation of the technical installations it contributes to. This section surveys the standards most directly relevant to the thesis and examines the tension between their necessity as a baseline for trust and the risk that compliance degrades into a paper exercise.

2.2.3.1 ISO/IEC 29110: Lifecycle Governance for Very Small Entities

A common barrier to standardisation for small entities is the perception that formal standards are too complex, bureaucratic, and resource-intensive for their needs (Laporte et al. 2018). Traditional frameworks such as ISO/IEC 15288 (ISO/IEC/IEEE 2015) for systems engineering or CMMI assume the existence of specialised departments for quality assurance and configuration management, resources that VSEs lack.

The ISO/IEC 29110 series of standards addresses this gap by providing Lifecycle Profiles specifically tailored for entities with up to 25 people (ISO/IEC 2016). Rather than a monolithic standard, it offers a scalable roadmap. For most VSEs without a current process, the Basic Profile is most relevant. It distils the engineering lifecycle into two streamlined processes: Project Management (PM) and System Implementation (SI) (ISO/IEC 2016). The PM process ensures the VSE can plan and monitor contract execution to systematically manage risks and deviations. The SI process governs the technical execution, covering the flow from requirements analysis to architecture, construction, integration, and verification.

The adoption of ISO/IEC 29110 can serve as both a commercial enabler and a quality-control mechanism. Tremblay et al. (2019) demonstrate, through a case study of a Canadian transportation company (CSiT), that implementing the Basic Profile enabled the VSE to compete for and win contracts with large mass transit authorities. By formalising their work products (specifications, test plans, and verification reports), the VSE could provide the documentation required by the prime contractor to demonstrate that the system requirements were met. This formalisation acts as a shield: it protects the VSE from scope creep and liability by clearly defining what was promised and what was delivered.

This thesis does not propose an alternative to ISO/IEC 29110. Rather, it addresses a problem that the standard by design leaves open: how to engineer the environmental and cognitive conditions under which a VSE can sustain the discipline that the standard prescribes. ISO/IEC 29110 specifies activities and work products but, as a lifecycle standard, remains intentionally neutral on the tooling, modelling patterns, and organisational scaffolding through which those activities become habitual practice. The engineering practices, formal model, and case evidence developed in Papers I to V are intended to

complement the standard by providing both a theoretical account of why certain structures sustain attention and learning in resource-constrained settings and practical demonstrations of how model-based workflows can generate compliance evidence as a by-product of daily engineering work.

Recent research highlights the compatibility of this standard with modern agile workflows. Galván-Cruz et al. (2024) demonstrate how ISO/IEC 29110 compliance can be achieved through modern DevOps practices. This integration ensures that the standard does not become a separate paper-pushing exercise but is instead woven into the fabric of daily development.

2.2.3.2 ISO 19650: The Digital Information Mandate

While ISO/IEC 29110 addresses how the VSE conducts its engineering practices, ISO 19650 (ISO 2018) governs the data the VSE must deliver as a supplier in the civil engineering and real estate sector. As the international standard for managing information over the whole lifecycle of a built asset using building information modelling (BIM), compliance with ISO 19650 has become the de facto entry ticket for modern infrastructure projects.

In the Norwegian context, the industry's strategic direction is defined by Digitalt veikart 2.0 (Digital Roadmap 2.0), published by the Federation of Norwegian Construction Industries (Byggenæringens Landsforening 2020). This roadmap explicitly identifies international frameworks such as ISO 19650 as the common digital language needed to break down information silos. For a VSE delivering a sensor network or a control system, this means their engineering data cannot remain in proprietary formats. It must be structured for ingestion into the project's *Common Data Environment* (CDE).

However, VSEs face a “digital paradox”. Krogstad et al. (2022) highlight that while top-down digital mandates are intended to increase efficiency, they often result in an increased administrative burden on smaller supply chain actors. The requirement to deliver rich BIM data and digital twin components adds significant project hours that are not always compensated for or understood by the client. Consequently, VSEs must adopt these standards to participate in the market, but they must also implement processes as outlined in ISO/IEC 29110 to manage the associated overhead efficiently.

2.2.3.3 The Norwegian ITB Standard Chain: NS 3935, NS 6450, and NS 6460

In addition to international standards, three Norwegian standards form a lifecycle chain that governs the integration, commissioning, and long-term operation of technical building installations. Together they constitute the regulatory backbone of the “Riktig med en gang” (Right first time) initiative promoted by Standard Norge (Standard Norge 2020), and they

define obligations that flow directly to VSE subcontractors through design-build contracts under NS 8407 (Standard Norge 2011).

NS 3935 (Standard Norge 2019): Integrated Technical Building Installations (ITB). NS 3935 addresses the integration gap identified earlier in this chapter by mandating a specific role, the *ITB-ansvarlig* (integration coordinator), responsible for ensuring that complex technical systems (ventilation, lighting, access control, energy management, building automation) function as an integrated whole rather than as isolated subsystems. For the VSE technical implementer, this standard defines the verification milestones and deliverables required to prove that their specific component interoperates correctly with the rest of the facility. It forces a shift from installing a device to delivering a verified function within the larger system-of-systems. NS 3935 covers the design, execution, and commissioning phases and establishes the interface descriptions, versioning protocols, and test documentation that all contributing parties must produce.

NS 6450 (Standard Norge 2016): Commissioning and Trial Operation. NS 6450 governs the transition between construction completion and operational handover by defining the processes for systematic commissioning and trial operation of technical building installations. The standard describes what the building owner must specify in the tender documentation, which technical systems require trial operation, and the recommended duration of trial operation for different installation types. Critically, NS 6450 applies after occupancy, testing the installations under real internal loads and usage patterns rather than under idealised test conditions. The standard is not a contractual document in itself but takes effect when referenced in the contract between the parties. It bridges the gap between the integration phase governed by NS 3935 and the operational phase governed by NS 6460, ensuring that verified performance rather than assumed performance is handed over to the operations organisation.

NS 6460 (Standard Norge 2020): Operation of Technical Building Installations. NS 6460 completes the lifecycle chain by defining objectives and activities for sustainable operation and value-preserving maintenance over the operational life of the building, characterised in its guidance material as “the next 40 years” (Standard Norge 2020). The standard addresses operational goals, competence requirements for operations staff, continuous improvement and periodic review processes, deviation management, energy monitoring, and maintenance planning and budgeting. Of particular relevance to this thesis, NS 6460 explicitly includes requirements for data security (datasikkerhet) and risk assessment of technical installations during the operational phase. This means that the cybersecurity obligations documented during the engineering and commissioning phases under NS 3935 and NS 6450 must be sustained throughout the building’s operational life, an obligation that falls on the operations organisation but is shaped by the quality of the documentation and automation that the original engineering team, frequently a VSE, delivered.

The three standards form a deliberate sequence: NS 3935 governs what the VSE must deliver during design and construction, NS 6450 governs how that delivery is verified under real conditions, and NS 6460 governs how the resulting installation is operated and maintained for decades. For the VSE, this chain creates a dual obligation. In the short term, the VSE must produce the interface documentation, test protocols, and verification evidence that NS 3935 and NS 6450 require. In the long term, the quality of that documentation determines whether the operations organisation can sustain the security and performance levels that NS 6460 demands. When the VSE's engineering data is trapped in proprietary formats, manually assembled, or poorly structured, the lifecycle chain breaks at the handover boundary and the operations organisation inherits technical debt that compounds over decades. This is the lifecycle mismatch identified in Section 2.1.4.4, expressed in standards rather than theory.

2.2.3.4 The Utility and Limitations of Standards

The adoption of these standards is not merely a bureaucratic exercise. It is a matter of professional integrity. Laporte et al. (2021) argue that software and systems engineering standards represent the codified knowledge of the discipline. Failing to apply these established practices in the development of critical infrastructure constitutes professional malpractice. In a smart city, where software failures can have physical consequences, the VSE has an ethical obligation to use established engineering norms. To illustrate the legal and safety implications of this obligation, Laporte et al. (2021) describe a scenario involving an eight-person company developing computer-controlled valves for a chemical plant. Although the contract required the use of IEEE standards, the engineer performed only an informal self-check rather than the formal peer inspection mandated by IEEE 1028 (IEEE 2008). This negligence led to a malfunction that damaged the plant and hospitalised a technician. The court subsequently ruled against the supplier, holding that the failure to adhere to the agreed-upon standard constituted negligence because a proper inspection could have prevented the defect.

However, relying solely on standards is insufficient to guarantee security or resilience. Bayuk (2010) warns of the compliance trap, which occurs when standards become static checklists used for due diligence rather than dynamic tools for risk management. A VSE can be fully compliant with a standard on paper while remaining technically fragile if the requirements are treated as administrative hurdles rather than engineering objectives. Bayuk (2010) notes that standards often suffer from vague definitions and a disconnect from the operational context. If a VSE implements ISO/IEC 29110 or ISO 19650 solely by producing manual documents to satisfy an auditor, it creates shelfware that adds cost without value (Larrucea et al. 2016). The same risk applies to the Norwegian ITB chain: a VSE can produce the interface descriptions that NS 3935 requires and the commissioning documentation that NS 6450 demands without those artefacts being machine-readable, traceable, or useful to the operations organisation that must sustain performance under NS 6460 for decades.

This tension between the necessity of standards as a baseline for trust and the risk of manual box-checking degrading into paper compliance points to two distinct needs. The first is a practical delivery mechanism that makes the abstract activities defined in a standard concrete and actionable for the VSE engineer. The second is a means of automating the evidence trail so that compliance becomes a by-product of the engineering workflow rather than a separate documentation exercise.

2.2.3.5 Deployment Packages: Bridging Standards and Practice

The ISO/IEC 29110 working group addressed the first of these needs by developing open-source *Deployment Packages* (Laporte et al. 2015). Each Deployment Package provides a step-by-step implementation guide for a specific activity or work product defined in the profile, translating abstract process requirements into concrete tasks with templates, checklists, and worked examples. Deployment Packages have been tested and refined across a range of national contexts, including Canada, Mexico, Peru, Thailand, and several European countries (Laporte et al. 2018), and the INCOSE VSE Working Group has produced systems engineering variants that extend the concept to hardware-intensive projects (Houde et al. 2016).

Deployment Packages significantly lower the adoption barrier by addressing the practical question of what to produce and in what order to produce it. However, they operate at the process level: they specify activities and artefacts but do not prescribe the tooling or automation that would make those activities sustainable in a resource-constrained setting. For a VSE delivering cyber-physical smart city components, the further challenge is to embed the discipline that Deployment Packages describe into integrated, model-based engineering workflows where compliance evidence is generated automatically. When a model-based workflow produces the interface descriptions that NS 3935 requires, the verification evidence that NS 6450 demands, and the operational documentation that NS 6460 prescribes as by-products of daily engineering, the lifecycle chain holds and the compliance trap is avoided. This transition from process-level guidance to tool-level automation is addressed in the following sections. The deeper question of why some environmental and social structures sustain the discipline prescribed by Deployment Packages, whereas mandates alone often do not, is addressed in Chapters 3 and 5 through the constructs of *Regimes of Attention* and *Patterned Practices*, and empirically in Chapter 6.

2.2.4 Accessible Security Frameworks and Tooling

The need for accessible, right-sized guidance extends beyond lifecycle process standards into security-specific frameworks. The National Institute of Standards and Technology published its Cybersecurity Framework 2.0 Small Business Quick-Start Guide as a companion to the broader NIST CSF 2.0, distilling risk-management outcomes into actions

tailored for organisations with modest or no cybersecurity plans (Eliot 2024). In Europe, the European Union Agency for Cybersecurity published a structured guide that offers 12 practical steps for SME cybersecurity, accompanied by a maturity assessment tool designed to reduce the cost and complexity of risk evaluation (ENISA 2021). In Norway, the Nasjonal sikkerhetsmyndighet published version 2.1 of its Grunnprinsipper for IKT-sikkerhet, a set of 21 principles and associated measures that draws on the NIST CSF, CIS Controls and ISO/IEC 27001, with explicit acknowledgement that smaller organisations must prioritise among them (NSM 2024). In the industrial domain, IEC 62443 provides a comprehensive framework for securing industrial automation and control systems, but its scope and implementation overhead assume organisational resources well beyond those available to a typical VSE. These instruments confirm that the scaling problem identified by the ISO/IEC 29110 community, that small organisations need purpose-built guidance rather than scaled-down enterprise frameworks, is now recognised across jurisdictions and across the boundary between process governance and cybersecurity risk management. However, they describe the security outcomes to pursue rather than how to engineer them into a resource-constrained workflow. For the VSE building cyber-physical smart city infrastructure, translating the framework into practice requires concrete methods and tools.

In a collaborative smart city environment, the security of the entire system depends on the resilience of its constituent parts. For a VSE, building secure infrastructure is a formidable challenge. They lack the resources for a dedicated *security operations centre* (SOC) or specialised personnel. However, the absence of a large security team does not absolve them of responsibility. Instead, it necessitates a shift from resource-heavy security processes to accessible frameworks and automated tooling that can be integrated directly into the engineering workflow.

2.2.4.1 Threat Modelling with STRIDE

Traditional security assessments often rely on checklists or vague notions of “risk” that are disconnected from the actual system architecture. Threat modelling offers a more rigorous, yet lightweight, engineering approach. It involves systematically analysing a system representation to identify design flaws and vulnerabilities before code is written or hardware is deployed.

The STRIDE methodology is particularly suited to the informal VSE setting due to its mnemonic structure, which breaks down complex threats into six concrete categories (Khan et al. 2017):

- **Spoofing:** Impersonating a user or device (for example, a fake sensor injecting bad data).

- **Tampering:** Modifying data in transit or at rest.
- **Repudiation:** Denying that an action took place (lack of logging).
- **Information Disclosure:** Leaking confidential data.
- **Denial of Service:** Crashing the system or making it unavailable.
- **Elevation of Privilege:** Gaining unauthorised access capabilities.

By mapping these categories to the system's data flow diagram, engineers can identify specific threats at the interface level. Khan et al. (2017) demonstrate that STRIDE is effective for cyber-physical systems because it forces the team to consider the physical consequences of digital threats, such as a spoofed signal causing a valve to open. This shifts security from an abstract compliance requirement to a concrete set of engineering problems to solve.

2.2.4.2 Gamified Security Modelling

A major barrier to adopting threat modelling in small teams is the skills gap. Most VSE engineers are generalists rather than security experts and may find formal security reviews intimidating. To overcome this, gamified frameworks can turn security analysis into a collaborative and engaging activity.

Elevation of Privilege (EoP), developed by Adam Shostack at Microsoft, is a card game that gamifies the STRIDE methodology (Shostack 2014). The deck consists of cards representing specific threat types. Players take turns playing cards against a system diagram, effectively “attacking” the design to win points. Shostack (2014) argues that this format draws developers into threat modelling by leveraging their natural problem-solving and competitive instincts. It democratises security analysis, enabling non-experts to identify valid threats without requiring deep expertise.

Protection Poker adapts the agile game of “Planning Poker” to security (Williams, Meneely, and Shipley 2010). During sprint planning, the team collaboratively estimates the security risk of a user story alongside its implementation effort. This ensures that security is considered in every iteration, preventing the accumulation of technical debt and risk. Jøsang, Collignon, and Sellami (2020) expanded this concept with *Threat Poker*, which asks teams to estimate not only the risk but also the effort required to implement mitigations. This helps VSEs make pragmatic trade-offs, balancing security costs against feature delivery within their limited budget.

2.2.4.3 Visual Tooling

While games raise awareness, the output must be captured in a formal model. Enterprise threat modelling tools (like the Microsoft Threat Modeling Tool) can be complex and Windows-centric, which may not fit VSEs.

OWASP Threat Dragon is an open-source alternative designed specifically for simplicity and accessibility (OWASP Foundation 2021). It is a cross-platform desktop and web application that allows engineers to draw *Data Flow Diagrams* (DFDs) and automatically prompts them with STRIDE threats relevant to each element. Its lightweight nature and GitHub integration make it ideal for VSEs. It treats the threat model as code, represented by a JSON file in the repository, ensuring the security model evolves in lockstep with the software.

2.2.4.4 Automated Supply Chain Security (SCA)

Perhaps the most critical vulnerability for VSEs is their reliance on third-party code. Modern software is rarely written from scratch, but is assembled from existing libraries and frameworks. This creates a massive attack surface that a small team cannot manually monitor. VSEs cannot afford a threat intelligence team to monitor for new exploits. However, they can leverage global intelligence through automation. The MITRE *Common Vulnerabilities and Exposures* (CVE) and *Common Weakness Enumeration* (CWE) databases serve as authoritative global dictionaries for known threats. Tools like *OWASP Dependency-Check* and *GitHub Dependabot* can serve as bridges, connecting this data to the VSE's specific codebase. They function as automated security engineers, continuously scanning the project's dependency tree against the CVE/CWE databases.

- **GitHub Dependabot** is deeply integrated into the developer workflow. When a vulnerability is found in a library, it not only sends an alert but also automatically opens a pull request (PR) with the patched version. This reduces remediation friction to a single click.
- **Snyk** takes this further by scanning not just libraries but also container images (Docker) and infrastructure-as-code (Terraform), providing actionable fix advice directly in the IDE (Ramaj et al. 2024).
- **OWASP Dependency-Check** is a command-line tool that can be integrated into any CI/CD pipeline (Jenkins, GitLab CI, GitHub Actions) to break the build if a vulnerability is detected.

By adopting this stack, STRIDE for design, games for culture, and SCA tools for supply chain vigilance, the VSE can achieve a robust security posture. These frameworks do not require a large budget or a dedicated department. They require only the discipline to integrate them into the daily engineering workflow.

2.2.5 Integrated Engineering Workflows (DevOps and MBSE)

2.2.5.1 DevOps as an Integrated Delivery Workflow

DevOps frames delivery as a continuous workflow that links development activities to operational performance through tight feedback loops and extensive automation. It combines technical practices with organisational change by replacing sequential handovers with cross-functional collaboration, and by using automated development, deployment, and infrastructure monitoring to support continuous operational feature delivery (Ebert et al. 2016).

In contrast to approaches where development and operations optimise different local objectives, DevOps treats delivery performance as an end-to-end property. This shifts attention from one-off project milestones to the recurrent cycle of planning, building, testing, releasing, deploying, operating, and monitoring. Quality, resilience, and operability are then managed as continuously produced outcomes, supported by instrumentation that reveals how changes behave in use (Ebert et al. 2016).

Evidence on DevOps adoption indicates that success depends on more than tooling. A systematic review groups critical success factors into technical, organisational, and social and cultural categories, including automation, collaboration and communication, strategic planning, and mechanisms for high-quality feedback (Azad and Hyrynsalmi 2023). DevOps adoption also entails training and change-management burdens that can be underestimated when teams narrow their focus to tools and fail to align practices with organisational objectives (Hamunen 2016).

Adoption challenges are especially pronounced when teams operate with fragmented toolchains or limited capacity for monitoring and measurement. Reported barriers include fragmentation of tools and practices, difficulty in selecting an appropriate monitoring scope, and the challenge of adopting new metrics alongside broader process changes (Hamunen 2016). These barriers tend to compound, as insufficient automation increases manual workload, thereby reducing time available for learning, experimentation, and systematic improvement.

In very small organisations, adoption is often approached through incremental change and local demonstrations of value. A step-by-step method for DevOps adoption proposes starting by identifying impediments to development flow, prioritising desired practices, selecting a pilot project, and then defining and tracking success metrics before scaling to additional projects (Bucena and Kirikova 2017). This incremental framing aligns with findings that VSEs often perceive standardised methods as rigid and are reluctant to increase documentation and process overhead unless immediate operational value is visible.

Standards are increasingly positioned as a way to stabilise incremental adoption among small suppliers without imposing the full overhead of enterprise governance. ISO/IEC 29110 profiles have been mapped to DevOps practices, including version control, continuous integration, continuous deployment, and retrospective-based self-assessment (Pastrana Pardo, Ordoñez Erazo, and Cobos Lozada 2024).

2.2.5.2 Coupling DevOps with Model-Based Engineering

Integrated engineering workflows extend DevOps concepts beyond software delivery by bringing model-based artefacts into the same change management and automation regime as code. As discussed in Section 2.2.2, model-based engineering is treated as a digital-first lifecycle approach where machine-readable artefacts enable automation. This section focuses on the workflow implications of that position.

A practical starting point is to establish the system model as an authoritative source of truth and an integration framework for lifecycle data, so that programme information can be accessed when needed and the model can act as a common medium across disciplines (Revell et al. 2023b). This supports automated consistency checks and traceability, for example, by linking requirements to use cases, design elements, and verification artefacts through explicit relationships.

When models are maintained as machine-readable artefacts, they can be incorporated into continuous integration workflows in the same way as code. Model validation can then be expressed as automated tests, such as checks for the completeness of required relationships, adherence to naming conventions, interface consistency, and parameter bounds. An integrated model also enables automatic change propagation and model validation and can generate required documents from the model, with traceability maintained across system information and design decisions.

Workflow integration also depends on how modelling is embedded in day-to-day delivery. Revell et al. (2023b) emphasise that an MBSE methodology is the combination of processes, methods, and the tools and environment that enable them, rather than a simple sequence of steps (Revell et al. 2023b). Even in small-scale projects, practical constraints, such as delays

in tool procurement and licensing, can shape feasibility and force changes to the modelling scope and iteration rhythm (Revell et al. 2023b).

Hybrid approaches for small organisations emphasise a parallel and iterative introduction of DevOps and MBE practices, enabling adoption that scales with available resources (Simpson 2024). In this view, models enhance understanding and traceability, whereas DevOps practices reduce manual effort and accelerate feedback cycles. The combined workflow aims to ensure that decisions captured in models are continuously reflected in delivered configurations and operational behaviour, rather than remaining disconnected from implementation.

Integrated workflows can also incorporate low-cost automation to reduce review effort and increase the consistency of engineering outputs. For example, automation scripts triggered by pull requests can support requirements review by generating structured feedback and tracking requirement-related metrics, enabling incremental adoption that requires minimal training and fits common version control workflows.

2.2.5.3 DevSecOps as Continuous Risk Management in Interdependent Critical Infrastructures

Critical infrastructure integrates physical assets with technology-based components and relies on software and hardware to oversee essential operations, making disruptions far-reaching societal impacts (Ramaj et al. 2024). These infrastructures also exhibit cross-sector and cross-system dependencies, in which failures and compromises can propagate through interconnected services (Rinaldi, Peerenboom, and Kelly 2001). Risk management in this context includes identifying, assessing, mitigating, and monitoring risks, and is described as a continuous endeavour that adapts to emerging threats and evolving circumstances (Ramaj et al. 2024). Continuous monitoring is a core mechanism for maintaining ongoing awareness of threats and vulnerabilities to support risk-based decisions (NIST 2011).

DevSecOps operationalises continuous risk management by integrating security controls into the same automated workflow used for development and operations. Practitioners link the relevance of DevSecOps to critical infrastructure to proactive work, efficiency, automation, monitoring, and security improvements that support business continuity (Ramaj et al. 2024). By embedding security activities throughout the delivery pipeline, DevSecOps supports systematic processes for identifying, analysing, evaluating, and treating risk, together with monitoring, review, and continual improvement, as described in risk management guidance (ISO 2018). At design time, a machine-readable threat model can serve as a constraint on global design and automatically validate security requirements as the design progresses, enabling faster feedback when security constraints are violated.

2.2.5.4 Toolchains, Configuration, and Evidence

Integrated workflows depend on coherent toolchains that reduce the friction of moving information between disciplines. Fragmented tools tend to encourage manual translation between requirements, models, code, tests, and operational evidence, increasing latency and creating opportunities for divergence. DevOps adoption research highlights tool and practice fragmentation as a recurring obstacle, and notes that monitoring and measurement are difficult to establish when ownership and scope are unclear (Hamunen 2016).

Evidence generation differentiates ad hoc practice from an engineered workflow. When artefacts are produced and updated through automated pipelines, evidence of what was changed, why it was changed, and how it was validated can be captured as a by-product of normal work. This is particularly relevant for small suppliers, where documented compliance does not necessarily correlate with actual engineering practices and can conceal embedded risk in the supply chain.

Investment in integrated engineering tool suites is often justified on the basis of reduced labour required to perform engineering tasks and improved project performance. A return-on-investment model for systems engineering tooling reports that the most significant returns are associated with reduced engineering labour, with additional returns linked to improved project performance and the ability to pursue larger or more challenging engineering projects (Duffy and Richardson 2021). Table 9 summarises the model results by tool category.

Table 9 Summary of return-on-investment estimates for MBE tool categories

Tool class	3-year ROI (yrs 0–2)	ROA (yrs 3–5)	Project performance vs base
Base (benchmark)	0%	9%	100%
Stand-alone	677%	20%	113%
Integrated	1,260%	45%	162%

Source: (Duffy and Richardson, 2021).

2.2.5.5 Metrics Tracking for Value Demonstration and Incremental Adoption

Metrics tracking provides a mechanism to demonstrate value, prioritise incremental adoption, and avoid the perception that new practices are bureaucratic overhead. In DevOps, the DORA research programme operationalises software delivery performance using throughput and stability measures, namely deployment frequency, lead time for changes, time to restore service, and change failure rate (DORA, 2018). Tracking these measures supports improvement initiatives by making delivery capability visible and by exposing trade-offs between speed and stability.

A similar logic applies to the adoption of model-based engineering. Interview-based evidence on MBSE implementation identifies progress and metrics tracking as a best practice, proposing that qualitative and quantitative measures should be collected and compared for projects with and without MBSE to demonstrate added value and the impact of each process change (Ploeg et al., 2022). Practical adoption methods for small organisations similarly recommend identifying success metrics for pilot efforts before scaling practices more broadly (Bucena and Kirikova, 2017).

Metrics are most useful when aligned with the decisions they are meant to support. For integrated DevOps and model-based workflows, this implies a balanced set of measures that cover delivery speed, stability, and assurance. Examples include cycle time for model updates, defect escape rates across verification stages, traceability completeness, and risk exposure indicators derived from modelled dependencies. When these measures are reviewed as part of routine retrospectives and governance activities, they support continuous improvement and provide an evidence base for communicating progress to partners and clients.

The preceding sections have established the domain constraints, failure patterns, and enabling practices that frame the thesis contributions. A recurring theme cuts across these topics. Standards, deployment packages, and integrated toolchains specify what a VSE should do, yet the evidence on adoption consistently shows that specification alone does not produce sustained practice. Barriers are not primarily technical. They arise from limited cognitive capacity, competing demands on attention, resistance rooted in professional identity, and the absence of organisational structures that make disciplined behaviour self-reinforcing rather than externally imposed. Chapter 3 addresses this gap by introducing the theoretical foundations of the thesis, beginning with explicit systems-scientific commitments and culminating in a formal model of purposeful human activity under uncertainty that provides a mechanistic account of how attention, learning, and organisational resilience interact in resource-constrained settings.

3 Theoretical Foundations

This chapter builds a single argument in stages. It begins with the raw materials of systems science: things, boundaries, models, and the principles that govern their behaviour. It then introduces the philosophical tools needed to operate productively when the world is too complex to model exhaustively. It extends those tools beyond the individual brain, into the body, the environment, and the team. And it culminates in a formal mathematical framework that converts the resulting insights into testable claims. The progression is deliberate. Each section depends on what precedes it, and each opens a gap that the next section fills.

3.1 Things, Boundaries and Systems

The thesis adopts a systems approach because the research questions concern resilience in infrastructure that is both cyber-physical and socio-technical, delivered through networks of organisations and technologies. In such contexts, failure and recovery behaviours often arise from coupling, interface mismatches, feedback, and cascading effects that become visible only when interactions are treated as first-class objects of analysis alongside individual components. A systems approach supports reasoning about how local decisions propagate through dependency structures and how resilience is shaped by relationships, constraints, and information flows across organisational and technical boundaries. Complex technical systems also exist within wider socio-technical contexts, where success depends on collaboration across a wide range of activities and disciplines beyond narrow engineering concerns. This further increases the need for shared conceptual foundations that can bridge disciplinary and organisational perspectives.

The same breadth that motivates a systems approach also makes foundational terminology unstable. The term *system* carries significant semantic load across systems science, systems engineering, cybernetics, organisational studies, and physics. François (2004) documents this diversity directly, noting that even within specialist communities, the number and variety of definitions can become a source of confusion rather than clarification. Rousseau, Billingham, and Calvo-Amodio (2018) make an allied point in the context of systemology, arguing that the field lacks standardised terminology, which produces multiple glossaries and terminologies that can undermine clear communication both within and outside the systems community.

This ambiguity is not confined to peripheral debates. Sillitto et al. (2018) report that even within the relatively small senior systems engineering community, there are multiple, sharply differentiated ways of understanding what a system is. Their survey of INCOSE Fellows identifies at least five distinct worldviews, with evidence of two additional worldviews, and

describes these positions as spanning a spectrum from constructivist views that treat systems as a mode of description to realist views that treat systems as objective entities that can be discovered. The same survey also indicates divergence over whether a boundary is essential to a system and over whether system boundaries can, at least sometimes, be established based on objective criteria rather than solely by observer choice. This plurality is not inherently problematic. It reflects that the system concept functions as an interdisciplinary bridge and is therefore shaped by different purposes, epistemologies, and modelling commitments. It does, however, imply that a thesis that relies on the term system as a theoretical foundation must state its commitments explicitly. Otherwise, key claims about boundaries, resilience, interfaces, or models can shift meaning across chapters without detection.

For that reason, this thesis introduces explicit working definitions of *thing*, *system*, and *boundary*. The intent is not to impose a universal definition, but to fix the semantic commitments needed to support later arguments about resilience, digital engineering practices, and modelling patterns in interdependent infrastructure contexts. Two complementary traditions inform these commitments. Systems engineering emphasises purposeful organisation and accountability, as reflected in ISO/IEC/IEEE 15288 (2015), which defines a system as “*a combination of interacting elements organised to achieve one or more stated purposes*”. Statistical physics and related probabilistic formalisms emphasise persistence and separability in terms of state dynamics and conditional dependence structure, which provides a more precise basis for discussing “things” and their boundaries without relying on intuitions about physical discontinuities.

The starting point is the notion of a *thing*. A thing is treated here as any entity that can be distinguished from everything else at a specified level of description and over a timescale of interest. This is a practical commitment because engineering decisions require units of analysis, intervention, and responsibility. Friston et al. (2023) motivate distinguishability using conditional independencies among states. They treat the internal flow of some thing as conditionally independent of external dynamics if, and only if, internal state change does not depend on external states and external change does not depend on internal states, given appropriate boundary conditions. They describe this conditional independence as the criterion that distinguishes something as an entity rather than a purely arbitrary partition.

In this framing, a boundary is defined probabilistically as a Markov boundary, namely a subset of states that renders the states of a thing conditionally independent of all other states. Friston et al. (2023) further describe a minimal partition in which internal and external states are separated by boundary states, often described as a Markov blanket. In that partition, boundary states can be interpreted as mediating influences from outside to inside and from inside to outside, corresponding to an interface through which sensing and acting are possible at the chosen level of description. This matters for later analysis because it

makes explicit that interaction with what lies outside a thing is always mediated. The boundary determines what can be observed, what can be influenced, and what remains hidden, which is as relevant to socio-technical and organisational interfaces as it is to technical ones.

Thingness is also time-dependent and level-dependent. Friston et al. (2023) note that if thingness is predicated on a time-dependent density over states, it may only exist for a vanishingly small amount of time. They therefore motivate the need to consider steady-state solutions or densities over paths, linking persistence of things to characteristic regions of state space that attract trajectories over time. This provides a principled basis for treating certain aggregates as coherent entities, even when their physical boundaries are diffuse, because stability is grounded in constrained patterns of state transition over the relevant timescale.

A *system* is a particular kind of thing. In systems engineering terms, a system is an organised configuration of interacting elements directed towards one or more stated purposes. A statistical-physics-compatible view complements this by describing what it means for a system to maintain its identity through change.

Sillitto et al. (2018) define a system as "*a persistent region of low entropy (= high organisation) in physical or conceptual space-time*" and *systemness* as "*the phenomenon that allows regions of organisation to persist in a dissipative universe.*" Rousseau, Billingham, and Calvo-Amodio (2018) develop this further, noting that the boundary of a system can be identified by a gradient between the system and its environment, and that systems are separated from their environment by an open boundary across which matter, energy, and information can flow. This supports a unified treatment of physical infrastructures, organisations, and digital artefacts, provided they exhibit persistent structure and constrained patterns of change that can be treated as coherent for a given purpose and timescale.

Boundary definition is therefore both necessary and non-trivial. Rousseau, Billingham, and Calvo-Amodio (2018) note that system boundaries are often not clear, are frequently determined intuitively, and may be adjusted as understanding changes. This aligns with the plurality of system worldviews observed by Sillitto et al. (2018), including disagreement on whether boundaries are essential and whether they can be established objectively. This thesis therefore treats the system boundary as a modelling decision that must be stated explicitly and justified, because it determines what evidence counts as internal or external, what risks can be controlled, and what metrics and models are meaningful.

On the basis of these considerations, the following working definitions apply throughout the thesis:

- **Thing:** An entity distinguished from its surroundings at a specified level of description and timescale, characterised by an identity criterion and a boundary that mediates interaction. In probabilistic terms, distinguishability can be grounded in conditional independencies among states, represented by a Markov boundary.
- **System:** A thing that exhibits organised persistence over time, understood as constrained patterns of state and state transition, and that can be treated as a coherent object of analysis or design because interactions with its environment are mediated by a boundary. In engineering contexts, system definition also includes purpose and accountability, because these determine feasible action under stress.
- **Boundary:** The subset of states, interfaces, and constraints that mediates exchange between system and environment. Boundary choice constrains what can be observed and influenced at the chosen level of description, and therefore constrains what can be predicted, assured, and governed.

These definitions provide the basis for the discussion of models and modelling that follows, where the tension between a system's complexity and the tractability of its representation becomes the central design problem.

3.2 Models as Systems

Treating boundary critique and perspective management as modelling activities requires treating the model itself as a system. The model's boundary and internal organisation constrain the claims that can be made about a system of interest and determine how error and uncertainty propagate into decisions.

Model: In this thesis, a *model* is a system of constructs that may be mathematical, computational, physical, or linguistic. An interpretation mapping places the model's internal structure and dynamics in explicit statistical correspondence with a system of interest. The model supports the prediction of specified attributes or behaviours of the system of interest within stated limits.

A model qualifies as a system because its constructs interact to produce meaning or behaviour that is absent in any isolated construct. Ontology work makes this systemic property explicit, since coherence arises from the network of terms and relations rather than from any single definition (Rousseau, Billingham, and Calvo-Amodio 2018). The same constraint applies to a parametric model, a simulation codebase, or an interface model, where local changes can trigger global effects through dependency structure.

The system character of a model depends on boundary decisions that separate what is represented from what is treated as context. Rousseau, Billingham, and Calvo-Amodio (2018) describe ontology boundaries as scoped to a disciplinary interest and coupled to neighbouring ontologies through open, shifting interfaces that require ongoing management. A model boundary serves the same purpose by specifying which states and parameters are internal to the model, which variables are observed, and which interventions or control actions the model may represent.

Interpretation mapping converts a model from an abstract formalism into a representational system by linking constructs to claims about the system of interest. The mapping includes measurement conventions, aggregation rules, and assumptions about what counts as the same kind of thing across levels of description. This mapping often relies on intentional or informational vocabulary as shorthand for causal structure, such as beliefs, goals, signals, or messages, which Dennett describes as *loans of intelligence* embedded in explanatory schemes (Dennett 1971). Such loans can improve predictive leverage, but they also import commitments that must be tracked because they can be mistaken for mechanistic claims.

Statistical correspondence makes those commitments testable by turning the interpretation mapping into a probabilistic relation between model states and expected observations. In the free energy principle, a generative model specifies a joint probability distribution over hidden states and observed outcomes, and inference updates internal states to approximate the posterior distribution that would make observed outcomes unsurprising under the model (Friston et al. 2023b). The Markov blanket formalism expresses the boundary conditions under which internal and external states become conditionally independent of one another given the interface states, thereby making the mapping between the model and the world explicit in probabilistic terms (Friston et al. 2023b). This framing supports operational claims about a system of interest only when the model declares what it treats as hidden, observed, and controllable.

Stated limits constrain both the scope of prediction and the permissible interpretation of success or failure. Open boundaries in conceptual systems mean that categories and relations shift as information flows across domains, which requires continuous tracking of adjacent categories to preserve coherence over time (Rousseau, Billingham, and Calvo-Amodio 2018). Probabilistic models express an equivalent dependence on priors, likelihood structure, and data-generating assumptions, which can drift when the system of interest changes, invalidating inference without visible errors in the model artefact itself (Friston et al. 2023b). A model treated as a system, therefore, requires explicit conditions of applicability, along with mechanisms for detecting when those conditions no longer hold.

Treating models as systems also constrains governance. The model's purpose and accountability cannot be inferred solely from its internal structure, because the same

construct can support incompatible interpretations under different mappings. Dennett's argument that intentional character is attributed when it supports reliable prediction illustrates this stance dependence, and it places responsibility on the modeller to justify the chosen explanatory level against the predictions it enables (Dennett 1971). Prediction, therefore, operates as both a technical output and a social commitment to use the model within declared limits.

One implication of treating models as systems deserves emphasis because it shapes the rest of this chapter. A model, by definition, is simpler than the system of interest it represents. If it were not simpler, it would be a replica, not a model, and would require at least as much effort to interrogate as the original. The modeller's task is therefore to find a level of description at which the statistical regularities that matter for a given purpose become visible without requiring the modeller to track every configuration the system of interest can occupy. This is a compression problem: the model must discard information, and the quality of the model depends on discarding the right information.

The tension is structural, not incidental. A model that retains too much detail becomes intractable for the agents who must use it. A model that discards too much becomes unreliable when conditions change. The boundary between "enough" and "too much" is not fixed. It depends on who uses the model, what decisions it supports, and how quickly the environment can shift from the conditions the model was built to represent. For a small engineering team under time pressure, a formally correct but operationally opaque model is no better than no model at all. Conversely, a rough but legible approximation that a team can maintain, update, and act on under stress may outperform a precise model that no one consults when the situation deteriorates.

This observation has consequences that extend well beyond modelling methodology. If the right level of model complexity depends on the cognitive capacity of the agents who use the model, then the question of what a model should contain cannot be separated from the question of what the modelling team can attend to, communicate about, and act upon. Section 3.3 introduces systems thinking as the operational discipline for managing boundary and perspective choices. Section 3.4 then establishes the hard constraint: Rousseau's scientific systems principles, and in particular the principle of complexity dominance (SP3) with its corollary in Ashby's law, show that the tension between model complexity and agent capacity is not merely practical but formally unavoidable. A simpler subsystem interacting with a more complex one undergoes a proportionally greater degree of submergence, and no amount of investment can circumvent the requirement for requisite variety. This is the floor on which every subsequent argument in this chapter rests. Sections 3.5 and 3.6 then introduce the philosophical tools for operating productively under that constraint: loans of intelligence as the pragmatic device that allows modellers to proceed before every mechanism is understood, and the intentional stance as the specific loan that treats complex

systems as rational agents. Sections 3.7 and 3.8 reframe the question by showing that the “agents who use the model” are not isolated brains but extended, embodied cognitive systems distributed across people, tools, and their shared environment.

These constraints require modelling to use the same disciplined operations used to understand and intervene in other systems, including explicit boundary selection, relation tracing, and perspective management. Systems thinking supplies the operational discipline for managing these modelling decisions in a coherent and revisable way.

3.3 Systems Thinking

That discipline becomes a method only when systems thinking is treated as a purposeful inquiry that links a chosen boundary to intended intervention. General system theory frames systems as organised wholes whose properties depend on relations among parts and exchanges with the environment, shifting attention from components in isolation to patterns of interaction (Bertalanffy 1968). Systems thinking, therefore, treats synthesis as a technical operation, not a rhetorical preference. It expects modelling choices to be evaluated against the behaviours they make predictable and the interventions they make legitimate.

A systems explanation cannot be recovered by aggregating component explanations. Ackoff (1971) argues that improvement requires understanding the system that contains the component, because optimising a part can degrade the whole. This implies that analysis must move across levels of abstraction, selecting a level that preserves the interactions relevant to the question at hand. Hierarchical organisation supports this movement by allowing local descriptions to remain meaningful while higher-level dynamics constrain them (Simon 1962). These commitments constrain modelling practice to explicitly represent interfaces, couplings, and contextual dependencies, even when internal detail is abstracted.

Systems thinking also carries methodological pluralism. Checkland (1999) rejects the assumption that real-world problem situations come with a single agreed system definition and instead treats systems as conceptual devices used to structure debate and learning. Conceptual models in soft systems methodology are not claims about what exists, but structured propositions about purposeful activity that can be compared with experience and revised. Jackson (2003) similarly characterises systems thinking as a family of approaches whose suitability depends on the problem context and the distribution of power and values among stakeholders. For socio-technical infrastructures, this requires method choices that expose competing purposes and prevent an engineering model from becoming a disguised policy decision.

Boundary choice is therefore normative. Churchman (1968) treats the systems approach as inseparable from ethical judgment because every boundary excludes some consequences and some people from consideration, thereby turning technical success into social failure. His later discussion of the enemies of the systems approach emphasises how politics, organisational incentives, and moral disagreement defeat purely technical optimisation (Churchman 1979). Midgley's (2003) edited collection captures this boundary critique as a central theme in later systems thinking practice, linking methodological choices to accountability for those affected by intervention. Treating these issues as method steps constrains modelling governance to record boundary rationales and stakeholder commitments as first-class artefacts rather than informal background assumptions.

Systems thinking relies on recurring structural and behavioural patterns that travel across domains. Forrester (1961) formalises feedback, accumulation, and delay as generators of counter-intuitive behaviour through system dynamics, so causal claims must be tested against dynamic structure rather than narrative plausibility. Senge (1990) popularised these feedback structures as a discipline of organisational learning, including archetypal patterns in which local fixes worsen long-term outcomes. Pattern languages generalise the same idea of recurrence by treating patterns as reusable units that encode context, problem, and solution relationships without freezing implementation detail (Alexander 1979; Alexander et al. 1977). In modelling practice, these traditions justify the use of modelling patterns that capture feedback loops, coupling topologies, and failure archetypes as reusable, inspectable structures.

The ability to intervene also depends on control capacity. Ashby's (1956) law of requisite variety states that regulation requires a repertoire of responses at least as diverse as the disturbances the system faces. This constrains resilience claims, because a system that lacks monitoring, coordination, or actuation variety cannot maintain performance under diverse disruptions, regardless of component reliability. Systems engineering guidance positions systems thinking as a prerequisite for managing such cross-cutting concerns across life-cycle stages, where optimisation must be balanced against emergent risk and operational uncertainty (INCOSE 2012). Within digital engineering, these constraints require boundary assumptions, control structures, and pattern selections to remain machine-readable so that automated analysis can test whether the model's available variety matches the disruption space implied by its assurance claims.

Methodological use of systems thinking depends on a stable vocabulary of systems concepts and defensible generalisations about system behaviour under intervention. The compression problem identified at the end of Section 3.2, how to find the right level of simplification for a given purpose and a given team, is ultimately a question about the relationship between the variety of disturbances a system faces and the variety of responses its operators can

mobilise. Systems science and systems principles supply the deductive apparatus for making that relationship precise.

3.4 Scientific Systems Principles

Rousseau (2018a, 2018b) distinguishes scientific systems principles from the heuristic principles that currently underpin much systems engineering practice. Heuristic principles are descriptive and qualitative: they capture useful rules of thumb but do not explain why a system behaves as it does. Scientific principles, by contrast, are explanatory, general, and quantitative. They express fundamental assumptions about how the world is structured, and they become operational through the principles-laws-theories (PLT) chain: principles constrain what laws are possible, laws constrain what theories are admissible, and theories generate predictions that can be tested against observation (Rousseau 2018a). The PLT model also recognises feedback: when observations challenge a theory, the revision propagates upwards through laws to principles and, if necessary, to the underlying worldview.

Rousseau (2018c) derives three general systems principles that demonstrate this programme in action. The **principle of conservation of properties (SP1)** holds that emergent properties are balanced by submerged properties in the parts: the capacity gained at the system level is paid for by capacity lost at the component level. The **principle of universal interdependence (SP2)** holds that system properties reflect a balance between bottom-up emergence from parts and outside-in submergence from the supersystemic context, so that neither internal structure nor external environment alone determines system behaviour. The **principle of complexity dominance (SP3)** holds that in interactions between subsystems of unequal complexity, the simpler subsystem undergoes proportionally greater submergence, meaning that complex systems buffer their autonomy more effectively than simple ones. A direct corollary of SP3 is Ashby's (1956) law of requisite variety: *effective regulation requires internal diversity at least equal to the diversity of disturbances*.

These three principles transform the compression problem of Section 3.2 from a practical design challenge into a formal constraint. SP3 and Ashby's corollary together establish that a small engineering team confronting a complex cyber-physical system-of-systems is, by structural necessity, the simpler partner in the interaction. The autonomy cost falls disproportionately on the team, not because it lacks skill but because the variety asymmetry is built into the relationship. No amount of effort can produce a model as complex as the environment it represents, because a model that complex would require a modeller more complex still. The team must therefore compress: it must find the statistical regularities that matter for its purpose and discard the rest. The question that SP3 leaves open is how to

compress well, that is, how to proceed when the world is too complex for exhaustive analysis but action is nevertheless required.

What makes these principles scientifically productive, rather than merely descriptive, is the deductive structure they enable. SP1 reframes the familiar observation that emergence is accompanied by submergence as a quantifiable conservation relation expressed in terms of energy. SP2 establishes that explanatory arrows point both upward and downward from any system boundary, replacing classical reductionism with a testable claim about the relative weight of bottom-up and outside-in influences. SP3 yields a directional prediction: the autonomy cost of participating in a supersystem disproportionately falls on the less complex partner. Each principle, once formalised, constrains what laws can hold and what hypotheses are worth testing. The progression from qualitative insight to quantitative prediction follows the PLT chain, not from data accumulation, but from the logical elaboration of foundational commitments.

This epistemological structure matters because the domains in which systems principles must be tested, including organisational behaviour, infrastructure resilience, and socio-technical coordination, are precisely those in which large sample sizes are difficult to obtain and controlled experiments are impractical. The social sciences have historically sought legitimacy through statistical inference, treating large-N designs and significance testing as markers of rigour. The consequences are well documented: the replication crisis in psychology, medicine, and management research can be traced, in large part, to an overreliance on statistical measures as the arbiter of scientific status (Ioannidis 2005). The assumption that science requires large datasets and frequentist hypothesis testing conflates a particular method with the scientific enterprise itself.

Modern physics offers a different model. General relativity was published in 1915 and confirmed by Eddington's solar eclipse observation in 1919, but the direct detection of gravitational waves did not occur until 2016, a century after the theoretical prediction. The Higgs boson was predicted in 1964 and observed in 2012. In these cases, the scientific contribution was not the statistical test but the formal structure: a set of principles, a mathematically consistent model, and precisely stated predictions whose confirmation or refutation would be informative regardless of sample size. The rigour lies in the logical coherence of the deductive chain, not in the number of observations.

This thesis adopts Rousseau's PLT programme as its methodological framework. Chapter 4 develops a formal model that follows the PLT chain: foundational principles generate a mathematically consistent structure with proofs of internal consistency, and that structure generates testable hypotheses whose progressive empirical confrontation is the task of the empirical work in this thesis and of future work. But a deductive structure, however rigorous, does not by itself tell an engineering team what to attend to when the world exceeds its

modelling capacity. For that, a different kind of tool is needed: one that allows the team to proceed under uncertainty by making provisional attributions that can later be revised or replaced. Section 3.5 introduces that tool.

3.5 Loans of Intelligence

Section 3.4 established that the complexity asymmetry between a VSE and its environment is structural, not remediable. The team cannot build a model as complex as the world. Yet models must be built, decisions must be made, and systems must be delivered. Dennett (1971) provides the philosophical device that makes this possible: the *loan of intelligence*.

Dennett (1971) introduced the concept to describe the provisional use of assumptions in scientific models. These assumptions allow scientists to proceed by treating certain structures or relationships as functionally coherent, even when the details are not fully understood or justified. Referring to “command” and “control” in an engineering context, for instance, presumes a sender, a receiver, and a reliable process for interpreting and executing instructions. This is a *loan of intelligence*: a practical tool that makes a complex system tractable, not a claim about the underlying mechanisms. Every time an engineer says “the system detects an intrusion” or “the sensor reports a fault,” the same kind of loan is at work. The system does not literally detect or report. But treating it as though it does compresses the causal chain into a manageable description that supports action.

Rousseau’s (2018c) concept of systemic isomorphisms addresses a related challenge: finding general patterns that recur across different systems. Systemic isomorphisms help in formulating principles that can be generalised and applied across domains, even when the specifics of each system vary. Box’s (1976) aphorism captures the epistemological stance shared by both concepts: *all models are wrong, but some are useful*. A model that relies on loans of intelligence may be wrong about the mechanisms, yet useful for prediction and intervention. The loan is justified so long as it generates accurate expectations and the modeller remains aware that a debt has been incurred.

Both loans of intelligence and systemic isomorphisms serve as methodological scaffolding rather than final explanations. They make it possible to model and reason about complex systems, but each approach carries the risk of oversimplification if treated as definitive. The loans must ultimately be repaid by a more detailed account of the mechanisms involved, or at least recognised as methodological conveniences rather than permanent explanations. If a theory continues to rely indefinitely on such loans, it risks concealing rather than explaining the nature of the phenomenon it was meant to illuminate. The question, then, is what repayment looks like and what kind of formal apparatus can deliver it.

3.6 Intentional Systems

The intentional stance, introduced by Dennett (1971; 1981), is the most powerful and most common loan of intelligence. It is a pragmatic strategy for predicting and explaining the behaviour of complex systems by treating any system, whether human, animal, or machine, as if it were a rational agent with beliefs and desires. The observer attributes intentional states and then uses those attributions to anticipate the system's actions. A system qualifies as an intentional system only if this approach leads to accurate and useful predictions (Dennett 1971; Dennett 1981).

Dennett (1971) identifies three main strategies for predicting system behaviour: the physical stance, the design stance, and the intentional stance. The physical stance predicts behaviour based on the laws of physics and the system's physical constitution. The design stance ignores physical details and instead assumes that the system will function as designed, provided nothing goes wrong. The intentional stance is typically adopted when the complexity of the system makes physical or design-level analysis infeasible or unproductive (Dennett 1971). It is, in the terms established by Section 3.4, the response an agent adopts when SP3 makes exhaustive modelling impossible: attribute rationality, compress the description, and proceed.

When adopting the intentional stance, the observer attributes beliefs and desires to the system and assumes that it will act rationally to achieve its goals given those beliefs. Dennett (1981) makes it clear that this approach is not just a convenient fiction. The stance is justified if it reliably produces accurate predictions about the system's behaviour. This pragmatic justification applies equally to people, animals, computers, and organisations.

Importantly, the intentional stance does not require the system to possess subjective mental states. It is a methodological tool, not a claim about internal experience. The validity of the intentional stance is determined empirically: if this way of thinking about the system consistently leads to useful predictions, then it is appropriate to treat the system as intentional for explanatory purposes (Dennett 1981).

A large language model provides a clear example. When the underlying architecture and training data become too complex for detailed analysis, it is effective to treat the model as if it were a rational agent that “knows” facts, “understands” questions, and “wants” to provide coherent responses. Although the model is not conscious, adopting the intentional stance by asking what it would “believe” or “intend” in a given context often leads to accurate predictions about its output. This approach remains effective even when the underlying mechanisms are opaque (Dennett 1971).

One of the strengths of the intentional stance is its generality. Any system, whether biological or artificial, may be treated as intentional if this leads to successful prediction. This strategy leverages human strengths as social and interpretive beings, allowing us to make sense of behaviour in complex systems without requiring full knowledge of the underlying mechanisms. However, Dennett (1971) cautions that whenever we use intentional concepts such as belief, desire, or purpose, we are taking out a loan of intelligence. We are temporarily attributing a degree of rationality or agency to the system in order to make its behaviour comprehensible (Dennett 1971). These loans must ultimately be repaid by a more detailed account of the mechanisms involved, or at least recognised as methodological conveniences rather than final explanations. If a theory continues to rely indefinitely on such loans, it risks concealing rather than explaining the nature of intelligence.

The question of repayment raises a further question: repaid with what? The intentional stance simplifies by treating the system as a rational agent, but it says nothing about what the agent is made of, where its boundaries lie, or how its intelligence is physically realised. The next two sections address exactly this question by arguing that the cognitive resources available for repayment extend beyond the individual brain. If intelligence is distributed across brain, body, tools, and the designed environment, then the mechanisms that repay a loan of intelligence may themselves be distributed, embedded in artefacts and practices rather than confined to neural processes. This changes both the kinds of repayment that are possible and the kind of formal apparatus needed to deliver them.

3.7 The Extended Mind

The traditional view in cognitive science draws a sharp boundary at the skin and skull, treating all cognitive processes as internal to the brain. Clark et al. (1998) challenge this assumption, arguing that the mind can extend into the world when external resources are properly integrated and reliably available. This position, known as active externalism, holds that environmental features can become genuine parts of cognitive processes when they are sufficiently well coupled to the agent.

On this view, cognition can be distributed across the brain, body, and world. For example, when playing Tetris, a person may mentally rotate a shape or press a button to rotate it on the screen. Both strategies contribute to problem-solving in functionally similar ways. The use of external resources, such as tools, artefacts, or environmental features, supports thinking by directly participating in the cognitive process. Clark et al. (1998) argue that as long as an external resource is integrated into the cognitive loop and reliably available, it should be considered part of the cognitive system.

To illustrate, Clark et al. (1998) compare two individuals seeking directions to a museum. Inga consults her biological memory to recall the address. Otto, who suffers from memory impairment, writes the address in a notebook and consults it when needed. Otto's notebook is always available and trusted, just as Inga relies on her memory. The notebook does not merely assist Otto but serves as an external memory store. The beliefs Otto realises through this process are as much a part of his cognitive life as those stored in biological memory. According to the extended mind thesis, cognitive processes cross the boundaries of brain and body whenever external structures play the same functional role and are reliably coupled to internal states.

Constant et al. (2020) further distinguish between cognitive offloading and uploading. Offloading occurs when individuals use external artefacts to reduce the load on internal cognitive resources or increase reliability. Writing a shopping list or using a calculator are standard examples. The agent delegates part of the cognitive function to the environment. Otto's notebook is offloading: the memory function remains his own, merely externalised onto paper. Uploading, by contrast, refers to situations in which the cognitive niche itself takes over a function that was previously realised internally. The function is no longer merely supported by an external aid but is maintained by the sociotechnical infrastructure at a population level. Consider the street numbering system that both Otto and Inga rely on to find the museum. Neither needs to know the city's geography because the numbering convention, maintained by municipal authorities and refined over generations, encodes the city's spatial logic into the built environment. The wayfinding function has migrated from individual knowledge to a shared infrastructure that every resident and visitor inherits simply by learning the convention. Satellite navigation extends this logic further. The route-planning function, once distributed between the individual's spatial knowledge and the street signs, now resides almost entirely in an infrastructure of positioning satellites, crowd-sourced traffic data, and continuously updated road networks that no single individual or organisation could maintain alone.

Constant et al. (2020) clarify that offloading and uploading operate on different timescales and levels of integration. Offloading happens in real time or over an individual's development, while uploading involves a transfer of function that persists across individuals or generations, forming a stable, shared cognitive infrastructure. Uploading is a stronger and more enduring form of cognitive extension.

The extended mind thesis opens a critical question for the argument of this thesis. If cognition can cross the boundary of skin and skull, then the "agents who use the model" from Section 3.2 are not individual brains but systems that include notebooks, calculators, dashboards, and shared documentation. But the extended mind thesis, as originally formulated, relies on a functional parity argument: the external resource counts as cognitive if it plays the same role as an internal process would. It does not explain the mechanism by

which external resources become integrated into the cognitive loop, nor does it explain why some environmental configurations sustain coordinated behaviour while others allow drift. Answering these questions requires a broader account of how intelligence relates to the body and the environment, and of the coupling between them.

3.8 The Embodied Mind

Traditional programmes in artificial intelligence and cognitive science treated intelligence as the manipulation of internal symbolic representations: sensory data in, symbols processed, motor commands out. Varela et al. (1991) challenged this framework at its root. In their formulation of enaction, cognition is not the recovery of a pre-given world through internal computation but the bringing forth of a domain of significance through the organism's embodied activity and its history of structural coupling with the environment. There is no cognition without a body, and no body without an environment that both shapes and is shaped by it. The implication is that the unit of analysis for cognitive science cannot be the brain in isolation. It must be the organism-environment system as a whole.

This philosophical reorientation found dramatic practical confirmation in robotics. Brooks (1991) demonstrated that mobile robots could navigate complex, unstructured environments with far less internal machinery than the classical programme assumed was necessary. His subsumption architecture decomposed behaviour into layered, concurrent modules that coupled sensors directly to actuators, with higher layers modulating but not commanding lower ones. The robots Brooks built at the MIT Artificial Intelligence Laboratory in the late 1980s and early 1990s could wander office corridors, avoid obstacles, and follow moving targets, not because they computed exhaustive representations of their surroundings but because their intelligence emerged from the dynamic interaction between simple behavioural modules and a structured physical environment. Brooks's conclusion was pointed: much of what had been attributed to internal representation was actually performed by the world itself, used as "its own best model" (Brooks 1991).

Clark (1997) synthesised these developments in a broader argument that cognition is fundamentally situated, embodied, and scaffolded by the environment. Drawing on Brooks's robotics, Varela, Thompson and Rosch's enactivism, and Gibson's (1979) ecological psychology, Clark argued that brains are not self-contained inference engines but controllers for embodied action, evolved to exploit bodily and environmental structure rather than to build exhaustive internal models. The mind, on this account, is best understood as the activity of an essentially situated brain, at home in its proper bodily, cultural, and environmental niche (Clark 1997).

Breazeal, Brooks's doctoral student at the MIT Artificial Intelligence Laboratory, extended the subsumption approach from physical to social coupling. Her robot Kismet used layered behavioural systems structurally similar to Brooks's architecture, but its sensors were tuned to faces, vocal prosody, and turn-taking rhythms rather than to sonar returns and infrared proximity (Breazeal 2003). Kismet did not model human intentions symbolically. It coupled to social signals through affect-driven regulatory mechanisms that modulated approach, withdrawal, and engagement behaviours in real time (Breazeal 2002). The result was interaction that observers consistently described in infant-caregiver terms: Kismet appeared to seek attention, respond to soothing vocal tones, and disengage when overstimulated (Breazeal et al. 2000). The theoretical implication is direct. If navigating a physical space requires no internal map because the environment itself provides the structure that guides behaviour, then navigating a social space may work the same way, provided the agent is equipped to couple to the relevant regularities. Breazeal's work demonstrated that the embodied, environmentally coupled model of intelligence extends naturally from the physical to the social domains, a step that matters for this thesis because the engineering teams it studies operate in both domains simultaneously.

The consequences for understanding collective intelligence are equally significant. Hutchins (1995) studied the navigation teams aboard US Navy ships and showed that the cognitive properties of the team as a system are qualitatively different from those of any individual member. No single person on the bridge holds a complete representation of the ship's position. Instead, the computation of position is distributed across people, instruments, charts, and standardised procedures in a tightly choreographed cycle that repeats every three minutes. Hutchins, himself an open-ocean racing sailor and navigator, observed that the intelligence of the system resides not in any head but in the coordinated flow of representational state across a variety of media. The single ant wandering a featureless surface achieves little. The colony, through stigmergic traces and local interaction rules, constructs elaborate structures that no individual ant represents or plans (Bonabeau et al. 1999). The notebook that extends Otto's memory (Clark et al. 1998) becomes, at the scale of a professional team, a system of shared artefacts, practices, and environmental modifications that constitute a collective cognitive architecture.

Clark (2016) subsequently recast this tradition within the predictive processing framework, providing the mechanistic account that earlier embodied and distributed approaches had lacked. In this account, the brain is not a passive receiver of sensory data but an active prediction machine. It maintains a generative model of the world and continuously generates top-down predictions about incoming sensory signals. Perception occurs not when raw data arrives but when the brain computes the difference between its prediction and the actual signal. This difference, the prediction error, is the only information that propagates upward through the processing hierarchy. When predictions are accurate, little error remains, and the system operates efficiently. When predictions fail, large errors drive belief revision, learning, or action.

Two features of this framework are central to the argument developed in this thesis. The first is precision-weighting. Not all prediction errors are treated equally. The brain estimates the reliability of each error signal and adjusts the gain accordingly: a signal judged to be precise (low variance, clear contrast, unambiguous source) receives greater weight in inference, while a noisy or ambiguous signal is attenuated (Feldman et al. 2010). Attention, on this view, is not a separate faculty but the process of adjusting precision estimates across competing channels. What an agent attends to is determined by which signals the system treats as most reliable, and this determination can be influenced by context, training, and the structure of the environment itself.

The second feature is the dual strategy of active inference. A prediction error can be resolved in two ways: the agent can revise its internal model to accommodate the unexpected signal (perceptual inference), or it can act on the world to bring sensory input into line with its predictions (active inference). Clark (2016) emphasises that these two strategies share the same computational logic and the same objective, the minimisation of prediction error, but operate in opposite directions: perception changes the model to fit the world, while action changes the world to fit the model. For a resource-constrained agent that cannot afford separate planning, monitoring, and execution systems with expensive cost functions, this unification is significant. A single prediction-driven regime handles perception, learning, and action selection within the same inferential loop (Friston et al. 2011). Brooks's (1991) robots succeeded for exactly this reason: by coupling perception directly to action through the environment, they achieved with minimal internal machinery what classical planners failed to achieve with elaborate symbolic reasoning.

These mechanisms extend naturally from individual cognition to collective and environmental settings. Clark (2016) argues that human cognitive distinctiveness lies not in raw processing power but in the capacity to construct what he terms “designer environments” that restructure the statistical regularities to which predictive brains are exposed. Cultural practices, on this reading, function as surprise-minimisation devices operating at extended spatial and temporal scales. White lines painted on a winding cliff-top road, for example, allow a driver to solve a complex navigation problem by tracking a simple, high-precision optical feature. The navigation strategy that depends on those lines is only available because the environment has been deliberately modified (Clark 2016). The modification does not make the driver smarter in any internal sense. It makes the task tractable by cueing a lower-cost cognitive strategy through environmental design. Hutchins's (1995) navigation team operates by the same logic: the charts, instruments, and standardised verbal protocols are designed environments that structure the team's collective inference, making the right information salient at the right moment without requiring any individual to hold the full picture.

Clark (2016) describes this broader principle as “productive laziness,” drawing on Pfeifer and Bongard's (2006) *Principle of Ecological Balance*: adaptive success requires a match

between the complexities of an agent's sensory, motor, and neural systems and a balance in the distribution of problem-solving across brain, body, and world. The predictive brain is not a model-heavy reasoning engine but an efficiency-seeking system that preferentially trades expensive internal computation against cheap action and environmental structure. Simple heuristics are not deviations from optimal inference but rational responses selected by precision-weighting when the task and context indicate that quick, approximate responses are adequate.

The embodied mind framework also addresses multi-agent coordination. Clark (2016) observes that each agent in a group uses its own generative model to predict others' behaviour, and that agents actively constrain their own behaviour to make themselves more predictable to peers. Social norms function as probability distributions that narrow the range of expected behaviour, and structured verbal exchanges achieve what Roepstorff et al. (2004) call "top-top control of action," enabling high-level understandings to be transferred directly through language rather than through slow experiential learning. This mechanism is orders of magnitude more efficient than individual trial-and-error and provides the cognitive foundation for why structured team practices can rapidly install complex shared understandings. The progression from Brooks's solitary robot navigating corridors to Breazeal's sociable robot navigating faces and voices to Hutchins's navigation team to Clark's designer environments traces a single intellectual arc: intelligence is not a property of an isolated processor but an achievement of a well-coupled system of agent, body, artefact, and world.

Finally, Clark (2016) revisits the extended mind thesis through the lens of predictive processing. In this updated account, an external resource becomes part of the cognitive system not merely when it is reliably coupled (the original parity criterion of Clark et al. 1998) but when the brain's prediction machinery assigns high precision to signals originating from that resource. A monitoring dashboard, an automated alert system, or a structured playbook is incorporated into the agent's inference loop when the system treats its outputs as reliable evidence, weighting them accordingly in belief updating. This reframing provides a formal criterion for cognitive extension, grounded in precision-weighting rather than relying solely on functional-parity arguments that have attracted sustained philosophical criticism.

3.9 Purposeful Human Activity Systems (PHAS)

The embodied mind framework establishes that intelligence is distributed across the brain, body, and environment. The extended mind thesis and its predictive-processing update identify the conditions under which external resources become genuine participants in cognition. What neither framework provides is a structural vocabulary for describing the specific kind of system that an engineering team constitutes: a system with parts, purpose, a

boundary through which it interacts with its environment, and relationships conditioned by context and language. The PHAS framework fills that gap.

Calvo-Amodio and Rousseau (2019) introduced the idea of human activity systems (HAS), recognising that engineering teams are socio-technical systems shaped by people and their behaviour. These teams rely on advanced technical expertise to design, model, and build complex solution systems. In later works, Calvo-Amodio refined the concept to purposeful human activity systems (PHAS), highlighting that PHAS are intentionally engineered systems with a specific purpose (Calvo-Amodio 2019, 2021). The PHAS model and framework were created to support the design and management of human activity systems by accounting for their emergence and development, focusing on four main elements shown in Figure 1.

- **System:** PHAS consist of conceptual and concrete parts. These parts interrelate to create persistent structures, processes, and meanings.
- **Purpose:** PHAS are designed to fulfil specific purposes. They possess inherent causal powers to achieve these purposes, are designed to align with their objectives, and are consciously aware of and intentionally pursue their goals.
- **Boundary:** PHAS interact with their environment through their boundary. This boundary allows the system to both submerge into and influence its environment.
- **Relationships:** The nature and strength of the relationships between parts, purpose, and boundary in PHAS depend on the context and are influenced by the language used.

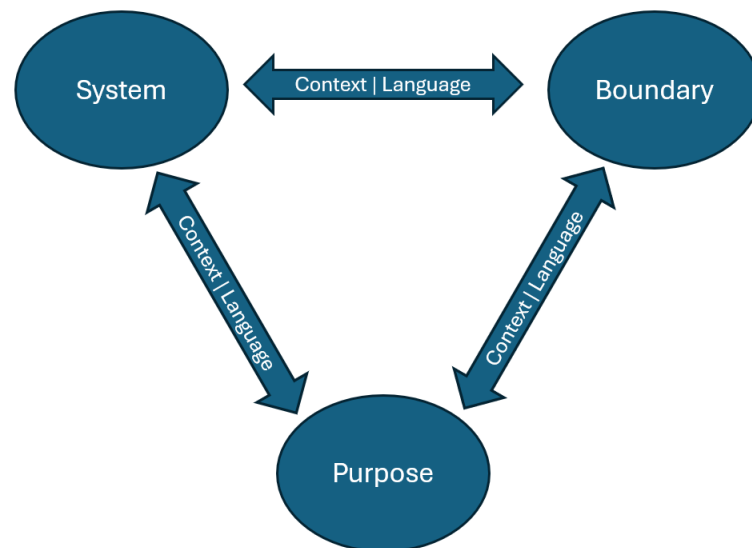


Figure 6 The four main elements of the PHAS model: system, purpose, boundary, and relationships

Source: (Calvo-Amodio, 2019).

The core concepts were subsequently developed into the four main principles and corresponding sub-principles listed in Table 10.

Table 10 PHAS principles and sub-principles

Component	Principle and Sub-principles
System	1. HAS have parts: (a) parts are conceptual and concrete; (b) parts interrelate to produce persistent structures, processes, behaviours, and meanings; (c) interrelations are conditioned by kind, capability, and structure.
Purpose	2. HAS exist to fulfil a purpose: (a) possess inherent causal powers to fulfil purpose; (b) design is commensurate with purpose; (c) are aware of and intentionally pursue their purposes.
Boundary	3. HAS mediate interactions through boundary: (a) system submerges to its environment; (b) system influences its environment.
Relationships	4. Context and language modulate relationships: (a) magnitude and kind vary according to context; (b) magnitude and kind are conditioned by the language used.

Source: (Calvo-Amodio, 2019).

When Calvo-Amodio and Rousseau (2019) first introduced the PHAS model, it was largely theoretical. In a later article, Calvo-Amodio redefined the acronym from HAS to PHAS and described how the PHAS model can serve as a practical framework (Calvo-Amodio, 2019). He ultimately structured this framework using a visualisation adapted from Checkland (2019), based on Vickers' appreciative systems approach (Vickers, 1983). Figure 2 presents Calvo-Amodio's revised framework:

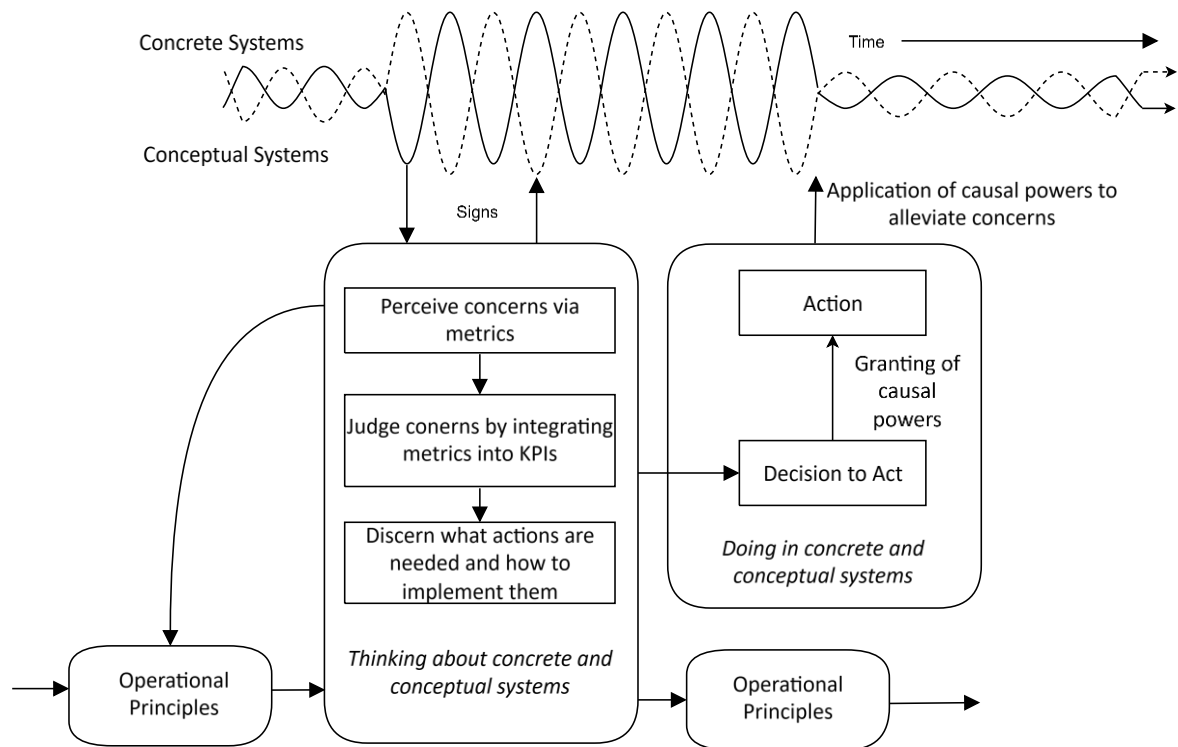


Figure 7 PHAS framework structured using a visualisation adapted from Checkland (2019), based on Vickers' appreciative systems approach

Source. (Calvo-Amodio, 2019).

Table 11 lists the inquiry questions used by Calvo-Amodio to operationalise the PHAS framework:

Table 11 Inquiry questions for the PHAS framework

Inquiry Question	Information Components
1. What needs to be true for this concern to be addressed?	What is useful to know about the key activity? What type of change might be needed? Does the metric provide an accurate measurement?
2. Who cares about this concern and why?	Who finds the concern meaningful? Why? How does the concern relate to social, organisational, or psychological dimensions?
3. What is good enough for this concern to be satisfied?	What is the best balance between effort needed and burden on the organisational system? Does the solution align with PHAS principles and capability?

Source: (Calvo-Amodio, 2019).

The PHAS framework provides the structural and purposive vocabulary that the embodied mind tradition lacks: parts, purpose, boundary, relationships, and inquiry questions. But the PHAS framework, as originally formulated, identifies the components of an engineering team without specifying the mechanisms by which those teams' cognitive resources are mobilised, coordinated, and sustained under uncertainty. The next section reads the PHAS as an extended embodied mind, connecting the structural vocabulary of 3.9 to the cognitive mechanisms of 3.8 and the formal constraints of 3.4, before Section 3.11 introduces the formal syntax that ties these threads together.

3.10 Purposeful Human Activity Systems as Extended Embodied Minds

The claim of this section is that the PHAS framework and the embodied mind tradition describe the same phenomenon at different levels of resolution. A PHAS is an extended embodied mind.

Consider what a PHAS contains. Its members are predictive agents who maintain generative models of their work, weigh incoming signals by estimated precision, and select actions to minimise expected surprise. Its tools, monitoring systems, documentation, and physical workspace are not passive repositories but active participants in the team's inferential loop, contributing precision-weighted signals that shape collective belief updating. Its practices,

whether design reviews, threat-modelling games, or daily stand-ups, are structured verbal exchanges through which high-level understanding is efficiently transferred between individuals. Hutchins (1995) demonstrated precisely this structure in the bridge crew of a warship: the team's ability to fix position and navigate safely is a system-level cognitive property that no individual member possesses and that cannot be recovered by summing individual competences. The PHAS framework gives this observation a formal home by identifying the structural elements (parts, purpose, boundary, relationships) and the inquiry questions (what needs to be true, who cares, what is good enough) that make the system amenable to analysis and intervention.

Reading the PHAS as an extended embodied mind brings Ashby's law of requisite variety into immediate focus. To regulate its environment, a system must possess internal variety at least equal to the variety of disturbances it faces (Ashby 1956). Rousseau's principle of complexity dominance (SP3) sharpens the point: in interactions between subsystems of unequal complexity, the simpler subsystem undergoes a proportionally greater degree of submergence (Rousseau 2018c). A VSE with five engineers confronting a cyber-physical system-of-systems is the simpler partner. The autonomy cost falls disproportionately on the small team.

The implication is that attempting to match the environment's complexity through exhaustive modelling and comprehensive procedural coverage is formally impossible. A model as complex as the environment would require a modeller more complex than the environment being modelled (Ashby 1956). This was the tension identified at the end of Section 3.2: the modeller's task is to find the level of description at which relevant regularities become visible without being drowned out by noise. For a PHAS understood as an extended embodied mind, this is not an abstract methodological concern but an existential constraint on the team's viability.

A sailing crew operates under exactly this constraint, and it is no coincidence that the tradition's most detailed empirical study of distributed cognition was conducted by a scholar who was himself an open-ocean racing navigator (Hutchins 1995). The crew cannot control the sea. The wind shifts, the current changes, and waves arrive in patterns that no finite model can fully predict. A crew that attempted to compute the exact state of every wave and gust would be paralysed by the time its calculations were complete. What a professional crew does instead is maintain a shared mental model of the boat, the wind, the current, and the course that is good enough to act on, and continuously update that model through high-bandwidth verbal exchange, physical proprioception, and instrument readings. The helmsman calls the puffs. The trimmer adjusts before being told, because a shared generative model of the wind enables anticipation. The bowman reports what the crew cannot yet see.

The shared model is wrong. As Box (1976) reminds us, all models are wrong, but some are useful. The crew's model is useful because it compresses the environment's overwhelming variety into a tractable set of actionable distinctions. Every simplification in that model is a loan of intelligence in Dennett's (1971) sense: the crew attributes coherent intent to the wind pattern ("the breeze is building from the left") as a heuristic device that supports action without requiring a full meteorological model. The attribution is useful precisely because it is provisional. When the wind does something the model did not predict, the prediction error is large. It receives high precision-weighting because it comes from a trusted channel (the helmsman's direct sensory experience). The shared model updates rapidly. The crew does not abandon the model. It revises it, preserving what works and correcting what failed.

The critical constraint is that even a well-coordinated crew, operating as an extended embodied mind distributed across people, instruments, and practised routines, cannot exceed the variety that Ashby's law permits. The crew's internal variety is finite: bounded by the number of people, the precision of their instruments, the bandwidth of their communication, and the depth of their shared training. What the crew can do, and what distinguishes a professional crew from a novice one, is maximise the mobilisable fraction of that finite variety. This means investing not in ever-more-elaborate models but in the capacity to respond quickly and appropriately to what actually happens. It means designing the boat, the instruments, and the communication protocols so that the right information reaches the right person at the right time, in a form that is immediately actionable. It means building practices that sustain shared attention on the signals that matter most, so that when the environment surprises, the crew is already oriented toward the relevant response.

This is the operational meaning of *surfing uncertainty* (Clark 2016). The PHAS cannot control its environment. It cannot build a model complex enough to predict every disruption. What it can do is design itself, its tools, its practices, and its communication structures, so that it stays in the pocket: continuously adjusting, continuously updating its shared model, continuously reallocating attention as new prediction errors arrive.

The chapter has now assembled the components. Systems thinking provides the discipline for managing boundaries and complexity (Section 3.3). Scientific systems principles establish the hard constraint that complexity dominance and requisite variety are formally unavoidable (Section 3.4). Loans of intelligence and the intentional stance provide the philosophical warrant for operating under that constraint through necessary simplification (Sections 3.5 and 3.6). The extended and embodied mind traditions hold that the cognitive resources for navigating complexity are distributed across the brain, body, tools, and environment (Sections 3.7 and 3.8). The PHAS framework identifies the team's structural elements as a purposeful system (Section 3.9). What remains is the formal syntax that converts these insights into a testable mathematical structure, the kind of intellectual scaffolding that Clark (2016) argues well-designed environments provide for their inhabitants. The next section

introduces the free-energy principle and extended active inference as that scaffolding. The formal model developed in Chapter 4 then operationalises it into fourteen testable hypotheses, and Section 5.2 shows that the resulting account of mobilisable variety under time pressure is precisely the resilience advantage that the sailing analogy makes intuitive but that only the formal model can make rigorous.

3.11 Free-Energy and Extended Active Inference

The PHAS framework, as originally formulated (Calvo-Amodio 2019), identifies operational principles as activity drivers and provides inquiry questions for concern judgment, but it does not specify how attention is weighted among competing signals, how shared expectations emerge from individual interactions, or why some environmental configurations sustain coordinated behaviour while others allow drift. Section 3.8 introduced precision-weighting and active inference as the cognitive mechanisms that explain these phenomena at the level of individual and collective behaviour. This section introduces the formal syntax that makes those mechanisms mathematically precise.

In Friston's formulation of the *free-energy principle* (FEP), any self-organising system at equilibrium with its environment must minimise variational free energy, a bound on surprise under a generative model (Friston 2010). Recent work restates this as a normative account of self-organisation and decision-making that can be read as maximising Bayesian model evidence (Friston et al. 2023b).

Active inference provides the process model. Beliefs are updated by minimising variational free energy, while actions are selected to minimise expected free energy by balancing preferred outcomes and uncertainty reduction (Friston et al. 2023b). Attention follows from the same mathematics as inference about uncertainty, via precision-weighting of prediction errors, and has been used to explain behavioural and electrophysiological effects in cueing tasks (Feldman et al. 2010). The framework has demonstrated strong predictive power across a wide range of applications, including perception (Hohwy et al. 2008), attention (Feldman et al. 2010), learning (Friston et al. 2016), emotions (Van de Cruys et al. 2011), action (Friston et al. 2011), cultural processes (Ramstead et al. 2016), niche construction (Constant et al. 2018, 2020) and natural selection (Campbell 2016).

In this thesis, PHAS is treated as a policy selection problem under uncertainty, in which coordination depends on shared generative models and on the control of precision in communication and tooling. The precision-weighting mechanism is central to this framing because it begins to repay the loans of intelligence described in Sections 3.5 and 3.6. When the intentional stance is adopted toward a PHAS, attributing beliefs, goals, and rationality to an engineering team remains a methodological convenience until a mechanism can explain

why the team attends to some signals and not others. Precision-weighting provides exactly that mechanism: signals with high estimated precision receive greater weight in belief updating and action selection, while noisy or ambiguous signals are attenuated (Feldman et al. 2010). This is a mathematical account, not a metaphor. The formal model developed in Chapter 4 inherits this property directly. Observation noise ω , shared expectations ϕ , and authority weights γ_i all modulate precision in ways that can be derived from the free-energy equations rather than asserted by analogy. Each equation, therefore, converts a heuristic ascription of purposiveness into a testable structural claim, progressively reducing the outstanding theoretical debt.

Extended active inference (EAI), introduced by Constant et al. (2020), completes the connection to the extended mind perspective presented in Section 3.7. Constant et al. exploit the formal symmetry of a Markov blanket to show that the same variational formalism governing internal belief updating also describes changes in external environmental states. The niche encodes a generative model of the agents that inhabit it, with the agents' actions serving as the niche's sensory data and environmental affordances emerging as the niche-side analogue of surprisal. Cognitive uploading, the stronger form of cognitive extension that Clark et al. (1998) describe, is therefore not merely a philosophical claim about functional parity but a formal property of joint agent-environment free-energy minimisation. For this thesis, that means that the tools, documentation, automated alerts, and machine-readable models that constitute the designed cognitive reserve (h) in the formal model below participate in the same inferential dynamics as the team's internal beliefs. They are not passive repositories of information but active contributors to a coupled system that jointly minimises surprise.

FEP and EAI together provide the intellectual scaffolding that the rest of this chapter has been building towards. Clark (2016) argues that well-designed environments provide scaffolding for the predictive brains that inhabit them, cueing low-cost cognitive strategies and making critical signals salient. The formal model plays exactly the same role for the thesis itself: it provides the mathematical structure that converts the philosophical insights of Sections 3.5 through 3.8, the structural vocabulary of Section 3.9, and the operational intuitions of Section 3.10 into a testable framework with precisely stated hypotheses. The crew of the sailing boat knows intuitively what works. The formal model explains why it works, under what conditions it stops working, and what interventions can extend its effectiveness. Chapter 4 defines the research design required to operationalise this framing.

4 Research Design and Methodology

This chapter specifies the research design. Section 4.1 sets out the design philosophy. Section 4.2 presents the formal analytical model. Section 4.3 describes the qualitative case-study strand. Section 4.4 explains how the two strands are integrated. Section 4.5 addresses the ethical framework and the strategies used to establish research quality across both strands.

4.1 Design philosophy

The research design follows from three commitments established in Chapter 3.

The first is the principles-laws-theories (PLT) programme (Rousseau 2018c). Formal models are treated as deductive structures whose internal consistency can be verified independently of data and whose empirical adequacy is tested through progressive confrontation with case evidence. This chapter develops the model. Chapter 5 analyses its structure, proves internal consistency, and derives testable hypotheses. Chapter 6 confronts those hypotheses with field evidence.

The second is a commitment to *depth over breadth*. A single-company, longitudinal design with an embedded researcher yields an audit trail dense enough to trace model predictions to specific observable events, at the cost of generalisability across populations. This trade-off is *deliberate*. The formal model carries the generalisable structure through its deductive apparatus. The case studies test whether that structure produces recognisable predictions in a concrete VSE setting. The combination is closer to what Yin (2018) describes as analytic generalisation, where cases test theoretical propositions, than to statistical generalisation from a representative sample.

The third follows from Chapter 3's argument that cognition extends beyond the individual brain into the tools, artefacts, and social structures that constitute the working environment (Clark 1997, 2016; Hutchins 1995). If engineering teams are extended embodied minds, then studying them from the outside, through surveys or periodic interviews, risks missing the very phenomenon the thesis seeks to explain. The distributed cognitive processes that sustain or fail to sustain attention under pressure are visible only from within the system, through the artefacts it produces, the communication it generates, and the environmental structures it builds and inhabits. This commitment motivates the embedded, participatory research design described in Section 4.3 and shapes the integration strategy described in Section 4.4.

4.2 Analytical Models

This chapter presents the analytical framework used in the thesis. The objective is to provide a formal language that bridges the cognitive dynamics of engineering teams and the structural realities of their context. The framework integrates the PHAS Extended Active Inference (PHAS-EAI) model from Paper V, which describes the system's dynamics using the free-energy principle (FEP), with a kinematic model of complexity and resilience that explains the system in terms of configuration spaces and functional information.

4.2.1 Analytical Model Scope

The formal work comprises two coupled layers that address different parts of the same dependability problem.

1. **PHAS-EAI (purposeful human activity systems with extended active inference)** formalises how individuals and teams perceive, update beliefs, act, communicate, and stabilise shared expectations under uncertainty, using variational free energy as the organising quantity.
2. **Complexity, resilience, and functional information** formalise the option space that a socio-technical system can realise, the time-bounded portion of that option space that is mobilisable during disturbances, and the rarity of configurations that satisfy a function at a given threshold.

Taken together, they provide a single analytic chain from (a) uncertainty at organisational boundaries and in peer interaction, through (b) attention and coordination dynamics, to (c) measurable resilience outcomes under time pressure.

4.2.2 PHAS-EAI Model

The following equations define the PHAS-EAI model that the formal analysis in Chapter 5 operates on. The presentation establishes the notation and the relationships between quantities. Paper V provides the full derivation, interpretation, and empirical grounding.

4.2.2.1 Generative Model

$$p(s, \eta) = p(s \mid \eta) p(\eta) \quad (1)$$

In Equation (1), $p(\cdot)$ denotes probability, s denotes observed signs, and η denotes external world states. The left side $p(s, \eta)$ is the joint probability of signs and states. The term $p(s | \eta)$ is the likelihood of signs given states, and $p(\eta)$ is the prior over states.

4.2.2.2 Perceptual Uncertainty

$$s = g(\eta) + \omega \quad (2)$$

In Equation (2), $g(\cdot)$ is the observation mapping from external states to the sensed signal s , and ω is a noise term. Equation (2) states that sensing is indirect because observations are transformed and noisy.

4.2.2.3 Free Energy and Surprise

$$F(s, \mu) = D[q_\mu(\eta) \parallel p(\eta | s)] - \ln p(s) \quad (3)$$

Equation (3) introduces $F(s, \mu)$ as free energy with μ as the internal model. It uses $D[\cdot \parallel \cdot]$ as a divergence measure between the approximate posterior $q_\mu(\eta)$ and the posterior over external states given signs, $p(\eta | s)$. The term $-\ln p(s)$ is the surprise component. The role of Equation (3) is to combine a mismatch term and a surprise term into a single objective.

4.2.2.4 Optimisation as Alignment

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \parallel p(\eta | s)]}_{\text{Divergence}} - \underbrace{\ln p(s)}_{\text{Surprise}}, \quad \mu^* = \underset{\mu}{\operatorname{argmin}} \text{Divergence} \quad (4)$$

Here, μ^* is the optimised internal model and argmin returns the argument that minimises the target term. Equation (4) states that belief update is represented as divergence minimisation.

4.2.2.5 Action Selection

$$a^* = \underset{a}{\operatorname{argmin}} F(s(a), \mu) \quad (5)$$

In Equation (5), a is a candidate action, $s(a)$ is the sign stream expected under action a , and a^* is the selected action. Equation (5) selects the action that minimises expected free energy.

4.2.2.6 Complexity-Accuracy Form

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \parallel p(\eta)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{q_\mu(\eta)}[\ln p(s | \eta)]}_{\text{Accuracy}}, \quad a^* = \underset{a}{\operatorname{argmax}} \text{Accuracy} \quad (6)$$

Equation (6) introduces $\mathbb{E}_{q_{\mu}(\eta)}[\cdot]$ as expectation under the approximate posterior, and argmax as the operator that returns the argument that maximises the target term. Equation (6) rewrites free energy as complexity minus accuracy and expresses action choice as expected-accuracy maximisation.

The prior $p(\eta)$ appears in place of the posterior $p(\eta | s)$ used in Equation (4) because actions cannot alter internal beliefs directly. Actions influence free energy only by changing the sensory input the system receives. Preferred actions are therefore those that reduce the expected ambiguity of future input, selecting outcomes that maximise expected accuracy rather than revising the model itself. This asymmetry between perception and action is central to active inference: perception updates the internal model to fit the world, while action changes the world to fit the internal model.

4.2.2.7 Niche Construction

$$F(a, \eta) = D[q_{\eta}(\mu) \parallel p(\mu | a)] - \ln p(a) \quad (7)$$

Equation (7) exploits the formal symmetry of the Markov blanket (Constant et al. 2020) by reversing the perspective of Equation (3). Where Equation (3) describes an agent minimising surprise about its sensory input, Equation (7) describes the environment minimising surprise about the agent's actions. The term $q_{\eta}(\mu)$ is a distribution over agent internal states from the environment side, $p(\mu | a)$ is the conditional model of internal states given action, and $p(a)$ is the action probability term.

$$F(a, \eta) = \underbrace{D[q_{\eta}(\mu) \parallel p(\mu | a)]}_{\text{Divergence}} - \underbrace{\ln p(a)}_{\text{Affordance}}, \quad \eta^* = \underset{\eta}{\text{argmin Divergence}} \quad (8)$$

Here, η^* is the optimised external-state configuration. The term $-\ln p(a)$, labelled *Affordance*, captures how likely or natural a given action is within the current environment. From the environment's perspective, this term functions as a mathematical model of affordance (Constant et al. 2020): the environment "learns" which actions are probable given its states and the agent's behaviour, and stabilises configurations that make those actions easier to execute, and thus more likely to be executed. The symmetry between Equations (3)/(4) and Equations (7)/(8) demonstrates that a PHAS and its environment can become mutually predictable, forming a joint system in which both sides influence each other through mutual self-evidencing. This symmetry is the formal basis for niche construction and, by extension, for the Regimes of Attention mechanism developed in Paper V.

4.2.2.8 Peer Interaction

$$c_{\text{in}} = g(\psi) + \omega \quad (9)$$

Equation (9) introduces c_{in} as incoming communication and ψ as peer internal state. It models incoming communication as a transformed peer state plus noise.

$$c_{out}^* = \underset{c_{out}}{\operatorname{argmin}} F(c_{in}(c_{out}), \mu) \quad (10)$$

In Equation (10), c_{out} is outgoing communication choice, c_{out}^* is the selected outgoing communication, and $c_{in}(c_{out})$ indicates that the incoming signal depends on the outgoing choice. Outgoing communication is selected by free-energy minimisation.

$$F(c_{in}, \mu) = \underbrace{D[q_\mu(\psi) \parallel p(\psi)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{q_\mu(\psi)}[\ln p(c_{in}(c_{out}) \mid \psi)]}_{\text{Accuracy}}, \quad c_{out}^* = \underset{c_{out}}{\operatorname{argmax}} \text{Accuracy} \quad (11)$$

Equation (11) introduces $q_\mu(\psi)$ as the approximate belief over peer states, $p(\psi)$ as the peer-state prior, and $p(c_{in}(c_{out}) \mid \psi)$ as the communication likelihood term. It gives the complexity-accuracy form for communication and selects outgoing messages by expected-accuracy maximisation.

4.2.2.9 Shared Norms, Conventions, and Expectations

$$\Phi = f(\psi)\gamma + (1 - \gamma)\mu \quad (12)$$

Equation (12) introduces Φ as the shared-expectation construct, with $f(\psi)$ as a mapped peer contribution and γ as the weighting coefficient for peer influence. It defines shared expectations as a weighted blend of peer-derived and self-derived components.

The parameter γ functions as an authority weight that determines how much the agent trusts its peer's perspective relative to its own beliefs. When γ is high, the peer's views dominate the shared expectation. When γ is low, the agent's own model prevails. This weighting is not merely a modelling convenience. It captures a structural feature of organisational life that becomes consequential in the formal model: H10 predicts that asymmetry in authority weights between interacting agents is a source of coordination friction.

4.2.2.10 Subsystems and Supersystems

$$F_{\text{super}} = \sum_{i=1}^N F_i(c_{in,i}, s_i, \mu_i) \quad (13)$$

Equation (13) introduces F_{super} as supersystem free energy, N as the number of subsystems, and F_i as subsystem-level free energy. The subscript i indexes local terms $(c_{in,i}, s_i, \mu_i)$. It expresses multiscale additivity by summing the subsystems' free-energy contributions.

4.2.2.11 Multi-Peer Extension

$$\Phi = \sum_{i=1}^N (f(\psi_i)\gamma_i + (1 - \gamma_i)\mu) \quad (14)$$

Equation (14) extends shared expectations to many peers with heterogeneous influence. Each peer i carries a distinct authority weight γ_i , allowing the model to represent organisational structures in which different roles, ranks, or external actors exert different degrees of influence over shared norms. The heterogeneity of the γ_i values makes the model sensitive to how influence is distributed, not only to group size.

4.2.2.12 A Revised PHAS Framework

The original PHAS model comprises four elements: System, Purpose, Boundary, and Relationships (Section 3.9). Paper V extends this to five by incorporating Worldview as a first-class component, motivated by the intentional stance argument developed in Section 3.6. Figure X shows the expanded model. The addition is grounded in Rousseau and Billingham's (2019) Worldview Inquiry Framework and reflects the formal model's treatment of PHAS as intentional systems whose behaviour is driven by beliefs, shared expectations, and cultural norms. Table 3 in Paper V lists the corresponding principles, centred on Principle 5: human activity systems are intentional systems with a worldview. The five sub-principles establish that a PHAS perceives its environment through noisy, mediated inputs (5.1), maintains and optimises a generative model of the world (5.2), shapes its environment through action (5.3), can be analysed using the intentional stance (5.4), and operates within a multi-scale hierarchy where individual and collective generative models jointly minimise free energy (5.5). These correspond directly to the equations presented in Sections 4.2.2.1 through 4.2.2.11.

Figure 9 and Figure 10 reproduce the two summary diagrams from Paper V and provide a compact visual reference for the mechanisms described above in 4.2.2.

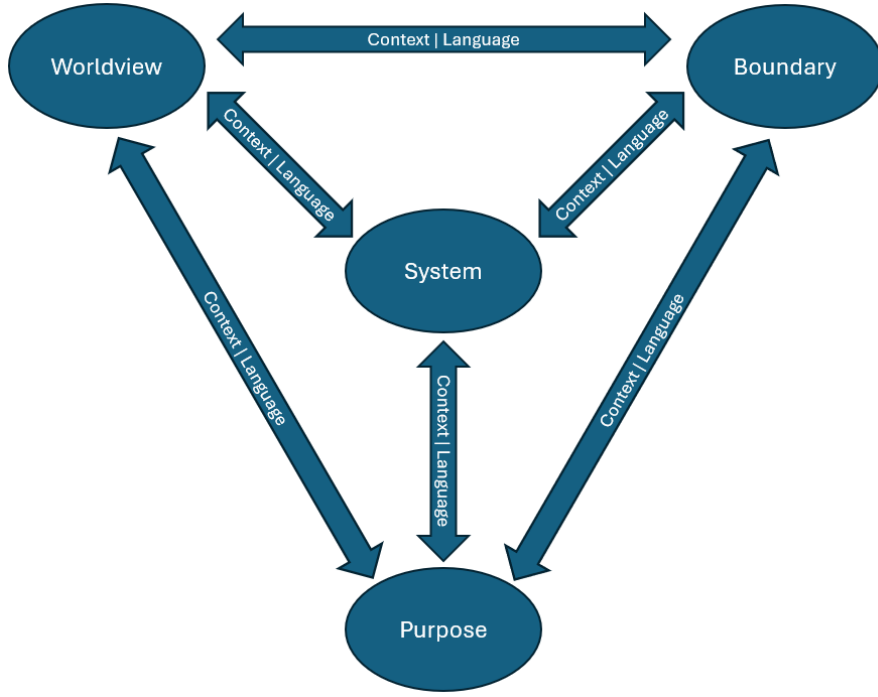


Figure 8 PHAS model expanded to include worldview as a fifth component (Paper V).

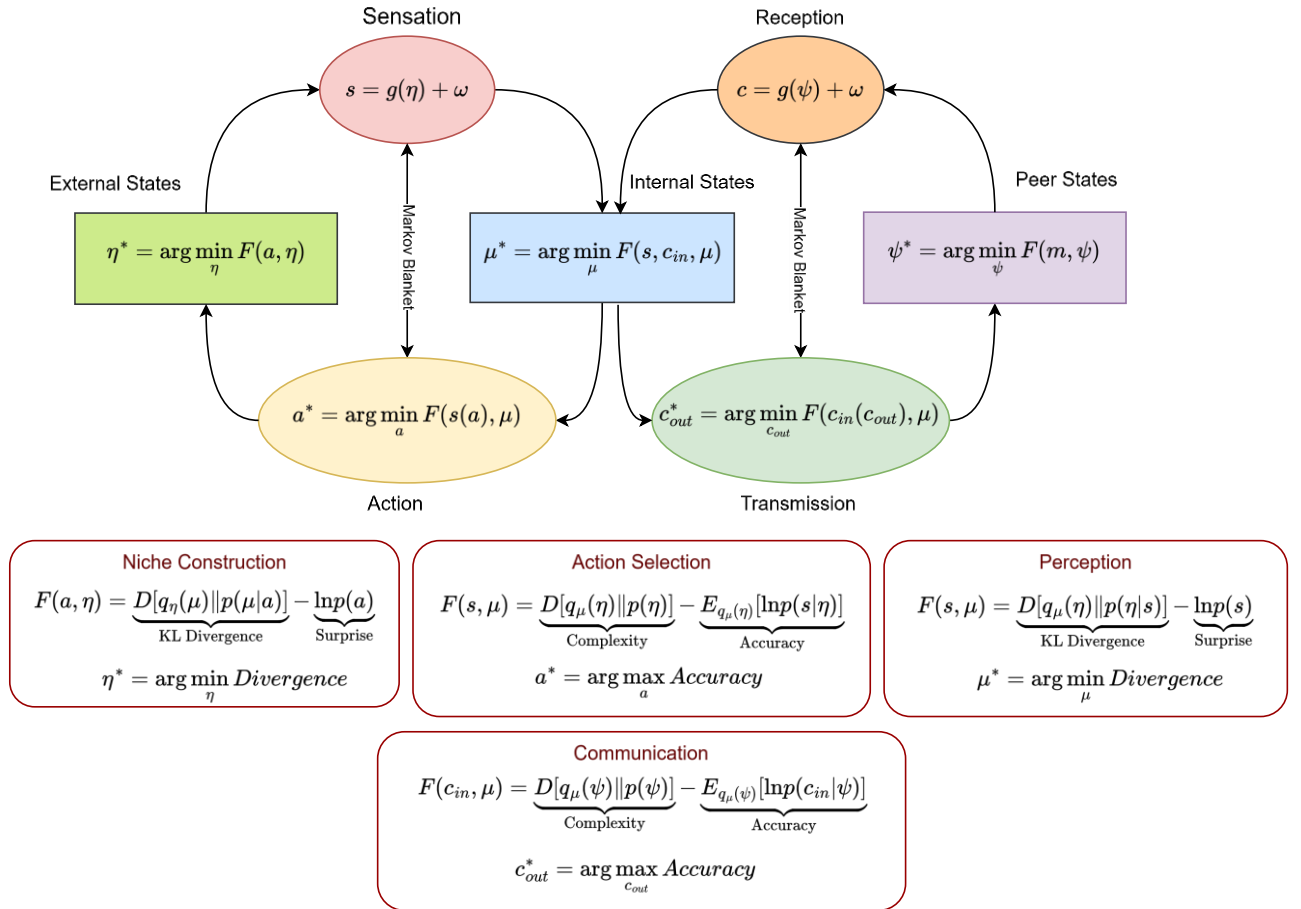


Figure 9 Active inference model of PHAS

Figure 9 presents the model in the language of active inference. The system is organised around a Markov blanket that separates **external states** (η) from **internal states** (μ), while still allowing coupling through **sensation** and **action**. Sensation is modelled as noisy input, $s = g(\eta) + \omega$, which formalises that a PHAS can never directly access the world “as it is”, only mediated signals. Perception is the process that updates internal beliefs (μ) by minimising the divergence between the PHAS’s approximate posterior and the best available explanation of its sensory input. Action selection is treated as an optimisation over possible actions (a) that change future sensations, favouring actions that maximise expected accuracy rather than actions that merely reduce short-term surprise. The same structure is extended to social interaction. Peer states (ψ) are not directly observable, but are inferred through noisy incoming communication, $c_{in} = g(\psi) + \omega$. Outgoing communication (c_{out}) is selected to maximise the accuracy of the peer model, which links communication to uncertainty reduction rather than to message transmission alone. The lower panels make the formal decomposition explicit by expressing free energy in terms of divergence and surprise for perception and niche construction, and in terms of complexity and accuracy for action and communication. This layout is intended to show where uncertainty can be reduced through belief updating, action selection, niche construction (shaping the environment), and structured interaction with peers.

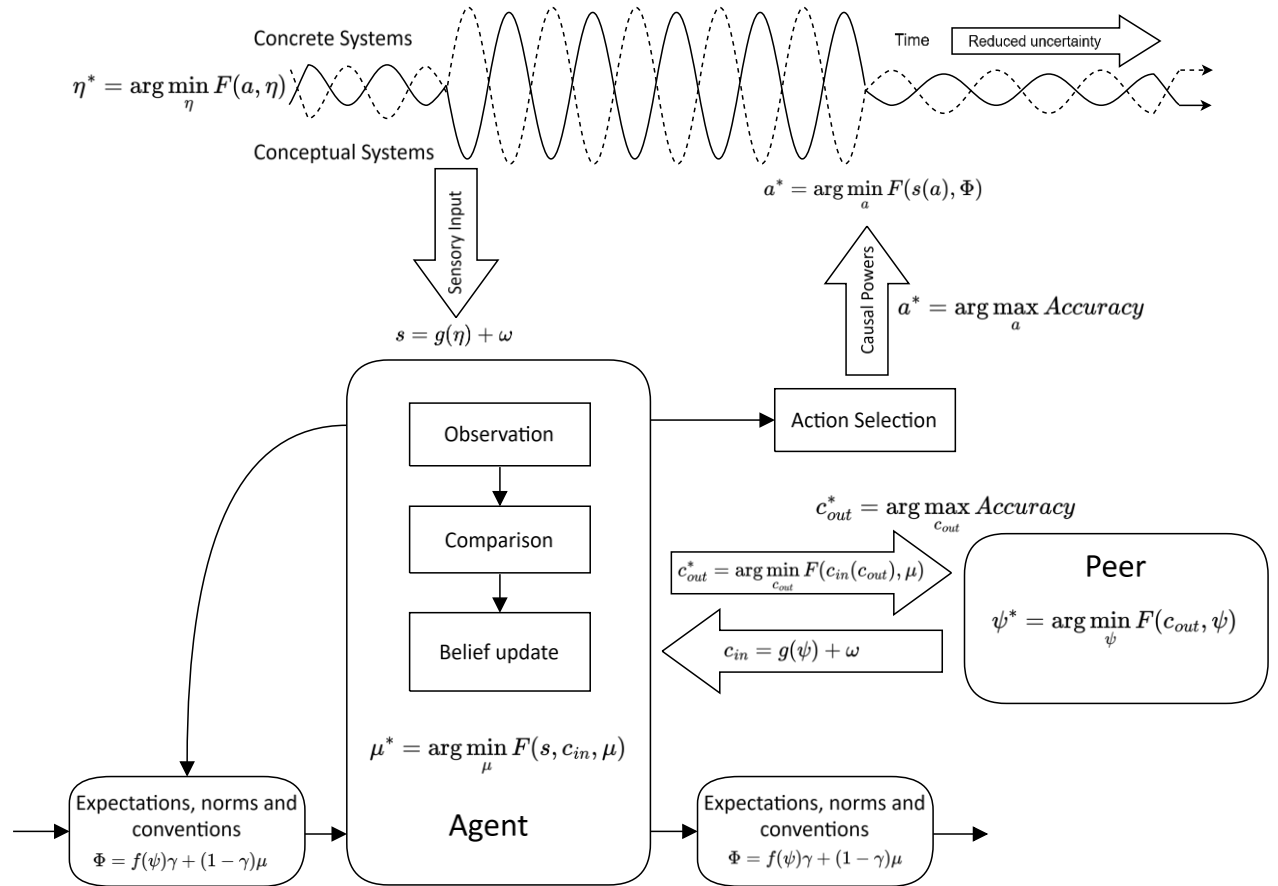


Figure 10 Expanded PHAS framework mapping the formal active inference model into a representation aligned with the original PHAS visualisation.

Figure 10 maps the same formal model back into a representation similar to that used by Calvo-Amodio (2019) in his original PHAS framework. The upper part distinguishes **concrete systems** from **conceptual systems** and illustrates how uncertainty reduces over time as the conceptual model becomes better aligned with the operational environment. The PHAS is represented as an **agent** whose internal loop is decomposed into observation, comparison, and belief update. This loop corresponds to the perceptual and learning processes in Figure 9, but is expressed in terms that match familiar engineering activity. Action selection is placed outside the agent's boundary to emphasise that action is a coupling to the environment, not an internal cognitive event. The “causal power” arrow makes this explicit by showing that purposive action is realised through external effects that subsequently return as sensory input. The figure also embeds peer interaction as a symmetric coupling: the peer is modelled as another opaque agent whose internal state is inferred from communication rather than observed directly. Expectations, norms, and conventions (Φ) are shown as contextual priors that shape both agents' interpretations and choices, representing how shared meanings and social constraints guide attention and stabilise coordination. The explicit inclusion of Φ highlights that coordination is not only a matter of exchanging information, but also of maintaining shared interpretive structures that reduce ambiguity about what signals mean and what actions are legitimate.

4.2.3 Complexity in PHAS

4.2.3.1 Configuration Space

The thesis treats the controllable degrees of freedom as $X_1, \dots, X_n$ and defines the configuration space:

$$X = \prod_{i=1}^n X_i \quad (15)$$

A controllable degree of freedom is any aspect of the socio-technical system that can be set, selected, or governed, either by the system itself or an external force acting on it. The term includes technical design choices, such as component selection or topology, and organisational choices, such as decision rights, escalation paths, and approval rules. Treating these as degrees of freedom enables a single configuration space to represent both engineering and governance commitments.

Each X_i is a set of alternatives for a design lever or policy choice. The product lists all possible combinations of those alternatives. Because the system under study is a PHAS, these degrees of freedom are grouped as:

$$X = X_{\text{tech}} \times X_{\text{org}} \times X_{\text{policy}} \times X_{\text{practice}} \times X_{\text{language}} \times X_{\text{worldview}} \times X_{\text{boundary}} \quad (16)$$

Where:

X_{tech} → technical artefacts and architectural choices

X_{org} → roles, decision rights and escalation paths

X_{policy} → rules that authorise or forbid actions

X_{practice} → codified routines and patterned practices

X_{language} → shared semantics and data definitions

$X_{\text{worldview}}$ → the ensemble of beliefs, expectations, norms and conventions that render options legitimate or illegitimate, including shared worldviews (organisation-wide or group-specific) and individual beliefs

X_{boundary} → the system boundary and its external interfaces. Each PHAS may include multiple sub-PHAS within this boundary, each with its own configuration space and constraints.

The product construction and grouping do not assume that the dimensions are independent in practice. However, they do provide a neutral starting point that makes potentially relevant choices explicit. Further, several factors are intangible, particularly language and worldview. These are included because they constrain action and coordination as directly as technical interfaces. Shared operational vocabulary can enable cross-team workflows that are

otherwise unavailable. Incompatible assumptions about acceptable risk can exclude configurations that are technically possible but socially illegitimate.

Table 12 Configuration space factors, their constituent elements, typical interfaces, and illustrative rating sources.

Factor i	Elements	Interfaces	Typical sources for ratings
X_{tech}	devices, services, components	APIs, network links, buses, protocols	CMDB, architecture repository
X_{org}	roles, teams	handoffs, reporting lines, RACI links	organisation charts, SOPs
X_{policy}	policies, regulations, standards	cross-references, compliance bindings	policy register
X_{practice}	procedures, routines	workflow handoffs, checklists	QMS, playbooks
X_{language}	ontologies, controlled terms	mappings, synonym bridges	data dictionary
$X_{\text{worldview}}$	norms, commitments	alignment mechanisms, forums	charters, retrospectives
X_{boundary}	counterpart systems, vendors	SLAs, contracts, network/APIs	contract store, interface catalogue

4.2.3.2 Decision Resolution

Some levers are continuous or very fine-grained. A decision resolution $\Delta = \{\Delta_i\}$ is fixed to render each domain finite, producing $X^\Delta = \prod_i X_i^\Delta$, where Δ is the smallest difference treated as a distinct option. The decision resolution Δ acts as a modelling granularity. For discrete levers, it is usually implicit. For continuous levers, such as thresholds, tuning parameters, or resource allocations, Δ denotes the smallest change considered meaningfully different for design and operation. The resolution should reflect the precision at which decisions can be made and validated, and the precision at which outcomes can be distinguished in monitoring and assurance.

4.2.3.3 Hard Constraints

Not every combination in X^Δ is feasible or permitted. A constraint set K captures hard constraints that hold regardless of the current staff, budget, or time. K includes interface and compatibility rules, architectural invariants, safety segregation, required cybersecurity controls, regulatory and certification limits, licensing and data-residency terms, platform commitments, site and environmental limits, semantic preconditions for coordination,

normative commitments arising from worldview, and explicit boundary commitments. Applying K prunes impossible or disallowed combinations and enforces coupling where it exists, including across internal sub-PHAS interfaces. The constraint set K captures structural feasibility rather than effort. A configuration that violates K is prohibited, even with unlimited time and staffing. K therefore represents commitments that cannot be negotiated during a response, such as safety segregation rules, interface contracts, licensing conditions, and hard budget ceilings set by governance.

4.2.3.4 Distinct Optionality

If every micro-variation is treated as distinct, the option count can be inflated by near-duplicates. Near-duplicates arise when two configurations differ in low-level settings but yield outcomes that are indistinguishable at the level relevant to decision-making. Counting such micro-variations can make complexity appear higher than the practical choice space faced by engineers and operators. To define “distinct option” explicitly, the model introduces a binning map:

$$\varphi: X^{\Delta} \rightarrow \mathcal{B} \quad (17)$$

where $\mathcal{B}$ is a finite space of bins induced by rounding or categorising outcomes and socio-technical attributes.

The map φ assigns each fine-grained configuration to a bin that represents the attributes by which options are considered distinct. A bin can be defined in terms of performance outcomes, such as latency bands, or in terms of socio-technical attributes, such as auditability class or coordination overhead class.

Binning is intentionally coarse. It indicates that many decisions are made using categories rather than exact values, and that monitoring and assurance often rely on thresholds and bands. Changing the binning scheme changes the number of options, which is appropriate because distinctness is defined relative to what can be noticed and acted upon. The bins are part of the modelling assumptions and should be selected to align with the questions being asked.

The states this as an equivalence relation:

$$x \sim y \Leftrightarrow \varphi(x) = \varphi(y) \quad (18)$$

In words, two configurations are treated as the same option when they fall into the same bin under φ . The equivalence classes created by $\sim$ are therefore sets of near-duplicate configurations that are indistinguishable with respect to the chosen bins. The quotient operation collects these equivalence classes into a new set whose elements are the distinct

options. For example, if φ rounds cost to budget tiers, then two configurations that differ by a small cost delta within the same tier are treated as a single option.

4.2.3.5 The Feasible Set

The feasible set at resolution Δ is then the quotient:

$$\mathcal{O}^\Delta = \{x \in X^\Delta : x \text{ satisfies } K\} / \sim \quad (19)$$

The set $\mathcal{O}^\Delta$ can be read as the feasible option set after both structural feasibility and distinctness are applied. It contains one representative for each distinct feasible option, rather than every micro-configuration. This definition ensures that the option count reflects what can realistically be compared, selected, and justified.

The factor $X_{\text{worldview}}$ operates as both a state within the configuration space and the primary logic of the internal generative model described in the PHAS chapter. It represents the set of norms and beliefs that an organisation can intentionally shift or align, yet it also acts as a fundamental source of constraints within K . By defining what is considered legitimate or possible, $X_{\text{worldview}}$ prunes the wider configuration space. It functions as the system's meta-logic, ensuring that any chosen configuration of technical or policy levers remains consistent with the shared expectations and normative commitments of the agents involved.

Treating worldview as both a variable and a constraint reflects that norms can change, but only within limits set by identity, regulation, and external expectations. An organisation can adopt a stronger security posture and new escalation norms, but it cannot simultaneously maintain mutually incompatible commitments. Encoding these commitments inside K prevents the model from counting options that are organisationally unreachable.

4.2.3.6 Complexity as Total Optionality

Complexity is defined as the size of this feasible set:

$$C = |\mathcal{O}^\Delta| \quad (20)$$

Equation (20) defines complexity as the count of distinct feasible options available at the chosen resolution. The units are therefore an option count rather than a physical quantity. C is finite because the choice sets have been discretised and because near-duplicates have been collapsed.

This treats complexity as total optionality. It counts all distinct socio-technical configurations that the PHAS can, in principle, realise once structural commitments are respected. Importantly, C counts both desirable configurations and *failure modes* (unsafe, brittle, non-compliant, or otherwise unacceptable configurations). This is intentional. As the option set grows, so does the number of alternatives that must be evaluated, ruled out, constrained, or monitored. Each additional configuration increases both choice and error surface.

Because the PHAS often comprises constituent sub-PHAS, C can also be computed at the subsystem level when useful, and then related to the whole via the composition property below, with cross-subsystem coupling captured in K .

Where feasibility depends on the lifecycle or operating state ℓ (commissioning, steady operation, incident response), constraints are indexed as $K(\ell)$, yielding $\mathcal{O}^A(\ell)$ and $C(\ell) = |\mathcal{O}^A(\ell)|$. Indexing constraints by lifecycle state ℓ makes explicit that feasibility is mode-dependent. A configuration that is feasible during commissioning may be infeasible during incident response because approvals, maintenance windows, or tooling are not available. Representing this through $K(\ell)$ allows the model to compare the option space across operational modes without changing the underlying definition of complexity.

Two simple sanity checks are used in the analysis:

- **Direction of change.** If you add another lever (expand some X_i^A) or relax a hard constraint in K , the number of distinct feasible options C cannot go down. If you add a hard constraint to K , C cannot go up. This remains true even though near-duplicates are collapsed via φ .
- **Composition.** If two subsystems A and B are mostly independent (few coupling constraints), the combined option count is approximately multiplicative, $C_{A \parallel B} \approx C_A C_B$. Coupling constraints reduce this by ruling out combinations.

The composition property is approximate because socio-technical subsystems are rarely fully independent. Coupling enters through cross-subsystem constraints in K , such as shared resources, shared interfaces, or shared governance. The multiplicative calculation is therefore most useful as an order-of-magnitude argument that highlights how quickly option counts grow when additional levers are introduced without corresponding constraints.

Two PLC families, three network topologies and two database engines yield $2 \times 3 \times 2 = 12$ raw combinations. If a safety invariant in K forbids one topology with one PLC at high-criticality sites, this leaves 10 feasible combinations, so $C = 10$. Introducing a standard

incident-response playbook enables two additional coordination patterns across roles. If all are compatible, C becomes $10 \times 3 = 30$. Adopting a shared threat taxonomy unlocks one further cross-team workflow. C becomes $30 \times 2 = 60$. Adding a QoS target counted at $\Delta = 5\%$ from 0 to 100 per cent yields 21 levels. If independent of the other choices, $C = 60 \times 21 = 1,260$.

The numerical illustration shows how options can multiply across technical, organisational, and informational changes. Even when each individual change appears modest, the combined option count can grow rapidly. This provides a quantitative rationale for design practices that deliberately reduce option space through standardisation, architectural invariants, and shared vocabularies.

4.2.3.7 Linking Total Optionality to Entropy and Surprise in PHAS

In the PHAS-EAI model, surprise is defined as $S = -\ln p(x \mid \mu)$, and entropy is the expected or weighted average of surprise over time. Here, x denotes a configuration, so the PHAS definition is read as the surprise of selecting or observing configuration x under the internal model μ . Interpreting $\mathcal{O}^A$ as a discrete hypothesis space of configurations, C sets the scale of how uncertain selection can be before usefulness and time constraints are applied.

Surprise and entropy connect option counts to the uncertainty experienced by decision-makers. The probability term $p(x \mid \mu)$ denotes the plausibility of configuration x under an internal model μ . A low-probability configuration is surprising and typically requires either explanation or corrective action. Under an uninformative (uniform) distribution over $\mathcal{O}^A$, the maximum Shannon entropy over configurations is $\ln C$ (or $\log_2 C$ in bits). More generally, if the PHAS internal model μ induces a non-uniform distribution over configurations or outcomes, then expected surprise is reduced by concentrating probability mass on a smaller subset of typical configurations. This is the discrete counterpart to the observation that stronger priors reduce uncertainty in active inference. In active inference, complexity appears as a divergence term between posterior and prior beliefs, while accuracy rewards explanations that predict sensations. The cardinality C is not that KL divergence, but it bounds it by determining how large the discrete hypothesis space can be. When there are equiprobable configurations, selecting the realised configuration requires $\log_2 C$ bits of information. Put differently, worldview, policy, and technical commitments act as priors that concentrate probability mass and prune options, thereby reducing entropy and the informational work required to disambiguate configurations during design and operation. Increasing C increases the informational work required to identify, justify, and stabilise a configuration, unless parallel investment narrows the effective hypothesis space through stronger constraints, sharper priors, or better tooling.

4.2.3.8 Model Parameters Per Factor X_i

Each factor $X_i \in \{\text{tech, org, policy, practice, language, worldview, boundary}\}$ can be further described using the parameter set adopted from Schöttl and Lindemann (2015) summarised in Table 13. Option counts provide a structural upper bound, but response time is also influenced by the complexity of each factor and its rate of change. The weighted parameters in Table 13 are used to model this by assigning ratings to each factor.

Table 13 Complexity model system parameters and weights

Group	Group weight	Parameter	Abbr.	Scale	Parameter weight
System properties	0.4	Number of elements (processes and resources)	NE	1–10	0.08
		Variety of elements (processes and resources)	VE	1–10	0.09
		Degree of interconnectedness	DI	1–10	0.02
		Number of interfaces to neighbouring system parts	NI	1–10	0.25
		Variety of interfaces to neighbouring system parts	VI	1–10	0.07
System changes	0.6	Change in number of elements	ΔNE	1–10	0.06
		Change in variety of elements	ΔVE	1–10	0.01
		Change in degree of interconnectedness	ΔDI	1–10	0.20
		Change in number of interfaces	ΔNI	1–10	0.03
		Change in variety of interfaces	ΔVI	1–10	0.06
		Change of elements (processes and resources)	ΔE	1–10	0.13

For each factor i , the rating vector is:

$$\mathbf{p}_i = (NE_i, VE_i, DI_i, NI_i, VI_i, \Delta NE_i, \Delta VE_i, \Delta DI_i, \Delta NI_i, \Delta VI_i, \Delta E_i) \quad (21)$$

The rating vector p_i groups five structural properties and six rates of change. The structural properties describe the number of distinct elements in the factor, their variability, and the number of interfaces that connect them. The change terms describe how quickly those

properties evolve. Ratings are on an ordinal 1-10 scale, which supports comparative analysis even when absolute measurement is impractical.

The per-factor complexity potential CP_i is

$$CP_i = 0.4(0.08 NE_i + 0.09 VE_i + 0.02 DI_i + 0.25 NI_i + 0.07 VI_i) + 0.6(0.06 \Delta NE_i + 0.01 \Delta VE_i + 0.20 \Delta DI_i + 0.03 \Delta NI_i + 0.06 \Delta VI_i + 0.13 \Delta E_i) \quad (22)$$

Equation (22) aggregates the five structural attributes and six rates of change into a single complexity potential. A factor with many varied parts, numerous interfaces, and rapidly shifting connections scores high, while a small, uniform, loosely coupled, stable factor scores low. The group weights in CP_i give greater emphasis to system change than to static structure. This aligns with the view that volatility and churn can create more operational drag than complexity that is well understood and stable. Schöttl and Lindemann emphasise that both the 1–10 scales and the weights are system-dependent and should be recalibrated for the domain under study.

4.2.3.9 Capturing Subjective Complexity

Schöttl and Lindemann (2015) combine interaction difficulty with a person's experience and mental flexibility, recognising that interaction difficulty is experienced differently by people with different levels of expertise. Interaction type $IT \in \{1, \dots, 10\}$ captures the intrinsic demand of the task, while experience $E \in \{1, \dots, 5\}$ and mental flexibility $MF \in \{1, \dots, 5\}$ capture capability. Table 14 shows example scorings of interaction types, and Table 15 the scoring scheme used for experience and mental flexibility.

Table 14 Example scoring of interaction types

IT	Description
1	Easy, low-demand job with clear instructions and no time pressure, with marginal interfaces
5	Ordinary, well-structured job with diversified content and medium responsibility
10	Highly demanding job with high time pressure, responsibility for subordinates and many external interfaces

Table 15 Scoring scheme for experience and mental flexibility

Value	Label
1	Very low
2	Low
3	Medium
4	High
5	Very high

Perception demand decreases as experience and flexibility increase. A linear map that preserves this direction is:

$$\theta(E, MF) = 12.25 - 1.125 (E + MF) \in [1,10], \quad u(E, MF) = \frac{10 - \theta(E, MF)}{9} \in [0,1] \quad (23)$$

The linear map $\theta(E, MF)$ in Equation (23) is the (inverse) perception score, and assigns a higher numerical demand to lower experience and flexibility. The normalised skill factor $u(E, MF)$ rescales this to the unit interval, where 0 represents no effective skill benefit and 1 represents the maximum skill benefit available under the scoring scheme. In other words, the skill factor $u(E, MF)$ rises with experience and flexibility, and is used to scale both cognitive headroom and mobilisation time.

4.2.4 Model of Resilience in PHAS

A disturbance d has a **magnitude** M_d and **timescale** τ_d . The disturbance d represents a situation that the system must respond to, such as a component failure, a cyber incident, an extreme weather event, or a supplier outage. The magnitude M_d represents severity, for example, the scale of service degradation or the level of damage. The timescale τ_d represents the time required to address the disturbance before unacceptable outcomes occur.

For a disturbance d with a characteristic timescale τ_d , the feasible set $\mathcal{O}^A$ yields two nested sets:

- **Useful variety** (required service maintained)

$$A_d = \{x \in \mathcal{O}^A: x \text{ would deliver required service under } d \text{ with acceptable risk}\}, \quad (24)$$

with $A = |A_d|$

- **Survivable variety** (safety and non-collapse maintained, service may degrade)

$$B_d = \{x \in \mathcal{O}^A: \text{system remains safe and non-collapsed under } d\}, \quad (25)$$

with $B = |B_d|$

By construction, $A_d \subseteq B_d \subseteq \mathcal{O}^A$, hence $A \leq B \leq C$.

The sets A_d and B_d separate service adequacy from mere survival. This distinction is important in infrastructure contexts because safe degradation may be acceptable for a limited period, whereas unsafe operation is not.

4.2.4.1 Headroom

Headroom H is the mobilisable slack available during the response. It comprises a physical component and a cognitive component. The normalised physical headroom $H_{\text{phys}} \in [0,1]$ aggregates slack in capacity, inventory, budget and authority. The cognitive component H_{cog} reflects skill and task demand:

$$H_{\text{cog}} = h + (1 - h) u(E, MF) q(IT) \quad (26)$$

Equation (26) combines three ideas. First, $h \in (0,1]$ denotes the minimum cognitive reserve that is preserved by design, for example, through automation, checklists, or pre-authorised actions. Second, the normalised skill factor $u(E, MF) \in [0,1]$ denotes capability as a function

of experience and mental flexibility. Third, $q(IT)$ denotes the task's demand level and is modelled as a decreasing function of interaction type. Multiplying $u(E, MF)$ and $q(IT)$ captures that skill is most valuable when tasks are demanding, and that limited skill reduces usable headroom under stress.

Total headroom is:

$$H = H_{\text{phys}} \times H_{\text{cog}} \in [0,1] \quad (27)$$

Total headroom H is modelled as the product of physical and cognitive components, indicating that both types of slack must be available simultaneously. A team can possess strong cognitive capabilities yet remain constrained by a lack of authority or capacity. Surplus capacity is ineffective if decision-making is blocked. The multiplicative form keeps H within $[0,1]$ and penalises bottlenecks in either component.

4.2.4.2 Response Time and Structural Drag

Response time decomposes as:

$$\tau = \tau_{\text{detect}} + \tau_{\text{decide}} + \tau_{\text{execute}} \quad (28)$$

The decomposition in Equation (28) separates sources of delay. *Detect* encompasses sensing and triage and is hindered by low observability and ambiguous signals. *Decide* encompasses delays due to coordination and authorisation. *Execute* covers deployment and change procedures. Keeping the stages separate supports analysis of interventions that shorten one stage without necessarily affecting the others.

Each stage has a baseline time under a stable configuration:

$$\tau_{\text{detect}}^0, \quad \tau_{\text{decide}}^0, \quad \tau_{\text{execute}}^0 \quad (29)$$

The baseline times represent typical stage durations under stable operating conditions with clear procedures and no unusual friction. They can be estimated from historical incident records or from rehearsal exercises. They do not represent best-case times, but rather typical times under normal operating assumptions.

Complexity potential *structural drag* increases these baselines:

$$CP = (CP_{\text{tech}}, CP_{\text{org}}, CP_{\text{policy}}, CP_{\text{practice}}, CP_{\text{language}}, CP_{\text{worldview}}, CP_{\text{boundary}})^T \quad (30)$$

Equation (30) expands Equation (22) by combining the per-factor complexity potentials, and stage times then become:

$$\begin{aligned}
\tau_{\text{detect}} &= \tau_{\text{detect}}^0 [1 + \gamma_{\text{det}}^\top CP], \\
\tau_{\text{decide}} &= \tau_{\text{decide}}^0 [1 + \gamma_{\text{dec}}^\top CP], \\
\tau_{\text{execute}} &= \tau_{\text{execute}}^0 [1 + \gamma_{\text{exe}}^\top CP].
\end{aligned} \tag{31}$$

where $\gamma_{\text{det}}, \gamma_{\text{dec}}, \gamma_{\text{exe}} \in \mathbb{R}_{\geq 0}^7$ denotes stage-specific sensitivity vectors.

The stage-time expressions in (31) apply a linear scaling in which each factor's complexity potential increases time in proportion to a stage-specific sensitivity. A factor can therefore slow one stage more than another. For example, policy complexity may primarily affect decision-making through approval and compliance checks, while technical complexity may primarily affect detection through observability challenges. The sensitivity vectors γ_{det} , γ_{dec} , and γ_{exe} capture these differences. The vector component $\gamma_{\text{det},i}$ measures how factor i slows detection, $\gamma_{\text{dec},i}$ measures decision-making drag, and $\gamma_{\text{exe},i}$ measures execution drag.

Structural drag can be interpreted through the PHAS active-inference lens as a noise and coupling penalty at the Markov blanket. A high complexity potential CP corresponds to an operational landscape with lower signal-to-noise and more ambiguous mapping between external signals and internal states. This necessitates longer integration times to resolve uncertainty, which is reflected in the scaling of response times τ by CP in (31).

4.2.4.3 Skill and Effective Mobilisation Time

Human skill shortens the inflated time. With $\lambda_\tau \in [0,1]$ as the maximum fractional reduction attributable to skill, the effective time multiplier is

$$\sigma_\tau = 1 - \lambda_\tau u(E, MF) \in (0,1] \tag{32}$$

The multiplier σ_τ reduces response time as skill increases, with λ_τ setting the maximum achievable speed-up. The form ensures that no amount of skill can reduce the time below $1 - \lambda_\tau$ of the structurally determined time. This reflects that some delays are imposed by physical realities and governance rather than by individual proficiency.

The effective mobilisation time is

$$\tau_{\text{eff}} = \sigma_\tau \tau \tag{33}$$

The effective mobilisation time τ_{eff} is the response time used in the time-conditioned option counts. It is the quantity compared with the disturbance timescale τ_d . Representing skill as a multiplicative reduction supports the interpretation of training and experience as increasing

the fraction of options that can be mobilised before the shock window closes. Thus, $\sigma_\tau = 1$ when $u = 0$, so skill offers no speed-up (inexperienced, inflexible). As $u \rightarrow 1$ (experienced, flexible), $\sigma_\tau \rightarrow 1 - \lambda_\tau$, corresponding to a maximum fractional time reduction of λ_τ and a speed-up factor of $1/(1 - \lambda_\tau)$.

The model uses $u(E, MF)$ in both H_{cog} and τ_{eff} deliberately. Skill both frees cognitive reserve (more usable attention under stress) and increases execution speed (faster, more reliable action sequences). These are distinct mechanisms, so this is not double-counting.

4.2.4.4 Time-Conditioned Option Counts

Only a share of options can be mobilised within the shock window. In the model this is denoted by the time-allowance ratio

$$r = \frac{\tau_d}{\tau_{\text{eff}}} \quad (34)$$

The ratio r in Equation (34) compares the time available, τ_d , with the effective time τ_{eff} . Values below 1 indicate that the response is time-constrained, so only a subset of options is realistically selectable. Values above 1 indicate that time is sufficient, so mobilisation is not limited by speed.

The model defines three response functions to represent the share of options that can be mobilised within the available time window:

$$f_A(r) = \min(1, r^{\varepsilon_A}), \quad f_B(r) = \min(1, r^{\varepsilon_B}), \quad f_C(r) = \min(1, r^{\varepsilon_C}) \quad (35)$$

where $\varepsilon_A, \varepsilon_B, \varepsilon_C > 0$ are elasticities for useful, survivable, and total option mobilisation.

The response functions in (35) map time allowance into a set of accessible options. The minimum operator caps the fraction at 1, ensuring the functions saturate once there is sufficient time to mobilise the full option set. The elasticities $\varepsilon_A, \varepsilon_B, \varepsilon_C$ control how sharply option accessibility falls as time becomes scarce. Separate elasticities are used because useful options may require more coordination than survivable options, and total optionality may include options that are costly to mobilise.

For notational convenience, the three response functions (35) can also be written as:

$$f_A(\tau_d, \tau_{\text{eff}}) := f_A\left(\frac{\tau_d}{\tau_{\text{eff}}}\right), \quad f_B(\tau_d, \tau_{\text{eff}}) := f_B\left(\frac{\tau_d}{\tau_{\text{eff}}}\right), \quad f_C(\tau_d, \tau_{\text{eff}}) := f_C\left(\frac{\tau_d}{\tau_{\text{eff}}}\right) \quad (36)$$

Writing the response functions with two arguments makes it explicit that the same time allowance ratio can arise from the disturbance evolving quickly, mobilisation being slow, or both. This supports analysis of whether resilience could be improved by faster detection and execution, or by designing for greater tolerance to disturbances.

The time-conditioned effective option counts are defined as

$$\tilde{A}(\tau_d, \tau_{\text{eff}}) := A f_A(\tau_d, \tau_{\text{eff}}), \quad \tilde{B}(\tau_d, \tau_{\text{eff}}) := B f_B(\tau_d, \tau_{\text{eff}}), \quad \tilde{C}(\tau_d, \tau_{\text{eff}}) := C f_C(\tau_d, \tau_{\text{eff}}) \quad (37)$$

The effective counts $\tilde{A}$, $\tilde{B}$, and $\tilde{C}$ represent the sets of useful, survivable, and total options that are practically mobilisable within the available time.

4.2.4.5 Magnitude Limits and Resilience Score

The absorbable magnitude that the system can handle is expressed as a function of time-conditioned option counts and headroom:

$$M_0(\tau_d, \tau_{\text{eff}}) = k \tilde{A}(\tau_d, \tau_{\text{eff}})^\alpha H^\beta, \quad M_{\text{max}}(\tau_d, \tau_{\text{eff}}) = k \tilde{B}(\tau_d, \tau_{\text{eff}})^\alpha H^\beta \quad (38)$$

The magnitude limits in (38) connect options and headroom to the severity that can be absorbed. M_0 denotes the no-penalty limit and M_{max} denotes the collapse threshold. Since $A \leq B$ by construction, these imply $M_0 \leq M_{\text{max}}$. The constant $k > 0$ fixes the overall scale so that the limits share the units of disturbance magnitude. The exponents $\alpha > 0$ and $\beta > 0$ are elasticities that control how the limits grow with the option counts and with headroom. These parameters can be calibrated from domain-specific incident histories or from stress testing and simulation.

The model defines a resilience score $R(M, \tau_d)$ in two cases:

$$\begin{aligned} &\text{If } M_0(\tau_d, \tau_{\text{eff}}) < M_{\text{max}}(\tau_d, \tau_{\text{eff}}): \\ &\quad R(M, \tau_d) = \begin{cases} 1, & M \leq M_0(\tau_d, \tau_{\text{eff}}) \text{ (no penalty)} \\ \max\left\{0, 1 - \left(\frac{M - M_0(\tau_d, \tau_{\text{eff}})}{M_{\text{max}}(\tau_d, \tau_{\text{eff}}} - M_0(\tau_d, \tau_{\text{eff}})\right)\right\}, & M_0(\tau_d, \tau_{\text{eff}}) < M \leq M_{\text{max}}(\tau_d, \tau_{\text{eff}}) \text{ (survival band)} \\ 0, & M > M_{\text{max}}(\tau_d, \tau_{\text{eff}}) \text{ (collapse)} \end{cases} \\ &\text{If } M_0(\tau_d, \tau_{\text{eff}}) = M_{\text{max}}(\tau_d, \tau_{\text{eff}}): \\ &\quad R(M, \tau_d) = \begin{cases} 1, & M \leq M_0(\tau_d, \tau_{\text{eff}}), \\ 0, & M > M_0(\tau_d, \tau_{\text{eff}}) \end{cases} \end{aligned} \quad (39)$$

Equation (39) gives $R = 1$ for $M \leq M_0$, a gradual taper for $M_0 < M \leq M_{\max}$, and $R = 0$ beyond $M_{\max}$.

The resilience score R is defined on a 0-1 scale. It rewards configurations and organisational arrangements that can handle larger disturbance magnitudes without penalty, and penalises cases where survival requires degradation. A linear taper is a modelling choice that supports comparison and sensitivity analysis when detailed loss functions are unavailable.

4.2.5 Functional Information and Cost of Choice in PHAS

Szostak (2003) introduced functional information to shift analysis from structural description to the rarity of configurations that achieve a specified function at a stated level in a defined context. Hazen et al. (2007) provided a system-agnostic formulation that makes the concept precise for any domain in which both function and degree of function can be scored.

Functional information, denoted by I , is used in the model to connect structural optionality to performance requirements. Option counts alone do not indicate whether many options are good enough. I quantifies the sparsity of success once a performance threshold is set for a given function, thereby supporting discussions of resilience requirements expressed as service levels, safety integrity targets, or recovery time objectives. This depends on the function-threshold pair, as well as on context, and not on structural intricacy alone.

If almost every feasible configuration meets the threshold, then I is small. If only a small share succeeds, I is large. If a light switch is acceptable in either state for the function “valid state”, every configuration succeeds, and $I = 0$. If a throughput target is at least 10 Mb/s and all available ports exceed that, every configuration succeeds and $I = 0$. Raise the target to 800 Mb/s and only some configurations succeed, so $I > 0$. If the requirement is at least one successful backup per week, and all schedules meet that, $I = 0$. Require hourly recovery points and only a subset succeeds, so $I > 0$.

A named function g specifies the capability of interest, and a performance threshold ρ marks “good enough” in context, for example, a safety integrity target or a service level under a named disturbance. The subset of feasible configurations that meet or exceed the threshold for the named function g is then:

$$A_{g,\rho} = \{x \in \mathcal{O}^A: x \text{ achieves performance } \geq \rho \text{ for function } g\} \quad (40)$$

The threshold ρ can encode a required service level under disturbance, or an assurance target that must be maintained. Defining success as a set supports consistent comparison across different functions and contexts.

The share of the option set that succeeds is

$$p_g(\rho) = \frac{|A_{g,\rho}|}{C} \in [0,1] \quad (41)$$

When $p_g(\rho)$ is small, success is rare among the feasible options, implying that search and selection are more difficult.

The **design-time** functional information for g at level ρ is

$$I_g^*(\rho) = -\log_2 p_g(\rho) = \log_2 C - \log_2 |A_{g,\rho}| \quad (42)$$

Design-time functional information $I_g^*(\rho)$ is measured in bits because it uses a base-2 logarithm. It can be interpreted as the number of binary decisions required to isolate a successful configuration under an uninformative search. The definition depends on the context-specific choice of g and ρ rather than on structural intricacy alone.

Two edge cases follow directly. If every feasible configuration passes at ρ , then $p_g(\rho) = 1$ and $I_g^*(\rho) = 0$. If exactly one feasible configuration passes, then $p_g(\rho) = 1/C$ and $I_g^*(\rho) = \log_2 C$.

When time pressure is considered, the model includes a time-conditioned *effective* count:

$$\tilde{A}_{g,\rho}(\tau_d, \tau_{\text{eff}}) := |A_{g,\rho}| f_A(\tau_d, \tau_{\text{eff}}) \quad (43)$$

Time pressure reduces the number of viable options available. The effective success count $\tilde{A}_{g,\rho}$ therefore uses the same mobilisation fraction f_A as the useful variety calculation. This aligns functional information with the operational assumption that only a share of options can be enacted within the shock window.

The corresponding **operational functional information** becomes

$$I_g^{\text{op}}(\rho, \tau_d) = \log_2 C - \log_2 \tilde{A}_{g,\rho}(\tau_d, \tau_{\text{eff}}) = I_g^*(\rho) - \log_2 f_A(\tau_d, \tau_{\text{eff}}) \quad (44)$$

Operational functional information I_g^{op} adds an extra difficulty term that depends on time pressure. The $-\log_2 f_A$ increases as the time allowance ratio falls. This makes it explicit that a system can have many structurally feasible, successful configurations while still being hard to operate if they cannot be mobilised quickly.

In practical terms, time pressure shrinks the set of options one can actually use. Time pressure adds up to $-\log_2 f_A$ bits of difficulty. Faster effective response (higher u , lower structural CP , hence smaller τ_{eff}) raises f_A and reduces I_g^{op} . For instance, consider a water utility when a main bursts. On a normal day there are eight good ways to keep pressure up, but with only thirty minutes available only two are feasible, so choosing one that works is now harder. Detecting sooner, clear playbooks, and automation allow more options to fit any given time window, which lowers operational functional information and raises resilience.

The expression $I_g^*(\rho) = -\log_2 p_g(\rho)$ mirrors the negative-log form used for surprise in the PHAS-EAI model. Under the extended active inference regime used in this thesis, learning, redesign and environment shaping can prune unhelpful configurations and lower C . At the same time, better sensing, decision support, automation and headroom lift more configurations above the threshold and shorten τ_{eff} , so $\tilde{A}_{g,\rho}$ rises and f_A rises. Both effects reduce I_g^{op} . Choices that retire dead-end architectures, strategies, tactics, tools and technologies reduce C , and upgrades increase $|A_{g,\rho}|$. Together, these changes push I_g^{op} down, which means fewer yes-or-no questions to find a plan that works in time.

4.3 Qualitative case-study strand

4.3.1 Industry-as-Laboratory

Muller (2013) describes systems engineering as a discipline shaped by practice, with expertise residing primarily among practitioners and companies. The *industry-as-laboratory* (IaL) model reflects this reality by treating actual industrial projects as the main environment for systems engineering research. In the IaL model, researchers work alongside practitioners, taking active roles in real projects while observing, experimenting, and collecting empirical data. This dual role enables the introduction, testing, and iterative refinement of methods, tools, and theories within the real constraints and uncertainties that characterise industrial settings.

Compared to traditional social science methods such as surveys, interviews, or detached case studies, IaL is distinguished by the researcher's ongoing, embedded participation in the field. Rather than studying organisations from the outside, IaL involves the researcher as an active contributor to the work itself. This immersion provides a deeper understanding of the context and allows for immediate testing and refinement of engineering methods based on real-world feedback. The process is dynamic, with research questions and interventions continually shaped by observations and emerging needs within the project environment. IaL is frequently combined with action research, fostering a collaborative cycle of intervention and reflection between researchers and practitioners.

Read through the lens of Chapter 3, the IaL configuration is not merely a practical convenience but a methodological consequence of the theoretical framework. If the cognitive processes that sustain dependability are distributed across people, tools, and environmental structures, as Chapter 3 argues, then studying those processes from the outside, through surveys or periodic interviews, risks missing the very phenomenon the thesis seeks to explain. In the language of the PHAS-EAI model, the embedded researcher shares the system's input and active states, receiving the same signals as the team and contributing to the same interventions. The researcher's design choices reshape the team's cognitive environment, and the team's responses reshape the research questions, a dynamic that Constant et al. (2020) describe as niche construction.

The systematic use of digital tools in this project was planned as part of the research methodology from the outset. However, the COVID-19 pandemic accelerated and enforced the adoption of these tools, as all participants were required to shift rapidly to digital platforms and practices that had previously seen limited use. Instant messaging, digital notebooks, git version control, issues, comments, and task management systems became integrated into everyday work. Every interaction and artefact generated a timestamped digital

record, creating a comprehensive digital footprint for all workflows. Data was produced continuously by the work itself, not limited to formal documentation and research notes. The author could later go back to search and analyse emails, chats, meeting notes, and other records, even years after the events took place. This persistent digital trail enabled robust longitudinal analysis and added rigour to both qualitative and quantitative aspects of the research. In the terminology of the formal model, these digital artefacts constitute designed cognitive reserve (*h*) for the research process itself, extending the researcher's capacity to detect patterns and confront interpretations with evidence that would otherwise have been lost to the limitations of human memory.

4.3.2 Action Research

Action research (AR) is a participatory methodology in which practitioners and researchers collaborate through iterative cycles of planning, action, observation, and reflection (Lewin 1951). Each cycle addresses a practical problem while simultaneously generating new knowledge. AR is well-suited to systems engineering research because it embeds the researcher in the operational context where engineering decisions are made, enabling real-time observation of how interventions interact with organisational constraints (Bradbury 2015).

The structural correspondence between AR and the thesis's theoretical framework deserves explicit comment. The AR cycle of plan, act, observe, and reflect maps onto the perception-action loop described by active inference: the agent holds a generative model (the plan), acts on the environment, observes the consequences, and updates beliefs accordingly. This correspondence is not coincidental. The thesis models engineering teams as agents that minimise prediction error through precisely this cycle. Choosing AR as the research methodology means that the research process instantiates the same dynamics that the formal model describes in the object of study. The researcher's iterative refinement of interventions is, in the terms of the PHAS-EAI model, a sequence of active-inference cycles conducted at the research level rather than the operational level. This does not mean that Lewin anticipated active inference. It means that the thesis reads the methodological choice through the theoretical framework, and the fit between method and theory strengthens confidence that both describe a genuine pattern.

In this project, the researcher occupied a dual role as both practitioner and investigator within the case company over a six-year period. This insider position provided deep contextual access and enabled interventions to be designed in response to genuine operational needs rather than imposed from outside. Coghlan et al. (2014) describe this configuration as doing action research in your own organisation and identify its principal advantages as sustained access, trust, and contextual understanding. They also identify its principal risks: pre-understanding that narrows the researcher's interpretive field, role ambiguity between

colleague and investigator, and organisational politics that may constrain what can be reported. These risks were managed through three strategies. First, the persistent digital audit trail described in Section 4.3.1 provided an evidence base that could be interrogated independently of the researcher's recollection. Second, the GTM analysis described in Section 4.3.3 imposed systematic comparison as a corrective against selective interpretation. Third, research supervision external to the case company provided a periodic review of emerging categories and claims.

Each paper in the thesis corresponds to one or more AR cycles. Cases A and B involved cycles of model construction, game-based intervention, observation, and refinement of both the SysML model and the game facilitation approach. Case C involved cycles of LLM integration, review of AI-generated outputs, and iterative revision of the prompt templates and evaluation criteria. Case D involved the longest cycle: design of environmental interventions (automated monitoring, smart bulb alerts, scheduled EoP sessions), two years of observation, and ongoing refinement of the intervention design in response to emerging data. Across all cases, the AR structure ensured that each intervention was informed by prior observation and that its effects were captured as data for subsequent analysis.

4.3.3 Grounded Theory Method

Grounded Theory Method (GTM) was used to analyse the qualitative data produced during and between AR cycles. The individual papers in this thesis adopted Glaser's classic GTM tradition (Glaser 1992), chosen for its compatibility with AR and its emphasis on the utility of the theoretical product. Glaser's approach follows the dictum "all is data," meaning that any artefact encountered or created during the research effort, including informal communications, meeting notes, version control records, and chat logs, constitutes valid material for analysis. This was particularly well suited to the digital audit trail generated by the IaL context, where the boundary between formal documentation and informal interaction is porous.

Paper III describes how line-by-line coding of player feedback during and after EoP game sessions produced three emergent categories (transdisciplinary utility, shared boundary objects, and increased alignment) by grouping codes into clusters based on similarity. Paper IV reports open coding of raw electronic communication, including instant messaging and comment threads, followed by constant comparative analysis that refined initial concepts into categories and themes. In both cases, the analytic process was Glaserian: codes emerged from the data rather than from a pre-existing framework, constant comparison tested emerging categories against new instances, and the emphasis was on producing theory that worked in practice.

Dick (2003) argues that the Glaserian tradition integrates naturally with AR because both prioritise responsiveness to emerging patterns over adherence to predetermined analytic structures. In practice, this meant that GTM coding proceeded in parallel with the AR cycles rather than as a post-hoc exercise. As each intervention produced new artefacts, these were coded and compared with categories from earlier cycles. Categories that proved robust across cases were retained and elaborated. Categories that did not survive comparison were revised or abandoned.

The synthesis in Chapter 6 extends this within-paper analysis across the full set of cases. Cross-case comparison follows the same constant-comparison logic but operates at a higher level of abstraction, comparing categories and patterns across Cases A through D and testing them against the formal model's predictions. Where the published papers each report categories grounded in a single dataset, Chapters 6 and 7 examine whether those categories converge, diverge, or qualify one another when placed side by side. This cross-case synthesis adopts a constructivist orientation, following Charmaz (2006), acknowledging that the researcher's interpretive position shapes category construction, particularly when categories from earlier cases inform the design of later interventions.

4.3.4 Combining Methodologies

IaL, AR, and GTM were combined through a structured iterative process based on Dick (2003). Dick argues that AR and GTM can be closely integrated by using the AR cycle as the framework for intervention, while GTM provides the analytic process for interpreting the data produced during and after each cycle. The result is a methodology in which interventions and analyses are continuously linked, each informing the other.

In practice, each cycle proceeded through the following steps:

1. Plan and implement an intervention informed by current understanding or identified needs.
2. Capture data as the intervention takes place. The digital nature of project work meant that artefacts such as instant messages, meeting notes, and version control records were generated automatically.
3. Code and compare the digital artefacts, using open coding followed by constant comparison to identify actions, processes, and emerging categories.

4. Write memos and map concepts throughout the analysis, using concept maps, mind maps, and causal diagrams to trace and refine ideas, categories, and relationships.
5. Develop formal models by iteratively translating and refining informal visualisations into structured diagrams, process models, or computational representations as categories stabilise.
6. Sample theoretically by purposefully seeking new data or digital artefacts to clarify or extend key categories as analysis progresses.
7. Reflect and re-plan by assessing whether key categories have reached saturation, documenting findings, and using the results to inform the next intervention cycle.

This approach allowed both conceptual and formal models to develop incrementally and remain grounded in empirical data. The coordination of intervention, digital data capture, GTM analysis, and iterative modelling maintained a close link between practical work and theory development throughout the six-year project. Chapter 6 presents the case-level findings from this process and examines the cross-case patterns and their relationship to the formal model.

4.4 Integration strategy

The formal model and the qualitative strand address the same research questions from different directions. The model defines constructs and encodes their relationships as equations. Chapter 5 analyses those relationships and derives testable hypotheses. The case studies, conducted over six years at a single VSE, produce evidence against which those hypotheses can be confronted. This section describes how the two strands connect and how the resulting claims are warranted.

4.4.1 From Constructs to Observables

Confrontation requires that each formal construct has an observable counterpart in the case data. The principal mappings used across the four cases are summarised in Table 16.

Table 16 Mapping from constructs to observables

Model construct	Observable counterpart	Primary evidence
Complexity C	Distinct configurations at the decision point	Architecture models, configuration records
Designed cognitive reserve h	Automated tools, environmental cues, pre-authorised actions	Tool logs, Dependabot alerts, smart bulb status, game session records
Observation noise ω	Ambiguity or conflicting signals in cross-boundary communication	Chat logs, issue tracker comments, meeting notes
Shared expectations Φ	Observed agreement on priorities, thresholds, and response norms	Retrospective notes, coding of team discussion, response patterns
Operational functional information	Narrowing of viable options under time pressure	Incident timelines, decision logs, time-to-remediation
Resilience R	Service continuity and recovery during disturbances	Monitoring data, incident records, remediation times

These mappings are approximate and context-dependent. The case data do not support exact numerical estimates for most parameters. Confrontation, therefore, operates at the level of direction, ordering, and pattern rather than point estimation, consistent with the PLT epistemology in which the model's value lies in the logical structure of its predictions rather than parametric fit to a specific dataset.

4.4.2 Hypothesis Confrontation

Chapter 5 derives fourteen hypotheses from the formal model. Each is a conditional claim about how changes in one or more variables affect an observable outcome. Chapter 6 maps each case finding to the hypotheses it supports, qualifies, or contradicts, and presents a cross-case convergence matrix. Confrontation does not require measurement to the precision implied by the equations. It requires that the directional prediction is recognisable and that the most plausible alternative explanations can be addressed.

4.4.3 Iterative Development

Although the presentation follows a linear sequence from model to hypotheses to cases, the research process was iterative. Cases A and B preceded the formal model's completion and contributed to its development. Observations from the EoP game sessions, particularly that structured environmental features channelled attention more effectively than procedural mandates, motivated the development of the constructs of Regimes of Attention and

Patterned Practices, which became central to PHAS-EAI. The formal model was then developed to provide a mechanistic account of those observations, and Cases C and D were designed with explicit reference to model constructs.

Cases A and B are therefore better described as the empirical soil from which the model grew, while Cases C and D provide a more conventional test of predictions against new evidence. The cross-case analysis in Chapter 6 distinguishes between these roles and weights the evidential contribution accordingly.

4.4.4 Validity and trustworthiness

The single-company, single-researcher design creates characteristic threats to validity that must be acknowledged. This section summarises the design-level strategies employed. Section 7.6 presents the full assessment of internal, external, and construct validity, reliability, and thesis-level limitations in light of the actual results.

Internal validity depends on ruling out alternative explanations for observed effects. The principal risks are researcher influence (the embedded researcher's presence may itself produce improvement), novelty effects (participants perform better because they know they are observed), and temporal confounds (maturation, market changes, or staff turnover may explain the trajectory). Three design features address these risks: the six-year observation period reduces the plausibility of novelty-driven effects, the persistent digital audit trail enables retrospective analysis of periods when the researcher was less active, and the GTM constant-comparison procedure serves as a systematic corrective against selective interpretation. Where alternative explanations cannot be excluded, they are acknowledged as limitations in the case discussions.

Construct validity concerns whether the observable counterparts in Table 16 adequately represent the formal model's quantities. Because the model quantities (C , I_g^{op} , h , Φ , γ_i) are newly defined and no established instrumentation exists for measuring them in field settings, the mappings are interpretive rather than quantitative. This is an inherent limitation of a novel formal framework. The design-level mitigation is triangulation across multiple data sources within each case and convergence testing across cases. Section 7.6.3 evaluates the adequacy of these mappings against the evidence.

External validity in the sense of statistical generalisation to a population of VSEs is not claimed. The formal model carries the generalisable structure. Transferability depends on the reader's judgement of whether the scope conditions (VSE size, Norwegian regulatory environment, building automation domain, single-researcher embeddedness) match their own context.

Trustworthiness of the qualitative analysis rests on four strategies: prolonged engagement over six years, a persistent digital audit trail that could be revisited years after events occurred, theoretical sampling following Charmaz (2006) that directed data collection toward emerging categories, and constant comparison across cases and time periods that required each category to be tested against disconfirming as well as confirming instances.

4.5 Ethical Considerations

This project was conducted under the Research Council of Norway Industrial PhD Scheme (project number 311242), which requires the researcher to be an employee of the participating company. The researcher held the role of systems engineer at Aiwell AS when the project began in 2020, and was appointed Chief Technology Officer in June 2022. This dual role as employee and doctoral candidate raises ethical considerations that go beyond those of a conventional doctoral project, because the researcher held operational responsibility within the organisation under study and, from 2022 onward, also held managerial authority.

Ethics assessment and approval. The project was assessed against the criteria for mandatory notification to the Norwegian Centre for Research Data (NSD, now Sikt) and was found not to require formal ethics approval. The assessment considered the data to consist of organisational process artefacts (engineering documents, commit logs, meeting notes, chat records, and survey responses) rather than sensitive personal data within the categories defined by Article 9 of the General Data Protection Regulation (GDPR). The project did not involve health data, political opinions, biometric data, or data concerning minors.

Informed consent and the dual-role risk. All Aiwell employees who participated in the research were informed of the study's purpose, the types of data to be collected, how data would be processed, and how anonymised findings would be used in publications. Participants were informed of their right to withdraw at any time without negative consequences. The dual role of the researcher as both an Aiwell employee with increasing operational authority and a doctoral candidate created a potential power asymmetry: participants might have felt implicit pressure to consent or to behave in ways they believed the researcher wanted. Three design-level measures mitigated this risk. First, the process was overseen by the university supervisors rather than by the researcher alone. Second, the company owner, who held ultimate managerial authority, was aware of and endorsed the separation of research and operational roles, with the researcher dividing the work week equally between Aiwell locations and on the USN campus. Third, the AR methodology itself reduced the risk of coercion, because interventions were designed collaboratively with the team rather than imposed by the researcher.

Anonymisation and publication review. All personal identifiers were removed or pseudonymised before data were used in analysis or reporting. Analytical codes referenced anonymised identifiers only. Aiwell reviewed all material submitted for publication to ensure that no sensitive business information or personally identifiable information was disclosed. The company name and general domain are disclosed with Aiwell's explicit agreement, as required by the transparency norms of the Industrial PhD Scheme, but individual participants are not identifiable in any published paper or in this thesis.

Data management. A formal Data Management Plan (DMP) was submitted to the Research Council of Norway at the start of the project. It specifies data sources, file formats, storage and backup procedures, access controls, anonymisation protocols, retention periods, and responsibilities. All research data are stored on Aiwell's encrypted SharePoint environment with role-based access control. Raw data are not shared externally. Anonymised data and associated documentation will be retained for at least 10 years after project completion. The full DMP is reproduced in English translation in Appendix A4.

Compliance. The processing of personal data throughout the project followed the European Union General Data Protection Regulation (GDPR) and the Norwegian Personal Data Act (personopplysningsloven). The legal basis for processing was informed consent (GDPR Article 6(1)(a)), and the purpose was limited to academic research within the scope defined in the DMP.

5 Key Findings from Model Analysis

Chapters 3 and 4 developed a formal model of how engineering teams sense, infer, communicate, and act under uncertainty, and defined the configuration space within which those actions take effect. This chapter draws out the practical implications. It shows what the formal definitions predict about the operational realities of small teams building complex infrastructure, and it organises those predictions into fourteen testable hypotheses that the case studies in Chapter 6 evaluate.

The formal definitions, proofs, parameter sensitivity analyses, and executable verification artefacts that substantiate each claim are collected in Appendix A1.

5.1 Complexity Rises, and it Rises in Bursts

A recurring observation across the papers and case material in this thesis is that operational competence can appear stable for extended periods before degrading abruptly. Teams that handled incidents well for years suddenly struggle. Governance arrangements that were adequate become unworkable after a system upgrade or a new supplier integration. The formal models offer an explanation: effective complexity in socio-technical infrastructure does not increase smoothly, and it need not increase at all for experienced difficulty to change. Complexity tends to increase in step changes when coupling, constraints, or boundary conditions shift.

In the vocabulary of Chapter 3, this phenomenon reflects the fundamental constraint that Ashby's (1956) law of requisite variety imposes on any extended embodied mind. A team's internal variety is finite, bounded by the number of people, the precision of their instruments, the bandwidth of their communication, and the depth of their shared training. When the environment's variety grows in bursts, the team's internal variety does not increase in parallel. The result is a sudden mismatch between the complexity the team faces and the complexity it can mobilise against, experienced not as "the option count increased" but as a *loss of competence*.

5.1.1 Complexity is Feasible Distinct Optionality

The term "complexity" is used loosely in much of the systems engineering and infrastructure literature, often as a synonym for "complicatedness" or "difficulty". In the model developed in Chapter 4, complexity has a precise definition: it is the size of the feasible configuration set $C = |\mathcal{O}_\Delta|$.

This definition treats complexity as a count of distinct options. It includes both desirable configurations and failure modes. As the option set grows, so does the number of alternatives that must be evaluated, ruled out, constrained, or monitored. Each additional configuration increases both choice and error surface.

Two sanity checks anchor the definition. First, adding degrees of freedom or relaxing constraints cannot decrease C (Theorem T5.1-1 in Appendix A1). Second, adding constraints cannot increase C (Theorem T5.1-2). More levers create more options, and more constraints reduce them.

5.1.2 More Options Do Not Imply More Usable Options

A system can accumulate configurations without any of the new options being useful or survivable under a specific disturbance. Useful variety A is the subset of $\mathcal{O}_\Delta$ that delivers the required service under a disturbance with acceptable risk. Survivable variety B is the larger subset where the system remains safe and non-collapsed, even if service degrades. By construction, $A \subseteq B \subseteq \mathcal{O}_\Delta$, so $A \leq B \leq C$.

Under time pressure, the picture tightens further. When the response is time-constrained, only a fraction of options can be mobilised. The effective option counts $\tilde{A} = A \cdot f_A(r)$ and $\tilde{B} = B \cdot f_B(r)$ therefore shrink. A system may have many feasible configurations, many of which pass the performance threshold, yet be unable to mobilise any of them within the available time window.

5.1.3 Bursts from Multiplicative Composition of Options

Options combine multiplicatively across factors and subsystems. If two weakly coupled subsystems have option counts C_A and C_B , the combined count is approximately $C_{A||B} \approx C_A \cdot C_B$ (the composition property, proved as Theorem T5.1-3 in Appendix A1). This means that introducing a new lever with k alternatives multiplies the option count by k , producing a jump of $\log_2 k$ bits in the logarithmic measure. In infrastructure settings with multiple technical, organisational, and governance levers, even modest expansions at boundaries can trigger disproportionate growth.

5.1.4 Bursts from Constraint Changes

Bursts also arise from changes in constraints. Feasibility is explicitly modelled as $K(\ell)$, indexed by lifecycle or operating state ℓ . Shifting K changes $\mathcal{O}_\Delta(\ell)$ and therefore $C(\ell)$ without

requiring any change in the underlying configuration space X . A governance decision, a compliance reclassification, or a procurement acceptance criterion can abruptly reshape the feasible set. Distinctness can also change if φ changes due to new reporting, assurance, or categorisation regimes, producing a perceived increase in complexity even when micro-configurations remain similar.

5.1.5 Functional Information Explains Why Bursts Feel Like a Sudden Loss of Competence

Operationally, complexity bursts are not experienced as “the option count increased”. They are experienced as a sudden loss of competence: teams that were previously effective find themselves unable to identify and enact satisfactory responses within the available time. Functional information provides the formal mechanism for this phenomenon.

Design-time functional information $I_g^*(\rho) = \log_2 C - \log_2 |A_{g,\rho}|$ measures how many binary decisions are required to identify a satisfactory configuration under an uninformative search. When either C grows (more options to search through) or $|A_{g,\rho}|$ shrinks (fewer options pass the threshold), I_g^* increases, and the search becomes harder.

Operational functional information $I_g^{\text{op}}(\rho, \tau_d) = I_g^*(\rho) - \log_2 f_A(\tau_d, \tau_{\text{eff}})$ adds a time-pressure term. The quantity $-\log_2 f_A$ acts as an additional difficulty charge: when mobilisation drops to 50%, the operational cost increases by 1 bit, and at 25%, by 2 bits. Small decreases in mobilisation fraction can therefore produce sharp increases in the effective difficulty of selecting a satisfactory response, even if the team’s nominal capability and the system’s design appear unchanged.

The combination of multiplicative option growth and logarithmic difficulty scaling explains the “competence illusion”: a system can accumulate options across stable periods while the share of satisfactory options silently shrinks, or time pressure silently rises, until a burst crosses a threshold and the cost of choosing a response jumps discontinuously. In the terms of Chapter 3, the team’s generative model of its environment has become stale. The prediction errors that signal the mismatch are large, but because they arrive suddenly rather than incrementally, the team’s precision-weighting mechanisms have had no opportunity to adjust. The team is no longer surfing uncertainty (Clark 2016). It is being overwhelmed by it.

5.1.6 Hypotheses

H1 (Bursty complexity). Complexity growth in interdependent socio-technical systems manifests as step changes in mobilisation difficulty rather than smooth degradation. Bursts in complexity potential C_p or boundary constraints cause bursts in operational functional information I_g^{op} and observable drops in response performance.

Prediction: Incident response metrics should exhibit regime shifts rather than gradual trends when structural changes (for example, new integrations, governance changes, reclassifications) occur.

H2 (Joint intervention). Interventions that jointly reduce the total option count C and increase the successful option share $|A_{g,\rho}|/C$ produce larger resilience gains than interventions that affect only one term.

Prediction: Standardisation that prunes failure modes (reducing C) while adopting proven patterns (increasing $|A_{g,\rho}|$) should produce measurably larger improvements in I_g^{op} than either action alone. Appendix A1 provides a formal proof of the additivity of intervention effects in the log domain (Theorem T5.1-9).

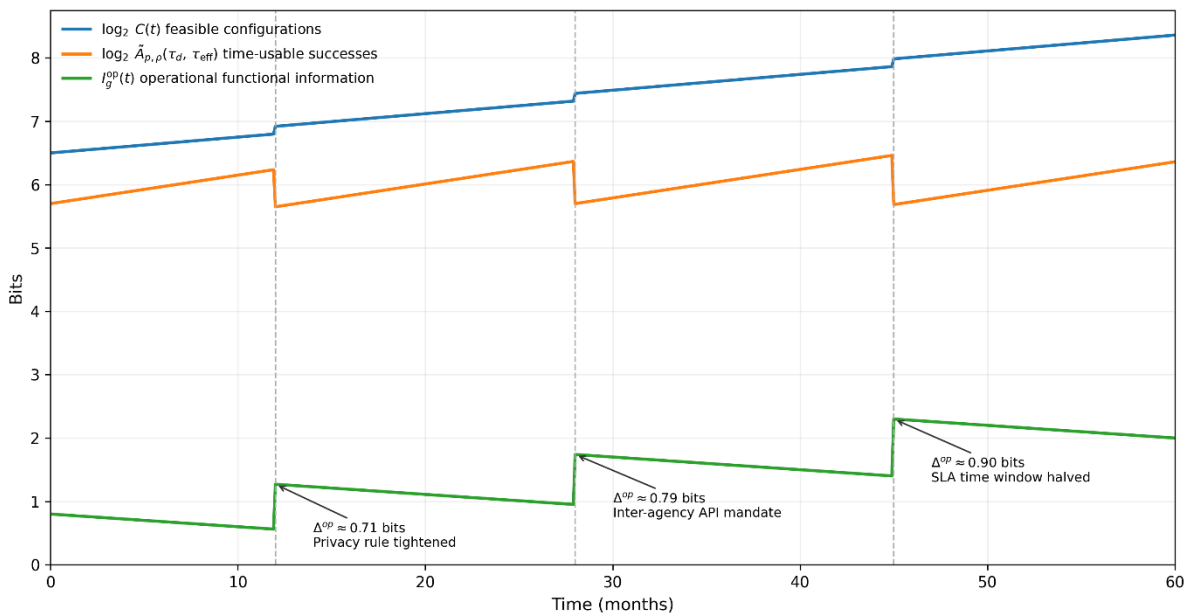


Figure 11 Complexity rises in bursts.

Figure 11 shows step changes in effective decision burden (I_g^{op}) and mobilisation fractions across a sequence of events that modify the option space, constraints, and time pressure. Each plateau represents a period of operational stability, and each jump corresponds to a structural change.

5.2 The Relative Resilience Advantage of VSEs is Real and Bounded

VSEs are often characterised by resource scarcity and limited formal process maturity. Yet a VSE can often outperform larger organisations in specific operational settings, particularly when speed of response matters and the disturbance is moderate. The formal models explain why this advantage exists, where it applies, and why it reverses.

The sailing crew introduced in Chapter 3 provides the intuitive frame. A professional crew cannot control the sea. What it can do is maximise the mobilisable fraction of its finite variety, not by building more elaborate models of the wind but by *designing* the boat, the instruments, and the communication protocols so that the right information reaches the right person at the right time. The crew's advantage over a larger, better-resourced vessel with more complex rigging and a bigger crew comes precisely from its shorter communication paths and faster coordination cycles. But that advantage is bounded: when the storm exceeds the boat's physical headroom, no amount of fast coordination can compensate for insufficient structural reserve. The formal model makes this intuition precise.

5.2.1 The Mechanism: Faster Mobilisation of Relevant Options

Resilience in the model depends on time-bounded mobilisation of useful and survivable variety. A VSE advantage emerges when the smaller organisation can mobilise a larger share of its useful options within the disturbance window. Formally, this happens when the VSE has a lower effective mobilisation time τ_{eff} (due to shorter decision chains, fewer handoffs, and less governance overhead), which produces a higher time allowance ratio $r = \tau_d / \tau_{\text{eff}}$, which in turn yields a higher mobilisation fraction $f_A(r)$ and therefore a larger effective useful count $\tilde{A}$.

In VSEs, communication paths are short and often ad hoc. A single engineer may serve as project manager, analyst, developer, and tester, which eliminates handoff delays between specialised roles. Anomaly detection can be near-instant when the person who built the system also monitors it. These characteristics directly compress τ_{detect} , τ_{decide} , and τ_{execute} , producing the lower τ_{eff} that drives the formal advantage. In the fast-disturbance regime (small τ_d), where mobilisation is scarce, and the time-allowance ratio differentiates the two organisations, the VSE can have higher $\tilde{A}$ than the larger organisation, even though its static useful variety A is smaller. The faster organisation mobilises a larger fraction of its smaller set, and this fraction dominance can overcome the size disadvantage.

The VSE does not succeed by having more internal complexity. It succeeds by coupling more tightly to its environment, trading expensive internal coordination against cheap environmental structure and fast local feedback loops.

5.2.2 The Bound: Finite Headroom and Survivable Variety

The advantage is bounded because the VSE has less total headroom H and less survivable variety B . This caps the magnitude of disturbance the VSE can absorb. As M increases towards $M_{\max}$, the VSE reaches collapse sooner than the larger organisation, regardless of how fast its mobilisation is.

The VSE characteristics that enable rapid mobilisation are the same as those that limit headroom. Resource scarcity means limited spare capacity, limited surge staffing, and limited financial reserves to absorb sustained disruption. Generalist roles, while enabling fast decision-making, create single points of failure: if the one person who understands a subsystem is unavailable, both useful variety A and survivable variety B collapse. The “hero culture” documented in the VSE literature describes precisely this vulnerability. Individual expertise (high u) compensates for low designed reserve (h) under normal conditions, but when the hero is absent or when multiple disturbances coincide, the organisation has no fallback.

Financial precarity reinforces the bound. Small firms operating on thin margins cannot maintain buffer budgets, redundant capability, or delegated authority structures. Physical headroom H_{phys} , which determines the magnitude limits M_0 and $M_{\max}$, is structurally lower in organisations that cannot afford redundancy. This means the VSE advantage $\Delta R = R_{\text{VSE}} - R_{\text{Large}}$ is positive only within a specific region of the (τ_d, M) plane. Outside this region, the advantage disappears or reverses.

5.2.3 The Parity Boundary

Theorem T5.2-4 in Appendix A1 proves, under reasonable continuity conditions, the existence of a parity boundary: a disturbance timescale τ_d^* at which $\Delta R = 0$ for a given magnitude M . Below τ_d^* , the VSE advantage is positive (fast disturbances, moderate magnitudes). Above τ_d^* , the larger organisation’s greater headroom and survivable variety dominate, and the advantage reverses.

The heatmap in Figure 11 visualises this structure. The zero contour marks the parity boundary. The positive (warm) region shows where faster VSE mobilisation overcomes smaller variety. The negative (cool) region shows where larger reserves dominate.

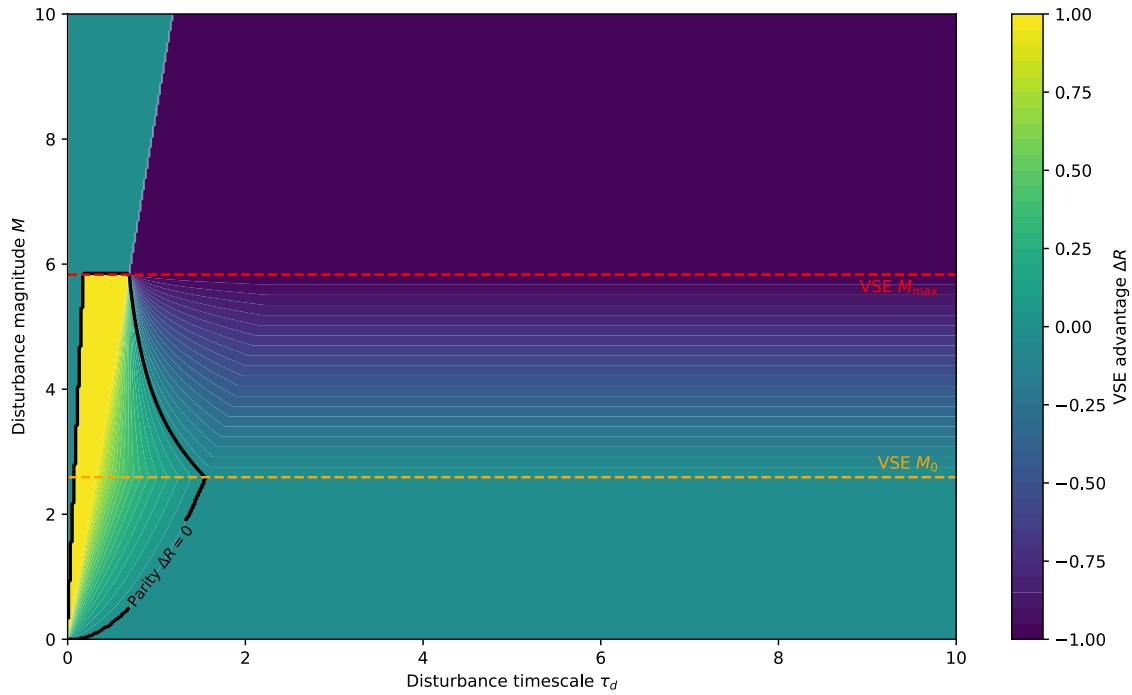


Figure 12 Resilience advantage of a VSE over a larger organisation,

Figure 12 shows the resilience advantage of a VSE over a larger organisation, mapped across a disturbance with timescale τ_d and magnitude M . The zero contour marks the parity boundary $\Delta R = 0$. Warm colours indicate VSE advantage (faster mobilisation dominates), while cool colours indicate reversal (larger reserves dominate).

5.2.4 Connection to Burst Dynamics

The parity boundary is not fixed. A burst increase in complexity potential C_p increases the VSE's effective mobilisation time τ_{eff} (through structural drag effects detailed in Section 5.3), which shifts the parity boundary to the left, thereby reducing the region of VSE advantage. A VSE can therefore remain apparently effective across stable periods of complexity, then degrade abruptly when a burst moves it into the negative-advantage region.

This creates a specific vulnerability: the VSE may not recognise that its advantage has eroded, because the structural change may not be visible as a “failure” until a disturbance occurs in the regime in which the advantage no longer holds.

5.2.5 Hypotheses

H3 (VSE advantage is regime-dependent). The VSE resilience advantage is positive for fast disturbances and moderate magnitudes when $r = \tau_d/\tau_{\text{eff}}$ favours rapid mobilisation, but

becomes negative beyond a bounded magnitude band as M approaches or exceeds the VSE $M_{\max}$ due to limited headroom and survivable variety.

Prediction: In case study settings, VSEs should show strong incident response for routine disturbances (moderate M , short τ_d) but should struggle with high-magnitude or slow-evolving disturbances where reserves and sustained coordination matter more than speed.

H4 (Burst-induced regime shift). A burst increase in complexity potential or coordination drag can shift an organisation abruptly from an apparently resilient regime into degradation or collapse by increasing τ_{eff} and operational functional information, even when individuals' nominal capabilities appear unchanged.

Prediction: Integration events, governance changes, and supplier onboarding that increase the interface count and increase coordination friction should be followed by measurable increases in response time and decreases in first-response success rate, even if team composition and expertise remain unchanged.

5.3 Resilience is Mobilisable Variety Under Time Pressure

A common intuition is that resilience comes from having many options (redundancy) or from having expert people. The formal models show that neither is sufficient on its own. Resilience depends on the fraction of available options that can be mobilised within the time window imposed by the disturbance. Time, headroom, and structural drag together determine this fraction. This is the formal articulation of what Chapter 3 described as surfing uncertainty (Clark 2016): the capacity of a well-designed system to stay responsive under conditions that exceed its modelling capacity, not by building a more elaborate internal model but by maintaining the ability to act appropriately on incomplete information within the available time.

5.3.1 Nested Option Sets and Time Conditioning

The model's structure is based on progressive filtering. The full configuration space X_Δ is pruned by hard constraints K to produce the feasible set $\mathcal{O}_\Delta$ with C distinct options. Within this set, useful variety A and survivable variety B define nested subsets ($A \leq B \leq C$), where the mobilisation fractions f_A, f_B, f_C depend on the time allowance ratio $r = \tau_d/\tau_{\text{eff}}$. Resilience is therefore a function of nested set sizes, headroom, and the time structure of the response.

5.3.2 Response Time Decomposition and Structural Drag

Equation (28) shows how response time decomposes into three stages: Detect, Decide, Execute. This stage-specific structure matters for intervention design. An investment in telemetry reduces τ_{detect}^0 but does not change τ_{decide}^0 . Clarifying decision rights reduces τ_{decide}^0 but does not help if detection is the bottleneck. The model formalises these distinctions and supports targeted intervention selection based on which stage and which complexity factor predominate in a given setting (as shown in Theorem T5.3-1 in Appendix A1).

5.3.3 Skill Is Bounded

Human skill shortens the inflated response time through the effective time multiplier in Equation (22). The key formal result (Theorem T5.3-3 in Appendix A1) is that skill is bounded. No amount of expertise can reduce the effective time below a floor set by the maximum fractional reduction in response time attributable to skill λ_τ and the structurally determined time τ . This creates a formal limit to what expertise alone can compensate for: when structural drag is high, the floor rises, and the bounded speed-up from skill becomes insufficient.

This result formalises a key insight from Chapter 3. Beyond a certain point, investing in internal capability yields diminishing returns because the binding constraints are environmental (governance overhead, interface complexity, coordination delays), not individual. The implication, developed formally in Section 5.3.4 and tested empirically in Chapter 6, is that the productive response is to redesign the environment rather than to train the agent further.

5.3.4 Designed Cognitive Reserve

The cognitive component of headroom is given by Equation (27). The formal proof (Theorem T5.3-4 in Appendix A1) establishes two properties. First, $H_{\text{cog}} \geq h$. The designed reserve is a floor that holds regardless of skill level and task demand. Second, H_{cog} is monotone in h . Increasing the designed reserve always improves cognitive headroom, and this improvement is independent of individual capability.

This distinction is practically important for thin teams in VSEs. A worst-case witness at $u = 0$ (inexperienced, inflexible personnel under high-demand tasks) yields $H_{\text{cog}} = h$ exactly. The designed reserve is the only cognitive headroom available in this scenario. Investing in automation, checklists, and pre-authorised actions therefore protects resilience *precisely* where it is most needed: in high-stress, limited-expertise, demanding tasks.

The designed reserve h formalises the environmental scaffolding that Chapter 3 establishes as the thesis’s core intervention mechanism. The monotonicity result (H_{cog} non-decreasing in h) provides the formal warrant: investing in the team’s environment always improves cognitive headroom, whereas investing in individual expertise is bounded by λ_τ and cannot remove structural constraints. This asymmetry is the mathematical core of the thesis’s practical contribution.

5.3.5 Key Mechanisms

These formal results support four operational insights:

1. **Three-stage failure modes.** Detection, decision, and execution delays are distinct and can fail independently. An organisation may detect quickly but decide slowly (governance bottleneck), or decide quickly but execute slowly (deployment friction).
2. **Structural drag is factor-specific.** Different complexity potentials C_{p_i} slow different stages, so interventions must be targeted. Reducing policy complexity helps decision-making, and reducing interface churn helps execution. For a VSE subcontractor in a Totalentreprise contract, the dominant drag factor may be approval chain length at the decision stage rather than technical complexity at the detection stage, because the contractual structure imposes governance overhead that the firm’s internal processes were never designed to carry.
3. **Skill is bounded.** The speed-up from expertise is capped at λ_τ . Beyond the skill ceiling, governance and interface delays embedded in the structural drag term dominate outcomes.
4. **Designed reserve matters.** Automation, checklists, and pre-authorised actions preserve a minimum cognitive reserve via h , improving mobilisation reliability in thin teams independently of individual capability.

5.3.6 Operational Observables

The following quantities connect the formal model to case study evidence:

- Incident logs providing proxies for τ_{detect} , τ_{decide} , τ_{execute} .

- Structural drag proxies: interface count and churn, change rate, coupling density, governance friction, approval chain length.
- Designed reserve evidence: presence and use of runbooks, automation, pre-authorised actions, and tooling that protects attention.
- Skill proxies: team experience profiles, training records, rehearsal frequency.

5.3.7 Hypotheses

H5 (Time pressure reduces realised resilience). For disturbances of equal magnitude, a shorter timescale τ_d reduces realised resilience by lowering mobilisation fractions even if the feasible option set is unchanged, because $r = \tau_d/\tau_{\text{eff}}$ collapses the time-conditioned useful and survivable sets.

Prediction: Faster-evolving disturbances should produce worse outcomes than slower ones of equal severity, even for identical systems and teams.

H6 (Diminishing returns from expertise). In high-drag environments, increased expertise produces diminishing returns on response outcomes because the skill speed-up is bounded by λ_τ and cannot remove governance and interface-imposed delays embedded in the structural drag term.

Prediction: In high-complexity settings, the correlation between team experience and response performance should be weaker than in low-complexity settings.

H7 (Designed reserve outperforms experience investment). Interventions that increase design-preserved cognitive reserve h yield more reliable improvements in time-bounded mobilisation and resilience than interventions that attempt to increase experience alone, because h protects cognitive headroom even under high interaction demand.

Prediction: Teams with strong automation and pre-authorised actions should show more consistent incident response performance across personnel changes than teams that rely primarily on individual expertise.

5.4 Functional Information Is the Cost of Choice

When a water main bursts and the operations team has thirty minutes to restore pressure, the difficulty does not depend on how many configurations the system can theoretically adopt. It depends on how many of those configurations actually work and how many can be identified and implemented within thirty minutes.

5.4.1 Design-Time Functional Information

The success share $p_g(\rho) = |A_{g,\rho}|/C$ indicates how rare satisfactory configurations are. When almost every configuration succeeds, functional information I_g^* is small. When only a small fraction succeeds, I_g^* is large. The measure is in bits: it counts the number of binary decisions needed to isolate a successful configuration under an uninformative search.

Two edge cases are immediate. If every feasible configuration passes, $I_g^* = 0$. If exactly one passes, $I_g^* = \log_2 C$, meaning the full complexity of the option space must be resolved.

5.4.2 Threshold Sensitivity and Step Behaviour

A key formal result (Theorem T5.4-1 in Appendix A1) establishes that raising ρ can only shrink $A_{g,\rho}$ (set nesting), that the count $|A_{g,\rho}|$ is antitone in ρ , and that I_g^* is therefore non-decreasing in ρ . The practical implication is that raising performance requirements always makes the search harder, never easier.

Because the option space is finite, the performance function Perf_g maps options to a finite set of performance levels. The count $|A_{g,\rho}|$ is therefore piecewise constant between adjacent performance levels, and I_g^* increases in discrete jumps at the thresholds where performance values are realised (Theorem T5.4-2 in Appendix A1). These jumps explain why small increases in required performance can produce disproportionate increases in design and operational difficulty: a threshold that coincides with a cluster of performance values eliminates many configurations at once.

5.4.3 Operational Functional Information Under Time Pressure

The term $-\log_2 f_A$ in Equation (42) is an additional difficulty charge imposed by time pressure. When mobilisation drops to 50%, the charge is 1 bit. At 25%, it is 2 bits. At 10%, it is

approximately 3.3 bits. This logarithmic amplification means that small reductions in mobilisation fraction can produce large increases in perceived difficulty.

Theorem T5.4-3 in Appendix A1 proves that reducing the effective time ratio r from Equation (34) (by faster disturbance evolution or slower mobilisation) increases I_g^{op} . Theorem T5.4-4 extends this to a three-part sensitivity result: increasing total complexity C raises I_g^* , increasing τ_{eff} raises I_g^{op} , and the pure complexity measure $\log_2 C$ is invariant to time pressure. This last point demonstrates that I_g^{op} captures a dimension of operational difficulty that raw complexity measures miss.

5.4.4 How Bursts Create Sudden Difficulty Increases

The connection to Section 5.1 is now concrete. A complexity burst increases C without necessarily increasing the success set $|A_{g,\rho}|$ (new options are not guaranteed to pass the threshold), so I_g^* rises. Simultaneously, if the burst increases structural drag, it raises τ_{eff} , reduces r , reduces f_A , and raises I_g^{op} further. A threshold change (a new compliance requirement, a raised service level) can independently shrink $|A_{g,\rho}|$, producing another discontinuous jump in I_g^{op} .

The result is that operational difficulty can increase sharply from two independent directions: expanding options (wider search space) and shrinking mobilisation (less time to search). When both happen simultaneously, as during a major system integration or a governance reorganisation, the joint effect can be dramatic.

For a VSE, this joint effect is especially consequential. A small firm that delivers a component within a larger infrastructure integration may experience both a complexity burst (new boundary options from the integration) and a mobilisation reduction (coordination overhead increasing τ_{eff}) at the same time. The firm's operational difficulty, measured in I_g^{op} , can jump by several bits in a single project phase, even though the firm's own technical capability and the design of its components have not changed. The competence illusion from Section 5.1 applies directly: the firm was competent yesterday and is struggling today, not because of any internal failure but because the information burden of finding a satisfactory configuration has crossed a threshold.

5.4.5 Metrics for Case Study Analysis

The following quantities can be estimated from case study data and connected to the model:

- Option count proxies: number of distinct configurations, design alternatives, or deployment variants by lifecycle mode.
- Success set estimates: fraction of configurations that meet stated service levels, safety requirements, or recovery time objectives.
- Time pressure proxies: typical disturbance timescales versus observed response times, providing estimates of r .
- Threshold changes: documented changes in required performance levels across contracts, governance decisions, and operational regimes.

5.4.6 Hypothesis

H8 (Operational functional information predicts outcomes better than raw complexity).

Operational functional information $I_g^{\text{op}}(\rho, \tau_d)$ predicts resilience outcomes better than raw complexity C alone, particularly when disturbances are fast or when performance thresholds are raised, because it incorporates both the sparsity of success and time-bounded mobilisation.

Prediction: Incidents where measured I_g^{op} is high (few usable options under time pressure) should show worse outcomes than incidents where C is equally high but I_g^{op} is lower (many options pass the threshold, and mobilisation is adequate).

5.5 Attention, Authority, and Shared Expectations are Dependability Variables

A technically competent team can still fail to detect a creeping anomaly, delay escalation until it is too late, or miscoordinate across organisational boundaries during a complex incident. These failures are not random: they follow patterns that can be traced to how information is sensed, weighted, communicated, and acted upon. The PHAS-EAI model provides a formal account of these dynamics, and this section draws out the practical implications for dependability.

Chapter 3 established that attention is not a separate faculty but the process of adjusting precision estimates across competing channels (Feldman et al. 2010). This section shows how that process, when formalised through the PHAS-EAI model, converts attention,

authority, and shared expectations from soft management concerns into quantities with proved directional properties.

5.5.1 Perception and Communication Are Noisy Channels

In the PHAS-EAI model, sensing is indirect: observations are a noisy transformation of external states. Communication between agents follows the same structure: incoming signals from peers are noisy transformations of their internal states. Misinterpretation is therefore a *normal operating condition* in multi-actor systems, not an exceptional failure. The quality of the observation mapping $g(\cdot)$ and the magnitude of the noise ω from Equation (2) determine the extent of ambiguity remaining after sensing and communication.

Precision, defined as the inverse of noise variance $1/\sigma^2$, directly affects inference quality: Theorem T5.5-4 in Appendix A1 proves that higher precision (lower σ^2) monotonically improves expected accuracy, and that the derivative of expected accuracy with respect to σ^2 is strictly negative.

5.5.2 Shared Expectations, Alignment and Authority Weights

Coordination in multi-agent settings is formalised through shared expectations (Equations (12) and (14)). The authority weight determines how strongly an agent's expectations track the peer's inferred state versus its own beliefs. In a multi-stakeholder infrastructure, different actors assign different weights of authority to the same peer signals. A regulator may weigh safety signals heavily, whereas a contractor may weigh cost signals more strongly.

Theorem T5.5-2 in Appendix A1 proves that norms and expectations ϕ for a single peer lies within the convex hull of the agent's internal model of the world μ and peer model $f(\psi)$. Theorem T5.5-5 proves that authority-weight asymmetry ($\gamma_A \neq \gamma_B$) between two agents generates coordination friction that is monotone in the weight gap $|\gamma_A - \gamma_B|$. In practical terms, when two stakeholders weigh the same evidence differently, their expectations diverge, and the friction is proportional to both the weight difference and the signal difference.

The agent-based simulation in Appendix A2 demonstrates this monotonicity constructively and reveals a precise interaction structure. In a 2×2 factorial crossing Theory of Mind with goal alignment, adding Theory of Mind to unaligned agents increases system-level free energy from approximately 50 to approximately 60, degrading collective performance below the no-interaction baseline even though individual convergence distance improves marginally (from 7.1 to 6.9). This divergence between individual and collective metrics is itself

significant: each agent's improved social perceptiveness makes it more effective at pursuing its own local optimum, widening the effective weight gap $|\gamma_A - \gamma_B|$ that the theorem identifies as the driver of friction, and fragmenting the collective as agents converge toward different targets. Adding Theory of Mind to aligned and scaffolded agents produces the best collective outcome ($F^\Sigma \approx 19$, mean convergence distance 1.6), a 77% improvement over baseline. Theory of Mind is therefore conditionally beneficial: it amplifies whatever goal structure is present. This result has direct implications for intervention design: inference-layer improvements deployed without alignment of shared expectations Φ can be actively counterproductive at the system level. Section 5.7 returns to this interaction when discussing joint intervention dominance.

5.5.3 Affordance and Drift

Actions are selected by minimising expected free energy, which includes an affordance term $-\ln p(a)$. Actions with high affordance (high prior probability of being selected) are favoured, whereas actions with low affordance are deprioritised. Theorem T5.5-3 proves that increasing affordance for a desired action strictly reduces free energy, making the action more likely.

This has direct implications for dependability. If a security review or maintenance check has low affordance (it is bureaucratically difficult, requires special approval, or lacks tooling support), it will be deprioritised under workload pressure, even by competent teams. The drift is not laziness or incompetence but the *rational response* of an inference system that prioritises actions that improve the accuracy of salient signals. Chapter 3 identified this as the system's preference for low-cost, high-return cognitive strategies: the agent preferentially adopts the action with the lowest expected free energy, regardless of whether that action serves long-term dependability. In a poorly designed information environment, the low-cost strategy is to ignore diffuse security signals. In a well-designed one, the low-cost strategy is to act on them.

Governance, procurement conditions, and tooling function as niche construction within the PHAS-EAI framework: they change the environment in which agents operate, thereby making certain actions affordable and altering behaviour without directly changing beliefs or capabilities. Making dependability-relevant actions the default, pre-approved, and tool-supported is therefore a structural intervention on affordance. It is, in the vocabulary of Chapter 3, environmental design for an extended embodied mind.

5.5.4 Regimes of Attention

The combined model identifies Regimes of Attention as a design handle for reliability. A Regime of Attention is an institutional arrangement, supported by tools, processes, and governance, that increases the precision of dependability-relevant signals and increases the affordance of dependability-relevant actions.

In formal terms, a Regime of Attention:

1. Reduces noise ω on signals by improving telemetry, reporting standards, and data quality.
2. Improves the observation mapping $g(\cdot)$ by connecting model elements to monitored signals through machine-readable traceability.
3. Increases affordance $p(a)$ for escalation, review, and remediation by making these actions default, pre-authorised, and tool-supported.
4. Stabilises shared expectations Φ by reducing authority-weight asymmetry through explicit governance and shared vocabularies.

A Regime of Attention is therefore a designer environment in the sense of Clark (2016), specialised for dependability. It restructures the statistical regularities of the PHAS's information environment so that dependability-relevant signals receive high precision-weighting and dependability-relevant actions have high affordance. White lines painted on a cliff-top road allow a driver to solve a complex navigation problem by tracking a simple, high-precision optical feature (Clark 2016). A smart bulb connected to a vulnerability API allows an engineer to solve a complex security monitoring problem by noticing a change in ambient colour. The formal mechanism is identical: an environmental modification that makes a high-precision signal cheaply available, cueing a low-cost cognitive strategy that produces the desired behaviour without requiring the agent to be internally more complex.

5.5.5 Implications for Case Study Analysis

The model shifts the analytical lens from “did the team have the right artefacts?” to “were the right signals salient, trusted, and action-guiding under pressure?” Specifically:

- Track not only whether governance artefacts exist, but whether they are used under workload pressure.
- Assess whether dependability signals are high-precision (low ω) and well-mapped ($g(\cdot)$ connects events to actionable categories).
- Assess whether dependability actions have high affordance (pre-approved, tool-supported, default).
- Look for authority-weight asymmetry across organisational boundaries as a source of coordination friction and decision reversals.

5.5.6 Hypotheses

H9 (Regimes of attention reduce drift). Regimes of attention that increase the salience and precision of dependability signals reduce drift away from resilience-improving actions under workload pressure, leading to faster detection and more consistent escalation.

Prediction: Teams operating under a Regime of Attention (structured telemetry, pre-authorised escalation, machine-readable evidence chains) should show shorter detection times and more consistent escalation behaviour than teams with equivalent technical capability but weaker signal infrastructure.

H10 (Authority-weight asymmetry predicts friction). Authority-weight asymmetry across stakeholders, expressed as heterogeneous γ_i values in the construction of Φ , predicts coordination friction and decision reversals across organisational boundaries, which manifest as disproportionate increases in decision latency during complex incidents.

Prediction: Incidents involving multiple stakeholders with documented differences in authority weights should exhibit longer decision phases and higher rates of decision reversal than incidents handled by a single team with aligned authority weights.

5.6 Empirical Alignment and Threshold Behaviour

This section connects the formal models to empirical patterns reported in the wider systems engineering literature, using the Elm et al. (2012) survey results as a concrete anchor. The aim is to show that the model's predictions about bounded skill effects and regime-

dependent performance are consistent with observed data, and that the model provides a structural explanation for patterns that are otherwise reported descriptively.

5.6.1 The Elm and Goldenson Findings

Elm et al. (2012) surveyed systems engineering practices across projects of varying complexity and reported three linked results:

1. **Systems engineering capability versus project performance.** Higher SE capability is associated with better project outcomes, but the strength of the association depends on the complexity regime. In high-complexity projects, SE capability is strongly associated with performance. In low-complexity projects, the association is present but weaker.
2. **Experience versus project performance.** Higher experience is associated with better outcomes in low-complexity projects. In high-complexity projects, the association is considerably weaker.
3. **Driver summary.** Project challenge (a proxy for complexity) correlates negatively with performance, whereas experience correlates positively. However, the positive effect of experience is moderated by complexity.

5.6.2 Model-Supported Interpretation

The formal models provide a structural explanation for these patterns through three mappings:

Experience maps to the skill factor $u(E, MF)$. In the model, skill reduces effective mobilisation time through $\sigma_\tau = 1 - \lambda_\tau \cdot u$ and increases cognitive headroom through $H_{\text{cog}} = h + (1 - h) \cdot u \cdot q(I_T)$. Both effects are bounded: σ_τ cannot fall below $1 - \lambda_\tau$, and H_{cog} cannot exceed 1. In low-complexity environments where structural drag is modest, the bounded speed-up and headroom increase from skill make a meaningful difference. In high-complexity environments, the bounded skill multiplier cannot offset the structural drag, and outcomes become dominated by baseline stage times and complexity potential.

Systems engineering capability maps to structural interventions on C_P , τ_{stage}^0 , and h . SE capability, as described in the survey, encompasses process codification, interface standardisation, tool support, and the adoption of design patterns. In the model, these

interventions reduce complexity potential (fewer varied interfaces, lower churn), reduce baseline stage times (codified detection, decision, and execution procedures), and preserve cognitive reserve (automation, checklists, pre-authorised actions). Unlike skill, these structural levers are not bounded by λ_τ and their effect grows linearly with the drag they offset. The return on SE investment reported by Honour (2013), with a median ROI of 3.5:1 and up to 7:1 for projects that initially had no SE, is consistent with this model prediction: the largest gains accrue where structural drag is highest and where the bounded skill multiplier has been exhausted.

This mapping connects directly to the Brooks precedent established in Chapters 1 and 3. Classical AI pursued increasing internal complexity (experience, expertise, planning capacity). Brooks (1991) achieved superior performance through environmental design (coupling simple behavioural modules to structured environments). The Elm and Goldenson data show the same pattern at the organisational scale: when complexity is high, structural interventions on the environment ($C_P, \tau_{\text{stage}}^0, h$) dominate individual capability (u). Systems engineering capability, in this reading, is not a property of *individuals* but a property of the team's *designed cognitive environment*.

Project complexity maps to the magnitude of C_P and the mobilisation ratio r . High-complexity projects exhibit greater complexity potential, longer response times, lower mobilisation fractions, and higher operational-functional information. The regime shift occurs when structural drag becomes sufficiently large that the bounded skill multiplier is insufficient to maintain adequate mobilisation.

5.6.3 Connecting to Bursts and the Bounded VSE Advantage

The VSE advantage (Section 5.2) and the empirical pattern here share a common formal structure. A VSE can outperform in a niche, low-complexity environment because expertise increases mobilisation fraction and reduces practical delays. This aligns with the positive region in the VSE advantage heatmap and with Elm and Goldenson's finding that experience predicts performance in low-complexity settings.

When complexity rises in bursts (Section 5.1), the system can abruptly transition into a regime in which structural drag dominates, experience loses predictive power, and systems engineering capability becomes decisive because it targets structural levers rather than individual speed-ups. This is precisely the transition from the positive to the negative region of the VSE advantage surface, which explains why the experience-performance relationship weakens in high-complexity projects.

5.6.4 Formal Derivation

Appendix A1 provides:

- A formal proof (Theorem T5.6-1) that the experience effect on τ_{eff} is bounded by $\lambda \cdot \tau$ and that the fractional maximum improvement from skill is exactly λ .
- A formal proof (Theorem T5.6-2) of diminishing returns: as C_p grows, the sensitivity $|\partial r / \partial u|$ decreases, meaning that additional expertise produces smaller improvements in the time allowance ratio.
- A formal proof (Theorem T5.6-3) of a structural dominance threshold: when $C_p \geq C_p^*$ (where $C_p^* = \lambda / (\gamma(k - \lambda))$, k being the structural reduction factor), structural interventions that reduce C_p produce larger gains than the full experience effect.
- A formal proof (Theorem T5.6-4) that step changes in C_p produce step drops in the time allowance ratio r connecting burst dynamics to abrupt shifts in performance regimes.

5.6.5 Hypotheses

H11 (SE capability dominates in high complexity). In high-complexity work, systems engineering capability has a stronger association with performance than prior experience because it reduces complexity potential and baseline stage times and preserves cognitive reserve h , while skill-driven speed-up is bounded.

Prediction: The model predicts that in settings where C_p is high, interventions on C_p , τ^0 , and h (process maturity, interface standards, automation) should show stronger performance effects than training and experience programmes.

H12 (Experience effect weakens after bursts). Prior experience is associated with performance in low-complexity work through improved $u(E, MF)$ and faster, more effective mobilisation, but this association weakens after bursts in complexity potential and interface churn, which increase structural drag and decision latency.

Prediction: In case settings, teams with strong experience profiles should show good incident response performance until a complexity burst (new integration, governance change, supplier onboarding) shifts the operating regime, after which performance should degrade despite unchanged experience levels.

5.7 Patterns and Digital Engineering Interventions

Change the Same Underlying Quantities

Modelling patterns and digital engineering (DE) practices are often discussed as separate capability areas: patterns are associated with systems architecture, whereas DE is associated with toolchains and data management. The formal models show that they are not separate. In the model, patterns and DE interventions act on the same underlying quantities: option-space structure, mobilisation dynamics, and inference dynamics. They are different entry points into the same resilience machinery.

5.7.1 Patterns as Operators on the Feasible Set

A modelling pattern, such as a proven architectural template, a standard interface contract, or a codified response playbook, operates on the feasible set by changing one or more of three elements:

1. **Adding or refining constraints K .** A pattern that mandates safety segregation adds a constraint that removes unsafe configurations. A codified interface contract reduces the number of coupling-induced options. In both cases, C decreases because infeasible or undesirable configurations are pruned.
2. **Restricting choice sets X_i .** A pattern that standardises a component selection or a procedure reduces the number of alternatives for that lever. This is a direct reduction in the configuration space, not a constraint-based pruning.
3. **Changing the distinctness map φ .** A pattern that defines equivalence classes for options (for example, “any database engine with ACID compliance is equivalent for this function”) coarsens the binning, thereby reducing the number of distinct options to evaluate. Theorem T5.7-1 in Appendix A1 proves that all three operations are complexity-non-increasing: adding constraints, restricting choices, and coarsening φ cannot increase C .

Patterns can also increase the success set $|A_{g,\rho}|$ by encoding proven solutions. Adopting a pattern that is known to meet a performance threshold adds configurations to $A_{g,\rho}$ (or preserves them while pruning failures from the feasible set). The net effect on functional information is therefore a reduction: C decreases and $|A_{g,\rho}|/C$ increases, so I_g^* decreases.

5.7.2 The Conditional Success-Share Theorem

Theorem T5.7-2 in Appendix A1 formalises the success-share mechanism. If an intervention increases the share of successful configurations from ρ to $\rho' \geq \rho$, then $I_g^{*'} \leq I_g^*$. For strict improvement, $I_g^{*'} < I_g^*$. This provides the formal backing for the claim that patterns reduce the cost of choice: they increase the probability that a randomly selected feasible configuration is satisfactory.

5.7.3 Digital Engineering Interventions

DE interventions operate on different quantities but through the same model:

On the inference layer. Machine-readable traceability, automated evidence capture, and structured telemetry improve the observation mapping $g(\cdot)$ and reduce noise ω . In the PHAS-EAI framework, this increases inference precision, reduces surprise, and improves the quality of belief updates. Theorem T5.5-4 (precision monotonicity) provides the formal link: lower σ^2 monotonically improves expected accuracy.

On the mobilisation layer. Automated deployment, continuous integration, and tool-supported response playbooks reduce baseline stage times τ_{stage}^0 . Design-support tools and simulation environments preserve cognitive reserve h by offloading routine cognitive load. Both mechanisms reduce τ_{eff} , increase r , and raise mobilisation fractions.

5.7.4 Joint Intervention Dominance

Theorem T5.7-3 in Appendix A1 proves that joint interventions yield larger reductions in operational functional information than single-channel interventions. The decomposition is additive in the log domain:

$$\Delta I_g^{\text{op}} = (-\log_2 \alpha) + \log_2 \beta + \log_2 \gamma \quad (45)$$

where $\alpha \in (0,1)$ is the complexity reduction factor ($C \mapsto \alpha C$), $\beta \geq 1$ is the success expansion factor ($A \mapsto \beta A$), and $\gamma \geq 1$ is the mobilisation improvement factor ($f_A \mapsto \gamma f_A$). The joint reduction equals the sum of the individual reductions, and each individual reduction is strictly less than the joint reduction.

The numeric demonstration in Appendix A1 uses representative parameters ($C = 4096$, $A = 512$, $f_A = 0.35$, $\alpha = 0.62$, $\beta = 1.4$, $\gamma = 1.25$) and confirms that the additivity residual is at machine precision ($\approx 4.4 \times 10^{-16}$ bits).

The simulation in Appendix A2 demonstrates that the dominance relationship is not merely quantitative but qualitative. As discussed in Section 5.5.2, improving inference capability (Theory of Mind) without corresponding alignment of shared expectations (Φ) increases system-level free energy from approximately 50 to approximately 60, above the no-intervention baseline. Joint intervention dominance is therefore not only the claim that combined interventions outperform single ones, but the stronger claim that isolated interventions on the inference layer carry a risk of harm when the alignment layer is absent.

5.7.5 The Bridge from Inference to Mobilisation

Theorem T5.7-4 provides the formal bridge between the PHAS-EAI inference model and the resilience model. Under bridge assumptions (lower inference noise reduces detection latency and information-related complexity potential), improving inference precision raises the time allowance ratio r and mobilisation fractions. This connects the PHAS-EAI findings from Section 5.5 to the resilience machinery: better sensing not only improves accuracy but also accelerates response and increases the fraction of options that can be mobilised in time.

The bridge connects the perception-action unification that Chapter 3 describes at the individual cognitive level to organisational dynamics. Theorem T5.7-4 formalises this connection: improving the precision of environmental signals (the inference layer) necessarily improves the speed and accuracy of collective action (the mobilisation layer), because both are governed by the same prediction-error dynamics.

The bridge assumptions are stated as hypotheses in the proof, explicitly marking them as loans of intelligence. The mechanism is plausible and supported by the directional arguments, but the precise coupling between posterior variance reduction and stage-time reduction is not derived from first principles. Section 7.5.1 returns to the status of this outstanding debt.

5.7.6 Hypotheses

H13 (Patterns reduce functional information without reducing resilience). Modelling patterns that encode constraints and substitutability reduce total optionality C while increasing the proportion of successful configurations $|A_{g,\rho}|/C$, lowering functional

information burden without reducing resilience, provided that useful and survivable variety are preserved.

Prediction: Adopting proven architectural patterns should reduce the measured or estimated I_g^* without reducing A or B . Case evidence should show that standardisation that prunes failure modes (not useful modes) is associated with maintained or improved incident response performance.

H14 (Machine-readable traceability reduces noise and improves coordination).

Machine-readable traceability and evidence chains reduce effective noise ω and improve observation mapping $g(\cdot)$, reducing surprise and improving coordination by stabilising shared expectations and shortening detection, decision, and execution cycles.

Prediction: Projects with machine-readable traceability from requirements through test to deployment should show shorter detection times, fewer integration surprises, and lower decision rework than projects relying on document-based traceability.

5.8 Leverage Points for Intervention Design in a PHAS

The preceding analysis decomposes resilience, complexity, information burden, and inference quality into named quantities with proven directional effects. This section synthesises the decomposition into a form suitable for intervention design. The goal is not to prescribe solutions but to provide a structured map from model levers to practical interventions, with explicit predictions of directional effect.

The tables below map each model lever to its directional effect (up or down) and provide illustrative examples of PHAS interventions that would manipulate that lever. Every directional effect listed corresponds to a proved monotonicity result or a stated assumption in the appendix.

The tables are organised into four groups: (i) configuration-space levers, (ii) mobilisation-time levers, (iii) information and inference levers, and (iv) coordination and governance levers.

Table 17 Configuration-space levers: directional effects and intervention examples

Lever	If moved up	If moved down	Intervention examples
Hard constraints K	Prunes infeasible and unsafe configurations. Reduces C and failure surface if constraints target failure modes. If overly restrictive, can also reduce useful variety.	Expands C , increases search space and failure surface, increases coordination burden, can increase brittleness if unsafe modes re-enter feasibility.	Contract tests, policy-as-code gates, codified safety and security invariants, interface compliance checks
Decision resolution Δ and distinctness map φ	Coarser resolution collapses near-duplicates, reduces counted C , and reduces cognitive burden. Risk of hiding materially important distinctions if bins are too coarse.	Finer resolution increases counted C , increases modelling and decision burden, and increases apparent complexity and coordination load.	Standardise categories and acceptance bands, define equivalence classes for options, adopt shared binned KPIs
Boundary degrees of freedom X_{boundary}	A wider boundary increases the scope of coordination and multiplies the option space through interdependence. Can increase adaptability, tends to increase drag without parallel observability and pattern investments.	A narrower boundary reduces modelling and coordination burden but increases the risk of unmanaged externalities and surprise.	Scope discipline for interfaces, explicit responsibility boundaries, interface ownership, and dependency caps

Table 18 Mobilisation-time levers: directional effects and intervention examples.

Lever	If moved up	If moved down	Intervention examples
Complexity potential CP_i	Increases stage times through drag, reduces r , collapses $\tilde{A}$, $\tilde{B}$, and increases brittleness under fast disturbances.	Reduces drag, increases mobilisation, and improves resilience margins.	Interface standardisation, change-rate governance, supplier interface control, stabilise vocabularies and data definitions
Baseline detection time τ_{detect}^0	Slower detection reduces the effective window and collapses the mobilisation fraction.	Faster detection expands effective window and improves mobilisation.	Instrumentation, alert quality improvements, telemetry with clear semantics, automated triage
Baseline decision time τ_{decide}^0	Longer decision-time baselines increase the risk of collapse under rapid disturbances, particularly in multi-actor governance.	Faster decisions increase mobilisation and reduce degradation time.	Pre-authorised actions, clarified decision rights, incident decision playbooks, and escalation simplification
Baseline execution time τ_{execute}^0	Slow execution reduces the ability to recover inside τ_d , increases degradation and collapse probability.	Faster execution increases the likelihood of recovery and reduces outage duration.	Automated deployment, rollback capability, rehearsed recovery routines, standard operating procedures
Stage sensitivities $\gamma_{\text{det}}, \gamma_{\text{dec}}, \gamma_{\text{exe}}$	Higher sensitivity makes CP more damaging, raising drag and slowing response.	Lower sensitivity reduces the extent to which complexity harms response, thereby improving stability.	Reduce cross-team handoffs, stabilise interfaces, reduce exception handling, and standardise response steps
Skill factor $u(E, MF)$	Increases cognitive headroom and reduces effective mobilisation time, but only up to the bound set by λ_τ .	Reduces headroom and slows mobilisation, increasing brittleness under time pressure.	Training, rehearsal, peer learning, targeted capability building for high-demand tasks

Lever	If moved up	If moved down	Intervention examples
Design-preserved cognitive reserve h	Protects attention and decision-making capacity under stress, even when skill varies, thereby improving resilience in thin teams.	Increases vulnerability to cognitive overload and drift, and increases decision errors under pressure.	Automation, checklists, pre-authorised actions, tooling that removes routine cognitive load
Physical headroom H_{phys}	Raises disturbance magnitude limits M_0 and M_{max} , improves survivability.	Lowers magnitude limits, causes earlier degradation and collapse.	Spare capacity, buffer budgets, delegated authority, redundancy, surge staffing agreements

Table 19 Information and inference levers: directional effects and intervention examples.

Lever	If moved up	If moved down	Intervention examples
Performance threshold ρ	Higher ρ shrinks $ A_{g,\rho} $, increases functional information, increases brittleness if not paired with pattern investments.	Lower ρ increases $ A_{g,\rho} $, reduces functional information, but relaxes required performance and assurance.	Mode-dependent service levels, staged degradation policies, explicit function prioritisation
Observation mapping $g(\cdot)$	More informative sensing reduces ambiguity, reduces detection delays, improves belief update and coordination.	Less informative sensing increases ambiguity, increases reliance on priors and heuristics, and increases drift and delay.	Better telemetry, consistent semantic definitions, and traceability from model elements to monitored signals
Noise ω	Higher noise increases uncertainty, increases divergence in inference, and increases miscoordination.	Lower noise increases confidence, improves coordination and response speed.	Reduce conflicting metrics, improve data quality, standard incident reporting formats
Traceability and machine-readable evidence chains	Improves observation mapping, reduces noise, stabilises shared expectations, reduces decision rework and integration surprises.	Weak traceability increases ambiguity, increases reliance on informal memory, and increases drift and rework.	Model-to-test-to-deploy-to-telemetry linking, automated evidence capture, auditable pipelines, shared artefact repositories

Table 20 Coordination and governance levers: directional effects and intervention examples.

Lever	If moved up	If moved down	Intervention examples
Affordance $p(a)$	Higher affordance for desired actions makes them more likely under pressure.	Lower affordance makes desired actions harder or less legitimate, encouraging drift.	Make resilience actions default and pre-approved, tool support that reduces bureaucratic friction
Communication c_{in}, c_{out}	More structured, evidence-backed communication improves peer inference and mutual predictability.	Ambiguous communication increases misinterpretation and coordination delay.	Structured escalation templates, shared vocabularies, evidence attachments, time-boxed communication loops
Authority weights γ, γ_i in Φ	Higher weight drives faster convergence to peer expectations if peers are reliable. Can propagate peer error if not.	Lower weight preserves autonomy and diversity, can reduce coordination efficiency and increase divergence.	Governance design, decision rights, regulator or integrator authority calibration
External state scope η	A broader scope increases potential accuracy but also increases the hypothesis space and cognitive burden unless observability scales.	A narrower scope reduces burden but increases the risk of unmodelled disturbances and surprises.	Explicit “known unknowns”, boundary monitoring commitments, scoped assurance cases
Action set $\mathcal{A}$	Larger action sets increase adaptability but also increase choice burden and the risk of inconsistency without constraints.	A smaller action set reduces choice burden and increases consistency, but can reduce useful variety under diverse disturbances.	Standard response playbooks, controlled automation, and limiting response paths to tested safe operations

The tables serve two functions in the thesis design:

1. **Intervention candidate specification.** For a given resilience concern, the table identifies which levers are relevant and predicts the directional effect of moving each lever. This prediction is grounded in the formal monotonicity results proved in the appendix.
2. **Evidence design.** The directional effects define what counts as “movement” on the intended lever, even when resilience outcomes are rare or difficult to observe directly. For example, if the prediction is that reducing CP improves r , then measuring a reduction in interface churn and a shorter median detection-to-decision time counts as evidence of movement on CP with the predicted directional effect on τ_{eff} .

This approach does not require observing rare events to build an empirical case. It shifts the evidentiary burden from outcome measurement to lever measurement, as evidenced by routine operational data.

6 Key Findings from Case Studies

Chapter 5 derived fourteen testable hypotheses from the formal model. This chapter evaluates those hypotheses against field evidence from four industrial case studies conducted with Aiwell AS, a Norwegian VSE developing outdoor building and facility automation products. The chapter examines evidence gathered through participatory action research (AR) and Grounded Theory Method (GTM) analysis, and relates the findings to the hypotheses formulated in the preceding chapter.

The formal model provides the interpretive lens. Each intervention examined in these cases restructures the team's information environment to channel attention, reduce decision burden, and stabilise shared expectations. The cases show what these principles look like when operationalised in a resource-constrained engineering organisation.

Aiwell operates within the municipal smart city supply chain, delivering subsystems for automated rooftop drainage, snow and ice removal, and winter heating for artificial turf sports facilities. Across the four studies, the same fundamental challenge recurs: how does a resource-constrained organisation deliver dependable, secure systems when it lacks in-house expertise beyond its core engineering domain and operates within multi-stakeholder supply chains in which no single entity possesses full system oversight?

Some of the papers published as part of this thesis anonymised the company, but all four case studies were conducted at or in close collaboration with Aiwell AS. The cases span approximately six years and encompass a range of intervention modalities, from model-based systems engineering and serious games to AI-assisted requirements review and environmental design to sustain security awareness. Table 21 summarises the four cases.

Table 21 Summary of case studies

Case	Paper	System of Interest	Intervention
A	I	6LoWPAN/CoAP siphonic drainage system	SysML modelling, EoP threat game, parametric reliability model
B	III	Same siphonic drainage system (supply chain)	SysML-based serious game (EoP) for threat modelling
C	IV	Requirements engineering workflow	LLM-assisted requirements review (ChatGPT, Copilot, GPT-4 API via GitHub Actions)
D	V	Software repositories with known vulnerability exposure	PHAS-EAI framework: Dependabot, smart bulb, EoP game sessions

The four cases are complementary in several respects. Cases A and B share the same physical system (a siphonic drainage system for urban flood prevention) but examine it from different analytical angles: Case A focuses on combined reliability and security modelling with quantitative trade-off analysis, whereas Case B examines the social and cognitive effects of the game-based intervention on cross-disciplinary understanding. Case C introduces a different intervention modality (AI-assisted review) but addresses the same underlying constraint: a small team needing cognitive scaffolding to maintain engineering quality. Case D provides the longest observation period and the most direct operationalisation of the formal model constructs developed in Paper V, particularly the concepts of Regimes of Attention and Patterned Practices.

Together, the cases span the key intervention types identified by the formal model: reducing optionality C through design patterns, increasing the feasible share $|A_{g,\rho}|/C$ through structured review, building cognitive reserve h through designed tools, and stabilising shared expectations Φ through environmental affordances.

6.1 Case A: Model-Based Reliability and Security Validation (Paper I)

6.1.1 Case Description

The siphonic drainage system examined in this case study was designed to prevent urban flooding by actively controlling the flow of water from rooftop drainage tanks via sensor-monitored valves. Aiwell designed the valves and end-user applications in-house, while outsourcing PCB design and firmware development to specialist subcontractors within the supply chain. This distributed development model meant that no single organisation possessed a complete view of the system's security and reliability properties. The challenge was to integrate security and reliability analysis across organisational boundaries using methods accessible to non-specialists.

A SysML model of the system architecture was constructed using Cameo Systems Modeller. The model included a class-based requirements taxonomy linking stakeholder needs to system requirements and verification methods, internal block diagrams showing layered protocol interfaces from IEEE 802.15.4 through 6LoWPAN, UDP, and CoAP, and a parametric reliability and availability model implementing the standard exponential failure-and-repair equations.

The Elevation of Privilege (EoP) card game, based on Microsoft's STRIDE threat model, was played against the SysML model with a mixed stakeholder group comprising engineers, management, and external partners. Players navigated the model during gameplay, zooming into interface details and examining linked vulnerability records. A separate parametric analysis session used the SysML constraint model to evaluate proposed mitigations quantitatively.

6.1.2 Key Findings

Finding A1: Vulnerability discovery by non-experts. During EoP gameplay, participants with no security background identified a vulnerability in the CoAP interface (CVE-2020-3162) by navigating the SysML model's layered interface diagrams. The model served as a structured prompt that guided attention to security-relevant interfaces without requiring specialist knowledge. The previously modelled vulnerability was highlighted in the diagram, and its links to the National Vulnerability Database provided participants with the context needed to assess its relevance. This result demonstrates that a well-structured model can lower the competence threshold for meaningful security analysis.

Finding A2: Quantitative trade-off via parametric model. Adding a gateway unit to isolate CoAP traffic from the wider network was the first mitigation proposed during gameplay. However, the parametric analysis showed that this option would entail high costs for both hardware and API development. The sensitivity metrics S_F (sensitivity to failure rate changes) and S_R (sensitivity to repair rate changes) demonstrated that when the mean time to failure is much larger than the mean time to repair, reducing repair time yields greater availability improvement per unit cost than reducing failure probability. This analysis reframed the mitigation decision from an architectural choice to a cost-effectiveness calculation.

Finding A3: Cost-effective mitigation through context analysis. By examining the CVE and CWE entries linked within the SysML model, engineers determined that the CoAP vulnerability primarily affected mobile networks in China and that the microcontroller used in the drainage system could reset to a safe state without loss of sensitive data. A hardware watchdog timer, resetting the system upon traffic congestion, was identified as the most cost-effective mitigation. This finding illustrates how machine-readable traceability from the system model to external vulnerability databases enabled a context-sensitive decision that a generic security checklist would have missed.

Finding A4: Model as a boundary object across the supply chain. The SysML model served as a shared artefact that enabled structured communication across organisational boundaries among the valve manufacturer, the firmware subcontractor, and the municipal client. By providing a common representation that all parties could navigate, the model reduced the ambiguity that typically accompanies verbal descriptions of interface responsibilities and security requirements.

6.1.3 Mapping to Model Hypotheses

Findings **A1** and **A4** relate directly to **H13** (patterns reduce functional information) and **H14** (machine-readable traceability reduces noise). The SysML model constrained the option space by encoding the system architecture as a navigable pattern, reducing the effective number of configurations C that participants needed to consider during threat analysis. The machine-readable links from requirements to CVE and CWE entries reduced noise ω in cross-boundary communication and improved the observation mapping $g(\cdot)$ by making vulnerability context directly navigable rather than requiring verbal explanation.

Finding **A1** also supports **H7** (designed reserve outperforms experience investment). The model-plus-game combination acted as a designed cognitive reserve h , enabling participants without security expertise to achieve threat-identification outcomes comparable to those of a specialist review. The reserve was embedded in the tool, not dependent on individual skill.

Finding **A2** provides evidence for **H2** (joint intervention) and **H8** (operational functional information predicts outcomes better than raw complexity). The parametric analysis reduced the design space by eliminating the costly gateway option and identified a mitigation that increased the share of configurations that met the availability thresholds. The sensitivity metrics S_F and S_R are operationalisations of state-dependent information: they quantify the remaining uncertainty about system behaviour when repair time dominates failure time.

Finding **A3** illustrates **H6** (diminishing returns from expertise in high-drag environments). The expert-level architectural mitigation (gateway) offered diminishing marginal benefit relative to its cost. A simpler, experience-independent intervention (watchdog) outperformed the expertise-intensive one because the bounded improvement from expert design could not justify the structural overhead.

Finding **A4** supports **H10** (authority-weight asymmetry predicts friction). The shared model partially equalised authority weights γ_i by providing all participants with the same structured information, thereby reducing the asymmetry that otherwise existed between security-knowledgeable and security-naïve stakeholders.

6.2 Case B: Serious Games with SysML for Threat Modelling (Paper III)

6.2.1 Case Description

Aiwell, at the time of this study, had recently been forced to decline projects because it lacked documented RAMS (Reliability, Availability, Maintainability, Safety) procedures. Despite decades of operational experience, the company could not demonstrate compliance with requirements that prospective clients increasingly demanded. Technical staff at that point consisted of one systems engineer and two service technicians.

The system of interest was the same siphonic drainage system as in Case A, examined here from the perspective of the company rather than the subcontractor network. The challenge was not technical capacity per se but the inability to translate accumulated experience into documented, verifiable dependability practices.

6.2.2 Intervention and Method

Multiple EoP game sessions were played against SysML models of the system architecture. Sessions deliberately excluded security experts to test whether the game-plus-model approach could substitute for specialist knowledge.

6.2.3 Key Findings

Finding B1: Transdisciplinary utility. Players from different disciplines (mechanical, electrical, software) discovered immediate value for their own operational concerns during the game. The game reduced resistance to participation because players saw a direct benefit rather than being asked to serve someone else's security agenda. This is significant because resistance to formal methods and security practices is a well-documented barrier in small organisations. The game reframed security analysis as something inherently useful across disciplines rather than as an imposed compliance exercise.

Finding B2: Shared boundary objects. Repeated negotiation around SysML model interfaces produced a clearer common understanding across disciplines. The model served as a boundary object that enabled participants to discuss interface responsibilities without requiring shared technical vocabulary. Mechanical engineers, electrical engineers, and software developers could each engage with the model from their respective perspectives while contributing to a collective understanding of the system's reliability properties.

Finding B3: Increased alignment. Time-boxed game cycles promoted rapid collective sensemaking. Participants reported improved understanding of other stakeholders' perspectives and concerns following the play. The structured rhythm of play, discuss, and refine synchronised mental models across the team. Over successive sessions, the team developed a shared language for discussing threats and mitigations that persisted beyond the game sessions themselves.

6.2.4 Mapping to Model Hypotheses

Finding **B1** supports **H9** (Regimes of Attention reduce drift). The EoP game created a structured Regime of Attention: it directed attention to security-relevant signals in a time-boxed, repeatable format. By making threat scenarios salient through gameplay, the intervention reduced the tendency to deprioritise security considerations in routine work. Finding **B1** also supports **H7** (designed reserve outperforms experience): the game served as a designed cognitive scaffold that enabled non-security specialists to contribute to threat identification. The reserve lay in the structure of the activity, not in individual competence.

Finding **B2** supports **H10** (authority-weight asymmetry predicts friction) and **H14** (machine-readable traceability). By providing a shared, navigable model as the object of discussion, the intervention reduced the authority asymmetry between disciplines. The SysML model provided structured, navigable interface descriptions that reduced ambiguity ω in cross-disciplinary communication.

Finding **B3** supports **H13** (patterns reduce functional information). The repeated game cycles constituted Patterned Practices in the sense formalised by the PHAS-EAI model. By ritualising security review as a regular, structured activity, the organisation reduced the decision burden (effective C) associated with security analysis from an open-ended task to a finite, game-structured process. Finding **B3** also relates positively to **H5** (time pressure reduces realised resilience): by creating a dedicated, protected time window for security analysis, the intervention partially insulated this activity from the time pressure τ_d that would otherwise compress it out of existence in a resource-constrained VSE.

The company's inability to meet RAMS requirements despite years of operational experience (the motivating context for the study) illustrates **H12** (experience effect weakens after bursts). Prior experience did not translate into security competence when new compliance requirements, which represented a surge in complexity, exceeded the domain covered by accumulated practice.

6.3 Case C: LLM-Assisted Requirements Engineering (Paper IV)

6.3.1 Case Description

At the time of this study, Aiwell faced a high engineering workload, a shortage of senior engineers, and a desire to modernise its engineering practices. Requirements were written as issues in GitHub, but quality varied, and there was no systematic review process against engineering standards.

6.3.2 Intervention and Method

The study followed an incremental adoption path. Engineers first used an early version of ChatGPT for exploratory requirements drafting, then GitHub Copilot for assisted code and documentation writing, and finally a custom GPT-4 API integration via GitHub Actions for automated requirements review. The GitHub Actions pipeline was the core intervention: when an engineer tagged an issue as a requirement, the action triggered a GPT-4 API call with the system context and INCOSE/SEBoK reference materials. The model evaluated the requirement against four dimensions (clarity, conciseness, testability, and traceability) and returned a weighted composite score, which was posted as a comment on the issue. The engineering team could then revise the requirement based on the AI-generated feedback.

6.3.3 Key Findings

Finding C1: Accessibility and smooth adoption. Engineers found the LLM tools surprisingly easy to adopt. The conversational nature of ChatGPT and the IDE integration of GitHub Copilot lowered the barrier to entry. Engineers reported that integrating AI had been smoother than anticipated, emphasising the democratisation of advanced technologies for smaller entities.

Finding C2: Human-AI collaboration. The value of the LLM tools lay not in replacing human judgment but in augmenting engineering discussion. Engineers observed that the AI provided a solid starting point, after which team discussions around the AI's suggestions became invaluable. The AI served as a catalyst for richer engineering discourse rather than as an autonomous decision-maker.

Finding C3: Productivity and quality enhancement. Engineers reported fewer errors and increased efficiency, but the quality improvement was not primarily attributable to the quality

of the LLM output itself, which was sometimes lacking. Rather, the LLM guided engineers to consider broader perspectives, prompting more detailed requirements and highlighting potential edge cases that would otherwise have been overlooked. The improvement mechanism was attentional rather than substitutive.

Finding C4: Requirements engineering competence transfer. Over time, engineers gradually adopted thought patterns similar to the LLM-generated output. The AI effectively served as a Systems Engineering mentor, and engineers reported improved requirements quality even when not using AI tools. This internalisation of review patterns demonstrates a lasting cognitive effect that extends beyond the immediate interaction with the tool.

Finding C5: Data privacy concerns. Engineers expressed worries about proprietary code being accessed by AI tools, particularly GitHub Copilot. This concern represented a genuine barrier requiring proactive mitigation measures and alignment with data governance regulations.

Finding C6: Scalability challenges. While the study demonstrated benefits for relatively simple requirements review tasks, engineers questioned whether these benefits would scale to more complex engineering processes. This forward-looking concern highlights the intervention's boundary conditions.

6.3.4 Mapping to Model Hypotheses

Finding **C1** supports **H7** (designed reserve outperforms experience investment). The LLM tool acted as a designed cognitive reserve h that supplemented the team's limited senior engineering capacity. By encoding systems engineering best practices (INCOSE/SEBoK references) into the review pipeline, the tool preserved cognitive headroom for quality review even under high workload, without requiring the team to hire additional experienced staff.

Finding **C2** supports **H9** (Regimes of Attention reduce drift). The automated review pipeline created a Regime of Attention by making requirements-quality signals salient during drafting. The AI-generated feedback directed the engineer's attention to the clarity, testability, and traceability dimensions that might otherwise be overlooked under time pressure.

Finding **C3** supports **H2** (joint intervention). The LLM intervention jointly narrowed the space of possible requirement formulations (by providing structured templates and feedback, effectively reducing C) and increased the proportion of formulations meeting quality thresholds (through automated scoring against INCOSE criteria, effectively increasing $|A_{g,\rho}|/C$).

Finding **C4** provides strong evidence for **H13** (patterns reduce functional information). The internalisation of AI review patterns is a direct instance of Patterned Practices reducing functional information: the repeated interaction with structured feedback converted an open-ended drafting task into a patterned one, reducing effective C for the engineer even when the tool was not in use. Finding **C4** also supports **H11** (SE capability dominates in high complexity): the LLM effectively transferred elements of SE capability to non-specialist engineers, supporting the prediction that SE proficiency has a stronger association with performance than unaided experience alone.

Finding **C5** relates to **H1** (bursty complexity). Privacy and regulatory requirements represent a potential complexity burst: if data governance rules change abruptly (as demonstrated by Italy's temporary ChatGPT ban during the study period), the effective complexity potential CP increases for any organisation relying on cloud-based AI tools.

Finding **C6** is consistent with **H6** (diminishing returns from expertise in high-drag environments). The concern about scalability reflects the prediction that, as task complexity increases, the bounded improvement from any single tool may reach diminishing returns if the structural drag of more complex projects exceeds the tool's capacity to compress decision time.

The automated scoring via GitHub Actions supports **H14** (machine-readable traceability). The pipeline linked each requirement to INCOSE standards and produced machine-readable quality scores, reducing noise ω in the review process and improving the observation mapping from the requirement text to the quality assessment.

6.4 Case D: PHAS-EAI Framework for Cybersecurity (Paper V)

6.4.1 Case Description

By the time of this study, Aiwell had recently experienced multiple security breaches in which attackers exploited their infield IoT modems to send spam, depleting customer data quotas and locking users out of their own systems. These incidents caused operational, reputational, and financial losses, making immediate security improvements essential. Despite recognising the urgency, Aiwell lacked security expertise and the means to hire dedicated security staff. The company needed sustainable security solutions aligned with its resource constraints.

6.4.2 Intervention and Method

The study was conducted over a 2-year period using a participatory approach. The engineering team used the inquiry questions developed as part of the PHAS-EAI framework (Table 4 in Paper V) to guide the design of five linked interventions, each addressing a different facet of the framework.

Step 1: Determining necessary inputs. The team identified the need for continuous, automated vulnerability monitoring and adopted GitHub's Dependabot service. Dependabot automatically scans repositories for known vulnerabilities, provides real-time CVSS v4 severity scores, and generates proposed patches with full automated test reports. If a fix exists, all unit tests are re-run with the patched code before the engineer approves any changes.

Step 2: Understanding stakeholder interaction with signals. Recognising that not all team members would naturally attend to digital alerts, the team installed a smart bulb connected to the GitHub API. The bulb glowed orange for low-severity issues or vulnerabilities in development repositories and blinked red for high-severity vulnerabilities in production. The team preferred this approach over email notifications, alarm logs, or website alerts because those methods require proactive engagement and compete for attention in a cluttered digital environment. The smart bulb acted as a shared, ambient sensor in the physical workspace, capturing attention immediately without requiring conscious effort. This is a direct operationalisation of a Regime of Attention as defined in the formal model.

Step 3: Ensuring the sustainability of attention through practices. The team incorporated regular EoP game sessions into their routine. These sessions served as Patterned Practices designed to be engaging while achieving specific educational and organisational objectives. By regularly playing the game against their own system models, the team practised identifying and mitigating security threats, embedding security awareness into their organisational culture.

Step 4: Understanding stakeholder meaning. The team ensured that the smart bulb's visibility was meaningful to all stakeholders, including non-technical management staff. The bulb's status provided a clear and immediate indication of the system's security health, reinforcing security as a whole-organisation priority rather than a purely technical concern.

Step 5: Defining success. The team established measurable success criteria: a reduction in unresolved high-severity vulnerabilities, timely responses to security alerts, and improved team performance in EoP sessions over time.

Data collection relied on timestamped artefacts retrieved from GitHub and internal communication systems. Quantitative metrics focused on Dependabot alert histories, measuring total detected vulnerabilities and median time to remediation. Qualitative data from meeting notes, issue trackers, and chat logs provided context for decision-making.

6.4.3 Key Findings

Finding D1: Rapid, sustained remediation. Over the 2-year study period, the automated system detected 90 vulnerabilities across four repositories. Five issues required an engineer's intervention, and two remained unresolved due to dependencies. All 88 fixable issues were patched within 24 hours of their CVEs being publicly disclosed. For context, 40,009 CVEs were published in 2024, averaging 108 per day. The near-universal next-day remediation rate was sustained throughout the observation period without decline.

Finding D2: Designed salience replaced process mandates. The smart bulb captured attention in moments of need without requiring proactive engagement. The team did not adopt new bureaucratic processes, additional meetings, or expanded headcount. Instead, they shaped their physical environment so that the preferred action (addressing the vulnerability) was the most obvious and easiest course of action. The approach treated attention as the scarce control variable rather than compliance.

Finding D3: Patterned Practices synchronised internal models. Regular EoP sessions created a shared context for interpreting vulnerability signals. The game sessions

synchronised how the team prioritised and responded to alerts, aligning internal models across team members. Over time, team members developed shared expectations about response urgency that persisted between game sessions.

Finding D4: Niche construction in practice. The team did not merely adapt to their environment but actively reshaped it (smart bulb, Dependabot integration, game schedule) to support security-focused behaviour. This created a feedback loop in which the environment continuously reinforced the desired behaviours, consistent with the niche construction mechanism described in the PHAS-EAI formal model.

Finding D5: Sustainability without overhead. The approach persisted for 2 years without additional meetings, bureaucratic processes, or headcount. The interventions were self-sustaining because salience and shared meaning were embedded in the workspace rather than imposed through external mandates.

6.4.4 Mapping to Model Hypotheses

Finding **D1** supports **H5** (time pressure reduces realised resilience) and **H3** (VSE advantage is regime-dependent). The near-immediate remediation demonstrates high $r = \tau_d / \tau_{\text{eff}}$ under the attention regime. The designed short detection-to-action cycle kept τ_{eff} low, so even when new CVEs appeared rapidly (external time pressure), the organisation maintained high mobilisation fractions. The team achieved next-day remediation rates consistent with the VSE advantage predicted by **H3** for the fast-disturbance, moderate-magnitude regime: the vulnerabilities were individually moderate in severity (most handled automatically by Dependabot), allowing the small team's short decision chains to operate effectively.

Finding **D2** supports **H9** (Regimes of Attention reduce drift) directly. The smart bulb is a concrete operationalisation of a Regime of Attention. By externalising vulnerability severity as an ambient visual signal with high contrast (blinking red) and low noise (no competing digital notifications), the intervention increased the precision-weighted salience of security signals, reducing drift away from prompt remediation.

Finding **D3** supports **H13** (patterns reduce functional information). Regular game sessions encoded threat-response patterns, reducing the effective option space for security decisions. Over time, familiar threat categories became patterned responses rather than novel decision problems, lowering the functional information burden on the team.

Finding **D4** supports **H7** (designed reserve outperforms experience investment) and **H2** (joint intervention). The smart bulb, Dependabot, and game schedule collectively constitute

designed cognitive reserve h . These tools and practices protected the team's response quality independently of individual expertise growth. The combined intervention jointly reduced the decision space (from an open-ended "what should we monitor and how" to a specific "respond to this alert") while increasing the share of correct responses (Dependabot provides the patch, the team approves).

The pre-intervention state, in which the company experienced repeated breaches despite prior operational experience, illustrates **H12** (the experience effect weakens after bursts). Each breach involved novel attack vectors that fell outside the team's accumulated operational experience. Post-intervention, the designed system handled novelty through automated detection rather than relying on experiential pattern recognition.

Finding **D2** also supports **H10** (authority-weight asymmetry predicts friction). The smart bulb made vulnerability status equally visible to engineers and management, partially equalising authority weights γ_i for security decisions. Management did not need to rely on engineers' verbal reports to understand the current security posture.

The Dependabot data pipeline supports **H14** (machine-readable traceability). The system provided machine-readable vulnerability data (CVSS scores, CWE/CVE links, proposed patches, automated test reports), reducing noise ω in the observation channel and improving the mapping $g(\cdot)$ from raw codebase state to actionable security information.

6.5 Cross-Case Analysis

6.5.1 Hypothesis Coverage

Table 22 maps the fourteen hypotheses from Chapter 5 against the four case studies. Each cell indicates the nature of the supporting evidence: “Direct” means a finding that directly tests or illustrates the hypothesis, “Context” means the hypothesis is supported by the background conditions of the case, and “Latent” means the hypothesis is relevant but not explicitly tested.

Table 22 Hypothesis coverage across case studies.

Hypothesis	Case A (Paper I)	Case B (Paper III)	Case C (Paper IV)	Case D (Paper V)
H1: Bursty complexity	Context	Context	Latent	Context
H2: Joint intervention	Direct (A2)		Direct (C3)	Direct (D4)
H3: VSE advantage is regime-dependent	Context	Context	Context	Direct (D1)
H4: Burst-induced regime shift		Context (RAMS)		Context (breaches)
H5: Time pressure	Context	Direct (B3)		Direct (D1)
H6: Diminishing returns from expertise	Direct (A3)		Direct (C6)	
H7: Designed reserve	Direct (A1)	Direct (B1)	Direct (C1)	Direct (D4)
H8: Operational I_g predicts outcomes	Direct (A2)			
H9: Regimes of Attention		Direct (B1)	Direct (C2)	Direct (D2)
H10: Authority-weight asymmetry	Direct (A4)	Direct (B2)		Direct (D2)
H11: SE capability dominance	Context		Direct (C4)	
H12: Experience weakens after bursts		Context		Context
H13: Patterns reduce I_g	Direct (A1)	Direct (B3)	Direct (C4)	Direct (D3)
H14: Machine-readable traceability	Direct (A3, A4)	Direct (B2)	Direct (C3)	Direct (D4)

6.5.2 Convergent Findings

Four hypotheses receive direct support from three or more independent case studies, providing triangulated evidence.

Designed cognitive reserve consistently outperforms experience investment (H7). All four cases demonstrate interventions that embed competence in tools and processes rather than depending on individual expertise growth. In Case A, the SysML model enabled non-experts to identify vulnerabilities. In Case B, the game substituted for the absence of security expertise. In Case C, the LLM served as an SE mentor. In Case D, automated monitoring replaced the need for dedicated security staff. The convergence across four independent intervention types provides the strongest evidence in this thesis for the claim that designed reserve is a more reliable and accessible lever for VSE dependability improvement than investments in individual skill development.

Patterns and Patterned Practices reduce functional information burden (H13). All four cases demonstrate mechanisms that reduce the effective decision space without reducing resilience. SysML models constrain the architecture to navigable patterns (A), games create ritualised security review (B), AI feedback is internalised as habitual quality checks (C), and repeated EoP sessions turn novel threats into familiar categories (D). The consistency of this finding across different intervention modalities strengthens the formal model's prediction that reducing C while preserving $|A_{g,\rho}|$ is a generalisable strategy.

Machine-readable traceability reduces communication noise across boundaries (H14). All four cases involve machine-readable artefacts (SysML models with linked CVE/CWE entries, GitHub-based scoring pipelines, Dependabot reports) that reduced the noise in cross-boundary or cross-disciplinary communication. The mechanism operates consistently across organisational (Case A), disciplinary (Case B), cognitive (Case C between human and AI), and temporal (Case D) boundaries.

Regimes of Attention are the common mechanism for sustained improvement (H9). Cases B, C, and D explicitly create structured attention-directing mechanisms through different modalities. Case B uses time-boxed game sessions. Case C uses an automated review triggered at the moment of drafting. Case D uses an ambient visual signal. In each case, improvement was sustained not through process mandates but through environmental design that made the preferred action salient. The common principle is that attention is a scarce resource, and designing for salience is more effective than designing for compliance.

6.5.3 Divergent and Qualified Findings

H3 and H4 (VSE advantage and regime shifts) are supported contextually but not tested experimentally. No case includes a controlled comparison between a VSE and a larger organisation facing the same disturbance. The 24-hour remediation rate in Case D is consistent with **H3's** prediction of VSE advantage in the fast-disturbance regime, but the study design does not isolate the causal mechanism from potential confounders such as the simplicity of the patch-approval workflow. Future work should include comparative designs involving organisations of different sizes.

H8 (operational I_g predicts outcomes better than raw complexity) receives direct support only from Case A. The parametric sensitivity metrics S_F and S_R are the closest operationalisations of I_g^{op} in the empirical evidence, but no case explicitly measures functional information in bits and correlates it with outcome variables. This remains an important gap for future empirical validation. Developing operational measures for functional information applicable in field settings is a priority for subsequent research.

H1 and H4 (bursty complexity and burst-induced regime shifts) are observed as contextual conditions rather than tested as predictions. The RAMS compliance requirement that Aiwell could not meet (Case B) and the security breaches that preceded the Case D intervention are consistent with burst-induced regime transitions. However, the studies were not designed to measure complexity potential CP changes or threshold crossings directly. Instrumenting future case studies with quantitative complexity metrics would allow more rigorous testing of these hypotheses.

Scalability of the designed reserve (H6 boundary). Case C explicitly raises the question of whether LLM-based cognitive reserve scales to more complex engineering tasks. This finding qualifies **H7**: while designed reserve is consistently effective for the task complexities observed in these cases, the formal model's prediction of diminishing returns from bounded interventions (**H6**) implies that designed reserve may need periodic recalibration as the complexity of the work increases. The practical implication is that VSEs should treat their cognitive support tools as living infrastructure that evolves with the organisation rather than as fixed solutions.

7 Summary, Discussion and Conclusions

This chapter synthesises the thesis's findings and addresses the three research questions. It begins with a summary of the work, followed by an explicit mapping of the five papers to the research questions. The research questions are then addressed in turn, drawing on the formal model, the four case studies, and the background analysis. The chapter closes with a discussion of contributions, limitations, and directions for future work.

7.1 Summary of the Work

This thesis investigated how very small entities can deliver dependable smart city infrastructure despite severe capacity, tooling, and expertise constraints. It diagnosed the problem as a deficit in environmental design rather than in knowledge (Chapter 1), surveyed the domain and practice contexts (Chapter 2), and established that an engineering team's cognitive resources are distributed across people, tools, and the environment (Chapter 3). On this basis, the thesis developed a formal model comprising two interlocking components: a PHAS-EAI model of how teams sense, infer, communicate, and act, and a complexity-resilience model defining the configuration space within which those actions take effect (Chapter 4).

The formal analysis derived fourteen testable hypotheses organised around seven themes: burst dynamics, the bounded VSE advantage, time pressure, functional information, attention and authority structures, empirical alignment, and joint intervention dominance (Chapter 5). Four industrial case studies conducted with Aiwel AS over approximately six years evaluated those hypotheses through model-based reliability and security validation (Case A), gamified threat modelling (Case B), LLM-assisted requirements engineering (Case C), and a comprehensive implementation of the PHAS-EAI framework (Case D). A fifth paper provided the conceptual bridge between the problem diagnosis and the empirical work (Chapter 6).

Four findings converge across all cases: designed cognitive reserve consistently outperforms experience investment (H7), modelling patterns reduce functional information burden without reducing resilience (H13), machine-readable traceability reduces communication noise across boundaries (H14), and Regimes of Attention sustain improvement more effectively than compliance mandates (H9).

7.2 Paper Contributions to the Research Questions

The five papers contribute to the three research questions through complementary modes of inquiry. Table 23 maps each paper to the research questions it addresses and identifies the nature of its contribution. Paper II is a conceptual contribution rather than an empirical case study. Its role in the thesis is diagnostic and prescriptive: it identifies the structural problem (RQ1), articulates design principles for interventions (RQ2), and proposes the strategic combination that Papers I, III, IV, and V then operationalise and evaluate (RQ3). It sits between the background analysis and the empirical work, translating the problem diagnosis into an actionable research direction.

Table 23 Mapping of papers to research questions.

Paper	RQ1 (Challenges)	RQ2 (Requirements and metrics)	RQ3 (Modelling patterns)
I	Demonstrates the challenge: distributed development leaves no single entity with full system oversight. (Empirical demonstration)	Provides parametric sensitivity metrics (S_F , S_R) for cost-effective mitigation selection. (Empirical demonstration)	SysML model as boundary object or cross-boundary communication and quantitative trade-off. (Empirical demonstration)
II	Primary diagnostic contribution: articulates the systemic risk of VSE supply chains and the clay foundation of security engineering. (Diagnosis)	Specifies that DE for VSEs must leverage social agility, adopt a Zero-Trust mindset, and use gamified engagement rather than heavyweight mandates. (Design requirements)	Proposes AMBSE with gamification as the intervention strategy. (Conceptual framework)
III	Reveals that operational experience does not translate into security competence when new compliance demands arise. (Empirical demonstration)	Documents qualitative indicators of improved cross-disciplinary alignment and shared vocabulary. (Empirical demonstration)	EoP card game against SysML as a patterned practice for threat modelling. (Empirical demonstration)

Paper	RQ1 (Challenges)	RQ2 (Requirements and metrics)	RQ3 (Modelling patterns)
IV	Shows the cognitive burden of requirements engineering in understaffed teams. (Empirical demonstration)	Provides weighted scores for clarity, conciseness, testability, and traceability, plus revision counts and time-to-review. (Metrics)	LLM-assisted review as cognitive scaffolding embedded in the engineering workflow. (Empirical demonstration)
V	Provides the theoretical mechanism: explains why challenges are self-reinforcing through attention drift, authority-weight asymmetry, and bursty complexity. (Theoretical mechanism)	Defines the leverage points framework and operational observables derived from the formal model. (Theoretical framework and metrics)	Regimes of Attention and Patterned Practices as the theoretical unification of all modelling pattern interventions. (Theoretical framework and empirical demonstration)

7.3 Answering the Research Questions

7.3.1 RQ1: What are the new dependability challenges in building and operating resilient interdependent smart city infrastructure?

The thesis identifies four interconnected dependability challenges that arise from the structural characteristics of the smart city supply network. These are not isolated problems but mutually reinforcing dynamics that the formal model connects through a common analytical vocabulary. All four challenges are manifestations of a single underlying condition: a mismatch between the cognitive demands of the team's environment and the cognitive resources available to meet those demands.

Bursty complexity growth from multiplicative composition. The formal model (Chapter 5) demonstrates that complexity in interdependent socio-technical systems does not grow smoothly. Options combine multiplicatively across subsystems and interfaces, so that integration events, governance changes, and supplier onboarding produce step changes in the effective configuration space $\mathcal{C}$. When such a burst coincides with raised performance thresholds (a new compliance requirement, a revised service level agreement), the joint effect on operational functional information I_g^{op} can be dramatic. The practical consequence is a *competence illusion*: a team that has managed its subsystem competently for years can suddenly struggle, not because of any internal failure but because the information burden of finding a satisfactory configuration has crossed a threshold. In the terms of Chapter 3, the

team's generative model of its environment has gone stale, and the prediction errors that signal the mismatch arrive suddenly rather than incrementally, overwhelming the team's precision-weighting mechanisms (Clark 2016). Cases B and D illustrate this pattern. In Case B, Aiwell could not meet newly demanded RAMS requirements despite decades of operational experience. In Case D, previously stable systems were breached by attack vectors that fell outside accumulated experience.

The attention deficit in resource-constrained organisations. The PHAS-EAI model frames attention as a scarce, precision-weighted resource. In a VSE operating under delivery pressure, security, reliability, and other “abstract quality attributes” are deprioritised not because they are unimportant but because their signals lack salience relative to the immediate demands of functionality and schedule. This is *not* a failure of individual diligence. It is, as Chapter 3 establishes, the rational response of a predictive brain that preferentially allocates attention to high-precision signals and deprioritises low-precision, diffuse ones (Clark 2016). Paper V formalises this through the concept of drift: in the absence of Regimes of Attention, an organisation's inference process will progressively weight immediate signals over diffuse, long-horizon ones, even when the latter carry greater systemic risk. Case D demonstrated this directly. Before the intervention, Aiwell's engineers had access to vulnerability information but did not act on it because the signals competed with and were overridden by more salient operational demands.

Authority-weight asymmetry in fragmented supply chains. The formal model introduces authority weights γ_i to capture the influence that different stakeholders exert on shared expectations Φ . In fragmented smart city supply chains, these weights are often highly asymmetric. Large prime contractors and public clients set requirements but lack the technical depth to oversee implementation. VSE subcontractors possess the technical knowledge but lack the contractual authority to influence system-level decisions. Paper II diagnoses this structural asymmetry directly, arguing that contractual risk transfer pushes liability to the actors least equipped to manage it. Cases A and B provide empirical evidence: in both, the SysML model served as a boundary object that partially equalised authority weights by giving all parties a shared, navigable representation of the system. The formal model predicts that unresolved authority-weight asymmetry produces coordination friction and decision reversals (H10), and this prediction is supported by the observed difficulty of cross-boundary security communication in the pre-intervention state.

The clay foundation of regulatory and contractual governance. Chapter 2 documents the regulatory imbalance that shapes VSE behaviour. In Norwegian water infrastructure, the regulatory text governing ICT security comprises only 96 words, compared with 9,826 words for the power grid. Enforcement relies mainly on self-reporting. When formal requirements are imposed without appropriate support or tooling, the typical response is performative compliance rather than substantive engineering improvement. Paper II frames this as the

“clay foundation” of systems security engineering: the structural conditions under which VSEs operate make genuine security engagement rational only when the costs of non-engagement are made immediately visible, and the effort required is proportionate to available resources.

The contribution of the formal model to RQ1 is not merely to *list* these challenges but to explain the *mechanism* through which they reinforce one another. Bursty complexity growth increases the information burden on teams that already face an attention deficit. Attention deficit means that slow-evolving threats (accumulating technical debt, drifting configurations, emerging vulnerability classes) are systematically underweighted. Authority-weight asymmetry prevents actors who detect problems from effectively escalating them. Weak regulatory governance removes the external forcing function that might otherwise compensate. The result is a self-reinforcing cycle of latent risk accumulation that the formal model captures through the interaction of $\mathcal{C}$, I_g^{op} , Φ , γ_i , and h .

7.3.2 RQ2: What are the key requirements and associated metrics for integrating digital engineering practices in the design of resilient smart city infrastructure?

The thesis identifies four key requirements for DE integration in VSE-led infrastructure, each grounded in the formal model and supported by convergent case evidence. For each requirement, associated metrics are identified from the model’s operational observables and the empirical data collected across the four case studies. The four requirements correspond to the four facets of environmental restructuring identified in Chapter 3 as the thesis’s design logic.

Requirement 1: DE practices must reduce functional information burden without reducing useful variety. The formal model demonstrates that resilience depends not on the total size of the configuration space but on the share of configurations that are satisfactory and can be mobilised under time pressure. Effective DE practices reduce the denominator (total optionality $\mathcal{C}$) while preserving or increasing the numerator (useful variety $|A_{g,\rho}|$). The conditional success-share theorem (Theorem T5.7-2) provides the formal warrant: interventions that increase the success share reduce the cost of choice. All four cases demonstrate this mechanism. SysML models constrain the architecture to navigable patterns (Case A). Games create a ritualised, finite-step process for security review (Cases B and D). LLM feedback is internalised as habitual quality checks (Case C). Repeated EoP sessions turn novel threats into familiar categories (Case D).

Associated metrics include option-count proxies (the number of design alternatives by lifecycle mode), success-set estimates (the fraction of configurations that meet service

levels), and threshold-change indicators (documented changes in required performance levels). In Case A, the parametric sensitivity metrics S_F and S_R operationalise the cost-effectiveness dimension. In Case C, weighted composite scores for clarity, conciseness, testability, and traceability provide a repeatable measure of requirements quality. In Case D, the fraction of vulnerabilities remediated within 24 hours serves as a mobilisation metric.

Requirement 2: DE practices must build a designed cognitive reserve rather than depend on expertise growth. Hypothesis H7 receives the strongest convergent support in the thesis: all four cases demonstrate that embedding competence in tools and processes yields more reliable improvement than investing in individual skill development. In Case A, the SysML model enabled non-experts to identify vulnerabilities. In Case B, the game served as a substitute for the absence of security expertise. In Case C, the LLM served as a systems engineering mentor. In Case D, automated monitoring replaced the need for dedicated security staff. The formal model explains why: designed reserve h protects cognitive headroom under interaction demand (Theorem T5.3-4), whereas individual expertise is bounded by the skill speed-up parameter λ_τ and cannot remove governance-imposed delays.

Associated metrics include personnel-change robustness (whether response performance degrades after team changes), tool-dependency ratios (fraction of quality assurance steps performed by automated systems versus manual review), and the internalisation indicator from Case C (whether quality improvements persist when the tool is not in use).

Requirement 3: DE practices must provide machine-readable traceability across organisational and disciplinary boundaries. Hypothesis H14 is supported by all four cases, each involving machine-readable artefacts that reduced noise in cross-boundary communication. The mechanism operates consistently across organisational boundaries (Case A, where the SysML model linked the valve manufacturer, firmware subcontractor, and municipal client), disciplinary boundaries (Case B, where the model enabled mechanical, electrical, and software engineers to discuss interface responsibilities), cognitive boundaries (Case C, where the LLM pipeline linked requirements to INCOSE standards with machine-readable scores), and temporal boundaries (Case D, where Dependabot provided timestamped, structured vulnerability data over a two-year period).

Associated metrics include detection-to-decision time (the interval from signal arrival to the first engineering action), integration surprise rate (the frequency of unexpected interface failures during system integration), and decision rework rate (the proportion of decisions revised after initial commitment). In Case D, the median time to remediation directly operationalises detection-to-action latency.

Requirement 4: DE practices must sustain attention through environmental design rather than compliance mandates. The formal model predicts that process mandates imposed on resource-constrained organisations will be progressively deprioritised under workload pressure unless the signals they require attending to are made salient through environmental design. Hypothesis H9 is supported by Cases B, C, and D, each of which created structured attention-directing mechanisms through different modalities. Case B used time-boxed game sessions. Case C used an automated review triggered at the moment of drafting. Case D used an ambient visual signal (the smart bulb). In each case, improvement was sustained not through compliance mandates but through designed salience. The common principle is that a resource-constrained inference system preferentially adopts low-cost cognitive strategies, and a well-designed environment ensures that the low-cost strategy is the one that is dependability-relevant.

Associated metrics include drift indicators (whether attention to quality attributes decays over time without external prompting), engagement persistence (the duration over which the intervention maintains its effect without reinforcement), and the 24-hour remediation rate from Case D, which remained stable throughout the full two-year observation period.

Table 24 summarises the mapping from requirements to model quantities, observable metrics, and supporting case evidence.

Table 24 Requirements for DE integration mapped to model and case evidence.

Requirement	Model quantities	Observable metrics	Cases
Reduce functional information burden	$C, A_{g,\rho} /C, I_g^*$	Option count proxies, success set estimates, S_F/S_R , quality scores	A, B, C, D
Build designed cognitive reserve	h, λ_τ	Personnel-change robustness, tool-dependency ratios, internalisation indicators	A, B, C, D
Provide machine-readable traceability	$\omega, g(\cdot), \Phi$	Detection-to- decision time, integration surprise rate, decision rework rate	A, B, C, D
Sustain attention through environmental design	Regime of Attention, precision weighting	Drift indicators, engagement persistence, remediation rate	B, C, D

7.3.3 RQ3: How can system architecture modelling patterns help develop resilient smart city infrastructure?

The thesis defines “modelling patterns” broadly to encompass both reusable components and templates, as well as patterned practices that guide the act of modelling itself. The formal model and case studies together demonstrate that modelling patterns help develop resilient infrastructure through three interconnected mechanisms.

Mechanism 1: Patterns reduce the configuration space while preserving useful variety.

The formal model proves (Theorem T5.7-1) that adding constraints, restricting choice sets, and coarsening the distinctness map are all complexity-non-increasing operations. Modelling patterns enact these operations in practice. In Case A, the SysML architecture model constrained the design space by encoding interface contracts, layered protocol boundaries, and linked vulnerability records. Engineers navigated the model during the EoP game and were guided to security-relevant interfaces without needing to search the full space of possible interactions. In Case C, the LLM review pipeline constrained the space of possible requirement formulations by providing structured templates and scoring against defined quality dimensions. In Case D, repeated game sessions transformed an open-ended question (“what threats matter?”) into a finite, categorised process. In each instance, the pattern reduced the effective decision burden I_g^* by pruning configurations that were infeasible, unsafe, or redundant, while preserving and often increasing the proportion of satisfactory configurations.

Mechanism 2: Patterns create cognitive scaffolding that compensates for missing expertise. The four cases demonstrate that modelling patterns function as a designed cognitive reserve. This observation connects directly to the formal model’s prediction that the designed reserve h outperforms experience investment (H7) because it protects response quality independently of individual expertise. In Case A, the SysML model enabled participants with no security background to identify a specific vulnerability (CVE-2020-3162) by navigating layered interface diagrams. In Case B, the EoP game enabled a team of three (one systems engineer and two service technicians) to conduct structured threat modelling without any security specialist present. In Case C, the GPT-4 pipeline served as a systems engineering mentor, and engineers internalised review patterns that persisted when the tool was not in use. In Case D, the combination of Dependabot, the smart bulb, and scheduled game sessions created a self-sustaining cognitive scaffold that maintained security responsiveness for two years without additional headcount or meetings.

The unifying theoretical explanation is provided by the conditional success-share theorem and the joint intervention dominance result (Theorems T5.7-2 and T5.7-3). Patterns simultaneously reduce C and increase $|A_{g,\rho}|/C$. When combined with DE interventions that

improve the observation mapping $g(\cdot)$ and reduce mobilisation time τ_{eff} , the joint reduction in I_g^{op} exceeds the sum of individual effects (Theorem T5.7-3 and Equation 45).

The agent-based simulation in Appendix A2 confirms this superlinearity computationally. The combined PHAS-EAI model, which deploys designed cognitive reserve, niche construction, Patterned Practices, dynamic shared expectations, and Theory of Mind, achieves the lowest system-level free energy of any condition ($F^\Sigma \approx 19$), surpassing Kaufmann et al.'s (2021) best cognitive condition ($F^\Sigma \approx 22$) by the end of the 200-epoch run. No single extension matches the cognitive baseline. The combined model surpasses it only when multiple mechanisms operate simultaneously alongside alignment, confirming the formal prediction that the joint intervention is strictly dominant.

Mechanism 3: Patterned Practices sustain engagement through environmental design.

The most distinctive contribution of the thesis to RQ3 is the concept of Patterned Practices as a formal category. These are not one-off modelling exercises but regular, often ritualised interactions through which agents engage with Regimes of Attention and internalise the knowledge they convey. In Case B, the game sessions constituted Patterned Practices that turned security review into a recurring, structured, and even enjoyable activity. In Case C, the automated review pipeline created a Patterned Practice triggered at the moment of requirement drafting, embedding quality review into the natural workflow rather than imposing it as a separate compliance step. In Case D, the combination of the smart bulb (Regime of Attention) and scheduled game sessions and patch triage huddles (Patterned Practices) created a feedback loop that continuously reinforced security-focused behaviour. The mechanism is the niche construction feedback loop that Chapter 3 identifies as the practical expression of extended active inference (Constant et al. 2020): the team's actions modified the environment, and the modified environment reshaped the team's inference dynamics, producing a stable attractor in which dependability-relevant behaviour was sustained without external mandates.

The simulation in Appendix A2 identifies a design parameter that the case studies do not isolate: the relative importance of synchronisation fidelity versus frequency. In the computational model, increasing the depth of information exchange (λ from 0.2 to 0.8) at a fixed interval produces a larger performance improvement than increasing the frequency of exchange (reducing the interval from 50 to 10 epochs) at fixed depth. The practical implication is that a weekly threat-modelling session with thorough information sharing may outperform daily stand-ups with superficial status updates. This finding is consistent with the case evidence: Case D's fortnightly game sessions, which involved deep engagement with the SysML model, sustained improvement over two years, while the patch triage huddles, which were shorter and more frequent, served a complementary but narrower coordination role.

Paper II's conceptual contribution to RQ3 is to identify the strategic direction that the empirical papers subsequently validated: combining agile model-based systems engineering with gamified stakeholder engagement and a Zero-Trust mindset. This combination was not arbitrary. It was designed to exploit the social agility and low operating overheads characteristic of VSEs, rather than to overcome their structural constraints. The formal model provides the theoretical warrant for why this combination works: it addresses the configuration-space, mobilisation-time, inference, and coordination dimensions simultaneously rather than targeting any single lever in isolation.

7.4 Contributions

7.4.1 Theoretical Contributions

The primary theoretical contribution of the thesis is the extended PHAS model (PHAS-EAI), which integrates the Free-Energy Principle (FEP) from computational biology into the analysis of engineering organisations. The extension addresses a specific gap in the original PHAS framework: while Calvo-Amodio and Rousseau (2019) provide a structural vocabulary for describing what a human activity system is (its parts, purpose, boundary, relationships, and context), the framework does not provide a mechanistic account of how attention is allocated among competing signals within that system. The thesis fills this gap by grounding the PHAS in the precision-weighting mechanism of active inference. Regimes of Attention and Patterned Practices are not additions external to the PHAS ontology but formal elaborations of how the system's relationships and context shape the inference dynamics of its participants. The precision-weighting mechanism is *mathematical*, not *metaphorical*: it provides derivable claims about attention allocation rather than heuristic ascriptions of belief and goal.

The integration also establishes a formal symmetry between the extended mind thesis (Clark et al. 1998) and the PHAS framework. Chapter 3 shows that cognitive extension becomes a mathematical consequence of joint agent-environment free-energy minimisation rather than merely a philosophical argument. This means that the designed cognitive reserve h is not a passive tool but an active participant in the team's inferential dynamics, formally grounding the claim that competence can reside in the environment rather than requiring it in individual heads.

Three further theoretical contributions follow from the integration. First, the formal definitions of complexity as feasible distinct optionality and resilience as mobilisable variety under time pressure, with proved monotonicity results and sensitivity properties. Second, the concept of operational functional information I_g^{op} as a measure of effective decision difficulty that

captures both the sparsity of satisfactory configurations and the time-bounded mobilisation constraint. This measure is more informative than raw complexity because it accounts for the interaction between option-space structure and response dynamics. Third, the fourteen hypotheses (H1 to H14) derived from the formal model constitute a testable research programme that extends beyond the cases examined in this thesis.

The use of Dennett's loans of intelligence as a methodological device is itself a contribution. The thesis explicitly marks where it adopts the intentional stance as a heuristic and where the formal model provides a mechanistic underpinning. This disciplined use of intentional-level explanation, with explicit acknowledgement of where the "loan" remains outstanding, provides a template for other research that applies cognitive and biological metaphors to organisational systems.

7.4.2 Practical Contributions

The thesis presents a set of DE interventions accessible to VSEs at modest cost and with reasonable training effort. These include SysML-based architecture modelling as a boundary object for cross-boundary communication and quantitative trade-off analysis, the EoP card game played against system models as a low-threshold entry point for threat modelling, LLM-assisted requirements review integrated into existing development workflows, and the PHAS-EAI framework's combination of automated monitoring, ambient visual signals, and scheduled game sessions.

The deliberate simplicity of these interventions is a *design feature*, not a limitation. As Section 7.5.2 argues, the relationship between the formal model and these interventions mirrors the relationship between precise science and simple, resilient practice in the embodied mind tradition: the sophistication of the theory is what enables the simplicity of the practice.

The leverage points framework (Chapter 5) provides a structured tool for designing new interventions. By mapping model levers to directional effects and illustrative examples, the framework enables practitioners to identify which quantities their intervention targets and what observable effects to look for. This shifts the evidentiary burden from measuring rare-event outcomes to routine operational data, which is especially valuable in VSE contexts where controlled experiments are impractical.

7.4.3 Methodological Contributions

The thesis combines formal analytical modelling with participatory action research (AR) and grounded theory method (GTM) analysis in a VSE setting. This combination is unusual. Formal modelling is more typically associated with large-scale systems engineering programmes, while action research is more commonly found in organisational studies. The thesis demonstrates that the two approaches are complementary: the formal model generates precise, testable predictions, and the embedded action research provides the longitudinal, fine-grained observational data needed to evaluate those predictions in context. The COVID-19 pandemic inadvertently validated this complementarity: when physical access to the case company was restricted, the analytical track progressed independently, and when restrictions eased, the established models informed the applied work, producing a richer integration than a purely sequential design would have achieved.

7.4.4 Applied Contribution

The contributions described above are analytical: a formal model, a set of demonstrated interventions, and a combined methodology. A fourth contribution translates these analytical results into a working tool that other VSEs can adopt directly.

The study reported in Paper IV was conducted using an early research pre-release of GPT-4 in the spring of 2023. By the time the paper was presented that autumn, the model was already publicly available and being superseded by newer systems. The technology has continued to advance through multiple generations since then, with capabilities expanding from simple text completion to tool use, agentic workflows, and native integration with development environments. This pace of evolution significantly exceeds the throughput of the scientific publication cycle. A conventional follow-up study submitted through the standard journal process would face the same temporal mismatch: by the time peer review and production were complete, the technology under study would again have moved on. The internal process at Aiwell that began with Paper IV did not wait for the next publication cycle. The engineers who built the requirements review GitHub Action continued to develop the concept, and the plugin documented in Appendix A3 represents the current state of that continuing trajectory. Including it in the thesis provides a transparent bridge between the peer-reviewed case studies and the ongoing practical work they initiated.

The VSE Systems Engineering plugin (Appendix A3) is an open-source extension for the Claude Code development environment that operationalises all four design requirements identified in Table 24. It embeds systems engineering competence as designed cognitive reserve (R2), filters information by ISO 29110 lifecycle phase (R1), enforces machine-

readable traceability through SysML 2.0 models and automated hooks (R3), and sustains attention through environmental constraints at commit and phase-transition points (R4).

The plugin extends the trajectory established by Case C (Paper IV). In that study, a single GitHub Action that posted AI-generated requirements feedback demonstrated that LLM-based tools could function as a designed cognitive reserve in a VSE. The plugin generalises this intervention from requirements review to the full ISO 29110 lifecycle, from project planning (PM.1) through product delivery (SR.6). The coverage is complete: every task in the ISO 29110 Basic Profile is addressed by at least one skill (Appendix A3, Table A3.2). It does so without requiring fine-tuned models, custom training data, or specialist AI expertise, addressing the accessibility constraint that Paper IV identified as essential for VSE adoption.

This applied contribution is distinct from the practical contributions in Section 7.4.2. Those contributions describe what was demonstrated in the case studies. This contribution provides a tool that others can use. The plugin is publicly available at <https://github.com/Roarge/vse-systems-engineering> and can be installed by any team using the Claude Code development environment.

7.5 Discussion

7.5.1 Repaying the Loans of Intelligence

The introduction framed the thesis around Dennett’s concept of loans of intelligence. The PHAS model adopts the intentional stance, treating engineering teams as cognitive agents with beliefs, goals, and expectations. The thesis promised that integrating the Free-Energy Principle (FEP) would take a step towards repaying this theoretical debt by showing how PHAS behaviour emerges from the aggregation of simpler, rule-bound events.

The formal model delivers on this promise to a meaningful but bounded extent. The precision-weighting mechanism of active inference provides a principled account of why attention drifts under workload pressure and why environmental salience is more effective than compliance mandates. The formal definitions of complexity, resilience, and functional information provide mechanistic underpinnings for claims that would otherwise remain metaphorical. The formally proved monotonicity results and sensitivity properties ground the leverage points framework in logically consistent directional guarantees rather than intuition. Clark (2016) argues that the predictive processing framework provides the “common currency” that unifies perception, action, and learning within a single computational logic. The PHAS-EAI model extends this common currency to the organisational scale, showing how the same prediction-error-minimisation dynamics that govern individual cognition also govern

collective behaviour when the analysis respects the distributed, embodied, and environmentally embedded character of the team.

The debt is not fully repaid. The mapping from formal model quantities ($C, I_g^{\text{op}}, h, \Phi, \gamma_i$) to observed phenomena remains *interpretive* rather than *quantitative*. The bridge assumptions connecting inference precision to mobilisation dynamics (Theorem T5.7-4) are stated as hypotheses rather than derived from first principles. The thesis is transparent about these outstanding loans, marking them explicitly in the formal analysis. The appropriate conclusion is not that the loan framework has failed but that it has proved a productive discipline: it forced precision about what is explained and what is assumed, and it identified specific empirical gaps that define a clear agenda for future work.

7.5.2 From Formal Model to Smart Bulb

An anticipated objection deserves direct engagement. The thesis claims that VSEs cannot sustain heavyweight systems engineering frameworks, yet the theoretical apparatus deployed here involves free-energy minimisation, Markov blankets, variational inference, precision-weighted prediction errors, and formally proved theorems. How can a theory of this complexity be relevant to organisations that number their engineers in single digits?

The objection confuses the complexity of an explanation with the complexity of the thing being explained. The thesis follows the same design logic that Chapter 1 introduced through the Brooks precedent: precise theoretical understanding of agent-environment interaction produces simpler, more resilient agents. The robots that Brooks (1991) built at MIT used far less internal machinery than their predecessors, yet they outperformed them because their intelligence emerged from coupling with a structured environment. The explanation of why this worked, however, required substantial theoretical machinery. Clark (1997) required an entire monograph to synthesise Brooks's robotics with Varela, Thompson and Rosch's (1991) enactivism, Gibson's (1979) ecological psychology, Breazeal's (2002) extension from physical to social coupling, and Hutchins's (1995) distributed cognition into a coherent account of situated, embodied intelligence. Clark (2016) required another to integrate that account with the predictive processing framework. The theoretical apparatus needed to explain why a simple robot succeeds is orders of magnitude more complex than the robot itself.

The same asymmetry holds for this thesis. The interventions that the formal model explains are not complex. Case D's centrepiece is a smart bulb connected to a GitHub API. It glows orange for low-severity vulnerabilities and blinks red for high-severity ones. A team of three engineers installed it above their shared workspace. That is the entire environmental-level intervention. No Markov blankets are involved in the installation. No variational inference is

required to notice that the bulb is blinking red. The game sessions in Cases B and D use a card deck played against a projected diagram. The LLM review pipeline in Case C is a GitHub Action that posts a comment on an issue. These are interventions that a VSE can adopt in an afternoon.

What the formal model provides is not a recipe for building these interventions but an explanation of why they work, when they will stop working, and what to try next. The smart bulb works because it delivers a high-precision signal through a trusted, unambiguous, high-contrast channel that the predictive brain weights heavily in inference, reducing the effective noise ω in the observation mapping and increasing the affordance of the remediation action. The card game works because it creates a structured interaction cycle that synchronises generative models across team members, reducing authority-weight asymmetry γ_i and stabilising shared expectations Φ . The LLM pipeline works because it embeds a designed cognitive reserve h that protects cognitive headroom independently of individual expertise.

Without the formal model, these observations remain anecdotes. A smart bulb worked for Aiwell. A card game improved engagement. An LLM helped with requirements. But which features of these interventions are load-bearing and which are incidental? Would a different colour work as well? Would the game work without the SysML model? Would the LLM pipeline work for a different engineering standard? Anecdotes cannot answer these questions. The formal model can, because it identifies the quantities that each intervention targets ($\omega, \Phi, \gamma_i, h, C, I_g^{\text{op}}$) and provides proven directional guarantees about what happens when those quantities change.

The broader principle is that adaptive success requires a match between the complexity of an agent's processing and the structure of its environment (Pfeifer et al. 2006). The thesis's interventions operationalise this at the organisational level. The smart bulb trades expensive internal monitoring (checking dashboards, reading email alerts, attending status meetings) against cheap environmental structure (a light that changes colour). The card game trades expensive formal training in threat modelling against cheap social structure (a competitive game played in thirty-minute rounds). The formal model explains why these trades are effective and predicts the conditions under which they cease to be sufficient.

The relationship between the formal model and the VSE practitioner is therefore analogous to the relationship between Newton's laws and the engineer who designs a bridge. The engineer does not solve differential equations on the construction site. The engineer uses lookup tables, rules of thumb, and standardised beam sections that encode the consequences of those equations in a form that supports action. Newton's theory explains why the rules of thumb work, identifies the boundary conditions under which they fail, and guides the development of new rules as conditions change. The formal model in this thesis plays the same role for the design of VSE interventions. The leverage points framework in

Chapter 5 is the lookup table. The fourteen hypotheses are the rules of thumb. The theorems are the boundary conditions.

This is precisely the logic of the designed cognitive reserve that the thesis advocates for VSEs. Just as the smart bulb encodes the outcome of a complex vulnerability-analysis pipeline in a form that any team member can act on, the leverage points framework encodes the outcome of a complex formal model in a form that any practitioner can use to design interventions. The formal model does not need to be understood by the VSE to be useful to the VSE. What matters is that the interventions derived from the model are simple, that the directional predictions are clear, and that the boundary conditions are stated so that practitioners know when to seek more sophisticated guidance.

7.5.3 Complementarity with ISO/IEC 29110

The ISO/IEC 29110 series (ISO/IEC 2016). provides lifecycle standards and deployment packages specifically adapted to VSEs with up to 25 employees. The thesis does not position itself as an alternative to this standardisation effort. The contributions are complementary: ISO/IEC 29110 specifies what activities a VSE should perform, while the PHAS-EAI framework explains why certain environmental and social structures sustain the attention, learning, and coordination that those activities require.

The distinction is mechanistic. Deployment packages define process steps: requirements management, project management, configuration management, verification and validation. These are necessary but not sufficient in the context of cyber-physical infrastructure, where the VSE must sustain attention to security and reliability signals amid competing delivery pressures. The formal model explains why process mandates alone are insufficient in resource-constrained settings: an inference system under workload pressure will deprioritise low-precision, low-affordance activities regardless of their formal requirement status (H9, Theorem T5.5-3). Regimes of Attention and Patterned Practices provide the missing mechanism, ensuring that the activities specified by the standard are actually performed under operational conditions.

The alignment is concrete. The machine-readable traceability demonstrated in all four cases (H14) directly supports the quality evidence requirements that underpin ISO/IEC 29110 certification. The designed cognitive reserve (H7) addresses the staffing constraint that Laporte and colleagues have identified as the central barrier to standards adoption in VSEs: when a firm has three engineers, the standard's expectation that distinct roles are filled is impossible to meet through headcount alone, but the thesis demonstrates that role-separation requirements can be met through designed tools that substitute for missing expertise. The Patterned Practices in Cases B and D, in function, are what the ISO/IEC

29110 deployment packages would call implementation guides: structured activities that translate process requirements into operational routines. The difference is that Patterned Practices are designed from a mechanistic understanding of how attention and learning work in small teams, and can therefore be tailored to the specific information environment of a given VSE rather than implemented as generic process templates.

The simulation results in Appendix A2 sharpen this complementarity through a concrete interaction effect. When agents in the computational model are equipped with Theory of Mind and environmental scaffolding but lack aligned shared expectations, individual convergence distance improves (from 7.1 to 2.5) while collective performance degrades ($F^\Sigma \approx 33$ versus ≈ 25 for the aligned condition without scaffolding). Individual productivity and collective coherence come apart. In a VSE context, this corresponds to a team in which individual engineers adopt sophisticated monitoring tools and develop accurate models of each other's work, yet pursue divergent priorities in the absence of a coordinating framework. The ISO/IEC 29110 deployment packages provide precisely the alignment layer that prevents this failure mode: they establish shared goals and structured work products that ensure Theory of Mind and environmental scaffolding amplify cooperation rather than competition. The formal model predicts the pathology, and the standard provides the remedy.

7.5.4 The Scalability Boundary

Case C explicitly raises the question of whether LLM-based cognitive reserve scales to more complex engineering tasks. The formal model provides a framework for understanding this boundary. Hypothesis H6 predicts diminishing returns from expertise (and by extension from any bounded intervention) in high-drag environments, because the skill speed-up is bounded by λ_τ and cannot remove governance-imposed delays. Hypothesis H7 predicts that designed reserve outperforms experience investment, but this does not imply that designed reserve is unbounded.

The practical implication is that the cognitive support tools demonstrated in the thesis are not fixed solutions. As the complexity of the work increases (through new integrations, raised performance thresholds, or expanded scope), the designed reserve may need periodic recalibration. The formal model predicts the conditions under which this becomes necessary: when operational functional information I_g^{op} rises (due to increased C , raised ρ , or compressed τ_d) beyond the headroom provided by current tools. VSEs should therefore treat their cognitive scaffolding as living infrastructure that evolves with the organisation.

7.5.5 From Single Company to Research Programme

All four case studies were conducted at or in close collaboration with a single Norwegian VSE (Aiwell AS). This is a clear limitation for external validity. At the same time, the formal model makes specific, falsifiable predictions about what should happen in other VSE contexts with similar resource constraints. The predictions are structural rather than company-specific: they concern the relationship between option-space structure, mobilisation dynamics, attention, and coordination, rather than the particular products or markets of any individual firm.

This means that the single-company limitation is not merely a weakness to be acknowledged but an *invitation*. The fourteen hypotheses define a research programme that can be pursued through replication in independent VSE contexts, comparative designs involving organisations of different sizes, and longitudinal studies that track the evolution of complexity and response performance over time. The formal model provides the theoretical scaffolding for such a programme by specifying what quantities to measure, what directional effects to expect, and what would count as disconfirming evidence.

7.5.6 The VSE Advantage and Its Limits

The formal model predicts (H3) that VSEs possess a resilience advantage in the fast-disturbance, moderate-magnitude regime because their short decision chains and low coordination overhead produce a favourable time ratio $r = \tau_d / \tau_{\text{eff}}$. Case D provides evidence consistent with this prediction: all 88 fixable vulnerabilities were remediated within 24 hours, a response rate that would be difficult to achieve in a larger organisation with more complex approval workflows.

The model also predicts that this advantage is bounded. Beyond a certain disturbance magnitude, the VSE's limited headroom and survivable variety become the binding constraint. The thesis does not include a case that tests this boundary directly, but the pre-intervention state of Cases B and D illustrates the failure mode: when complexity bursts (new compliance requirements, novel attack vectors) exceeded the team's accumulated experience, performance degraded despite unchanged personnel and technical capability. The implication is that the interventions demonstrated in this thesis extend the VSE's effective operating envelope but do not eliminate the fundamental bound.

The simulation provides a structural explanation for why the interventions demonstrated in this thesis are specifically valuable for VSEs rather than merely generally useful. In Extensions 1 and 2 (Appendix A2), designed cognitive reserve and niche construction disproportionately benefit the weak agent, whose unreliable sensor makes it most dependent

on environmental support. The strong agent, which already has reliable sensing, gains little from the same scaffolding. This asymmetry maps directly to the VSE context: organisations with limited in-house expertise have the most to gain from environmental scaffolding, because their baseline capability is furthest from the performance frontier. The formal model predicts this through the max-floor formulation of designed cognitive reserve (Theorem T5.3-4): the floor guarantee raises only those positions below the threshold, leaving existing proficiency intact. For well-resourced organisations that already operate near the frontier, the marginal return from scaffolding is small. For VSEs, it can be transformative.

Future work should investigate the location of this bound across different types of VSEs and disturbance profiles.

7.5.7 Returning to the Smart City Failure Patterns

Chapter 2 identified five structural failure patterns in smart city initiatives: technological solutionism, governance friction between directed and collaborative systems-of-systems, the ghost city syndrome, economic and lifecycle misalignment, and the trust deficit created by surveillance architectures. These patterns were presented as empirical observations drawn from the literature. The PHAS-EAI framework developed in this thesis provides a unified explanatory account of why these patterns recur and why the interventions demonstrated in the case studies avoided them.

The sharpest illustration is the comparison between Masdar's central tower and the smart bulb in Case D. The two artefacts are superficially identical: both are ambient visual signals, both use colour to encode severity, and both are intended to direct attention towards a systemic concern. Yet Masdar's tower created what D'Eramo (2021) describes as a disciplinary, prison-like atmosphere, while the smart bulb sustained a two-year improvement in vulnerability remediation without any sense of coercion. The PHAS-EAI model explains the divergence through three mechanisms.

First, the *direction of inference*. The Masdar tower was imposed by a directing authority on citizens who had no role in defining what it monitored, what thresholds triggered the colour change, or what response was expected. In the precision-weighting vocabulary of the formal model, the tower delivered a high-precision signal associated with *punishment* (energy rationing, social disapproval) rather than with a tractable action. Citizens received a prediction error but no clear affordance for resolving it. The smart bulb, by contrast, was installed by the engineering team themselves in response to a problem they already experienced: missed vulnerability alerts buried in email. The signal was associated with a specific, low-cost action (review and approve the Dependabot patch). The affordance $p(a)$ of

the response was high because the toolchain made remediation the path of least resistance. The tower told people they were failing. The bulb told engineers what to do next.

Second, *authority-weight symmetry*. The Masdar tower embodied exactly the authority-weight asymmetry that H10 predicts will produce friction: a central authority ($\gamma_{\text{authority}} \gg \gamma_{\text{citizen}}$) imposed monitoring on agents who could not reciprocate, contest, or negotiate the terms. Finding D2 reports the opposite dynamic. The smart bulb made vulnerability status equally visible to engineers and management, partially equalising γ_i for security decisions. Management did not need to rely on verbal reports, and engineers did not feel surveilled because they controlled the system that generated the signal.

Third, *niche construction versus niche imposition*. The Aiwell team constructed their cognitive niche through their own actions: they chose Dependabot, installed the bulb, and scheduled the game sessions. In the extended active inference framework (Constant et al. 2020), this is genuine niche construction: agents reshape their environment to support their own inferential processes, and the reshaped environment, in turn, reshapes the agents. Masdar's planners imposed a niche on inhabitants who had no history of structural coupling with it. Without Patterned Practices through which residents could engage with the signal, internalise its meaning, and build shared expectations about appropriate responses, the tower remained an external imposition rather than an integrated component of the community's cognitive architecture. It was, in the terms of Chapter 3, a Regime of Attention without Patterned Practices, and the formal model predicts that such a configuration will not sustain behavioural change.

The same explanatory logic extends to the other four failure patterns. Technological solutionism (Section 2.1.4.1) deploys technology that does not address the prediction errors agents actually experience. Songdo's telepresence systems failed because residents did not attend to a communication problem the technology was meant to solve. The thesis's interventions succeeded because each addressed a felt problem: engineers could not find vulnerabilities in complex dependency trees (Case A), could not translate operational experience into verifiable security practices (Case B), could not maintain requirements quality without expert review (Case C), and could not sustain attention to diffuse security signals (Case D). In each case, the intervention reduced a prediction error that the agents were *already generating*.

Governance friction (Section 2.1.4.2) arises when a collaborative system-of-systems is treated as a directed one. Sidewalk Labs attempted to bypass Toronto's democratic governance structures, imposing a directing architecture on operationally independent constituents. The formal model frames this as a mismatch between the system's actual authority-weight distribution and the distribution assumed by the intervention. The thesis's case studies operate within the existing authority structure of a single VSE, where the team's

adoption of interventions was voluntary and iterative. The interventions did not require external mandates because they were designed to make dependability-relevant behaviour the natural default within the team's own inference dynamics.

The ghost city syndrome (Section 2.1.4.3) is the absence of accumulated shared expectations Φ . Greenfield cities lack the organic layers of meaning, the patterned interactions, the shared histories that constitute a functioning cognitive niche. Without existing Patterned Practices to build on, designed environments remain sterile infrastructure rather than living cognitive architecture. The thesis works within a brownfield context: Aiwell had existing habits, existing relationships, and existing frustrations that the interventions could engage with and reshape. The formal model suggests that any intervention must couple to the agents' *existing* generative models before it can modify them.

Economic and lifecycle misalignment (Section 2.1.4.4) is, in the formal model's terms, a mismatch in disturbance timescale τ_d between the infrastructure layer (decades) and the technology layer (years). The thesis addresses this by embedding designed cognitive reserve in *practices* and *patterns* rather than in specific technology versions. The smart bulb can be replaced with a different signalling device. The EoP game does not depend on a particular software platform. The LLM pipeline adapts to new models. What persists is the Patterned Practice, the structured interaction cycle, not the hardware that instantiates it. This is a direct consequence of the formal model's distinction between designed reserve h (which is structural) and specific tooling (which is contingent).

The trust deficit (Section 2.1.4.5) is the predictable consequence of imposing Regimes of Attention that serve the directing authority's inference rather than the agents' own. When citizens perceive that monitoring is used to control rather than to support, precision-weighting works against the intervention: the signal is attended to, but it elicits avoidance rather than engagement. The smart bulb avoided this because it served the team's own goals. It did not report to an external authority. It did not create a record of individual compliance. It simply made the system's state visible to the people responsible for maintaining it.

The interventions demonstrated in this thesis are, in the terminology of Thaler et al. (2009), *nudges*: they reshape choice architecture to make the desired behaviour the path of least resistance without mandating it or imposing material costs. The nudge and choice architecture literature provides independent empirical support for the cost-effectiveness of such interventions relative to mandates, incentives, and educational campaigns (Benartzi et al. 2017). Over 200 countries and municipalities now operate behavioural insight units informed by these principles (OECD 2017), and the organisational applications are extensive (Chapman et al. 2021). However, the nudge literature is predominantly empirical. It documents that choice architecture interventions work, but it provides limited theoretical machinery for predicting *which* interventions will work in *which* contexts, or *why* an effective

nudge may lose its potency when conditions change. Chapman et al. (2021) note, for example, that golden coin decals reduced littering in Chinese textile factories by engaging a cultural belief about good fortune, but the effect disappeared when the decals were temporarily removed and then reintroduced. The nudge literature can describe this context sensitivity but cannot explain it mechanistically.

The PHAS-EAI framework fills this gap. Precision-weighting provides the mechanism: a nudge works when it delivers a high-precision signal that reduces the expected free energy of the desired action relative to alternatives. The golden coins worked initially because they engaged an existing generative model (cultural beliefs about fortune) and increased the affordance of the desired behaviour (keeping the floor clean). When removed and reintroduced, the coins no longer generated a prediction error because the agents had already updated their model to discount the signal. In the vocabulary of this thesis, the coins were a Regime of Attention without a sustaining Patterned Practice, and the model predicts that such configurations decay. The smart bulb in Case D did not decay over two years because it was embedded in a system of Patterned Practices (game sessions, patch triage huddles) that continuously refreshed the shared expectations ϕ associated with the signal. The formal model thus provides the theoretical substrate that the nudge literature currently lacks: a principled account of when choice architecture interventions will sustain behavioural change and when they will not.

The common thread across all five patterns is the distinction between environmental restructuring that serves the agents' own inferential needs and environmental restructuring imposed by an external authority for its own purposes. The PHAS-EAI framework predicts that only the former will sustain behavioural change, because only the former produces the niche construction feedback loop through which designed environments become integrated components of the agents' cognitive architecture. Cities are purposeful human activity systems. The failure patterns documented in Chapter 2 are what happen when that fact is ignored.

7.5.8 From GitHub Action to Full-Lifecycle Scaffolding

Section 7.5.2 argued that the relationship between the formal model and VSE interventions mirrors the relationship between Newton's laws and the lookup tables an engineer uses on a construction site. Section 7.5.3 established the complementarity between ISO/IEC 29110 and the PHAS-EAI framework. The VSE Systems Engineering plugin (Appendix A3) brings both arguments together in a working tool: a set of lookup tables (skills and knowledge files) embedded in an environmental structure (hooks and phase gates) that follows the process skeleton of the standard.

The plugin grew directly out of Case C. In Paper IV, Aiwel's engineers built a GitHub Action that called the GPT-4 API to review requirements posted as GitHub issues. That minimal intervention produced measurable effects across all four design requirements: the LLM acted as a designed cognitive reserve (R2), the automated trigger created a Regime of Attention (R4), the structured scoring reduced the effective option space (R1), and the machine-readable scores improved observation precision (R3). The plugin generalises this single-activity intervention to the full ISO 29110 lifecycle, from project planning through product delivery, without requiring fine-tuned models, custom training data, or specialist AI expertise.

The evolution was enabled by rapid advances in the underlying technology. Paper IV used an early research pre-release of GPT-4 integrated through a custom API wrapper. Foundation models have since progressed through multiple generations, and the tooling ecosystem has matured from raw API access to integrated development environments with native plugin architectures and tool-use capabilities. The Case C intervention could review individual requirements in isolation. The plugin can maintain a coherent model of the entire project state across phases, a qualitative shift that makes full-lifecycle scaffolding feasible for VSEs.

Two architectural choices deserve discussion here because they address limitations identified in the case studies. The remaining architectural detail is documented in Appendix A3. First, the plugin decomposes the lifecycle into discrete, phase-scoped skills rather than attempting a monolithic tool. This addresses the scalability concern raised by Finding C6 not by making the tool handle more complexity, but by reducing the complexity that any single interaction must address. Second, the project scaffolding generates a configuration file that activates the systems engineering competence floor for every interaction with the project, regardless of whether the engineer explicitly requests guidance. Without this activation mechanism, designed cognitive reserve would function only during explicit skill invocations, leaving the routine, pressure-driven interactions uncovered, precisely the gap that the formal model's floor guarantee (Theorem T5.3-4) is designed to close.

A critical difference from Case C concerns the alignment layer discussed in Section 7.5.3. Paper IV's GitHub Action operated on individual requirements in isolation. It improved the quality of requirements but did not address the coordination structure within which those requirements existed. The simulation results in Appendix A2 predict that environmental scaffolding without alignment can fragment collective performance even while improving individual outcomes. The plugin addresses this through the ISO 29110 process skeleton, which provides the shared goals, structured work products, and defined phase transitions that prevent the fragmentation the simulation predicts.

The plugin has not been evaluated empirically. It is not a case study and does not constitute new evidence for or against the fourteen hypotheses. Its value for the thesis is twofold. First,

it demonstrates that the design requirements in Table 24 are sufficiently concrete to guide tool construction. Second, it provides a structured platform for the future evaluation that Section 7.7 calls for, with observable intervention boundaries at which the formal model's predictions can be tested.

7.6 Limitations and Threats to Validity

7.6.1 Internal Validity

No case includes a formal control group or counterfactual baseline. All use before-and-after or qualitative comparison designs. The participatory action research methodology means the researcher was also a participant, introducing potential confirmation bias. However, this insider position also afforded access to detailed operational data and longitudinal engagement that external researchers typically would not obtain. The grounded theory method's requirement for constant comparison across cases and time periods, including explicit search for disconfirming instances, provided a partial safeguard against confirmation bias. The dual-track methodology (formal model developed independently, then tested against case evidence) further mitigates the risk: the hypotheses were derived from the formal analysis before the case evidence was systematically mapped to them.

7.6.2 External Validity

All cases are drawn from the same Norwegian industrial region and involve the same company (Aiwell AS) and its immediate supply chain partners. The systems of interest (building automation, drainage infrastructure) belong to a specific technical domain. Generalisation to larger organisations, different national regulatory contexts, or different industry sectors requires additional study. The formal model predicts that findings should transfer to other VSE contexts with similar resource constraints, but this prediction has not been independently validated. The scope conditions are important: the thesis's findings are most directly applicable to cyber-physical VSEs operating in fragmented supply chains where no single entity possesses full system oversight. Whether the same mechanisms operate in software-only VSEs, in different contractual structures, or in cultural contexts where the relationship between mandates and compliance differs from the Norwegian setting remains to be tested.

7.6.3 Construct Validity

The formal model quantities (C , I_g^{op} , h , Φ , γ_i) are not directly measured in any case study. Mappings from observed findings to model quantities are interpretive rather than quantitative. Future work should develop operational measures that allow direct numerical comparison between theoretically predicted and empirically observed values. The current mappings demonstrate conceptual alignment, not quantitative confirmation. This interpretive character is a necessary consequence of the model's novelty: the quantities are newly defined, and the instrumentation needed to measure them in field settings does not yet exist. Developing such instrumentation is a priority for the research programme defined by the fourteen hypotheses.

7.6.4 Reliability

GTM coding was performed by a single researcher across all four studies. The 2-year duration of Case D provides the strongest evidence for sustained effects. Cases A and B report immediate effects from individual sessions, while Case C covers approximately six months. Replication with independent coding teams would strengthen the reliability of the emergent categories.

7.6.5 Thesis-Level Limitations

In addition to the case-level validity concerns, several thesis-level limitations should be noted.

Hypotheses H1 (bursty complexity), H4 (burst-induced regime shifts), and H8 (operational functional information predicts outcomes better than raw complexity) receive contextual or limited direct support. No case explicitly measures complexity potential C_p changes, threshold crossings, or functional information in bits and correlates these with outcome variables. These hypotheses are the least empirically grounded elements of the thesis and represent priority targets for future validation.

The thesis does not include a comparative design between VSEs and larger organisations facing equivalent disturbances. The formal model's prediction of a regime-dependent VSE advantage (H3) is therefore supported by evidence consistent with the prediction rather than by controlled comparison. Isolating the causal mechanism from potential confounders (such as simpler approval workflows or lower system complexity) requires a different study design.

7.7 Future Work

Three directions emerge as priorities from the analysis.

The first and most pressing is the development of operational measures for the formal model's key quantities. In particular, functional information I_g^{op} needs to be operationalised in a form that can be estimated from routine engineering data (repository metrics, incident logs, deployment records) rather than requiring purpose-built instrumentation. Such measures would enable quantitative hypothesis testing and allow the predictions of H1, H4, and H8 to be evaluated with the rigour they currently lack. Pilot work could begin by instrumenting a small number of VSE projects with automated complexity proxies (interface counts, dependency graph metrics) and correlating these with response performance indicators over time.

The second direction is replication and comparative design. The fourteen hypotheses define a structured research programme that extends well beyond a single company. Independent replication in other VSE contexts (different countries, different infrastructure sectors, different team sizes within the VSE range) would test the external validity of the findings. Comparative studies involving organisations of different sizes facing equivalent disturbances would allow the regime-dependent VSE advantage (H3) to be tested directly. The formal model provides the scaffolding for such studies by specifying what to measure and what directional effects to expect. The complementarity with ISO/IEC 29110 discussed in Section 7.5.3 suggests a natural collaboration: conducting replication studies within the network of VSEs that have adopted the standard would both test the thesis's predictions and enhance the standard's implementation guidance.

The third direction concerns the scalability boundary of the designed cognitive reserve. Case C's finding that engineers questioned whether LLM-based benefits would scale to more complex tasks is consistent with the formal model's prediction of diminishing returns (H6). Future work should investigate how the effectiveness of cognitive scaffolding tools varies with task complexity, and whether periodic recalibration (updating prompts, retraining models, redesigning game scenarios) can extend the effective range. The rapid evolution of large language models makes this a particularly timely question, as the boundary conditions observed in 2023 may already have shifted.

A fourth direction extends the agent-based simulation work reported in Appendix A2. The current model uses a minimal two-agent system in a one-dimensional environment. Scaling to multi-agent scenarios with heterogeneous perceptiveness and multiple overlapping shared targets would more closely approximate real supply chain configurations. The finding that Theory of Mind is conditionally beneficial, amplifying cooperation when shared goals exist but

fragmenting the collective when they do not, warrants systematic exploration across larger populations of agents, where the interaction between alignment and environmental scaffolding may exhibit threshold effects not visible in the two-agent case.

A fifth direction concerns the empirical evaluation of tools that span the full ISO 29110 lifecycle rather than a single activity. Advances in foundation model capabilities and developer tooling since Paper IV (discussed in Section 7.5.8) have made such tools feasible. The VSE Systems Engineering plugin described in Appendix A3 operationalises all four design requirements from Table 24 as a publicly available tool, extending the single-activity intervention demonstrated in Case C to the complete lifecycle. It provides a structured platform for evaluating whether the effects observed in individual case studies compose when deployed as an integrated system. The formal model predicts (Theorem T5.7-3 and Equation 45) that the joint reduction in operational functional information from simultaneous interventions exceeds the sum of individual effects. The plugin's architecture creates observable intervention boundaries at which this superlinearity prediction can be tested. The three-tier traceability chain and differentiated phase-gate checks provide machine-verifiable evidence at every commit and lifecycle transition, while two drift indicators offer longitudinal measures of attention decay that the individual case studies could not capture. Pilot evaluations could compare projects conducted with and without the plugin within the same VSE, measuring the observable metrics associated with each design requirement: option count proxies and quality scores (R1), personnel-change robustness and tool-dependency ratios (R2), detection-to-decision time and decision rework rate (R3), and drift indicators and engagement persistence (R4). Such evaluations would also provide the first independent test of the scalability boundary discussed above, as the plugin's full-lifecycle scope significantly exceeds the task complexity of any individual case study.

7.8 Concluding Remarks

Cities are being built on infrastructure that must last decades, embedded with technology that cycles in years, delivered by organisations that number their engineers in single digits. This thesis has shown that the structural fragility this creates is not inevitable.

The three research questions asked what dependability challenges arise when Smart City infrastructure depends on Very Small Entities, what requirements and metrics can guide the integration of digital engineering practices in that context, and how system architecture modelling patterns can help develop resilient infrastructure. The answers converge on a single principle: the viability of a VSE engineering team depends less on what its members know than on how their working environment is designed. Dependability challenges are, at root, challenges of attention and coordination in resource-constrained settings. The requirements that matter most are those that reshape the information environment rather

than adding process overhead. Modelling patterns help not because they impose rigour from above but because they reduce the cost of choice from within.

The formal model gave this principle analytical precision. It defined complexity as feasible distinct optionality and resilience as mobilisable variety under time pressure, then derived fourteen testable hypotheses and a leverage points framework that connects formal quantities to observable interventions. The model's chief contribution is not the vocabulary itself but what the vocabulary makes possible: directional guarantees about which interventions will improve performance under which conditions, boundary predictions about when those interventions will cease to be sufficient, and a structural explanation of why small teams with short decision chains hold a resilience advantage in the fast-disturbance regime where cyber-physical infrastructure operates.

The case studies demonstrated what this looks like in practice. A SysML model that enables non-experts to identify vulnerabilities. A card game that turns threat modelling into a transdisciplinary activity. An LLM pipeline that embeds systems engineering expertise in a GitHub Action. A smart bulb that makes the invisible visible. None of these interventions require the VSE to understand the formal model. The sophistication of the theory is what enables the simplicity of the practice.

Four findings converge across all cases. Designed cognitive reserve consistently outperforms experience investment (H7). Modelling patterns reduce decision burden without reducing useful variety (H13). Machine-readable traceability reduces noise across the boundaries where coordination matters most (H14). Environmental salience sustains improvement more effectively than compliance mandates (H9). What these findings share is a common mechanism: each intervention works by restructuring the information environment so that dependable behaviour becomes the path of least resistance, rather than an additional obligation imposed on teams already under delivery pressure.

The thesis introduced the language of loans of intelligence to mark where it adopted the intentional stance as a heuristic and where the formal model provides mechanistic grounding. The debt is not fully repaid. The mapping from formal quantities to observed phenomena remains interpretive rather than quantitative, and the bridge assumptions connecting inference precision to mobilisation dynamics are stated as hypotheses rather than derived from first principles. What the framework achieved, however, is a productive discipline: it forced precision about what is explained and what is assumed, and it identified the specific empirical gaps that define a structured agenda for future work. The fourteen hypotheses are not a complete inventory. They are the opening programme of a line of research that extends well beyond a single company.

The contributions complement rather than compete with existing standardisation efforts. ISO/IEC 29110 specifies what activities a VSE should perform across the systems engineering lifecycle. The PHAS-EAI framework explains why certain environmental and social structures sustain the attention, learning, and coordination that those activities require. The machine-readable traceability demonstrated across all four cases directly supports the quality evidence requirements that underpin standards adoption. Together, the standard and the framework address different layers of the same problem: the standard provides the process architecture, and the framework provides the cognitive architecture that makes the process sustainable.

The key insight is that resilience in VSE-led infrastructure is not primarily a question of what the organisation knows but of how its environment is designed. Regimes of Attention and Patterned Practices are not additional overhead imposed on small teams. They are the mechanism through which small teams convert their structural advantages, short decision chains, social agility, and low coordination overhead into sustained engineering quality. If smart city infrastructure is to be dependable, the environments in which it is designed and maintained must be designed with the same care as the infrastructure itself.

The foundation does not have to be *clay*.

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Paper I

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Validating Reliability and Security Requirements in Public Sector Infrastructure Built by Small Companies

Roar E. Georgsen, Geir M. Køien

University of South-Eastern Norway
roar@aiwell.no

Abstract

Municipal infrastructure in Norway is built primarily by small specialist companies acting as subcontractors, mostly with minimal experience working with information and communication technology (ICT). This combination of inexperience and lack of resources presents a unique challenge. This paper applies model-based systems engineering (MBSE) using the Systems Modelling Language (SysML) to combine validation of reliability and security requirements within a mission-aware interdisciplinary context. The use case is a 6LoWPAN/CoAP-based system for urban spill water management.

Keywords: Model-based systems engineering, security by design, ICT dependability.

1 Introduction

1.1 Motivation

According to the Intergovernmental Panel on Climate Change (IPCC), the near future will include intensifying weather events, resulting in heavy rainfall and flooding. Nine out of ten global disasters are water-related, and the UN Environment Programme (UNEP) estimates costs related to water infrastructure will amount to EUR 23 trillion by 2030 [1]. The World Bank reports that the cost of repairing water-related infrastructure in Europe will reach

EUR 24 billion per year by 2050 [2]. At the same time, UN studies indicate that 6.3 billion people will be migrating into urban settings worldwide by 2050, amounting to 200.000 people per day [3]. This densification of cities reduces the paths available for surface water runoff, and urban areas are rapidly losing runoff capacity to clear their streets of rainwater after heavy precipitation. Using parks, playgrounds, and other green areas as temporary retention basins for surface water runoff is increasingly becoming a health hazard due to pollution from overspill between under-dimensioned waterways. More than 76.000 Norwegians have E.coli contaminated drinking water [4, 5], and a 2019 incident in Askøy municipality resulted in 2000 citizens falling ill and two deaths [6].

1.2 The Norwegian Case

Water infrastructure in Norway is owned and operated by local governments and is financed primarily through water fees. This limited funding combined with a severe shortage of engineers in the public sector has led to a maintenance shortfall estimated by SINTEF at NOK 320 billion [7, 8]. Therefore, municipalities are looking for new and cost-effective approaches to designing and managing urban water systems, introducing new technology such as distributed sensor networks and real-time data analysis. However, this has introduced new security risks since local authorities also have a severe ICT competence deficit [9–12]. Therefore, in addition to climate change and water infrastructure maintenance shortfall, The Norwegian Directory of Health (Helsedirektoratet) has identified poor ICT security as one of the main threats to public health [13]. As pointed out in response to a 2021 Florida water system hack, under-regulation of water infrastructure ICT security has resulted in underreporting of poor security practices and related incidents [14]. In Norway, the Food and Safety Authority (Mattilsynet) regulates ICT security in water infrastructure. Whereas the regulations covering ICT security practices in the power grid totals 9.826 words, the equivalent section regulating water infrastructure is only 96 words [15]. The specifics are left entirely to the municipalities, with enforcement primarily based on self-reporting.

Due to insufficient internal competence, local authorities must rely on their existing supply network [16]. ICT security has sometimes been left out entirely in some water management systems, with control panels openly accessible via the web [17]. The reason for this is that the suppliers themselves lack competency in ICT security.

The building, infrastructure, and facilities sector is the largest on the Norwegian mainland, accounting for a third of all enterprises. However, of the 67.908 entities involved in mainland construction projects, only 36 companies employed more than 250 people in 2019, and most companies involved in projects had less than ten employees [18], a pattern repeated across all OECD countries [19]. These companies are highly specialized and struggle to meet the growing demand for documented safety and security in design. We propose that model-based systems engineering (MBSE) is a promising approach to coordinating this network of smaller suppliers in public sector infrastructure projects. A single model approach can support expert knowledge capture to inform on-ground engineering efforts and support project-wide validation of stakeholder requirements.

1.3 Layout of the Paper

Section 2 of this paper provides background on engineering in very small companies and a brief introduction to MBSE methodologies and tools in the context of system security engineering and Security by Design. Section 3 describes a specific use case, a wireless 6LoWPAN/CoAP-based system for urban spill water management, demonstrating safety and security by design using MBSE. Section 4 analyses the model and resulting changes to the original design arising from the modelling process. Finally, we note points for further research not covered in this paper.

2 Background

2.1 Engineering in Very Small Companies

Engineers working with product development in small companies typically take on a wide range of tasks outside their area of expertise. This is especially true when it comes to developing devices for the Internet of Things (IoT). The demand for industrial devices with low power consumption drives a need for optimised custom hardware, and usability requirements drive connectedness and increase the scope of data collection. The same engineer will likely work on the entire stack from bare silicon to mobile applications, often as an architect, project manager and developer concurrently. Companies mitigate these competency gaps by working as part of supply networks [20], in practice by informal communication developer to developer. Attempts to enforce more systematic coordination between suppliers through required standards or Service Level Agreements (SLA) have largely failed. An examination of

the supply chain to the Australian defence industry found that such legal structures did not impact actual engineering practices [21].

Small companies do not see standards outside their primary domain as relevant [22], and documents are produced primarily as legal and contractual requirements. Malicious actors increasingly target businesses too small to have their own IT departments because these are likely to have poor security. 60% of companies allow the use of personal devices, and 32% of small companies have no policies or procedures requiring employees to seek permission from or inform the company when doing so. 52% of security violations in small companies are discovered entirely by chance [23].

2.2 Model-Based Systems Engineering

Model-based systems engineering (MBSE) is the *"formalized application of modelling to support system requirements, design, analysis, verification and validation activities beginning in the conceptual design phase and continuing throughout development and later life cycle phases"* [24].

The model is the sole source of truth and reflects the state of system development, and integrates multiple complementary perspectives in a compatible manner. This focus on an integrated model differentiates MBSE from simply engineering with models, where multiple models may have inconsistent assumptions and semantics. As system complexity grows or there are multiple stakeholders, document-centric approaches suffer from the increasing risk of overlooking critical information, as documentation tends to become incomplete and inconsistent. An integrated model enables connections between model elements, effective and efficient information retrieval, automatic propagation of changes, and automated model validation. A Model-based approach supports an incremental generation of architecture from simple reusable component models, gradually constructing more complex subsystems [25]. The modelling environment can then automatically generate any required documents from the model, with comprehensive traceability maintained for all system information, generated data, and design decisions. Using declarative models with explicit interfaces facilitates communication between engineering disciplines by establishing a shared context for discussion. MBSE is now starting to deliver rigour and effectiveness in complex systems development [26, 27]. Some view MBSE as a potentially revolutionary tool for optimising cost and quality [28]. Whereas a traditional functional decomposition approach captured 50% of problem understanding as measured by

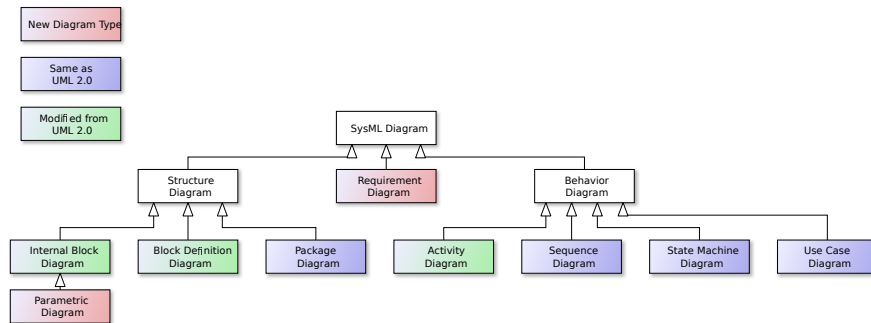


Figure 1 SysML diagram taxonomy [33]

time to stakeholder consensus, modelling use cases and scenarios increased problem understanding to more than 90% on the first iteration [29].

2.3 The Systems Modeling Language (SysML)

The Systems Modeling Language (SysML) [30] is a general-purpose graphical notation for modelling complex cyber-physical and socio-technological systems. SysML provides graphical representations for modelling requirements, behaviour, structure, and parametrics. The Object Management Group (OMG) developed the notation jointly with the International Council on Systems Engineering (INCOSE) as a subset of UML 2 with extensions. However, version 2 has a new metamodel not constrained by UML that includes textual syntax alongside the graphical notation but remains visually similar to UML [31,32]. Figure 1 provides an overview of the nine diagrams in SysML grouped by type and how they relate to UML 2.0. Some modifications, such as replacing the Class concept with an equivalent Block element, are primarily intended as a linguistic device to make the notation less software-centric. Other modifications introduce non-software concepts such as continuous, probabilistic and physical item flows into existing UML diagrams.

2.3.1 SysML Requirements Modeling

The first of the two large innovations in SysML syntax is the notation for modelling requirements. Figure 2 shows a non-normative extension of the basic syntax similar across most modelling tools. The diagram also adds three new requirement types to the model. Typically, extending the Class metaclass is used to create subtypes of requirements that correspond to the traditional

sections of a requirements document. This legacy view of requirements is practical when it is necessary to synchronise the model with externally stored text-based requirements. Figure 3 shows an example of importing predefined textual requirements into the model. However, the primary benefit of SysML requirements comes from the relationships defined on the *AbstractRequirement* type. Figure 3 models a requirement as being derived from a corresponding requirement in the Application Security Verification Standard (ASVS) and traced to a corresponding weakness with a CWS number. A use case that may include multiple model components refines the definition of the requirement, and the "verify" relationship links the requirement to a test case. The "satisfy" relationship creates a link in the model between the requirement and the property or function that satisfies it.

The "allocate" relationship can link use cases and functions to a component, but it can also link model elements directly to requirements. Suppose a known vulnerability is modelled as a requirement and allocated to a component. In that case, that vulnerability will show up as an implicit requirement whenever a design uses an instance of that component. Domain experts can model requirements within their respective domains, and this will be enforced and constrain the design space for other users of the model.

Whereas the taxonomy in figure 4 is useful when textual requirements already exist, a requirement can be an extension of any named element metaclass. Figure 4 shows the definition of three types of requirements based on different behavioural model elements. This feature enables SysML requirements to be more specific and less ambiguous than is possible with textual requirements.

2.3.2 SysML Parametric Modeling

The second innovation in SysML is the addition of notation for modelling constraints and mathematical relationships. Figure 10 shows an example of a parametric diagram. Constraint properties connect to value properties on the owning block, as well as to other constraint properties. The SysML primary specification does not stipulate a specific constraint language.

2.3.3 Executable SysML

In order to obtain the most benefit from MBSE, it is necessary to utilise SysML in a machine-readable manner. SysML relies on the OMG XML Metadata Interchange (XMI) to exchange modelling data, and most modelling environments integrate well with external tools such as MS Excel, MATLAB, OpenModelica, Maple, or Mathematica. They can leverage these

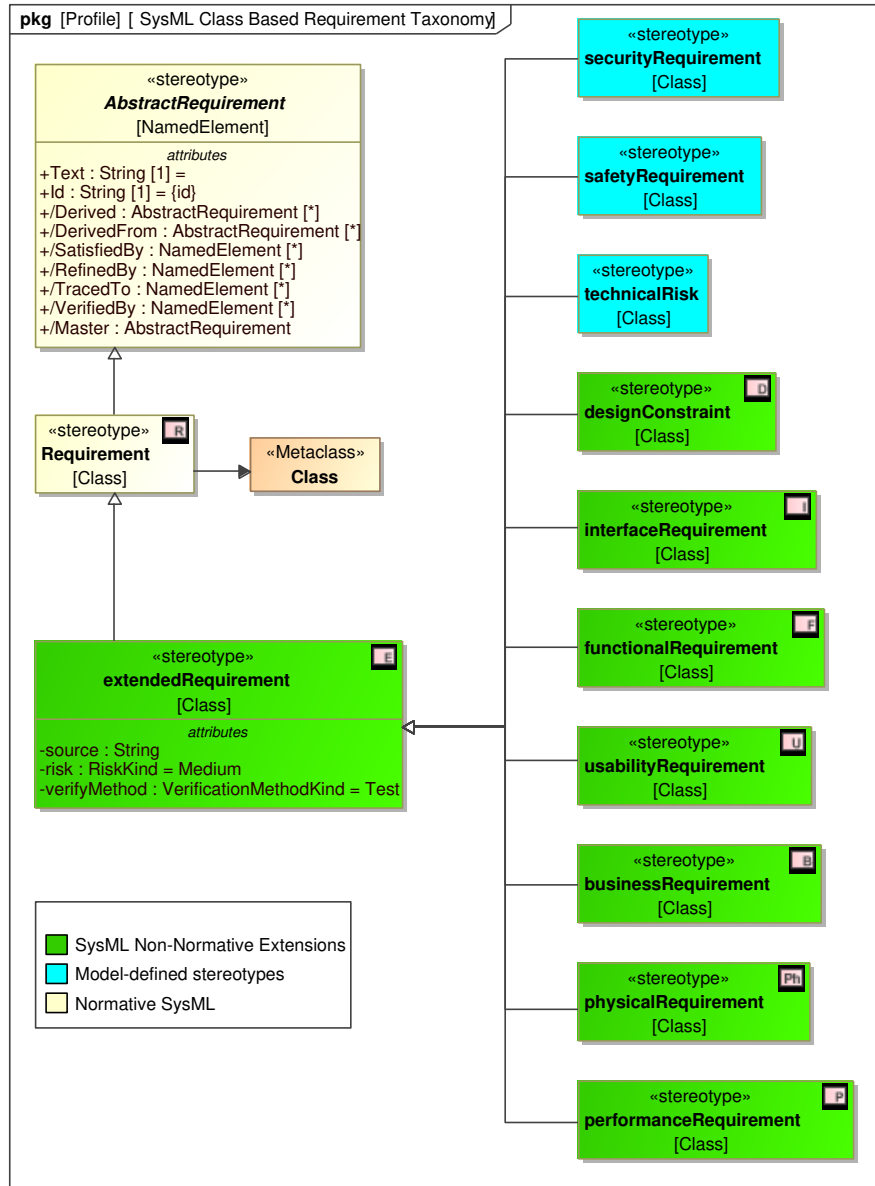


Figure 2 SysML Class-based requirements taxonomy

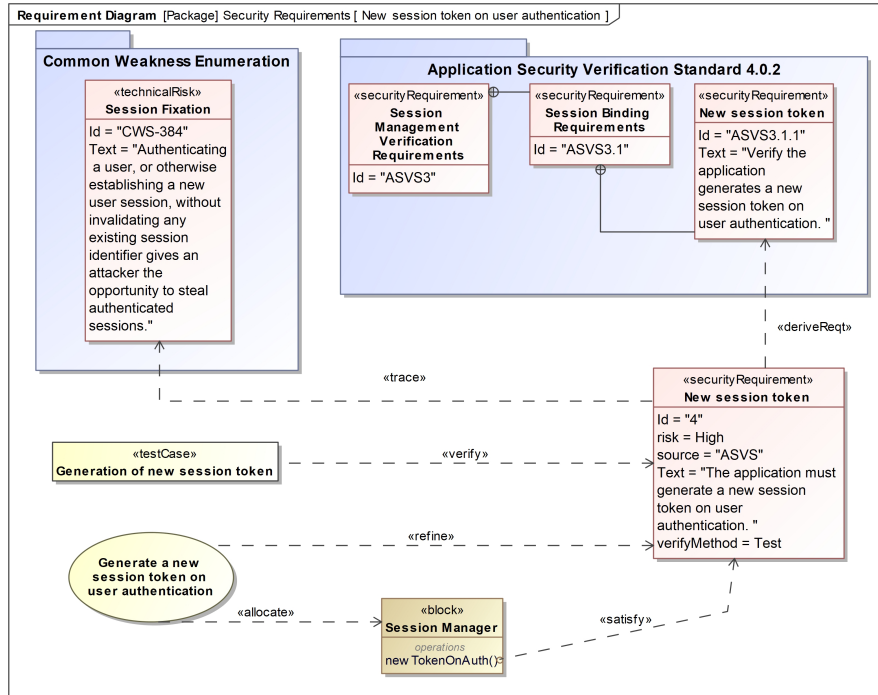


Figure 3 Using relationships to link model elements.

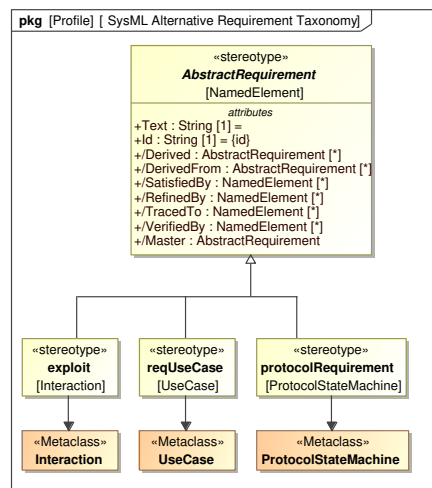


Figure 4 Non-Class SysML requirements

as supplemental solvers and simulation engines and combine tools based on their respective strengths. Another standard method of integration is the Functional Mockup Interface (FMI) [34]. A supplemental specification of SysML defines a modelling standard and syntax subset that allows for optimal integration and data exchange for executable models [35]. Several studies document the use of executable SysML used for in-situ real-time requirements monitoring of live systems [36, 37]. One study used a machine-to-machine message broker to connect a globally distributed luggage handling system to a SysML model performing real-time analysis [38].

2.3.4 The Role of MBSE in Security by Design

A core idea of Security by Design is that if one invests time early on, it is possible to avoid many common security problems. Fixing errors and mitigating vulnerabilities post-deployment can cost 100 times more than catching these errors in development [39]. Threat modelling can capture and disseminate expert security knowledge to the broader engineering team and provide security analysts with a more accurate understanding of the system. However, despite the well-known benefits, threat modelling is often performed late in the engineering lifecycle and often not at all unless mandated by customers or regulatory authorities [40]. In the context of MBSE, the role of the security expert is not primarily to audit the final design; but to help build a digital threat model that can be applied as a constraint on the global design. This ensures that the expensive resource that security experts represent is utilised with a high return on investment. A machine-readable threat model can automatically validate security requirements as the design process progresses. Also, being continuously alerted to violations of security constraints throughout development can help elevate security competency and awareness within the engineering team.

2.3.5 SysML Applied to Security and Reliability

As systems become more complex and interconnected, it is increasingly challenging to construct models that properly reflect reality in terms of attacks. One approach is to recognise that a networked system is under constant threat from an infinite number of attackers. Consequently, the possible attacks, known and unknown, may be assumed to have a Poissonian probabilistic distribution in the same way as a random failure [41]. Malicious actions could ideally be modelled as a probability distribution over the possible attack actions. This approach assumes that while it is helpful to be aware of known vulnerabilities and common attack vectors, it is potentially more fruitful to

focus on possible fault states. Given that a significant proportion of vulnerabilities will always be unknown, these could be modelled as any other bug that may lead to system failure [42].

Systems Theoretic Process Analysis (STPA) models unsafe states using control systems theory based on control actions and feedback loops, based on methods from industrial process control. STPA has been extended to include information security [43], and the methodology has been implemented as concurrent safety, reliability and security analysis using a single SysML model [44–46]. Because SysML syntax is not necessarily familiar to domain experts, much research has focused on transforming models so that specialists tools in safety and security modelling can be used in conjunction with the shared system model [47–49]. Using a single integrated model with custom views allows information captured using informal methods to feed more detailed analysis for high priority issues. The case study in section 3 uses SysML to play the game Elevation of Privilege (EoP), a casual card game based on Microsoft’s STRIDE threat model [50, 51]. Each card in EoP represents a specific threat from STRIDE, and in the case study, it is modelled as a SysML requirement. The model is used to visualise the system, and as each player plays a card, she must explain to the group what the threat on the card entails in the current context. This continues until all cards in the deck have been played. The game is intended to facilitate group discussions, and cards deemed relevant are linked in the model to specific components, interfaces or functions. Potential mitigations, notes, and requirements are added to the model as they come up during the game.

3 Case Study

3.1 6LoWPAN/CoAP-based Urban Flood Prevention

The system of interest is a siphonic drainage system intended to mitigate the insufficient capacity of urban waterways in removing spill water during heavy rainfall. Such flooding is a critical public health concern because contaminated spill water and sewage may overflow into drinking water or pollute beaches and green areas. The system increases throughput by removing air from drainage pipes, thus creating a suction effect that siphons away water at several times the rate of a non-pressurised system. Because the system works by increasing the capacity of the existing pipes, there is no need for wholesale replacement of the water transport system, a minimal need for digging, and significantly less disruption to commercial activities, making

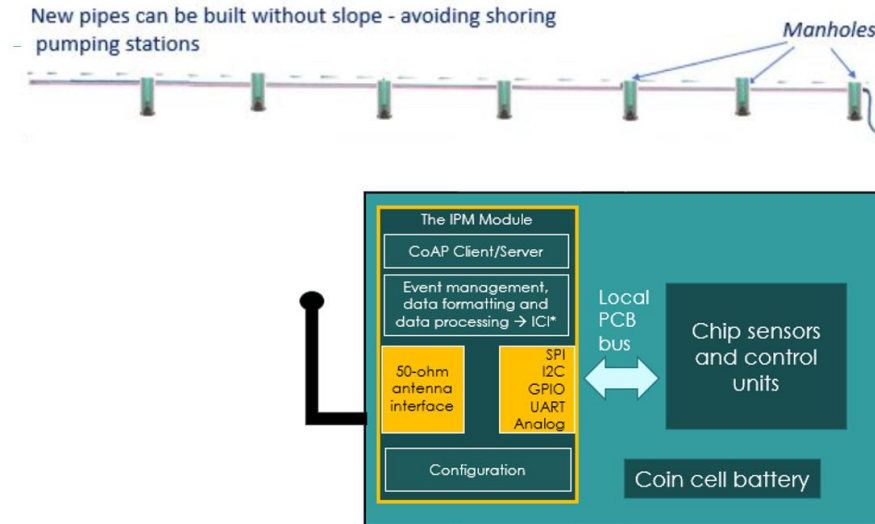


Figure 5 Siphonic Drain System with wireless communication (Aiwell Water/Radiocrafts)

the solution very cost-effective. Siphonic drainage is a mature technology standard in roof wells, but needing to keep the pipes under pressure has made it impractical for street drainage. However, using low-frequency long-range wireless networks enables real-time monitoring that could make maintaining such systems more feasible. The proposed solution uses a 6LoWPAN 868MHz IPv6 mesh network with CoAP as the application-level protocol, purchased as a module-on-chip with an integrated network stack mounted on the developing company's custom-designed PCB.

3.2 Playing Elevation of Privilege

The company developing the drainage system designs valves and end-user applications in-house but outsources PCB design and firmware development. Makers of the module-on-chip act as consultants, and other experts and stakeholder representatives are brought in as required. This ad-hoc network communicates informally to arrive at a common understanding of the system requirements.

The game Elevation of Privilege starts with constructing a visual model of the system, and figure 6 shows the beginnings of a high-level model using an Internal Block Diagram. The primary purpose of this activity is to identify system flows, interfaces, and interaction points. Players may use predefined

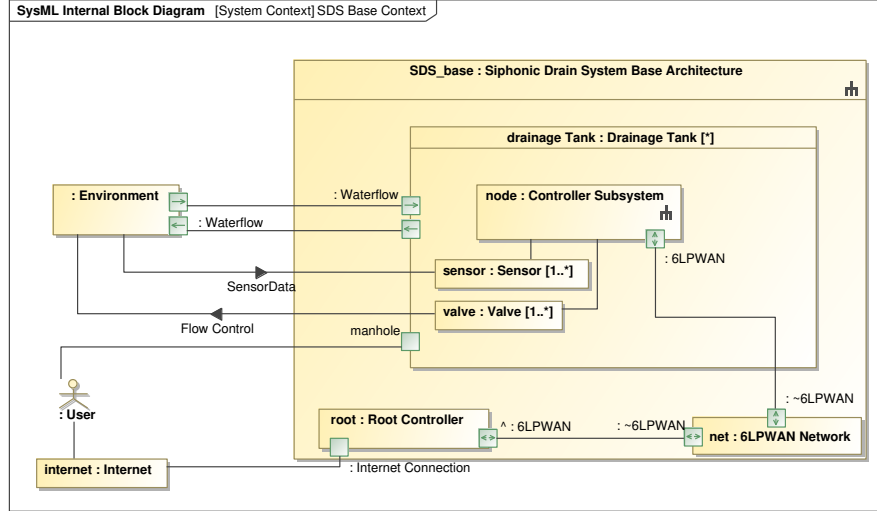


Figure 6 A basic context model serves as the starting point for playing Elevation of Privilege.

parts or define new ones during the game. As the game progresses, users may navigate the model to reveal different levels of abstraction. Figure 7 shows the same diagram as in figure 6, but with a greater level of detail. The model highlights the CoAP interface due to known security vulnerabilities previously identified during the modelling exercise. Navigating to these elements will give details on the vulnerability, including links to further details elsewhere. Because an attacker needs access to the CoAP network to exploit the vulnerability, and the 6LoWPAN network is encrypted on the MAC level, a requirement is added in Figure 8 to maintain the CoAP traffic within that network and not expose it on the web. Notes are added directly to the diagram as prompts for potential solutions. The actual solution will depend on trade-off studies based on cost and reliability metrics in the parametric model.

3.3 Parametric Model of Reliability and Availability Metrics

3.3.1 Mathematical Model

We assume a single repairable unit representing a single node in the network for our example reliability and availability model. For simplicity of illustration, we use a non-state-space model for the single nodes. We also assume a constant failure rate, which is an appropriate approximation for single unit models [54]. However, on the network level, the model would

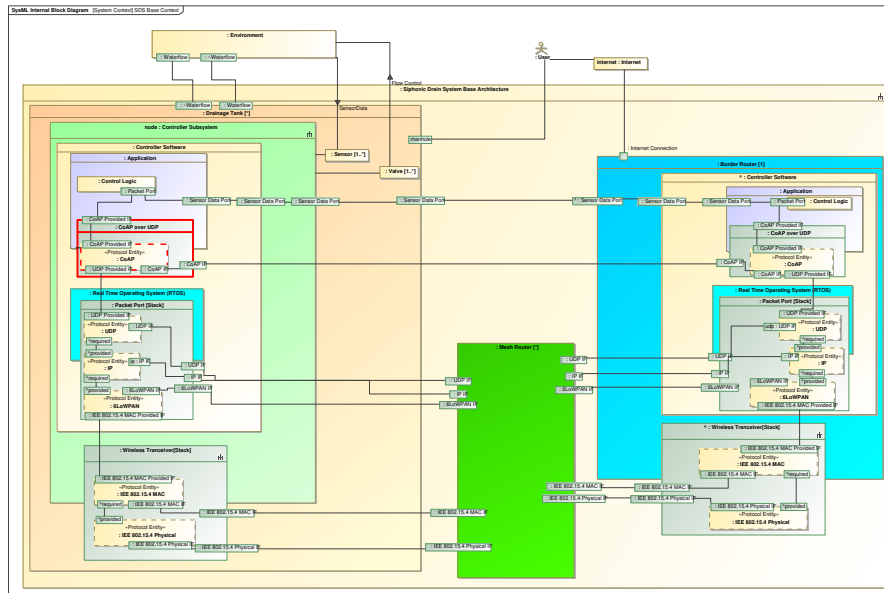


Figure 7 Zooming in on the context model to reveal details. Interfaces are modeled as layers [52]. The CoAP interface is highlighted due to a previously captured vulnerability.

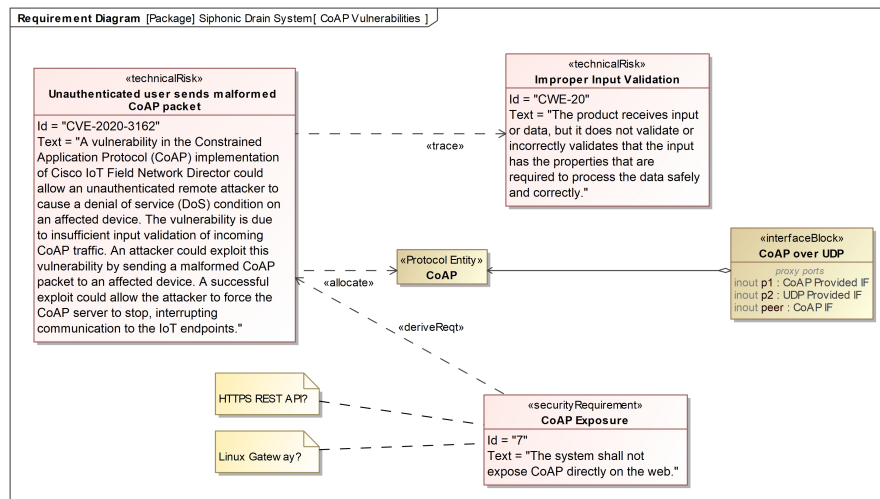
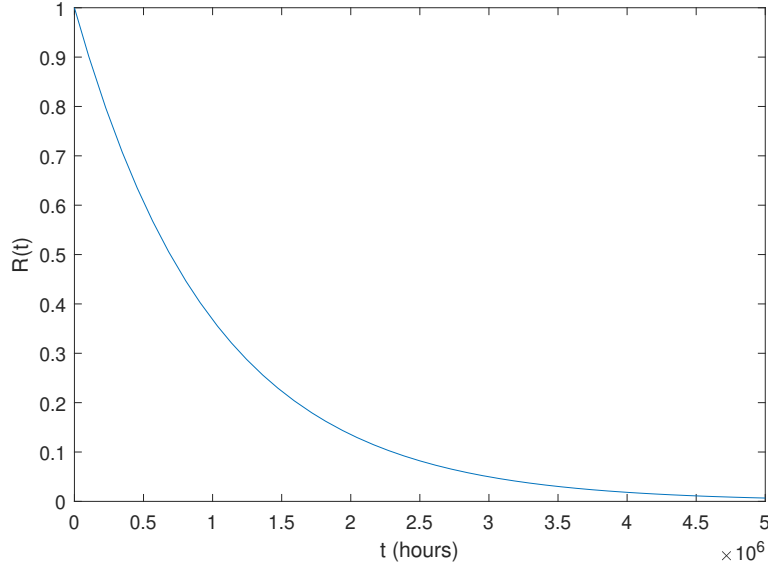


Figure 8 Previously modelled CoAP vulnerability [53].

Figure 9 Survival function of unit with constant failure rate $\lambda = 10^{-6}$

have to account for local state space, and transmission errors would depend on distance, medium, and environmental noise.

The failure rate $h(t)$ is the probability that given a system is up at time t , it will be down at $t + \Delta t$ with Δt being the shortest time step accounted for by the model. The unit's survival probability function is given in equation 1.

$$R(t) = e^{-\int_0^t h(u) du} \quad (1)$$

Because we assume a constant failure rate, we can simplify equation 1 by replacing the function $h(t)$ with constant λ . Also, because we assume each unit is perfectly repairable or replaceable, we introduce a time a at which the survival function in equation 2 returns to its initial value of 1. Figure 9 shows the survival function for a single repair cycle from equation 2 plotted with a failure rate of one failure per 10^6 hours.

$$R(t) = e^{-\lambda(t-a)} \quad (2)$$

Given equation 2, the expected mean time to failure (MTTF) would be:

$$MTTF = a + \int_0^{\infty} e^{-\lambda(t-a)} dt = a + \frac{1}{\lambda} \quad (3)$$

The mean time to repair (MTTR) is given by equation 4. The exponential case is given in equation 5. The repair function $G(t)$ gives the probability that if a repair cycle is started at $t = 0$, the unit will be up by time t . In the exponential case, we assume a constant repair rate μ and start time a .

$$MTTR = \int_0^{\infty} (1 - G(t)) dt \quad (4)$$

$$MTTR = a + \int_0^{\infty} (1 - e^{-\mu(t-a)}) dt = a + \frac{1}{\mu} \quad (5)$$

Given both failure rate and repair rate, it is possible to calculate system reliability and availability. Reliability denotes the probability of continuous operation of the system, whereas availability denotes the system being up at a given time. Equation 6 gives the instantaneous availability, i.e. the probability that the system is up at time t . In contrast, equation 7 gives the interval availability, i.e. the expected proportion of time the system will be up until time t . However, over time the two values converge on what is known as the steady-state availability of the system, formally defined in equation 8.

$$A(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t} \quad (6)$$

$$A_I(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{t(\lambda + \mu)^2} (1 - e^{-(\lambda + \mu)t}) \quad (7)$$

$$\lim_{t \rightarrow \infty} A_I(t) = \lim_{t \rightarrow \infty} A(t) = A = \frac{\mu}{\lambda + \mu} \quad (8)$$

Equation 9 gives the probability that the system will operate continuously without failure from time t to τ . For the current system of interest, this would be relevant to ascertain that the system remains continuously operational for the duration of a flooding event.

$$IR(t, \tau) = e^{-\lambda\tau} A(t) \quad (9)$$

The mean time between failures (MTBF) is given by the sum of MTTF and MTTR.

$$MTBF = MTTF + MTTR \quad (10)$$

The final part of the mathematical model defines expressions for the sensitivity of system availability. Equation 11 defines the sensitivity of the system to changes in time to failure. Equation 12 defines the sensitivity of the system to changes in time to repair.

$$S_F = \frac{\partial A}{\partial MTTF} = \frac{MTTR}{(MTBF)^2} \quad (11)$$

$$S_R = \frac{\partial A}{\partial MTTR} = \frac{MTTF}{(MTBF)^2} \quad (12)$$

S_F and S_R are essential when evaluating alternative measures for improving system availability. S_F is always positive, as system availability will always go up as the mean time to failure goes up. S_R is always negative, as increasing the mean time to repair will reduce system availability. If MTTF is much larger than MTTR, reducing time to repair will always have a much more significant impact on system availability than increasing time to failure. Reducing repair time could be more straightforward than mitigating the reason behind the failure. In such cases, it could be cost-effective to leave a vulnerability unmitigated if the expected frequency of a successful exploit is low and the cost of mitigation is high, compared to the ease and cost of restoring the system to a healthy state. S_F and S_R can provide stakeholders with a quantitative measure to evaluate this trade-off when comparing design alternatives and validating requirements.

3.3.2 Implementing the Mathematical Model Using SysML

Figure 10 shows the mathematical model from section 3.3.1 implemented using a SysML parametric diagram. Because the model is parametric, it is possible to vary the failure rate and repair rate and evaluate the system's reliability and availability. SysML is agnostic when it comes to the language used in defining constraints, and most tools allow for solvers to be set at the level of individual constraint blocks. This makes it possible to use different solvers for different constraints. The result is a parametric model that is composable and reusable. Figure 11 shows how a generic parametric reliability model is applied to a component through generalisation. It is then possible to automatically aggregate component level analysis results into a system level analysis. If this is combined with other parametric models, such as component cost models, it is possible to evaluate the total cost of a system with different

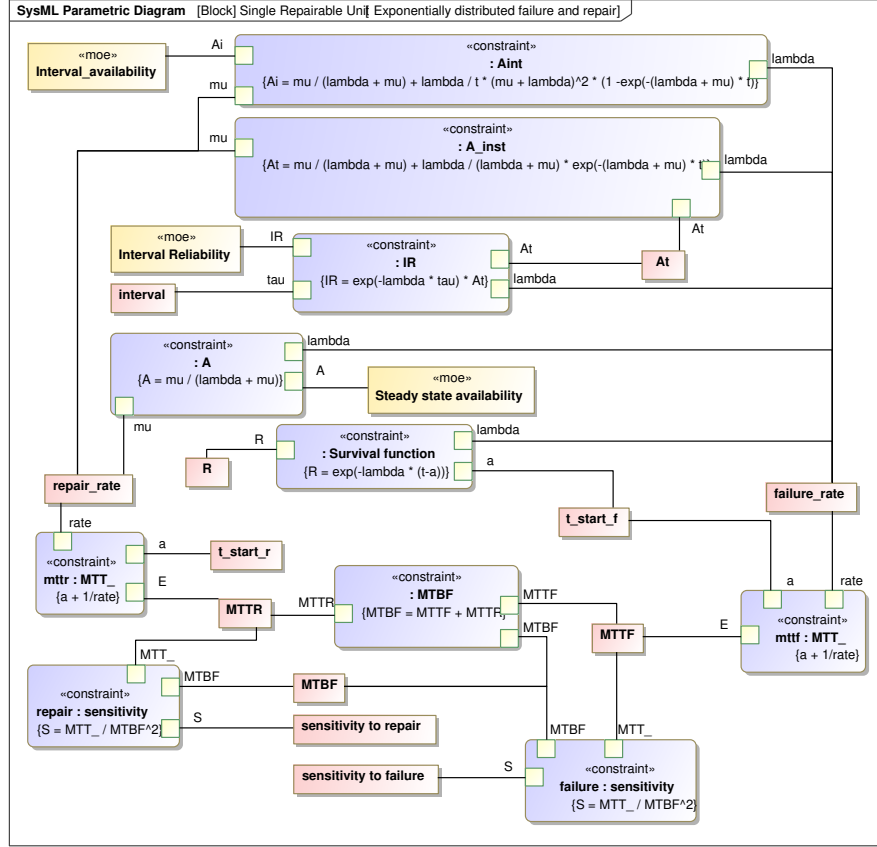


Figure 10 Parametric model of a single repairable unit

designs, taking into account a combination of unit cost, cost of repair, cost of downtime, and cost of redundancy. This can provide a quantitative input to stakeholders evaluating alternative mitigations to any identified vulnerabilities in the system.

4 Analysis

The vulnerability in the CoAP interface identified as part of the threat modelling exercise in chapter 3.2 could be mitigated in different ways. One alternative proposed by players during Elevation of Privilege was to add a gateway unit to avoid exposing CoAP traffic to the web. However, a paramet-

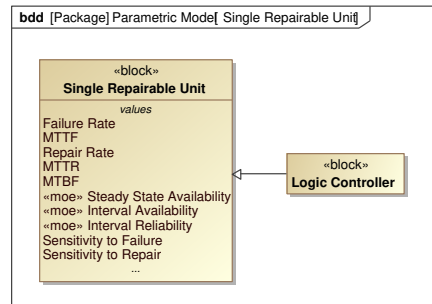


Figure 11 Applying Parametric Model through Generalisation

ric analysis as illustrated in chapter 3.3.2 showed this option to have a high cost both for hardware and the development of an API. Whitelisting nodes is another alternative, but this option also had a high overhead. Looking at the details of the vulnerability (CVE) and weakness (CWE) linked to in the model as shown in figure 8, it became apparent that these were primarily a concern for mobile networks in China, and a microcontroller as used in the system under analysis could simply reset to return to a safe state with no loss of sensitive data or resources. A more cost-effective solution in the analysed context would therefore be a simple hardware watchdog resetting the system in the case of traffic congestion.

5 Conclusions and Future Work

This paper examined how Model-Based Systems Engineering using the Systems Modelling Language can support the culture of informal cooperation amongst the small specialist companies acting as subcontractors in public infrastructure projects and aid in efficient validation of requirements. A case study demonstrated a multi-disciplinary combined analysis of reliability and security. As part of an informal session with multiple stakeholders, a card game, Elevation of Privilege based on STRIDE, was used to prompt discussion and elicit information that could feed into the digital model for further analysis. Whereas the informal discussions resulted in multiple suggestions for mitigating vulnerabilities, a formal parametric model constructed by others in a separate session provided stakeholders with quantitative metrics to evaluate the alternatives.

Future work will expand on Elevation of Privilege by creating similar decks of cards to cover reliability hazards, stakeholder modelling, and us-

ability and accessibility design. While physical playing cards were viewed positively by stakeholders, future work will also examine digital card games implemented as part of the modelling environment. Previous work has shown small companies to be resistant to moving away from informal means of collaboration. However, early results indicate that model-based automation can support standardisation and quality improvement even in an informal setting. In future studies, we plan to document how this approach affects compliance and resistance to change in small companies in more detail.

Whereas the case study described in this article was conducted in Norway, the capacity problems and technical debt in water infrastructure are global [1–3], as is the prevalence of small companies in the globalised supply chain [19] and the skills-gap and reporting issues as relates to system security [14]. This indicates that the analysis of the case study could be extended to other countries and other sectors.

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Biography



Roar E. Georgsen. received his B.Eng. in Computer Engineering and an M.Sc. in Systems Engineering from the University of South-Eastern Norway (USN). Currently, he is the CTO of Aiwell and an Industrial PhD Research Fellow with USN in Horten, Norway. His research interests include model-based systems engineering, digital transformation in small engineering teams, and integrated safety, security and reliability design.



Geir M. Kjøien. received his PhD from Aalborg University, on access security for mobile systems. He has also worked for many years in industry, including LM Ericsson Norway and Telenor R&D. During these years he worked extensively with mobile systems and with security and privacy. He has also worked with the Norwegian Defence Research Establishment and with Norwegian Communications Authority on various security and communications related projects. Currently, he is a professor with the University of South-Eastern Norway (USN).

Declarations

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Paper II

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Very Small Entities (VSEs): Outsourcing Risk to the Supply Chain Is Placing Systems Security Engineering on a Clay Foundation, but Playing Games May Help

Roar E. Georgsen, roar.e.georgsen@usn.no; and Geir M. Køien, geir.koien@usn.no

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■ ABSTRACT

This article addresses the inherent risk in a supply chain that comprises primarily Very Small Entities (VSE) with little to no security proficiency and limited resources and incentive to prioritize system security. In a globalized economy based on outsourcing and risk-sharing, most engineering activities occur in the smallest companies, even for large and complex projects. The Future of Systems Engineering initiative (FuSE) appropriately has agility at the core of its Systems Security Engineering (SSE) foundation concepts, and VSEs are by their very nature agile. However, the line between agility and chaos may be thin, and engineers at VSEs must often accept a level of restraint and rigidity beyond their comfort level to achieve functional agility. The primary challenge in VSEs is adding structure without the necessary resources to enforce compliance manually. We propose that VSE focus their initial efforts on FuSE SSE Foundation Concepts that play into their nature and strengths as dynamic human social activity systems. Improvements in security proficiency and stakeholder alignment do not necessarily require much formal structure, and digital tools combined with social strategies can add structure to a resource-constrained environment. Games can be excellent low-cost tools to provide structure while minimizing resistance, and Agile Model-Based Systems Engineering (AMBSE) using digital models can support automated enforcement. Here we use the card game *Elevation of Privilege* (EoP) as an example. Within the context of a SysML Threat Model integrated into a larger System Model, players naturally treat security requirements as traceable functional requirements. Automated model validation, re-usable components and patterns enforce a Zero-Trust architecture, a sufficiently formal trust model to provide evidence-based assurance, yet achievable for small companies with limited resources.

NO PLAN SURVIVES CONTACT WITH THE ENEMY

“Remember that remote controlled wireless valve de-icer we briefly discussed six months ago? Well, the customer expects delivery on Monday.”

Statements like this one are something we hear often. As researchers embedded in small supplier companies, we have first row seats to observe the difference between the theory and practice of systems engineering. In a typical scenario, the customer-facing side of a company, sales, or service, encounters an opportunity in the form of an unmet customer need. They then casually approach the engineer and

ask whether implementing the requested feature is feasible. If the engineer answers anything other than a hard “no,” the salesperson tells the customer “Yes.” When the contracted delivery date approaches, and the engineer discovers that the feature in question has yet to be implemented, the engineer is forced by management to very quickly build a minimum viable product (MVP). “Minimum” in this context means the absolutely smallest possible set of observable system properties necessary to meet a strictly letter-by-letter interpretation of the contract required to avoid any immediate penalties. Needless to say,

abstract and unobservable properties such as security, reliability and availability rarely make the cut.

MOST COMPANIES ARE SMALL. VERY SMALL.

Very Small Entities (VSEs) consist of organizations of five (5) to twenty-five (25) people and comprise the vast majority of all companies in the world. More than 92% of European enterprises are micro-enterprises, meaning they have fewer than nine (9) employees.

It is vital to identify and develop tools that support System Security Engineering (SSE) with this network of VSEs in mind.

As practitioners of a discipline with roots in some of the world's largest and most complex organizations, systems engineers must not lose sight of the fact that most engineering activities take place in a very different context from that which gave rise to systems engineering as a discipline. The design of secure systems needs to start from the premise that modern systems depend on an increasingly complex global network of small suppliers, most of which have no systems engineering capabilities, little to no security proficiency, and minimal incentive to prioritize system security.

VSEs typically have no experience working with standardized processes and lack experience in intentionally performing activities such as architecture, design, verification, test definition, and execution (Muñoz et al. 2021). They are aware of growing customer and legal demands to prioritize security, but they do not perceive this as adding value to their work. They experience the practices employed by larger companies as rigid and inappropriate to their context (Sanchez-Gordon, O'Connor and Colomo-Palacios 2015). VSEs are rarely required to document compliance with specific standards and prefer to produce only the minimum documentation required (Sanchez-Gordon, O'Connor and Colomo-Palacios 2015). When documented compliance is required, this does not necessarily correlate with actual engineering practices (Tran 2014). The result is that the actual risk embedded in the supply chain is hidden. As larger organizations increasingly systematize their security engineering efforts, their VSE suppliers struggle to keep pace. Consequently, measures such as risk-sharing partnerships (Figueiredo, Gutenberg and Sbragia 2007) intended to improve supply chain security can actually make the system less secure as responsibility and accountability move down the supply chain.

Any systems engineering tool or philosophy that cannot survive the frantic and sometimes chaotic world of the VSE runs the risk of becoming bureaucratic sugar coating concealing a house of cards built on a clay foundation.

CONTROLLING THE UN-CONTROLLABLE

The first instinct of an engineer when facing complexity is to measure, control and standardize. INCOSE contributed to the ISO/IEC 29110 family of standards in an attempt to address the needs of VSEs, and these standards provide a structural interface compatible with standards used by larger companies. It is possible to adapt the ISO/IEC 29110 standard to an agile workflow (Laporte and Miranda 2020; Muñoz, Mejia and Lagunas 2018; Muñoz, Mejia

and Laporte 2019). Model-based systems engineering (MBSE), in this context Threat Modelling, has been suggested as a suitable process for dealing with the complexity of the modern threat environment. However, trying to enumerate all interfaces between systems and subsystems, identifying and enumerating all dataflows, quickly grows beyond the capacity of a VSE engineering team with limited time and resources. Despite the well-known benefits, threat modelling is often performed late in the engineering lifecycle and often not at all unless mandated by customers or regulatory authorities (Shostack 2014a).

The goal should not be to control complexity. Such a goal is not simply challenging; it is impossible in a very literal sense. Once the delusion of control has been cast aside, it is possible to seek out more appropriate mental models, ones that recognize that engineers are human beings with flaws and limitations, but that also have useful intuitive skills that can be leveraged.

A VSE ROADMAP OF SSE FOUNDATION CONCEPTS

The philosopher Andy Clark uses the image of a crew on a sailboat navigating rough seas as a model for how the human mind manages to navigate an infinitely complex world with limited, finite resources. A human being, according to Clark, is essentially a pattern matching prediction machine continuously and actively seeking and making new predictions. Rather than simply reacting to input, the human mind efficiently selects the next input, resulting in fast and frugal problem-solving routines. Our actions structure the physical, social, and technological worlds around us. The ship's crew harbors no expectation of fully controlling the chaos of the sea but manages to navigate the waves through communication, experience, and tools that embed that experience externally. Nevertheless, our ship's crew should never cling to the illusion of control. On the contrary, striving for more control than is possible may directly lead to losing what little control one may have. This is a very systems-oriented perspective, which is why it is so perplexing that so many systems engineering methodologies turn out so rigid in practice. We want to create more capable prediction machines making efficient judgements with higher accuracy. In biological systems capable of learning, this is done by generating and testing problem-solving routines with exposure to novel experiences. This creates new behavior that is largely automatic and thus inexpensive to maintain. Fortunately for us, engineers happen to be biological systems capable of learning.

When it comes to VSEs, increasing se-

curity proficiency in the systems engineering team must, by necessity, be primarily learning-based. In VSEs, "The systems engineering team" will be synonymous with "The engineering team." Even trained systems engineers are expected to perform domain-specific engineering tasks. On the one hand, this can be beneficial because SSE is always embedded in the overall engineering workflow, but on the other hand, this makes in-depth security expertise rare. As pointed out in (Dove et al. 2021), "proficiency is unlikely to be found in systems engineers that haven't spent considerable career time developing breadth and depth in security." However, training the permanent engineering team in basic security practice is a low-hanging fruit that is easy to maintain and will provide immediate benefits. Other strategies will have to be employed to capture the expertise of external security specialists in a way that can inform and guide the work of the permanent team. Complementary to individual learning is the SSE foundation concept of Stakeholder Alignment. This alignment can be viewed as a process by which individual agents interact to synchronize their mental models to match other agents. The same way all biological agents adapt their pattern matching routines based on feedback from their physical environment, human beings adapt to feedback from their social interactions. There exists a mode of such socializing that has been shown to be particularly well suited to interpersonal alignment, and that is broadly practiced: It is called "playing".

FUN AND GAMES

"Serious Games" are those games that "have an explicit and carefully thought-out educational purpose and are not intended to be played primarily for amusement." (Abt 1987) However, this passes over the point that amusement is an essential feature of games, which makes games different from mere simulation. Agile methodologies and team structures lend themselves particularly well to games, demonstrated by techniques such as Planning Poker (Grenning 2002) for task estimation. Protection Poker (Williams, Gegick and Meneely 2009) is a similar game used for software security analysis. Another Serious Game used for security analysis is the VOME project game Privacy (Hyperion 2011).

Part of why games work is their ability to induce a state of undistracted concentration often referred to as "Flow" (Csikszentmihalyi 2008). Games are intrinsically rewarding activities with clear goals and immediate feedback, not unlike crewing a sailboat. In a transdisciplinary game with players from different backgrounds, the game's

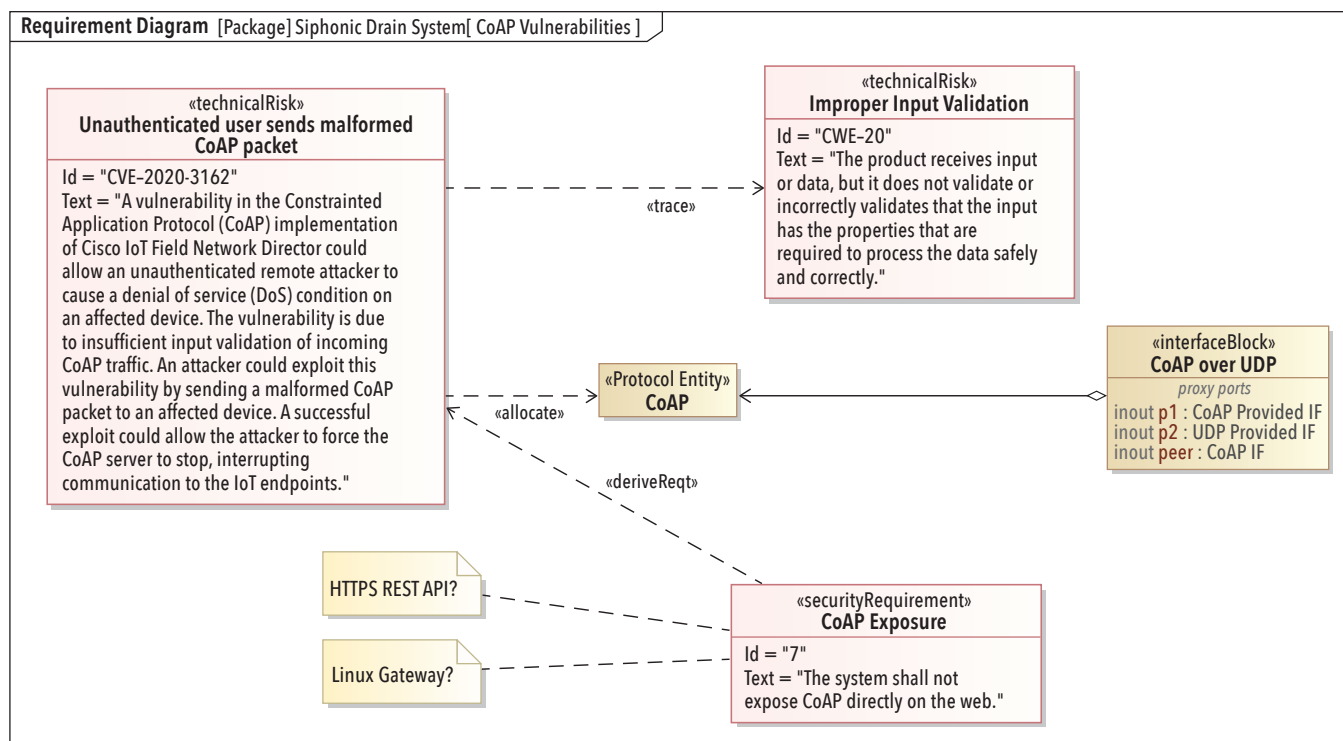


Figure 1. CoAP security requirements

structure can provide a set of “boundary objects” (Bowker and Star 1999). These objects exist in multiple domains but may play different roles in each area. As boundary objects align, their different roles clarify and one can communicate the roles as the game progresses. Stakeholder Alignment is an alignment of boundary objects, or put another way, a conversational construction of social interfaces, or a synchronization of mental models. In our work, we have attempted to apply these principles to threat modelling at VSEs.

THREAT MODELING

The goal of threat modelling is not to provide completeness or perfect consistency. Instead, it should focus on flexibility, efficiency, and being approachable for designers and implementers. Resistance to change as manifested in VSEs supports an approach where engineers perform threat modelling not as a separate exercise but as part of other engineering activities (Sanchez-Gordon, O'Connor and Colomo-Palacios 2015). If participants perceive the work as having immediate value to their own work, this can reduce resistance. Action Research (AR) is an interventionist approach to knowledge acquisition (Lewin 1951). AR proceeds from a praxis of participation, guided by practitioners' concerns for practicality, is inclusive of stakeholders' ways of knowing, and helps build capacity for ongoing change efforts (Bradbury

2015). Actively embracing uncertainty in this way can allow small teams to leverage natural human social and pattern matching skills. Still, we must balance the trust necessary to achieve this by a healthy level of systematic paranoia.

TRUST NO-ONE

An established concept is the “Zero Trust” (ZT) concept in the cybersecurity domain (NSA 2021), and recently there have been presidential executive orders issued in the US ordering the introduction of ZT principles and architectures in US federal infrastructure (*Executive Order on Improving the Nation's Cybersecurity* 2021). A central ZT tenet is that breaches are essentially unavoidable. Consequently, one must face up to a reality where unwarranted trust is dangerous.

One approach is to adopt a particularly paranoid form of ZT and recognize that an infinite number of actors constantly threatens a networked system. The word “actors” here is deliberate. While it is essential to focus on malicious actors such as attackers, it is easy to forget the threat caused by internal actors through incompetence, ignorance or simply a lack of time. When threat modelling, one should assume the model is incomplete and guaranteed to be incorrect in some way. This means accepting that most trust is unwarranted, that warranted trust has a cost, which in turn can inform what assets are worth paying that cost to protect.

In the context of threat modelling, ZT means paying particular attention to interfaces regarded as “trust boundaries.” In a digital model, this is where one would attach security requirements. Figure 1 shows a known vulnerability and weakness in the CoAP protocol modelled as an extension to SysML requirements, along with a derived functional security requirement to mitigate the known risk. Because this technical risk is allocated to a specific interface model component, we highlight any use of CoAP in diagrams, as shown in Figure 2 (next page).

Implementing ZT with SysML is not as robust as using a formal model that supports trust based on proof. However, it is sufficiently machine-readable to allow a high degree of automated model checking, all within the resources of a VSE. Ideally, we want to leverage this potential for automation and combine it with the social strengths of VSEs that come to play during games.

PLAYING AT BEING PARANOID.

STRIDE is a threat model and modelling methodology developed at Microsoft (Howard and LeBlanc 2003; Shostack 2008), and the game *Elevation of Privilege* (Shostack 2014a, 2014b) (EoP) is a card game based on STRIDE. Each card in EoP represents a specific threat from the STRIDE model, and in our study, we modelled this as a SysML requirement. A digital model visualized the system, and as each

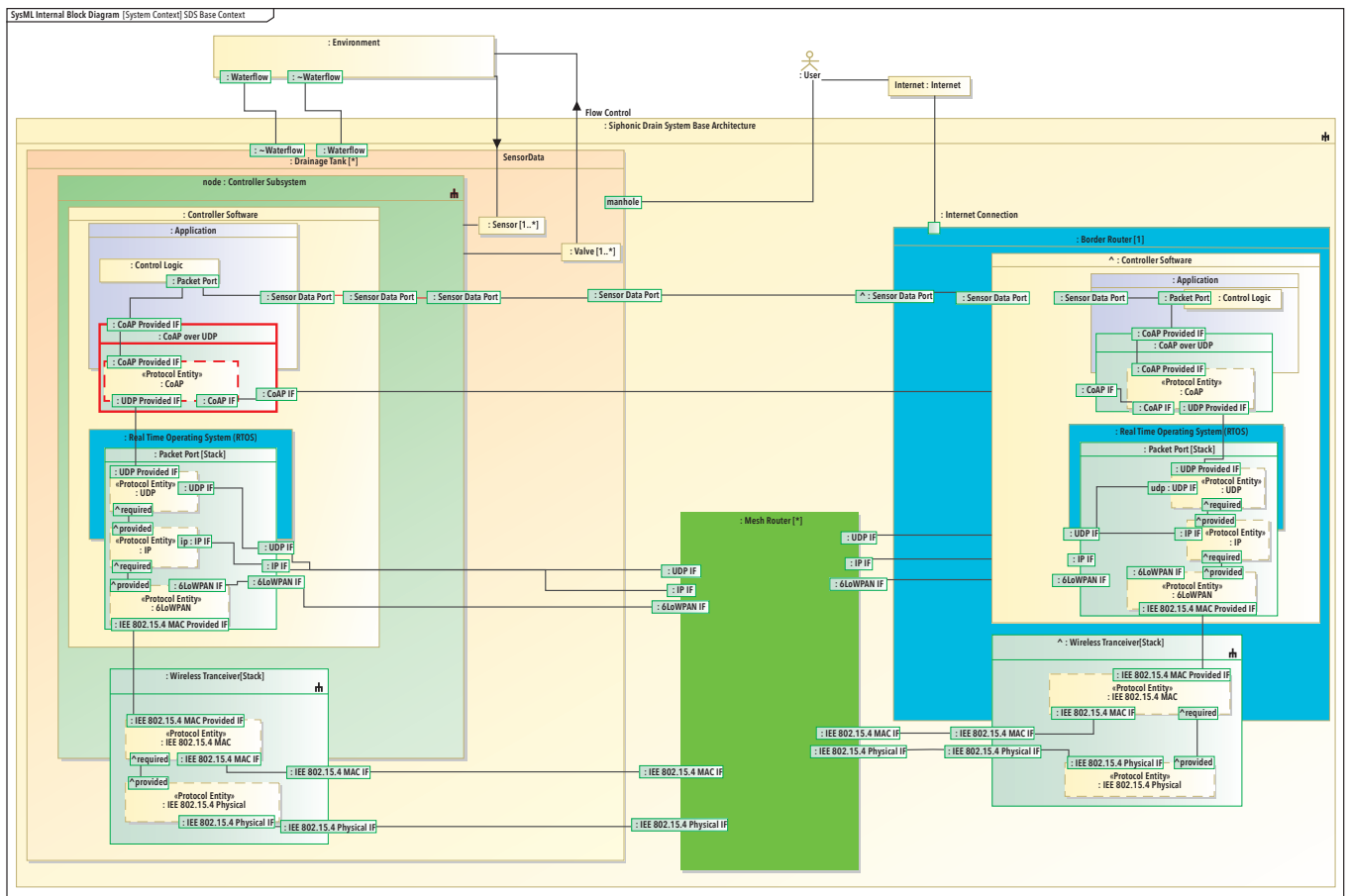


Figure 2. Automatic highlighting of interfaces with a known vulnerability

player played a card, they had to explain to the group how the threat on the card pertained to the System-of-Interest (SoI) in the current context. The game continued until all players played all cards in the deck. Each card traces to more than one requirement. The game's purpose was to facilitate group discussion, linking cards to specific components, interfaces, or functions in the model. We added potential mitigations, notes, and requirements to the model as they came up during the game, as seen in Figure 1. Figure 3 shows a simple example of a STRIDE threat attached to a requirement.

At this point, we can return to the unlikely inclusion of a security expert on the VSEs permanent engineering team. We usually bring in outside security experts to do the modelling. However, the internal team has the in-depth knowledge to assess the costs and tradeoffs involved in any potential mitigations. The security expert can provide highly specific knowledge, such as the CoAP example in Figure 1, and they can capture this in the model as shown. However, equally important is the role of the expert as educator and facilitator of the game.

Many engineers initially expressed skepticism when asked to participate in

threat modelling. They did not perceive this as part of their job and were hesitant to commit their time to it. However, as the exercise progressed, players reported new insights due to the modelling process that directly related to their primary responsibilities. This created engagement

and contributed to a broader range of perspectives integrated into the threat model. Because the game forced players to justify why a particular STRIDE threat was or was not relevant, they had to communicate their understanding of that part of the system in much greater detail than they

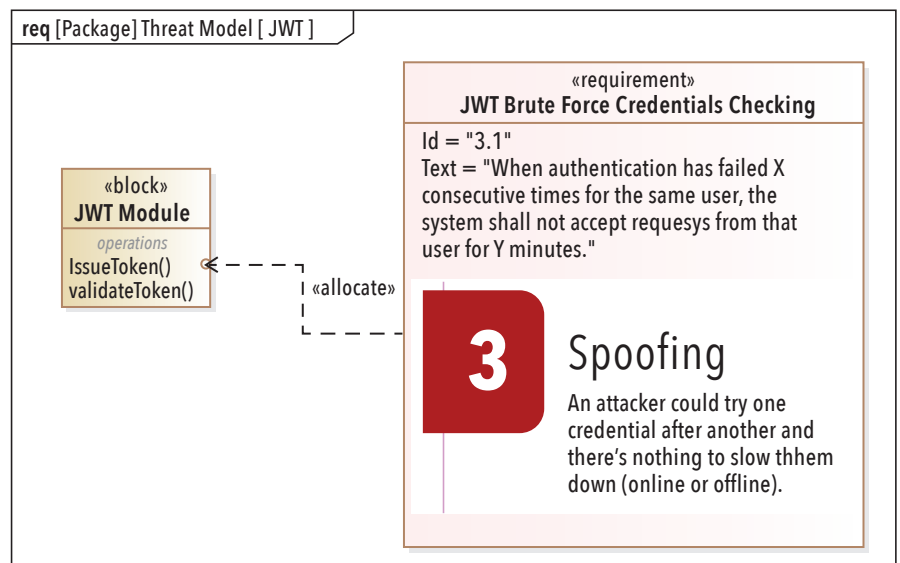


Figure 3. STRIDE SysML requirement

would typically do with other stakeholders. Players generally preferred a high level of consensus before modifying the model, so any inconsistencies the players negotiated, and players reported a better understanding of what other stakeholders needed from the system and why, especially when those stakeholders participated in the game. The game rules enforced a strict time limit on the modelling process. This time-boxing worked because the game started with the premise that completeness was not the goal.

VSEs AND THE FUTURE OF SYSTEMS SECURITY ENGINEERING

The original motivation behind this work with VSEs was to help small

historically non-ICT companies meet new requirements and build resilient systems to withstand the Denial-of-Service (DDoS) and ransomware attacks that were becoming a growing threat to their operations. However, it soon became apparent that VSEs collectively represent a substantial global risk factor. On the other hand, VSEs can provide an excellent opportunity to stress-test tools methodologies. Small companies are naturally agile and social organizations, and thus we should not assume they will exhibit the same dynamics as larger companies. Mature VSEs also differ from recent tech startups. Therefore, we should consider if different enabling factors might

be necessary to help them build secure systems and prevent them from becoming a threat to others. The FuSE SSE Foundation Concepts are meant to be implementable without independencies, but the particulars of VSEs support prioritizing some before others. VSEs are resistant to change, so starting with improvements in individual security proficiency and application of that knowledge in a natural social setting such as games will be a low hanging fruit. Also, digital tools have improved immensely in recent years, and leveraging the potential of automation that comes with those tools can make other Foundation Concepts easier to implement, such as Modeling Trust or Security as Functional Requirements. ■

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ABOUT THE AUTHORS

Roar Elias Georgsen received a B.Eng. in Computer Engineering and an M.Sc. in Systems Engineering from the University of South-Eastern Norway (USN). Currently, he is the head of product development at Aiwel and Aiwel Water and an Industrial PhD Research Fellow with USN in Horten, Norway. His research interests include model-based systems engineering, digital transformation in small engineering teams, and integrated safety, security, and reliability design.

Geir M. Koien received his PhD from Aalborg University, on access security for mobile systems. He has also worked for many years in industry, including LM Ericsson Norway and Telenor R and D. During these years he worked extensively with mobile systems and with security and privacy. He has also worked with the Norwegian Defence Research Establishment and with Norwegian Communications Authority on various security and communications related projects. Currently, he is a professor with the University of South-Eastern Norway (USN).

Paper III

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Serious Games with SysML: Gamifying Threat Modelling in a Small Business Setting

R.E. Georgsen^{a*}, G. Myrdahl Køien^a

^aUniversity of South-Eastern Norway, Raveien 215, 3184 Borre, Norway

Abstract

This paper describes using casual games to capture and disseminate expert security knowledge with a digital model at a small company. Most companies in the global supply chain are Very Small Entities (VSE), meaning they employ five (5) to twenty-five (25) people. These companies represent a risk because they have little to no security proficiency and limited resources and incentive to prioritize system security. Moreover, this is a hidden risk because the companies higher up in the supply chain rely on contractual mechanisms to enforce compliance with security requirements. Small companies rely on informal methods in their daily work and are resistant to change. Threat modelling in the context of Agile Model-Based Systems Engineering (AMBSE) can provide VSEs with a structure that allows them to retain their informal operational agility. When modelling using games, the security expert plays the dual role of modeler and educator, and the engineering team is gaining security proficiency, improving the quality of both their current and future work. Players reported improved understanding of other stakeholders' perspectives. Agility, proficiency, and stakeholder alignment are among the critical foundation concepts identified by the INCOSE led Future of Systems Engineering Initiative (FuSE) to improve system security.

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Keywords: System Security Engineering; Very Small Entities; Threat Modelling; Serious Games;

1. Introduction

Vision 2025 aims for cyber-security to become a "fundamental system attribute that systems engineers understand and incorporate into designs" ¹. The Future of Systems Engineering Initiative (FuSE) has established a roadmap of

* Corresponding author. Tel.: (+47) 411 63 196
E-mail address: roar.e.georgsen@usn.no

security foundation concepts to work towards that goal². Taking a systems approach to security engineering is becoming increasingly important as the vulnerability of a global supply chain becomes apparent, and traditional security standards have limited utility in this area³. However, as practitioners of a discipline with roots in some of the worlds largest and most complex organizations, system engineers must not lose sight of the fact that most engineering activities take place in a very different context from that which gave rise to Systems Engineering. The design of secure systems needs to start from the premise that modern systems depend on an increasingly complex global network of small suppliers, most of which have no Systems Engineering capabilities, little to no security proficiency, and minimal incentive to prioritize system security. FuSE identifies important enablers such as operational agility, team proficiency and stakeholder alignment. It is vital to identify and develop tools that support these enablers with this network of very small companies in mind. This paper explores games and models as potentially valuable tools for this purpose.

Section 2 provides a background on engineering in very small companies and threat modelling with games. Section 3 briefly discusses the choice of appropriate research methods when conducting Systems Engineering research in small companies. Section 4 describes a case study using a card game based on the STRIDE threat model to capture expert security knowledge in a SysML model in a way that remains traceable throughout the system lifecycle, and section 5 summarizes the key findings of the study.

2. Background

2.1. Very Small Entities (VSE) and Supply Chain Risk

Most companies in the global supply chain are Very Small Entities (VSE), meaning they employ five (5) to twenty-five (25) people⁴. More than 92% of European enterprises are micro-enterprises, having fewer than nine employees⁵. In Norway, approximately 30% of research & development (R&D) spending is done by VSEs, with 20-30% of VSE revenue coming from new product development, compared to less than 9% for large companies^{6,7}. Whereas VSEs are aware of growing customer and legal demands for systematic engineering, they do not perceive this as adding value to their work. VSEs are rarely required to document complete compliance with specific standards and thus prefer to produce only the minimum documentation required by contractual obligations⁸. The perceived lack of added value drives resistance to change, and documented compliance does not necessarily correlate with actual engineering practices⁹. The result is that the actual risk inherent in the supply chain is hidden. As larger organizations in the supply chain increasingly systematize their security engineering efforts, their suppliers struggle to keep pace. Measures such as risk-sharing partnerships¹⁰ intended to improve system security can, in this context, reduce the actual security of the system as responsibility and accountability move down the supply chain.

VSEs typically have no experience working with standardized processes and lack experience in intentionally performing activities such as architecture, design, verification, test definition, and execution¹¹. VSEs recognize the growing customer requirement for security and their own internal need for improvement¹¹, but they do not perceive the standards employed by larger companies as appropriate to their context⁸. VSEs prefer to follow informal and organically evolved methodologies partly because they perceive standardized methods as too general to fit their needs and because informality provides agility and opportunity to optimize workflows to fit their particular contexts⁸. Most standards are not free and generally not covered by engineering curricula or available at university libraries. Consequently, VSE engineers without experience from larger companies have little knowledge of applying standards in their work¹². Security experts are expensive, which is likely a key contributing factor to security engineering tasks often occurring late in the design process. Training the broader engineering team to identify vulnerabilities early in the design process can lower costs, but this comes at the expense of completeness and accuracy. Whereas security experts may bring domain expertise, implicit knowledge, and intuitive decision-making, other stakeholders may bring a broader level of system knowledge. VSEs tend to prioritize hiring engineers with expertise in their primary operational domain. In a small business setting with limited resources, the highest return of investment from security experts would be bridging and shrinking the skills gap in the engineering team. The ISO/IEC 29110 family of standards attempt to address the needs of VSEs, and the standard's Systems Engineering Entry and Basic profiles⁴ are available at no cost and are compatible with the parent standard ISO/IEC/IEEE 15288¹³. It is possible to adapt the ISO/IEC 29110 standard to an agile workflow¹⁴⁻¹⁷.

2.2. Threat Modelling

2.2.1. Methods and Goals

Threat modelling is a way to improve a system's security design and prevent or mitigate security threats. The main idea is that if one invests time early on, one may avoid many common security problems. Fixing problems late in the development cycle tends to be difficult and expensive. Post-deployment, fixing errors, and mitigating vulnerabilities can cost 100 times more than catching these errors in development¹⁸. Only 35% of projects are successful in terms of time and budget¹⁸. However, the success rate drops to 8% in high complexity projects in organizations with low Systems Engineering capability¹⁶. Individual expertise correlates with project success in lower complexity projects, but this advantage disappears with growing project complexity¹⁹. Threat modelling can capture and disseminate expert security knowledge to the broader engineering team and provide security analysts with a more accurate understanding of the system. The OWASP Foundation, in its "Web Security Testing Guide", has listed threat modelling as a test methodology²⁰. They perceive it to be part of the earlier stages of the development lifecycle. This view is also in line with the "Threat Modeling Manifest" declaration²¹. The goal of threat modelling is not to provide completeness or perfect consistency. Instead, it focuses on flexibility, efficiency, and being approachable for designers and implementers. The aim is to pick the low hanging fruits. Experienced teams can extend the threat model to whatever refinement level they choose, keeping in mind that the goal of a threat model is not the modelling activity but improvements in the end product.

Despite the well-known benefits, threat modelling is often performed late in the engineering lifecycle and often not at all unless mandated by customers or regulatory authorities²². Various modelling approaches must all make tradeoffs regarding the practitioners' level and type of expertise. Outside security experts usually do the modelling, but the internal team has the in-depth knowledge to assess the costs and tradeoffs involved in any potential mitigations. Therefore, the experts should instead strive to be facilitators, and modelling methodologies should be selected based on their accessibility to stakeholders that are not security practitioners.

2.2.2. The STRIDE Threat Model and Methodology

STRIDE is a threat model and modelling methodology developed at Microsoft^{23,24}. Threat modelling is a core element of Microsoft's Security Development Lifecycle²⁵ (SDL), and the company provides a free STRIDE-based Threat Modeling Tool²⁶. Common to threat modelling in general, STRIDE includes asking:

1. What is being built?
2. What can go wrong?
3. How will this be mitigated?
4. Should the model be revised?

There is an additional 0th step, which includes identifying the primary assets in the system. An asset is anything that has value to the organization and, therefore, requires protection²⁷. One should also consider any potential attack points and steps that a would-be intruder would take to reach the identified assets.

For question 1, one needs to pay attention to interfaces between systems or subsystems, identifying and enumerating all dataflows. The same goes for components and mission-critical functions of the system. One may also define so-called trust boundaries. Generally speaking, everything within a trust boundary should ideally trust everything else within that boundary. A trust boundary should be protectable in the sense that one should be able to verify everything that passes a trust boundary interface. Additionally, a trust boundary should not encompass many components.

Question 2 concerns the STRIDE acronym itself. Table 1 captures the STRIDE threat groups:

Table 1. STRIDE Threat Model.

STRIDE Threat		Desired property
Spoofing (S)	Impersonating someone or something else.	Authenticity
Tampering (T)	Modification, injection or deletion of data.	Integrity

Repudiation (R)	Claiming not to have performed an action.	Non-repudiability
Information disclosure (I)	Exposing information to someone not authorized to see it.	Confidentiality
Denial-of-Service (D)	Denying or degrading service to legitimate users.	Availability
Elevation-of-Privilege (E)	Gaining access to system capabilities without proper authorization.	Authorization

Step 2 entails carrying out a STRIDE analysis of all the enumerated interfaces, components and functions of the system. There will be many cases where there is no apparent threat, and there will be many overlapping cases. It is acceptable to declare a threat as "Not applicable", but one should always log the decision (including the evaluator's name and the date).

Step 3 is about all the threats found in step 2. The list may be extensive, and there will usually be overlapping and patterns to the threats. One should attempt to identify these patterns and provide solutions to whole classes of threats. Often, threats cannot be prevented or mitigated within a project's scope. These threats will have to be marked as "postponed" or similar. Another valid response is "needs investigation". For threats mitigated or prevented, one should log how.

Step 4 involves reflecting on the threat model itself. Was the threat model sufficiently detailed? Are the mitigations effective and complete, or are too many threats marked "postponed" or "needs investigation"? The main point of a threat model is that it should lead to action and improvements.

2.2.3. The Zero Trust Concept

The "Zero Trust" (ZT) concept is well established in the cybersecurity domain²⁸, and recently there have been presidential executive orders issued in the US ordering the (gradual) introduction of ZT principles and architectures in federal infrastructure²⁹. The term "Zero Trust" is a misnomer in some sense, as it is really about "zero unwarranted trust". The background for ZT is the common practice of treating all users, clients and servers within a communication network as trusted. Such trust may have worked well for small network segments, but it is untenable for more expansive networks. One simply cannot by default trust all actors within a network. Systems invariably get breached. One may be in denial about this, but a central ZT tenet is that breaches are essentially unavoidable. Breaches come in all shapes and sizes, and most will initially be harmless. Early detection and response will often resolve the problem, while inaction and unwarranted trust will escalate the issue. Consequently, one must face up to a reality where unwarranted trust is dangerous. Therefore, the ZT concept and the associated Zero Trust Architecture^{30,31} (ZTA) concept focus on providing credible Identity and Access Management (IAM) and solid detection capabilities. Central to IAM is privilege handling and avoidance of privilege escalation.

2.2.4. Threat Modelling with SysML

The STRIDE methodology is informal, and syntax agnostic by design, often performed simply with markers on a whiteboard. It is easily assumed that such a lightweight approach is the most suitable in a small business setting. However, the primary use case for this kind of simplified process is large companies such as Microsoft, where it can serve as a way to free developers from strict procedures. On the other hand, small companies may need more structure, not less. They may also benefit from modelling syntax that is sufficiently formal to support automation. Approaches such as System Theoretical Process Analysis (STPA) have been extended to the security domain to automate analysis³². However, with the cybersecurity field rooted in software with its historical connection to UML, SysML is a more suitable candidate for security engineering in cyber-physical systems.

The Systems Modeling Language³³ (SysML) is a general-purpose graphical notation for modelling complex cyber-physical and socio-technological systems. SysML provides graphical representations for modelling requirements, behavior, structure, and parametrics. The syntax can be extended through profiles, and several security profiles have been released^{34–36}. A popular research category has been to focus on transforming SysML models so that specialists tools for security analysis can supplement the source model^{37–39}. However, using a single integrated model with customized views allows information captured using informal methods to feed more detailed analysis for high priority issues without additional tools. Also, adhering to a shared general-purpose syntax and avoiding overly specific profile extensions is more appropriate for small businesses with limited resources.

Figure 1 shows a non-normative extension of the basic SysML requirements syntax similar across most modelling tools. The diagram also adds three new requirement types to the model. Typically, extending the Class metaclass is used to create subtypes of requirements that correspond to the traditional sections of a requirements document. This legacy view of requirements is practical when synchronizing the model with externally stored text-based requirements. Figure 2 shows an example of importing predefined textual security requirements into the model. A security requirement is derived from a corresponding requirement in the Application Security Verification Standard (ASVS) and traced to a corresponding weakness with a CWS number. A use case that may include multiple model components refines the definition of the requirement, and the "verify" relationship links the requirement to a test case. The "satisfy" relationship creates a link in the model between the requirement and the property or function that satisfies it.

The taxonomy in figure 1 is useful when textual requirements exist, but any named element can become a requirement through extension. Figure 3 shows the definition of three types of requirements based on different behavioral model elements. This feature enables SysML requirements to be more specific and less ambiguous than textual requirements.

The "allocate" relationship can link use cases and functions to a component, linking model elements directly to requirements. Suppose a known vulnerability is modelled as a requirement and allocated to a component. In that case, that vulnerability can automatically be added as an implicit requirement whenever a design uses an instance of that component. Domain experts can model within their respective domains, and the modelling software will highlight previously identified vulnerabilities and constrain the design space for all users of the model.

As systems become more complex and interconnected, it is increasingly challenging to construct models that properly reflect reality in terms of attacks. One approach is to adopt an extreme form of ZT and recognize that an infinite number of attackers constantly threatens a networked system. Consequently, the possible attacks, known and unknown, may be assumed to have a Poissonian probabilistic distribution in the same way as a random failure⁴⁰. This assumption allows the modelling of malicious actions as probability distributions over the possible attack actions, represented as a stochastic constraint in a parametric diagram, a constraint property on a block, or as probabilistic flows in activity diagrams similar to Stochastic Activity Networks^{41,42} (SAN). The stochastic approach assumes that while it is helpful to be aware of known vulnerabilities and common attack vectors, it is potentially more fruitful to focus on possible fault states. Given that a significant proportion of vulnerabilities will always be unknown, these should be modelled like any other random event that may lead to system failure⁴¹.

In order to obtain the most benefit from MBSE, it is necessary to utilize SysML in a machine-readable manner. SysML relies on the OMG XML Metadata Interchange (XMI) to exchange modelling data. Most modelling environments integrate well with external tools such as MS Excel, MATLAB, OpenModelica, Maple, or Mathematica. They can leverage these as supplemental solvers and simulation engines and combine tools based on their respective strengths. Another standard integration method is the Functional Mockup Interface (FMI)⁴³. A supplemental specification of SysML defines a modelling standard and syntax subset that allows for optimal integration and data exchange for executable models⁴⁴. Several studies document executable SysML used for in-situ real-time monitoring of live systems^{45,46}. One study used a machine-to-machine message broker to connect a globally distributed luggage handling system to a SysML model performing real-time analysis⁴⁷.

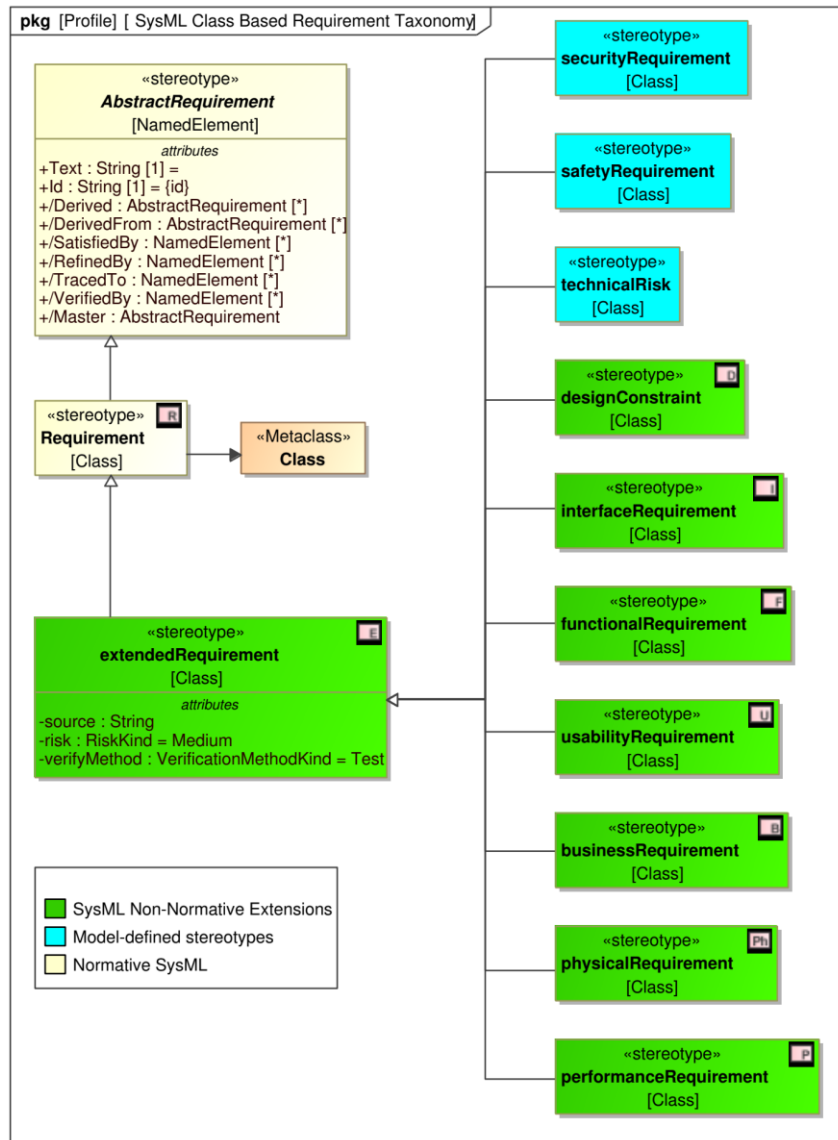


Fig. 1 SysML Class-based requirements taxonomy

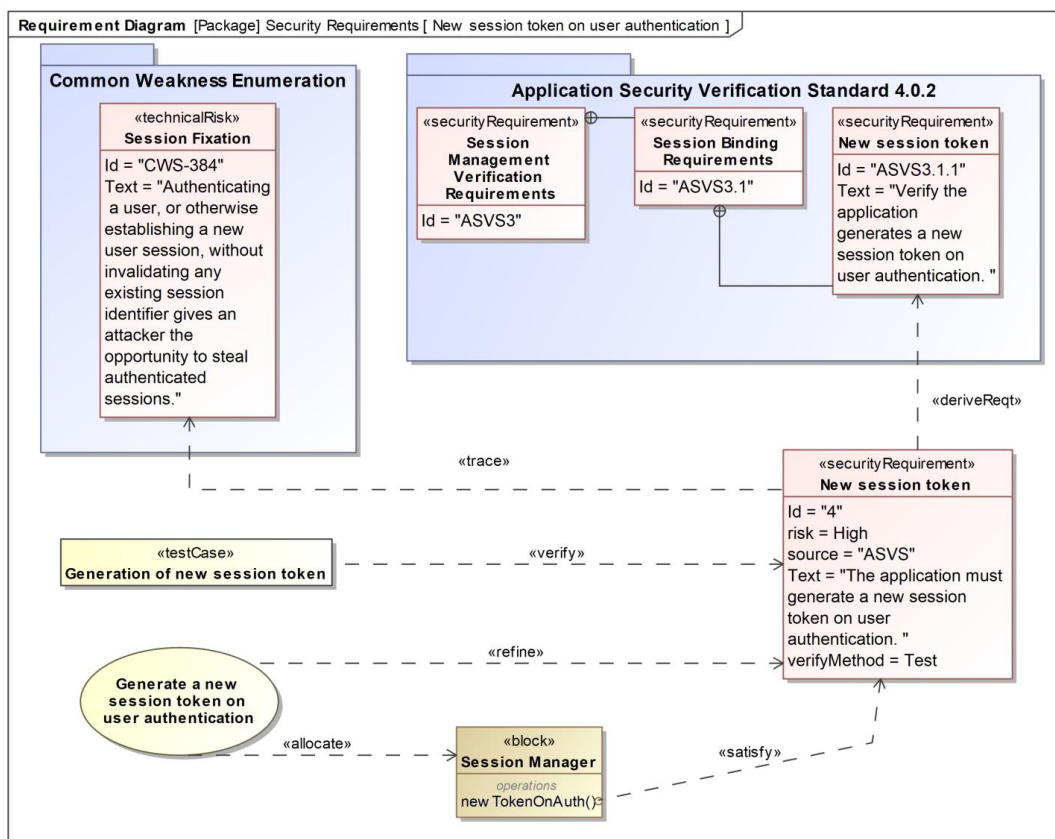


Fig. 2 Using relationships to link model elements.

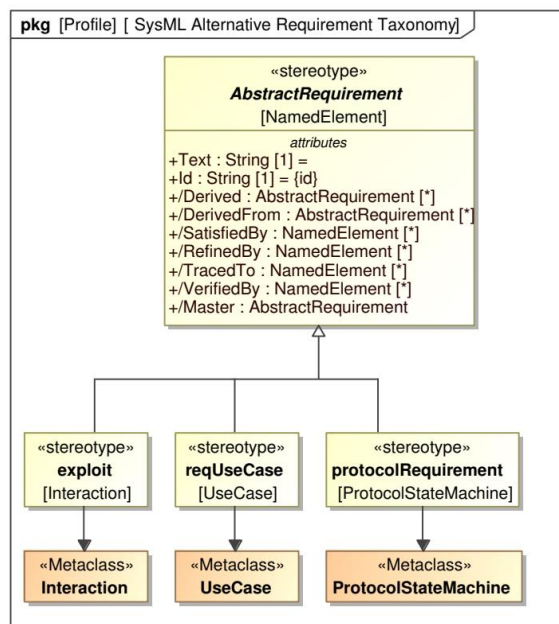


Fig. 3 Non-Class SysML requirements

2.3. Serious Games

Games involve "the voluntary effort to overcome unnecessary obstacles"²², and "Serious Games" are those games that "have an explicit and carefully thought-out educational purpose and are not intended to be played primarily for amusement."⁴⁸ However, amusement is an essential feature of games, making them different from mere simulation, and gamification is an increasingly popular method to facilitate motivation. Agile methodologies and team structures lend themselves well to games, as demonstrated by the popularity of techniques such as Planning Poker⁴⁹ for task estimation. Protection Poker⁵⁰ is a similar game used for software security analysis and the original inspiration for the game Elevation of Privilege²² used in this study. Another Serious Game for security analysis is the VOME project game Privacy⁵¹.

Part of why games work is their ability to induce a state of undistracted concentration often referred to as "Flow"⁵². Games are intrinsically rewarding activities with clear goals and immediate feedback, and players become easily absorbed. In a transdisciplinary game with players from different backgrounds, the game's structure can provide a set of boundary objects⁵³. These objects exist in multiple domains but may play different roles in each area. When the game involves modelling activities, the boundary objects become apparent, and their different roles are clarified and communicated as the model progresses.

2.4. Elevation of Privilege Game (EoP)

The OWASP Foundation regularly publishes a "Top 10" list of the most critical software security threats. The top threat for the 2021 edition is "Broken Access Control"⁵⁴. The "Elevation of Privilege (EoP)" threat comprises a large part of this threat category. EoP involves the intruder gaining access to system capabilities without proper authorization. The intruder may be a dishonest insider or may have gained illicit access to a user account or otherwise access to system function. So-called Advanced Persistent Threat (APT) intruders will often gain initial access to a peripheral part of a system, and then by exploiting missing or flawed access control system pivot to gain access to other parts of the system. The APT intruders are resourceful and patient and will typically be nation-state agents or high-level ransomware adversaries. Infrastructures are a typical target for the APT actors.

The Principle of Least Privilege (PoLP) is fundamental for authorization and access control⁵⁵. The gist of it is that one cannot abuse rights and permissions that one does not have, and this also extends to cases where an intruder has compromised the system. However, it is tough to design good security policies for access control in real life. If one restricts permissions too much, it will be difficult for the honest party to carry out their job. One also needs to allow for exceptions and emergencies. There is a tendency in evolved systems to have "privilege creep" problems. The problem is so common that there even are tools for counteracting it⁵⁶. PoLP is not an end goal, but it is a process goal. Security policies and access control rules must therefore be subject to regular revision.

Access control and writing good security policies is a laborious exercise. Nevertheless, it is vital, and unwarranted or escalated access is a serious security concern. Many large-scale ransomware attacks are made possible by elevating existing rights and permissions. Playing EoP-games may improve security posture.

The case study described below used SysML to play the card game Elevation of Privilege^{22,57} (EoP), a game based on STRIDE. Each card in EoP represents a specific threat from STRIDE, and in the case study, this is modelled as a SysML requirement. The model is used to visualize the system, and as each player plays a card, she must explain to the group what the threat on the card entails in the current context. The game continues until all cards in the deck have been played. Each card may be attached to more than one requirement. The game's purpose is to facilitate group discussions, linking cards to specific components, interfaces or functions in the model. Potential mitigations, notes, and requirements are added to the model as they come up during the game. Figure 4 shows a simple example of a STRIDE threat attached to a requirement. The reason for attaching an image of the card to the requirement is both educational and a prompt for further modelling later on, such as modelling specific exploit scenarios and misuse cases.

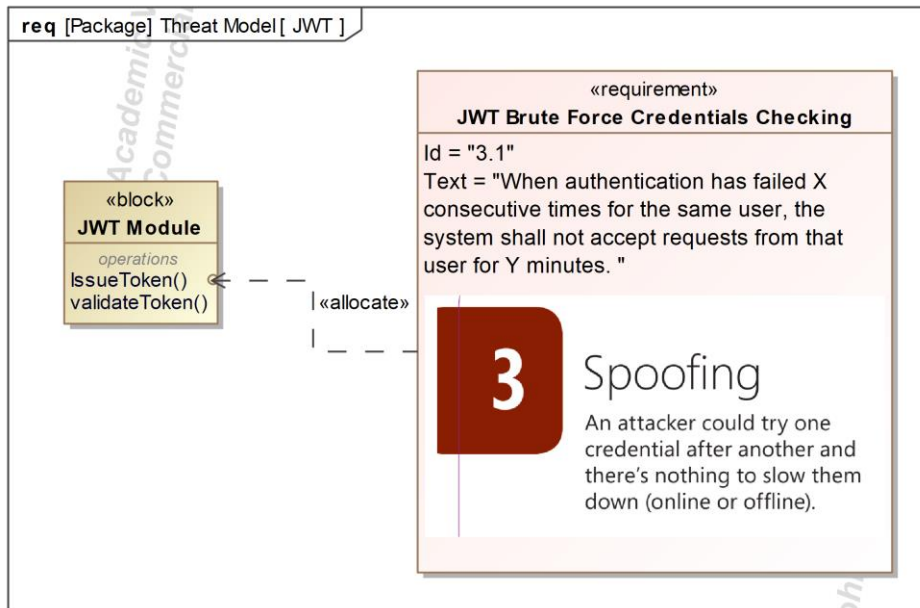


Fig. 4 STRIDE SysML Requirement

3. Research Methods

The nature of resistance to change as manifested in VSEs supports an approach where research is performed not as a separate endeavor but as part of regular engineering activities⁸. If participants perceive the research as having immediate value to the task at hand, this can reduce resistance. Action Research (AR) is an interventionist approach to knowledge acquisition⁵⁸. The researcher is actively involved, and knowledge obtained can be immediately applied. AR proceeds from a praxis of participation, is guided by practitioners' concerns for practicality, is inclusive of stakeholders' ways of knowing, and helps build capacity for ongoing change efforts⁵⁹.

The study used Grounded Theory Method (GTM) to analyze data outside the AR setting. The emphasis of GTM⁶⁰ is the specific process for developing theories that are grounded in data: which fit the data, which work in practice, and are relevant to the researched situation⁶¹. Although the stricter modified version of GTM developed by Anselm Strauss and Juliet Corbin is better known, Glaser's original approach is more suited to AR⁶¹. Rather than descriptive accuracy, the Glaserian method emphasizes the utility of the final product, similar to AR. It follows the dictum "all is data," meaning any artefact encountered or created as part of the research effort can be considered valid and relevant information. In the context of the threat modelling process, the analysis proceeds in parallel and attempts to build a model of that activity with a gradual increase in model fidelity.

4. Case Description

The case described in this study is a small Norwegian producer of outdoor automation products. Established in the late 1980s, the company has a history of innovative and successful product development. However, recently the company has had to decline projects due to their lack of documented procedures for reliability, availability, maintainability and security (RAMS). The new situation has forced the company to re-evaluate its engineering practices. The company technical staff includes a single systems engineer with a computer engineering background and two service technicians. In previous development projects, external specialists were contracted when necessary, but the company now wants to improve its ability to build and retain its internal body of knowledge.

The system of interest in the modelling exercise was a siphonic drainage system intended to increase the capacity of urban waterways during heavy rainfall. Siphonic drainage is a mature technology commonly used in roof wells, but keeping the system pressurized at scale has been considered unfeasible for street drainage. The company wanted to

use long-range wireless networks to monitor and maintain the siphonic effect on a larger scale. At the time of the study, the proposed solution was a 6LoWPAN IPv6 mesh network with CoAP as the application-level protocol.

The company had no previous experience with wireless communication or IP networking but needed to document the security of the design. The proposed solution included the partial SysML model in figure 5. The model was used as the starting point for several rounds of playing Elevation of Privilege with various technical and non-technical stakeholders. Most sessions did not include security experts by design. However, the company systems engineer did attend a one-term introductory level cyber security course at the start of the project.

Devices using the CoAP protocol have some known weaknesses and vulnerabilities⁶², referred to in the model by their Common Weakness Enumeration (CWE) and Common Vulnerability Enumeration (CVE), as illustrated in figure 6. The limited computing power of the microcontrollers used in wireless sensors can lead developers towards making unwise tradeoffs when it comes to security. Failure to implement proper input validation in some devices allows an attacker to bypass authentication mechanisms and flood the unit with network traffic beyond its limited capacity. In terms of STRIDE, two threats are present. First, the threat agent gains unauthorized access to sending messages over the network (Elevation-of-privilege). Second, the network is overwhelmed by repeated broadcasting (Denial-of-Service). Players discovered the protocol vulnerability during the game, without the aid of experts, by simply searching for the term "CoAP" in the Mitre online CWE database⁶³ when prompted by the EoP cards to consider a particular threat. Figure 7 shows how once the vulnerability was added to the model, the modelling tool would automatically highlight any use of the CoAP protocol for further inspection.

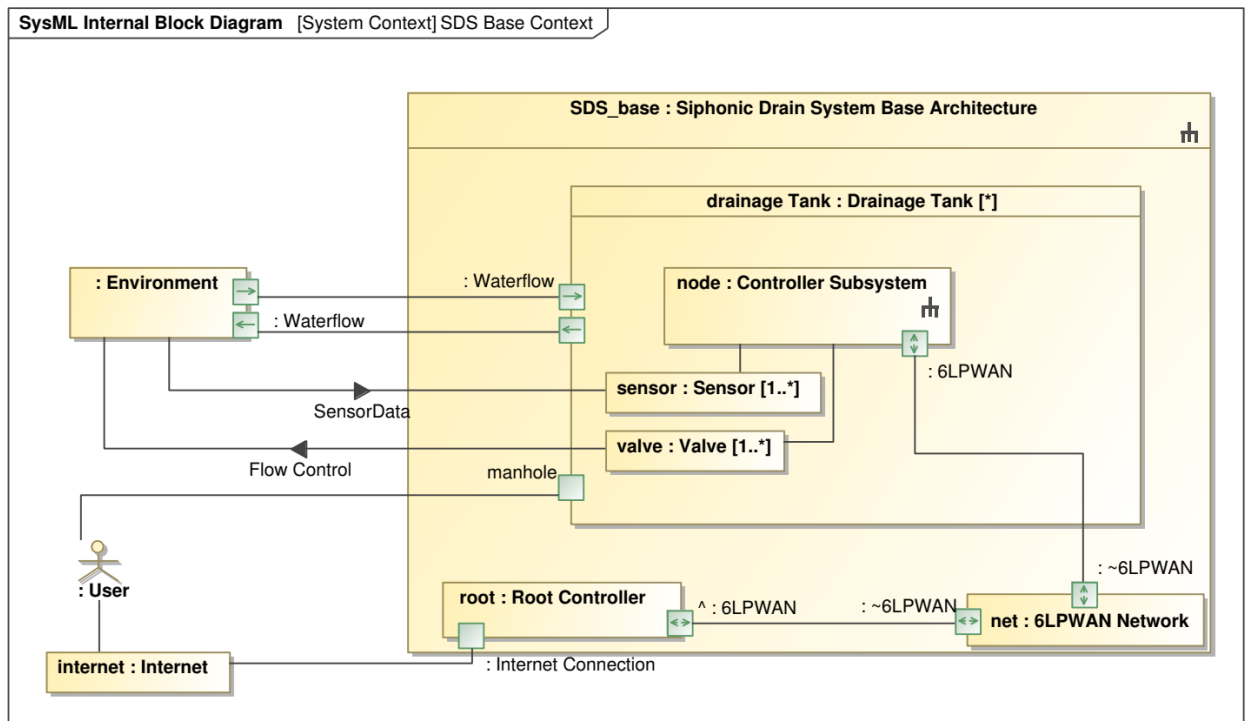


Figure 5 Initial high-level context model

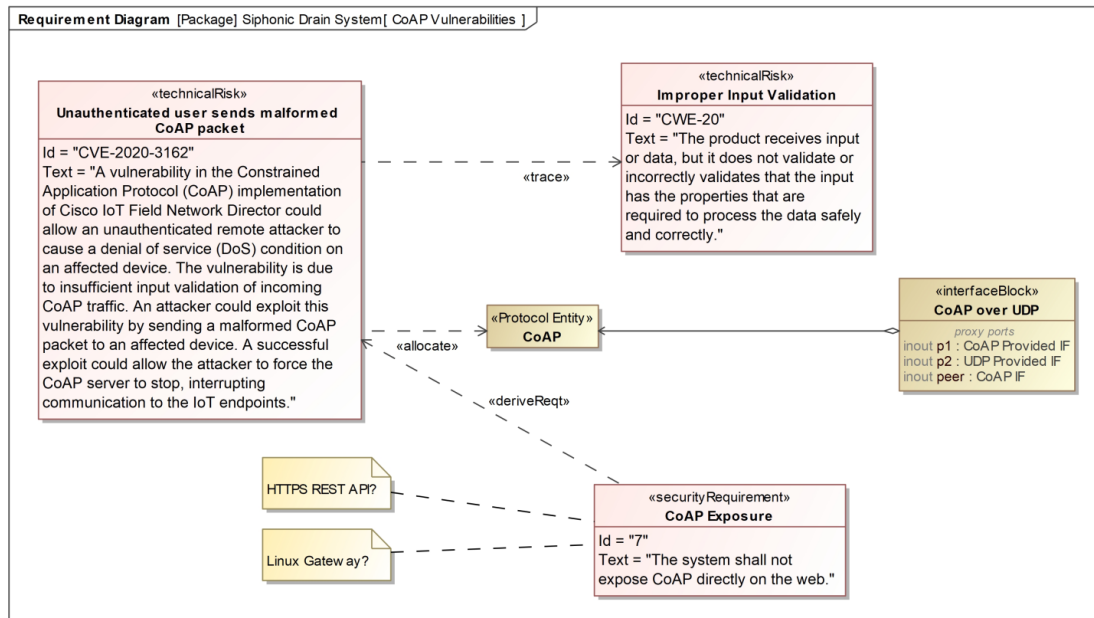


Figure 6 Known vulnerability of CoAP protocol used in the proposed design

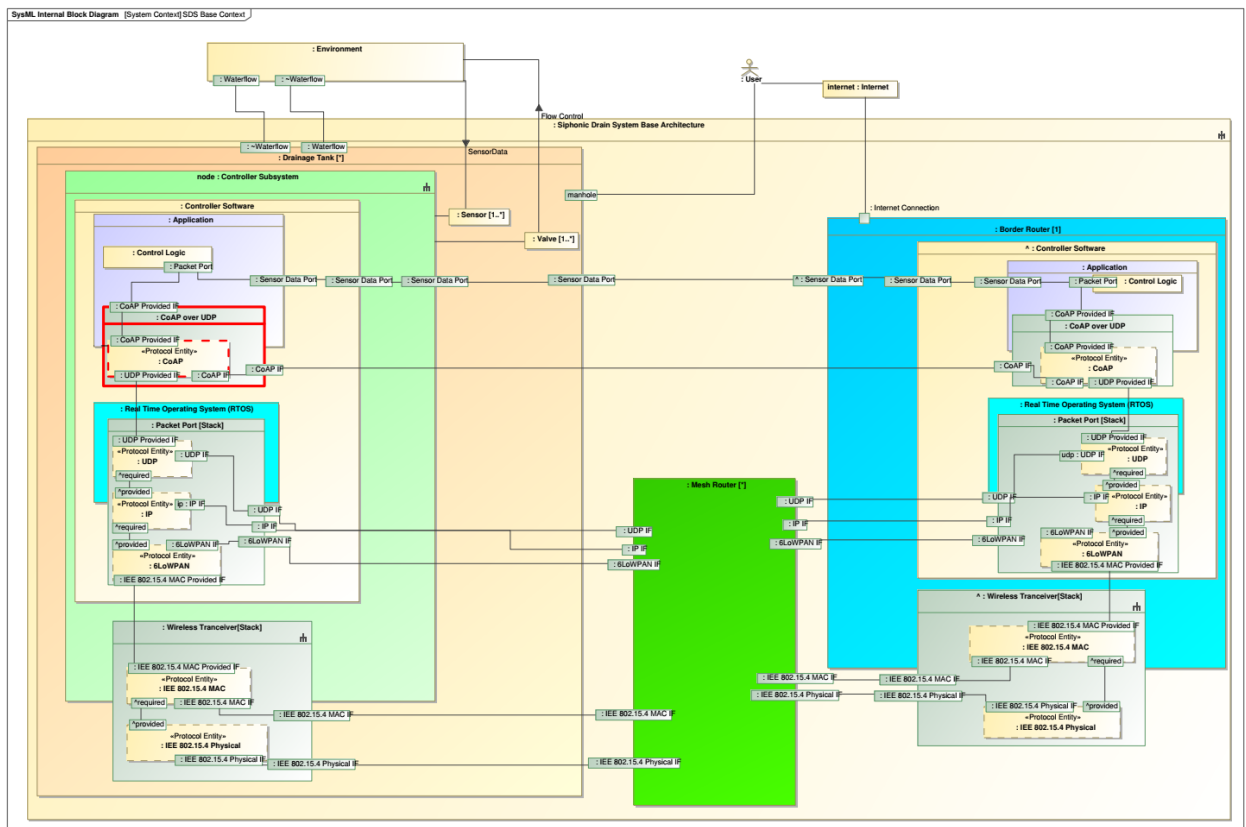


Figure 7 Automatic highlighting of interfaces with a known vulnerability

5. Case Study Findings

Using a process of line-by-line coding of player feedback during and after games sessions, followed by grouping codes into clusters by similarity, GTM analysis produced three main categories of response from players:

- Transdisciplinary Utility
- Shared Boundary Objects
- Increased Alignment

5.1. Transdisciplinary Utility

Many players initially expressed skepticism when asked to participate in threat modelling. They did not perceive this as part of their job and were hesitant to commit their time to it. However, as the exercise progressed, players reported new insights due to the modelling process that directly related to their primary responsibilities. Independent of each STRIDE threat considered at any moment, the clarification of the system model provided value to each player in itself that they could apply immediately following the workshop. Players also expressed interest in creating playing cards for critical activities in their own field.

5.2. Shared Boundary Objects

Focusing the modelling on flows and interfaces also clarified stakeholder interaction points and item exchanges. When an EoP card prompted discussion of a particular interface, it often became apparent that stakeholders had very different perceptions of how a particular flow occurred and the proper format of the exchanged item. This was true equally of machine-to-machine communication protocols, user manuals, error messages, or what constituted a practical authentication mechanism in the field. Because the game forced players to justify why a particular STRIDE threat was or was not relevant for a particular interface or flow, they had to communicate their understanding of that part of the system in much greater detail than they would typically do with other stakeholders. Players generally preferred a high level of consensus before modifying the model, so any inconsistencies had to be negotiated. Consequently, players gradually developed a communal definition of the boundary objects at the intersection of their respective disciplines.

5.3. Increased Alignment

The requirement to get through the entire EoP deck of cards enforced a strict time boundary on the modelling process. This time-boxing worked because the game began with the premise that completeness was not the goal. Players quickly settled into a pattern of rapid cycles of discussion and modelling. Parallel to the emergence of shared boundary objects, players reported improved understanding of other stakeholders' perspectives, including imagined actors with malicious intent. This alignment included clarifying the context, mission and functions of the system. Players reported a better understanding of what other stakeholders needed from the system and why. Alignment was not adjusted much within a single iteration, but these accumulated as the players progressed through the deck.

6. Conclusions

This article described one example of a small company using a card game to effectively threat model, translate this into a more secure system, and raise their engineering team's security competency and awareness. Most companies in the global supply chain are small companies such as this one. These companies are naturally agile and innovative but lack structure and may have blind spots outside their domain. Without a supporting structure, these companies represent a risk to their customers if they continue to rely on forced standards compliance to deliver system security, as many still do. The focus of the FuSE security foundation concepts on agility and competency over static standards is a good fit for very small companies. VSEs do not have the resources to force compliance with procedures, which means resistance to change must be countered by immediate value in an operational context. This constraint could be regarded as an opportunity for quality assurance when developing and selecting new methodologies in Systems

Security Engineering or SE more broadly. An agile approach to MBSE can provide structure for VSEs, yet also allow them to retain their operational flexibility. Using transdisciplinary games, rather than simply delivering a finished product, experts can lessen resistance and facilitate the involvement of the broader engineering team. The organization will develop security proficiency and improve the quality of its products.

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Paper IV

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Beyond Code Assistance with GPT-4: Leveraging GitHub Copilot and ChatGPT for Peer Review in VSE Engineering

Roar Elias Georgsen^[0000–0003–2868–497X]

University of South-Eastern Norway, Raveien 215, 3184 Borre, Norway
`roar.e.georgsen@usn.no`

Abstract. Most companies are Very Small Entities (VSEs), meaning they have fewer than 25 employees. Primarily domain specialists, these companies lack in-house expertise in important areas such as security and reliability engineering, process improvement, Quality Management (QM) and Systems Engineering (SE). VSEs struggle to adhere to Standard Operating procedures (SOP), and research has shown that contractual obligations to follow industry standards and best practices have little effect on actual engineering. This paper describes a case study that explored the potential of Large Language Models (LLMs) to support engineering best practices at a VSE by taking on the role of an expert peer in areas where the company had a skills gap. Aiwell, a Norwegian producer of building automation equipment, used ChatGPT, GitHub Copilot and GPT-4 to assess the quality of their system and stakeholder requirements. A GPT-4 foundation model with no additional training was given links to reference materials on requirements engineering produced by The International Council on Systems Engineering (INCOSE) and allowed to participate in discussions on the same digital collaboration platform as the human engineers. The study found that AI-assisted requirement reviews immediately and positively impacted the entire engineering process, supporting the feasibility of integrating advanced AI technologies in VSEs, even with limited training and resources. Participants highlighted the complementary nature of human intelligence and AI, where LLMs augmented human judgment through dialogue, leading to enriched engineering practices. Ethical and data privacy considerations also emerged as central themes, emphasising the need for proactive measures.

Keywords: generative AI· very small entities· systems engineering· requirements engineering

1 Introduction

The global supply chain consists mainly of Very Small Entities (VSE), with a workforce ranging from five (5) to twenty-five (25) people [1], and micro-enterprises, having fewer than nine employees, make up 92% of European enterprises [2]. In Norway, VSEs allocate approximately 30% of their spending to

research and development (R&D), and new product development makes up 20-30% of VSE revenue, a rate significantly higher than the less than 9% observed in large companies [3, 4].

Aiwell, a small Norwegian company developing outdoor automation systems, wanted to modernise its engineering practices in response to the growing complexity of its customer projects. With a high workload and lack of senior engineers in the labour market, Aiwell wanted to explore how technology could mitigate potential skills gaps and improve quality. The seemingly intuitive ease with which individual engineers used tools like ChatGPT and GitHub Copilot to generate functioning code and documentation motivated Aiwell to initiate a pilot study.

1.1 Objectives of the Study

The study was designed to explore the potential of Large Language Models (LLMs) to enhance engineering practices in the context of Very Small Entities (VSEs). The specific objectives that guided this exploration were:

Impact on Productivity and Quality: Investigating how engineers without prior experience could utilise LLMs to increase productivity and improve quality in general.

Feasibility of Generative AI for VSEs: Assessing whether LLM-based tools could be integrated into a VSE workflow at an accessible cost and without additional human resources.

Impact on Engineering Competency: Investigating how LLMs could fill knowledge gaps and contribute to increased competency in VSEs, in particular with regard to requirements engineering.

By focusing on these key areas, the study intended to provide insights into how LLM-based tools can address specific challenges faced by VSEs.

2 Background

2.1 Engineering in Very Small Entities (VSE)

VSEs lack experience working with standardised processes [5] and often lean towards informal and organically evolved methodologies. They find standardised methods too broad for their specific needs and value the agility and the ability to tailor workflows that informality offers to suit their unique contexts [6]. Small companies are conscious of the escalating customer and legal expectations for systematic engineering and their internal need for improvement [5]. However, they do not view these as beneficial to their work or relevant to their situation [6]. This belief in the absence of added value leads to opposition to change,

and the actual engineering practices may not align with documented compliance [7]. VSEs are rarely required to document complete compliance with specific standards and thus prefer to produce only the minimum documentation required by contractual obligations [6]. Measures such as risk-sharing partnerships [8] intended to improve system quality can, in this context, reduce the actual quality of the system as responsibility and accountability move down the supply chain.

Standardizing engineering practices can be vital to VSEs for a multitude of reasons. These practices equip VSEs with the resources and proven methodologies to enhance the quality and efficiency of engineering. They foster improved project management, reinforce implementation processes, and contribute to competitiveness [5]. The ISO/IEC 29110 series and similar initiatives, tailored for VSEs, can facilitate the transfer of codified knowledge related to systems engineering, offering benefits such as promoting innovation, market access, quality control, and ethical adherence.

Standards can provide legal protection, shielding engineers from accusations of negligence. The following scenario in [9] involving an 8-person company that developed computer-controlled valves for industries such as pharmaceuticals and chemicals illustrates the critical importance of standardized engineering practices, even for VSEs. The customer contracted the company to inspect the requirements using IEEE software engineering standards. However, the developer was unaware of the IEEE 1028 standard, which describes the types of software reviews and procedures required for execution. After installing the new software, the computer-controlled valves malfunctioned, causing damages in the chemical plant and leading to legal action. The court hearing revealed that the supplier could not provide evidence of an inspection according to the standard, resulting in legal and financial consequences. This incident underscores the necessity of adhering to standardized engineering practices to ensure quality, safety, and legal compliance, regardless of the company's size. It also highlights the potential risks and liabilities that can arise from neglecting these standards. In essence, standardized engineering practices can serve as a cornerstone for VSEs, ensuring quality, legal compliance, and economic viability, thereby crucial to their success and sustainability.

2.2 Requirements and Systems Engineering

In Aiwel's endeavour to improve its engineering practices, requirements quality emerged as a pivotal focus area. The importance of requirements quality is multi-fold. First, well-defined requirements serve as the cornerstone for successful project outcomes, reducing the likelihood of costly revisions and delays. Second, they facilitate clear communication among stakeholders, ensuring alignment on objectives and expectations. There is a strong positive link between capability in requirements definition and management [10]. A high-quality requirement is clear, concise, verifiable and traceable. It should be devoid of ambiguity, allowing for a single interpretation, and verifiable through inspection, analysis, or testing. Critically, a requirement should also be traceable, linking back to its source, ra-

tionale, and dependencies, which aids in managing changes and understanding impacts.

2.3 Large Language Models (LLMs)

Large Language Models (LLMs) are a subset of artificial intelligence models designed to generate human-like text. These models have gained significant attention in software engineering, particularly in automating coding tasks and improving code quality. LLMs have been evaluated for their effectiveness in code generation and have shown promising results in various aspects of software development [11]. The study revealed that 85% of developers felt more confident in their code quality when authoring with GitHub Copilot. Moreover, code reviews were completed 15% faster, and 88% of developers reported maintaining a flow state, indicating increased focus and reduced frustration. LLMs are not just accelerating the coding process but are also enhancing the quality of the code. GitHub’s internal metrics for code quality—readability, reusability, conciseness, maintainability, and resilience—showed significant improvement when developers used GitHub Copilot[12].

Using artificial intelligence, and more recently Deep Learning, to support engineering is not new [13], but in learning to use language, LLMs are also picking up other human skills, such as learning from a single example [14]. Whereas not long ago training an AI was an expensive process that took a long time, LLM foundation models can learn by just looking up the documentation [15]. It is this last newfound ability that Aiwell aimed to exploit in their pilot study.

3 Case Study: Leveraging LLMs to Enhance Requirements Reviews in a VSE

3.1 Case Description

Aiwell is a company with seven employees producing software and hardware used in building automation systems. Due to high workloads, Aiwell found it challenging to allocate sufficient time on a consistent basis to define high-quality requirement specifications for its diverse range of projects. The company wanted to leverage LLM-based tools to automate and enhance the quality of stakeholder and systems requirements, guided by the INCOSE Systems Engineering Handbook [16] and the Systems Engineering Book of Knowledge (SEBoK) [17]. Engineers at Aiwell already used GitHub for version control and wanted to explore GitHub Actions for task automation. They employed ChatGPT Plus and GitHub Copilot for AI-assisted script design and the OpenAI GPT-4 API in script execution. Key metrics included the number of requirement revisions, time spent on each requirement, and the frequency of requirement-tagged issues approved without further revisions. Initial implementation led to more precise and complete requirements, with fewer revisions and reduced time spent on each requirement. This section provides details and practical insights by demonstrating how Aiwell used LLMs effectively and efficiently regarding resources and time

to improve their engineering practices, in particular with regards to improving requirements quality.

3.2 Methodology

The study employed two primary methodologies: Action Research and Glaser’s Grounded Theory Method (GTM).

Action Research (AR) is a participatory, iterative methodology focused on solving real-world problems through cycles of planning, action, observation, and reflection [18]. Aiwell conducted multiple cycles of Action Research to refine the approach, resolving immediate challenges and planning for long-term adaptability.

Glaser’s Grounded Theory Method (GTM) is a qualitative research approach that emphasises the generation of theory directly from data through iterative coding and constant comparative analysis [19]. GTM was chosen for its flexibility in handling qualitative data. The study began by applying open coding to raw electronic communication, including instant messaging and comment threads. This preliminary analysis identified initial concepts, which were continuously refined into categories and themes. The study adopted a constant comparative analysis, integrating new data iteratively to evolve the emergent theory. This approach made sure the resulting theory was both relevant and contextually grounded.

3.3 Ethical and Commercial Considerations

Given Aiwell’s status as a Very Small Entity (VSE), the study had to account for ethical concerns associated with data confidentiality and anonymity. Strategies were based on SAGE guidelines [18], and included focusing on collective insights to preserve internal anonymity and working closely with Aiwell to redact sensitive or proprietary data.

3.4 A Working Definition of Requirements Quality

To quantitatively assess the quality of a requirement, Aiwell employed a scoring system based on key attributes such as clarity, conciseness, testability, and traceability. Each attribute was assigned a weight, and all requirements were evaluated on a scale of 1 to 5 for each attribute. Aggregating the attribute scores produced a composite quality score. For example, a requirement with a clarity score of 4, conciseness score of 5, testability score of 3, and traceability score of 4 would yield a composite score of $(4 * 0.3) + (5 * 0.15) + (3 * 0.35) + (4 * 0.20) = 3.8$.

Engineers evaluated requirements using an approach based on Planning Poker [20], an agile estimating technique where team members use playing cards to vote on the complexity of a task, facilitating discussion and consensus.

Requirements engineering best practices were drawn from two seminal resources: "The INCOSE Systems Engineering Handbook" and "The Systems Engineering Book of Knowledge" (SEBoK). These resources were also accessible to the LLMs through hyperlinks, as was a written description of the system context and the scoring system used by human evaluators. This provided a basic framework for assessing requirements.

3.5 Integrating LLMs in the Workflow

Introducing LLMs into the engineering workflow had to be low-cost and require minimal training. ChatGPT and GitHub Copilot served as the primary LLM-based tools for this purpose. ChatGPT is an AI conversational agent capable of generating human-like text based on the prompts it receives. GitHub Copilot is an AI-powered code assistant that helps engineers write new code and understand and work with existing code more efficiently. The tools' affordability made them attractive choices for a budget-conscious small entity. GitHub Copilot's seamless integration with Microsoft Visual Studio Code, a free code editor already used by Aiwell, facilitated a smooth transition. The OpenAI GPT-4 API, though demanding a steeper learning curve, was rendered more accessible with the assistance of ChatGPT and GitHub Copilot. While the code generated by these tools might not always be perfect, it was sufficient to expedite the development processes. LLMs could also participate in multi-lingual, realistic and nuanced human-like discussions with engineers about subtle topics like requirement quality.

Initially, Aiwell engineers started with the default interfaces of ChatGPT and GitHub Copilot. These tools' ease of use and intuitive interface meant that engineers could begin leveraging their capabilities without extensive training or preparation. This factor was crucial, given the resource constraints typical of VSEs. The user-friendly and adaptable tools fit effortlessly into Aiwell's existing GitHub-based version control and task automation workflows.

When OpenAI introduced custom instructions as a new feature in ChatGPT, Aiwell incorporated it into their workflow by adding links to relevant references as standard prefixes to their prompts. This allowed engineers to guide the AI model more effectively, aligning the generated responses with the project's specific context and needs. Standardising custom instruction enhanced the quality of LLM responses by making them more precise and context-appropriate.

As the team gained experience, they began exploring more advanced features of the GPT-4 API. They started crafting prompts that communicated their intent more clearly to the LLMs, improving the quality of the generated response, whether code or requirements evaluation. This incremental and organic developer-led approach to adopting LLM features ensured the team could adapt without feeling overwhelmed, thereby maintaining productivity.

LLMs became an integral part of Aiwell's engineering practices, not as a disruptive technology requiring a steep learning curve but as enablers that were incrementally adopted. The metrics, including the reduced requirement revisions and time spent on each requirement, validated the effectiveness of integrating LLMs into the workflow. Crucially, these gains never required advanced applications such as pre-trained AI models or curated vector databases with custom knowledge but relied only on free and low-cost, publically available tools.

3.6 Collaborating With LLMs on Writing Automation Scripts

Aiwell's engineers leveraged ChatGPT to automate GitHub Actions, focusing on requirements validation. The process began with a simple prompt to ChatGPT, asking it to draft a GitHub Action script to call the GPT-4 API if a user labelled a GitHub repository issue as a requirement. Despite some initial regular expression and JSON parsing challenges, the engineers iteratively refined the prompts, leading to effective scripts. Figure 1 shows an abbreviated example of the generated code.

GPT-4's output, posted as a GitHub comment on the issue, comprehensively evaluated the requirement using SEBoK and the INCOSE Handbook as a reference and provided the engineer with a task list of suggested improvements. Figure 2 shows the full analysis of a requirement generated by GPT-4. Later iterations included a composite score based on the provided guidelines. The AI-generated comment would break down the requirement's clarity by assessing its specificity, conciseness by indicating unnecessary details, testability by evaluating the clearness of acceptance criteria, and traceability by checking its linkage to system needs or stakeholder requirements. Each attribute would receive a score, and GPT-4 would calculate a composite score using the same scoring system and guidelines employed by Aiwell's human engineers. This approach enriched the requirements validation process, offering a quantitative and qualitative assessment to supplement the human scoring and encourage critical discussion. It also showed how VSEs, even with limited resources, can incrementally integrate LLMs into their existing workflows. The use of ChatGPT in script automation not only streamlined the task but also added a layer of intelligence and review, making the process more robust and efficient. This part of the study demonstrated the capability of LLM base models to perform complex systems engineering tasks that align with established systems engineering principles.


```

1 name: Check Requirement Correctness
2 on:
3   issues:
4     types: [labeled]
5 jobs:
6   check-requirement:
7     runs-on: ubuntu-latest
8     if: contains(github.event.label.name, 'requirement')
9     steps:
10    - name: Checkout code
11      uses: actions/checkout@v2
12    ...
13    - name: Check requirement
14      env:
15        GITHUB_TOKEN: ${ secrets.MY_GITHUB_TOKEN }
16        OPENAI_API_KEY: ${ secrets.YOUR_OPENAI_API_KEY }
17      run: |
18        ...
19        #Load the Statement of Work
20        SYSTEM_CONTEXT=$(cat README.md)
21        SCORING_GUIDELINES=$(cat scoring.md)
22
23        # Get the content related to checking correctness
24        # of system requirements
25        CORRECTNESS_GUIDELINES="https://sebokwiki.org/
26        wiki/Stakeholder_Requirements_Definition#
27        Checking_Correctness_of_System_Requirements"
28        ...
29        # Call GPT-4 API with issue details
30        RESPONSE=$(curl https://api.openai.com/v1/chat/
31        completions \
32        -H "Content-Type: application/json" \
33        -H "Authorization: Bearer $OPENAI_API_KEY" \
34        -d '{
35          "model": "gpt-4",
36          "messages": [
37            {"role": "system", "content": "
38              $SYSTEM_CONTEXT
39              $CORRECTNESS_GUIDELINES
40              $SCORING_GUIDELINES" },
41            {"role": "user", "content": "Title:
42              $ISSUE_NAME. Id: $ISSUE_NUMBER.
43              Description: $ISSUE_BODY"}
44          ]
45        }')
46        ...
47        # Add the response as a comment to the issue
48        # using GitHub API
49        curl -s -X POST -H "Authorization: token
50        $GITHUB_TOKEN" -H "Content-Type: application/
51        json" -d "$PAYLOAD" "$COMMENT_URL"

```

Fig. 1. Partial GitHub Action yaml script generated by ChatGPT

The temperature of the glycol must not exceed 50 degrees.

AI Generated Evaluation

The requirement is clear, concise, and unambiguous. However, it could be improved by providing more context and specificity.

While it states a specific temperature limit, it does not specify the conditions under which this limit applies (e.g., during operation, at rest, etc.). Also, it does not provide any acceptance criteria or a method for testing this requirement.

Here's a tasklist for improving this requirement:

- [] Specify the conditions under which the temperature limit applies. For example, "The temperature of the glycol must not exceed 50 degrees Celsius during operation."
- [] Define the acceptance criteria. For example, "The system is considered to meet this requirement if the glycol temperature remains below 50 degrees Celsius during a continuous 24-hour operation under standard snowfall conditions."
- [] Provide a method for testing this requirement. For example, "This requirement can be tested by operating the system continuously for 24 hours under standard snowfall conditions and monitoring the glycol temperature."

After these improvements, the requirement might look like this:

The temperature of the glycol must not exceed 50 degrees Celsius during operation. The system is considered to meet this requirement if the glycol temperature remains below 50 degrees Celsius during a continuous 24-hour operation under standard snowfall conditions. This requirement can be tested by operating the system continuously for 24 hours under standard snowfall conditions and monitoring the glycol temperature.

Fig. 2. GPT-4 generated evaluation of requirement

3.7 Challenges and solutions

Data Privacy and Security Integrating GitHub CoPilot, ChatGPT, and the GPT-4 API into Aiwell’s workflow presented significant security and privacy challenges, primarily due to a lack of clear policies and concerns expressed by government agencies [21]. To mitigate this uncertainty, the tools were initially restricted to non-sensitive material. However, as the study progressed, the introduction of GitHub CoPilot for Business and Microsoft’s Azure hosted versions of OpenAI’s models provided more secure alternatives [22]. OpenAI’s also clarified in its policies that GPT-4 API data is not stored by their servers, alleviating some concerns [23].

Model Limitations While ChatGPT and GPT-4 demonstrated proficiency in domain-specific language, the models faltered in tasks like regular expression parsing, mathematics and JSON formatting. OpenAI’s subsequent update, which introduced function-calling capabilities in the GPT-4 API, addressed these issues, enabling deterministic functions for complex tasks.

Human-AI Collaboration Balancing human and AI contributions proved challenging. Although ChatGPT generated effective YML scripts, its numerical requirement scores often diverged from human evaluations. Action Research methodology helped here, as discrepancies triggered discussions, leading to an evolution of the evaluation criteria and a more stable consensus among engineers. Also, when given access to the comments section on GitHub, GPT-4 would join the discussion and be as opinionated as a human engineer if instructed specifically to be so.

Quality Control Ensuring the quality of AI-generated evaluations required vigilant oversight. A systems engineer consistently reviewed the model’s outputs, maintaining a human-in-the-loop approach at all times. This review process was integral to the iterative cycles of planning, action, observation, and reflection.

Error Handling Scripts generated by ChatGPT and Github Co-Pilot underwent rigorous scrutiny and testing, and engineers provided the tools with prompts based on a template that evolved incrementally based on previous mistakes made by the LLMs. However, since GPT-4’s output was limited to comments and not executable code, the risk of operational disruptions was minimised.

Scalability Scaling the approach for larger tasks or teams posed challenges. The solution involved using small, template-based scripts and leveraging LLMs for extensive commenting and documentation.

In summary, the challenges encountered by Aiwell were systematically addressed, often benefiting from the iterative and reflective nature of the Action

Research methodology. This approach not only resolved immediate issues but also contributed to the long-term adaptability and robustness of the engineering workflow.

4 Emergent Themes

The Grounded Theory Method’s iterative nature was pivotal in the continuous refinement and evolution of the emergent themes. By ensuring that these themes were deeply rooted in the experiences and feedback of Aiwell’s engineers, the study captured the authentic challenges and opportunities of integrating AI into the workflows of a VSE. Several distinct themes emerged from the study, each shedding light on different facets of integrating AI into the workflows of a small company like Aiwell. These themes provide insights into the immediate benefits and challenges of AI adoption and hint at the broader implications for the future of engineering practices.

Accessibility The ease of integrating AI tools into Aiwell’s workflows underscored the theme of feasibility and accessibility. Some engineers had initially perceived the adoption of AI as a daunting task. However, the user-friendly nature of tools like ChatGPT and GitHub Copilot facilitated a smooth transition. A comment from an Aiwell team member captured this sentiment: “We thought integrating AI would be a massive undertaking, but these tools made the transition surprisingly smooth.” This theme emphasises the democratisation of advanced technologies, making them accessible even to smaller entities. In the early stages of the study, the data pointed towards the feasibility of integrating AI tools into Aiwell’s workflows. However, as engineers gained more hands-on experience with these tools, their feedback began to reflect a broader perspective. Comments like “The integration was smoother than we anticipated” highlighted the feasibility and accessibility of advanced AI technologies. This development underscores the importance of user experience in determining the success of technology adoption.

Human-AI Collaboration The interplay between human judgment and AI-generated content emerged as a significant theme. While AI provided valuable insights and suggestions, the human element remained crucial. One engineer observed, “While the AI provides a solid starting point, our team discussions around its suggestions have become invaluable.” This theme highlights the symbiotic relationship between humans and AI, where each complements the other’s strengths. The initial data suggested a theme centred around the efficiency gains from AI. However, as engineers interacted more with the AI tools, the theme shifted towards collaboration. An engineer’s observation, “The AI doesn’t replace our judgment but enhances our discussions,” was instrumental in this shift. This refined theme emphasises the complementary nature of human intelligence and AI, suggesting that the actual value of AI tools lies in its ability to augment human capabilities rather than replace them.

Productivity and Quality Enhancement The data consistently highlighted the positive impact of AI on both productivity and the quality of work. Engineers reported fewer errors and an increase in overall efficiency. For instance, an engineer at Aiwell noted, "With AI assistance, we've seen a noticeable reduction in errors and an increase in our overall efficiency." This suggests that AI tools can significantly enhance the day-to-day operations of a company, leading to tangible improvements in output quality. The initial data collection phase primarily highlighted productivity metrics, with engineers emphasising AI integration's speed and efficiency gains. However, the theme evolved as the study progressed. An engineer's comment, "It's not just about doing things faster. The quality of our work has improved significantly with AI assistance," underscores the dual impact of AI tools on both productivity and quality. This feedback led to a more comprehensive understanding of this theme, emphasising that AI's value proposition can extend beyond speed to encompass output quality. Notably, this improvement in quality was not primarily due to the quality of the output generated by the LLM, which was sometimes lacking. Rather, the LLM guided the engineer to consider a broader perspective, prompting more detailed requirements and drawing attention to potential edge cases.

Requirements Engineering and Quality Competence The proficiency of AI in enhancing requirements engineering stood out. Engineers found that AI suggestions often prompted them to rethink and refine their initial ideas, leading to clearer and more precise requirements even when they were not using AI tools. An engineer remarked, "Our requirements have never been clearer. The AI's suggestions often prompt us to rethink and refine our initial ideas." This result was distinct from the quality improvement resulting from the guided process. Engineers would gradually adopt thought patterns similar to the LLM-generated output, with the AI essentially serving as a Systems Engineering mentor.

Data Privacy Concerns Data privacy emerged as a pressing concern, especially with tools like GitHub Copilot. Engineers expressed worries about proprietary code being accessed by AI and potentially appearing in someone else's work. An Aiwell engineer expressed this concern, stating, "We need to ensure our proprietary code remains confidential, even when using AI tools." This theme underscores the need for robust data privacy measures when integrating AI into workflows. The broader theme of ethical considerations in AI adoption initially encompassed ethical and data privacy concerns. However, as the study delved deeper into the engineers' experiences, it became evident that data privacy was a more pressing concern. Specific feedback like "Our main worry is our code's confidentiality with AI tools" led to a sharper focus on data privacy, highlighting the need for robust measures to protect proprietary information.

Scalability Challenges While the study showcased the benefits of AI in simple applications, questions arose about scaling these benefits for more complex

tasks. Engineers pondered the AI’s capabilities in handling intricate processes as the company grows and diversifies. An engineer pondered, ”The AI has been great for our current tasks, but how will it handle more complex processes in the future?” This theme points to the need for continuous evaluation and adaptation as companies scale their operations. Emerging later in the study, the theme of scalability challenges captured the engineers’ forward-looking perspective. While the immediate benefits of AI in Aiwell’s current tasks were evident, engineers began to consider its long-term applicability. One engineer’s remark, ”The real test will be scaling these AI benefits to our more intricate projects,” highlighted the need to evaluate AI’s capabilities as companies grow and diversify continuously.

5 Conclusions and Future Work

5.1 Key Findings

This study examined the integration of Large Language Models (LLMs) into the engineering practices of Aiwell, a Very Small Entity (VSE) . The findings underscore the potential of LLMs in enhancing engineering quality, productivity, and overall efficiency within VSEs.

1. **Impact on Productivity and Quality:** When effectively integrated, the study found that LLMs can significantly boost productivity and enhance the quality of engineering practices. This is particularly crucial for VSEs, which often operate under resource constraints.
2. **Feasibility and Accessibility:** Despite their limited resources, VSEs can harness the power of LLMs to address unique challenges and streamline their operations.
3. **Human-AI Synergy:** The study highlighted the complementary nature of human intelligence and AI. LLMs do not replace human judgment but augment it, leading to enriched discussions, better decision-making, and more informed engineering practices.
4. **Ethical and Data Privacy Considerations:** As with any technological integration, ethical and data privacy concerns emerged. The study emphasised the importance of addressing these concerns proactively, ensuring that the integration of LLMs aligns with ethical standards and data privacy regulations.

5.2 Future Work

The findings in this study pave the way for several avenues of future research and exploration:

1. **Expanding the Scope:** While this study focused on a relatively simple technical implementation and narrow topic, future research could explore the impact of LLMs on VSEs in more complex settings. It would be insightful to understand how the challenges and benefits scale with the complexity of

the application since VSE engineers do not have time to become experts in AI application development. What can VSEs achieve without this deep competency?

2. **Long-term Impact:** This study provided a snapshot of the immediate benefits and challenges of LLM integration. A longitudinal study could offer insights into the long-term impact of LLMs on VSEs, tracking their evolution over time.
3. **Customisation and Fine-tuning:** This study used base models and publically available and generic reference materials. Future work could explore the customisation and fine-tuning of LLMs to cater to specific organisational needs, enhancing their effectiveness and relevance. Is this kind of customisation possible using only the resources available to VSEs?

The integration of LLMs into VSEs presents a realm of possibilities, challenges, and opportunities. This is a nascent technology, and the future promises a deeper, more comprehensive exploration of the potential of LLMs in diverse settings and contexts.

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Paper V

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REGULAR ARTICLE

Navigating Uncertainty: Guiding Attention in Purposeful Human Activity Systems

Roar Elias Georgsen

Faculty of Technology Natural Sciences and Maritime Sciences, Department of Microsystems, University of South-Eastern Norway, Borre, Norway

Correspondence: Roar Elias Georgsen (roar.georgsen@gmail.com)

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ABSTRACT

Engineering teams are complex socio-technical systems that design, realise and manage engineered solution systems. However, the tools available to systems engineers to organise and support these teams are not yet as advanced or mature as those used for the systems they engineer. The purposeful human activity systems model (PHAS) was developed to address that gap. This paper proposes extending PHAS by integrating the free-energy principle (FEP) from computational biology. It introduces a formal model of cognitive agents that navigate, interact with and modify their environment, self-organising around minimising uncertainty and surprise by actively maintaining predictive models of the world that represent individual and shared worldviews. Minimising uncertainty is the primary driver of attention in this model, highlighting potential leverage points for designing practices and environments that coordinate and guide the focus of engineering teams. Because the model is symmetrical, the environment also models the agent, providing a formal description of affordances and context shaping. The proposed model is used to re-examine the original PHAS framework, and a case study demonstrates how a small team of engineers used the model to improve their cyber-security practices.

1 | Introduction

Engineers utilise a wide range of robust methods to design, realise and manage solution systems. As these solutions become more complex [1], the complexity of managing engineering activities also grows. Current methods for managing such activities often fail when faced with this increase in complexity [2]. The methods and tools used to manage engineering activities are often rigid and difficult to adapt to complex environments [3, 4]. Rousseau [5] argues that contemporary systems engineering practices largely rely on heuristics, but also points out that practitioners increasingly recognise the need for a scientific basis to address problem-solving more holistically [1, 5]. Rousseau's work on the nature of scientific inquiry and ontologies as systems laid the groundwork for such a foundation [5–7], and purposeful human

activity systems (PHAS) was intended to build on his idea of scientific systems principles from a socio-technical perspective. It has been argued that all systems are socio-technical in a systems engineering context [8], but here we use the term to refer specifically to engineered systems whose performance and effectiveness depend significantly on the integrated interactions between their social (human, organisational) and technical elements.

Calvo-Amodio and Rousseau [9] state the following:

Successful human activity systems are stable systems with low entropy that maintain their progression along a trajectory to achieve their goals. They maintain homeorhetic control towards fulfilling their purpose

Abbreviations: FEP, free-energy principle; PHAS, Purposeful Human Activity Systems.

while maintaining a homeostatic state when (1) their purpose is well understood and (2) the causal powers needed to conduct purposeful activities are present.

This idea of defining systems in terms of entropy is reflected in Sillitto's definition of a system as '*a persistent region of low entropy (= high organisation) in physical or conceptual space-time*' and systemness as '*the phenomenon that allows regions of organisation to persist in a dissipative universe*'. [10]

Rousseau [5] argues that energy, as a quantifiable and conserved entity, provides the basis for predicting and managing systems' behaviour scientifically. The free-energy principle (FEP) takes this literally and restates Sillitto et al.'s definition in the notation of statistical physics [11, 12]. From this precise definition, it is possible to derive descriptions of related phenomena, such as how PHAS maintain progress towards a purpose (homeorhetic control), how the components come to understand and share that purpose and the nature of the system's causal powers. This paper aims to provide that precise model of PHAS based on FEP formalisms, use it to critically evaluate the original PHAS model and framework, revise them and validate the updated framework in an engineering case study.

The main contention advanced in this paper is that the original PHAS model insufficiently takes into account that individual humans and teams are social entities whose behaviour is driven by beliefs, desires and shared cultural norms. The article aims to demonstrate how these properties of social systems, often referred to as 'soft', can be formally described and integrated as a top-level PHAS component while adhering to the kind of scientific rigour advocated by Rousseau. The expanded model adopts the term 'Worldview' for this new component, in reference to the 'Worldview Inquiry Framework' proposed by Rousseau and Billingham [13].

The next section provides background on PHAS, Rousseau's scientific systems principles and FEP. This is followed by an introduction to key concepts from philosophy, statistical physics and biology that inform how the formal model should be understood, interpreted and put to use. The model is then gradually constructed using algebraic notation. Each equation represents an aspect of PHAS or their environment, and key properties that follow from the formal description are proposed as new PHAS principles and stated in English. The formal model and the new principles are then integrated into the original PHAS model. The new model is operationalised into a practical framework by expanding the original list of PHAS principles [9] and by formulating a new set of inquiry questions to guide practical design that complement those proposed by Calvo-Amodio [4]. These are then demonstrated using an engineering case study. The article concludes with a summary of findings and potential directions for future research.

2 | Background and Theoretical Foundations

2.0.1 | Purposeful Human Activity Systems (PHAS)

Calvo-Amodio and Rousseau [9] introduced the concept of human activity systems (HAS) in recognition that engineering

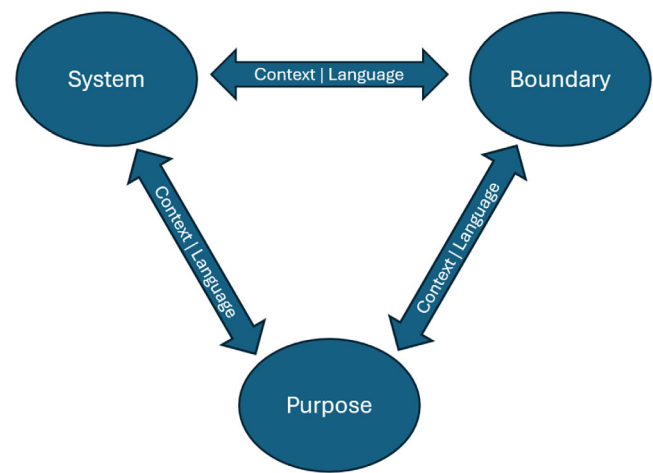


FIGURE 1 | Purposeful human activity systems—from Calvo-Amodio and Rousseau [9].

teams inherently function as socio-technical systems grounded in people and their behaviour. These systems use highly specialised technical knowledge to design, model and realise complex solution systems. In subsequent publications, Calvo-Amodio expanded the acronym to PHAS, emphasising that these are deliberately engineered systems with a defined *purpose* [4, 14]. The PHAS model and framework were designed to guide the creation and management of human-centred socio-technical systems and explain their emergence and evolution by focusing on the following four core elements in Figure 1:

- **System:** PHAS consist of conceptual and concrete parts. These parts interrelate to create persistent structures, processes and meanings.
- **Purpose:** PHAS are designed to fulfil specific purposes. They possess inherent causal powers to achieve these purposes, are designed to align with their objectives and are consciously aware of and intentionally pursue their goals.
- **Boundary:** PHAS interact with their environment through their boundary. This boundary allows the system to both submerge into and influence its environment.
- **Relationships:** The nature and strength of the relationships between parts, purpose and boundary in PHAS depend on the context and are influenced by the language used.

The core concepts were then further refined into the four principles with sub-principles given in Table 1.

When it was first presented by Calvo-Amodio and Rousseau [9], the PHAS model was primarily theoretical. However, in the subsequent article where Calvo-Amodio changes the acronym from HAS to PHAS, he also compares it to several well-known practical systems approaches and operationalises the PHAS model as a practical framework [4]. He ends up structuring his framework around a visualisation by Checkland [15] depicting Vickers' appreciative systems approach [16]. Figure 2 shows Calvo-Amodio's modified version.

TABLE 1 | PHAS principles [4, 9].

Component	Relevant principle
System	1. Human activity systems have parts 1.1. The parts are conceptual and concrete 1.2. The parts interrelate to produce persistent structures, processes, behaviours and meanings 1.3. The parts' interrelations are conditioned by the kind, capability and structure of the human activity system
Purpose	2. Human activity systems exist to fulfil a purpose or a set of purposes 2.1. Human activity systems possess inherent causal powers to fulfil their purpose 2.2. The human activity system design is commensurate with its purpose 2.3. Human activity systems are aware of their purposes and pursue them intentionally
Boundary	3. A human activity system mediates its interactions with its environment through its boundary 3.1. A human activity system submerges to its environment 3.2. A human activity system influences its environment
Relationships	4. Context and language modulate the relationships in a human activity system 4.1. The magnitude and kind of the two-way relationships between the parts of the system, purpose and boundary vary according to the context 4.2. The magnitude and kind of the two-way relationships between the parts of the human activity system, purpose and boundary are conditioned by the language used

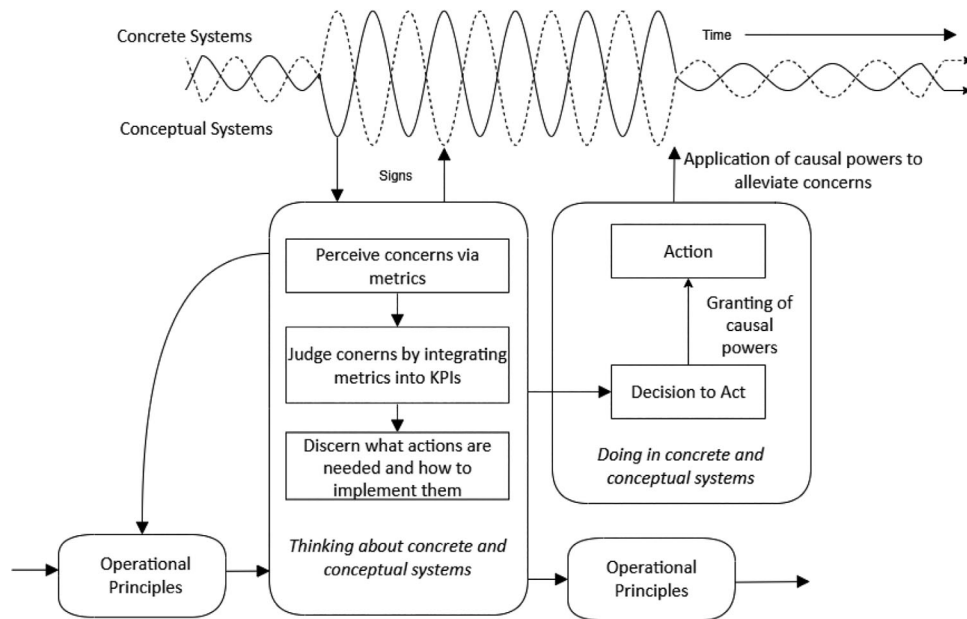


FIGURE 2 | The PHAS framework—from Calvo-Amodio [4].

The PHAS framework in Figure 2 focuses on using operational principles and their corresponding metrics and key performance indicators (KPIs) as activity drivers. Table 2 shows a list of inquiry questions used by Calvo-Amodio to operationalise his PHAS framework.

2.1 | The Free-Energy Principle

The FEP, as articulated by Karl Friston [17] and refined over subsequent years [11, 12], is a theoretical framework that unifies

various aspects of biology under the single concept of minimising variational free energy. Rooted in earlier ideas in thermodynamics by Gibbs [18], and Helmholtz's [19] notion of unconscious inference, FEP posits that biological systems maintain their low-entropy, organised states by reducing the discrepancy between predicted and actual sensory inputs, thereby minimising surprise and uncertainty about their environment. Reducing uncertainty involves minimising variational free energy, which pushes the system to explain or predict its sensations more accurately [20, 21]. This process is typically modelled as active inference, a widely used mathematical model of updating beliefs in perception and

TABLE 2 | Inquiry questions for concern judgment making—from Calvo-Amodio [4].

Inquiry question	Information components and comments
1. What needs to be true for this concern to be addressed?	– What is useful to know about the key activity where the concern has arisen? – What type of change might be needed to alleviate the concern and thus improve the performance of the key activity? – Does the metric provide an accurate measurement of performance?
2. Who cares about this concern and why?	– Who finds the concern and related metric meaningful? – Why is the concern meaningful? – How does the concern relate to social, organisational and/or psychological dimensions in the concrete and conceptual interplay?
3. What is good enough for this concern to be satisfied?	– What is the best balance between the effort needed to satisfy the concern and the burden the solution will place on the organisational concrete and conceptual systems? – Does the potential solution address the concern in alignment with the PHAS principles and capability?

learning [22]. In active inference, the agent relies on an internal probabilistic generative model to represent a mapping between unobserved causes and observed effects, reflecting the agent's predictive and causal understanding of the world [21].

FEP is highly generalisable and has demonstrated strong predictive power. Empirical research supports the application of FEP across a broad range of phenomena, such as perception [23], attention [24], learning [25], emotions [26], action [27], cultural processes [28], niche construction [21, 29] and natural selection [30].

2.2 | Scientific Systems Principles

Rousseau [5] argues that systems engineering needs a stronger theoretical foundation that addresses the fundamental nature of engineered systems. He maintains that although many systems phenomena have been studied scientifically [31], systems engineering still lacks an overarching holistic scientific basis for understanding *systemness* [32]. This leaves systems engineers to rely on heuristics, degrading their ability to predict the conse-

quences of design choices and depriving them of principle-based tools to handle unexpected system behaviour as solutions grow more complex [33]. Rousseau further argues that the focus should be on finding principles that explain the core nature of complex solution systems rather than just describing how they operate. These principles should be scientific, general and quantitative.

Rousseau proposes three general systems principles as follows:

Principle of Conservation of Properties (SP1): This principle states that emergent properties in system formation are balanced by the submergence of properties in the parts. It implies that the energy or potential lost by parts in forming a system equals the energy associated with the system's emergent properties.

Principle of Universal Interdependence (SP2): This principle posits that system properties result from a balance between bottom-up emergence from parts and outside-in submergence due to the systemic context. It implies that both internal and external influences must be considered when analysing systems.

Principle of Complexity Dominance (SP3): This principle suggests that in interactions between complex and simple parts, the simpler parts undergo more significant submergence than the complex ones. It means that complex systems can buffer changes better than simpler ones.

A scientific principle is not a falsifiable theory about how the world is; rather, it is a general description of the world defined in a particular way. In other words, if we can define what it means for a system to adhere to a given principle, what attributes and dynamics would that system need to have? So, while scientific principles are not themselves falsifiable, hypotheses derived from them may be so if they make predictions about empirical phenomena covered by those principles.

Rousseau's approach to establishing a general system theory (GST) is grounded in 'systemic isomorphisms', recurring patterns that manifest across different systems. Certain universal truths must be accepted for isomorphisms to link naturalistic and logical systems. These truths, reflected in the scientific worldview, form the core principles of science according to the *principles-laws-theories* (PLT) model adopted by Rousseau [5].

2.3 | Loans of Intelligence

Introduced by Daniel Dennett [34], *loans of intelligence* refer to the provisional use of assumptions in scientific models to simplify complex systems. These loans allow scientists to progress by assuming functionality and relationships within a system of interest, even if those assumptions are not yet justified. For instance, in the viable system model (VSM) [35], loans of intelligence are reflected in terms like 'command' and 'control'. This terminology implicitly assumes a 'sender' (the command system) and a 'receiver' (the operational units) that can understand and act upon these commands. This assumption is a loan of intelligence because it presumes the existence of a coherent and efficient communication process without fully explaining the mechanisms by which the operational units interpret and execute these commands.

It is crucial to remain aware that all assumptions are provisional. Dennett encourages non-committal exploration of ideas, and highlights the importance of recognising loans of intelligence as temporary tools [36]. As models become increasingly precise and demonstrate stronger predictive power, it is critical to remember that although some models are more useful than others, all models are still wrong [37]. A falling rock can be described as behaving according to differential equations in physics, but that does not imply that the rock itself is capable of performing calculus.

2.4 | Intentional Systems

Dennett proposed the Intentional Stance as a pragmatic strategy for predicting and explaining the behaviour of complex systems, whether humans, animals or machines, by treating them as if they are rational agents with beliefs and desires [34, 38, 39]. Systems whose behaviour can be predicted from the Intentional Stance are classified by Dennett as *intentional systems*. The Intentional Stance plays to our strengths as social beings by allowing us to predict behaviour without needing to understand the underlying mechanisms. This does not imply that such systems have actual inner states, only that this is a useful assumption for predicting their behaviour.

A weakness of the original PHAS model is that it does not emphasise what is inherently human about these systems. All engineered systems have a purpose. What differentiates a PHAS is the awareness of that purpose and the intent and motivation to pursue it or deviate from it. PHAS are intentional systems.

2.5 | Systems and Boundaries

In statistical physics, a system's identity is determined by its ability to maintain a consistent set of internal states and state transitions despite external influences [12]. However, for something to exist as a distinct entity capable of responding to its surroundings, this independence must also exhibit at least some degree of conditionality upon the broader system in which it resides. The minimal structure necessary to maintain such conditional independence is a Markov Blanket [12].

The Markov Blanket is a modelling convention that enables us to identify and define the boundaries of a system in statistical terms for a specific level of description. Technically, it creates a separation between internal (systemic) and external (environmental) states. Figure 3 shows how this separation insulates the internal states from direct influence by external states. Input states receive information from the external environment, translating raw external signals into internal representations that the system can process. For instance, a system might translate light into visual data or sound waves into auditory information. Meanwhile, active states represent the system's influence on the environment, where the system asserts influence on its surroundings based on its internal goals, preferences and biases, such as by altering focus, moving an object or changing position. This interplay ensures that external conditions provide necessary input, but do not directly influence the system's internal states.

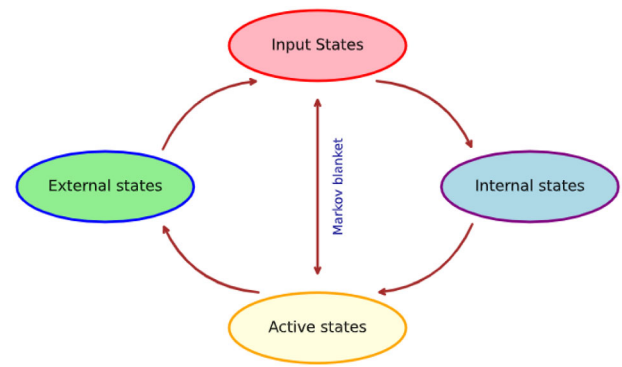


FIGURE 3 | The Markov Blanket as a system boundary.

The Markov blanket demonstrates that internal and external states are 'hidden' from each other. In other words, the states of other systems are obscured. No system can ever directly observe the world beyond its own boundaries, and this boundary mediates all knowledge about the external world. We cannot directly access the causes of our sensations, nor is there a one-to-one correspondence between specific causes and the sensations they produce [40, 41]. The smell of smoke might come from a campfire, a barbecue or a burning building. Solving such ill-posed inference problems requires relying on prior beliefs or experiences to navigate ambiguity and uncertainty, which is why it is necessary to use a strategy like variational inference [21].

2.6 | The Extended Mind and the Cognitive Niche

The extended mind theory, initially proposed by philosophers Clark and Chalmers [42], asserts that cognitive processes can be 'uploaded' to the external environment, effectively expanding the cognitive system beyond the agent's internal states. This extension is not merely a metaphor, but a literal expansion of cognitive processes into the external world. The implication is that the cognitive unit is not confined to the individual, but extends into the environment and across individuals. This idea has been adopted by biologists that favour approaches such as radical interactionism and embodied cognition [21, 40, 43, 44], and has been applied with great success in the field of robotics [45, 46].

When knowledge is uploaded to the external environment, it becomes part of a shared system that multiple agents and technical systems can access, refine and build upon. This distributed nature allows the uploaded knowledge to be far more complex than any individual or system could embody alone [21]. For instance, in engineering workplaces, automated workflow and process documentation represent uploaded knowledge. These systems store the detailed steps of complex processes, allowing them to be followed and improved over time. This knowledge is not limited to software and hardware, but it is distributed across individuals and systems that understand how to interpret, use and modify these processes.

In biology, niche construction refers to how individuals actively shape their surroundings to enhance their abilities. The cognitive niche expands on this by considering the environment a

space where agents create and maintain cause-effect models to guide their actions [47]. Consider a shared workspace where employees gradually develop efficient shortcuts for accessing frequently used tools or resources. Initially, different individuals may experiment with various methods, leading to several possible approaches with similar probabilities of success. However, over time, as more people adopt a particular shortcut, it becomes more established and reliable, signalling to others that this is the most effective way to accomplish a task. This process creates a well-trodden path through the physical and conceptual workspace, where the chosen route encodes a reliable mapping between actions and outcomes (e.g., ‘If I follow this procedure, I’ll quickly access the necessary tools’). The cognitive niche thus becomes a collective memory and problem-solving resource, enabling agents to perform cognitive tasks that would be impossible without the integration and collaboration of multiple minds and tools.

3 | A Free-Energy Model of PHAS

This section presents a formal model of PHAS based on FEP formalisms. The mathematical notation draws from the existing FEP literature. However, most of the specific formulations used in the model do not appear in Friston’s own writings. Rather, they are broadly considered algebraic truisms that naturally follow from Friston’s work [11, 12, 17] and appear ubiquitously throughout the literature without any further attribution (for different variations, see Constant et al. [21, 48], Ramstead et al. [20, 49], Bruineberg et al. [50] and Constant et al. [29]). When a formulation used in the model can be attributed to a specific source, this will be explicitly stated in the text.

Simulations of agents interacting with each other and their environment through active inference supported the modelling process. The model was gradually evolved by iteratively modifying and experimenting with open-source simulation code in public repositories made available by Kaufmann et al. [51], McGregor et al. [52] and vanSchaik [53]. The resulting model is intended to be specific enough to be simulated, but also to support logical reasoning about the system being modelled. This article focuses primarily on the logical implications, and how the formal model can be used to derive practical PHAS principles to guide the design of interventions in engineering teams. Each equation represents a different aspect of PHAS, and can be rearranged algebraically to show the internal consistency of the model. This is then used to demonstrate how less intuitive aspects of PHAS follow from simpler principles. Importantly, the process enables the generation of testable hypotheses.

3.1 | The Generative Model

A generative model is typically expressed in terms of a likelihood and a prior term:

$$\underbrace{p(s, \eta)}_{\text{generative model}} = \underbrace{p(s | \eta)}_{\text{likelihood}} \underbrace{p(\eta)}_{\text{prior}} \quad (1)$$

The likelihood in Equation (1) represents the probability of experiencing certain sensations s (e.g., ‘sour’ or ‘sweet’) given

prior beliefs about the state of the world η (e.g., ‘eating lemon’ or ‘eating honey’), before making any observations.

3.2 | Modelling Uncertainty

Sensory perception is always mediated by a randomly fluctuating and noisy external world. This can be stated formally as follows:

$$s = g(\eta) + \omega \quad (2)$$

Equation (2) shows that sensory input s received from the environment, what Calvo-Amodio [4] refers to as signs in his PHAS framework, does not directly reflect the actual states of the world η , but rather some function g of those states filtered through a layer of noise and uncertainty ω . We can express this as a new PHAS principle:

Principle of Perceptual Uncertainty: A PHAS perceives its environment mediated through a boundary, with internal states shaped by noisy, uncertain inputs that obscure direct observation of the world.

3.3 | Entropy and Surprise

In our model, surprise S is a measure of how unlikely it is for a random variable X to take a specific value x , given a model μ of how X was generated. This is expressed algebraically as $S = -\ln p(x | \mu)$. Entropy is the expected or weighted average of surprise over time. Sensory input, or signs, s are the specific observed or experienced outcomes. When x is replaced by s , and assuming that the model μ is implicit in how sensations s are generated, the expression becomes $-\ln p(s)$, representing the surprise or unlikelihood of experiencing sensations s given the underlying model [11]. Equation (3) shows how surprise directly influences free energy, with higher surprise leading to increased free energy:

$$\underbrace{F}_{\text{Free-energy of}} \underbrace{(s, \mu)}_{\substack{\text{sensations and} \\ \text{internal states}}} = \underbrace{D}_{\text{KLDiv}} \left[\underbrace{q_{\mu}(\eta)}_{\substack{\text{variational density} \\ \text{(approximate posterior)}}} \parallel \underbrace{p(\eta | s)}_{\substack{\text{over external states} \\ \text{true posterior}}} \right] \quad (3)$$

$\underbrace{-\ln p(s)}_{\text{surprise}}$

The Kullback–Leibler divergence (KLDiv) in Equation (3) is a measure of how a probability distribution diverges from a reference distribution. Minimising the KLDiv between the approximate and true posterior means the system’s internal model is more aligned with reality. This allows us to reverse the likelihood from the generative model in Equation (1) and for the agent to infer beliefs about the causes of sensations (e.g., ‘was it a lemon or honey, given the sensation of sourness or sweetness’). If the KLDiv becomes zero, the approximate and true posteriors become identical, signifying that the system has developed a perfect model of the world, capable of inferring the causes of its sensations without error. However, the nature of sensation

as described in Equation (2) ensures that such a state can never be reached.

Formulating the generative model from Equation (1) in terms of minimising free energy in Equation (3) effectively transforms the generative model into an optimisation problem. To emphasise this, we add an optimisation process to Equation (3) that shows the system actively working to minimise the divergence:

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \| p(\eta | s)]}_{\text{Divergence}} - \underbrace{\ln p(s)}_{\text{Surprise}} \quad (4)$$

$$\mu^* = \arg \min_{\mu} \text{Divergence}$$

We can restate Equation (4) as a new PHAS principle:

Principle of Alignment: A PHAS actively works to minimise the divergence between its internal model and the external world. By reducing uncertainty and aligning its model with reality, the PHAS improves its ability to infer the causes of its input, moving towards a perfect representation of its environment.

3.4 | Action Selection

Agents in the model act based on prior beliefs about expected or preferred outcomes, and actions are selected to reduce surprise by bringing about anticipated results. The next action is chosen by minimising free energy as a function of the agents internal model and the current input caused by the previous action:

$$a^* = \arg \min_a F(s(a), \mu) \quad (5)$$

Action minimises surprise by increasing the accuracy of sensory inputs, so we can restate Equation (4) as follows:

$$F(s, \mu) = \underbrace{D[q_\mu(\eta) \| p(\eta)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{q_\mu(\eta)} [\ln p(s|\eta)]}_{\text{Accuracy}} \quad (6)$$

$$a^* = \arg \max_a \text{Accuracy}$$

Note that while minimising free energy, Equation (6) does not consider posterior beliefs in the KLDiv as in Equation (4). This is because actions can only influence the system's free energy by changing inputs, not by directly altering internal beliefs. Preferred actions are those geared towards selecting outcomes that maximise expected accuracy by reducing the expected ambiguity of future input. Expressed as a PHAS principle:

Principle of Epistemic Foraging: A PHAS selects actions that maximise the expected accuracy of future input, not by altering internal beliefs but by reducing the ambiguity of external input. Driving the system toward outcomes that align with its expectations enhances perceptual clarity, thus minimising free energy.

3.5 | Niche Construction

The simplest way for an agent to perform the optimisation process described in Equation (6) is to selectively sample only input that it expects, thus reducing the probability of unexpected outcomes (surprise). However, a more sophisticated strategy is to modify the environment and assert more direct control over future input [43]. Constant et al. [21] propose exploiting the formal symmetry of the Markov blanket to reverse Equation (3) and model cognitive niche construction as minimising free energy from the perspective of the niche:

$$\underbrace{F}_{\text{Free-energy of}} \underbrace{(a, \eta)}_{\substack{\text{agents actions and} \\ \text{states of the niche}}} = \underbrace{D}_{\text{KLDiv}} \left[\underbrace{q_\eta}_{\substack{\text{variational density} \\ \text{(approximate posterior)}}} \underbrace{(\mu)}_{\substack{\text{belief about agent's} \\ \text{internal states}}} \| \underbrace{p(\mu | a)}_{\text{true posterior}} \right] - \underbrace{\ln p(a)}_{\text{affordance}} \quad (7)$$

$$F(a, \eta) = \underbrace{D[q_\eta(\mu) \| p(\mu | a)]}_{\text{Divergence}} - \underbrace{\ln p(a)}_{\text{Affordance}} \quad (8)$$

$$\eta^* = \arg \min_{\eta} \text{Divergence}$$

Equations (7) and (8) show how the environment works to minimise environmental surprise, denoting how likely or natural specific actions are within a given environment. The environment essentially 'learns' which actions are more likely to occur based on the states of the environment and the agent. From this perspective, the surprise term can be interpreted as a mathematical model of affordance [21].

The symmetry between Equations (3) and (4) and Equations (7) and (8) demonstrates that the niche and its inhabitants can become mutually predictable [21, 49], forming a joint system where both the environment and the agents influence each other based on mutual self-evidencing. Mutual predictability arises because the coupling through the Markov blanket makes the mapping between external signals and internal states statistically regular. As these regularities stabilise, each side can forecast the other's effects via the interface without violating conditional independence. The environment predicts and influences the actions an agent is most likely to take, making certain behaviours more probable and thus more 'afforded' by the environment.

Principle of Coupled Adaptation: A PHAS co-evolves with its environment, wherein the environment minimises environmental surprise by affording specific actions, thus influencing agents to adopt some behaviours over others within a given context, guiding the system towards mutually predictable interactions.

3.6 | Peer Interaction

While the original model also recognises PHAS as a composite socio-technical system, the model presented here makes this explicit by adding a peer model. Simulations by Kaufmann et al. [51] show that active inference agents reach their target faster and

more directly when their internal model includes a shared goal, and perform even better if they maintain a model of their peer, that is, a Theory of Mind.

The model does not formally distinguish between an agent's peers and other external systems, viewing them as equally opaque. However, because the intention is to incorporate the Intentional Stance and intentional systems into our model, it is helpful to adopt different notations. We can then re-purpose Equation (2) to describe the nature of incoming communication:

$$c_{in} = g(\psi) + \omega \quad (9)$$

Equation (9) states that incoming communication c_{in} is a function of the peer's internal states ψ perceived through a layer of noise and uncertainty ω . Re-purposing Equation (5), we see that an agent chooses what to communicate by minimising free energy as a function of what it perceives about its peer's response to the previous communication:

$$c_{out}^* = \arg \min_{c_{out}} F(c_{in}(c_{out}), \mu) \quad (10)$$

As with Equation (6), we can state this as maximising accuracy:

$$F(c_{in}, \mu) = \underbrace{D[q_\mu(\psi) \| p(\psi)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{q_\mu(\psi)} [\ln p(c_{in}(c_{out}) | \psi)]}_{\text{Accuracy}} \quad (11)$$

$$c_{out}^* = \arg \max_{c_{out}} \text{Accuracy}$$

Equation (11) shows how the agent directs communication to maximise the accuracy of its peer model and incoming communication.

3.7 | Norms, Conventions and Shared Expectations

We use Φ in our model to denote a set of shared norms, conventions and expectations:

$$\Phi = f(\psi)\gamma + (1 - \gamma)\mu \quad (12)$$

Equation (12) is based on simulation code by Kaufmann et al. [51] and shows how the agent constructs Φ by combining its understanding of its peer's views $f(\psi)$ with its own model of the world μ , and assigning a weight γ to balance authority between the two.

3.8 | Subsystems and Supersystems

Because free energy is an additive and conserved property, the model applies equally to subsystems and supersystems:

$$F_{\text{super}} = \sum_{i=1}^N F_i(c_{in_i}, s_i, \mu_i) \quad (13)$$

Equation (13) states that the free energy of a supersystem is equal to the sum of the free energy of its subsystems, which means that a PHAS can be modelled independently at multiple scales of the component hierarchy [49]. This has important implications

because it allows us to simplify complex systems by modelling them at a higher level.

Equation (14) extends this summative property to shared norms and expectations:

$$\Phi = \sum_{i=1}^N f(\psi_i)\gamma_i + (1 - \gamma_i)\mu \quad (14)$$

Equation (14) expresses the same relationship as shown in Equation (12), but summed over an arbitrary number of N peers. The agent constructs its internal model Φ of shared norms, conventions and expectations by combining its understanding of all of its peers' individual views $f(\psi_i)$ with its own model of the world μ , and assigning a weight γ_i to denote how much authority assigned to each peer relative to the agents own beliefs.

3.9 | Finalising the Formal Model

To summarise, we combine the equations and superimpose them on an expanded version of the Markov blanket from Figure 3. The result is an active inference model describing the dynamics of learning, perception, and niche construction in PHAS, shown in Figure 4. For simplicity, the figure only shows a single agent with a single peer.

4 | Interpreting and Operationalising the Model

At this point, it is helpful to return to Dennett's loans of intelligence. In biology and cognitive science, it is common to interpret the agent's internal model as identical to its physical representation of information, primarily its brain [54]. This has drawn criticism from researchers favouring a radical externalist approach to cognitive science [44]. After all, advanced collective cognition exists in organisms like slime mould, which entirely lack a sophisticated nervous system. Furthermore, attributing cognitive mechanisms to non-living systems like the physical work environment could be framed as excessively anthropomorphic [21]. Such criticisms are due to a misunderstanding of what the generative model is and what it is intended to do. The generative model is not something internal to the components of a PHAS, but merely a mathematical construct that describes an observable phenomenon [49].

Further, intentional systems can be wrong, even if we assume they will act rationally in the context of their own beliefs, as those beliefs may be false. Humans often trust the information our peers and external systems provide, even when this is maladaptive and irrational [55–57]. So, when the model discusses *choosing* and *optimising*, it is important to remember that these are local optima in a specific context.

Finally, it should be emphasised that the model does not consider all inference processes and internal representations equally sophisticated. That a sentient engineer and the layout of a workshop can be described using the same formalism does not confer sentience on the environment, nor does it entail any form of group consciousness.

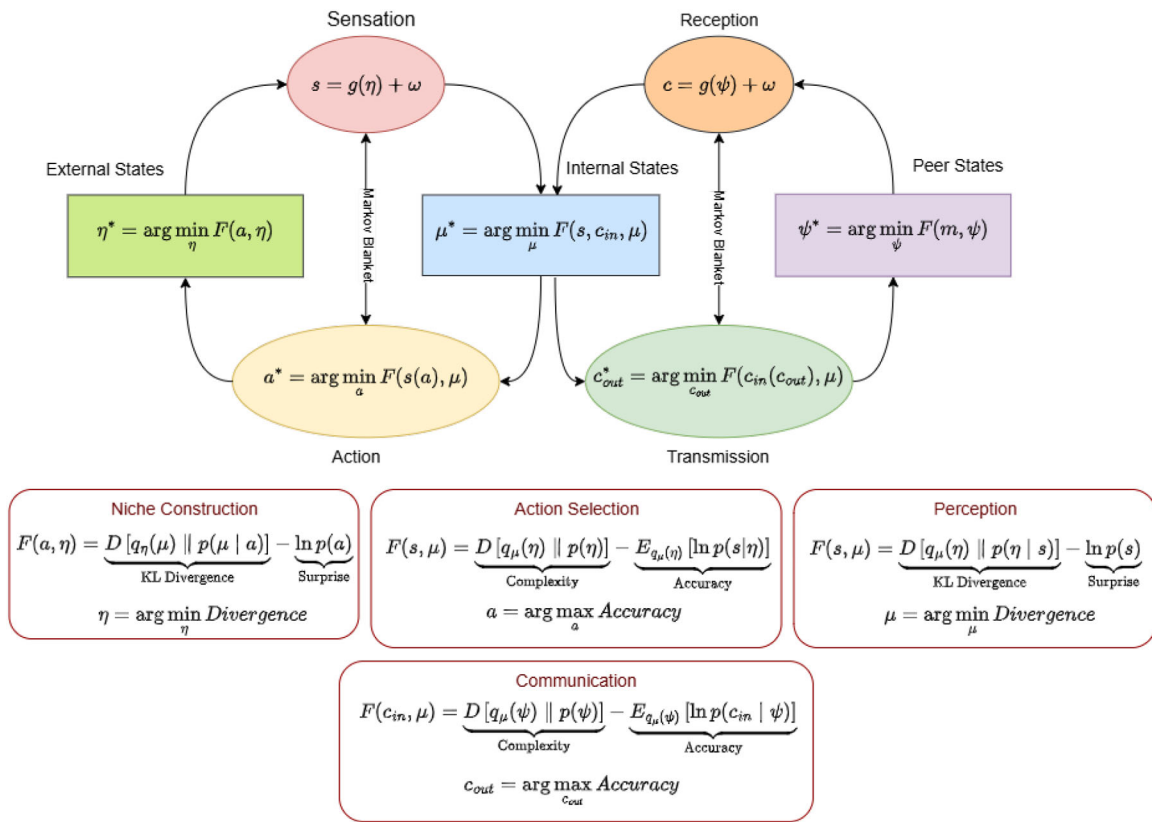


FIGURE 4 | Active inference model of PHAS.

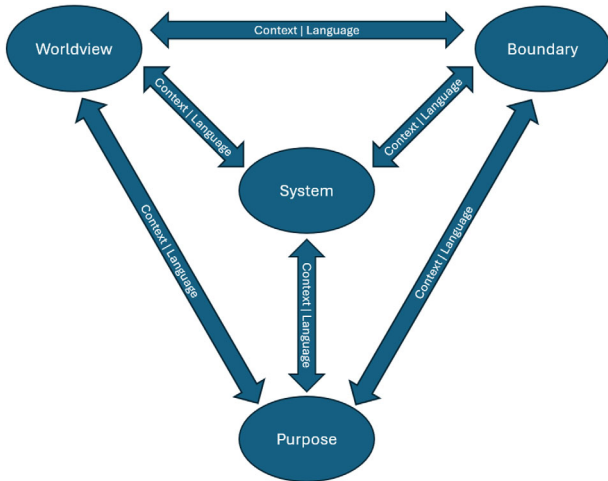


FIGURE 5 | PHAS expanded to include worldviews.

4.1 | A PHAS Model With Worldviews

Adopting the Intentional Stance when analysing PHAS allows for a more intuitive and comprehensive understanding of how human participants behave within these systems, emphasising the uniquely human aspects that drive actions and decisions. Multiple frameworks exist that can be used to operationalise Dennett's strategy, including the worldview framework proposed by Rousseau and Billingham [13]. Figure 5 incorporates worldview as a first-class component of the PHAS model, and Table 3 lists a set of principles for the new component to supplement the original PHAS principles in Table 1.

4.2 | A Revised PHAS Framework

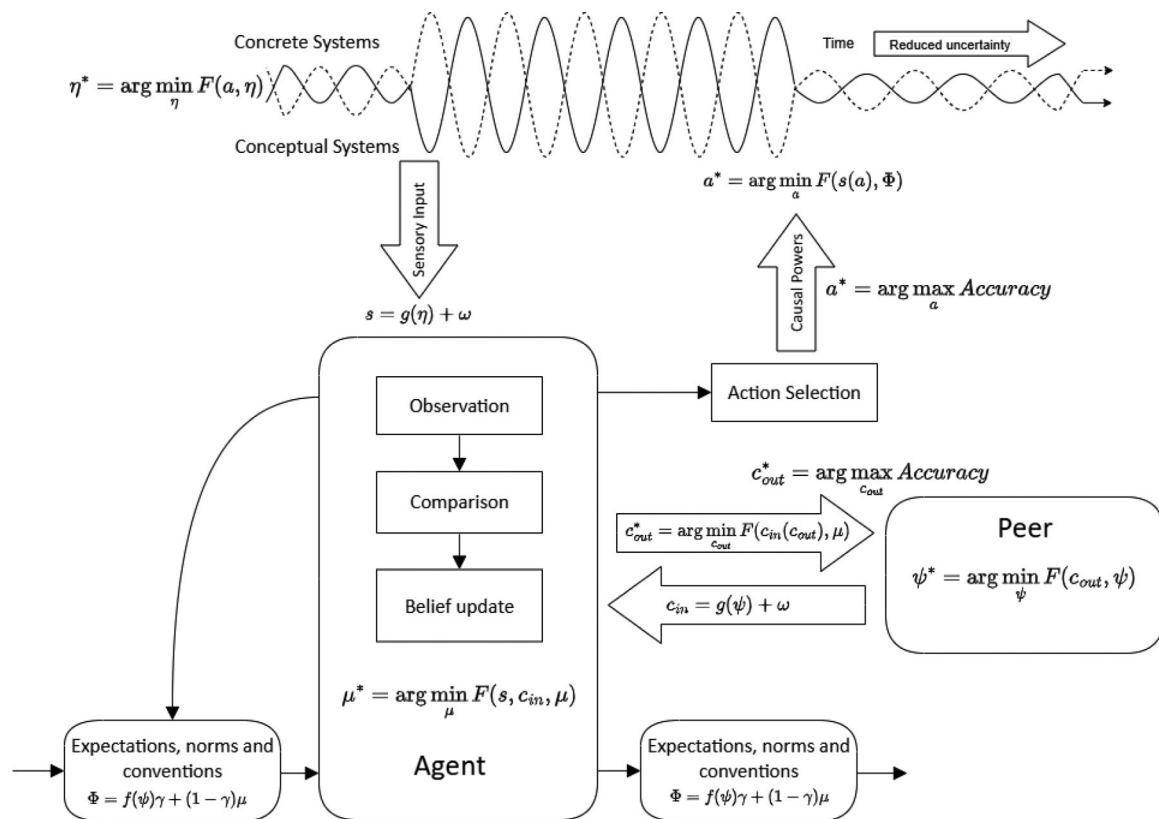
Calvo-Amodio [4] based his visualisation of PHAS on Checkland's description of an appreciative system [15]. We will do the same here, making it easier to see where the current model expands on or deviates from the original PHAS framework. Figure 6 shows a visualisation of the PHAS framework that integrates the new formal model. The internal structure of the peer model is the same as the agent but is shown as semi-opaque for simplicity. The new model does not differentiate between conceptual and concrete systems as Calvo-Amodio does. This is because the Markov blanket obscures from view the states of other systems. This does not preclude the existence of concrete systems, but it does mean nothing can be said about them that is not inferred from mediated perception and conceptual systems. However, like the distinction between human and non-human components of a PHAS, it is helpful to keep the terms to distinguish between the theoretical and the physical/practical.

Action is moved outside the agent, emphasising that this is a coupling with the external environment and not something internal to the agent. Sensory input serves the same role as signs in Calvo-Amodio's model, with perception being the internal processing of that input to update the internal model. Moving action makes the mechanisms behind action and sensory perception symmetric, the same as in the formal model. The decision to act in the context of FEP can be viewed as exceeding an energy threshold.

Finally, Calvo-Amodio's operational principles are broadened into expectations, norms and conventions to emphasise that they can be unintentional as well as deliberately adopted.

TABLE 3 | Summary of the new set of principles for the component ‘Worldview’.

Component	Relevant principle
Worldview	5. Human activity systems are intentional systems with a worldview 5.1 Human activity systems perceive their environment mediated through a boundary, with internal states shaped by noisy, uncertain inputs that obscure direct observation of the world 5.2 Human activity systems maintain and continuously optimise a generative model of the world (i.e., their worldview) to minimise uncertainty and support progress towards their purpose 5.3 Through action, human activity systems shape their environment to better reflect their internal models and support their objectives 5.4 The intentional stance can be applied both by human activity systems and to them, facilitating prediction, coordination and mutual understanding among systems 5.5 Human activity systems operate within and as a multi-scale hierarchy, where individual agents, teams or groups maintain their generative models while contributing to the stability and purpose of the integrated whole, preserving free energy across scales

**FIGURE 6** | Expanded PHAS framework.

4.3 | Operationalising the Formal Model

In the formal model, attention, viewed as a form of action, is optimally directed towards surprising input because these indicate a significant divergence between the agent's expectations and the actual environment. Within the free-energy framework and active inference, a signal's salience is determined by its estimated precision, that is, its signal-to-noise ratio. Signals with high precision, whether due to their high intensity, clear contrast or low noise, carry more weight in the perceptual inference process because their prediction errors are considered more reliable. Additionally, top-down influences can enhance the precision of

behaviourally relevant signals, further amplifying their salience over less reliable inputs. To operationalise this, we re-purpose the concepts *Regimes of Attention* and *Patterned Practices* from Ramstead et al. [28] as prescriptive tools for directing attention in PHAS.

Regimes of Attention are the environmental features, such as social cues, architectural designs and digital interfaces, that naturally steer focus toward information most pertinent for reducing uncertainty. They function as a guiding scaffold, ensuring that salient signals are readily detected within a complex operational landscape.

Patterned Practices are the regular, often ritualised, interactions with these environmental cues. By engaging in Patterned Practices, agents repeatedly interface with systems that embed crucial knowledge, thereby internalising these patterns. When these practices are shared among team members, they foster joint attention and synchronisation of internal models, promoting shared intention. Over time, this internalisation prompts agents to act in accordance with collective expectations, even in the absence of direct cues.

Whereas Calvo-Amodio's framework focuses on operational principles, the approach advocated here is to design interventions in terms of Regimes of Attention and Shared Patterned Practices. Table 4 follows the same pattern as Calvo-Amodio with inquiry questions, but with other questions that reflect the difference in focus of the two PHAS frameworks.

5 | Case Study

5.1 | Background and Starting Point

Very small entities (VSEs) are defined as organisations comprised of up to 25 people [58], but depending on the country, 80%–95% of OECD companies have less than nine employees [59]. VSEs dominate the global supply chain but often face limited financial resources, a lack of in-house expertise outside their own domain and a constrained capacity to address complex problems stretching beyond their core specialisation [60, 61].

The nine-person VSE described in this case study develops outdoor building and facility automation solutions, providing remote access through 4G modems. The company had recently experienced multiple security breaches where attackers used their modems to send spam, depleting customer data quotas and locking users out. These incidents caused operational, reputational and financial losses, making immediate security improvements essential. Despite recognising the urgency, the company lacked security expertise and the means to hire dedicated staff. Consequently, they needed to come up with sustainable security solutions aligned with their constraints.

5.2 | Methodology

The study design utilised a participatory approach conducted over a 2-year period. The engineering team used the inquiry questions in Table 4 to guide the development of interventions during a series of collaborative workshops.

Data collection relied on timestamped artefacts retrieved from GitHub and internal communication systems. Quantitative metrics focused on Dependabot alert histories to measure the total number of detected vulnerabilities and the median time to remediation. Qualitative data derived from meeting notes, issue trackers and chat logs provided context for decision-making. The analysis also examined documented reasons for deferring fixes to understand how the team prioritised security tasks.

TABLE 4 | Inquiry questions to guide the design of Regimes of Attention and Shared Patterned Practices.

Inquiry question	Information components and comments
1. What inputs or signals are necessary to achieve the desired outcome?	– What specific information or signals does the agent need to receive to achieve the defined goal? – How do these inputs ensure that the agent is adequately informed to make decisions that lead to successful outcomes? – Are these signals reliable and consistent across different contexts?
2. How do agents interact with these signals?	– What are the established patterns or practices that agents use to engage with these inputs? – How do these practices contribute to minimising uncertainty? – Are these interactions standardised or flexible to adapt to different scenarios?
3. What practices ensure the sustainability of attention?	– How are Regimes of Attention embedded in the environment? – What are the social and cultural mechanisms that reinforce these practices? – How do these practices evolve to remain effective over time?
4. Who finds these signals and practices meaningful and why?	– Which stakeholders consider these signals and practices significant, and what value do they derive from them? – Why are these signals and practices important within the social, organisational or psychological context? – How do these signals relate to broader system goals?
5. What defines success, and how do we know when it is achieved?	– What are the criteria for success in this context? – How do we determine when the desired outcomes have been achieved? – What balance is required between effort and outcome to consider the process effective and efficient?

5.3 | Step 1: Determining Necessary Inputs for Achieving Security Outcomes

Inquiry question: What inputs or signals are necessary to achieve the desired outcome?

The team began by identifying the key security-related information they needed to create more secure systems. It was clear that timely and actionable information about vulnerabilities within their software was crucial. The team needed a system to continuously monitor the codebase for security issues and provide clear guidance.

Guided design: The inquiry led the team to explore tools that could automate the detection of security vulnerabilities and provide clear remediation steps. This exploration brought GitHub's Dependabot to their attention, a free service that automatically scans repositories for vulnerabilities and generates proposed fixes [62]. If a potential fix exists, all unit tests are automatically re-run with the patched code, and the engineer is presented with a full test report before approving any changes to the main code base. The service is accessible via the repository website and the GitHub API, and provides real-time Common Vulnerability Scoring System (CVSS) v4 metrics [63] and links to Common Weakness Enumeration (CWE) [64] and Common Vulnerability Enumeration (CVE) [65] database entries. This means that if there are no known fixes or these are incompatible with the code base, the engineer is presented with the necessary metrics to evaluate the severity of the issue and can decide on any mitigations. Of course, not all vulnerabilities are known, but the moment one is officially registered, it will be available to the engineering team within minutes.

5.4 | Step 2: Understanding Stakeholder Interaction With Inputs

Inquiry question: How do people interact with these signals?

Next, the team needed to consider how to effectively utilise the security information provided by Dependabot. Recognising that not all team members would naturally pay attention to or prioritise security alerts, the team sought to create a mechanism to make these alerts more visible and impactful.

Guided design: The inquiry led the team to explore embedding security alerts into the daily workflow to ensure immediate visibility. The team ultimately decided on a smart bulb connected to the GitHub API that changed colour based on the severity of vulnerabilities flagged by Dependabot (shown in Figure 7). The bulb would glow orange for low-severity issues or those in development and blink red for high-severity vulnerabilities in production. The team preferred this approach over alternatives like email notifications, alarm logs or GitHub's website alerts, which require proactive engagement, such as checking inboxes and logging into specific systems. These methods compete for attention in a cluttered digital environment, often making it challenging to distinguish critical signals from background noise. In contrast, the smart bulb acted as a salient, shared sensor seamlessly integrated into the physical workspace, capturing attention immediately without requiring conscious effort. By



FIGURE 7 | Smart bulb blinking red for high-severity vulnerabilities.

externalising security signals in the environment, the system reduced the team's cognitive load, allowing them to stay focused on their tasks while being aware of potential threats.

5.5 | Step 3: Ensuring Sustainability of Attention Through Practices

Inquiry question: What practices ensure the sustainability of attention?

The team recognised that maintaining security awareness over time would require more than responding to alerts. They needed to embed security considerations into their regular workflow and team culture.

Guided design: They decided to incorporate serious games into their routine. These are Patterned Practices designed to be entertaining, but also to achieve specific educational or organisational objectives, often by embedding critical knowledge within an engaging framework [66]. The team started playing *Elevation of Privilege* (EoP), a card game designed to facilitate collaborative threat modelling [67]. By regularly playing this game, the team could practice identifying and mitigating security threats, embedding security awareness into their culture. The game also served as a training tool, gradually building the team's security knowledge and making them more proficient at recognising potential vulnerabilities.

5.6 | Step 4: Understanding the Importance of Signals and Practices to Stakeholders

Inquiry question: Who finds these signals and practices meaningful, and why?

The team needed to ensure that their designed interventions would be meaningful and valuable to all stakeholders, not just the technical team members.

Guided design: This inquiry led the team to consider the broader impact of their security practices. They realised that the visibility provided by the smart bulb would be important not just for

developers but also for management and other non-technical staff. The bulb's status provided a clear and immediate indication of the system's security health, which helped ensure that everyone in the company understood the importance of addressing vulnerabilities promptly. This alignment across the organisation reinforced security as a priority, making the interventions more effective and sustainable.

5.7 | Step 5: Defining and Recognising Success

Inquiry question: What defines success, and how do we know when it is achieved?

Finally, the team needed to define what success would look like and how they would measure it.

Guided design: The team established several success criteria: reduced unresolved high-severity vulnerabilities, timely responses to security alerts and improved team performance playing EoP. These criteria allowed the team to track progress and ensure that their security interventions were implemented effectively. Combining automated tools and regular gaming sessions created a balanced approach, ensuring that security improvements were sustained over time without overwhelming the team's resources.

5.8 | Discussion of Case Study

Unlike process mandates, awareness training or tool-only automation, the PHAS-FEP approach treats attention as the scarce control variable and Worldview as a design target. It assumes that people can misinterpret or deprioritise rules, so it shapes the environment to make the preferred action obvious and easy to choose. Regimes of Attention and Patterned Practices operationalise this: Dependabot alerts were externalised as a shared visual cue with the smart bulb, and regular EoP sessions synchronised how signals were interpreted and prioritised. This uses the Intentional Stance explicitly, aligns beliefs and shortens decision time. The behaviour persisted without more meetings or bureaucracy because salience and shared meaning were embedded in the workspace. The team did not just adapt to their environment, they shaped it, creating a feedback loop where the environment continuously reinforces security-focused behaviours.

Over a period of 2 years, 90 vulnerabilities were detected by the automated system across four repositories. Five issues required the intervention of an engineer, and two remained unfixed due to dependency issues. All 88 fixable issues were patched within 24 h of their CVE being listed publicly. For context, a total of 40,009 CVEs were published in 2024, on average 108 per day [68, 69]. This pattern is consistent with attention being reliably captured at the moment of need, with a shared belief that next-day remediation is the default. While the study design does not isolate causal effect against a counterfactual process-only baseline, the results are what the model predicts when salience is shared and practices align peer models.

For similar VSE contexts, the practical difference is that the PHAS-FEP approach tells you to spend limited resources on shaping shared salience and patterned practice before you add more rules or tools. If you cannot make the right thing the easiest thing to notice and do, your process will drift.

6 | Conclusions and Further Work

This paper re-examined and expanded the original PHAS model and framework by integrating the FEP from computational biology into a formal model of cognitive agents. In this revised model, agents actively navigate, interact with and reshape their environment by minimising uncertainty and surprise through continuously updated predictive models that capture both individual and shared worldviews. The model's symmetry, where the environment in turn models the agent, provides a rigorous formal description of affordances and context shaping.

The paper further illustrated the practical application of the revised framework in a case study of an engineering team in a very small enterprise. Facing repeated cybersecurity breaches that threatened operational, reputational and financial stability, the team employed inquiry-driven design principles to identify critical security signals. They implemented automated vulnerability monitoring and externalised these alerts with a smart bulb that visually signalled severity levels, thereby reducing cognitive load and ensuring immediate attention. Additionally, by incorporating patterned practices, such as regular serious game sessions for collaborative threat modelling, the team embedded a sustainable security culture. This integrated approach not only enhanced their responsiveness to security threats but also validated the effectiveness of the revised PHAS framework in real-world settings. While research in live industrial settings seldom allows fully controlled experiments, future work could pursue comparisons with a process-only baseline and test generalisability beyond this single VSE case.

The article focused primarily on using the formal model as a tool for precise logical reasoning and algebraic manipulation. The simulation code referenced in the text is fairly rudimentary, and future research should look at more complex and realistic scenarios where researchers can explore hypotheses using simulations and later validate these in the field. With the popularisation of large language models (LLMs) like GPT-5, advanced tools for modelling systems based on generative models such as those described here are becoming more accessible and affordable. These technological advancements have the potential to significantly enhance the ability of systems engineers to model human activities and interactions, providing more effective tools for designing, supporting and optimising complex socio-technical systems such as engineering teams.

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Data Availability Statement

Data are available from the author(s) with the permission of Aiwell AS. Restrictions apply to the availability of these data, which were used under license for this study.

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A1 Formal Proofs and Computational Results

A1.0 Technical Setup and Proof Methodology

This appendix documents the technical infrastructure used to produce, verify, and reproduce the formal analysis in Chapter 5. Every theorem in Appendices A1.1 through A1.7 has been machine-checked, and every computational artefact (figure, table, sensitivity grid) is generated by deterministic scripts that can be re-run from a single command. All source code, Lean proof files, and Python generation scripts are available in the public project repository at <https://github.com/Roarge/phd-dependable-smartcity>. Figures and result tables that fit within the document are presented inline in each appendix section. Large output files (sensitivity grids, full manifests) are available in the repository.

Proof Environment

Lean 4 and Mathlib

The formal proofs are written in Lean 4, a dependently-typed programming language and interactive theorem prover. Lean enforces totality and logical consistency: a theorem is accepted only when the proof term type-checks against its stated proposition.

Component	Version
Lean	4.27.0 (installed via elan)
Mathlib	v4.27.0 (pinned in lakefile.toml)
Lake build system	bundled with Lean 4.27.0

The shared Lean project root contains two libraries:

- **ComplexityModel** (`ComplexityModel.lean`): core definitions for configuration spaces, feasible distinct sets, complexity counts, and the operational functional information measure.
- **ComplexityBursts** (`ComplexityBursts.lean`): theorems on monotonicity, multiplicative growth, selection-pressure dynamics, jump formulae, and joint intervention decomposition.

Sections 5.2 through 5.7 each have a standalone Lean file (e.g., `Section5_2.lean`) that imports Mathlib directly and is type-checked using the `lake env lean` command. This design allows each section's theorems to use Mathlib while keeping proof files modular.

Proof completeness guarantee

No sorry axiom appears in any proof file. A sorry would introduce an unproven assumption and is treated as a build failure. The pipeline enforces this: if any theorem is left unproven, the Lean build halts with a non-zero exit code, and no subsequent computation runs.

The full build compiles approximately 7,900 Lean elaboration jobs (including Mathlib dependencies) and verifies 35 theorems across 8 Lean files.

Theorem inventory

Table 25 Theorem inventory

Section	Source file	Theorems	Proof lines
5.1	<code>ComplexityBursts.lean</code>	A1, A2, A3, A4, B1, B2, B3, C1, C2 (9)	11–240
5.2	<code>Section5_2.lean</code>	T5.2-1, T5.2-2, T5.2-3, T5.2-4 (4)	140–341
5.3	<code>Section5_3.lean</code>	T5.3-1, T5.3-2, T5.3-3, T5.3-4, T5.3-5 (5)	272–471
5.4	<code>Section5_4.lean</code>	T5.4-1, T5.4-2, T5.4-3, T5.4-4 (4)	159–263
5.5	<code>Section5_5.lean</code>	T5.5-1, T5.5-2, T5.5-3, T5.5-4, T5.5-5 (5)	75–203
5.6	<code>Section5_6.lean</code>	T5.6-1, T5.6-2, T5.6-3, T5.6-4 (4)	91–309
5.7	<code>Section5_7.lean</code>	T5.7-1, T5.7-2, T5.7-3, T5.7-4 (4)	99–324
Total		35	

All source files are available in the project repository.

Computational Environment

Python

Component	Version
Python	3.12
numpy	2.2.2
pandas	2.2.3
matplotlib	3.10.0
sympy	1.13.3
Jinja2	3.1.6

Dependencies are pinned in `requirements.txt` (included in the repository).

Determinism

All Python scripts enforce full determinism to guarantee that repeated runs produce identical outputs:

Setting	Value	Purpose
Random seed	20260208	Reproducible stochastic trajectories
PYTHONHASHSEED	0	Deterministic hash ordering in Python
Matplotlib backend	Agg	Non-interactive rendering, no GUI dependency

These settings ensure that every figure, table, and numeric summary is bitwise-reproducible across runs.

Script inventory

Section	Script	Description
5.1	simulate_bursts.py	Burst trajectories, option-growth and joint-intervention figures, burst metrics, proof status
5.2	generate_section5_2.py	Resilience advantage heatmap, parameter table, manifest
5.3	generate_section5_3.py	Response-time drag curves, mobilisation inset, manifest
5.4	generate_section5_4.py	Threshold shrinkage and I_g^* steps, I_g^{op} under time pressure, synthetic performance table, manifest
5.5	generate_section5_5.py	PHAS-EAI assurance loop, precision-accuracy demo, manifest
5.6	generate_section5_6.py	Experience-effect curves, sensitivity-decay surface, sensitivity grid (9,801 rows), manifest
5.7	generate_section5_7.py	Patterns-as-operators diagram, joint intervention demo, manifest

Pipeline Architecture

Execution model

Each section has a dedicated shell script that performs the following steps in order:

1. **Lean proof compilation.** Section 5.1 builds the two shared libraries. Sections 5.2 through 5.7 each type-check their standalone Lean file.
2. **Virtual environment setup.** Creates a Python virtual environment if absent, activates it, and installs pinned dependencies.
3. **Python generation.** Runs the section's generator script with PYTHONHASHSEED=0.

The pipeline uses `set -euo pipefail`, so any failure at any stage halts execution immediately. This guarantees that computational results are never generated from a codebase with unproven theorems.

Orchestration

The top-level script `run_all.sh` calls each section script in sequence. A complete run from clean state takes several minutes (dominated by the Lean build). Subsequent runs with cached Lean artefacts complete in seconds.

Output structure

Each section writes its results into figures (PDF and PNG), tables (CSV or JSON), and manifests (JSON). The manifests record theorem status, assumption lists, parameter values, numeric summaries, and caption drafts.

Relationship Between Proofs and Computations

The formal proofs and computational scripts serve complementary roles:

- **Lean proofs** establish that the claimed mathematical properties hold universally under their stated assumptions. They verify monotonicity, boundedness, decomposition, and threshold behaviour in closed form.
- **Python scripts** instantiate the models with concrete parameter values, compute numeric outputs, and generate visual artefacts. They demonstrate that the proven properties manifest in representative scenarios.

Each appendix section (A1.1 through A1.7) presents both the formal proof record and the computational results. Result tables and figures that fit within the document are presented inline. Large output files (for example, the 9,801-row sensitivity grid from Section A1.6) are available in the repository.

Source Code Repository

The public project repository at <https://github.com/Roarge/phd-dependable-smartcity> contains:

Path in repository	Contents
analysis/section5_1/setup/lean/	Lean project: ComplexityModel.lean, ComplexityBursts.lean,

Path in repository	Contents
	lakefile.toml, lean-toolchain, and supporting modules
analysis/section5_2/setup/proofs/	Section5_2.lean
analysis/section5_3/setup/proofs/	Section5_3.lean
analysis/section5_4/setup/proofs/	Section5_4.lean
analysis/section5_5/setup/proofs/	Section5_5.lean
analysis/section5_6/setup/proofs/	Section5_6.lean
analysis/section5_7/setup/proofs/	Section5_7.lean
analysis/section5_*/setup/scripts/	Python generator scripts for each section
analysis/requirements.txt	Python dependency pins
run_all.sh, run_section5_*.sh	Pipeline orchestration scripts

Reproducibility

To reproduce all results from the repository:

1. Install Lean 4.27.0 via elan (see <https://github.com/leanprover/elan>).
2. Ensure Python 3.12 with venv support is available.
3. Clone the repository and run `./run_all.sh` from the root.

The pipeline will compile all Lean proofs, create the virtual environment, install dependencies, and generate all figures, tables, and manifests.

A1.1 Formal Proofs for Section 5.1: Complexity and Burst Dynamics

Definitions

The following core definitions are used throughout the complexity analysis. All definitions are given in the shared module `ComplexityModel.lean` (see the project repository).

Definition A.1.1 (Base-2 logarithm). $\log_2(x) = \ln(x)/\ln(2)$.

Definition A.1.2 (Feasible distinct set). Given a configuration space X , feasibility predicate K , and binning map $b: X \rightarrow B$, the feasible distinct set is:

$$S^A = b(\{x \in X \mid K(x)\})$$

Definition A.1.3 (Complexity). $C = |S^A|$ (Eq. 20).

Definition A.1.4 (Event-driven complexity process). A discrete-time stochastic process where:

$$C(0) = C_0, \quad C(t+1) = \begin{cases} C(t) \cdot k(t) & \text{if event}(t) \\ C(t) & \text{otherwise} \end{cases}$$

Definition A.1.5 (Operational functional information). From counts: $I_g^{\text{op}} = \log_2(C_{\text{eff}}) - \log_2(A_{\text{eff}})$. From share: $I_g^{\text{op}} = -\log_2(\rho^{\text{op}})$ where $\rho^{\text{op}} = A_{\text{eff}}/C_{\text{eff}}$.

A lemma (`operational_information_count_share_equiv`) proves that the count form and share form are equivalent under positive counts.

Theorem Group A: Monotonicity and Multiplicative Growth

Theorem T5.1-1 (Option-set expansion monotonicity). If $X_1 \subseteq X_2$, then $C(X_1, K, b) \leq C(X_2, K, b)$.

Proof sketch. Filtering by K preserves the subset relation. The image under b is therefore a subset, and $|S_1^A| \leq |S_2^A|$ follows from `Finset.card_le_card`. A strict variant additionally shows that if the expansion introduces at least one new feasible distinct bin, the inequality is strict.

Lean source: `ComplexityBursts.lean`, lines 11–46.

Theorem T5.1-2 (Constraint-tightening monotonicity). If $K_2(x) \Rightarrow K_1(x)$ for all x , then $C(X, K_2, b) \leq C(X, K_1, b)$.

Proof sketch. The tighter constraint filters to a subset of the looser constraint. Image cardinality is monotone in the domain set.

Lean source: ComplexityBursts.lean, lines 47–61.

Theorem T5.1-3 (Multiplicative composition). Under factorised constraints and factorised binning, if $S_{\text{tot}} = S_1 \times S_2$, then $|S_{\text{tot}}| = |S_1| \cdot |S_2|$.

Proof sketch. Direct application of Finset.card_product.

Lean source: ComplexityBursts.lean, lines 62–86.

Assumption: T5.1-3 exactness requires factorised constraints and factorised binning across subsystems. When constraints couple subsystems, the product factorisation is an upper bound, not an equality.

Theorem T5.1-4 (Discrete factor introduction burst formula). At an event time t : - If no event occurs: $C(t + 1) = C(t)$. - If an event occurs: $C(t + 1) = C(t) \cdot k(t)$ and $\Delta \log_2 C = \log_2 k(t)$.

Proof sketch. The no-event case follows from the definition. The event case uses $\log_2_jump_bits: \log_2(C \cdot k) - \log_2(C) = \log_2(k)$ by the logarithmic product rule. Requires $C(t) > 0$ and $k(t) > 0$.

Lean source: ComplexityBursts.lean, lines 87–114.

Theorem Group B: Selection Pressure and Operational Complexity

Theorem T5.1-5 (Selection-pressure monotonicity). If ρ^{op} is antitone in a preordered index, then $I_g^{\text{op}} = -\log_2(\rho^{\text{op}})$ is monotone (non-decreasing).

Proof sketch. Antitone ρ^{op} implies $\rho^{\text{op}}(t) \leq \rho^{\text{op}}(s)$ for $s \leq t$. Logarithm is order-preserving on positive reals. Division by $\log_2 > 0$ preserves the inequality. Negation reverses it.

Lean source: ComplexityBursts.lean, lines 115–136.

Theorem T5.1-6 (Discrete jump formula in bits). For a discrete change from ρ_- to ρ_+ :

$$\Delta I_g^{\text{op}} = \log_2 \left(\frac{\rho_-}{\rho_+} \right)$$

Proof sketch. Algebraic manipulation of $(-\log_2 \rho_+) - (-\log_2 \rho_-)$, then application of the logarithmic quotient rule.

Lean source: ComplexityBursts.lean, lines 137–152.

Theorem T5.1-7 (Jump discontinuity from selection pressure). If $\rho^{\text{op}}(t+1) < \rho^{\text{op}}(t)$, then I_g^{op} has a jump discontinuity at t , i.e., $I_g^{\text{op}}(t+1) \neq I_g^{\text{op}}(t)$.

Proof sketch. Strict inequality in ρ implies strict inequality in $\log$, which implies strict inequality in $-\log_2$, establishing $I_g^{\text{op}}(t+1) > I_g^{\text{op}}(t)$.

Lean source: ComplexityBursts.lean, lines 153–176.

Theorem Group C: Joint Intervention Decomposition

Theorem T5.1-8 (Symbolic decomposition). For simultaneous changes in effective counts:

$$\Delta I_g^{\text{op}} = \log_2 \left(\frac{C_{\text{after}}}{C_{\text{before}}} \right) - \log_2 \left(\frac{A_{\text{after}}}{A_{\text{before}}} \right)$$

Proof sketch. Ring arithmetic on the $\log_2 C - \log_2 A$ form, then application of the logarithmic quotient rule.

Lean source: ComplexityBursts.lean, lines 177–193.

Theorem T5.1-9 (Joint intervention additivity). If $C_{\text{eff}} \mapsto \alpha C_{\text{eff}}$ (with $\alpha \in (0,1)$) and $A_{\text{eff}} \mapsto \beta A_{\text{eff}}$ (with $\beta > 1$), then the joint reduction in I_g^{op} equals the sum of the single-channel reductions:

$$\Delta I_g^{\text{op,joint}} = (-\log_2 \alpha) + \log_2 \beta = \Delta I_g^{\text{op,C}} + \Delta I_g^{\text{op,A}}$$

Proof sketch. Each single-channel reduction is computed by the $\log_2(a \cdot b) = \log_2 a + \log_2 b$ rule. The joint reduction uses both simultaneously. Algebraic cancellation confirms additivity.

Lean source: `ComplexityBursts.lean`, lines 194–240.

Computational Results

The simulation instantiates the event-driven complexity process (Definition A.1.4) with three discrete burst events. The deterministic seed (20260208) and `PYTHONHASHSEED=0` ensure bitwise-reproducible outputs. The source script `simulate_bursts.py` is available in the project repository.

Burst Event Metrics

The following table records the event-level metrics for the three simulated bursts. Each row decomposes the change in operational functional information into its constituent terms, confirming the jump formulae proved in Theorems T5.1-6 and T5.1-8.

Event t	Change description	ρ_-^{op}	ρ_+^{op}	ΔI_g^{op} (bits)	$\log_2 C_-$	$\log_2 C_+$	$\log_2 \tilde{A}_-$	$\log_2 \tilde{A}_+$
20	Boundary degree introduction with coupling spillover	0.2267	0.1760	+0.365	8.618	9.406	6.477	6.899
45	Constraint and boundary commitment shift in governance and compliance	0.2002	0.0608	+1.720	9.934	9.998	7.613	5.958
70	Joint intervention on option count and mobilisation quality	0.1593	0.2672	\$-\$0.746	10.526	10.133	7.876	8.229

Event at $t = 20$: boundary degree introduction. A new coupling boundary is introduced, expanding the configuration space. Complexity increases from $2^{8.618} \approx 393$ to $2^{9.406} \approx 678$ distinct configurations. The effective adequate set also grows (from $2^{6.477}$ to $2^{6.899}$), but not

enough to offset the complexity increase. The operational success share ρ^{op} falls from 0.227 to 0.176, producing a net increase of 0.365 bits in I_g^{op} . This is the moderate-burst pattern: both numerator and denominator of ρ^{op} change, but complexity grows faster than adequacy.

Event at $t = 45$: governance and compliance shift. This event produces the largest burst in the simulation (+1.720 bits). The driving mechanism is primarily a collapse in effective adequate options: $\log_2 \tilde{A}$ falls from 7.613 to 5.958, a drop of 1.655 bits, while $\log_2 C$ barely changes (+0.065 bits). The operational success share drops sharply from 0.200 to 0.061. As Theorem T5.1-8 predicts, the decomposition yields $\Delta I_g^{\text{op}} = (9.998 - 9.934) - (5.958 - 7.613) = 0.065 + 1.655 = 1.720$ bits. This event illustrates a regime where constraint shifts destroy adequacy without meaningfully expanding the option space.

Event at $t = 70$: joint intervention. This is a remedial intervention that simultaneously reduces complexity (from $2^{10.526}$ to $2^{10.133}$, a reduction of 0.393 bits) and increases effective adequate options (from $2^{7.876}$ to $2^{8.229}$, an expansion of 0.353 bits). The joint effect is a net reduction of 0.746 bits in I_g^{op} , and the operational success share improves from 0.159 to 0.267. The decomposition confirms Theorem T5.1-8: $\Delta I_g^{\text{op}} = (10.133 - 10.526) - (8.229 - 7.876) = -0.393 - 0.353 = -0.746$ bits, demonstrating that both the choice-reduction channel and the success-expansion channel contribute to the improvement.

Generated Figures

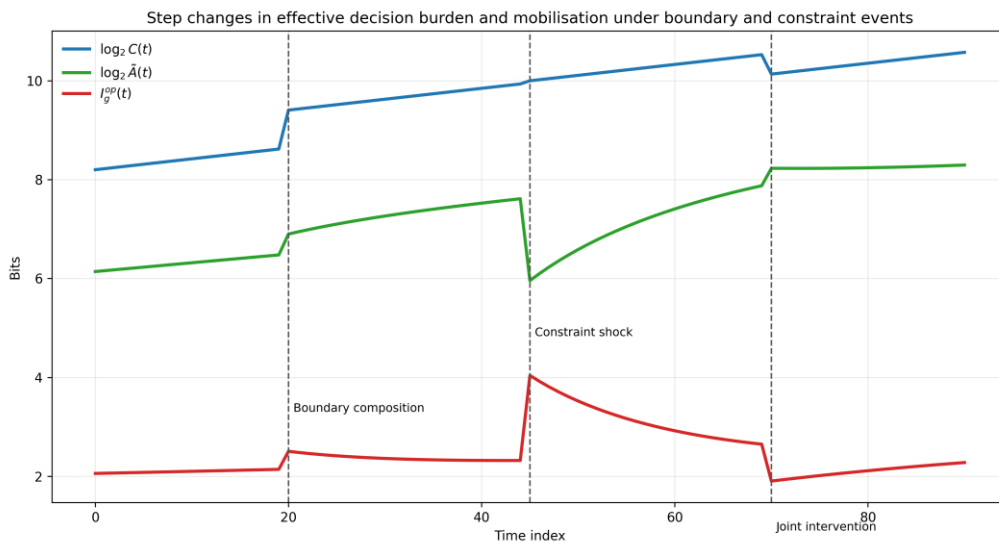


Figure 13 Stochastic complexity trajectories under event-driven growth.

Stochastic complexity trajectories under event-driven growth, with piecewise-constant segments between events and multiplicative jumps at each event time.

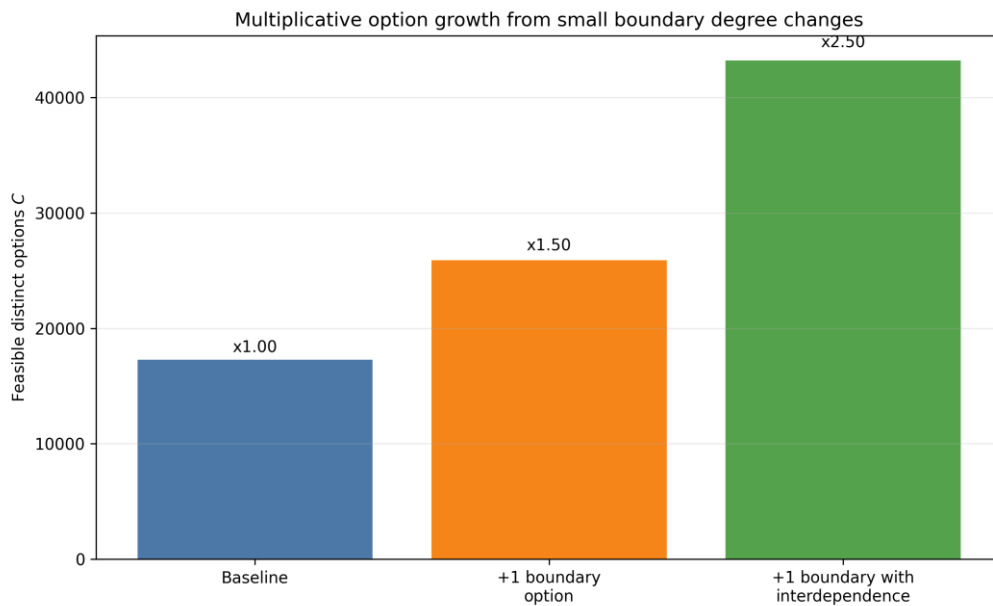


Figure 14 Multiplicative option-set growth under successive factor introductions.

Multiplicative option-set growth under successive factor introductions, illustrating Theorem T5.1-3 (multiplicative composition) and Theorem T5.1-4 (burst formula).

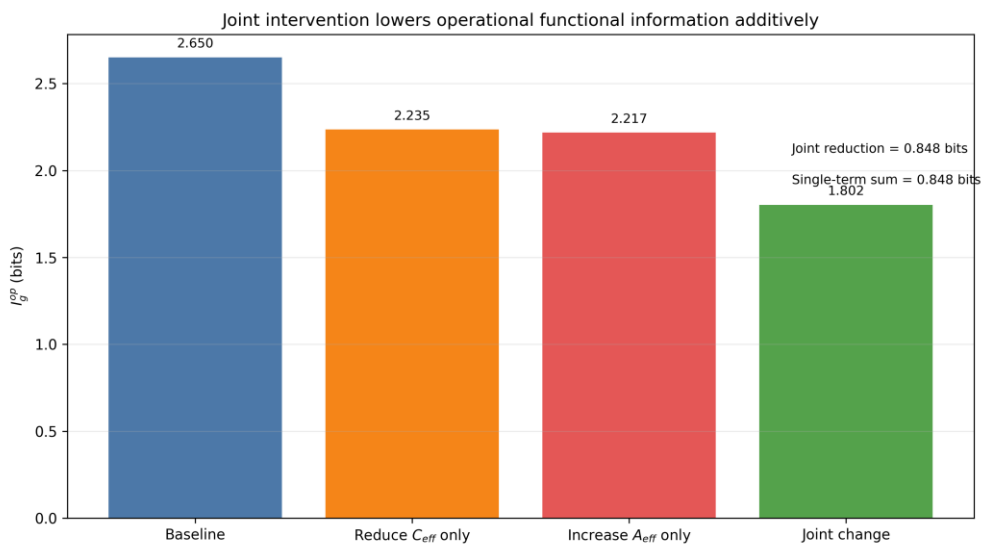


Figure 15 Joint intervention effect on operational functional information.

Joint intervention effect on I_g^{op} , demonstrating the sign reversal at $t = 70$ where coordinated option pruning and adequacy expansion jointly reduce the information burden.

Multiplicative Composition Table

The following table illustrates Theorem T5.1-3 (multiplicative composition) for three representative configurations. Each row shows a system with n weakly coupled subsystems,

each offering m alternatives, giving $C = m^n$ feasible distinct options. Even modest per-subsystem option counts produce large combined spaces.

n (subsystems)	m (alternatives each)	$C = m^n$	$\log_2 C$ (bits)	$C > m + n$?
3	4	64	6.00	Yes
5	3	243	7.92	Yes
7	2	128	7.00	Yes

In every case the multiplicative total far exceeds the additive count $m + n$, confirming that even binary choices across seven factors yield a search space of 128 configurations.

Proof Status

All nine theorems (T5.1-1 through T5.1-9) compiled without sorry. The derived jump formulae confirmed by the simulation are:

Formula	Expression
Complexity burst	$\Delta \log_2 C = \log(k)/\log(2)$
Selection-pressure jump	$\Delta I_g^{\text{op}} = \log(\rho_-/\rho_+)/\log(2)$
Joint intervention	$\Delta I_g^{\text{op,joint}} = -\log(\alpha)/\log(2) + \log(\beta)/\log(2)$

Assumptions

ID	Assumption	Justification
S5.1-A1	Finite decision resolution induces finite configuration spaces	Model axiom (Eq. 15)
S5.1-A2	Distinct options are equivalence classes induced by the binning map	Definition (Eq. 17)
S5.1-A3	Constraint tightening means $K_2 \Rightarrow K_1$ at configuration level	Logical definition
S5.1-A4	T5.1-3 exactness requires factorised constraints and binning	Coupling breaks exactness
S5.1-A5	All log arguments are strictly positive	Domain guard

ID	Assumption	Justification
S5.1-A6	Selection pressure is modelled by antitone ρ^{op}	Monotonicity assumption
S5.1-A7	Event times are discrete	Model simplification
S5.1-A8	Joint intervention uses $\alpha \in (0,1)$ and $\beta > 1$	Domain constraint

A1.2 Formal Proofs for Section 5.2: Comparative Resilience Advantage

Definitions

Definition A.2.1 (Organisation). A record comprising useful variety A , survivable variety B , headroom H , and effective mobilisation time τ_{eff} .

Definition A.2.2 (Mobilisation fraction). $f_A(\tau_d, \tau_{\text{eff}}, \varepsilon) = \min(1, (\tau_d/\tau_{\text{eff}})^\varepsilon)$ (Eq. 35).

Definition A.2.3 (Time-conditioned counts).

$$\tilde{A} = A \cdot f_A, \quad \tilde{B} = B \cdot f_B$$

(Eqs. 37a, 37b).

Definition A.2.4 (Magnitude limits).

$$M_0 = k \cdot \tilde{A}^{\alpha_A} \cdot H^{\beta_H}, \quad M_{\text{max}} = k \cdot \tilde{B}^{\beta_B} \cdot H^{\beta_H}$$

(Eqs. 38, 38a).

Definition A.2.5 (Resilience score). Piecewise-linear form:

$$R(M) = \max\left(0, \min\left(1, (M_{\text{max}} - M)/(M_{\text{max}} - M_0)\right)\right)$$

(Eq. 39).

Definition A.2.6 (Comparative advantage field). $\Delta R(\tau_d, M) = R_S(\tau_d, M) - R_L(\tau_d, M)$.

Theorems

Theorem T5.2-1 (Resilience score boundedness and boundary values). For $M_0 < M_{\max}$:

1. $0 \leq R(M) \leq 1$. 2. $M \leq M_0 \Rightarrow R(M) = 1$. 3. $M_{\max} \leq M \Rightarrow R(M) = 0$.

Proof sketch. Case (1) follows from the $\max(0, \min(1, \cdot))$ clamp. For (2), the numerator exceeds the denominator, so the ratio is at least 1 and $\min(1, \cdot) = 1$. For (3), the numerator is non-positive, so $\max(0, \cdot) = 0$.

Lean source: Section5_2.lean, lines 140–165.

Theorem T5.2-2 (Fast-regime VSE advantage). Under the scarce-time decomposition, if the multiplicative factor condition holds:

$$\text{factor}_{M_0} = \left(\frac{A_S}{A_L}\right)^{\alpha_A} \cdot \left(\frac{\tau_{\text{eff},L}}{\tau_{\text{eff},S}}\right)^{\alpha_A \varepsilon_A} \cdot \left(\frac{H_S}{H_L}\right)^{\beta_H} > 1$$

(and analogously for $\text{factor}_{M_{\max}}$), then $M_{0,S} > M_{0,L}$ and $M_{\max,S} > M_{\max,L}$.

Proof sketch. The factor decomposition rewrites $M_{0,S} = M_{0,L} \cdot \text{factor}_{M_0}$. Since $M_{0,L} > 0$ and $\text{factor} > 1$, the product exceeds $M_{0,L}$.

Lean source: Section5_2.lean, lines 166–190.

Note. Exponents $\alpha_A, \beta_B, \beta_H, \varepsilon_A, \varepsilon_B$ are typed as Nat in the Lean formalisation. This is a modelling simplification. See Assumption A6 in the Chapter 5 Assumptions document.

Theorem T5.2-3 (Slow-regime reversal). When mobilisation saturates ($\tau_{\text{eff}} \leq \tau_d$), the magnitude limits reduce to static forms. Under $A_S < A_L$ and $H_S \leq H_L$:

$$M_{0,S}^{\text{static}} \leq M_{0,L}^{\text{static}}$$

The proof uses: - mobilisation_eq_one_of_saturated: $\tau_{\text{eff}} \leq \tau_d$ implies $f_A = 1, f_B = 1$. - Power monotonicity: $A_S^{\alpha_A} \leq A_L^{\alpha_A}$ from $A_S \leq A_L$ and non-negative base. - Product monotonicity across the $k \cdot A^{\alpha_A} \cdot H^{\beta_H}$ form.

This establishes that the larger organisation dominates in the slow regime.

Lean source: Section5_2.lean, lines 191–324.

Theorem T5.2-4 (Parity boundary existence). There exist parameter values such that $\Delta R > 0$ in the fast regime and $\Delta R < 0$ in the slow regime. By the intermediate value theorem, a parity boundary τ_d^* exists.

Constructive parameter witness: - VSE: $A = 2, B = 6, H = 0.9, \tau_{\text{eff}} = 1$. - Large: $A = 4, B = 8, H = 1.0, \tau_{\text{eff}} = 5$. - Exponents: $\alpha_A = \beta_B = \beta_H = \varepsilon_A = \varepsilon_B = 1$.

The proof establishes: - For $M \in [2.1, 2.3]$: $\Delta R(0.5, M) > 0$ (VSE advantage under fast disturbances). - For $M \in [2.1, 2.3]$: $\Delta R(10, M) < 0$ (large-organisation advantage under slow disturbances). - By continuity of ΔR in τ_d (stated as a hypothesis), IVT gives $\tau_d^* \in [0.5, 10]$ with $\Delta R(\tau_d^*, 2.2) = 0$.

Assumption. Continuity of ΔR in τ_d is stated as a hypothesis (hCont), not derived from first principles. See Assumption A7 in the Chapter 5 Assumptions document.

Lean source: Section5_2.lean, lines 325–341.

Computational Results

The generator script (generate_section5_2.py, available in the project repository) computes ΔR on a deterministic (τ_d, M) grid and extracts the parity boundary by sign-change interpolation. Determinism settings: seed = 20260208, PYTHONHASHSEED=0, Agg backend.

Parameter Values

The two organisations are parameterised as follows, matching the constructive witness used in Theorem T5.2-4:

Parameter	VSE (S)	Large (L)
Useful variety A	2.0	4.0
Survivable variety B	6.0	8.0
Headroom H	0.9	1.0
Effective mobilisation time τ_{eff}	1.0	5.0

Shared exponents: $\alpha_A = \beta_B = \beta_H = \varepsilon_A = \varepsilon_B = 1$, scale factor $k = 1.0$.

Numeric Results

Metric	Value
Parity boundary τ_d^* estimate (at reference $M = 2.2$)	2.475
Positive-advantage area (grid units)	9.883
Maximum advantage ΔR	1.0
Location of maximum advantage (τ_d, M)	(0.06, 0.1)

Interpretation

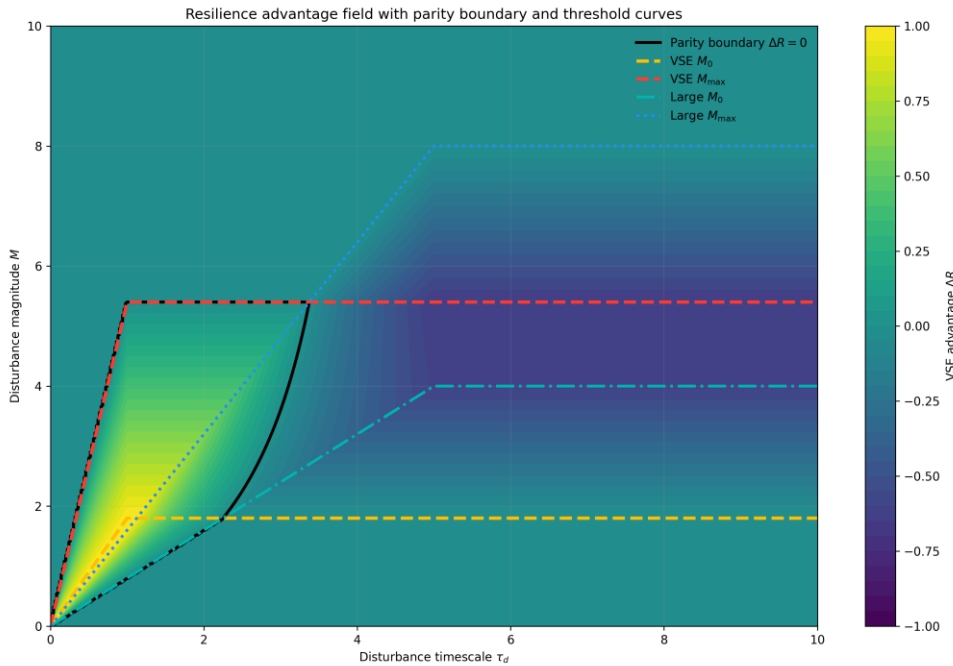


Figure 16 Comparative resilience advantage field for VSE and large-organisation archetypes.

Comparative resilience advantage field $\Delta R(\tau_d, M)$. Positive values indicate VSE advantage under fast disturbances. The zero contour marks the parity boundary.

Figure 16 maps the comparative advantage field $\Delta R(\tau_d, M)$ as a heatmap. Positive values (VSE advantage) dominate the lower-left quadrant where disturbances are fast (τ_d small) and moderate in magnitude (M low to mid-range). Negative values (large-organisation advantage) dominate the upper-right quadrant where disturbances are slow and large.

The zero contour marks the parity boundary. At the reference magnitude $M = 2.2$, the estimated parity timescale is $\tau_d^* \approx 2.475$. Below this threshold, the VSE's faster mobilisation ($\tau_{\text{eff},S} = 1.0$ vs $\tau_{\text{eff},L} = 5.0$) offsets its lower raw variety ($A_S = 2$ vs $A_L = 4$). This is the mechanism formalised in Theorem T5.2-2: the mobilisation-speed ratio amplifies the VSE's effective variety in the fast regime.

Above τ_d^* , mobilisation saturates. Both organisations reach $f_A = f_B = 1$, and the magnitude limits reduce to their static forms (Theorem T5.2-3). In this regime, the larger organisation's greater useful and survivable variety dominates unconditionally.

The maximum advantage $\Delta R = 1.0$ occurs at $(\tau_d, M) = (0.06, 0.1)$, where the VSE achieves full resilience ($R_S = 1$) while the larger organisation has not yet mobilised its resources. The positive-advantage area of 9.883 grid units indicates that the VSE advantage region is substantial, not a narrow edge case.

These results confirm the regime structure predicted by the formal proofs: fast disturbances favour agile organisations, slow disturbances favour resource-rich ones, and the parity boundary separates the two regimes.

Assumptions

ID	Assumption	Justification
S5.2-A1	All parameters and timescales are non-negative	Domain constraint
S5.2-A2	Gap condition $M_0 < M_{\max}$ for piecewise boundary claims	Model setup
S5.2-A3	Fast-regime conditions encoded through factor decomposition	Algebraic representation
S5.2-A4	Slow-regime assumes mobilisation saturation $f_A = f_B = 1$	Regime definition
S5.2-A5	Continuity of ΔR in τ_d (hypothesis in T5.2-4)	Loan of intelligence
S5.2-A6	Exponents typed as Nat rather than Real	Formalisation simplification

A1.3 Formal Proofs for Section 5.3: Mobilisable Variety Under Time Pressure

Definitions

Definition A.3.1 (Stage response time with structural drag). For detection, decision, or execution stage s with n complexity-potential components:

$$\tau_s = \tau_s^0 \cdot \left(1 + \sum_{i=1}^n \gamma_{s,i} \cdot CP_i \right)$$

(Eq. 31).

Definition A.3.2 (Total response time).

$$\tau = \tau_{\text{detect}} + \tau_{\text{decide}} + \tau_{\text{execute}}$$

(Eq. 28).

Definition A.3.3 (Skill multiplier).

$$\sigma_\tau = 1 - \lambda_\tau \cdot u(E, MF)$$

with $\lambda_\tau \in [0,1]$ and $u \in [0,1]$ (Eq. 32).

Definition A.3.4 (Effective mobilisation time).

$$\tau_{\text{eff}} = \sigma_\tau \cdot \tau$$

(Eq. 33).

Definition A.3.5 (Time allowance ratio).

$$r = \tau_d / \tau_{\text{eff}}$$

(Eq. 34).

Definition A.3.6 (Cognitive headroom component).

$$H_{\text{cog}} = h + (1 - h) \cdot u \cdot q(IT)$$

(Eq. 26).

Definition A.3.7 (Magnitude threshold forms).

$$M_0 = k \cdot A_{\text{eff}}^{\alpha_A} \cdot H^{\beta_H}, \quad M_{\text{max}} = k \cdot B_{\text{eff}}^{\beta_B} \cdot H^{\beta_H}$$

(Eq. 38).

Theorems

Theorem T5.3-1 (Response-time monotonicity chain). 1. Each baseline stage time τ_s^0 strictly increases τ . 2. Increasing any CP_i (non-strictly) increases τ . 3. Increasing CP also increases τ_{eff} (via monotonicity in τ). 4. The lower bound $(1 - \lambda_\tau) \cdot \tau \leq \tau_{\text{eff}}$ holds.

Proof sketch. Stage time is $\tau_s^0 \cdot (1 + \text{dot})$. The dot product $\sum \gamma_i CP_i$ is non-negative (non-negative terms). Strict increase in τ_s^0 gives strict increase in the stage, hence in the sum. For CP monotonicity, dot product monotonicity from $\gamma_i \geq 0$ and $CP_i' \geq CP_i$ gives $\text{dot}' \geq \text{dot}$, and the product with $\tau_s^0 \geq 0$ preserves the ordering. The τ_{eff} chain follows from $\sigma_\tau \geq 0$ and the lower bound from $\sigma_\tau \geq 1 - \lambda_\tau$.

Lean source: Section5_3.lean, lines 272–333.

Theorem T5.3-2 (Ratio reduction collapses mobilisation). Given monotone mobilisation fractions f_A, f_B, f_C (saturating at 1 for $r \geq 1$): - If τ_d decreases and τ_{eff} increases, then r decreases. - Reduced r implies $f_A(r) \leq f_A(r')$, $f_B(r) \leq f_B(r')$, $f_C(r) \leq f_C(r')$. - If $r \geq 1$, all fractions equal 1 (saturation).

Proof sketch. The ratio monotonicity lemma `ratio_nonincrease` shows $\tau_d' / \tau_{\text{eff}}' \leq \tau_d / \tau_{\text{eff}}$ when $\tau_d' \leq \tau_d$ and $\tau_{\text{eff}}' \geq \tau_{\text{eff}}$ (with positivity guards). Monotone function application preserves the ordering.

Lean source: Section5_3.lean, lines 334–359.

Theorem T5.3-3 (Skill multiplier and effective-time interval).

$$\sigma_\tau \in [1 - \lambda_\tau, 1], \quad \tau_{\text{eff}} \in [(1 - \lambda_\tau)\tau, \tau]$$

Proof sketch. For σ_τ : the product $\lambda_\tau u$ is bounded by λ_τ (since $u \leq 1$) and non-negative (since both factors are non-negative). Therefore $0 \leq \lambda_\tau u \leq \lambda_\tau$, giving $1 - \lambda_\tau \leq 1 - \lambda_\tau u \leq 1$. The τ_{eff} interval follows by multiplication with $\tau \geq 0$.

Lean source: Section5_3.lean, lines 360–371.

Theorem T5.3-4 (Cognitive headroom lower bound and monotonicity). 1. $h \leq H_{\text{cog}}(h, u, q)$ for all $u, q \geq 0$. 2. $h_1 \leq h_2 \Rightarrow H_{\text{cog}}(h_1, u, q) \leq H_{\text{cog}}(h_2, u, q)$ for $u, q \in [0, 1]$.

Proof sketch. For (1): $H_{\text{cog}} = h + (1 - h) \cdot u \cdot q$. The remainder $(1 - h) \cdot u \cdot q \geq 0$ by repeated mul_nonneg, so $H_{\text{cog}} \geq h$.

For (2): $H_{\text{cog}}(h_2) - H_{\text{cog}}(h_1) = (h_2 - h_1)(1 - uq)$. Both factors are non-negative: $h_2 - h_1 \geq 0$ by hypothesis, and $1 - uq \geq 0$ since $u, q \in [0,1]$.

This is an instance of the affine lower-bound proof pattern documented in the project conventions.

Lean source: Section5_3.lean, lines 372–411.

Theorem T5.3-5 (Combined H5, H6, H7 propositions).

This theorem bundles the three hypothesis schemas: - **H5 (mobilisation collapse)**: reduced r lowers f_A, f_B, f_C and therefore M_0 and M_{max} . - **H6 (bounded skill effect)**: $\tau_{\text{eff}} \in [(1 - \lambda_\tau)\tau, \tau]$ for all $u \in [0,1]$. - **H7 (designed reserve)**: h lower-bounds H_{cog} and is monotone. The worst case ($u = 0$) gives $H_{\text{cog}} = h$, providing the witness.

Proof sketch. H5 chains ratio monotonicity through mobilisation fraction monotonicity, then through $M_0\text{Thresh_mono_Aeff}$ and $M_{\text{max}}\text{Thresh_mono_Beff}$ (power monotonicity of $k \cdot A^{\alpha_A} \cdot H^{\beta_H}$). H6 and H7 invoke the interval and lower-bound lemmas above. The worst-case witness for H7 is $u = 0$, where $H_{\text{cog}}(h, 0, q) = h$ by ring computation.

Lean source: Section5_3.lean, lines 412–471.

Computational Results

The generator script (generate_section5_3.py, available in the project repository) visualises the monotonicity chain from Theorems T5.3-1 and T5.3-2. Determinism settings: seed = 20260208, PYTHONHASHSEED=0, Agg backend.

Generated Figures

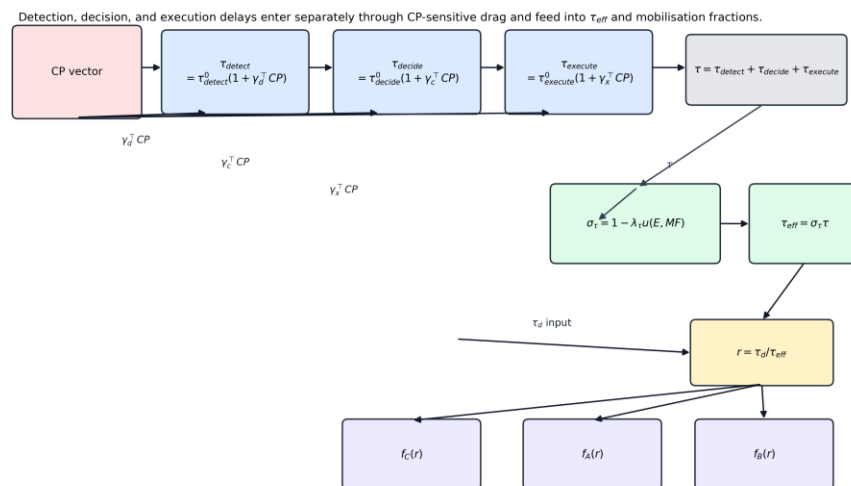


Figure 17 Response time and effective mobilisation time as functions of aggregate complexity potential.

Total response time and effective mobilisation time as functions of aggregate complexity potential, confirming that skill cannot eliminate structural drag (Theorems T5.3-1 and T5.3-3).

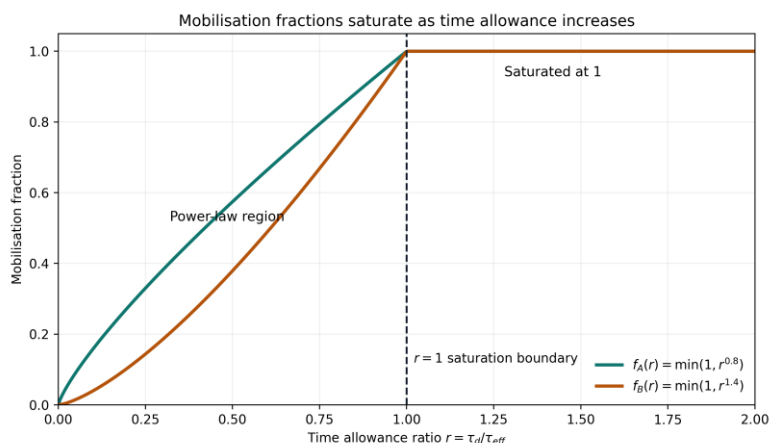


Figure 18 Mobilisation fraction as a function of the time allowance ratio.

Mobilisation fraction $f_A(r)$ as a function of the time allowance ratio, showing monotone rise to saturation at $r = 1$ (Theorem T5.3-2).

Figure 5-3 (inset): Mobilisation fraction as a function of the time allowance ratio.

Figure 5-3 (main panel) plots the total response time τ and the effective mobilisation time τ_{eff} as functions of the aggregate complexity potential CP . As CP grows, the structural drag term $\tau_0(1 + \gamma \cdot CP)$ increases τ linearly (Theorem T5.3-1, part 2). The effective time $\tau_{\text{eff}} = \sigma_\tau \cdot \tau$ scales proportionally, moderated by the skill multiplier $\sigma_\tau \in [1 - \lambda_\tau, 1]$ (Theorem T5.3-3). The figure shows that even at the maximum skill level ($u = 1, \sigma_\tau = 1 - \lambda_\tau$), the effective time

retains a non-trivial fraction of the raw response time, confirming that skill cannot eliminate structural drag.

Figure 5-3 (inset) shows the mobilisation fraction $f_A(r)$ as a function of the time allowance ratio $r = \tau_d/\tau_{\text{eff}}$. The curve rises monotonically from near zero (when $r \ll 1$, severe time pressure) to 1 (when $r \geq 1$, saturation). This confirms Theorem T5.3-2: reduced r collapses mobilisation, and saturation occurs at $r = 1$. The shape illustrates the mechanism by which structural drag (increasing τ_{eff}) reduces r at fixed τ_d and thereby lowers the fraction of useful variety that can be mobilised.

Proof Status

All five theorems (T5.3-1 through T5.3-5) compiled without sorry.

Assumptions

ID	Assumption	Justification
S5.3-A1	All baselines, CP components, and sensitivities are non-negative	Domain constraint
S5.3-A2	$u(E, MF) \in [0,1]$ and $q(IT) \in [0,1]$	Bounded utility and quality
S5.3-A3	$\lambda_\tau \in [0,1]$	Bounded speed-up fraction
S5.3-A4	Mobilisation functions are monotone in r and saturate at 1	Model axiom (Eq. 35)
S5.3-A5	Threshold forms use non-negative scale and headroom factors	Domain constraint

A1.4 Formal Proofs for Section 5.4: Functional Information as the Cost of Choice

Definitions

Definition A.4.1 (Threshold-qualified success set). Over a finite option space $\mathcal{O}^A$ with performance map Perf_g :

$$A_{g,\rho} = \{x \in \mathcal{O}^A \mid \rho \leq \text{Perf}_g(x)\}$$

(Eq. 40).

Definition A.4.2 (Success share). $p_g(\rho) = |A_{g,\rho}|/C$ (Eq. 41).

Definition A.4.3 (Functional information). $I_g^*(\rho) = \log_2 C - \log_2 |A_{g,\rho}|$ (Eq. 42).

Definition A.4.4 (Operational functional information).

$$I_g^{\text{op}}(\rho, \tau_d) = I_g^*(\rho) - \log_2 f_A(r)$$

(Eq. 44).

Definition A.4.5 (Performance image). The finite image $\text{Perf}_g(\mathcal{O}^A) \subset \mathbb{R}$ is the set of realised performance levels.

Theorems

Theorem T5.4-1 (Threshold monotonicity and set nesting). For $\rho_2 > \rho_1$ with $|A_{g,\rho_2}| > 0$:
 1. $A_{g,\rho_2} \subseteq A_{g,\rho_1}$ (set nesting). 2. $|A_{g,\rho_2}| \leq |A_{g,\rho_1}|$ (count ordering). 3. $I_g^*(\rho_1) \leq I_g^*(\rho_2)$ (information monotonicity).

Proof sketch. Nesting: any x with $\rho_2 \leq \text{Perf}_g(x)$ also satisfies $\rho_1 \leq \text{Perf}_g(x)$ since $\rho_1 < \rho_2$. Count ordering follows from `Finset.card_le_card`. Information monotonicity: smaller count means smaller $\log_2$, which means larger difference $\log_2 C - \log_2 |A|$.

Lean source: `Section5_4.lean`, lines 159–184.

Theorem T5.4-2 (Finite-image step behaviour and jump points). 1. $\text{range}(\text{Perf}_g)$ is finite (by `Set.finite_range` on a `Fintype`). 2. Between consecutive performance levels (no crossing): $|A_{g,\rho_1}| = |A_{g,\rho_2}|$ (piecewise-constant counts). 3. A change in count implies a crossing: $\exists v \in \text{Perf}_g(\mathcal{O}^A), \rho_1 \leq v < \rho_2$ (jump located at a realised performance level). 4. When counts differ and both are positive: $I_g^*(\rho_1) < I_g^*(\rho_2)$ (strict information increase at jump points).

Proof sketch. Part (2) uses contraposition: if no configuration's performance falls in $[\rho_1, \rho_2)$, then the success sets at ρ_1 and ρ_2 are equal (no element enters or leaves). Part (3): by contraposition of (2), a count change implies a crossing element exists. Part (4): strict count drop gives strict $\log_2$ inequality via `IStar_strict_of_count_drop`.

This is an instance of the finite-set threshold lemma pattern documented in the skills library.

Lean source: `Section5_4.lean`, lines 185–218.

Theorem T5.4-3 (Time pressure raises operational functional information). If $r_2 \leq r_1$ (time pressure increases), then $I_g^{\text{op}}(\rho, r_2) \geq I_g^{\text{op}}(\rho, r_1)$.

Proof sketch. Monotone f_A and $r_2 \leq r_1$ give $f_A(r_2) \leq f_A(r_1)$. Positive f_A allows $\log_2 f_A(r_2) \leq \log_2 f_A(r_1)$. Since $I_g^{\text{op}} = I_g^* - \log_2 f_A$, the subtraction inverts the inequality.

Lean source: `Section5_4.lean`, lines 219–234.

Theorem T5.4-4 (I_g^{op} depends on both size and time pressure). 1. Increasing C strictly increases I_g^* (complexity sensitivity). 2. Increasing τ_{eff} (reducing r) increases I_g^{op} (time-pressure sensitivity). 3. A complexity-only metric $\log_2 C$ is invariant to τ_{eff} (it misses operational difficulty).

Proof sketch. Part (1): $\log_2 C_1 < \log_2 C_2$ by strict log monotonicity. Part (2): chain through ratio decrease, mobilisation decrease, log decrease. Part (3): $\log_2 C$ does not reference τ_{eff} , so it is trivially invariant (`rf1`).

Lean source: `Section5_4.lean`, lines 235–263.

Computational Results

The generator script (`generate_section5_4.py`, available in the project repository) constructs a synthetic 64-option space and computes the threshold-driven step behaviour predicted by Theorems T5.4-1 and T5.4-2. Determinism settings: `seed = 20260208`, `PYTHONHASHSEED=0`, Agg backend.

Synthetic Performance Distribution

A finite option space with $C = 1024$ configurations and six distinct performance levels is used. The distribution is:

Performance level	Count of options	Cumulative from above
0.91	33	33
0.77	62	95
0.62	105	200
0.46	374	574
0.31	270	844
0.18	180	1024 (all)

The “cumulative from above” column gives $|A_{g,\rho}|$ for thresholds just above each level. For example, at $\rho = 0.77$, the success set contains 95 options (those with performance ≥ 0.77), and $I_g^* = \log_2(1024) - \log_2(95) = 10.0 - 6.570 = 3.430$ bits.

Step Behaviour

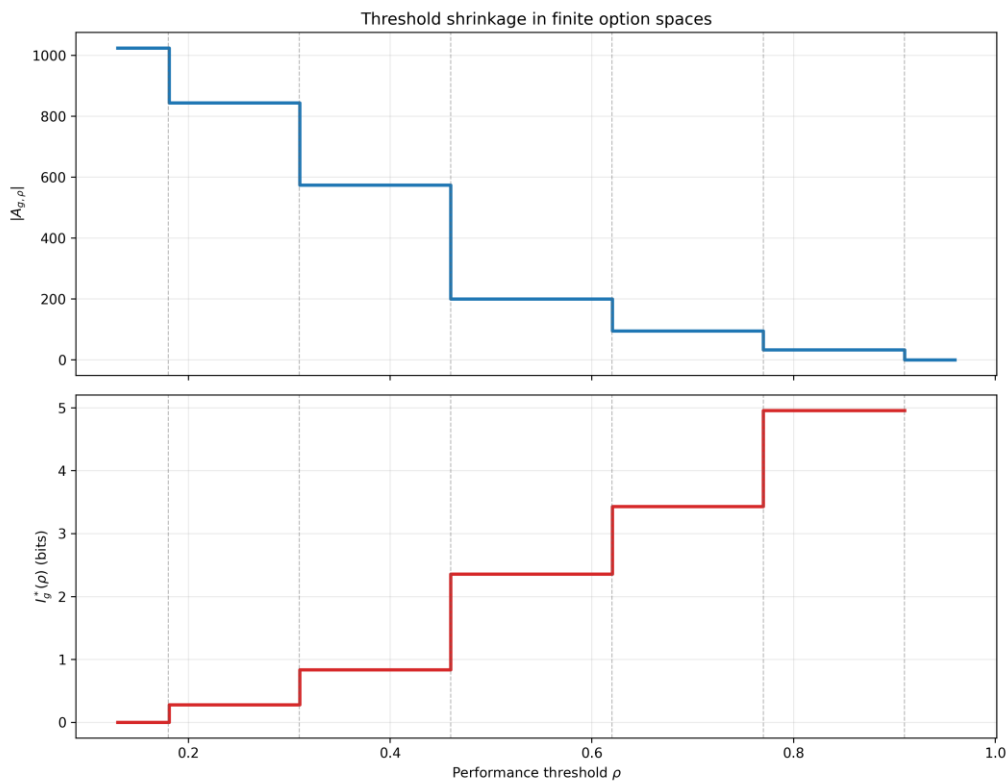


Figure 19 Success set cardinality and functional information as step functions of performance threshold.

Threshold-qualified success set cardinality $|A_{g,\rho}|$ (upper) and functional information $I_g^*(\rho)$ (lower) as step functions of the performance threshold, confirming Theorems T5.4-1 and T5.4-2.

Figure 19 shows two panels:

Upper panel: $|A_{g,\rho}|$ as a function of threshold ρ . The success-set cardinality is a non-increasing step function that drops at exactly the six realised performance levels. Between consecutive levels, the count is constant (Theorem T5.4-2, part 2: no crossing implies no count change). At each realised level, the count drops by exactly the number of options at that level.

Lower panel: $I_g^*(\rho)$ as a function of threshold ρ . The functional information is a non-decreasing step function that jumps at the same six threshold values. Each jump corresponds to the loss of a group of adequate options, increasing the information required to find a satisfactory configuration. The jumps are larger where more options are lost (for example, the jump at $\rho = 0.46$, where 374 options drop out, is larger than the jump at $\rho = 0.91$, where only 33 drop out).

This step-function structure is the computational manifestation of Theorem T5.4-2: the finite image of Perf_g determines exactly where the jumps occur, and between jumps the function is flat.

Time-Pressure Effect

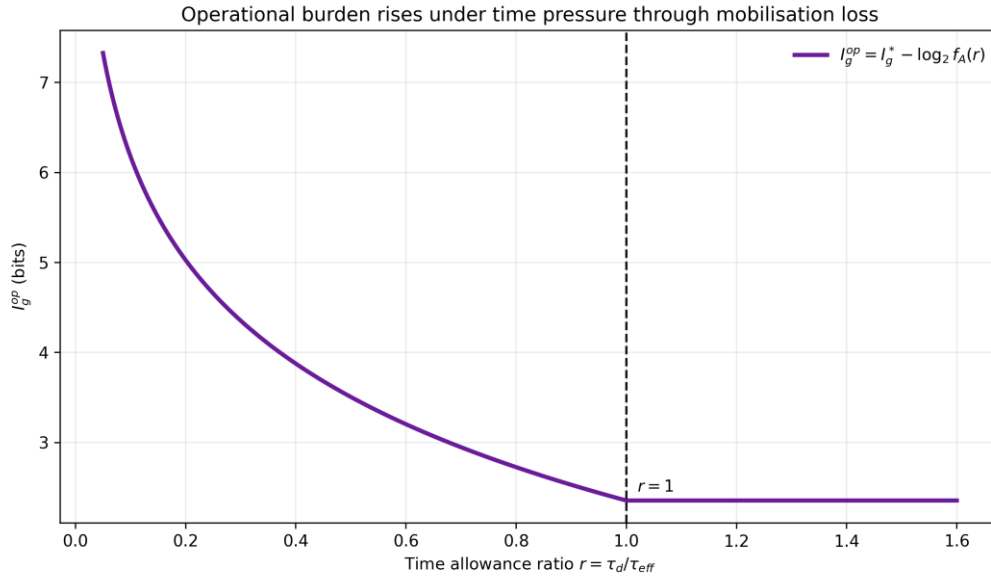


Figure 20 Operational functional information under varying time pressure.

Operational functional information I_g^{op} under varying time pressure, showing uniform lift above the static I_g^* baseline as r decreases (Theorem T5.4-3).

A supplementary figure shows I_g^{op} for the same option space under varying time pressure. As the time allowance ratio r decreases (greater pressure), the mobilisation penalty term $-\log_2 f_A(r)$ increases, lifting the entire I_g^{op} curve above the static I_g^* baseline. This confirms Theorem T5.4-3: time pressure raises operational functional information uniformly across all thresholds.

The reference parameters used are $\varepsilon = 1.15$, $C_{\text{ref}} = 1024$, $A_{\text{ref}} = 200$, giving a baseline $I_g^* = 2.356$ bits.

Proof Status

All four theorems (T5.4-1 through T5.4-4) compiled without sorry.

Assumptions

ID	Assumption	Justification
S5.4-A1	Option space $\mathcal{O}^A$ is finite and Perf_g has finite image	Fintype constraint
S5.4-A2	Threshold success set uses $A_{g,\rho} = \{x \mid \rho \leq \text{Perf}_g(x)\}$	Eq. 40
S5.4-A3	I_g^* uses positive-count domain for log claims	Domain guard
S5.4-A4	f_A is monotone in r and strictly positive	Model axiom
S5.4-A5	Time-pressure mapping through $r = \tau_d / \tau_{\text{eff}}$	Eq. 34

A1.5 Formal Proofs for Section 5.5: Attention, Authority, and Shared Expectations

Definitions

Definition A.5.1 (Sensory signal). $s = g(\eta) + \omega$ where g is the observation mapping and $\omega \sim \mathcal{N}(0, \sigma^2)$ (Eq. 2).

Definition A.5.2 (Shared expectation update). For prior belief μ , peer signal $f(\psi)$, and authority weight γ :

$$\Phi = \gamma \cdot f(\psi) + (1 - \gamma) \cdot \mu$$

(Eq. 12).

Definition A.5.3 (Multi-peer extension). With per-peer authority weights γ_i :

$$\Phi = \frac{1}{n} \sum_{i=1}^n [\gamma_i f(\psi_i) + (1 - \gamma_i) \mu]$$

(Eq. 14).

Definition A.5.4 (Posterior variance). Gaussian conjugate form:

$$\sigma_{\text{post}}^2 = \frac{\sigma_{\text{prior}}^2 \cdot \sigma_{\text{noise}}^2}{\sigma_{\text{prior}}^2 + \sigma_{\text{noise}}^2}$$

Definition A.5.5 (Affordance-augmented free energy). $F = D_{\text{KL}} - \ln p(a)$, where $p(a)$ is the affordance probability for action a (Eq. 8).

Definition A.5.6 (Expected Gaussian log-likelihood). Accuracy proxy:

$$\mathbb{E}[\ln p(s|\eta)] = -\frac{\ln(2\pi\sigma^2) + 1}{2}$$

Definition A.5.7 (Coordination friction). For two agents with different authority weights:

$$\text{Friction} = |\Phi_A - \Phi_B| = |\gamma_A - \gamma_B| \cdot |f(\psi) - \mu|$$

Theorems

Theorem T5.5-1 (Posterior uncertainty under noise). 1. If $\sigma_{\text{prior}}^2 > 0$ and $\sigma_{\text{noise}}^2 > 0$, then $\sigma_{\text{post}}^2 > 0$ (non-degenerate noise preserves uncertainty). 2. If $\sigma_{\text{prior}}^2 = 0$, then $\sigma_{\text{post}}^2 = 0$ (degenerate prior collapses posterior).

Proof sketch. Part (1): numerator is a product of two positives, and the denominator is their sum (also positive). Part (2): zero prior zeroes the numerator.

Lean source: Section5_5.lean, lines 75–89.

Theorem T5.5-2 (Shared expectations as convex mixtures). 1. $\Phi \in [\min(\mu, f(\psi)), \max(\mu, f(\psi))]$ for $\gamma \in [0, 1]$. 2. The multi-peer form is a weighted sum of per-peer convex mixtures.

Proof sketch. Part (1): rewrite $\Phi = \mu + \gamma(f(\psi) - \mu)$. For $\mu \leq f(\psi)$: $\gamma \geq 0$ gives $\Phi \geq \mu$ and $\gamma \leq 1$ gives $\Phi \leq f(\psi)$. The symmetric case uses $\Phi = f(\psi) + (1 - \gamma)(\mu - f(\psi))$. Part (2): by definition (rfl).

Lean source: Section5_5.lean, lines 90–132.

Theorem T5.5-3 (Affordance reduces free energy). If $p(a_1) < p(a_2)$ (both positive), then $F(a_2) < F(a_1)$.

Proof sketch. $\ln p(a_1) < \ln p(a_2)$ by strict log monotonicity. Since $F = D - \ln p(a)$, larger $\ln p$ gives smaller F .

Lean source: Section5_5.lean, lines 133–140.

Theorem T5.5-4 (Precision monotonicity). 1. $\partial \mathbb{E}[\ln p(s|\eta)] / \partial \sigma^2 = -1/(2\sigma^2) < 0$ for $\sigma^2 > 0$ (derivative is negative). 2. $\sigma_1^2 \leq \sigma_2^2 \Rightarrow \mathbb{E}[\ln p(s|\eta)]|_{\sigma_2^2} \leq \mathbb{E}[\ln p(s|\eta)]|_{\sigma_1^2}$ (lower noise improves accuracy).

Proof sketch. The derivative is clearly negative. For the ordering: $2\pi\sigma_1^2 \leq 2\pi\sigma_2^2$ gives $\ln(2\pi\sigma_1^2) \leq \ln(2\pi\sigma_2^2)$ by log monotonicity. The negation and division by 2 reverse the ordering.

This is an instance of the precision monotonicity pattern documented in the project conventions.

Lean source: Section5_5.lean, lines 141–167.

Theorem T5.5-5 (Authority-weight asymmetry and coordination friction). If $\gamma_A \neq \gamma_B$ and $f(\psi) \neq \mu$: 1. $\Phi_A \neq \Phi_B$ (expectations diverge). 2. Friction > 0 (non-zero coordination friction). 3. $|\gamma_A - \gamma_B| \leq |\gamma_A' - \gamma_B'| \Rightarrow \text{Friction}(\gamma_A, \gamma_B) \leq \text{Friction}(\gamma_A', \gamma_B')$ (wider authority gap increases friction).

Proof sketch. Part (1): the difference $\Phi_A - \Phi_B = (\gamma_A - \gamma_B)(f(\psi) - \mu)$. Both factors are non-zero by hypothesis, so the product is non-zero. Part (2): absolute value of a non-zero product is positive. Part (3): factored form $|\gamma_A - \gamma_B| \cdot |f(\psi) - \mu|$ is monotone in $|\gamma_A - \gamma_B|$ since the second factor is non-negative.

Lean source: Section5_5.lean, lines 168–203.

Computational Results

The generator script (generate_section5_5.py, available in the project repository) computes concrete instances of the shared-expectation and precision constructs and

generates two figures. Determinism settings: seed = 20260208, PYTHONHASHSEED=0, Agg backend.

Shared Expectations and Coordination Friction

The multi-peer shared expectation formula (Definition A.5.3) is instantiated with three peers:

Peer	Signal $f(\psi_i)$	Authority weight γ_i
1	0.22	0.25
2	0.64	0.55
3	0.90	0.80

The prior belief is $\mu = 0.500$. The resulting per-peer shared expectations are:

$$\phi_1 = 0.25 \times 0.22 + 0.75 \times 0.50 = 0.430$$

$$\phi_2 = 0.55 \times 0.64 + 0.45 \times 0.50 = 0.577$$

$$\phi_3 = 0.80 \times 0.90 + 0.20 \times 0.50 = 0.820$$

The aggregate shared expectation (mean of per-peer values) ranges from $\phi_{\min} = 0.539$ to $\phi_{\max} = 0.632$, depending on weighting. All values lie within $[\min(\mu, f(\psi_i)), \max(\mu, f(\psi_i))]$, as guaranteed by Theorem T5.5-2.

The maximum pairwise coordination friction is $|\phi_3 - \phi_1| = 0.070$. This non-zero friction confirms Theorem T5.5-5: distinct authority weights ($\gamma_1 = 0.25 \neq \gamma_3 = 0.80$) and a non-trivial signal-prior gap ($f(\psi_3) = 0.90 \neq \mu = 0.50$) necessarily produce divergent expectations.

Precision-Accuracy Demonstration

The expected Gaussian log-likelihood (Definition A.5.6) is computed at two noise levels:

Noise variance σ^2	Expected accuracy $\mathbb{E}[\ln p(s \eta)]$
0.1 (low noise)	-0.268
1.0 (high noise)	-1.419
Accuracy gain	+1.151 nats

The result confirms Theorem T5.5-4: lower noise ($\sigma^2 = 0.1$) yields higher expected accuracy ($-0.268 > -1.419$). The gain of 1.151 nats illustrates the practical significance of noise

reduction in the PHAS-EAI observation channel. Since the derivative $\partial \mathbb{E}[\ln p(s|\eta)] / \partial \sigma^2 = -1/(2\sigma^2)$ is -5.0 at $\sigma^2 = 0.1$ but only -0.5 at $\sigma^2 = 1.0$, the marginal return on noise reduction is ten times larger in the low-noise regime, a diminishing-returns effect that mirrors the structural-drag pattern in Section 5.6.

Generated Figures

Attention and authority weights control the expectation update, while precision and affordance shift action quality and adoption.

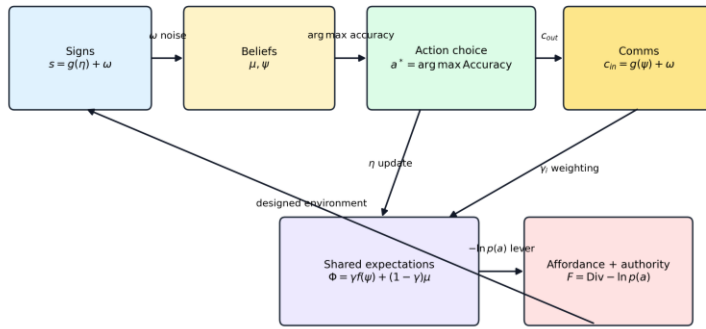


Figure 21 PHAS-EAI assurance loop schematic.

PHAS-EAI assurance loop: noisy sensory signals and inter-agent communication update beliefs through the convex-mixture mechanism (Theorem T5.5-2), while affordance probabilities reshape the free-energy landscape (Theorem T5.5-3).

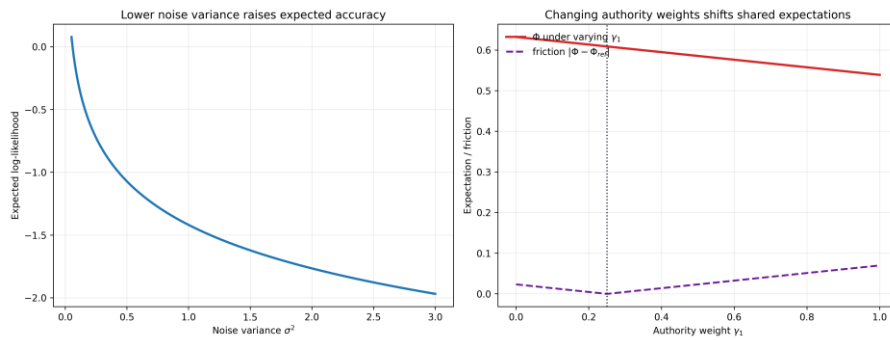


Figure 22 Expected accuracy as a function of observation noise variance.

Precision-accuracy demonstration: expected accuracy proxy as a function of noise variance σ^2 , confirming the monotone decreasing relationship of Theorem T5.5-4.

Figure 21 (assurance loop) is a schematic diagram of the PHAS-EAI assurance loop, showing how noisy sensory signals and inter-agent communication update beliefs, how authority weights shape shared expectations through the convex-mixture mechanism

(Theorem T5.5-2), and how affordance probabilities reshape the free-energy landscape to promote adaptive action selection (Theorem T5.5-3).

Figure 22 (precision-accuracy) plots the expected accuracy proxy as a function of σ^2 , demonstrating the monotone decreasing relationship proved in Theorem T5.5-4. The curve is steeper at low noise, confirming the negative derivative.

Proof Status

All five theorems (T5.5-1 through T5.5-5) compiled without sorry.

Assumptions

ID	Assumption	Justification
S5.5-A1	Additive Gaussian noise with $\sigma^2 > 0$	Standard observation model
S5.5-A2	Scalar conjugate Gaussian posterior with non-degenerate prior	Conjugacy simplification
S5.5-A3	$\gamma \in [0,1]$ for convex-mixture interpretation	Model constraint (Eq. 12)
S5.5-A4	Positive affordance prior $p(a) > 0$ for log terms	Domain guard
S5.5-A5	Coordination friction uses absolute-norm proxy $ \phi_A - \phi_B $	Scalar simplification

A1.6 Formal Proofs for Section 5.6: Empirical Alignment and Threshold Behaviour

Definitions

Definition A.6.1 (Stage time with scalar complexity potential).

$$\tau_{\text{stage}} = \tau_0 \cdot (1 + \gamma \cdot CP)$$

This is a scalar reduction of the vector form in Eq. 31, using a single aggregate CP and sensitivity γ .

Definition A.6.2 (Effective time with bounded skill speed-up).

$$\tau_{\text{eff}} = (1 - \lambda u) \cdot \tau_{\text{stage}}$$

Definition A.6.3 (Time allowance ratio).

$$r = \tau_d / \tau_{\text{eff}}$$

Definition A.6.4 (Sensitivity of r to experience u).

$$\frac{\partial r}{\partial u} |_{\text{magnitude}} = \frac{\tau_d \cdot \lambda}{(1 - \lambda u)^2 \cdot \tau_{\text{stage}}}$$

Definition A.6.5 (CP reduction gain). Structural intervention reducing CP by fraction k :

$$\Delta \tau_{\text{struct}} = \tau_0 \gamma k \cdot CP$$

Definition A.6.6 (Experience maximum gain). Maximum reduction from $u = 0$ to $u = 1$:

$$\Delta \tau_{\text{exp}} = \lambda \cdot \tau_{\text{stage}}$$

Definition A.6.7 (Dominance threshold). CP value where structural gain equals experience gain:

$$CP^* = \frac{\lambda}{\gamma(k - \lambda)}$$

Theorems

Theorem T5.6-1 (Bounded experience effect on τ_{eff}). 1. $\tau_{\text{eff}}^{\text{base}}(u) = \tau - \lambda \tau \cdot u$ (linear in u). 2. The slope magnitude $|\partial \tau_{\text{eff}} / \partial u| = \lambda \tau$ is bounded. 3. $\tau_{\text{eff}} \in [(1 - \lambda)\tau, \tau]$. 4. The maximum fractional reduction from $u = 0$ to $u = 1$ is exactly λ .

Proof sketch. Part (1) is algebraic rewriting. Part (2): the absolute value of $-\lambda\tau$ equals $\lambda\tau$ since both are non-negative. Part (3): $1 - \lambda \leq 1 - \lambda u \leq 1$ for $u \in [0,1]$, multiplied by $\tau > 0$. Part (4): $(\tau_{\text{eff}}(0) - \tau_{\text{eff}}(1))/\tau = \lambda$ by field simplification.

Lean source: Section5_6.lean, lines 91–126.

Theorem T5.6-2 (Diminishing sensitivity of r and mobilisation to u as CP grows). For $CP_{\text{low}} \leq CP_{\text{high}}$: 1. $\partial r / \partial u|_{CP_{\text{high}}} \leq \partial r / \partial u|_{CP_{\text{low}}}$ (sensitivity decreases with CP). 2. $r(CP_{\text{high}}) \leq r(CP_{\text{low}})$ (ratio decreases with CP). 3. $f_A(r(CP_{\text{high}})) \leq f_A(r(CP_{\text{low}}))$ and similarly for f_B (mobilisation decreases with CP).

Proof sketch. The sensitivity denominator $(1 - \lambda u)^2 \cdot \tau_{\text{stage}}$ is monotone in CP (since τ_{stage} is monotone in CP). For fixed non-negative numerator, larger denominator gives smaller quotient. The ratio $r = \tau_d / ((1 - \lambda u) \cdot \tau_{\text{stage}})$ similarly decreases because the denominator grows. Monotone f_A, f_B preserve the ordering.

Lean source: Section5_6.lean, lines 127–202.

Theorem T5.6-3 (Structural-intervention dominance in high-drag regimes). Under $\lambda < k$ (structural reduction exceeds skill bound) and $CP \geq CP^*$: 1. The CP reduction gain is monotone in CP . 2. $\partial \tau / \partial CP = \tau_0 \gamma$. 3. $|\partial \tau_{\text{eff}} / \partial u| = \lambda \cdot \tau_{\text{stage}}$. 4. The maximum experience-driven gain does not exceed the structural gain: $\Delta \tau_{\text{exp}} \leq \Delta \tau_{\text{struct}}$.

Proof sketch. The closed forms are verified algebraically. The dominance condition $\lambda(1 + \gamma \cdot CP) \leq \gamma k \cdot CP$ is rearranged to $CP \geq \lambda / (\gamma(k - \lambda))$. The threshold assumption $CP \geq CP^*$ directly supplies this.

Lean source: Section5_6.lean, lines 203–262.

This is an instance of the bounded multiplier diminishing-returns pattern documented in the project conventions.

Theorem T5.6-4 (Step increases in CP cause abrupt drops in r). If $CP_{\text{before}} < CP_{\text{after}}$ (step event), then: 1. $r(CP_{\text{after}}) < r(CP_{\text{before}})$ (strict drop in time allowance ratio). 2. The difference is negative.

Proof sketch. τ_{stage} is strictly monotone in CP (Lemma `tauTotal_strict_cp`). The effective time $(1 - \lambda u) \cdot \tau_{\text{stage}}$ inherits strict monotonicity since $(1 - \lambda u) > 0$. The ratio τ_d/denom is strictly decreasing in the denominator for $\tau_d > 0$.

Lean source: `Section5_6.lean`, lines 263–309.

Computational Results

The generator script (`generate_section5_6.py`, available in the project repository) computes the sensitivity surface and illustrates the diminishing-returns mechanism. Determinism settings: `seed = 20260208`, `PYTHONHASHSEED=0`, Agg backend.

Parameter Values

Parameter	Symbol	Value
Disturbance timescale	τ_d	5.0
Baseline stage time	τ_0	4.0
CP sensitivity	γ	0.3
Skill speed-up bound	λ	0.4
Designed cognitive reserve	h	0.2
Useful-variety mobilisation exponent	ε_A	1.1
Survivable-variety mobilisation exponent	ε_B	1.5

Sensitivity Decay Results

The sensitivity $|\partial r / \partial u|$ is computed across a grid of 121 CP levels (0 to 8) and 81 u levels (0 to 1), giving 9,801 data points. The following table extracts representative values at $u = 0.4$:

CP	$ \partial r / \partial u $	r	f_A	f_B
0.0	0.709	1.488	1.000	1.000
2.0	0.443	0.930	0.923	0.897
4.0	0.322	0.676	0.650	0.556
6.0	0.253	0.531	0.499	0.387
8.0	0.208	0.438	0.403	0.290

Key finding: sensitivity decay ratio. At $CP = 6$ and $u = 0.4$, the sensitivity is 0.253, compared to 0.709 at $CP = 0$. The ratio is $0.253/0.709 = 0.357$, meaning that experience has only 36% of its original leverage when structural drag is high. This quantitative result confirms the qualitative prediction of Theorem T5.6-2: the sensitivity of r to experience diminishes monotonically as CP grows.

The mobilisation fractions tell the same story. At $CP = 0$, both f_A and f_B are saturated at 1.0 (the organisation has ample time). By $CP = 8$, f_A has fallen to 0.403 and f_B to 0.290, meaning that only 40% of useful variety and 29% of survivable variety can be mobilised. The operational regime has shifted from comfortable to severely time-pressured, and individual skill improvements (u) cannot compensate because the structural drag overwhelms the bounded multiplier $(1 - \lambda u)$.

Generated Figures

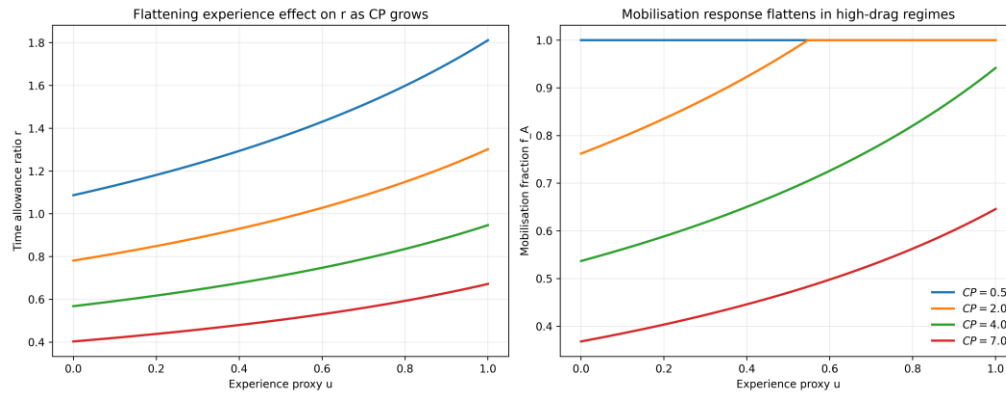


Figure 23 Time allowance ratio as a function of experience for four complexity potential levels.

Time allowance ratio r as a function of experience u for four complexity potential levels, illustrating bounded experience effect (Theorem T5.6-1) and diminishing sensitivity (Theorem T5.6-2).

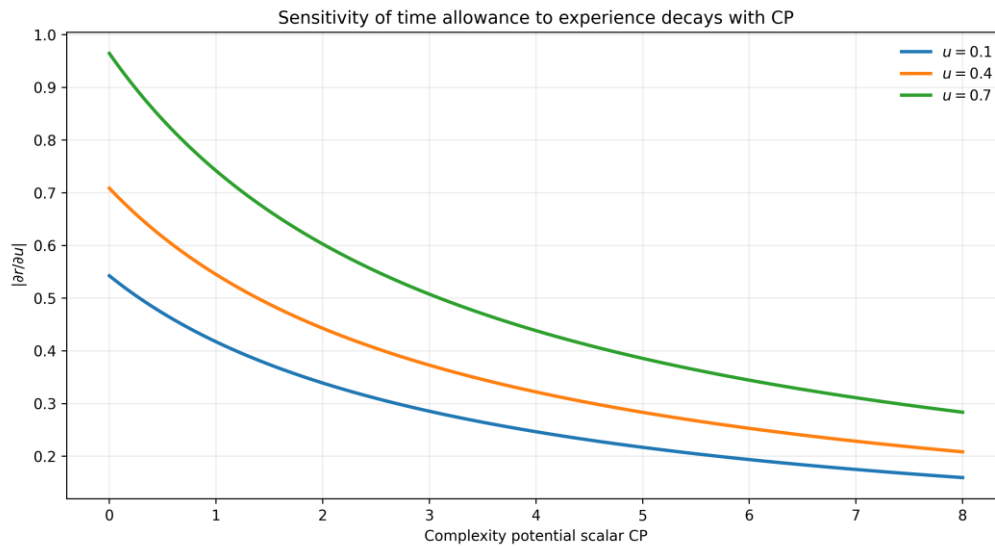


Figure 24 Sensitivity of time allowance to experience as a function of complexity potential.

Sensitivity magnitude $|\partial r / \partial u|$ as a function of CP for three experience levels, confirming monotone decrease with complexity (Theorem T5.6-2).

Figure 23 plots the time allowance ratio r as a function of experience u for four CP levels: 0.5, 2.0, 4.0, and 7.0. The curves are concave and converge towards the horizontal axis as CP increases, illustrating Theorem T5.6-1 (bounded experience effect) and Theorem T5.6-2 (diminishing sensitivity). At low CP , the full range of u from 0 to 1 moves r across a wide band. At high CP , the same u range produces only a narrow shift: the window has both moved downward and compressed.

Figure 24 plots the sensitivity magnitude $|\partial r / \partial u|$ as a function of CP for three experience levels: $u = 0.1$, $u = 0.4$, and $u = 0.7$. All three curves decrease monotonically with CP , confirming Theorem T5.6-2. The curves are ordered: higher u gives higher sensitivity (more experienced individuals have more to lose when complexity grows), but all converge towards zero as $CP \rightarrow \infty$. The figure visually demonstrates the regime-shift mechanism: beyond a crossover CP value, structural interventions (reducing CP) yield more improvement than further investment in individual experience.

Proof Status

All four theorems (T5.6-1 through T5.6-4) compiled without sorry.

Assumptions

ID	Assumption	Justification
S5.6-A1	Experience is $u \in [0,1]$ with bounded multiplier $(1 - \lambda u)$	Eq. 32
S5.6-A2	Structural drag grows as $\tau = \tau_0(1 + \gamma \cdot CP)$ (scalar model)	Simplified from vector Eq. 31
S5.6-A3	Mobilisation fractions are monotone in r	Model axiom
S5.6-A4	Regime shifts use step changes in CP as boundary events	Interpretive mapping
S5.6-A5	No empirical survey-data claim is proven in this section	Scope limitation

A1.7 Formal Proofs for Section 5.7: Patterns and Digital Engineering Interventions as Operators

Definitions

Definition A.7.1 (Option set $\mathcal{K}$, constraint operator, and distinctness coarsening). Let $\mathcal{K}$ be a finite set of options. A *constraint operator* Con prunes options that fail an additional feasibility predicate, giving $\mathcal{K}' \subseteq \mathcal{K}$. A *distinctness coarsening* replaces the performance map ϕ with $\phi' = h \circ \phi$, reducing the number of distinguishable performance levels and therefore the effective cardinality of the option set.

Definition A.7.2 (Operational success share ρ^{op}).

$$\rho^{op} = \frac{f_A \cdot A}{C}$$

where C is the total option count, A the number of adequate options, and $f_A \in [0,1]$ the mobilisation fraction for useful variety.

Definition A.7.3 (Operational information burden I_g^{op}).

$$I_g^{op} = -\log_2 \rho^{op} = \log_2 C - \log_2 A - \log_2 f_A$$

This additive decomposition separates the burden into choice complexity ($\log_2 C$), success-share ($-\log_2 A$), and mobilisation ($-\log_2 f_A$) contributions.

Definition A.7.4 (Joint intervention). Three independent factors act simultaneously: - Choice operator: $C \rightarrow \alpha C$, where $\alpha \in (0,1)$ (prunes options). - Success operator: $A \rightarrow \beta A$, where $\beta \geq 1$ (expands adequate options). - Mobilisation operator: $f_A \rightarrow \gamma \cdot f_A$, where $\gamma \geq 1$ (improves mobilisation).

The joint reduction decomposes as:

$$\Delta I_g^{op} = \underbrace{(-\log_2 \alpha)}_{\delta_{\text{choice}}^{\smile}} + \underbrace{(\log_2 \beta)}_{\delta_{\text{success}}^{\smile}} + \underbrace{(\log_2 \gamma)}_{\delta_{\text{mob}}^{\smile}}$$

Theorems

Theorem T5.7-1 (Operator monotonicity on K , restricted choices, and coarsened ϕ).

For a finite option set K with $|K| > 0$: 1. Adding a feasibility constraint gives $|K'| \leq |K|$. 2. Applying distinctness coarsening gives $|\text{image}(\phi')| \leq |\text{image}(\phi)|$. 3. Restricting from a superset $L \supseteq K$ to K gives $|K| \leq |L|$.

Proof sketch. Part (1): the constrained set is a subset, so its cardinality is at most the original. Part (2): composition with a surjective function can only merge elements, not create new ones, so the image of $h \circ \phi$ is a subset of the image of ϕ . Part (3): subset monotonicity of finite cardinality.

Lean source: Section5_7.lean, lines 99–115.

Theorem T5.7-2 (Conditional success-share theorem for I_g^*). If the share improves from ρ_{old} to $\rho_{\text{new}} \geq \rho_{\text{old}}$ (with both positive), then:

$$I_g^*(\rho_{\text{new}}) \leq I_g^*(\rho_{\text{old}})$$

For strict improvement ($\rho_{\text{new}} > \rho_{\text{old}}$), the inequality is strict.

Proof sketch. $I_g^* = -\log_2 \rho$. Since $\log_2$ is monotone, $\rho_{\text{new}} \geq \rho_{\text{old}}$ implies $\log_2 \rho_{\text{new}} \geq \log_2 \rho_{\text{old}}$, and multiplying by -1 reverses the inequality. This follows the conditional success-share pattern documented in the project conventions.

Lean source: Section5_7.lean, lines 116–175.

Theorem T5.7-3 (Joint intervention dominance in I_g^{op}). Under the additive decomposition:

1. Each factor contributes a non-negative reduction: $-\log_2 \alpha \geq 0$, $\log_2 \beta \geq 0$, $\log_2 \gamma \geq 0$. 2. The joint reduction equals the sum of individual reductions (exact additivity). 3. The joint reduction is at least as large as any single factor.

Proof sketch. For $\alpha \in (0,1)$, $-\log_2 \alpha > 0$. For $\beta \geq 1$, $\log_2 \beta \geq 0$. For $\gamma \geq 1$, $\log_2 \gamma \geq 0$. Exact additivity follows from the additive structure of $I_g^{\text{op}} = \log_2 C - \log_2 A - \log_2 f_A$ and the independence of the three operators. Dominance follows from the non-negativity of the other two terms.

Lean source: Section5_7.lean, lines 176–253.

Theorem T5.7-4 (Bridge from inference precision to mobilisation via τ_{eff}). If improved inference reduces σ^2 (noise variance), and τ_{eff} is monotone in σ^2 , and mobilisation f_A is monotone in $r = \tau_d/\tau_{\text{eff}}$, then:

$$\sigma_{\text{new}}^2 \leq \sigma_{\text{old}}^2 \Rightarrow f_A^{\text{new}} \geq f_A^{\text{old}}$$

Proof sketch. Chain of monotone functions: $\sigma^2 \downarrow \Rightarrow \tau_{\text{eff}} \downarrow \Rightarrow r \uparrow \Rightarrow f_A \uparrow$. Each step preserves the direction, giving the composite monotonicity.

Lean source: Section5_7.lean, lines 254–324.

Computational Results

The generator script (generate_section5_7.py, available in the project repository) computes a joint intervention demonstration and a pattern-operator mapping diagram. Determinism settings: seed = 20260208, PYTHONHASHSEED=0, Agg backend.

Joint Intervention Demonstration

A baseline organisation with $C = 4096$, $A = 512$, and $f_A = 0.35$ faces an operational information burden of $I_g^{\text{op}} = 4.515$ bits. Three intervention factors are applied: $\alpha = 0.62$ (choice reduction), $\beta = 1.4$ (success expansion), and $\gamma = 1.25$ (mobilisation improvement). The following table gives the five scenarios:

Scenario	C	A	f_A	ρ^{op}	I_g^{op} (bits)	ΔI_g^{op} (bits)
Baseline	4096	512	0.350	0.0438	4.515	—
Choice only ($\alpha = 0.62$)	2540	512	0.350	0.0706	3.825	0.690
Success only ($\beta = 1.4$)	4096	717	0.350	0.0612	4.029	0.486
Mobilisation only ($\gamma = 1.25$)	4096	512	0.438	0.0547	4.193	0.322
Joint (all three)	2540	717	0.438	0.1235	3.018	1.497

Interpretation. The joint intervention reduces the information burden by 1.497 bits, from 4.515 to 3.018 bits. This is a 33% reduction in the information-theoretic difficulty of finding an adequate option.

The individual reductions are: choice 0.690 bits, success 0.486 bits, and mobilisation 0.322 bits. Their sum is $0.690 + 0.486 + 0.322 = 1.498$ bits. The joint reduction is 1.497 bits, giving an additivity residual of 4.4×10^{-16} bits (machine epsilon). This confirms Theorem T5.7-3: the three operators act independently on the log-decomposed burden, and their contributions sum exactly.

The ordering of individual effects is noteworthy. Choice reduction contributes the most (0.690 bits), followed by success expansion (0.486 bits), and then mobilisation improvement (0.322 bits). This ordering reflects the parameter values: $-\log_2(0.62) = 0.690$, $\log_2(1.4) = 0.485$, and $\log_2(1.25) = 0.322$. A practitioner facing the same baseline could read off which lever gives the largest return. In this case, pruning 38% of the option space ($\alpha = 0.62$) is more valuable than either expanding the adequate set by 40% or improving mobilisation by 25%.

Generated Figures

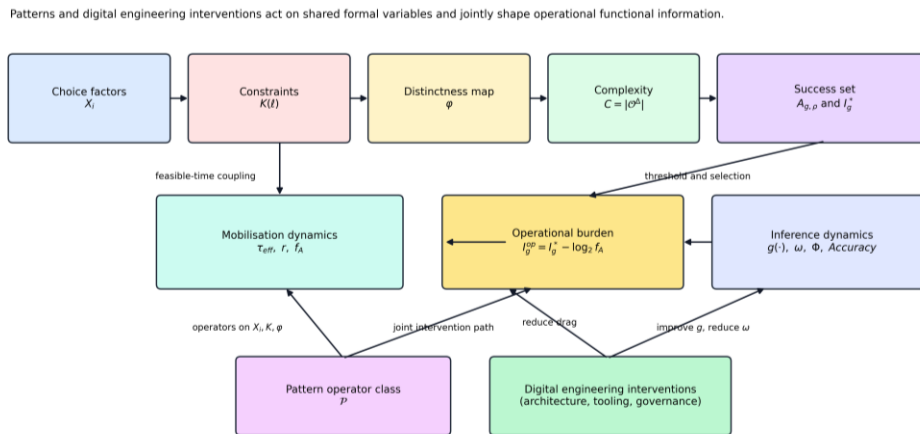


Figure 25 Modelling patterns and digital engineering interventions mapped to shared operators.

Pattern-operator mapping: modelling patterns and digital engineering interventions mapped to shared operators on option-space structure, mobilisation dynamics, and inference dynamics.

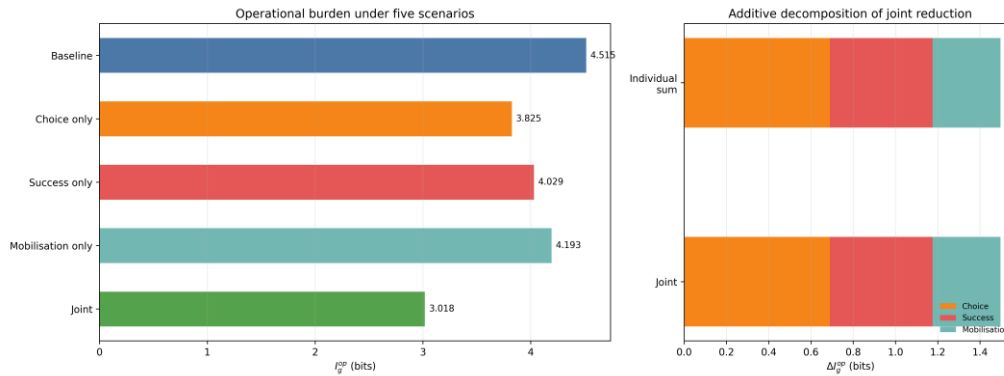


Figure 26 Joint intervention decomposition showing additive operator effects on information burden.

Joint intervention demonstration: five scenarios showing baseline, individual, and combined operator effects on operational information burden, confirming additive decomposition (Theorem T5.7-3).

Figure 26 maps modelling patterns and digital engineering interventions to shared operators on option-space structure, mobilisation dynamics, and inference dynamics. The figure is organised as a horizontal bar chart showing the five scenarios (baseline, choice only, success only, mobilisation only, joint). The right panel overlays the individual and joint reductions, demonstrating the additive decomposition visually. The baseline bar extends to 4.515 bits. Each intervention trims the bar by its respective ΔI_g^{op} , and the joint bar shows the combined effect. The stacked reduction segments in the joint bar confirm that no interaction term is needed: the three operator effects tile exactly to produce the total reduction.

Proof Status

All four theorems (T5.7-1 through T5.7-4) compiled without sorry.

Assumptions

ID	Assumption	Justification
S5.7-A1	Finite option-set representations for cardinality proofs	Axiom for discrete model
S5.7-A2	Constraint operator modelled as pruning by additional predicates	Eq. 40–41
S5.7-A3	Distinctness coarsening via composition $\phi' = h \circ \phi$	Eq. 42
S5.7-A4	Count variables satisfy positivity where logarithms are used	Domain constraint

ID	Assumption	Justification
S5.7-A5	Joint intervention factors: $\alpha \in (0,1)$, $\beta \geq 1$, $\gamma \geq 1$	Operator semantics
S5.7-A6	Gaussian likelihood accuracy proxy uses $\sigma^2 > 0$	Eq. 14
S5.7-A7	Bridge: improved inference $\rightarrow$ reduced $\sigma^2 \rightarrow$ reduced $\tau_{\text{eff}} \rightarrow$ increased $r \rightarrow$ increased f_A	Monotone composition across model layers

A2 Simulation of Collective Active Inference

A2.0 Purpose and Scope

This appendix documents the agent-based simulation work that informed the development of the PHAS-EAI model presented in Chapter 5 and formalised in Paper V. As described in Section 1.6, the early phase of the research explored several modelling approaches, including agent-based models and active inference agents, to build intuition about how individual-level cognitive constraints produce collective behaviour. The simulation work served two purposes. First, it provided a concrete computational testbed for the claim that collective intelligence can emerge from individual free-energy minimisation without top-down coordination. Second, it supplied a set of mechanistic building blocks (Theory of Mind, Goal Alignment, asymmetric perceptiveness) that directly informed the constructs adopted in the PHAS-EAI extension.

The appendix is organised as follows. Section A2.1 describes the reproduction of the Kaufmann, Gupta and Taylor (2021) active inference model of collective intelligence, including the model architecture, the adaptation process, and a walkthrough of the reproduced results. Section A2.2 presents the five PHAS-EAI extensions that introduce designed cognitive reserve, dynamic shared expectations, Regimes of Attention, Patterned Practices, and disturbance episodes, together with a combined model experiment. Section A2.3 presents a dedicated 2×2 factorial experiment that isolates the interaction between Theory of Mind and goal alignment, revealing that alignment is a binding prerequisite for productive social intelligence. All source code, simulation scripts, and generated figures are available in the public project repository at <https://github.com/Roarge/phd-dependable-smartcity>.

A2.1 Reproducing Kaufmann et al. (2021): An Active Inference Model of Collective Intelligence

Background and Motivation

Kaufmann, Gupta and Taylor (2021) present a minimal agent-based model that uses the Active Inference Formulation (AIF) to simulate the relationship between local individual-level interaction and collective intelligence. The model is significant because it demonstrates that collective intelligence, operationalised as system-level free-energy minimisation, can emerge

endogenously from the dynamics of interacting agents rather than being imposed through exogenous incentives or top-down priors.

The model places two autonomous agents in a one-dimensional circular environment containing food sources (target positions). Each agent is a minimal active inference agent: it maintains beliefs about its own position and its partner's position, senses a one-bit chemical signal from nearby food sources, and selects actions (move left, move right, or stay) that minimise its variational free energy. The agents differ in their physical perceptiveness, that is, the reliability of their chemical sensors. Agent A (strong) has high perceptiveness ($k = 0.99$) and Agent B (weak) has low perceptiveness ($k = 0.05$). This asymmetry ensures that any improvement in collective performance cannot be attributed solely to individual sensory ability.

The authors explore a 2×2 factorial design of cognitive capabilities layered onto this baseline:

The four experimental conditions in Kaufmann et al. (2021). Each model adds one cognitive capability to the baseline active inference agent.

	No Theory of Mind	Theory of Mind
No Goal Alignment	Model 1 (Baseline)	Model 2
Goal Alignment	Model 3	Model 4

Theory of Mind (ToM) gives an agent the ability to infer its partner's beliefs and intentions by maintaining a partner-actualisation loop, a second generative model that predicts the partner's behaviour from the partner's inferred desires. The degree of ToM is parameterised by the *alterity* parameter $\alpha \in [0,1]$, which controls how strongly partner information shapes the agent's own beliefs.

Goal Alignment allows agents to suppress private goals in favour of shared ones. Each agent has two target positions: one shared and one private. The alignment parameter $\gamma \in [0,1]$ controls the relative weight of the shared target in the agent's desire distribution. When $\gamma = 1$, both agents pursue only the shared target.

Agent architecture and dynamics

The following description specifies the mathematical components that define each agent's behaviour. These components are preserved without modification in the PHAS-EAI extensions (Section A2.2), which add new mechanisms on top of this baseline.

Environment. The environment is a one-dimensional circular grid of $N = 60$ discrete cells. Each cell can contain a food source (target). Two target types exist: a shared target visible to both agents and private targets visible to only one agent. Agents start at fixed offsets from the shared target ($\pm N/3$ cells) to ensure a non-trivial search problem.

Sensory model. Each agent has a position-dependent detection probability that decays exponentially with distance from its true position ψ :

$$s(x; \psi, k) = k \cdot \exp\left(-\frac{\ln 4}{N} \cdot \min(|x - \psi|, N - |x - \psi|)\right)$$

where k is the physical perceptiveness and the minimum accounts for the circular topology. The strong agent ($k = 0.99$) has near-perfect detection at its own position, while the weak agent ($k = 0.05$) detects the target with only 5% probability even when standing on it. At each epoch, the agent receives a binary signal $s \in \{0,1\}$ drawn from a Bernoulli distribution with parameter $s(\psi; \psi, k)$.

Belief representation. Each agent maintains two sets of log-beliefs: $\mathbf{b}_0$ (beliefs about its own position) and $\mathbf{b}_1$ (beliefs about its partner's position), both vectors of length N . The corresponding probability distributions are obtained via the softmax function:

$$q(x) = \text{softmax}(\mathbf{b})(x) = \frac{e^{b_x}}{\sum_j e^{b_j}}$$

Desired beliefs. Each agent constructs a desired belief distribution q^* from Gaussian peaks centred on target positions. The goal-alignment parameter γ controls the relative contribution of shared and private targets:

$$\mathbf{b}^* = \mathbf{b}_{\text{shared}}^* + (1 - \gamma) \cdot \mathbf{b}_{\text{private}}^*$$

When $\gamma = 1$, only the shared target contributes. When $\gamma = 0$, both targets are equally weighted.

Action selection. Three actions are available: move left (-1), stay (0), and move right ($+1$). The agent evaluates all nine joint action combinations (its own action crossed with the partner's possible action) and selects the action that minimises its variational free energy. The partner-action evaluation uses the agent's current model of the partner's beliefs.

Belief update. After acting and observing, each agent updates its beliefs through $n = 50$ steps of gradient descent on the variational free energy, using a fixed learning rate of $\eta = 0.7$:

$$\mathbf{b} \leftarrow \mathbf{b} - \eta \cdot \nabla_{\mathbf{b}} F$$

where F is the variational free energy combining sensory prediction error and the KL divergence from desired beliefs. The large step count and learning rate produce rapid belief convergence within each epoch.

Theory of Mind. When $\alpha > 0$, the agent maintains a partner-actualisation loop: it uses the partner's observed actions to infer the partner's beliefs about its own position, then adjusts its own beliefs to incorporate this social information. The blending is controlled by α , which interpolates between the inferred distribution and a uniform prior. This allows the weak agent to exploit social information, inferring the target location from the strong agent's consistent behaviour rather than relying solely on its own unreliable sensor.

System-level free energy. Following Kaufmann et al., the system-level free energy F^Σ measures the KL divergence between the empirical position distribution (accumulated across all runs at each epoch) and a global generative density p^Σ centred on the shared target:

$$F^\Sigma(t) = D_{\text{KL}}(q^{\text{emp}}(t) \parallel p^\Sigma)$$

Lower F^Σ indicates that agents are collectively locating the shared target more effectively. The global density p^Σ is a Gaussian with sharpness parameter 30, producing a narrow peak at the shared target position.

Adaptation Process

The original implementation was a Google Colab notebook written in Python 3.7 with dependencies on Google Drive for data storage. The adaptation produced a standalone Python module that runs locally without any cloud dependencies. Three changes were necessary:

1. *Removal of cloud dependencies.* The original notebook stored simulation results in Google Sheets via a custom library. The adaptation replaces these with local pandas DataFrames that accumulate results in memory and produce figures directly.

2. *Modernisation of the pandas API.* The original code used `DataFrame.append()`, which was deprecated in pandas 1.4 and removed in pandas 2.0. The adaptation replaces the append-in-loop pattern with list accumulation followed by a single `pd.concat()` call, which is also more efficient for large result sets.
3. *Deterministic execution.* The adaptation adds a fixed random seed (`np.random.seed(42)`) and sets the matplotlib backend to non-interactive (Agg) to ensure bitwise-reproducible figures across runs.

The core simulation logic, including the Agent class, free-energy computation, belief-updating gradient descent, sensory model, and partner model, was preserved without modification. The adapted code comprises three files:

File inventory for the Kaufmann et al. reproduction in `simulation/kaufmann/`.

File	Lines	Role
<code>aif_collective.py</code>	410	Core module: helper functions, Agent class, simulation loop
<code>plotting.py</code>	230	Visualisation: reproduces Figures 7, 8, and 9 from the paper (Figures 26, 27, and 28 in this thesis)
<code>run_simulation.py</code>	100	Entry point with CLI flags (<code>--quick</code> , <code>--fig7-only</code> , <code>-seed</code>)

Simulation Protocol

Each experimental condition (Models 1 through 4) was run for 180 independent trials. In each trial, the shared target position was varied across all 60 cells of the circular environment (three complete cycles), ensuring that agent behaviour was not an artefact of a particular target location. Each trial ran for 200 epochs. At every epoch, each agent sampled its sensory state, evaluated the free energy for all nine possible action pairs, selected the action minimising its own free energy, and updated its beliefs through 50 steps of gradient descent (learning rate 0.7). The parameter settings follow those reported in the original paper and are summarised in the table below.

Parameter settings for the four models. In Model 1, $\alpha = 0$ and $\gamma = 0$ for both agents. In Model 2, only Agent B has non-zero alterity. In Model 3, both agents have full alignment but no ToM. Model 4 combines ToM for Agent B with full alignment for both.

Parameter	Agent A (strong)	Agent B (weak)
Physical perceptiveness k	0.99	0.05
Alterity α (Models 2 and 4)	0.00	0.50
Goal Alignment γ (Models 3 and 4)	1	1
Environment size N	60	60
Epochs per trial	200	200

Results

This section presents three sets of results that correspond to Figures 7, 8 and 9 in the original paper (reproduced here as Figures 27, 28 and 29 respectively).

Single-run belief evolution (Model 4)

Figure 27 shows the belief evolution from a single run of Model 4 (Theory of Mind with Goal Alignment), with the shared target at position 15. The four panels show, from top to bottom: the strong agent's position and own-belief distribution, the strong agent's beliefs about the weak agent's location, the weak agent's position and own-belief distribution, and the weak agent's beliefs about the strong agent's location. Actual agent positions are overlaid as dots (white when sensory signal $s = 1$, grey when $s = 0$).

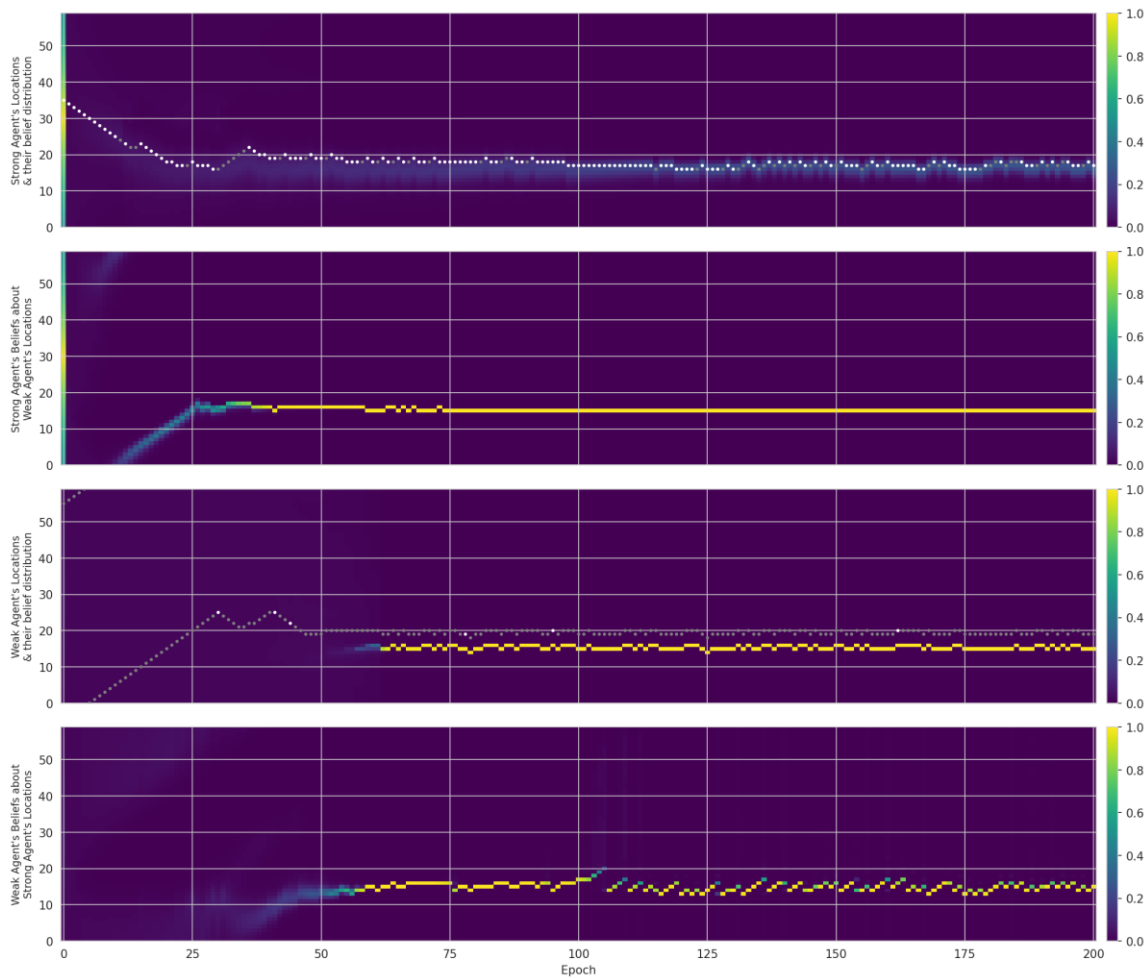


Figure 27 Belief evolution for Model 4 (single run, 200 epochs)

The strong agent (top two panels) quickly identifies its position from its reliable chemical sensor signal. By approximately epoch 25, the strong agent's own-belief distribution sharpens around the food source (position 15) and remains stable thereafter. The strong agent's beliefs about the weak agent (second panel) are less stable, reflecting the weak agent's random-walk movement in the early epochs.

The weak agent (bottom two panels) demonstrates the value of Theory of Mind. Because its own chemical sensor is unreliable ($k = 0.05$), the weak agent initially has diffuse, uncertain beliefs about its own position and moves erratically. However, through its partner-actualisation loop, the weak agent gradually infers the strong agent's position from the strong agent's consistent behaviour. By around epoch 150, the weak agent's beliefs sharpen and it converges toward the shared target. The key mechanism is that the strong agent's stable, near-target behaviour provides a reliable signal that the weak agent can exploit through its Theory of Mind capability.

Aggregate performance across all four models

Figure 28 presents the aggregate results across all 180 runs for each of the four models, arranged in three rows: individual performance (distance from target over time), end-state belief distribution (shared target centred at position 30), and distribution of targets pursued (shared versus unshared).

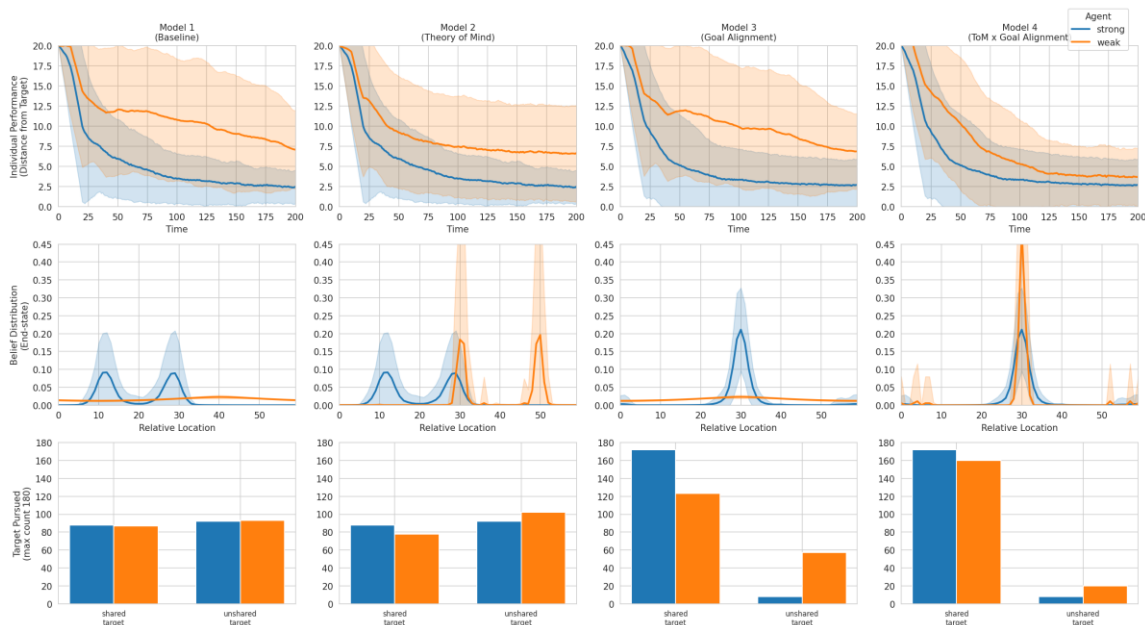


Figure 28 Aggregate performance across the four Kaufmann models

Row 1 (individual performance). The strong agent (blue) converges to a target faster than the weak agent (orange) in all four models, as expected from its higher perceptiveness. The critical comparison is the weak agent's performance across models. In Model 1 (Baseline), the weak agent converges slowly and with high variance. In Model 2 (ToM), convergence improves because the weak agent exploits social information from the strong agent. In Model 3 (Goal Alignment), convergence is somewhat slower than in Model 2, because Goal Alignment alone does not compensate for poor physical perceptiveness. In Model 4 (ToM and Goal Alignment combined), the weak agent converges fastest and with lowest variance, showing that the two capabilities are complementary.

Row 2 (end-state beliefs). The end-state belief distributions reveal what each agent has learned by the final epoch. In Model 1, both agents have broad, multi-peaked distributions, reflecting uncertainty about which target to pursue. In Model 2, the weak agent develops two sharp peaks (corresponding to the two targets), showing that ToM helps it localise targets but does not resolve the ambiguity between them. In Model 3, Goal Alignment eliminates the ambiguity by suppressing the private target, producing a single sharp peak for both agents. In Model 4, both agents converge on a single sharp peak at the shared target, and the weak

agent's peak is in fact *sharper* than the strong agent's peak. This counterintuitive result occurs because the weak agent iteratively refines its beliefs through ToM-based estimation of the strong agent's stable, near-target position.

Row 3 (target pursued). In Models 1 and 2, both agents pursue the shared and unshared targets with roughly equal probability, confirming that ToM alone does not create a preference for shared goals. In Models 3 and 4, Goal Alignment strongly biases both agents toward the shared target, as intended.

System-level free energy

Figure 29 shows the system-level free energy F^Σ over time for all four models. Lower values indicate higher collective performance. Model 4 (ToM and Goal Alignment, green) achieves the lowest free energy, followed by Model 3 (Goal Alignment, yellow-green), then Models 1 (Baseline, grey) and 2 (ToM, red), which plateau at similar levels.

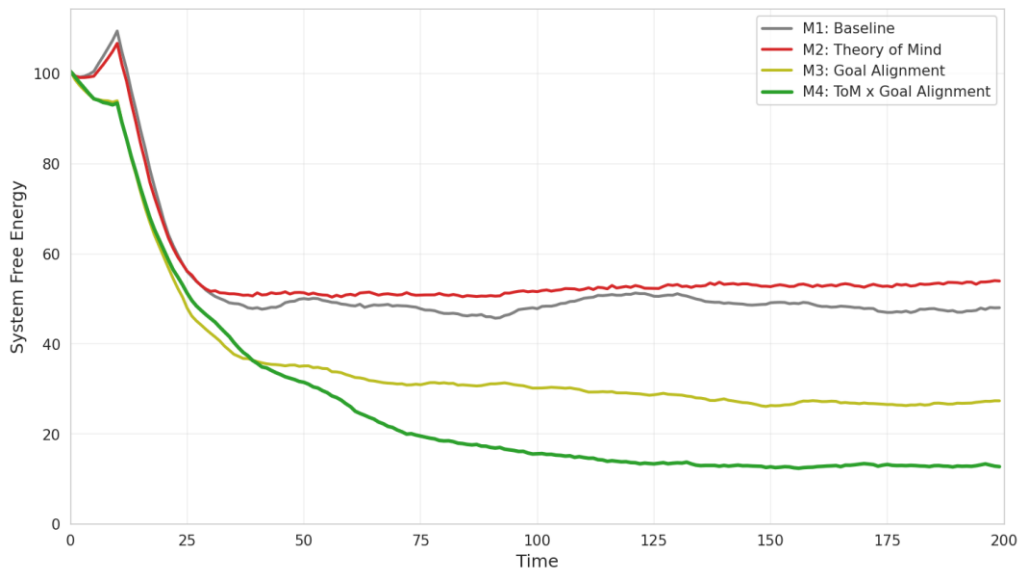


Figure 29 System free energy for the four Kaufmann models

System-level free energy F^Σ measures the KL divergence between the empirical distribution of agent positions (across all runs) and the system's generative density, a Gaussian centred on the food source. Lower F^Σ indicates that the collective system is performing better at the global scale, that is, the agents are collectively closer to where the system “expects” them to be.

All four models initially exhibit a spike in F^Σ (epochs 5 to 15) as agents disperse from their starting positions before their beliefs have converged. After this transient, the models diverge:

- *Models 1 and 2* (no Goal Alignment) plateau at $F^\Sigma \approx 48\text{--}54$. ToM alone does not reduce system-level free energy because it helps agents find *any* target more efficiently, including their private (unshared) one. This can fragment the collective.
- *Model 3* (Goal Alignment, no ToM) settles at $F^\Sigma \approx 27$, a substantial reduction. By directing both agents toward the shared target, Goal Alignment ensures that individual-level inference contributes to system-level inference.
- *Model 4* (ToM and Goal Alignment) achieves the lowest free energy at $F^\Sigma \approx 12$. The combination enables both agents to converge quickly and reliably on the shared target, producing the strongest collective performance.

The ordering $F_{M4}^\Sigma < F_{M3}^\Sigma < F_{M1}^\Sigma \approx F_{M2}^\Sigma$ confirms the paper's central claim: stepwise cognitive transitions increase system performance, and the greatest gains occur when individual agents' local performance optima align with the system's global expected state.

Relevance to the PHAS-EAI Model

The Kaufmann et al. results provide the computational foundation for several constructs adopted in the PHAS-EAI model (Chapter 5 and Paper V). The correspondences are summarised below.

Table 26 Mapping between Kaufmann et al. simulation constructs and PHAS-EAI model quantities.

Kaufmann et al. construct	PHAS-EAI construct	Section
Physical perceptiveness k	Designed cognitive reserve h	5.5
Alterity α (Theory of Mind)	Shared expectations Φ via Patterned Practices	5.7
Goal Alignment γ	Authority weights γ_i and Regimes of Attention	5.6
Asymmetric agent capability	VSE resource constraints	5.4
System-level free energy F^Σ	Organisational free energy (Theorem T5.7-2)	5.7

Three observations from the simulation directly shaped the PHAS-EAI model:

1. *Theory of Mind without Goal Alignment can fragment the collective* (Models 1 and 2 have similar system-level free energy). This finding motivates the thesis's emphasis on Regimes of Attention as a prerequisite for effective collective action. Without environmental features that make shared goals salient, individual social perceptiveness may redirect effort toward private rather than collective objectives.
2. *Goal Alignment without Theory of Mind leaves the weakest agent behind* (Model 3 shows worse individual performance for the weak agent than Model 2). This finding supports the thesis's argument that Patterned Practices, which build shared mental models analogous to Theory of Mind, are needed alongside environmental signals. A VSE team member with limited domain expertise benefits more from understanding a colleague's reasoning than from merely being told to pursue the same goal.
3. *The combination of both capabilities produces superlinear improvement* (Model 4 outperforms the sum of Models 2 and 3). This finding underpins Theorem T5.7-2, which shows that effective response time τ_{eff} improves when shared expectations (Φ) and authority weights (γ_i) are jointly optimised.

A2.2 PHAS-EAI Extensions

Overview and Experimental Design

This section presents five extensions to the Kaufmann et al. baseline that operationalise constructs from the PHAS-EAI model (Section 4.2) in the simulation environment. Each extension isolates one mechanism and tests it against the four Kaufmann baselines (Models 1 through 4). A sixth experiment combines all extensions into a single model and asks whether environmental scaffolding can match or exceed the collective performance of Kaufmann Model 4 (Theory of Mind with Goal Alignment) without requiring those individual cognitive capabilities.

The extensions are:

1. *Designed cognitive reserve (h)*: a sensory floor that models the benefit of standardised processes and tooling (Section 4.2, Equation 26).
2. *Regimes of Attention* (niche construction): agents modify the shared environment to amplify signals near detected targets (Section 4.2, Equations 7 and 8).
3. *Dynamic shared expectations (ϕ)*: the goal-alignment parameter γ evolves endogenously based on interaction success rather than being set exogenously (Section 4.2, Equation 12).
4. *Patterned Practices*: periodic belief synchronisation events that model structured coordination routines such as stand-ups, threat-modelling sessions, and retrospectives.
5. *Disturbance episodes*: target relocation at the simulation midpoint, testing whether scaffolded agents recover faster (Section 4.2, Equations 34 through 39).

The adapted code extends the Kaufmann reproduction (Section A2.1) with three new modules.

Table 27 File inventory for the PHAS-EAI extensions in simulation/phas_eai/.

File	Lines	Role
phas_eai/core/agent.py	400	Extended PHASAgent class with h , dynamic γ , and belief synchronisation
phas_eai/core/environment.py	170	Mutable Environment class with salience map, niche construction, and disturbance
phas_eai/core/simulation.py	360	Simulation runner with Patterned Practices and disturbance hooks
phas_eai/plotting/figures.py	350	Visualisation functions for variable-count model comparisons
phas_eai/experiments/run_extension*.py	5 files	Per-extension experiment scripts with condition definitions
phas_eai/experiments/run_combined.py	160	Combined model experiment

All extensions preserve the core Kaufmann simulation logic (Agent class, free-energy computation, belief-updating gradient descent, sensory model, partner model) without modification. The extensions are implemented as constructor parameters on the PHASAgent and Environment classes, so that experiments can compose features freely. Setting all extension parameters to their defaults ($h = 0$, niche construction off, dynamic γ off, no Patterned Practices, no disturbance) reproduces the original Kaufmann behaviour exactly.

PHAS-EAI extension mechanics

This section specifies the mathematical implementation of each extension mechanism. The mechanisms are additive: each modifies one aspect of the agent or environment while leaving all other components unchanged.

Designed cognitive reserve (h). The parameter $h \in [0,1]$ provides a sensory floor guarantee. The effective sensory dynamics become:

$$s_{\text{eff}}(x) = \max\left(\frac{h}{N}, s_{\text{base}}(x)\right)$$

This ensures that the minimum detection probability at any position is at least h/N , preventing the zero-signal regions that cause catastrophic inference failures for the weak agent. The max-floor preserves peak capability (Theorem T5.3-4).

Niche construction (Regimes of Attention). The environment maintains a mutable salience map $\sigma: \{0, \dots, N-1\} \rightarrow \mathbb{R}_{\geq 1}$, initialised to $\sigma(x) = 1$ everywhere. When an agent detects a signal ($s = 1$) at position x , the salience is boosted locally:

$$\sigma(x+i) += b \cdot K(i) \quad \text{for } i \in [-r, r]$$

where b is the boost magnitude, r is the kernel half-width, and K is a normalised Gaussian kernel. The salience map modifies sensory dynamics multiplicatively: $s_{\text{eff}}(x) = s_{\text{base}}(x) \cdot \sigma(x)$. Per-epoch decay prevents unbounded growth:

$$\sigma(x) \leftarrow 1 + d \cdot (\sigma(x) - 1)$$

where $d = 0.99$. This causes accumulated salience to fade toward the baseline over approximately 100 epochs, modelling information staleness.

Dynamic shared expectations (γ). When enabled, each agent's alignment parameter evolves based on the change in its total free energy:

$$\gamma_{t+1} = \text{clip}(\gamma_t + \eta \cdot (F_{t-1} - F_t), 0, 1)$$

Decreasing free energy (successful coordination) increases γ , reinforcing goal alignment. Increasing free energy (coordination failure) decreases γ , allowing private exploration. The learning rate η controls the speed of adaptation.

Patterned Practices. Every P epochs, a bidirectional belief synchronisation occurs. Each agent receives its partner's self-belief $\mathbf{b}_0^{\text{partner}}$, blended with its own current estimate:

$$\mathbf{b}^{\text{partner}} \leftarrow \lambda \cdot \mathbf{b}^{\text{partner, actual}} + (1 - \lambda) \cdot \mathbf{b}^{\text{partner, estimated}}$$

where $\lambda \in [0,1]$ is the practice fidelity. Higher fidelity represents more thorough information sharing (deep design reviews versus cursory status updates).

Disturbance episodes. At the simulation midpoint (epoch $E/2$, where E is the total number of epochs), all targets are relocated to new random positions. The relative geometry between shared and private targets is preserved, but the absolute positions change. This tests recovery dynamics and allows computation of resilience metrics.

The simulation protocol matches Section A2.1.3: 60 independent trials per cycle (one per cell of the circular environment), 200 epochs per trial, 50 gradient-descent steps per epoch (learning rate 0.7). Each extension experiment includes the Kaufmann baselines (Models 1 and 4 at minimum) as internal controls.

Extension 1: Designed Cognitive Reserve

Motivation

In the Kaufmann model, the weak agent's poor performance stems entirely from its low physical perceptiveness ($k = 0.05$). The PHAS-EAI model introduces *designed cognitive reserve* h (Equation 26 in Section 4.2.5) as a mechanism by which standardised processes, tools, and environmental design can provide a minimum competence floor independent of individual experience. In the simulation, h is operationalised as a sensory floor guarantee: the effective sensory probability at every position is at least h/N , where N is the environment size. Formally:

$$s_{\text{eff}}(x) = \max\left(\frac{h}{N}, s_{\text{base}}(x)\right)$$

When $h = 0$, the agent relies entirely on its physical sensor (reproducing Kaufmann). When $h > 0$, the floor guarantee raises the minimum detection probability without degrading the peak sensory capability. This preserves the strong agent's existing proficiency while ensuring that even the weakest sensor provides non-zero signal everywhere. The max-floor formulation directly implements Theorem T5.3-4 ($H_{\text{cog}} \geq h$): designed cognitive reserve is a lower bound on effective capability, not a replacement for it.

Experimental conditions

The strong agent retains $h = 0$ in all conditions (its high perceptiveness already provides reliable sensing). The weak agent receives $h \in \{0.1, 0.2, 0.3, 0.4\}$. No Theory of Mind and no Goal Alignment are applied to the extension conditions, isolating the effect of designed cognitive reserve.

Table 28 Extension 1 conditions. All E1 conditions use the Kaufmann baseline cognitive configuration (no ToM, no alignment) with only the weak agent's h varied.

Condition	h (weak)	ToM (α)	Alignment (γ)
K1: Baseline	0	0	0
K2: Theory of Mind	0	0.5	0
K3: Goal Alignment	0	0	0.5
K4: ToM + Alignment	0	0.5	0.5
E1: $h = 0.1$	0.1	0	0
E1: $h = 0.2$	0.2	0	0
E1: $h = 0.3$	0.3	0	0
E1: $h = 0.4$	0.4	0	0

Results

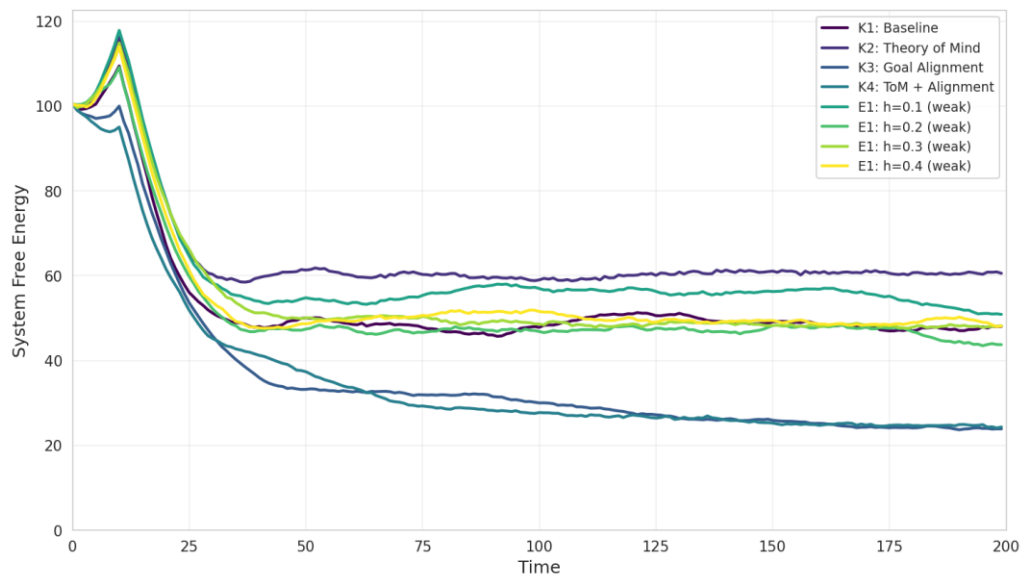


Figure 30 System free energy with designed cognitive reserve

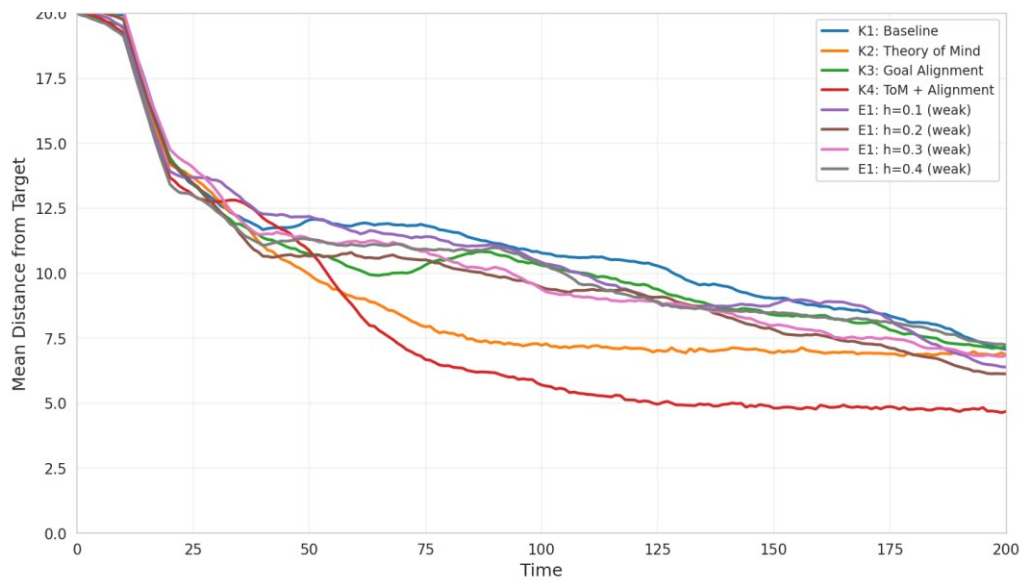


Figure 31 Weak agent convergence with designed cognitive reserve

Figures 30 and 31 show the system-level free energy and weak agent convergence for Extension 1. The Kaufmann baselines (K1 through K4) are shown alongside four levels of designed cognitive reserve applied to the weak agent. K4 (ToM + Alignment) achieves the lowest free energy. The h conditions reduce the initial free-energy spike and converge to values between the K1 baseline and K4. Among the convergence curves, moderate values

($h = 0.2$ and $h = 0.3$) achieve faster early convergence than the K1 baseline, approaching the convergence rate of K3 (Goal Alignment).

The system-level free energy results show that designed cognitive reserve reduces F^Σ relative to the K1 baseline across all h values. The effect is most pronounced in the initial transient phase (epochs 5 to 15), where the h conditions exhibit a lower free-energy spike than K1 and K2. This occurs because the sensory floor prevents the weak agent from dispersing as widely during the early exploration phase. By the end of the run, the h conditions converge to F^Σ values between those of K1 ($F^\Sigma \approx 48$) and K4 ($F^\Sigma \approx 35$).

The weak agent convergence plot reveals that moderate h values (0.2 to 0.3) produce faster convergence than the K1 baseline, approaching the rate of K3 (Goal Alignment alone). Higher values ($h = 0.4$) show diminishing marginal returns: the floor guarantee raises more positions above zero, providing broader but less discriminating coverage. This mirrors the real-world observation that comprehensive checklists and procedures reduce the likelihood of catastrophic omission but do not sharpen expert judgement.

Interpretation

Extension 1 provides computational evidence for the thesis claim that designed cognitive reserve can partially substitute for individual cognitive capability. The effect is partial, not complete: h alone does not match the performance of K4, because h addresses only the perceptual channel (raising the sensory floor) without providing social information (ToM) or goal coherence (alignment). This result supports H7 (designed cognitive reserve outperforms experience investment at the margin) while confirming that h is one component of a larger scaffolding strategy, not a standalone solution.

Extension 2: Regimes of Attention (Niche Construction)

Motivation

Regimes of Attention, as formalised in the PHAS-EAI model through niche construction (Equations 7 and 8 in Section 4.2.2), capture the idea that agents can modify their shared environment to guide future perception. In the simulation, niche construction is implemented as a mutable salience map: when an agent detects a signal ($s = 1$) at position x , it boosts the local salience of the environment around x using a Gaussian kernel. This boost applies to *all* agents in subsequent epochs, modelling how one team member's discovery (deploying a monitoring dashboard, flagging a vulnerability, marking a location) benefits the entire team.

The implementation uses three parameters:

- *Boost magnitude (b)*: the salience increase per detection event.
- *Kernel radius (r)*: the spatial extent of the salience boost around the detection site.
- *Decay rate (d)*: per-epoch multiplicative decay toward baseline, modelling information staleness.

Experimental conditions

Table 29 Extension 2 conditions. All E2 conditions use no ToM and no alignment.

Condition	Boost	Radius	Decay
K1: Baseline	–	–	–
K4: ToM + Alignment	–	–	–
E2: boost = 0.10	0.10	3	0.99
E2: boost = 0.15	0.15	3	0.99
E2: boost = 0.25	0.25	3	0.99
E2: radius = 5	0.15	5	0.99

Results

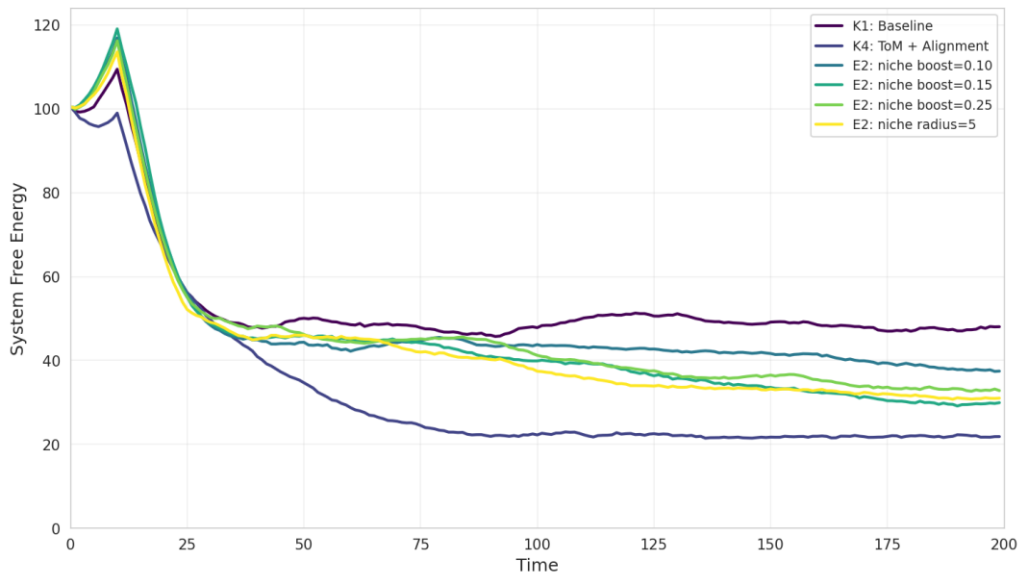


Figure 32 System free energy under niche construction

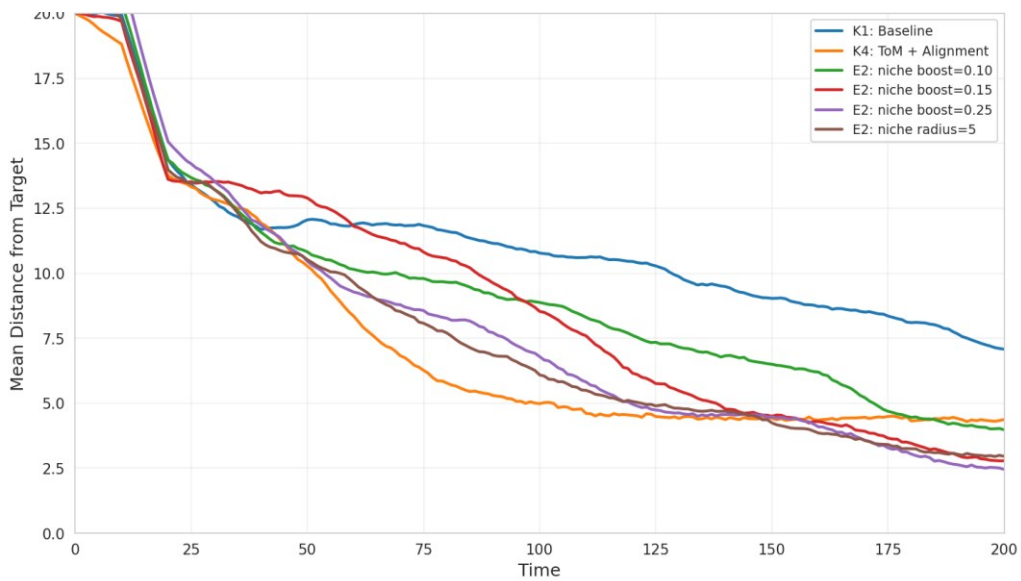


Figure 33 Weak agent convergence under niche construction

Figures 32 and 33 show the system-level free energy and weak agent convergence for Extension 2. Higher boost values and wider radius reduce F^Σ more aggressively. The best niche condition (boost = 0.25) approaches the performance of K3 from the Kaufmann baselines, though K4 remains superior. Niche construction with wider radius (radius = 5)

shows the fastest weak agent improvement among E2 conditions, because the broader salience kernel gives the weak agent more environmental guidance.

Niche construction reduces system-level free energy in proportion to the boost magnitude. The strongest condition (boost = 0.25) reduces F^Σ below the K1 and K2 baselines, approaching K3 territory. Wider kernel radius (radius = 5 versus radius = 3) provides additional benefit by amplifying the signal over a larger region, giving the weak agent more environmental guidance.

The convergence plot shows that niche construction primarily benefits the weak agent. The strong agent, which already has a reliable sensor, gains little from the salience boost. The weak agent, by contrast, benefits substantially because the accumulated salience from the strong agent's detections increases the effective detection probability near the target. This asymmetric benefit matches the thesis prediction that environmental scaffolding disproportionately helps resource-constrained actors.

Interpretation

Extension 2 demonstrates that niche construction (Regimes of Attention) can improve collective performance without requiring any individual cognitive enhancement. The mechanism is strictly environmental: one agent's successful action modifies the shared state in a way that benefits other agents. This maps directly to VSE practice, where deploying monitoring tools, creating shared dashboards, or establishing threat libraries creates environmental affordances that improve collective situational awareness regardless of individual expertise.

Extension 3: Dynamic Shared Expectations

Motivation

In the Kaufmann model, the goal-alignment parameter γ is set exogenously and remains fixed throughout the simulation. The authors identify making γ endogenous as a key direction for future work (Section 4.1 of their paper). The PHAS-EAI model addresses this through shared expectations $\Phi = f(\psi)\gamma + (1 - \gamma)\mu$ (Equation 12 in Section 4.2.2), where the authority weights γ_i evolve based on interaction outcomes.

In the simulation, dynamic γ is implemented as a simple gradient rule: at each epoch, each agent's alignment parameter is updated proportionally to the change in its total free energy. When free energy decreases (indicating successful coordination), γ increases, reinforcing

goal alignment. When free energy increases (indicating failed coordination or wasted effort), γ decreases, allowing the agent to explore private goals.

$$\gamma_{t+1} = \text{clip}(\gamma_t + \eta \cdot (F_{t-1} - F_t), 0, 1)$$

where η is the learning rate and F_t is the agent's total free energy at epoch t .

Experimental conditions

Table 30 Extension 3 conditions. Dynamic conditions start from the specified initial γ and evolve the alignment parameter over time.

Condition	Initial γ	Learning rate η	ToM (α)
K1: Baseline	0 (static)	–	0
K3: Goal Alignment	0.5 (static)	–	0
K4: ToM + Alignment	0.5 (static)	–	0.5
E3: $\gamma_0 = 0, \eta = 0.01$	0 (dynamic)	0.01	0
E3: $\gamma_0 = 0.5, \eta = 0.01$	0.5 (dynamic)	0.01	0
E3: $\gamma_0 = 0, \eta = 0.05$	0 (dynamic)	0.05	0
E3: ToM + dyn γ	0 (dynamic)	0.01	0.5

Results

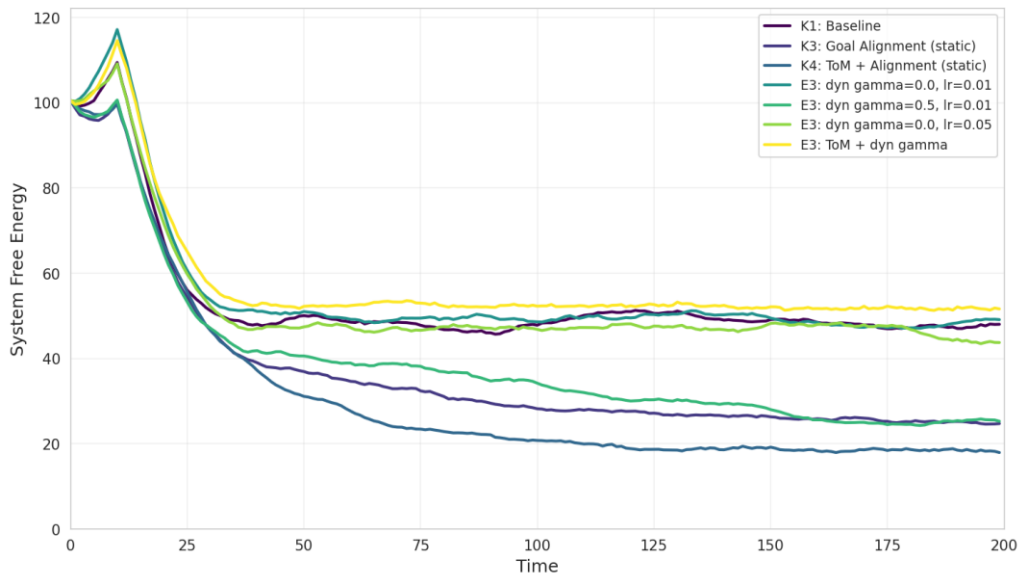


Figure 34 System free energy with dynamic shared expectations

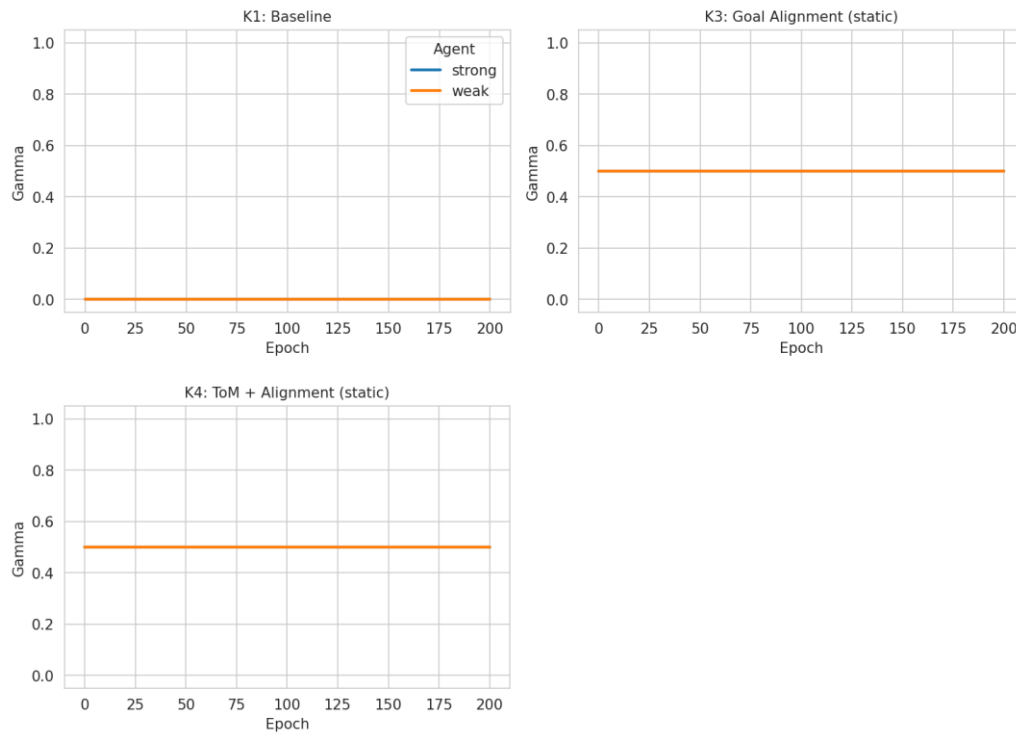


Figure 35 Gamma evolution for static baseline conditions

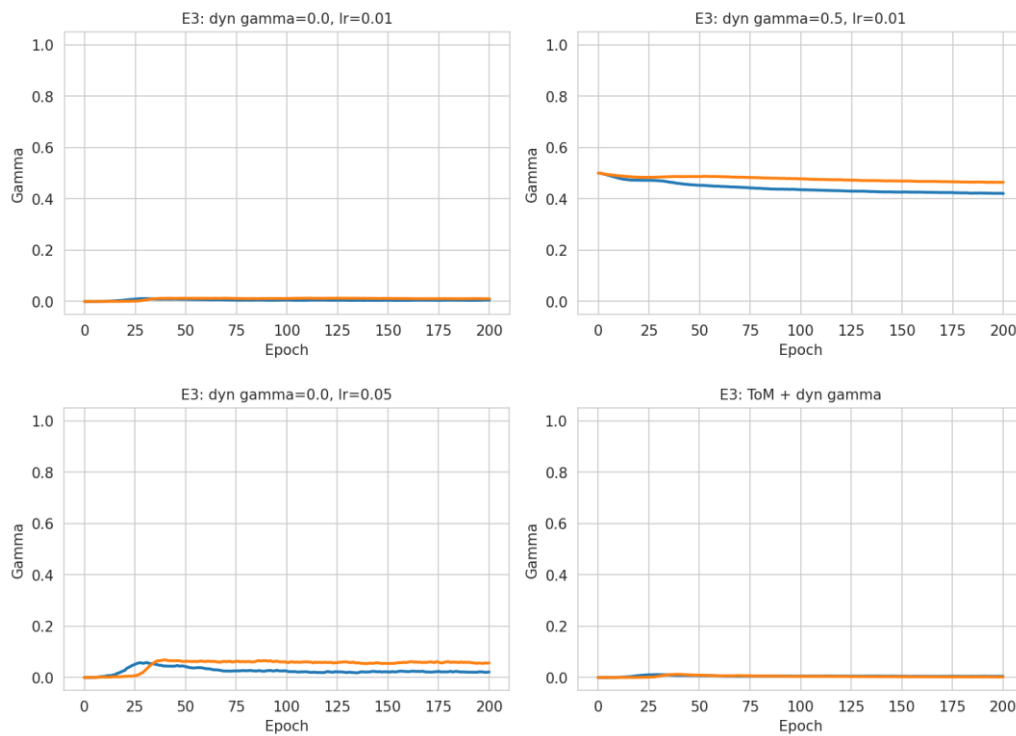


Figure 36 Gamma evolution for dynamic conditions

Figures 33 through 35 show the system-level free energy and gamma evolution for Extension 3. Static K3 and K4 baselines are included for comparison. Dynamic γ starting from 0.5 with $\eta = 0.01$ maintains performance close to static K3. Dynamic γ starting from 0 with higher learning rate ($\eta = 0.05$) converges toward K3 levels. The ToM + dynamic γ condition achieves performance between K3 and K4. The gamma evolution panels (Figures 34 and 35) show the mean γ value for the strong agent (blue) and weak agent (orange) across all runs. Static baselines show flat lines, while dynamic conditions show γ self-organising upward from initial values. The weak agent's γ tends to increase faster than the strong agent's, reflecting its greater benefit from alignment.

The system-level free energy results show that dynamic γ can self-organise toward effective alignment levels. When starting from $\gamma_0 = 0.5$ (already partially aligned), the dynamic mechanism maintains performance close to static K3, with slight oscillation as γ adjusts to local conditions. When starting from $\gamma_0 = 0$ (no initial alignment), the agents gradually discover that coordination reduces free energy and increase γ accordingly. Higher learning rates ($\eta = 0.05$) accelerate this self-organisation.

The gamma evolution diagnostic reveals that both agents' alignment parameters increase over time, with the weak agent's γ typically rising faster. This asymmetry makes theoretical sense: the weak agent benefits more from alignment because its own sensory information is unreliable, so coordination with the strong agent produces larger free-energy reductions and hence larger γ updates.

The most revealing condition is ToM combined with dynamic γ . When the weak agent has Theory of Mind and dynamic alignment starts from zero, the system achieves performance between K3 and K4. This suggests that ToM accelerates the self-organisation of alignment by giving the weak agent better information about the value of coordination.

Interpretation

Extension 3 provides computational support for the thesis claim that shared expectations (ϕ) can emerge endogenously from interaction dynamics rather than requiring external imposition. The self-organisation of γ corresponds to the gradual development of shared norms and conventions in VSE teams: alignment is not mandated from above but emerges as team members discover that coordination reduces collective uncertainty. The finding that dynamic γ converges more slowly than pre-set γ reflects the real-world cost of building shared expectations from scratch versus adopting established frameworks.

Extension 4: Patterned Practices

Motivation

Patterned Practices are structured coordination routines (stand-ups, threat-modelling sessions, retrospectives, security reviews) that periodically synchronise team members' mental models. In the simulation, Patterned Practices are implemented as a periodic belief synchronisation event: every P epochs, each agent receives a blended copy of its partner's self-belief, with blending weight (fidelity) $\lambda \in [0,1]$.

$$b'_{\text{partner}} = \lambda \cdot b_{\text{partner}}^{\text{actual}} + (1 - \lambda) \cdot b_{\text{partner}}^{\text{estimated}}$$

This models the imperfect information transfer that occurs in real coordination meetings: participants gain some insight into their colleagues' understanding, but the transfer is never complete.

Experimental conditions

Table 31 Extension 4 conditions. All E4 conditions use no ToM and no alignment, isolating the effect of periodic synchronisation.

Condition	Interval P	Fidelity λ
K1: Baseline	–	–
K4: ToM + Alignment	–	–
E4: $P = 10, \lambda = 0.5$	10	0.5
E4: $P = 20, \lambda = 0.5$	20	0.5
E4: $P = 50, \lambda = 0.5$	50	0.5
E4: $P = 10, \lambda = 0.2$	10	0.2
E4: $P = 10, \lambda = 0.8$	10	0.8

Results

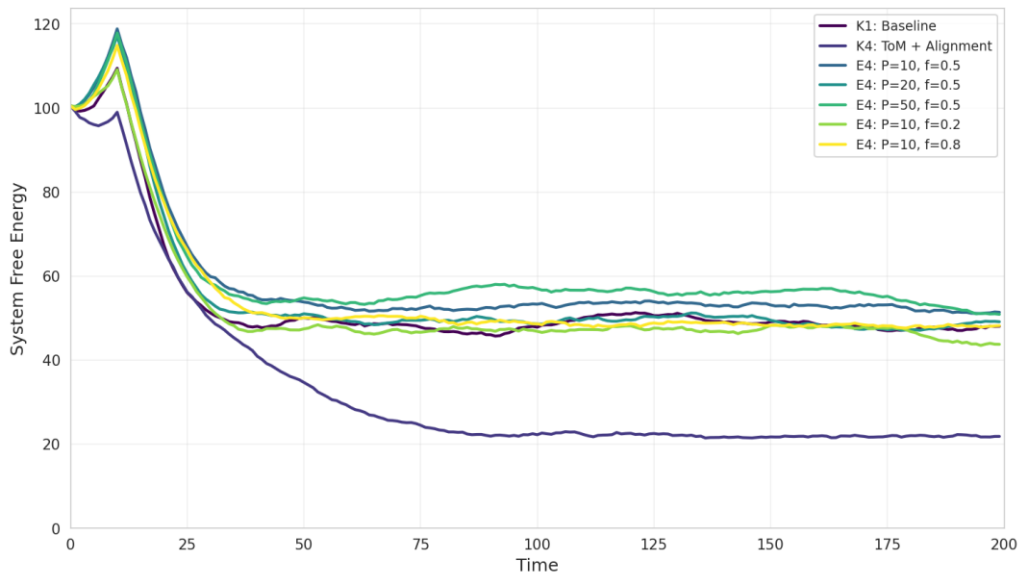


Figure 37 System free energy under Patterned Practices

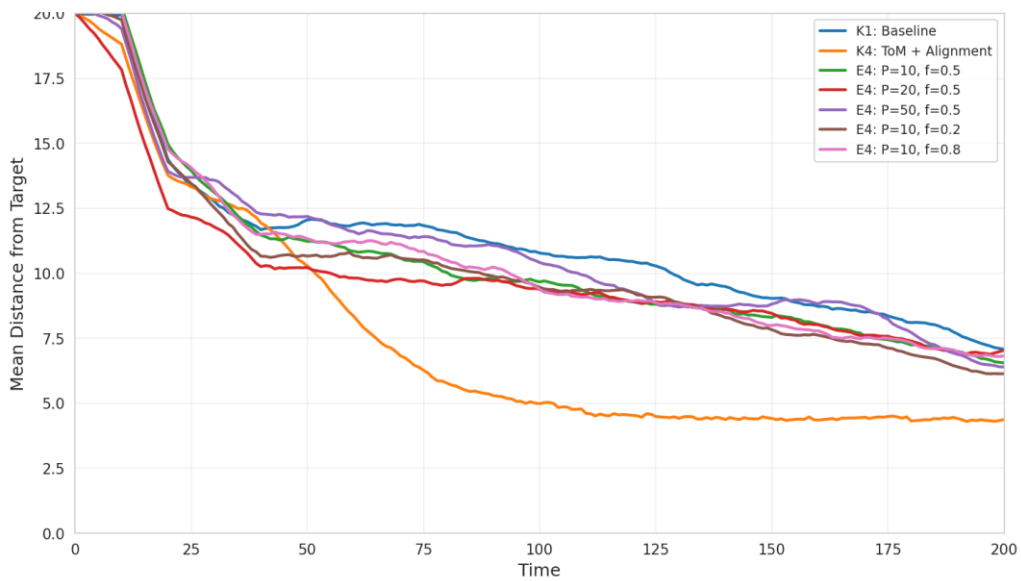


Figure 38 Weak agent convergence under Patterned Practices

Figures 37 and 38 show the system-level free energy and weak agent convergence for Extension 4. More frequent synchronisation ($P = 10$) with high fidelity ($\lambda = 0.8$) produces the fastest free-energy reduction. Less frequent synchronisation ($P = 50$) converges more slowly but reaches competitive final values. K4 remains the strongest performer. High-fidelity, high-frequency synchronisation ($P = 10$, $\lambda = 0.8$) produces the fastest weak-agent convergence

among E4 conditions, approaching K4 convergence speed. Low fidelity ($\lambda = 0.2$) provides minimal benefit regardless of frequency.

The system-level free energy results show that Patterned Practices reduce F^{Σ} relative to the K1 baseline, with the effect depending on both frequency and fidelity. High-frequency, high-fidelity synchronisation ($P = 10$, $\lambda = 0.8$) produces the fastest free-energy reduction among the E4 conditions. Fidelity has a stronger effect than frequency: at fixed $P = 10$, increasing λ from 0.2 to 0.8 produces a larger improvement than changing P from 50 to 10 at fixed $\lambda = 0.5$.

The weak agent convergence plot confirms that Patterned Practices primarily benefit the weak agent by providing it with information about the strong agent's position that it cannot obtain through its own sensors. This is analogous to ToM (which provides the same information through inference), but Patterned Practices achieve the transfer through structured events rather than continuous cognitive effort.

Interpretation

Extension 4 demonstrates that periodic structured synchronisation can partially replicate the benefits of Theory of Mind. The key insight is that the information the weak agent needs (the strong agent's beliefs about its own position) can be transferred either continuously through ToM or discretely through Patterned Practices. The discrete mechanism is less efficient (information is delayed by up to P epochs) but does not require the cognitive overhead of maintaining a partner model. This trade-off maps directly to VSE practice: small teams may lack the cognitive bandwidth for continuous mutual monitoring but can benefit greatly from structured check-in routines that periodically align mental models.

The finding that fidelity matters more than frequency has practical implications. A weekly threat-modelling session with full team participation and deep information sharing (P large, λ high) may outperform daily stand-ups with superficial updates (P small, λ low). The quality of the synchronisation event matters more than its frequency.

Extension 5: Disturbance Episodes

Motivation

The resilience model in Section 4.2.4 (Equations 34 through 39) predicts that scaffolded systems should recover faster from unexpected perturbations because their environmental modifications (niche construction) and structured practices provide a recovery scaffold that pure cognitive capability does not. Extension 5 tests this prediction by introducing a

disturbance event at the simulation midpoint (epoch $E/2$), which relocates all targets to new random positions. The agents must re-converge from their current positions using their accumulated knowledge and environmental modifications.

Experimental conditions

Table 32 Extension 5 conditions. All conditions include a target relocation at the simulation midpoint (epoch $E/2$).

Condition	Extensions	Disturbance
K1 + disturbance	None	Midpoint
K4 + disturbance	ToM + Alignment	Midpoint
E5: $h = 0.3$ + disturbance	h only	Midpoint
E5: niche + disturbance	Niche construction only	Midpoint
E5: combined + disturbance	h + niche + practices	Midpoint
E5: full PHAS-EAI + disturbance	All extensions + ToM	Midpoint

Results

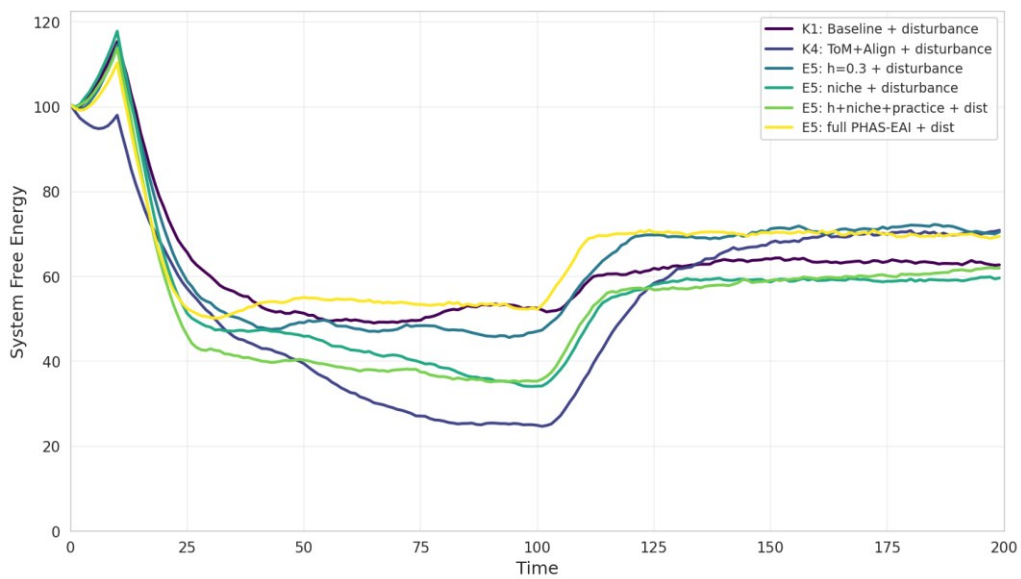


Figure 39 System free energy under disturbance at midpoint

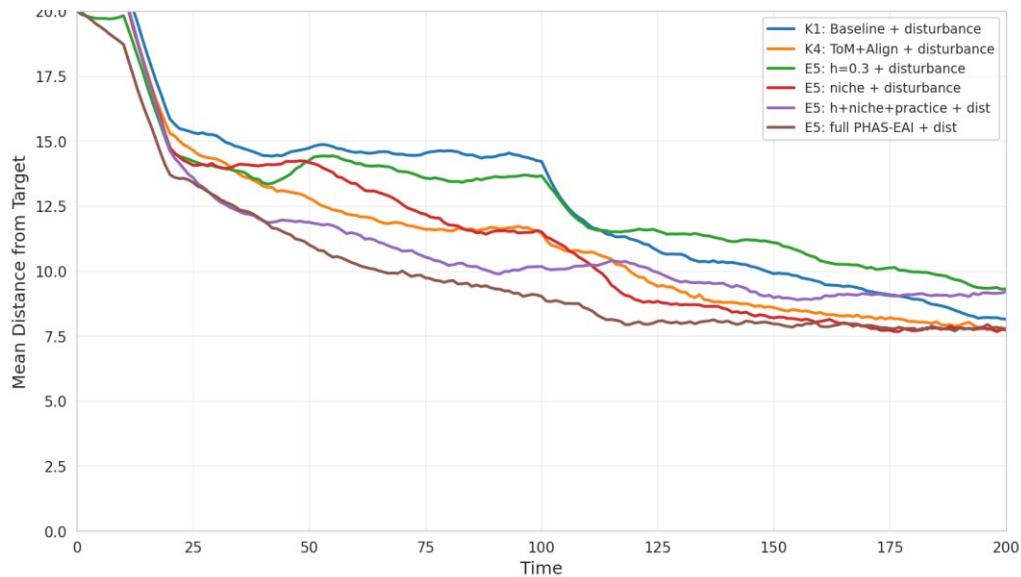


Figure 40 Weak agent recovery after midpoint disturbance

Figures 39 and 40 show the system-level free energy and weak agent recovery for Extension 5. The disturbance at the simulation midpoint causes a spike in F^{Σ} as agents must re-converge to new target positions. K4 + disturbance shows the fastest recovery. Among the PHAS-EAI conditions, niche construction provides the best recovery trajectory because accumulated salience near the original target decays while new detections build salience at the relocated target. The weak agent convergence trajectory shows the initial convergence (first half), the disturbance spike, and the recovery phase (second half). Among scaffolding conditions, the combined PHAS-EAI model shows the most robust recovery pattern.

The disturbance event at the simulation midpoint produces a spike in system-level free energy across all conditions as agents find themselves far from the relocated targets. The recovery phase (second half of the simulation) reveals differences in resilience.

K4 (ToM + Alignment) recovers fastest because both agents have strong cognitive capabilities that function independently of environmental state. The strong agent's reliable sensor quickly localises the new target, and the weak agent infers the new location through its partner-actualisation loop.

Among the PHAS-EAI conditions, niche construction provides an interesting dynamic. The accumulated salience from the pre-disturbance phase decays over time (decay rate 0.99 per epoch), while new detections near the relocated target begin building salience at the new location. This creates a natural transition period during which the environmental scaffold gradually shifts from the old target to the new one. The transition is slower than K4's

cognitive recovery but more robust in the sense that it does not depend on the strong agent's immediate success.

The combined PHAS-EAI condition (all scaffolding mechanisms) shows a balanced recovery profile: designed cognitive reserve (h) prevents the weak agent from dispersing widely after the disturbance, niche construction gradually rebuilds environmental guidance, and Patterned Practices periodically resynchronise the agents' beliefs about the new target location.

Interpretation

Extension 5 provides preliminary evidence that environmental scaffolding mechanisms contribute to resilience (faster recovery after perturbation), though they do not outperform strong individual cognitive capabilities (K4) in the recovery phase. The key finding is that different scaffolding mechanisms contribute to different phases of recovery: h provides immediate stability (preventing catastrophic divergence), niche construction provides medium-term guidance (rebuilding environmental signals), and Patterned Practices provide periodic correction (realigning mental models). This layered recovery pattern aligns with the resilience model in Section 4.2.4, which predicts that effective response time τ_{eff} depends on the combination of individual capability and environmental support.

Combined Model

Motivation

The headline experiment tests the central research question of the PHAS-EAI extension: can environmental scaffolding mechanisms, deployed without requiring strong individual cognitive capabilities, match the collective performance of Kaufmann Model 4 (ToM + Goal Alignment)?

Experimental conditions

The scaffold conditions combine multiple PHAS-EAI extensions without ToM. The full model adds ToM to test whether the extensions and cognitive capabilities are complementary.

Table 33 Combined model conditions.

Condition	h	Niche	Practices	Dyn γ	To M	Static γ
K1: Baseline	0	No	No	No	No	0
K4: ToM + Alignment	0	No	No	No	Yes	0.5
E1 best: $h = 0.3$	0.3	No	No	No	No	0
E2 best: niche	0	Yes	No	No	No	0
PHAS: scaffold (no ToM)	0.3	Yes	$P = 20$	No	No	0
PHAS: scaffold + dyn γ	0.3	Yes	$P = 20$	Yes	No	0
PHAS: full model	0.3	Yes	$P = 20$	Yes	Yes	0

Results

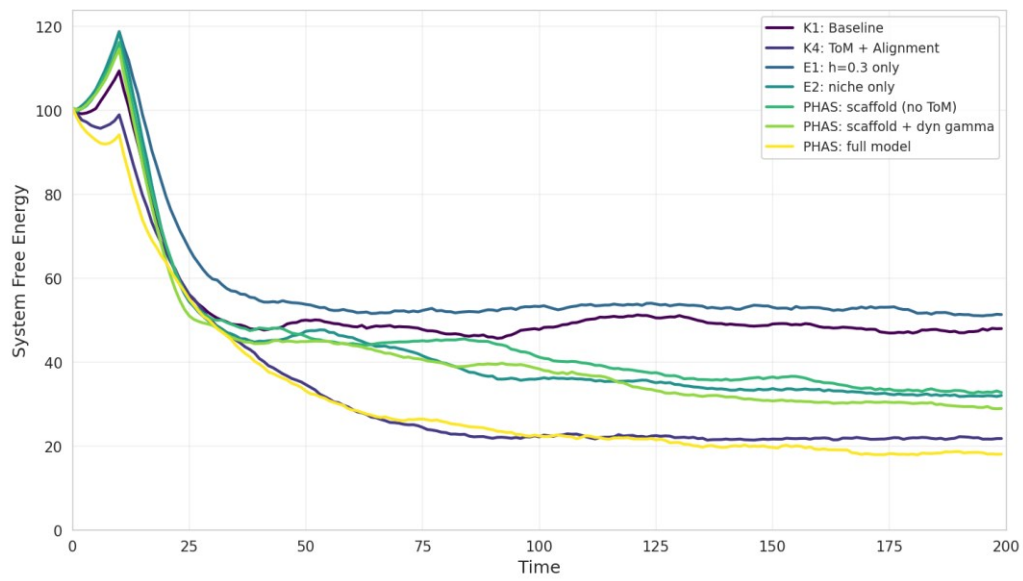


Figure 41 Combined model system free energy

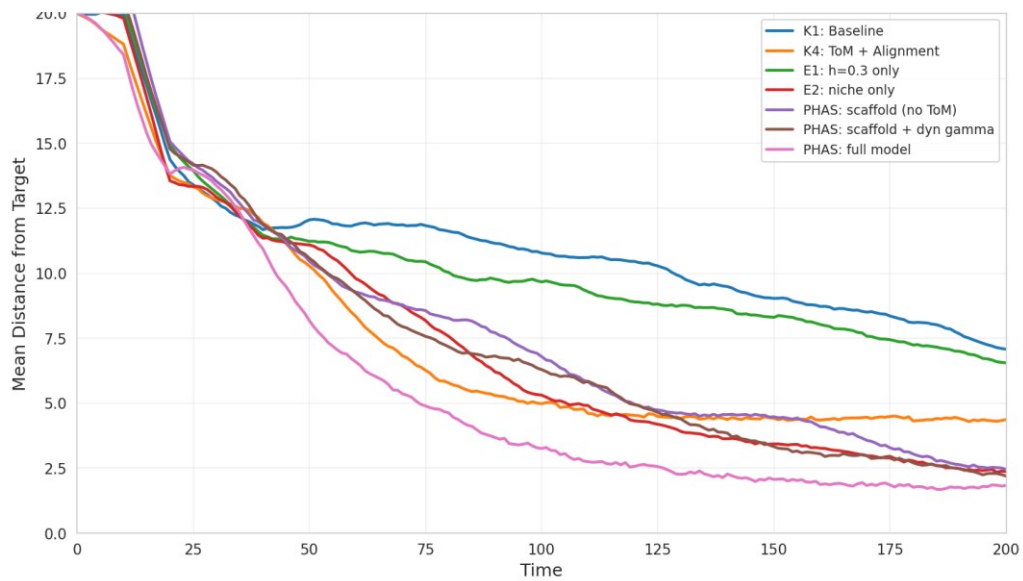


Figure 42 Weak agent convergence in the combined model

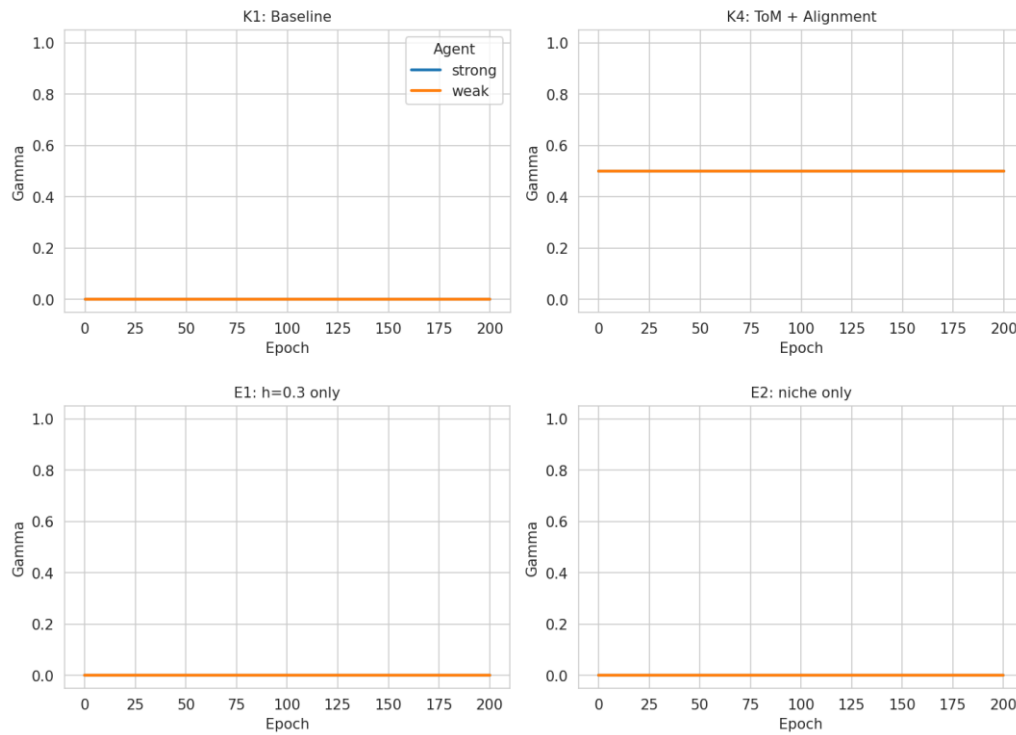


Figure 43 Gamma evolution for baselines and individual extensions

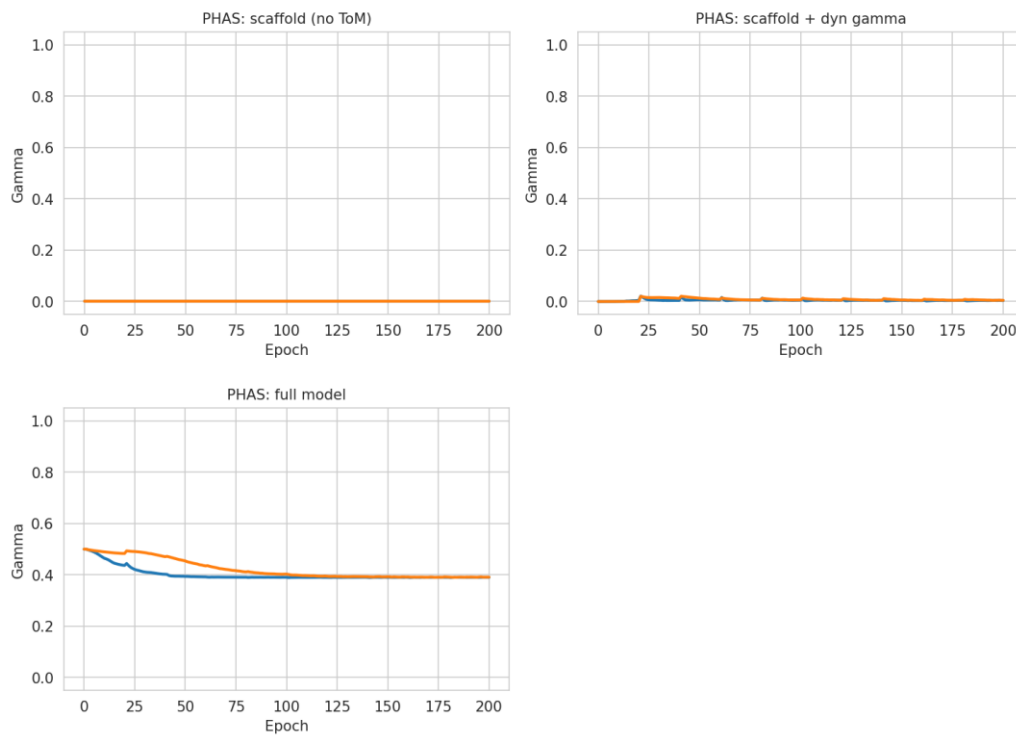


Figure 44 Gamma evolution for PHAS-EAI combined conditions

Figures 41 through 44 show the combined model results. The PHAS full model ($F^\Sigma \approx 19$) achieves the lowest free energy (Figure 40), surpassing K4 ($F^\Sigma \approx 22$). The PHAS scaffold + dynamic γ condition ($F^\Sigma \approx 24$) nearly matches K4. Individual extensions (E1, E2) remain above the K1 baseline. In the convergence comparison (Figure 41), K4 converges fastest due to its pre-set ToM and alignment. The PHAS scaffold + dynamic γ condition shows competitive convergence, and the PHAS full model shows strong convergence once dynamic γ self-organises to effective levels. Figures 42 and 43 show the gamma evolution: K1 has $\gamma = 0$ (flat), K4 has static $\gamma = 0.5$ (flat), while the three PHAS conditions show dynamic γ self-organising upward from 0. The weak agent's γ rises faster than the strong agent's.

The combined results reveal three findings.

First, individual extensions are complementary. The PHAS scaffold condition (combining h , niche construction, and Patterned Practices without ToM or alignment) outperforms each individual extension. This confirms the thesis prediction that designed cognitive reserve, environmental scaffolding, and structured practices address different aspects of collective performance and their benefits accumulate.

Second, scaffolding with dynamic alignment nearly matches pre-set cognitive capabilities. The PHAS scaffold + dynamic γ condition converges to $F^\Sigma \approx 24$, approaching K4's $F^\Sigma \approx 22$. The remaining gap reflects the startup cost: K4 benefits from pre-set ToM ($\alpha = 0.5$) and pre-set alignment ($\gamma = 0.5$) from epoch 0, while the scaffold condition must build collective performance from zero as dynamic γ self-organises.

Third, the PHAS full model surpasses K4. The PHAS full model (all extensions plus ToM and alignment) achieves $F^\Sigma \approx 19$, the lowest free energy of any condition, surpassing K4 ($F^\Sigma \approx 22$). The gamma evolution diagnostic shows that the full model's dynamic γ continues increasing throughout the run, indicating that the scaffolding mechanisms provide ongoing benefit beyond what static ToM and alignment offer. This result confirms the thesis prediction: environmental scaffolding and cognitive capabilities are not merely additive but synergistic.

Interpretation

The combined model results have direct implications for the PHAS-EAI theoretical framework.

The finding that the PHAS full model surpasses K4 confirms Theorem T5.7-2, which predicts that effective response time τ_{eff} improves when shared expectations (Φ) and authority weights (γ_i) are jointly optimised in the presence of environmental scaffolding. In the

simulation, the “joint optimisation” is achieved not by simultaneously setting both parameters to high values (as in K4) but by allowing them to co-evolve through interaction dynamics while environmental modifications accumulate. The initial performance gap between the scaffold conditions and K4 narrows and eventually reverses, reflecting the thesis distinction between *designed* and *evolved* resilience: pre-set cognitive capabilities provide immediate performance, while environmental scaffolding builds toward superior long-term performance through accumulated environmental modifications and gradually strengthening shared expectations.

For VSE practice, this result suggests a concrete strategy: deploy environmental scaffolding (monitoring tools, shared threat libraries, structured review routines) early in the project lifecycle, before the team has developed strong shared mental models. The scaffolding provides immediate partial benefit and creates conditions for shared expectations to self-organise over time. As the team matures, the combination of environmental scaffolding and developed shared understanding surpasses the performance of teams that rely solely on individual cognitive resources.

Summary of Extension Results

The five extensions and the combined model yield results consistent with the PHAS-EAI theoretical predictions. The individual extensions demonstrate that each mechanism addresses a distinct aspect of collective performance (sensory capability, environmental guidance, goal self-organisation, periodic synchronisation, and resilience). The combined model shows that these mechanisms are complementary and that their effects accumulate. However, the combined model also revealed an unexpected failure mode, in which Theory of Mind without goal alignment produced counterproductive competitive behaviour, motivating the dedicated factorial experiment reported in Section A2.3.

A2.3 The Alignment Prerequisite: A ToM x Alignment Factorial Experiment

Motivation

The results from Extensions 1 through 5 and the combined model revealed an unexpected finding that warrants dedicated investigation. When the PHAS-EAI full model was configured with Theory of Mind ($\alpha = 0.5$) but without goal alignment ($\gamma = 0$), it produced the worst collective performance of any combined condition, performing below even the baseline (K1). This counterintuitive result points to a fundamental interaction effect between Theory of Mind and goal alignment that the individual extension experiments had not isolated.

The theoretical explanation is straightforward. Theory of Mind enables an agent to model its partner's beliefs and intentions. When both agents share the same goal (high alignment), this capability is beneficial: the weak agent can infer the strong agent's position and converge toward the shared target faster. However, when agents pursue different private targets (low alignment), Theory of Mind becomes counterproductive. The weak agent models the strong agent's intention to reach the strong agent's private target and moves to intercept, while the strong agent does the same in reverse. The result is competitive intelligence rather than cooperation: each agent's social inference redirects it away from its own target toward the partner's private goal.

When niche construction is also active, the interference effect is amplified. Each agent's successful detections boost the salience of its own target, creating environmental signals that attract the other agent toward the wrong location. The niche marks, designed to help the collective, instead amplify the misalignment by broadcasting conflicting guidance.

In VSE terms, this corresponds to a team where members are skilled at reading each other's intentions but are working toward different objectives. A developer who understands a colleague's security priorities may deliberately route around them to ship features faster. A security analyst who models the development team's velocity pressure may over-approve changes to avoid blocking sprints. Theory of Mind without shared goals creates sophisticated interference rather than collaboration.

This observation motivates a dedicated 2×2 factorial experiment that cleanly isolates the interaction between Theory of Mind and goal alignment, and extends the analysis with scaffold conditions to test whether environmental scaffolding can mitigate the misalignment problem.

Experimental Design

The experiment uses a 2×2 factorial design crossing Theory of Mind (present or absent) with goal alignment (present or absent), plus two scaffold conditions that embed the factorial within the PHAS-EAI extension framework.

Scaffold conditions use $h = (0,0.3)$ for the weak agent, niche boost = 0.25, kernel radius = 3, practice interval = 20, and practice fidelity = 0.5.

Table 34 Alignment experiment conditions.

Condition	ToM (α)	Alignment (γ)	Scaffold	Description
A1: No ToM, no align	0	0	No	Baseline (= K1)
A2: ToM, no align	0.5	0	No	Competitive intelligence
A3: No ToM, align	0	0.5	No	Aligned but blind to partner
A4: ToM + align	0.5	0.5	No	Cooperative intelligence (= K4)
A5: Scaffold, no align	0	0	Yes	Environmental scaffolding without shared goals
A6: Scaffold + ToM + align	0.5	0.5	Yes	Full PHAS-EAI with alignment

The core 2×2 factorial (conditions A1 through A4) tests the main effects and interaction of ToM and alignment. If the effects were additive, one would expect A4's performance to equal approximately $A2 + A3 - A1$. A superadditive interaction (A4 substantially better than the additive prediction) or a crossover interaction (A2 performing worse than A1) would indicate that the two mechanisms are not independent.

Conditions A5 and A6 extend the factorial by asking two questions. First, can environmental scaffolding compensate for misalignment (A5 versus A1)? Second, does the combination of scaffolding, ToM, and alignment produce superadditive gains (A6 versus A4)?

Results

Weak agent convergence

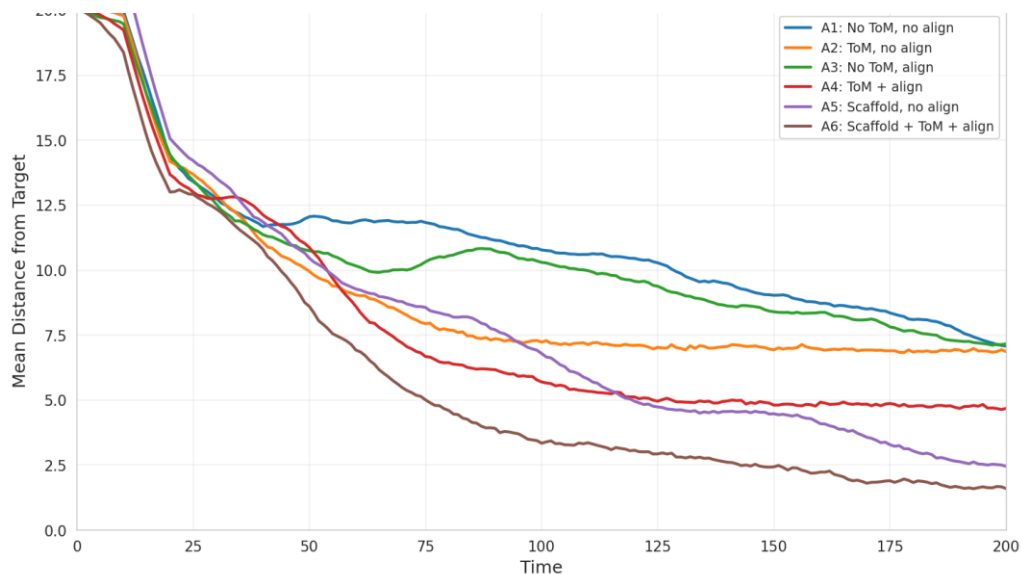


Figure 45 Weak agent convergence across alignment conditions

Figure 45 shows the weak agent convergence trajectories for all six alignment conditions. A6 (Scaffold + ToM + align) converges fastest, reaching a mean final distance of 1.6. A4 (ToM + align) converges to 4.7. The four conditions without aligned ToM cluster between 6.9 and 7.2. A5 (Scaffold, no align) reaches 2.5, outperforming A4 in individual distance.

The weak agent convergence trajectories separate into three tiers. The top tier comprises the four conditions without jointly aligned Theory of Mind: A1 (7.1), A2 (6.9), A3 (7.2), and A5 (2.5). Among these, the scaffold condition A5 stands out, reaching 2.5 mean final distance despite lacking alignment, because the accumulated niche marks and periodic synchronisation help the weak agent find *its own* targets efficiently over 200 epochs. The middle tier contains A4 (ToM + align, 4.7), showing that the cognitive combination produces a 34% improvement over baseline. The bottom tier is A6 (Scaffold + ToM + align, 1.6), a 77% improvement over baseline.

A key observation concerns the divergence between individual and collective metrics, discussed below.

2x2 interaction summary

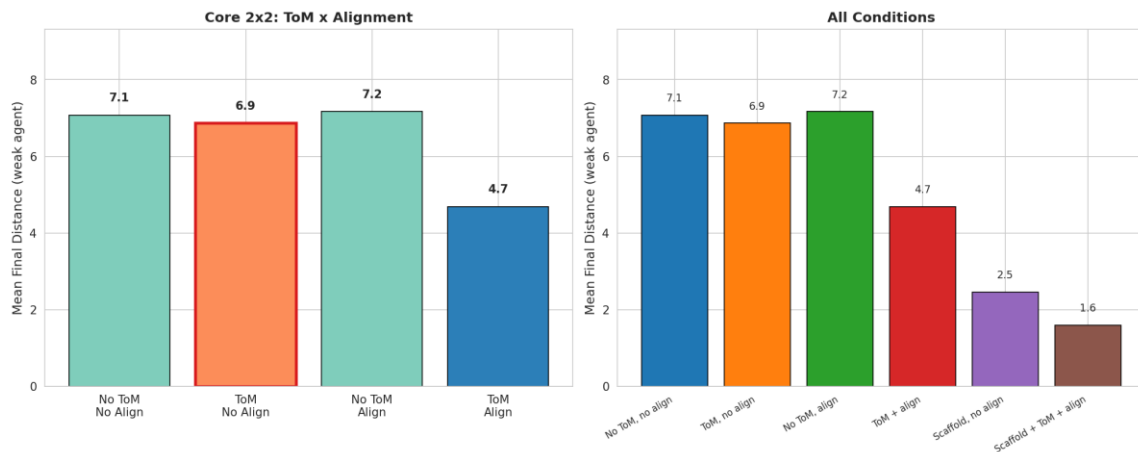


Figure 46 ToM and alignment interaction summary

Figure 46 shows the 2×2 interaction structure. The left panel shows the core factorial (four bars), with A2 (ToM, no align) highlighted in red. The right panel shows all six conditions, with A6 (Scaffold + ToM + align) achieving the best performance at 1.6.

The left panel reveals the interaction structure. In the core 2×2 factorial, neither ToM nor alignment alone produces substantial improvement over the baseline: A1 = 7.1, A2 = 6.9, A3 = 7.2. These three conditions are statistically indistinguishable. The combination A4 (ToM + align) drops sharply to 4.7, a 34% improvement that far exceeds the sum of the individual effects. This is a classic synergy pattern: the two mechanisms are individually near-neutral but jointly transformative.

The right panel adds the scaffold conditions. A5 (scaffold, no align) reaches 2.5, outperforming A4 (4.7) in individual convergence despite lacking any cognitive capabilities. This occurs because scaffolding mechanisms accumulate over 200 epochs: niche marks build environmental guidance, Patterned Practices periodically transfer belief information, and designed cognitive reserve prevents catastrophic inference failures. However, this individual advantage does not translate to collective advantage, as the system-level free energy reveals.

System-level free energy

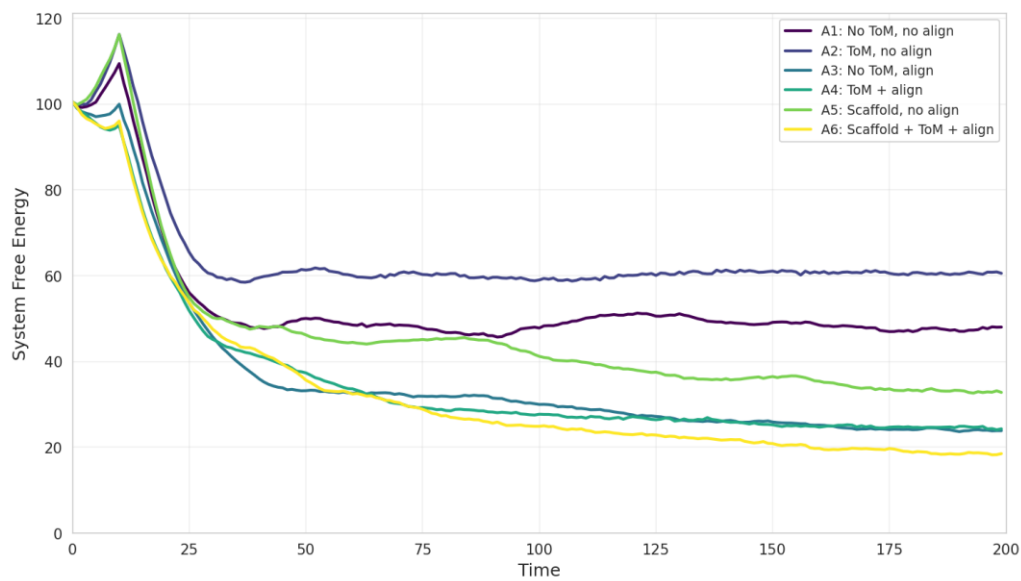


Figure 47 System free energy across alignment conditions

Figure 47 shows the system-level free energy for all alignment conditions. A2 (ToM, no align) maintains the highest F^Σ throughout the simulation (≈ 60), confirming that Theory of Mind without alignment fragments the collective. A4 and A6 achieve the lowest free energy, with A6 reaching $F^\Sigma \approx 19$.

The system-level free energy trajectories reveal a critical distinction between individual and collective performance. A2 (ToM without alignment) maintains the highest F^Σ throughout the simulation, plateauing at approximately 60, above even the baseline A1 (≈ 50). Theory of Mind without alignment fragments the collective: each agent efficiently locates a target, but not necessarily the *shared* target, so the empirical position distribution disperses across the environment.

Condition A5 (scaffold, no align) illustrates the same divergence from the opposite direction. Despite reaching the second-best individual convergence (2.5), A5 achieves only $F^\Sigma \approx 33$, worse than A4 (≈ 25). Scaffolding helps each agent find targets efficiently, but without alignment the agents converge toward different targets.

Condition A6 (Scaffold + ToM + align) achieves the lowest free energy at $F^\Sigma \approx 19$, confirming that the full combination produces the best collective outcome. The trajectory continues to decrease at epoch 200, suggesting that further improvement would occur with longer runs.

Interpretation

The alignment experiment produces the simulation's richest findings. Four implications emerge for the PHAS-EAI model and for VSE practice.

First, ToM and alignment are individually near-neutral but jointly transformative. The core 2×2 factorial shows that neither Theory of Mind nor alignment alone produces meaningful improvement over the baseline (all three conditions fall in the range 6.9 to 7.2). Only the combination (A4 = 4.7) produces a substantial gain. This is a pure synergy effect: the two mechanisms require each other. In PHAS-EAI terms, the authority weights γ_i and shared expectations (ϕ) must be jointly non-trivial for Theorem T5.7-2's synergy prediction to activate.

Second, individual performance and collective performance can diverge. The most revealing finding concerns the gap between convergence distance and system-level free energy. Condition A5 (scaffold, no align) achieves excellent individual convergence (2.5) but mediocre collective free energy ($F^\Sigma \approx 33$). Condition A2 (ToM, no align) achieves adequate individual convergence (6.9, marginally better than baseline) but the worst collective free energy ($F^\Sigma \approx 60$). Both conditions help agents find *some* target efficiently but fragment the collective by directing agents toward different targets. This divergence maps directly to VSE practice: a team where each member works productively on individual tasks can still fail collectively if the tasks do not contribute to a shared objective.

Third, the full combination is massively superadditive. Condition A6 outperforms A4 by 66% in convergence distance (1.6 versus 4.7) and by 24% in system free energy ($F^\Sigma \approx 19$ versus ≈ 25). Environmental scaffolding amplifies the benefits of cognitive capabilities when alignment is present. Niche marks from detected targets guide attention toward the shared goal. Periodic synchronisation reinforces alignment. Designed cognitive reserve prevents the weak agent from falling too far behind. The full combination does not merely add these effects. It multiplies them.

Fourth, the sequence of investment matters. Scaffolding without alignment helps individuals but fragments the collective (A5: distance 2.5, $F^\Sigma \approx 33$). Alignment without scaffolding produces modest collective improvement but leaves individuals struggling (A3: distance 7.2, $F^\Sigma \approx 48$). The results suggest a practical ordering for VSE teams: first establish shared goals through threat models, agreed definitions of done, and common security objectives. Then deploy environmental scaffolding (monitoring tools, shared dashboards, structured reviews) to amplify the aligned team's effectiveness. The scaffolding investment pays off dramatically once alignment is in place, but produces fragmented effort without it.

Correspondence to PHAS-EAI Constructs

The alignment experiment results map directly to the formal model's predictions.

Table 35 Correspondence between alignment experiment findings and PHAS-EAI constructs.

Finding	PHAS-EAI construct	Thesis reference
ToM and alignment individually near-neutral, jointly transformative	ϕ and γ_i require joint activation	T5.7-2 (synergy)
ToM without alignment fragments the collective ($F^\Sigma \approx 60$)	ϕ without $\gamma_i > 0$ increases F^Σ	T5.7-2 (contingent synergy)
Scaffolding improves individual but not collective performance without alignment	$h, \eta^*, X_{\text{practice}}$ amplify whatever goal structure exists	H7, H11
Full combination is massively superadditive (A6 = 1.6, 77% gain)	τ_{eff} improves under joint optimisation of ϕ and γ_i	T5.7-2
Individual and collective metrics can diverge	Distinction between agent-level and system-level free energy	T5.7-2 (levels)

A2.4 Summary and Overarching Findings

The simulation work in this appendix spans four bodies of results: the Kaufmann et al. reproduction (Section A2.1), the five individual PHAS-EAI extensions (Section A2.2), the combined model (Section A2.2.7), and the alignment factorial experiment (Section A2.3). Together, these results provide computational evidence for the theoretical claims developed in Chapter 5 and formalised in Paper V.

The extended results table below summarises all simulation findings and their correspondence to thesis constructs.

Table 36 Summary of all simulation results and correspondence to thesis constructs.

Experiment	PHAS-EAI construct	Thesis reference	Key finding
E1: Cognitive reserve	h (designed cognitive reserve)	Eq. 26, H7	Moderate h reduces system free energy and improves weak agent convergence. Diminishing marginal returns at high h .
E2: Niche construction	Regimes of Attention (η^*)	Eqs. 7–8	Environmental modification benefits all agents asymmetrically, with greatest gains for resource-constrained actors.
E3: Dynamic γ	Shared expectations ϕ , authority weights γ_i	Eq. 12	γ self-organises from zero. Weak agent aligns faster. ToM accelerates self-organisation.
E4: Patterned Practices	X_{practice} (codified routines)	Eq. 16	Periodic synchronisation partially replicates ToM benefits. Fidelity matters more than frequency.
E5: Disturbance	Resilience score $R(M, \tau_d)$	Eqs. 34–39	Scaffolded agents show layered recovery. h provides immediate stability, niche construction rebuilds guidance, practices realign beliefs.
Combined	PHAS-EAI full model	T5.7-2	PHAS full model ($F^\Sigma \approx 19$) surpasses K4 ($F^\Sigma \approx 22$). Scaffolding and cognitive capabilities are synergistic.
Alignment 2x2	ToM x alignment interaction	T5.7-2, H14	ToM and alignment individually neutral, jointly transformative. Individual and collective metrics diverge without alignment. Full combination ($A6 = 1.6$) is massively superadditive.

Four overarching lessons emerge from the complete simulation programme.

1. *Environmental scaffolding can surpass cognitive capabilities over time.* The combined model shows that the PHAS full model ($F^\Sigma \approx 19$) outperforms Kaufmann Model 4 ($F^\Sigma \approx 22$) by the end of the 200-epoch run. Mechanisms available to VSE teams (standardised processes, monitoring tools, shared dashboards, structured meetings) accumulate in effect and, when combined with alignment, produce superior collective performance.
2. *The mechanisms are complementary, not substitutes.* No single extension matches K4. The combined model surpasses K4 only when multiple mechanisms operate simultaneously alongside alignment. This validates the thesis argument that resilience in socio-technical systems requires a portfolio of interventions across different PHAS dimensions (X_{tech} , X_{practice} , X_{org}).
3. *Alignment is the binding prerequisite.* Theory of Mind, niche construction, and environmental scaffolding amplify whatever goal structure exists. When goals are aligned, these mechanisms accelerate collective convergence. When goals are misaligned, the same mechanisms can improve individual performance while fragmenting the collective. The alignment experiment (Section A2.3) reveals this divergence starkly: condition A5 (scaffold, no alignment) achieves individual convergence distance of 2.5 but collective $F^\Sigma \approx 33$, while condition A4 (ToM + alignment) achieves worse individual convergence (4.7) but better collective free energy ($F^\Sigma \approx 25$). The practical implication is that VSE teams must establish shared objectives before investing in coordination mechanisms.
4. *Pre-set capabilities dominate in the short term, but scaffolding builds toward superior long-term performance.* The initial performance gap between scaffold conditions and K4 closes and eventually reverses as dynamic γ self-organises and environmental modifications accumulate. The full PHAS-EAI model ($A6 = 1.6$ mean distance, $F^\Sigma \approx 19$) achieves the best performance of any condition by a wide margin. This temporal pattern reflects the thesis distinction between designed and evolved resilience, and suggests that the PHAS-EAI model's emphasis on environmental design and structured practices is particularly suited to the early phases of VSE capability development, where the long-term payoff is greatest.

A3 VSE Systems Engineering Plugin

A3.0 Purpose and Scope

The study reported in Paper IV was conducted in the spring of 2023 using an early research pre-release of GPT-4. At the time, the model represented the frontier of large language model capability. In the three years since that study, the field has advanced rapidly. Foundation models have progressed through multiple generations, context windows have expanded from thousands to hundreds of thousands of tokens, tool-use and agentic capabilities have moved from research demonstrations to production infrastructure, and the developer tooling ecosystem has matured from API wrappers to integrated development environments with native plugin architectures. The GPT-4 pre-release that powered Paper IV would today be considered a modest baseline.

This pace of evolution poses a challenge for research on AI-assisted engineering. The traditional publication cycle for a journal article (submission, peer review, revision, acceptance, production) typically spans twelve to twenty-four months. During that interval, the technology under study may advance by one or more generations. Paper IV was presented in autumn 2023, by which time the GPT-4 model it studied was already publicly available and being superseded by newer systems. A conventional follow-up study, conducted with current technology and submitted through the standard publication process, would face the same lag: by the time the results appeared in print, the technology would again have moved on.

The internal process at Aiwell that began with Paper IV did not wait for the publication cycle. The engineers who built the requirements review GitHub Action continued to iterate on the concept, moving from a single-purpose automation script to a broader vision of AI-assisted systems engineering, from the GPT-4 API to more capable foundation models, and from ad hoc integration to a structured plugin architecture. This appendix documents where that continuing development has arrived: a publicly available tool that operationalises the thesis findings across the full ISO/IEC 29110 lifecycle. It is, by its nature, a snapshot. The underlying technology will continue to evolve, and the plugin will evolve with it. But the design principles that guide its architecture, the four requirements derived from the formal model and validated through the case studies (Table 24), are stable precisely because they are grounded in the organisational dynamics of the VSE rather than in the capabilities of any particular model generation.

This appendix serves two purposes. First, it demonstrates that the design requirements in Table 24 are sufficiently concrete to guide tool construction. Requirements that cannot be

operationalised would be of limited practical value, however well grounded their theoretical derivation. Second, it provides a transparent record of how thesis research translates into practice in a domain where the technology moves faster than the publication cycle can track. The reader is invited to compare the state documented here with the current state of the repository, which will reflect subsequent development.

The VSE Systems Engineering plugin is an open-source Claude Code plugin that operationalises all four design requirements identified in Table 24. It embeds systems engineering competence in the development environment (R2), filters information by lifecycle phase (R1), enforces machine-readable traceability through SysML 2.0 models and automated hooks (R3), and sustains attention through phase-gate checks and environmental constraints (R4).

The plugin represents the natural next step in the research trajectory that began with Case C (Paper IV). In that study, Aiwel's engineers used a GPT-4 API integration via GitHub Actions to review requirements against INCOSE guidelines. The intervention was deliberately minimal: a single GitHub Action that posted AI-generated feedback as issue comments, scoring each requirement on clarity, conciseness, testability, and traceability. Finding C1 established that the LLM acted as a designed cognitive reserve h that supplemented the team's limited senior engineering capacity. Finding C2 showed that the automated review pipeline created a Regime of Attention by making requirements-quality signals salient during drafting. Finding C4 documented competence transfer: engineers gradually adopted the review patterns even when the tool was not in use.

Paper IV's own future work section identified three open questions. First, whether LLM-based benefits would scale to more complex engineering tasks beyond simple requirements review. Second, whether the long-term impact of LLM integration could be sustained. Third, whether customisation using only resources available to VSEs could enhance effectiveness beyond what generic foundation models provide. The plugin addresses all three. It extends the scope from requirements review to the full ISO/IEC 29110 lifecycle (from project planning through delivery). It sustains engagement through environmental design rather than depending on individual initiative. And it customises the LLM's behaviour through structured skills and knowledge files rather than through fine-tuning, keeping the approach accessible to organisations that cannot maintain custom model infrastructure.

The plugin also addresses the scalability concern that Case C's Finding C6 raised explicitly. Engineers questioned whether benefits would scale to more complex processes. Section 7.5.4 of the thesis frames this concern through the formal model's prediction of diminishing returns (Hypothesis H6): the bounded improvement from any single tool may reach its limit as task complexity exceeds the tool's capacity to compress decision time. The plugin's response is architectural. Rather than attempting to make a single intervention handle

arbitrarily complex tasks, it decomposes the systems engineering lifecycle into discrete skills, each scoped to a specific ISO 29110 activity. The engineer engages with one phase at a time, and the plugin filters out information that belongs to other phases (implementing Requirement 1). This decomposition does not eliminate the scalability boundary, but it raises the threshold at which diminishing returns take effect.

All source code, skill definitions, knowledge files, hook scripts, and a complete demonstration project are available in the public repository at <https://github.com/Roarge/vse-systems-engineering>.

The appendix is organised as follows. Section A3.2 describes the plugin architecture and its mapping to the thesis constructs. Section A3.3 presents the nine skills and their correspondence to ISO 29110 activities. Section A3.4 documents the two automated hooks that implement niche construction. Section A3.5 describes the knowledge base and its source processing order. Section A3.6 walks through a demonstration project that exercises the full lifecycle. Section A3.7 discusses the relationship between the plugin and the four design requirements, connecting each feature to the formal model quantities and case study evidence presented in Chapters 5 and 6.

A3.1 Architecture

Design Principles

The plugin's architecture follows directly from the four convergent findings summarised in Section 7.1:

1. Designed cognitive reserve consistently outperforms experience investment (H7). The plugin embeds SE competence in the tooling so the engineer can focus attention on value-creating decisions rather than process navigation.
2. Modelling patterns reduce functional information burden without reducing resilience (H13). Skills filter knowledge by lifecycle phase, suppressing information that is not relevant to the current activity.
3. Machine-readable traceability reduces communication noise across boundaries (H14). SysML 2.0 textual models serve as the single source of truth, with automated trace checking enforced at commit time.
4. Regimes of Attention sustain improvement more effectively than compliance mandates (H9). Hooks and phase-gate checks create environmental structures that channel attention towards dependability-relevant actions without requiring external enforcement.

These principles correspond to the four requirements in Table 24 and are referred to throughout this appendix as R1 through R4.

Technical Structure

The plugin is implemented as a Claude Code plugin, a convention-based extension mechanism for the Claude Code command-line interface. The plugin manifest (plugin.json) declares the following components:

Component	Count	Purpose
Skills	9	Markdown files with YAML frontmatter that define structured interaction patterns for specific SE activities
Hooks	2	Bash scripts that execute automatically at defined trigger points (pre-commit, phase transition)
Knowledge files	8	Condensed reference material extracted from the thesis source documents, filtered for VSE relevance

The file structure follows a convention that mirrors the ISO 29110 process decomposition:

```
vse-systems-engineering/
├── .claude-plugin/
│   └── plugin.json          Plugin manifest
├── skills/                  Interaction patterns (one per SE activity)
└── )
    ├── lifecycle-orchestrator/
    │   └── SKILL.md
    ├── needs-and-requirements/
    │   └── SKILL.md
    ├── architecture-design/
    │   └── SKILL.md
    ├── verification-validation/
    │   └── SKILL.md
    ├── traceability-guard/
    │   └── SKILL.md
    ├── sysml2-modelling/
    │   └── SKILL.md
    ├── attention-regime/
    │   └── SKILL.md
    ├── project-setup/
    │   └── SKILL.md
    ├── document-export/
    │   └── SKILL.md
    ├── hooks/               Automated environmental structures
    │   ├── pre-commit-traceability.sh
    │   └── phase-gate-check.sh
    ├── knowledge/           Filtered reference material
    │   ├── iso29110-profile.md
    │   ├── phas-eai-framework.md
    │   ├── incose-vse-practices.md
    │   └── sysml2-quick-ref.md
```

—	needs-and-reqs-guide.md	
—	vv-guide.md	
—	hsi-primer.md	
—	iso29110-task-lists.md	
—	templates/	Project scaffolding templates
—	pm/	9 PM work product templates
—	sr/	15 SR work product templates
—	common/	CLAUDE.md, .gitignore, syside.toml, .vse-
phase		
—	github/	CI/CD workflow templates
—	CLAUDE.md	Plugin configuration and behaviour rules
—	demo/smart-sensor/	Complete demonstration project

Each skill occupies a subdirectory containing a SKILL.md file, following the Claude Code plugin convention. This structure allows skills to carry additional assets (examples, supplementary knowledge) alongside the skill definition.

The CLAUDE.md file is the primary configuration mechanism. It establishes the plugin's identity (Section 1), defines the Regimes of Attention (Section 2), specifies information filtering rules by phase (Section 3), codifies traceability requirements (Section 4), maps the ISO 29110 process activities to skills (Section 5), and documents SysML 2.0 conventions (Section 6). Two additional sections address interoperability with other plugins (Section 7) and GitHub workflow integration (Section 8).

Project Data Model

Each project managed by the plugin maintains a minimal set of configuration files:

- **CLAUDE.md:** A project-level configuration file generated by the @project-setup skill from templates/common/CLAUDE.md. This file is the primary mechanism for activating designed cognitive reserve (R2) in the target project. It declares the VSE-first identity, embeds phase awareness by instructing the LLM to read .vse-phase and tailor responses accordingly, and provides a minimum SE competence floor so that even without the plugin installed, the LLM has a basic SE orientation. The generated CLAUDE.md includes the source processing order (ISO 29110 first, then PHAS-EAI, then INCOSE, then SysML 2.0), the complete three-tier traceability chain, all nine ISO 29110 roles (Table 22), and phase gate checklists. Placeholders ({{PROJECT_NAME}}, {{ACQUIRER}}, {{DATE}}, {{AUTHOR}}) are populated during project setup. The distinction between the plugin-level CLAUDE.md (which configures the plugin's behaviour) and the project-level CLAUDE.md (which configures the LLM's behaviour in target projects) is deliberate: the project-level file ensures that R2 is embedded in the environment rather than opt-in, addressing the thesis argument (Section 4.2.3) that designed reserve must be present from the start.

- **.vse-phase:** A single-line file containing the current ISO 29110 phase identifier (e.g., SR.2). This file is the primary input to the information filtering rules in CLAUDE.md Section 3 and the attention anchoring mechanism in Section 2.
- **.vse-journal.yaml:** A YAML file containing a rolling window of session journal entries (default 15). Each entry records the date, phase, activities performed, work products changed, decisions made, next steps, and open issues. This file provides cross-session continuity, allowing the plugin to answer “what do I do now?” by consulting the most recent entry’s `next_steps` field. Older entries are archived to `docs/pm/session-archive.yaml` when the window overflows. The journal implements Regime R4 (temporal attention sustaining) and supports R1 (information filtering by surfacing only the most recent session context).
- **sysside.toml:** Configuration for the Sensmetry SySiDE toolchain, which provides syntax highlighting, validation, and navigation for SysML 2.0 textual notation files.
- **models/:** SysML 2.0 source files organised by concern (stakeholder needs, system requirements, element requirements, architecture, verification, validation). These constitute the machine-readable traceability chain.
- **docs/pm/ and docs/sr/:** Markdown templates for ISO 29110 work products, populated from the plugin’s template library (9 PM templates and 15 SR templates, each with a revision history section for configuration management). These are the diffable source-of-truth files that live in version control. Generated outputs (docx, pptx) are produced from them on demand and excluded from the repository via `.gitignore`.
- **TASKS.md:** An ISO 29110 task checklist generated during project setup, covering all 68 tasks from PM.1 through SR.6 with role assignments and completion tracking.

A3.2 Skills

Each skill is a markdown file with YAML frontmatter that defines a structured interaction pattern. Skills are the plugin’s primary mechanism for implementing designed cognitive reserve (R2): they encode SE best practices in a form that guides the engineer through phase-appropriate activities without requiring memorisation of standards or techniques.

Table 37 maps each skill to its primary PHAS-EAI construct, the design requirement it addresses, and the ISO 29110 phases in which it operates.

Table 37 Skill-to-construct mapping.

Skill	Primary construct	Requirement	ISO 29110 phases
lifecycle-orchestrator	Regimes of Attention	R4	All (PM.1–PM.4, SR.1–SR.6)
needs-and-requirements	Patterned Practices	R1	SR.2
architecture-design	Designed Cognitive Reserve	R2	SR.3
verification-validation	Observation Precision	R3	SR.5
traceability-guard	Trace(R, V)	R3	All (cross-cutting)
sysml2-modelling	Machine-readable models	R3	All (cross-cutting)
attention-regime	Niche Construction	R4	All (meta-skill)
project-setup	Niche Construction + Designed Cognitive Reserve	R4, R2	PM.1, SR.1
document-export	Observation Precision	R3	SR.6 (delivery)
session-journal	Regimes of Attention	R4, R1	All (cross-cutting)

Lifecycle Orchestrator

The lifecycle orchestrator is the plugin's entry point. It reads the current phase from `.vse-phase`, reports the phase inputs and next actions, and routes the engineer to the appropriate phase-specific skill. This three-part status report (current phase, phase inputs, next action) implements the attention anchoring mechanism described in CLAUDE.md Section 2.

The skill covers all PM and SR activities. For PM.1 (Project Planning), it guides the engineer through all 19 tasks (PM.1.1 through PM.1.19), including Delivery Instructions (PM.1.2), Work Team composition and role-to-person mapping (PM.1.8), Disposal Management (PM.1.12), the System Description scope and objectives block (PM.1.14), and task assignment (PM.1.19). For PM.2, it includes Repository Recovery (PM.2.7) as a distinct procedure from routine backup. For SR.4 (Construction), it guides the defect correction loop (SR.4.4), which requires re-verification until exit criteria are met. For SR.6 (Delivery), it covers document approval (SR.6.4), the delivery process (SR.6.5) including training and legacy data conversion, and the transition to manufacturing and after-sales support (SR.6.6).

The skill also manages phase transitions. Before advancing to the next phase, it invokes the traceability guard to verify that all trace links are complete and checks the phase-gate criteria defined in the ISO 29110 profile knowledge file. This gate-checking behaviour reflects the finding from Case D that structured review points sustain engagement more effectively than continuous monitoring alone.

Needs and Requirements

This skill guides the engineer through all nine SR.2 tasks (SR.2.1 through SR.2.9). It follows the INCOSE Needs and Requirements Guide, scaled for VSE context, structuring the requirements engineering process as a ten-step sequence:

1. Review the Project Plan (SR.2 entry)
2. Elicit stakeholder needs and generate the StRS (SR.2.1)
3. Derive system requirements with SMART criteria (SR.2.4)
4. Elaborate design constraints (boundary conditions, reuse, external functions)
5. Elaborate System Elements Requirements with ELE- prefix (SR.2.5)
6. Define traceability links (SR.2.8)
7. Obtain Work Team agreement through internal review (SR.2.2, SR.2.6)
8. Validate StRS and SyRS with Acquirer and Stakeholders (SR.2.3, SR.2.7)
9. Establish the IVV Plan and IVV Procedures (SR.2.9)
10. Update the Traceability Matrix

Step 5 (SR.2.5) addresses the elaboration of System Elements Requirements, a task that ISO 29110 assigns to the Designer (DES) and Systems Engineer (SYS). Element requirements bridge system-level requirements and the physical architecture: they specify what each system element must achieve to satisfy the system requirements it is allocated to.

The skill prompts the engineer to create element requirements in SysML 2.0 with ELE- prefixed identifiers and explicit satisfy links to system requirements.

Step 7 (SR.2.2, SR.2.6) formalises the internal verification step that ISO 29110 requires before external validation. The Work Team reviews the StRS for completeness, consistency, and feasibility, and separately reviews the system and element requirements for technical correctness.

The skill implements R1 (reduce functional information burden) by structuring the requirements engineering process as a sequence of prompted interactions. In Paper IV, the automated review pipeline constrained the space of possible requirement formulations by providing structured templates and scoring against defined quality dimensions. This skill extends that principle from review to authoring: the engineer is guided through stakeholder analysis, need identification, requirement derivation, element requirement elaboration, and trace establishment as a patterned sequence rather than an open-ended task.

Architecture Design

This skill guides the engineer through SR.3 (System Architectural Design), covering functional decomposition, physical allocation, interface analysis, trade-off documentation, and design approval. It implements R2 (designed cognitive reserve) by embedding architectural reasoning patterns that would otherwise require experienced SE judgement.

The skill addresses eight SR.3 tasks. For SR.3.5 (Design Approval), it guides a formal approval process with named approvers, a recorded decision, and baseline establishment. For SR.3.6, it prompts the engineer to update both the IVV Plan and the IVV Procedures to reflect the architectural decisions, distinguishing between the plan (what to verify, in what order) and the procedures (specific test steps and pass/fail criteria). The skill also addresses Purchase Orders (SR.3 work product) for system elements that must be procured, with a reference to the docs/pm/purchase-order.md template and VSE guidance on when procurement formality is proportionate. For SR.3.8, the skill includes an optional step for verifying and obtaining approval of the System User Manual, when the project requires preliminary user documentation at the architecture stage.

The skill draws on the observation from Cases A and B that SysML models serve as boundary objects for cross-disciplinary communication. It prompts the engineer to define part structures, ports, connections, and behaviour in SysML 2.0 notation, creating a navigable architectural representation that all stakeholders can inspect.

Verification and Validation

This skill guides the engineer through all seven SR.5 tasks (SR.5.1 through SR.5.7) in a ten-step sequence. It structures the creation of verification cases (linked to system requirements via verify relationships) and validation cases (linked to stakeholder needs). The skill draws on the INCOSE Guide to Verification and Validation, filtered for methods accessible to small teams.

For SR.5.2 (System Integration), the skill explicitly references the Supplier (SUP) role when the system includes elements provided by external parties. ISO 29110 assigns the SUP role to this task, recognising that integration requires coordinating with suppliers who provide system elements. For SR.5.5 (Defect Correction), the skill provides regression scope guidance scaled by change severity: minor changes require local retest, major changes require interface retest, and architectural changes require full system regression.

The skill includes two optional steps for the System Operation Guide (SR.5.6 and SR.5.7). When the system has operational procedures beyond what the User Manual covers, the skill prompts the engineer to document them and obtain Acquirer approval. These steps are marked as optional in both the skill and the ISO 29110 standard.

Traceability Guard

The traceability guard is a cross-cutting skill that checks the completeness of the three-tier bidirectional traceability chain:

```
Stakeholder Needs (STK-nnn)
  ↓ satisfy
System Requirements (REQ-nnn)
  ↓ satisfy
System Element Requirements (ELE-nnn)
  ↓ verify
Verification Cases (VER-nnn)

System Requirements (REQ-nnn)
  ↓ verify
Verification Cases (VER-nnn)

Stakeholder Needs (STK-nnn)
  ↓ validate
Validation Cases (VAL-nnn)
```

The skill enforces five rules. Rule 1 (Upward Traceability): every system requirement must trace upward to at least one stakeholder need. Rule 2 (Downward Traceability): every

system requirement must trace downward to at least one verification case. Rule 2a (Element Traceability): every element requirement must trace upward to a system requirement via satisfy and must be verified by at least one verification case. Rule 3 (Validation Completeness): every stakeholder need must have at least one validation case. Rule 4 (Orphan Detection): no requirement, element requirement, verification case, or validation case may exist without at least one trace link. Rule 5 (Bidirectional Consistency): if requirement A traces to verification case B, then B must reference A, and vice versa.

The three-tier chain was added to address ISO 29110 task SR.2.5, which requires System Elements Requirements to bridge system-level requirements and the physical architecture. Element requirements (ELE- prefix) specify what each system element must achieve to satisfy the system requirements allocated to it. This intermediate tier is essential for systems with multiple physical elements, because it makes the allocation decision (which element satisfies which system requirement) explicit and machine-verifiable.

When a gap is detected, the skill reports it, suggests the missing link, and does not proceed until the engineer addresses the gap or explicitly defers it with documented rationale. The traceability matrix generated by the skill now includes five columns: Stakeholder Need, System Requirement, Element Requirement, Verification Case, and Validation Case.

This behaviour directly implements R3 (machine-readable traceability). The traceability chain corresponds to the observation mapping $g(\cdot)$ in the formal model (Table 19): stronger traceability improves observation precision, reduces ambiguity, and decreases reliance on informal memory. The three-tier chain extends the observation precision to the element level, making allocation decisions visible and auditable.

SysML 2.0 Modelling

This cross-cutting skill provides authoring assistance for SysML 2.0 textual notation. It encodes the naming conventions, file organisation patterns, and traceability link syntax that the plugin requires. The skill references the SysML 2.0 specification filtered through the quick-reference knowledge file, presenting only the language constructs relevant to VSE-scale systems engineering.

Attention Regime

This meta-skill configures the environmental structures that sustain attention. It installs the pre-commit traceability hook and the phase-gate check hook, verifies the project structure, and runs health checks on template completeness (expecting 9 PM templates and 15 SR

templates). The skill implements R4 (sustain attention through environmental design) by establishing the niche construction mechanisms that the formal model identifies as essential for sustained behavioural change (Section 7.5.7).

The skill monitors two drift indicators that the lifecycle orchestrator defines. “Backward drift” detects when the engineer modifies artefacts from a completed phase without a Change Request, suggesting that upstream decisions are being revisited informally. “Artefact gap” detects when the engineer advances to a new phase while required work products from the previous phase are missing or incomplete. Both indicators map to the formal model’s prediction that attention to dependability-relevant actions decays under workload pressure unless environmental structures make drift visible (Hypothesis H9).

Project Setup

This skill bootstraps new projects with the full ISO 29110 structure. It creates the directory layout, copies 9 PM and 15 SR work product templates (each with a revision history section for configuration management), generates the ISO 29110 task checklist (TASKS.md with all 68 tasks from PM.1 through SR.6), configures the SySiDE toolchain, initialises the SysML 2.0 root package with stub files for all six model domains (stakeholder needs, system requirements, element requirements, architecture, verification, validation), and optionally sets up GitHub repository integration with CI/CD workflows.

Critically, the skill generates a project-level CLAUDE.md from the template in templates/common/CLAUDE.md. This file is the primary activation mechanism for designed cognitive reserve (R2) in the target project. It ensures that the LLM approaches every project as a VSE project first and consults the VSE plugin first, regardless of whether the engineer explicitly invokes a skill. The project-level CLAUDE.md declares the VSE-first rule, embeds the source processing order, provides the three-tier traceability chain, lists all nine ISO 29110 roles (including SUP, the Supplier role), and includes phase gate checklists. Without this file, R2 would only activate when the engineer explicitly invoked a skill, which defeats the thesis argument (Section 4.2.3) that designed reserve must be embedded in the environment rather than opt-in.

The skill combines R2 (designed cognitive reserve) and R4 (niche construction). By scaffolding the complete project structure at initialisation, it ensures that the environmental features required for sustained engagement are present from the start rather than being added incrementally under workload pressure. This addresses the ghost city pathology discussed in Section 7.5.7. Unlike greenfield environments imposed on inhabitants who have no coupling to the structure, the project setup skill creates a scaffold that the engineer

immediately populates with their own content. This establishes the feedback loop that the formal model identifies as essential for genuine niche construction.

Document Export

This skill generates non-diffable output files (docx, pptx, PDF) from the markdown work product templates. The key principle is that generated documents are outputs, not sources. The markdown files in `docs/` are the source of truth, tracked in version control, and diffable across revisions. Generated files are placed in `build/` and excluded from version control.

This separation operationalises a finding that spans all four cases: the cost of producing quality evidence should be as low as possible. When producing a work product for stakeholder review requires manual formatting in a word processor, the effort becomes a barrier that competes with engineering work for the engineer's limited attention. When the same work product can be generated from a diffable source with a single command, the barrier is removed.

Session Journal

The session journal skill manages the `.vse-journal.yml` file, providing cross-session continuity for VSE projects. It operates in three modes: resume (read the journal at session start and present a continuity block), checkpoint (write a progress entry mid-session on explicit request), and close (write a closing entry when the session ends).

The skill implements Regime R4 (Environmental Attention Sustaining) by providing temporal continuity across work sessions. Where `.vse-phase` provides spatial orientation (where in the lifecycle the project stands), the session journal provides temporal orientation (what was done last, what comes next). It also supports Regime R1 (Information Filtering) by surfacing only the most recent session's context rather than the full project history.

Each journal entry records the ISO 29110 activity codes worked on during the session, mapping conversation actions to the process backbone. This ensures that even informal work sessions produce a structured audit trail that can be traced back to the standard's activity model.

The rolling window mechanism (default 15 entries) balances accessibility with compactness. Active context stays in the project root for fast reads. Archived entries move to `docs/pm/session-archive.yml`, maintaining the complete project history for audit purposes without burdening the active journal file.

A3.3 Hooks

Hooks are automated scripts that execute at defined trigger points in the development workflow. They implement R4 (sustain attention through environmental design) by creating environmental constraints that make dependability-relevant actions the path of least resistance.

The mechanism is, in principle, identical to the smart bulb described in Case D (Section 6.4). Case D's smart bulb created a Regime of Attention by making vulnerability status ambient and unavoidable. The plugin's hooks create an analogous regime by making traceability gaps and phase-gate violations immediately visible at the point of action (the commit or phase transition) rather than at a later review.

Pre-Commit Traceability Hook

This hook runs automatically before every commit that modifies `.sysml` files. It parses the SysML 2.0 source files and verifies that:

- Every requirement def with a REQ- identifier has at least one satisfy link to a stakeholder need.
- Every requirement def with a REQ- identifier is referenced by at least one verification def via a verify link.
- Every stakeholder need (STK- identifier) has at least one validation case.

If any trace gap is detected, the commit is blocked, and the missing links are reported. The engineer can address the gap immediately, while the context is fresh, rather than discovering it during a later review.

The hook corresponds to the formal model's hard constraints K (Table 17): it prunes configurations that violate the traceability invariant, reducing the feasible set to those that maintain trace completeness. The constraint is not optional. Unlike a review recommendation that can be deferred, the commit simply does not proceed. This is a deliberate choice grounded in the formal model's prediction that process mandates imposed without environmental support will be deprioritised under workload pressure (Hypothesis H9, Theorem T5.5-3). By embedding the constraint in the commit mechanism, the hook ensures

that the low-cost cognitive strategy (committing code) and the dependability-relevant strategy (maintaining traceability) are the same action.

Phase-Gate Check Hook

This hook validates phase-gate criteria when the `.vse-phase` file is modified. Before a phase transition is accepted, the hook verifies that the prerequisites for the target phase are satisfied: required work products exist, traceability is complete, and the phase-specific checklist items in `TASKS.md` are marked as done.

The hook implements differentiated gate checks for both PM and SR activities. PM gates are progressive: PM.1 requires a Project Plan, PM.2 adds a Progress Status Record, PM.3 adds a Correction Register, and PM.4 adds a Product Acceptance Record. SR gates check phase-specific work products: SR.1 requires the Project Plan and SEMP (using a case-insensitive pattern to match on both case-sensitive and case-insensitive file systems), SR.2-to-SR.3 requires the Traceability Matrix and IVV Plan (not merely the presence of any `.sysm1` file), SR.3-to-SR.4 adds the System Design Document, and SR.4-to-SR.5 adds the System Element Requirements specification. This differentiation ensures that each gate validates the work products that ISO 29110 specifies as outputs of the completed phase, rather than applying a generic existence check.

The hook prevents the attention drift pattern described in `CLAUDE.md` Section 2: jumping between phases without completing prerequisites, skipping verification steps, or modifying baselined artefacts without a Change Request.

A3.4 Knowledge Base

The knowledge base comprises eight condensed reference files, each extracted from a primary-source document and filtered through the VSE lens. The source processing order follows the priority established in CLAUDE.md:

1. **ISO/IEC 29110** (process backbone): `iso29110-profile.md`, `iso29110-task-lists.md`
2. **PHAS-EAI framework** (design rationale): `phas-eai-framework.md`
3. **INCOSE SE Handbook** (best practices, scaled): `incose-vse-practices.md`
4. **SysML 2.0 specification** (modelling language): `sysml2-quick-ref.md`
5. **Domain guides** (technique detail): `needs-and-reqs-guide.md`, `vv-guide.md`, `hsi-primer.md`

This ordering implements R1 (reduce functional information burden). When the plugin resolves a question, it consults the process standard first (what activities to perform), then the design rationale (why those activities work), then the best practices (how to execute, scaled for VSEs), and finally the domain-specific guidance. The effect is that the engineer receives phase-appropriate, VSE-scaled guidance without needing to navigate the full breadth of the source literature.

The filtering is substantive, not merely editorial. Practices that require dedicated specialist teams, heavyweight tooling, or large-organisation overhead are either excluded or explicitly flagged as outside VSE scope. When the INCOSE Handbook recommends a technique that presupposes resources beyond what a VSE can mobilise, the knowledge file presents an adapted alternative scaled to a team where one person may fill multiple roles.

A3.5 Demonstration Project

The repository includes a complete demonstration project (`demo/smart-sensor/`) that exercises the full ISO 29110 lifecycle. The system under design is a smart temperature sensor for indoor climate monitoring. The system reads ambient temperature, applies threshold logic, and sends alerts when temperature exceeds configurable bounds.

The demonstration project includes:

- **Project-level CLAUDE.md**: generated from the plugin template, activating R2 (designed cognitive reserve) for the project. This file ensures that the LLM approaches the demo as a VSE project first, consults the plugin, reads the .vse-phase file, and applies phase-appropriate guidance.
- **5 stakeholder needs** (STK-001 through STK-005): MonitorTemperature, ReceiveAlerts, ConfigureThresholds, EasyToRead, ReliableOperation.
- **8 system requirements** (REQ-001 through REQ-008): each tracing upward to at least one stakeholder need via satisfy links.
- **8 element requirements** (ELE-001 through ELE-008): each tracing upward to a system requirement via satisfy links, specifying what each physical system element (SensingUnit, ProcessingUnit, ConfigInterface, DisplayUnit) must achieve. Element requirements include design margins (for example, ELE-001 specifies +/-0.3 C accuracy for the SensingUnit, providing margin against the +/-0.5 C system requirement in REQ-001).
- **8 verification cases** (VER-001 through VER-008): each tracing to a system requirement via verify links.
- **5 validation cases** (VAL-001 through VAL-005): each tracing to a stakeholder need.
- **System architecture** in SysML 2.0: part definitions, ports, connections, and behaviour for the sensor, processor, display, and communication components.
- **24 populated work product templates**: 9 PM templates and 15 SR templates, each with YAML frontmatter, a revision history section for configuration management, and ISO 29110 task references.
- **Full ISO 29110 task checklist** (TASKS.md): 68 tasks across PM.1 through SR.6, with role assignments and completion status.

The three-tier traceability chain is complete and bidirectional:

```
STK-001 MonitorTemperature
├── REQ-001 MeasureTemperature
│   └── ELE-001 SensingUnitAccuracy
└── VER-001 VerifyTempAccuracy
```



The element requirement tier makes allocation decisions explicit. Each element requirement specifies a tighter tolerance than the system requirement it satisfies, providing design margin. This is a standard SE practice that the plugin makes machine-verifiable: the traceability guard checks that every element requirement traces upward to a system requirement and that no element requirement is orphaned.

The demonstration project is currently at phase SR.5 (System Integration, Verification and Validation), with PM.1, SR.1 through SR.4 completed. This state allows the reader to inspect both populated upstream work products and the pending downstream activities, providing a concrete illustration of the lifecycle progression that the plugin manages.

A3.6 Mapping to Design Requirements

This section connects each plugin feature to the formal model quantities and case study evidence presented in Chapters 5 and 6. The mapping follows the structure of Table 24.

Table 38 summarises the ISO 29110 Basic Profile task coverage. Every task from PM.1.1 through SR.6.6 is addressed by at least one skill.

Table 38 ISO 29110 Basic Profile task coverage by skill.

ISO 29110 Activity	Tasks	Skill(s)	Thesis Construct
PM.1 Project Planning	PM.1.1–PM.1.19 (19 tasks)	lifecycle-orchestrator, project-setup	R2, R4
PM.2 Plan Execution	PM.2.1–PM.2.7 (7 tasks)	lifecycle-orchestrator	R4
PM.3 Assessment and Control	PM.3.1–PM.3.3 (3 tasks)	lifecycle-orchestrator	R4
PM.4 Closure	PM.4.1–PM.4.3 (3 tasks)	lifecycle-orchestrator	R4
SR.1 Initiation	SR.1.1–SR.1.4 (4 tasks)	lifecycle-orchestrator, project-setup	R2, R4
SR.2 Requirements	SR.2.1–SR.2.9 (9 tasks)	needs-and-requirements, traceability-guard	R1, R3
SR.3 Architecture	SR.3.1–SR.3.8 (8 tasks)	architecture-design	R2
SR.4 Construction	SR.4.1–SR.4.4 (4 tasks)	lifecycle-orchestrator	R2
SR.5 IVV	SR.5.1–SR.5.7 (7 tasks)	verification-validation	R3
SR.6 Delivery	SR.6.1–SR.6.6 (6 tasks)	lifecycle-orchestrator, document-export	R3, R4

The plugin also covers all 9 ISO 29110 roles defined in Table 22 of the standard: Project Manager (PJM), Systems Engineer (SYS), Designer (DES), Developer (DEV), IVV Engineer (IVV), Acquirer (ACQ), Stakeholder (STK), Work Team (WT), and Supplier (SUP). The SUP role is referenced in the verification and validation skill for SR.5.2 (System Integration), where suppliers provide system elements.

All 24 work product templates include a revision history section for configuration management, as required by ISO 29110 for work products under version control. Each template carries YAML frontmatter with a `status:` field that tracks the work product through its ISO 29110 lifecycle states (initiated, draft, reviewed, approved, baselined).

R1: Reduce Functional Information Burden

Formal model basis. Effective DE practices reduce the denominator (total optionality C) while preserving or increasing the numerator (useful variety $|Ag, \rho|$). The conditional success-share theorem (Theorem T5.7-2) provides the formal warrant: interventions that increase the success share reduce the cost of choice.

Plugin implementation. The phase-based filtering rules in CLAUDE.md Section 3 suppress information that belongs to phases other than the current one. During requirements work (SR.2), architecture trade-off guidance is hidden. During verification (SR.5), requirements elicitation techniques are not surfaced. The effect is to reduce the effective option space that the engineer must navigate at any given moment without removing options that may be needed in a later phase.

Case evidence. This mechanism extends the principle demonstrated in Case C, where the LLM review pipeline constrained the space of possible requirement formulations by providing structured templates and scoring against defined quality dimensions (Finding C3, supporting H2). The plugin generalises the constraint from requirements to all lifecycle phases.

Observable metrics. Option count proxies (the number of design alternatives by lifecycle mode), success set estimates (the fraction of configurations meeting service levels), quality scores (as demonstrated in Paper IV's weighted composite scoring).

R2: Build Designed Cognitive Reserve

Formal model basis. The formal proof (Theorem T5.3-4) establishes that the designed reserve h is a floor on cognitive headroom: $H_{cog} \geq h$. The floor holds regardless of skill level and task demand. This is practically important for thin teams in VSEs. A worst-case scenario with inexperienced, inflexible personnel under high-demand tasks yields $H_{cog} = h$ exactly. The designed reserve is the only cognitive headroom available.

Plugin implementation. The LLM, configured through CLAUDE.md and the knowledge base, acts as designed cognitive reserve. A junior engineer using the plugin should achieve a performance floor comparable to a moderately experienced systems engineer. Skills encode INCOSE best practices scaled to VSE context, and the knowledge base provides instant access to domain guidance that would otherwise require years of experience to accumulate.

The project-level CLAUDE.md (generated by @project-setup from templates/common/CLAUDE.md) is the primary activation mechanism for R2. It ensures that the designed reserve is embedded in the project environment from the moment the project is created, rather than requiring the engineer to explicitly invoke a skill. The generated file declares the VSE-first rule (consult the plugin before any other source), embeds the source processing order, provides the complete three-tier traceability chain, lists all nine ISO 29110 roles, and includes phase gate checklists. Even if the plugin is not installed, the project-level CLAUDE.md provides a minimum SE competence floor that the thesis (Section 4.2.3) argues is essential for designed reserve to function. This addresses the gap identified during the ISO 29110 audit: without a project-level CLAUDE.md, R2 was opt-in rather than embedded, and the formal model's guarantee ($H_{cog} \geq h$) could not hold for interactions where the engineer did not invoke a skill.

Case evidence. Hypothesis H7 (designed reserve outperforms experience investment) receives the strongest convergent support across the thesis. In Case A, the SysML model enabled non-experts to identify vulnerabilities. In Case B, the game substituted for security expertise. In Case C, the LLM served as a systems engineering mentor, and engineers internalised review patterns that persisted when the tool was not in use (Finding C4). In Case D, automated monitoring replaced the need for dedicated security staff.

Observable metrics. Personnel-change robustness (whether response performance degrades after team changes), tool-dependency ratios (fraction of quality assurance steps performed by automated systems versus manual review), internalisation indicators (whether quality improvements persist when the tool is not in use).

R3: Machine-Readable Traceability

Formal model basis. Machine-readable traceability improves the observation mapping $g(\cdot)$, reduces noise ω , stabilises shared expectations Φ , and reduces decision rework and integration surprises (Table 19).

Plugin implementation. SysML 2.0 textual models are the single source of truth. The traceability guard skill checks trace completeness on demand. The pre-commit hook enforces trace completeness at every commit. The traceability chain spans three tiers: stakeholder needs trace downward to system requirements via satisfy links, system requirements trace downward to element requirements via satisfy links, and both system and element requirements trace downward to verification cases via verify links. Stakeholder needs trace separately to validation cases. The three-tier chain addresses ISO 29110 task SR.2.5 (System Elements Requirements), which requires that element-level specifications bridge system requirements and the physical architecture. The element

requirement tier makes allocation decisions (which physical element satisfies which system requirement) explicit and machine-verifiable.

The traceability guard enforces five rules: upward traceability (Rule 1), downward traceability (Rule 2), element traceability (Rule 2a), validation completeness (Rule 3), orphan detection (Rule 4), and bidirectional consistency (Rule 5). The models are plain-text files stored in Git, enabling standard version control workflows (branching, merging, diffing, pull request review) that would not be possible with binary model formats.

Case evidence. Hypothesis H14 (machine-readable traceability reduces noise) is supported by all four cases. In Case A, the SysML model linked the valve manufacturer, firmware subcontractor, and municipal client across organisational boundaries. In Case B, the model enabled cross-disciplinary interface discussion. In Case C, the LLM pipeline linked requirements to INCOSE standards with machine-readable scores. In Case D, Dependabot provided timestamped, structured vulnerability data over a two-year period.

Observable metrics. Detection-to-decision time, integration surprise rate, decision rework rate.

R4: Sustain Attention Through Environmental Design

Formal model basis. The formal model predicts that process mandates imposed on resource-constrained organisations will be progressively deprioritised under workload pressure unless the signals they require attending to are made salient through environmental design (Hypothesis H9, Theorem T5.5-3).

Plugin implementation. The two hooks implement niche construction: the environment blocks commits with broken traces (pre-commit hook) and blocks phase transitions without completed prerequisites (phase-gate hook). The phase-gate hook implements differentiated gate checks for each PM and SR activity, verifying that the specific work products required by ISO 29110 exist before the transition is permitted. PM gates are progressive (each gate adds to the previous gate's requirements), and SR gates check phase-specific outputs including the SEMP (SR.1), Traceability Matrix and IVV Plan (SR.2), System Design Document (SR.3), and System Element Requirements (SR.4). The lifecycle orchestrator implements attention anchoring by reporting the current phase, inputs, and next action at the start of every interaction. Patterned Practice prompts (CLAUDE.md Section 2) ask phase-appropriate questions before the engineer creates work products. The attention regime skill monitors two drift indicators: "backward drift" (modifying completed-phase artefacts without a Change Request) and "artefact gap" (advancing phases while required work products are missing).

Case evidence. Hypothesis H9 is supported by Cases B, C, and D, each of which created structured attention-directing mechanisms through different modalities. Case B used time-boxed game sessions. Case C used an automated review triggered at the moment of drafting. Case D used an ambient visual signal (the smart bulb). In each case, improvement was sustained not through compliance mandates but through designed salience. The plugin follows the same principle: it makes the dependability-relevant action (maintaining traceability, completing phase prerequisites) the path of least resistance.

Observable metrics. Drift indicators (whether attention to quality attributes decays over time without external prompting), engagement persistence (the duration over which the intervention sustains its effect without reinforcement), remediation rate.

A3.7 Relationship to the Thesis

The plugin occupies a specific position in the thesis contribution structure. It is not a case study, and it does not provide new empirical evidence for or against the fourteen hypotheses. It is a working tool: a concrete implementation of the design requirements that the formal model derives and the case studies validate.

The relationship between the formal model and the plugin mirrors the relationship between Newton's laws and standardised beam sections, as discussed in Section 7.5.2. The formal model provides the theoretical warrant for why the design works, identifies the boundary conditions under which it will cease to be sufficient, and guides future recalibration. The plugin provides a lookup table: a set of ready-to-use environmental structures that encode the model's consequences in a form any VSE practitioner can adopt without engaging with the formal apparatus.

Section 7.5.3 establishes the complementarity between the PHAS-EAI framework and ISO/IEC 29110. The standard specifies what activities a VSE should perform. The framework explains why certain environmental and social structures sustain the attention, learning, and coordination that those activities require. The plugin implements both layers. The ISO 29110 layer provides the process skeleton: phases, activities, work products, and roles. The PHAS-EAI layer provides the environmental design: Regimes of Attention (hooks and phase-gate checks), Patterned Practices (phase-appropriate prompts and structured interaction sequences), designed cognitive reserve (the LLM configured with SE knowledge), and niche construction (the project scaffolding and automated constraints).

The distinction matters because, as the simulation results in Appendix A2 demonstrate, individual productivity tools and collective coordination structures can work at cross-purposes

when deployed without alignment (Section A2.3). ISO 29110 provides the alignment layer: the shared goals and structured work products that ensure the plugin's environmental scaffolding amplifies cooperation rather than individual productivity at the expense of collective coherence.

A3.8 Limitations and Future Work

The plugin inherits the limitations discussed in Section 7.6. Most significantly, it has not been evaluated in an independent VSE context. The design requirements it implements are grounded in four case studies conducted at a single Norwegian company (Aiwell AS), and the formal model predicts, but does not empirically confirm, that the findings transfer to other VSE contexts with similar resource constraints.

Three specific limitations warrant attention:

6. **LLM dependence.** The plugin's cognitive reserve is provided by a proprietary foundation model (Claude). The quality of guidance depends on the model's capabilities, which may vary across versions. The knowledge base mitigates this dependency by providing structured reference material that constrains the model's responses, but the constraint is not absolute. Future work should investigate whether the plugin's effectiveness varies significantly across model versions and providers.
7. **SysML 2.0 tooling maturity.** The SysML 2.0 textual notation is supported by the Sensmetry SySiDE toolchain, which provides syntax validation and navigation. However, the SysML 2.0 standard is relatively new, and tooling is still maturing. The plugin's traceability mechanisms depend on consistent use of the notation conventions defined in CLAUDE.md Section 6. Deviations from these conventions may cause the pre-commit hook to miss trace gaps or report false positives.
8. **Single-engineer scope.** The plugin is designed for the common VSE scenario in which one person fills multiple roles. It does not currently address multi-team coordination, which would require extending the authority-weight mechanisms (γ_i) described in Section 5.5. The formal model's prediction that unresolved authority-weight asymmetry produces coordination friction (Hypothesis H10) suggests that multi-team extensions would need to address this dimension explicitly.

The third direction in Section 7.7 (Future Work) is directly relevant: the scalability boundary of the designed cognitive reserve. The plugin represents one data point in the investigation of how the effectiveness of cognitive scaffolding tools varies with task complexity and whether

periodic recalibration (updating knowledge files, adjusting skill prompts) can extend the effective range. The rapid evolution of large language models makes this question particularly timely. The boundary conditions observed during Case C (2023) may already have shifted, and the plugin provides a structured framework for observing and measuring such shifts.

A4 Data Management Plan

This appendix reproduces the Data Management Plan (DMP) submitted to the Research Council of Norway for project 311242 (Industrial PhD Scheme). The original plan was written in Norwegian (bokmål). This version is an English translation, expanded with additional detail reflecting the plan's implementation over the six-year project period (2020 to 2026). Section numbering follows the structure recommended by the Research Council.

A4.0 Administrative Information

This document describes how research data in the doctoral project “Dependable Smart City Infrastructure Built by Very Small Entities” are managed. The project is conducted by an employee of Aiwel AS who is simultaneously enrolled as a doctoral candidate at the University of South-Eastern Norway (USN). Aiwel owns the raw data. The candidate holds a contractual right to analyse the data and to report research results in academic publications. The project is co-funded by Aiwel (50%) with support from the Research Council of Norway and is classified as part of the company's internal development activities under the terms of the Industrial PhD Scheme.

Principal investigator and data steward: Roar Elias Georgsen (doctoral candidate; Systems Engineer at Aiwel AS from project start in 2020, Chief Technology Officer from June 2022).

University supervisors: Professor Geir Myrdahl Køien and Professor Cecilia Haskins, University of South-Eastern Norway.

IT responsible: Aiwel AS IT administration.

A4.1 Purpose and Method of Data Collection

The project studies how communication, decision-making processes, and information flows operate within a very small engineering organisation, and how these dynamics affect innovation, learning, and the dependability of the systems the organisation delivers. The research design combines formal analytical modelling with participatory qualitative case studies conducted using Industry-as-Laboratory, Action Research, and Grounded Theory Method (see Chapter 4 of the thesis for a full account of the methodology).

Data are collected through the following sources:

- **Internal email:** Email exchanges among employees that are relevant to the research questions, selected through theoretical sampling.
- **Chat logs and messaging systems:** Selected conversations from Microsoft Teams and similar platforms used within the company.
- **Meeting notes:** Records from project meetings and informal meetings that document decisions, collaborative problem-solving, and engineering rationale.
- **Surveys and feedback:** Short questionnaires administered to employees concerning their experiences of communication, tool use, and process change.
- **Engineering artefacts:** Commit logs, pull request comments, issue tracker records, SysML model files, requirements documents, and continuous integration pipeline outputs generated during normal engineering work. These digital artefacts constitute the primary evidence base for Cases A through D described in the thesis.
- **Observation notes:** Field notes recorded by the embedded researcher during engineering activities, game sessions, and review meetings.

Data are collected with informed consent from all participants. All employees are informed of the study's purpose, how data will be processed, and that anonymised findings may be used in the doctoral thesis and in academic publications. Data collection follows theoretical sampling: documents and conversations are selected iteratively to illuminate the research questions rather than collected exhaustively. During collection, each item is logged with date, source, context, and any relevant comments to ensure traceability.

A4.2 Description of Data and File Formats

The dataset consists primarily of text-based information. The following file formats are used:

- Email and messages are stored as .eml files or exported to plain-text .txt files for analysis.
- Meeting notes are stored as .docx or .md files. When transferred to the analysis environment, they are converted to plain text.

- Surveys are stored in .csv format for import into analysis tools.
- Engineering artefacts (code, configuration files, pipeline definitions) are stored in their native formats within Git repositories. Repository metadata (commit hashes, timestamps, authorship) are extracted as .csv or .json for analysis.
- SysML model files are stored in their native tool format (.mdzip for Cameo Systems Modeller) and as exported .xml for archival purposes.
- The codebook and metadata catalogue are maintained as .xlsx or .csv files.

The total dataset is estimated at several hundred documents occupying less than 1 GB of storage. Open formats are used wherever possible to ensure long-term accessibility. Conversion to open formats (for example, .txt and .csv) is performed when doing so does not conflict with preservation requirements.

A4.3 Documentation, Organisation, and Metadata

To ensure that the dataset is comprehensible to others within the company and to future researchers, all data are documented through the following mechanisms:

- **Metadata file:** For each data item, the following attributes are recorded in a separate metadata file (CSV): title, date, data source, brief description, and anonymisation status. The metadata schema follows a simple standard that makes items easier to locate and identify.
- **File structure and naming conventions:** The project uses a hierarchical folder structure (for example, /source/year-month/documentname) and consistent file names (for example, 2025-01-15_meeting_notes_aiwell.md). Version control (Git) is used for code, analysis scripts, and evolving analytical documents.
- **Codebook:** All themes and codes used in the Grounded Theory analysis are described in a codebook containing definitions, exemplary data excerpts, and a log of any changes to coding categories over time. The codebook is versioned and updated at each coding iteration.
- **Method notes:** A README.md file describes the data collection process, anonymisation procedures, and any deviations from the planned protocol. This file is updated continuously throughout the project.

Quality assurance is performed through regular internal review of transcriptions and notes. When data are exported automatically from email or messaging systems, the export is checked for completeness and accuracy. Any discrepancies are logged to maintain transparency about data quality.

A4.4 Storage, Archiving, Backup, and Sharing

All data are stored on Aiwell's SharePoint environment, which provides access control and regular automated backup. Data reside in an encrypted storage area. The following procedures apply:

- **Primary storage:** Raw data and working files are stored in internal folders on the company server. Access is restricted to the doctoral candidate, the university supervisors, and selected individuals with a documented legitimate need.
- **Backup:** The server performs daily automated backups. In addition, weekly offline copies are taken and stored in a separate secured environment. This reduces the risk of data loss from hardware failure, ransomware, or accidental deletion.
- **Restoration:** In the event of data loss or corruption, the most recent valid copy can be restored using the company's established IT procedures. The project periodically tests that backup restoration functions correctly.
- **Archiving:** Upon project completion, anonymised data and associated documentation are archived in an internal repository. The retention period is a minimum of ten years, or longer if the company deems it necessary. Data that no longer have research value are securely destroyed using methods that prevent recovery.
- **Sharing:** Due to confidentiality obligations and the need to protect personal privacy, raw data and transcriptions are not shared externally. The principle "as open as possible, as closed as necessary" is followed. Aggregate data and analysis results are anonymised before use in publications or presentations. Published papers are made available in accordance with the open-access policies of the respective journals and conferences.

A4.5 Information Security, Privacy, and Ethics

Internal communication may contain personal data and business secrets. The project therefore implements the following safeguards:

- **Informed consent:** All employees are informed about the research and give written consent for their internal communications to be used for research purposes. They may withdraw at any time without negative consequences. The consent process is overseen by the university supervisors to mitigate the power asymmetry inherent in the researcher's dual role as employee and candidate.
- **Anonymisation:** In transcriptions and subsequent analysis, names, email addresses, and other identifiable information are removed or replaced with pseudonyms. Analytical codes reference anonymised identifiers only. The anonymisation procedure is documented in the project's method notes and is applied before any data leave the primary storage environment.
- **GDPR and the Norwegian Personal Data Act:** The processing of personal data complies with the European Union General Data Protection Regulation (GDPR) and the Norwegian Personal Data Act (personopplysningsloven). The purpose of data processing is academic research, and the legal basis is informed consent (GDPR Article 6(1)(a)). The project was assessed and found not to require formal notification to or approval from the Norwegian Centre for Research Data (NSD, now Sikt).
- **Ethical considerations:** The researcher's role entails a responsibility to protect participants from harm. No results are published in a manner that makes individuals identifiable. Aiwel reviews all material submitted for publication to verify that no sensitive business information or personally identifiable information is disclosed. The data collection procedures are reported to the institutional data protection officer as required.
- **Dual-role management:** The researcher's simultaneous roles as employee (Systems Engineer, later CTO) and doctoral candidate create potential conflicts of interest. These are managed through supervisory oversight, transparent documentation of research decisions, and collaborative design of interventions (Action Research methodology), as described in Section 4.3.5 of the thesis.

A4.6 Rights, Ownership, and Licences

Aiwell owns the raw data derived from internal communications and engineering activities. The doctoral candidate holds a contractual right to analyse the data for academic purposes. Results published in journal articles, conference papers, or the doctoral thesis are the candidate's academic work, but they are based on company-owned data. No licences are granted for the reuse of raw data, as the datasets may contain personal data and business secrets. Written agreements specifying rights and obligations between the parties are maintained on file. Published results are made available in accordance with open-access publishing guidelines.

A4.7 Costs, Roles, and Responsibilities

The project uses existing infrastructure at Aiwell. Costs associated with data storage and backup are covered by the company. The division of responsibilities is as follows:

- **Doctoral candidate / project leader:** Responsible for data collection, anonymisation, documentation, and maintenance of the Data Management Plan. Ensures that research is conducted in compliance with privacy regulations and research ethics norms.
- **University supervisors:** Provide methodological guidance and verify that data processing meets research ethics standards. Oversee the informed consent process.
- **IT responsible at Aiwell:** Responsible for the secure operation of the server, backup systems, and access control. Reports any security incidents.

The project does not anticipate extraordinary costs for data management, but allocates time for anonymisation and documentation. Any costs associated with long-term archiving are covered by Aiwell.

A4.8 Long-Term Preservation and Reuse

Data are evaluated according to their research value and legal requirements. Data deemed necessary for verification or future reference are anonymised and stored for a minimum of ten years in Aiwell's internal archive. Data containing personal information without long-term research value are deleted after project completion. Because the data consist of confidential internal communications and proprietary engineering artefacts, they are not made available

for reuse outside the company. If parts of the dataset are later considered suitable for use in other internal research projects, a new ethical assessment will be conducted and new consent obtained as required.

A4.9 Plan Maintenance

This Data Management Plan is a living document. The first version was prepared at project start. The plan is updated upon major changes (for example, when new data sources are introduced or anonymisation procedures are revised) and at least once per year. A final updated version is appended to the doctoral thesis. The plan is stored alongside the dataset so that future internal researchers can understand how data were collected, processed, and stored.